



737

Quick Reference Handbook

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Normal Checklists

Chapter NC

PREFLIGHT

Oxygen..... Tested, 100%

Navigation transfer
and display switches NORMAL, AUTO

Window heatON

Pressurization mode
selector AUTO

Flight instruments Heading ____, Altimeter ____

Parking brakeSet

Engine start levers CUTOFF

BEFORE START

Flight deck door Closed and locked
(VP-BPI, VP-BPY) Do not use

Fuel __ LBS, Pumps ON
RA-73269 - RA-73321

Fuel __ KGS, Pumps ON

Passenger signs __

Windows Locked

MCP V2 ____, HEADING ____, ALTITUDE ____

Takeoff speeds V1 ____, VR ____, V2 ____

CDU preflight..... Completed

Rudder and aileron trim Free and 0

Taxi and takeoff briefing Completed

Anti-collision lightON



BEFORE TAXI

Generators On
Probe heat ON
Anti-ice ____
Isolation valve AUTO
RA-73269 - RA-73321
Engine start switches CONT
Recall Checked
Autobrake RTO
Engine start levers IDLE detent
Flight controls Checked
Ground equipment Clear

BEFORE TAKEOFF

Flaps ____, Green light
Stabilizer trim __ Units

AFTER TAKEOFF

Engine bleeds ON
Packs AUTO
Landing gear UP and OFF
Flaps UP, No lights



737 Flight Crew Operations Manual

DESCENT

Pressurization **LAND ALT** ____
Recall **Checked**
Autobrake ____
Landing data **VREF** ____, **Minimums** ____
Approach briefing **Completed**

APPROACH

Altimeters ____

LANDING

RA-73269 - RA-73321
Engine start switches **CONT** |
Speedbrake **ARMED**
Landing gear **Down**
Flaps ____, **Green light**

SHUTDOWN

Fuel pumps **OFF**
RA-73311 - RA-73321
(SB Changes RA-73269)
Probe heat **AUTO**
VP-BZV, VP-BZY
(SB Changes RA-73269)
Probe heat **OFF**
Hydraulic panel **Set**
Flaps **UP**
Parking brake ____
Engine start levers **CUTOFF**
Weather radar **Off**



SECURE

- IRSs OFF**
- Emergency exit lights OFF**
- Window heat OFF**
- Packs OFF**



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Emergency Descent.....0.1



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Intentionally
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Emergency Descent

Condition: One or more of these occur:

- Cabin altitude cannot be controlled
- A rapid descent is needed.

- 1 Announce the emergency descent. The pilot flying will advise the cabin crew, on the PA system, of impending rapid descent. The pilot monitoring will advise ATC and obtain the area altimeter setting.
- 2 Passenger signs ON
- 3 **Without delay**, descend to the lowest safe altitude or 10,000 feet, whichever is higher.
- 4 ENGINE START switches (both) CONT
- 5 Thrust levers (both) Reduce thrust to minimum or as needed for anti-ice
- 6 Speedbrake FLIGHT DETENT

If structural integrity is in doubt, limit speed as much as possible and avoid high maneuvering loads.

- 7  Set target speed to Mmo/Vmo.

▼ Continued on next page ▼

(VP-BZU - VP-BZY) Do not use

Caution! When gross weight is greater than 77,112 kgs, speed brake will autostow to the 50% flight detent if airspeed exceeds 320 knots. Do not override autostow function unless airspeed is less than 320 knots.

8 **When** approaching the level off altitude:

Smoothly lower the SPEED BRAKE lever to the DOWN detent and level off. Add thrust and stabilize on altitude and airspeed.

9 Crew oxygen regulators. Normal

Flight crew must use oxygen when cabin altitude is above 10,000 feet. To conserve oxygen, move the regulator to Normal.

10 ENGINE START switches (both)As needed

11 The new course of action is based on weather, oxygen, fuel remaining and available airports. Use of long range cruise may be needed.



Ditching

Condition: Airplane ditching and evacuation are needed.

- 1 Send distress signals. Determine position, course, speed, altitude, situation, intention, time and position of intended touchdown and transmit mayday. Report type of aircraft and request intercept.
- 2 Alert the cabin crew to prepare for ditching and seat passengers as far forward as possible.
(VQ-BUV) Do not use
- 3 Alert supernumeraries to prepare for ditching.
- 4 Burn off fuel to reduce touchdown speed and increase buoyancy.
- 5 Plan to touch down on the windward side and parallel to waves and swells.
- 6 Plan a flaps 40 landing unless another configuration is needed.
- 7 Set VREF 40.
- 8 Do **not** arm the autobrake.
- 9 Do **not** accomplish the normal landing checklist.
- 10 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

Pressurization. LAND ALT ____

Recall Checked

▼ Continued on next page ▼



▼ Ditching continued ▼

- AutobrakeOFF
- Landing data VREF 40 ____
- Approach briefing Completed

Approach Checklist

- Altimeters ____

Below 5000 feet

- LANDING GEAR AURAL WARN
circuit breaker (P6-3:D18) Pull
- This prevents the warning horn with gear retracted and landing flaps selected.
- The flight deck chime for an incoming call from the cabin crew is unavailable.

- (VQ-BUV) Do not use
LANDING GEAR AURAL WARN
circuit breaker (P6-3:D18) Pull
- This prevents the warning horn with gear retracted and landing flaps selected.

- Passenger signsON
- Engine BLEED air switches (both) OFF
- This allows the airplane to be depressurized with the outflow valve closed.
- Pressurization mode selectorMAN

▼ Continued on next page ▼

▼ Ditching continued ▼

Outflow VALVE switch Hold in CLOSE
until the outflow VALVE
indication shows fully closed

This prevents water from entering the airplane.

Note: The outflow valve takes up to 20 seconds to close.

APU switch OFF

GROUND PROXIMITY GEAR

INHIBIT switch GEAR INHIBIT

GROUND PROXIMITY TERR

INHIBIT switch TERR INHIBIT

Life vests, shoulder harnesses and seat belts . . . On

Confirm that passenger cabin preparations are complete.

Caution! Do not open aft entry or service doors as they may be partially submerged.

Transmit all pertinent information regarding final ditching position.

▼ Continued on next page ▼



▼ Ditching continued ▼

After Impact Procedure Review

Set both engine start levers to CUTOFF. This closes fuel shutoff valves to prevent discharge of fuel from ruptured fuel lines.

Open flight deck windows. This ensures no cabin differential pressure prevents the opening of the doors or emergency exits.

Start the evacuation.

Proceed to assigned ditching stations, launch rafts and evacuate the airplane as soon as practicable.

The airplane may stay afloat indefinitely if fuel load is minimal and no serious damage was sustained during landing.

Ditching Final

LANDING GEAR lever. UP and OFF
Flaps ____, Green light

At **500 feet**, advise the cabin crew that ditching is imminent.

At **50 feet**, advise the cabin crew to brace for impact.

Maintain airspeed at VREF. Flare the airplane to achieve the minimum rate of descent at touchdown. Maintain 200-300 fpm rate of descent until the start of the flare.

▼ Continued on next page ▼

▼ Ditching continued ▼

At flare, rotate smoothly to a touchdown attitude of 10-12°. Maintain airspeed and rate of descent with thrust.

At touchdown, reduce thrust to idle.



Intentionally
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Non-Normal Checklists

Chapter NNC

Airplane Gen., Emer. Equip., Doors, Windows

Section 1

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Intentionally
Blank

**AIR
STAIR**
AIRSTAIR
RA-73269 - RA-73312, RA-7321

Condition: The airstair door is not secure.

1 Choose one:

◆ Pressurization is **normal**:

The door is in a safe configuration.
Continue normal operation.


◆ Pressurization is **not** normal:

▶▶ **Go to step 2**

2 Don oxygen masks.

3 Establish crew communications.

4 Passenger signs ON

5 Choose one:

◆ Airplane **has** reached the planned cruise altitude:

▶▶ **Go to step 6**

◆ Airplane has **not** reached the planned cruise altitude:

Do **not** continue the climb.

Reset the FLT ALT indicator to the actual
airplane altitude.

▶▶ **Go to step 6**

6 LAND ALT indicator 9,000 feet

▼ Continued on next page ▼

▼ AIRSTAIR continued ▼

7 Choose one:

◆ Minimum safe altitude is **at or below 9000 feet:**

▶▶ **Go to step 8**

◆ Minimum safe altitude is **between 9000 feet and 13,000 feet:**

▶▶ **Go to step 10**

◆ Minimum safe altitude is **at or above 13,000 feet:**

▶▶ **Go to step 12**

8 Descend to 9000 feet.

9 Maintain a cabin differential pressure of 0 psi by limiting flight altitude to 9000 feet.

▶▶ **Go to step 15**

10 Descend to the minimum safe altitude.

11 LAND ALT indicator Select a higher altitude (maximum 13,000 feet) to maintain a cabin differential pressure of 0 psi

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

▶▶ **Go to step 15**

12 Descend to the minimum safe altitude.

▼ Continued on next page ▼

▼ AIRSTAIR continued ▼

13 Pressurization mode selector MAN

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

14  Outflow VALVE switch Move to OPEN until the outflow VALVE indication shows fully open to depressurize the airplane

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

15 Plan to land at the nearest suitable airport.

16 **When** the cabin altitude is at or below 10,000 feet:
Oxygen masks may be removed.



AUTO
UNLK

AUTOMATIC UNLOCK

Condition: The correct emergency access code is entered.

Objective: To deny unauthorized access to the flight deck before the door automatically unlocks.

1 FLT DK DOOR lock selector Rotate to DENY and hold for 1 second



CARGO DOOR

**FWD
CARGO**

**AFT
CARGO**

Condition: One or more cargo doors are not closed and secure.

1 Choose one:

◆ Pressurization is **normal**:

The door is in a safe configuration.
Continue normal operation.



◆ Pressurization is **not** normal:

►► **Go to step 2**

2 Don oxygen masks.

3 Establish crew communications.

4 Passenger signsON

5 Choose one:

◆ Airplane **has** reached the planned cruise altitude:

►► **Go to step 6**

◆ Airplane has **not** reached the planned cruise altitude:

Do **not** continue the climb.

Reset the FLT ALT indicator to the actual
airplane altitude.

►► **Go to step 6**

▼ Continued on next page ▼

▼ CARGO DOOR continued ▼

6 LAND ALT indicator 9,000 feet

7 Choose one:

◆ Minimum safe altitude is **at or below 9000 feet:**

▶▶ **Go to step 8**

◆ Minimum safe altitude is **between 9000 feet and 13,000 feet:**

▶▶ **Go to step 10**

◆ Minimum safe altitude is **at or above 13,000 feet:**

▶▶ **Go to step 12**

8 Descend to 9000 feet.

9 Maintain a cabin differential pressure of 0 psi by limiting flight altitude to 9000 feet.

▶▶ **Go to step 15**

10 Descend to the minimum safe altitude.

11 LAND ALT indicator Select a higher altitude (maximum 13,000 feet) to maintain a cabin differential pressure of 0 psi

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

▶▶ **Go to step 15**

▼ Continued on next page ▼

▼ **CARGO DOOR** continued ▼

12 Descend to the minimum safe altitude.

13 Pressurization mode selector MAN

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

14  Outflow VALVE switch. . . . Move to OPEN until the outflow VALVE indication shows fully open to depressurize the airplane

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

15 Plan to land at the nearest suitable airport.

16 **When** the cabin altitude is at or below 10,000 feet:
Oxygen masks may be removed.



Condition: The emergency locator transmitter is on.



**A
R
M
E
D**

EMERGENCY EXIT LIGHTS NOT ARMED

Condition: The emergency exit lights switch is not ARMED.

1 Choose one:

- ◆ EMER EXIT LIGHTS switch is **ON**:

Individual emergency exit light batteries supply a minimum of 10 minutes of lighting.



- ◆ EMER EXIT LIGHTS switch is **OFF**:

Emergency lighting is not available.



ENTRY DOOR**FWD
ENTRY****AFT
ENTRY**

Condition: One or more entry doors are not closed and secure.

- 1 Instruct the cabin crew to verify that the door handle is in the closed position or to move the handle to the closed position if possible.
- 2 Choose one:
 - ◆ Handle is in the **closed** position:
 - ▶▶ **Go to step 3**
 - ◆ Handle is **not** in the closed position:
 - Plan to land at the nearest suitable airport.
 - ■ ■ ■
- 3 Choose one:
 - ◆ Pressurization is **normal**:
 - The door is in a safe configuration.
 - Continue normal operation.
 - ■ ■ ■
 - ◆ Pressurization is **not** normal:
 - Plan to land at the nearest suitable airport.
 - ■ ■ ■

ENTRY DOOR**(VQ-BUV) Do not use****FWD
ENTRY**

Condition: Entry door is not closed and secure.

1 Choose one:

◆ Pressurization is **normal**:

The door is in a safe configuration.
Continue normal operation.

◆ Pressurization is **not** normal:

Plan to land at the nearest suitable airport.



EQUIP

EQUIPMENT DOOR

Condition: The equipment door is not closed and secure.

1 Choose one:

◆ Pressurization is **normal**:

The door is in a safe configuration.
Continue normal operation.



◆ Pressurization is **not** normal:

▶▶ **Go to step 2**

2 Don oxygen masks.

3 Establish crew communications.

4 Passenger signsON

5 Choose one:

◆ Airplane **has** reached the planned cruise altitude:

▶▶ **Go to step 6**

◆ Airplane has **not** reached the planned cruise altitude:

Do **not** continue the climb.

Reset the FLT ALT indicator to the actual airplane altitude.

▶▶ **Go to step 6**

6 LAND ALT indicator9,000 feet

▼ Continued on next page ▼

▼ EQUIPMENT DOOR continued ▼

7 Choose one:

◆ Minimum safe altitude is **at or below 9000 feet:**

▶▶ **Go to step 8**

◆ Minimum safe altitude is **between 9000 feet and 13,000 feet:**

▶▶ **Go to step 10**

◆ Minimum safe altitude is **at or above 13,000 feet:**

▶▶ **Go to step 12**

8 Descend to 9000 feet.

9 Maintain a cabin differential pressure of 0 psi by limiting flight altitude to 9000 feet.

▶▶ **Go to step 15**

10 Descend to the minimum safe altitude.

11 LAND ALT indicator Select a higher altitude (maximum 13,000 feet) to maintain a cabin differential pressure of 0 psi

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

▶▶ **Go to step 15**

12 Descend to the minimum safe altitude.

▼ Continued on next page ▼



▼ EQUIPMENT DOOR continued ▼

13 Pressurization mode selectorMAN

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

14  Outflow VALVE switch. Move to OPEN until the outflow VALVE indication shows fully open to depressurize the airplane

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

15 Plan to land at the nearest suitable airport.

16 **When** the cabin altitude is at or below 10,000 feet:
Oxygen masks may be removed.



**LOCK
FAIL**
LOCK FAIL

Condition: One or more of these occur:

- The FLIGHT DECK ACCESS SYSTEM switch is OFF
- The lock is failed.

Objective: To remove electrical power from the lock to prevent a possible overheat.

Do **if** conditions allow a crew member to leave the seat.

- 1  FLIGHT DECK ACCESS SYSTEM switch . . . OFF

Note: The door can be locked with the dead bolt.



**MAIN
CARGO**

MAIN CARGO DOOR

(VQ-BUV) Do not use

Condition: The main cargo door is not closed and secure.

1 Choose one:

◆ MAIN CARGO light is illuminated on **takeoff**:
PACK switches (both) OFF
Plan to land at the nearest suitable airport.
■ ■ ■ ■

◆ MAIN CARGO light is illuminated at a time **other than takeoff**:
▶▶ **Go to step 2**

- 2 Don oxygen masks.
- 3 Establish crew communications.
- 4 Passenger signs ON
- 5 Choose one:

◆ Airplane **has** reached the planned cruise altitude:
▶▶ **Go to step 6**

◆ Airplane has **not** reached the planned cruise altitude:
Do **not** continue the climb.
Reset the FLT ALT indicator to the actual airplane altitude.
▶▶ **Go to step 6**

▼ MAIN CARGO DOOR continued ▼

6 LAND ALT indicator9,000 feet

7 Choose one:

◆ Minimum safe altitude is **at or below 9000 feet:**

▶▶ **Go to step 8**

◆ Minimum safe altitude is **between 9000 feet and 13,000 feet:**

▶▶ **Go to step 10**

◆ Minimum safe altitude is **at or above 13,000 feet:**

▶▶ **Go to step 12**

8 Descend to 9000 feet.

9 Maintain a cabin differential pressure of 0 psi by limiting flight altitude to 9000 feet.

▶▶ **Go to step 15**

10 Descend to the minimum safe altitude.

11 LAND ALT indicatorSelect a higher altitude (maximum 13,000 feet) to maintain a cabin differential pressure of 0 psi

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

▶▶ **Go to step 15**

▼ Continued on next page ▼



▼ MAIN CARGO DOOR continued ▼

12 Descend to the minimum safe altitude.

13 Pressurization mode selector MAN

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

14  Outflow VALVE switch. . . . Move to OPEN until the outflow VALVE indication shows fully open to depressurize the airplane

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

15 Plan to land at the nearest suitable airport.

16 **When** the cabin altitude is at or below 10,000 feet:
Oxygen masks may be removed.



MID EXIT DOOR**(VP-BZU - VP-BZY) Do not use****LEFT MID
EXIT****RIGHT MID
EXIT**

Condition: One or more mid exit doors are not closed and secure.

1 Choose one:**◆ Pressurization is **normal**:**

The door is in a safe configuration.
Continue normal operation.

**◆ Pressurization is **not** normal:**

Plan to land at the nearest suitable airport.





OVERWING DOOR

- LEFT FWD OVERWING
- RIGHT FWD OVERWING
- LEFT AFT OVERWING
- RIGHT AFT OVERWING

Condition: One or more overwing doors are not closed and secure.

1 Choose one:

◆ Pressurization is **normal**:

The door is in a safe configuration.
Continue normal operation.



◆ Pressurization is **not** normal:

Plan to land at the nearest suitable airport.



- PASS OXY ON
- PASSENGER OXYGEN ON**

Condition: The passenger oxygen system is on.



- PASS OXY ON
- PASSENGER OXYGEN ON**

(VQ-BUV) Do not use

Condition: The DON OXYGEN sign and supernumerary horn are on.



SERVICE DOOR

**FWD
SERVICE**

**AFT
SERVICE**

Condition: One or more service doors are not closed and secure.

- 1 Instruct the cabin crew to verify that the door handle is in the closed position or to move the handle to the closed position if possible.
- 2 Choose one:
 - ◆ Handle is in the **closed** position:

▶▶ **Go to step 3**
 - ◆ Handle is **not** in the closed position:

Plan to land at the nearest suitable airport.

■ ■ ■ ■
- 3 Choose one:
 - ◆ Pressurization is **normal**:

The door is in a safe configuration.
Continue normal operation.

■ ■ ■ ■
 - ◆ Pressurization is **not** normal:

Plan to land at the nearest suitable airport.

■ ■ ■ ■

SERVICE DOOR**(VQ-BUV) Do not use****FWD
SERVICE**

Condition: Service door is not closed and secure.

1 Choose one:

◆ Pressurization is **normal**:

The door is in a safe configuration.
Continue normal operation.

◆ Pressurization is **not** normal:

Plan to land at the nearest suitable airport.





Intentionally
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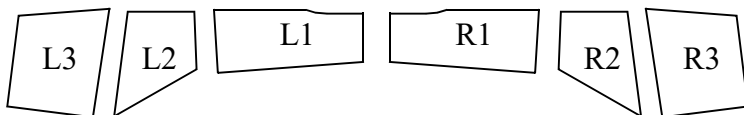
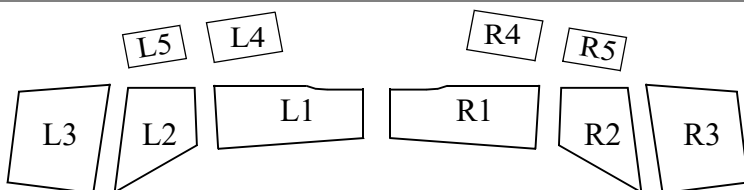
Window Damage - Forward (L1, L2, R1, R2)

Condition: A forward flight deck window has one or more of these:

- An electrical arc
- A delamination
- A crack
- Is shattered.

Objective: To remove electrical power, if needed, to prevent arcing. To reduce differential pressure and descend if the inner pane is shattered or cracked.

(VP-BPI, VP-BPY) Do not use



▼ Continued on next page ▼

▼ Window Damage - Forward (L1, L2, R1, R2) continued ▼

1 Choose one:

◆ Window is **delaminated** only:

Continue normal operation.


◆ Window is **arcing**:

▶▶ **Go to step 2**

◆ Window is **cracked** or **shattered**:

▶▶ **Go to step 5**

2 WINDOW HEAT switch
(affected window). OFF

Limit airspeed to 250 knots maximum below
10,000 feet.

3 Pull both WINDSHIELD AIR controls. This vents
conditioned air to the inside of the windshield for
defogging.

4 Continue normal operation.



5 Don seat belts and shoulder harnesses.

6 WINDOW HEAT switch
(affected window). OFF

Limit airspeed to 250 knots maximum below
10,000 feet.

▼ Continued on next page ▼

▼Window Damage - Forward (L1, L2, R1, R2) continued▼

7 Pull both WINDSHIELD AIR controls. This vents conditioned air to the inside of the windshield for defogging.

8 Choose one:

◆Damage is on the **outer** pane:

▶▶Go to step 9

◆Damage is on the **inner** pane:

▶▶Go to step 11

9 Continue normal operation.

10 Shoulder harnesses may be removed.



11 Don oxygen masks.

12 Establish crew communications.

13 Passenger signsON

14 Choose one:

◆Airplane **has** reached the planned cruise altitude:

▶▶Go to step 15

◆Airplane has **not** reached the planned cruise altitude:

Do **not** continue the climb.

Reset the FLT ALT indicator to the actual airplane altitude.

▶▶Go to step 15

▼ Continued on next page ▼

▼Window Damage - Forward (L1, L2, R1, R2) continued▼

- 15 LAND ALT indicator9,000 feet
- 16 Start a normal descent to below 14,000 feet or to the minimum safe altitude, whichever is higher.
- 17 Plan to land at the nearest suitable airport.
- 18 **When** cabin differential pressure is 2 psi or less:
Oxygen masks and shoulder harnesses may be removed.
- 19 Sustained flight below 10,000 feet is not recommended due to the greater risk of a bird strike.



Window Damage - Heated Side (L3, R3)

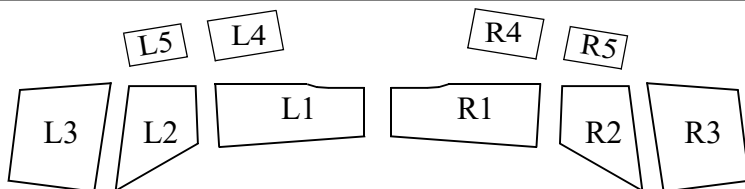
RA-73269, RA-73313, RA-73316 - RA-73318, RA-73321

Condition: A heated side flight deck window has one or more of these:

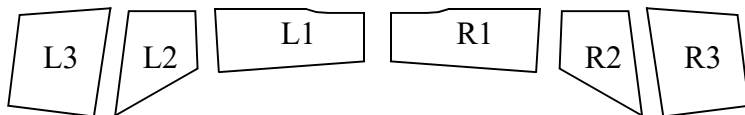
- An electrical arc
- A delamination
- A crack
- Is shattered.

Objective: To remove electrical power, if needed, to prevent arcing. To reduce differential pressure and descend if the inner pane is shattered or cracked.

(VP-BPI, VP-BPY) Do not use



RA-73269, RA-73313, RA-73316 - RA-73318, RA-73321



▼ Continued on next page ▼

▼ Window Damage - Heated Side (L3, R3) continued ▼

1 Choose one:

◆ Window is **delaminated** only:

Continue normal operation.



◆ Window is **arcng**:

►► **Go to step 2**

◆ Window is **cracked** or **shattered**:

►► **Go to step 5**

2 WINDOW HEAT switch
(affected window). OFF

Limit airspeed to 250 knots maximum below
10,000 feet.

3 Pull both WINDSHIELD AIR controls. This vents
conditioned air to the inside of the windshield for
defogging.

4 Continue normal operation.



5 Don seat belts and shoulder harnesses.

6 WINDOW HEAT switch
(affected window). OFF

Limit airspeed to 250 knots maximum below
10,000 feet.

▼ Continued on next page ▼



▼ Window Damage - Heated Side (L3, R3) continued ▼

- 7 Pull both WINDSHIELD AIR controls. This vents conditioned air to the inside of the windshield for defogging.
- 8 Choose one:
 - ◆ Damage is on the **outer** pane:
 - ▶▶ Go to step 9
 - ◆ Damage is on the **inner** pane:
 - ▶▶ Go to step 11
- 9 Continue normal operation.
- 10 Shoulder harnesses may be removed.
 - ■ ■ ■
- 11 Don oxygen masks.
- 12 Establish crew communications.
- 13 Passenger signs ON
- 14 Choose one:
 - ◆ Airplane **has** reached the planned cruise altitude:
 - ▶▶▶ Go to step 15
 - ◆ Airplane has **not** reached the planned cruise altitude:
 - Do **not** continue the climb.
 - Reset the FLT ALT indicator to the actual airplane altitude.
 - ▶▶ Go to step 15

▼ Continued on next page ▼

▼ Window Damage - Heated Side (L3, R3) continued ▼

- 15 LAND ALT indicator 9,000 feet
- 16 Start a normal descent to below 14,000 feet or to the minimum safe altitude, whichever is higher.
- 17 Plan to land at the nearest suitable airport.
- 18 **When** cabin differential pressure is 2 psi or less:
Oxygen masks and shoulder harnesses may be removed.
- 19 Sustained flight below 10,000 feet is not recommended due to the greater risk of a bird strike.



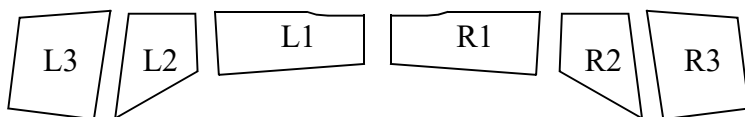
Window Damage - Unheated Side (L3, R3)

RA-73311, RA-73312, RA-73314, RA-73315, RA-73319

Condition: An unheated side flight deck window has one or more of these:

- A crack
- Is shattered.

Objective: To reduce differential pressure to 0 psi if both panes are shattered or cracked.



- 1 Don seat belts and shoulder harnesses.
- 2 Choose one:
 - ◆ Damage is **only** on the inner **or** outer pane:
 - ▶▶ **Go to step 3**
 - ◆ Damage is on **both** the inner **and** outer panes:
 - ▶▶ **Go to step 5**
- 3 Continue normal operation.
- 4 Shoulder harnesses may be removed.



▼ Continued on next page ▼

▼ Window Damage - Unheated Side (L3, R3) continued ▼

- 5 Don oxygen masks.
- 6 Establish crew communications.
- 7 Passenger signs ON
- 8 Choose one:

◆ Airplane **has** reached the planned cruise altitude:

▶▶ **Go to step 9**

◆ Airplane has **not** reached the planned cruise altitude:

Do **not** continue the climb.

Reset the FLT ALT indicator to the actual airplane altitude.

▶▶ **Go to step 9**

- 9 LAND ALT indicator 9,000 feet

- 10 Choose one:

◆ Minimum safe altitude is **at or below 9,000 feet**:

▶▶ **Go to step 11**

◆ Minimum safe altitude is **between 9,000 feet and 13,000 feet**:

▶▶ **Go to step 13**

◆ Minimum safe altitude is **at or above 13,000 feet**:

▶▶ **Go to step 15**

▼ Continued on next page ▼



▼ Window Damage - Unheated Side (L3, R3) continued ▼

- 11 Descend to 9000 feet.
- 12 Maintain a cabin differential pressure of 0 psi by limiting flight altitude to 9000 feet.
- ▶▶ **Go to step 18**
- 13 Descend to the minimum safe altitude.
- 14 LAND ALT indicator Select a higher altitude (maximum 13,000 feet) to maintain a cabin differential pressure of 0 psi

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

- ▶▶ **Go to step 18**
- 15 Descend to the minimum safe altitude.
- 16 Pressurization mode selector MAN

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

- 17  Outflow VALVE switch. Move to OPEN until the outflow VALVE indication shows fully open to depressurize the airplane

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

▼ Continued on next page ▼

▼ Window Damage - Unheated Side (L3, R3) continued ▼

18 **When** the cabin altitude is at or below 10,000 feet:

Oxygen masks may be removed.

19 Shoulder harnesses may be removed.



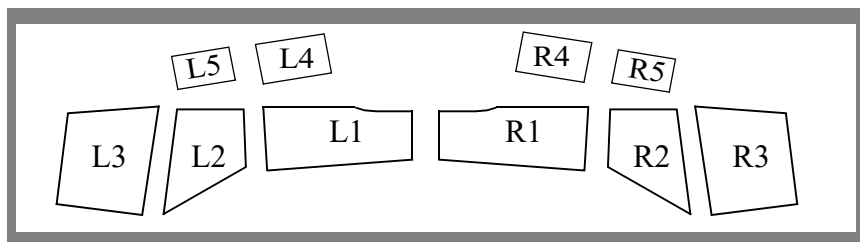
Window Damage - Upper (L4, L5, R4, R5)

(VP-BPI, VP-BPY) Do not use

Condition: An upper flight deck window has one or more of these:

- An electrical arc
- A delamination
- A crack
- Is shattered.

Objective: To remove electrical power, if needed, to prevent arcing. To reduce differential pressure and descend if the structural pane is shattered or cracked.



▼ Continued on next page ▼

▼ Window Damage - Upper (L4, L5, R4, R5) continued ▼

1 Choose one:

◆ Window is **delaminated** only:

Continue normal operation.



◆ Window is **arcng**:

►► **Go to step 2**

◆ Window is **cracked** or **shattered**:

►► **Go to step 5**

2 WINDOW HEAT switch
(affected window). OFF

Limit airspeed to 250 knots maximum below
10,000 feet.

3 Pull both WINDSHIELD AIR controls. This vents
conditioned air to the inside of the windshield for
defogging.

4 Continue normal operation.



5 Don seat belts and shoulder harnesses.

6 WINDOW HEAT switch
(affected window). OFF

Limit airspeed to 250 knots maximum below
10,000 feet.

▼ Continued on next page ▼

▼ Window Damage - Upper (L4, L5, R4, R5) continued ▼

7 Pull both WINDSHIELD AIR controls. This vents conditioned air to the inside of the windshield for defogging.

8 Choose one:

◆ Damage is on the **L4** or **R4** window:

▶▶ **Go to step 9**

◆ Damage is on the **L5** or **R5** window:

▶▶ **Go to step 10**

9 Choose one:

◆ Damage is on the **middle or outer** pane:

▶▶ **Go to step 13**

◆ Damage is on the **inner** pane:

▶▶ **Go to step 11**

10 Choose one:

◆ Damage is on the **outer** pane:

▶▶ **Go to step 11**

◆ Damage is on the **inner** pane:

▶▶ **Go to step 13**

11 Continue normal operation.

12 Shoulder harnesses may be removed.

**▼ Continued on next page ▼**

▼ Window Damage - Upper (L4, L5, R4, R5) continued ▼

13 Don oxygen masks.

14 Establish crew communications.

15 Passenger signs ON

16 Choose one:

◆ Airplane **has** reached the planned cruise altitude:

▶▶ **Go to step 17**

◆ Airplane has **not** reached the planned cruise altitude:

Do **not** continue the climb.

Reset the FLT ALT indicator to the actual airplane altitude.

▶▶ **Go to step 17**

17 LAND ALT indicator 9,000 feet

18 Start a normal descent to below 14,000 feet or to the minimum safe altitude, whichever is higher.

19 Plan to land at the nearest suitable airport.

20 **When** cabin differential pressure is 2 psi or less:

Oxygen masks and shoulder harnesses may be removed.

21 Sustained flight below 10,000 feet is not recommended due to the greater risk of a bird strike.



Window Open

Condition: A side window opens during takeoff or in flight.

- 1 Maintain the maneuvering speed for the existing flap setting until the window is closed.
- 2 The force needed to close the window increases with airspeed. It may not be possible to close the window at speeds above 250 knots.
- 3 Close and lock the window.
- 4 Choose one:
 - ◆ Window **locks and** the pressurization is **normal**:

Continue normal operation.

■ ■ ■ ■
 - ◆ Window does **not** lock **or** the pressurization is **not** normal:

Level off at the lowest safe altitude.

The airplane can fly unpressurized and land safely with the window open.

■ ■ ■ ■



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Intentionally
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CABIN ALTITUDE WARNING or Rapid Depressurization

**CABIN
ALTITUDE**

(If installed and operative)

Condition: One or more of these occur:

- A cabin altitude exceedance
- In flight, the intermittent cabin altitude/configuration warning horn sounds or a CABIN ALTITUDE light (if installed and operative) illuminates.

- 1 Don oxygen masks and set regulators to 100%.
- 2 Establish crew communications.
- 3 Pressurization mode selector MAN
- 4 Outflow VALVE switch Hold in CLOSE
until the outflow VALVE
indication shows fully closed
- 5 **If** cabin altitude is **uncontrollable**:
 Passenger signs ON
 PASS OXYGEN switch ON
**►► Go to the Emergency Descent checklist
on page 0.1**



▼ Continued on next page ▼



▼ **CABIN ALTITUDE WARNING or Rapid Depressurization**
continued ▼

6 **If** cabin altitude is **controllable**:
Continue manual operation to maintain correct cabin altitude.

When the cabin altitude is at or below 10,000 feet:

Oxygen masks may be removed.

7 **Checklist Complete Except Deferred Items**

Deferred Items

Note: Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

Descent Checklist

- Pressurization **Move outflow VALVE switch to OPEN or CLOSE as needed to control cabin altitude and rate**
- Recall Checked
- Autobrake ____
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

Approach Checklist

- Altimeters ____

▼ **Continued on next page ▼**

**▼ CABIN ALTITUDE WARNING or Rapid Depressurization
continued ▼**

At Pattern Altitude

Outflow VALVE switch Move to OPEN until
the outflow VALVE
indication shows fully open
to depressurize the airplane

Landing Checklist

ENGINE START switches CONT
Speedbrake ARMED
Landing gear Down
Flaps ____, Green light



AUTO FAIL **or** **Unscheduled Pressurization Change**

**AUTO
FAIL**
May or may not be illuminated

Condition: One or more of these occur:

- Automatic pressurization mode has failed
- The cabin altitude is uncontrollable.

Objective: To maintain control of cabin altitude.

- 1 Increasing thrust may ensure sufficient air supply to control cabin altitude.
- 2 Pressurization mode selector ALTN
- 3 Choose one:

◆ AUTO FAIL light is **extinguished and** cabin altitude is **controllable**:

Continue normal operation.



◆ AUTO FAIL light is **illuminated or** cabin altitude is **uncontrollable**:

▶▶ **Go to step 4**

- 4 Pressurization mode selector MAN

▼ Continued on next page ▼

▼ AUTO FAIL or Unscheduled Pressurization Change continued ▼

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

- 5  Outflow VALVE switch Move to OPEN or CLOSE as needed to control cabin altitude and rate

- 6 Choose one:

◆ Cabin altitude is **controllable**:

▶▶ **Go to step 11**

◆ Cabin altitude is **uncontrollable**:

▶▶ **Go to step 7**

- 7 Don oxygen masks and set regulators to 100%.

- 8 Establish crew communications.

- 9 Passenger signs ON

- 10 PASS OXYGEN switch ON

▶▶ **Go to the Emergency Descent checklist on page 0.1**



11 Checklist Complete Except Deferred Items

Deferred Items

Note: Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

▼ Continued on next page ▼



▼ **AUTO FAIL or Unscheduled Pressurization Change continued** ▼

Descent Checklist

Pressurization **Move outflow VALVE switch to OPEN or CLOSE as needed to control cabin altitude and rate**

Recall Checked

Autobrake ____

Landing data VREF ____, Minimums ____

Approach briefing Completed

Approach Checklist

Altimeters ____

At Pattern Altitude

Outflow VALVE switch Move to OPEN until the outflow VALVE indication shows fully open to depressurize the airplane

Landing Checklist

ENGINE START switches. CONT

Speedbrake ARMED

Landing gear Down

Flaps ____, Green light



**BLEED
TRIP OFF**
BLEED TRIP OFF

Condition: One or more of these occur:

- An engine bleed air overheat
- An engine bleed air overpressure.

- 1 WING ANTI-ICE switch OFF
- 2 TRIP RESET switch Push

The BLEED TRIP OFF light extinguishes if the bleed air temperature has cooled below limits.

- 3 Choose one:

◆ BLEED TRIP OFF light **stays illuminated**:

PACK switch (affected side) OFF

This causes the operating pack to regulate to high flow in flight with flaps up.

Avoid icing conditions where wing anti-ice is needed.



◆ BLEED TRIP OFF light **extinguishes**:

►► **Go to step 4**

- 4 WING ANTI-ICE switch As needed

Caution! Use of wing anti-ice above approximately FL350 may cause bleed trip off and possible loss of cabin pressure.





Cabin Temperature Hot

Condition: Flight deck or passenger cabin temperature is excessively hot. The temperature can cause incapacitation.

Objective: To regain temperature control. If unable to regain control, to descend and configure to provide alternate ventilation.

- 1 Choose one:
 - ◆**Flight deck** temperature is too high:
 - ▶▶ **Go to step 2**
 - ◆**Passenger cabin** temperature is too high:
 - ▶▶ **Go to step 8**
- 2 TRIM AIR switch OFF
- 3 **Wait** 1 minute.
- 4 Choose one:
 - ◆Air from the flight deck outlets is still too **warm**:
 - ▶▶ **Go to step 5**
 - ◆Air from the flight deck outlets is **cool**:
 - The flight deck temperature gradually cools.
 - ■ ■ ■
- 5 L PACK switch OFF
- 6 **Wait** 1 minute.

▼ Continued on next page ▼

▼ Cabin Temperature Hot continued ▼

7 Choose one:

◆ Air from the flight deck outlets is still too **warm**:

▶▶ **Go to step 14**

◆ Air from the flight deck outlets is **cool**:

The flight deck temperature gradually cools.



8 TRIM AIR switch OFF

9 **Wait** 1 minute.

10 Choose one:

◆ Air from the passenger cabin outlets is still too **warm**:

▶▶ **Go to step 11**

◆ Air from the passenger cabin outlets is **cool**:

The passenger cabin temperature gradually cools.



11 R PACK switch OFF

12 **Wait** 1 minute.

▼ Continued on next page ▼

▼ Cabin Temperature Hot continued ▼

13 Choose one:

◆ Air from the passenger cabin outlets is still too **warm**:

▶▶ **Go to step 14**

◆ Air from the passenger cabin outlets is **cool**:

The passenger cabin temperature gradually cools.



14 Start a descent to the lowest safe altitude, or 10,000 feet, whichever is higher. Use the speedbrakes to increase the rate of descent, if needed. Monitor cabin altitude and rate.

15 R RECIRC FAN switch AUTO

16 L RECIRC FAN switch OFF

17 Minimize the flight deck lighting intensity.

18 Open the flight deck door.

19 **During** daylight:

Use flight deck window shades, as needed.

Instruct the cabin crew to close cabin window shades.

RA-73269 - RA-73318, RA-73321

20 Instruct the cabin crew to dim cabin lighting.

▼ Continued on next page ▼

▼ Cabin Temperature Hot continued ▼

RA-73319

21 Advise the cabin crew that the cabin lighting will be extinguished, but passenger reading lights will continue to work.

22 CAB/UTIL switch OFF

23 IFE/PASS SEAT switch. OFF

24 **When** at level off:

Maintain 290 knots minimum. Flight deck and passenger cabin temperatures can increase rapidly at speeds below 290 knots.

25 Choose one:

◆ Airplane altitude is **at or below 10,000 feet:**

►► **Go to step 26**

◆ Airplane altitude is **above 10,000 feet:**

Don oxygen masks.

Establish crew communications.

►► **Go to step 26**

26 PACK switch (operating pack) OFF

27 Pressurization mode selector MAN

▼ Continued on next page ▼



▼ Cabin Temperature Hot continued ▼

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

- 28  Outflow VALVE switch. . . . Move to OPEN until the outflow VALVE indication shows fully open

This increases airplane ventilation.

- 29 Plan to land at the nearest suitable airport.





Intentionally
Blank

Cabin Temperature Hot

(VQ-BUV) Do not use

Condition: Flight deck or supernumerary area temperature is excessively hot. The temperature can cause incapacitation.

Objective: To regain temperature control. If unable to regain control, to descend and configure to provide alternate ventilation.

1 Choose one:

◆ **Flight deck** temperature is too high:

▶▶ **Go to step 2**

◆ **Supernumerary area** temperature is too high:

▶▶ **Go to step 8**

2 TRIM AIR switch OFF

3 **Wait** 1 minute.

4 Choose one:

◆ Air from the flight deck outlets is still too **warm**:

▶▶ **Go to step 5**

◆ Air from the flight deck outlets is **cool**:

The flight deck temperature gradually cools.



5 L PACK switch OFF

6 **Wait** 1 minute.

▼ **Continued on next page** ▼

▼ Cabin Temperature Hot continued ▼

7 Choose one:

◆ Air from the flight deck outlets is still too **warm**:

▶▶ **Go to step 14**

◆ Air from the flight deck outlets is **cool**:

The flight deck temperature gradually cools.



8 TRIM AIR switch OFF

9 **Wait** 1 minute.

10 Choose one:

◆ Air from the supernumerary area outlets is still too **warm**:

▶▶ **Go to step 11**

◆ Air from the supernumerary area outlets is **cool**:

The supernumerary area temperature gradually cools.



11 R PACK switch OFF

12 **Wait** 1 minute.

▼ Continued on next page ▼

▼ Cabin Temperature Hot continued ▼

13 Choose one:

◆ Air from the supernumerary area outlets is still too **warm**:

▶▶ **Go to step 14**

◆ Air from the supernumerary area outlets is **cool**:

The supernumerary area temperature gradually cools.



14 Start a descent to the lowest safe altitude, or 10,000 feet, whichever is higher. Use the speedbrakes to increase the rate of descent, if needed. Monitor cabin altitude and rate.

15 Minimize the flight deck lighting intensity.

16 Open the flight deck door.

17 **During** daylight:

Use flight deck window shades, as needed.

18 CAB/UTIL switch. OFF

19 **When** at level off:

Maintain 290 knots minimum. Flight deck and supernumerary area temperatures can increase rapidly at speeds below 290 knots.

▼ Continued on next page ▼

▼ Cabin Temperature Hot continued ▼

20 Choose one:

◆ Airplane altitude is **at or below 10,000 feet:**

▶▶ **Go to step 21**

◆ Airplane altitude is **above 10,000 feet:**

Don oxygen masks.

Establish crew communications.

▶▶ **Go to step 21**

21 PACK switch (operating pack) OFF

22 Pressurization mode selector MAN

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

23  Outflow VALVE switch Move to OPEN until the outflow VALVE indication shows fully open

This increases airplane ventilation.

24 Plan to land at the nearest suitable airport.





DUAL
BLEED

DUAL BLEED

Condition: The APU bleed valve is open and one of these occurs:

- BLEED 1 air switch is on
- BLEED 2 air switch is on and the ISOLATION VALVE is open.

Objective: To prevent possible backpressure of the APU.

- 1 Limit engine thrust to idle while the light is illuminated.
- 2 **After** engine start:
APU BLEED air switch OFF
■ ■ ■ ■

OFF

EQUIPMENT COOLING OFF

Condition: The equipment cooling supply or exhaust fan is failed.

- 1 EQUIP COOLING SUPPLY or EXHAUST switch (affected side) ALTN
- Note:** Illumination of the EQUIP COOLING SUPPLY or EXHAUST OFF light may be an indication of a pressurization problem. Ensure the pressurization system is operating normally.
- 2 No further action is necessary in flight if the equipment cooling OFF light does not extinguish.
■ ■ ■ ■

INOP

HIGH ALTITUDE LANDING INOPERATIVE

RA-73319

Condition: The high altitude landing system is inoperative.

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.


**OFF SCHED
DESCENT**

OFF SCHEDULE DESCENT

Condition: A descent is started before reaching the planned cruise altitude set in the FLT ALT indicator.

1 Choose one:

◆ **Landing** at airport of departure:

Continue normal operation.



◆ **Not** landing at airport of departure:

FLT ALT indicator Reset to actual
airplane altitude



PACK

PACK

Condition: One or more of these occur:

- The primary and standby pack controls are failed
- A pack overheat.

- 1 Temperature selectors (all) Select warmer temperature

This reduces the workload on the affected air conditioning pack.

- 2 TRIP RESET switch Push

If the PACK light illuminated as a result of the pack temperature exceeding limits, the light extinguishes if the pack temperature has cooled below limits.

▼ Continued on next page ▼

▼ PACK continued ▼

3 Choose one:

◆ **Both** PACK lights are **extinguished**:

Continue normal operation.



◆ **A single** PACK light **stays illuminated**:

ISOLATION VALVE switch CLOSE

PACK switch (affected side) OFF



◆ **Both** PACK lights **stay illuminated**:

Note: Both pack valves may have closed resulting in a gradual loss of cabin pressure and an eventual CABIN ALTITUDE warning.

►► **Go to step 4**

4 Descend to the lowest safe altitude, or 10,000 feet, whichever is higher. Monitor cabin altitude and rate.

5 **When** at level off:

Maintain 290 knots minimum. Flight deck and cabin temperatures may increase rapidly at speeds below 290 knots.

▼ Continued on next page ▼

▼ PACK continued ▼

6 Choose one:

◆ Airplane altitude is **at or below 10,000 feet**:

►► **Go to step 7**

◆ Airplane altitude is **above 10,000 feet**:

Don oxygen masks.

Establish crew communications.

►► **Go to step 7**

7 Pressurization mode selector **MAN**

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

8  Outflow VALVE switch. Move to **OPEN** until the outflow VALVE indication shows fully open

This increases airplane ventilation.

9 R RECIRC FAN switch **AUTO**

10 L RECIRC FAN switch **OFF**

11 Minimize the flight deck lighting intensity.

12 Open the flight deck door.

13 **During** daylight:

Use flight deck window shades, as needed.

Instruct the cabin crew to close cabin window shades.

▼ Continued on next page ▼

▼ PACK continued ▼**RA-73269 - RA-73318, RA-73321**

14 Instruct the cabin crew to dim cabin lighting.

RA-73319

15 Advise the cabin crew that the cabin lighting will be extinguished, but passenger reading lights will continue to work.

16 CAB/UTIL switch OFF

17 IFE/PASS SEAT switch. OFF

18 Plan to land at the nearest suitable airport.





PACK

PACK

(VQ-BUV) Do not use

Condition: One or more of these occur:

- The primary and standby pack controls are failed
- A pack overheat.

- 1 Temperature selectors (all) Select warmer temperature
- This reduces the workload on the affected air conditioning pack.
- 2 TRIP RESET switch Push
- If the PACK light illuminated as a result of the pack temperature exceeding limits, the light extinguishes if the pack temperature has cooled below limits.

▼ Continued on next page ▼

▼ PACK continued ▼

3 Choose one:

◆ **Both** PACK lights are **extinguished**:

Continue normal operation.



◆ **A single** PACK light **stays illuminated**:

ISOLATION VALVE switch CLOSE

PACK switch (affected side) OFF



◆ **Both** PACK lights **stay illuminated**:

Note: Both pack valves may have closed resulting in a gradual loss of cabin pressure and an eventual CABIN ALTITUDE warning.

►► **Go to step 4**

4 Descend to the lowest safe altitude, or 10,000 feet, whichever is higher. Monitor cabin altitude and rate.

5 **When** at level off:

Maintain 230 knots.

▼ Continued on next page ▼

▼ PACK continued ▼

6 Choose one:

◆ Airplane altitude is **at or below 10,000 feet**:▶▶ **Go to step 7**◆ Airplane altitude is **above 10,000 feet**:

Don oxygen masks.

Establish crew communications.

▶▶ **Go to step 7**

7 Pressurization mode selector MAN

Initially use momentary actuation of the
outflow valve switch to avoid large and rapid
pressurization changes.

8  Outflow VALVE switch. Move towards OPEN
or CLOSE to control
airplane temperature

Note: Moving towards OPEN increases airplane ventilation and reduces airplane temperature. Moving towards CLOSE reduces airplane ventilation and increases airplane temperature.

Do **not** fully close the outflow valve.

9 Minimize the flight deck lighting intensity, if needed.

10 Open the flight deck door.

▼ Continued on next page ▼

▼ PACK continued ▼

11 **During** daylight:

Use flight deck window shades, as needed.

12 CAB/UTIL switch OFF

13 Plan to land at the nearest suitable airport.

▼ Continued on next page ▼



▼ PACK continued ▼

Deferred Items

Note: Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

Descent Checklist

- Pressurization **Move outflow VALVE switch to OPEN or CLOSE as needed to control cabin altitude and rate**
- Recall Checked
- Autobrake _____
- Landing data VREF _____, Minimums _____

Approach Checklist

- Altimeters _____

At Pattern Altitude

- Outflow VALVE switch Move to OPEN until the outflow VALVE indication shows fully open to depressurize the airplane

Landing Checklist

- ENGINE START switches. CONT
- Speedbrake ARMED
- Landing gear Down

▼ Continued on next page ▼



▼ PACK continued ▼

Flaps _____, Green light



**WING-BODY
OVERHEAT****WING-BODY OVERHEAT**

Condition: An overheat from a bleed duct leak occurs.

Objective: To isolate the bleed duct leak.

1 ISOLATION VALVE switch CLOSE

2 Choose one:

◆ **Both** WING-BODY OVERHEAT lights illuminated:

TRIM AIR switch OFF

Note: Passenger cabin temperature control may be less accurate.



◆ **Only right** WING-BODY OVERHEAT light illuminated:

▶▶ **Go to step 3**

◆ **Only left** WING-BODY OVERHEAT light illuminated:

▶▶▶ **Go to step 8**

3 R PACK switch OFF

This causes the operating pack to regulate to high flow in flight with the flaps up.

4 BLEED 2 air switch OFF

5 WING ANTI-ICE switch OFF

This prevents possible asymmetrical ice buildup on the wings.

▼ Continued on next page ▼

▼ **WING-BODY OVERHEAT continued** ▼

6 Avoid icing conditions where wing anti-ice is needed.

7 Choose one:

◆ WING-BODY OVERHEAT light **extinguishes**:



◆ WING-BODY OVERHEAT light **stays illuminated**:

TRIM AIR switch OFF

Note: Passenger cabin temperature control may be less accurate.



8 L PACK switch OFF

This causes the operating pack to regulate to high flow in flight with the flaps up.

9 BLEED 1 air switch OFF

10 WING ANTI-ICE switch OFF

This prevents possible asymmetrical ice buildup on the wings.

11 Avoid icing conditions where wing anti-ice is needed.

▼ **Continued on next page** ▼

▼ WING-BODY OVERHEAT continued ▼

12 Choose one:

◆ WING-BODY OVERHEAT light **extinguishes:**◆ WING-BODY OVERHEAT light **stays illuminated:**▶▶ **Go to step 13**

13 Choose one:

◆ APU is **running:**▶▶ **Go to step 14**◆ APU is **not** running:

TRIM AIR switch OFF

Note: Passenger cabin temperature control may be less accurate.



▼ Continued on next page ▼

▼ **WING-BODY OVERHEAT continued** ▼

14 Choose one:

◆ APU BLEED air switch is **ON**:

APU BLEED air switch OFF

This stops the flow of bleed air from
the APU to the left side of the
pneumatic ducting.

▶▶ **Go to step 15**

◆ APU BLEED air switch is **OFF**:

APU switch OFF

▶▶ **Go to step 16**

15 Choose one:

◆ WING-BODY OVERHEAT light **extinguishes**:

▶▶ **Go to step 17**

◆ WING-BODY OVERHEAT light **stays illuminated**:

APU switch OFF

▶▶ **Go to step 16**

▼ **Continued on next page** ▼

▼ **WING-BODY OVERHEAT** continued ▼

16 Choose one:

◆ WING-BODY OVERHEAT light **extinguishes:**

▶▶ **Go to step 17**

◆ WING-BODY OVERHEAT light **stays illuminated:**

TRIM AIR switch OFF

Note: Passenger cabin temperature control may be less accurate.



17 ISOLATION VALVE switch. AUTO

18 BLEED 1 air switch ON

19 L PACK switch AUTO

20 WING ANTI-ICE switch As needed

21 Choose one:

◆ WING-BODY OVERHEAT light **stays extinguished:**

▶▶ **Go to step 22**

◆ WING-BODY OVERHEAT light **illuminates again:**

▶▶ **Go to step 23**

▼ **Continued on next page** ▼

▼ **WING-BODY OVERHEAT continued** ▼

22 Choose one:

◆ APU switch is **ON**:

The APU can be used during the rest of the flight, as an electrical source only, if needed.



◆ APU switch is **OFF**:

Do **not** start the APU for the rest of the flight.



23 ISOLATION VALVE switch CLOSE

24 L PACK switch OFF

25 BLEED 1 air switch OFF

26 WING ANTI-ICE switch OFF

27 Avoid icing conditions where wing anti-ice is needed.

28 The APU can be used during the rest of the flight, as an electrical source only, if needed.



**ZONE
TEMP**

ZONE TEMP

Condition: One or more of these occur:

- A zone duct overheat
- Flight deck temperature control is failed.

- 1 Temperature selector
(affected cabin) Select a cooler temperature

This prevents the trim air modulating valve from returning to an overheat condition.

- 2 TRIP RESET switch Push
- The ZONE TEMP light extinguishes if the duct temperature has cooled below limits.

- 3 Monitor duct temperature.
- If the duct temperature increases rapidly, set the TRIM AIR switch to OFF.

- 4 Choose one:

◆ Affected cabin temperature is excessively high:

►► **Go to the Cabin Temperature Hot checklist on page 2.8**



◆ Affected cabin temperature is normal:

▼ Continued on next page ▼

▼ **ZONE TEMP** continued ▼**(VQ-BUV) Do not use**

5 Choose one:

◆ Affected cabin temperature is excessively high:

**▶▶ Go to the Cabin Temperature Hot
checklist on page 2.14**

◆ Affected cabin temperature is normal:



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Non-Normal Checklists

Chapter NNC

Anti-Ice, Rain

Section 3

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**COWL
ANTI-ICE**
ENGINE COWL ANTI-ICE

Condition: An engine cowl anti-ice duct overpressure occurs.

Objective: To reduce cowl duct pressure by reducing thrust.

- 1 Reduce thrust only when flight conditions allow.
- 2 Autothrottle (if engaged) Disengage
- 3 Thrust lever
(affected engine) . . . Confirm . . . Retard until the
COWL ANTI-ICE
light extinguishes
- 4 Run the engine at reduced thrust to keep the light
extinguished.

Note: Do not use FMC performance predictions.

RA-73311, RA-73312, RA-73314, RA-73315, RA-73319

- 5 Transponder mode selector TA
This step prevents climb commands which can
exceed reduced thrust performance capability.

**RA-73269, RA-73313, RA-73316 - RA-73318,
RA-73321**

- 6 Transponder mode selector TA ONLY
This step prevents climb commands which can
exceed reduced thrust performance capability.



**COWL VALVE
OPEN****ENGINE COWL VALVE OPEN
OR TAI INDICATION**

Condition: One or both of the following occurs:

- An engine COWL VALVE OPEN light stays illuminated bright blue
- An amber TAI indication is shown.

1 Choose one:

◆ ENG ANTI-ICE switch is **ON**:

The cowl anti-ice valve is failed closed.

Avoid icing conditions.



◆ ENG ANTI-ICE switch is **OFF**:

The cowl anti-ice valve is failed open.

If TAT is **above 10°C**, limit thrust on the affected engine to 80% N1 if possible.



Ice Crystal Icing

Condition: Engine ice crystal or TAT probe icing is suspected. Ice crystal icing conditions exist when in visible moisture, and one or more of the following indications are present:

- Amber or red weather radar returns below the airplane
- Appearance of liquid water on the windshield at temperatures too cold for rain (the sound is different than rain)
- The autothrottle is unable to maintain the selected airspeed
- TAT indication stays near 0°C

(Additional items that can indicate ice crystal icing are listed in the Additional Information section.)

Objective: To exit the ice crystal icing conditions and reduce the operational effects of the icing.

Note: TAT probe icing can cause the reference N1 bugs to increase or decrease while flying at a constant altitude and airspeed.

Caution! Do not use engine anti-ice when TAT is above 10°C.

- 1 ENGINE START switches (both) CONT
- 2 ENG ANTI-ICE switches (both) ON
- 3 Minimize time above amber and red weather radar returns.

▼ Continued on next page ▼

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▼ Ice Crystal Icing continued ▼

- 4 If conditions allow, exit the ice crystal icing conditions laterally. Climbing or descending to exit ice crystal icing conditions is not recommended.
- 5 Choose one:
 - ◆ Autothrottle response **or** TAT indication is normal:
 - ▶▶ **Go to step 8**
 - ◆ Autothrottle is **unable** to maintain the selected airspeed and TAT indication stays near 0°C:
 - ▶▶▶ **Go to step 6**
- 6 Autothrottle (if engaged) Disengage
- 7 Thrust levers (both) Set to maintain
airspeed and
airplane flight path
- 8 When in ice crystal icing conditions, the following can be unreliable:
 - Reference N1 bugs and reference N1 readouts
 - TAS, TAT, SAT, ECON SPD, and LRC.
- 9 **When** ice crystal icing conditions are no longer present:
 - Use engine anti-ice normally.
 - The autothrottle can be re-engaged, if needed.



▼ Continued on next page ▼

▼ Ice Crystal Icing continued ▼

Additional Information

One or more of the following can indicate ice crystal icing:

- Light to moderate turbulence
- Static discharge around the windshield (St. Elmo's fire)
- Smell of sulfur
- Smell of ozone
- Humidity increase

An erroneous TAT indication can occur as a result of ice crystals blocking the sensor. The erroneous indication can last from one minute to more than 20 minutes. TAT normally should increase approximately 2°C per 1000 ft of descent.

Vibration indication due to fan hub icing can be 4 units or greater. Fan ice removal procedures have no effect on fan hub icing. When clear of clouds, fan hub ice evaporates and engine vibration decreases over time. Fan hub ice can remain into descent.

**ICE
DETECTOR**
ICE DETECTOR
(VP-BDU) Do not use

Condition: The ice detector system is failed.

- 1 Run engine anti-ice system as needed.





ICING
P-CANCEL

ICING

(VP-BDU) Do not use

Condition: Ice is detected.

- 1 ENG ANTI-ICE switches (both) ON
■ ■ ■ ■

PROBE HEAT

CAPT PITOT	L ELEV PITOT	L ALPHA VANE	TEMP PROBE
F/O PITOT	R ELEV PITOT	R ALPHA VANE	AUX PITOT

Condition: One or more probe heats are failed.

- 1 Avoid icing conditions.

Note: Flight in icing conditions may result in erroneous flight instrument indications.



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OFF**WINDOW HEAT OFF****(VP-BPI, VP-BPY) Do not use**

Condition: One of the following occurs:

- An overheat due to high ambient temperature
- A system failure
- Window temperature has reached normal limit.

1 Choose one:

◆On the **ground**:

WINDOW HEAT switches (all)ON

WINDOW HEAT TEST switchPWR

▶▶ **Go to step 2**◆**In flight**:▶▶ **Go to step 3**

2 Choose one:

◆**All WINDOW HEAT OFF lights are extinguished:**

Continue normal operation. Power has been removed from the window for overheat protection.



◆**Any WINDOW HEAT OFF light stays illuminated:**

The window heat system is inoperative.

▼ **Continued on next page** ▼

▼ **WINDOW HEAT OFF** continued ▼

3 Choose one:

◆ **All** window OVERHEAT lights are **extinguished**:

Continue normal operation. Electrical power has been temporarily removed to control window temperature.



◆ **Any** window OVERHEAT light **is illuminated**:

►► **Go to the WINDOW OVERHEAT checklist on page 3.10**



OVERHEAT**WINDOW OVERHEAT**

Condition: A window overheat occurs.

- 1 WINDOW HEAT switch (affected window) . . . OFF
- 2 **Wait** 2 - 5 minutes.
- 3 WINDOW HEAT switch (affected window) . . . ON
- 4 Choose one:

◆ Window OVERHEAT light **stays extinguished:**

Continue normal operation.



◆ Window OVERHEAT light **illuminates again:**

►► **Go to step 5**

- 5 WINDOW HEAT switch (affected window) . . . OFF
Limit airspeed to 250 knots maximum below 10,000 feet.
- 6 Pull both WINDSHIELD AIR controls. This vents conditioned air to the inside of the windshield for defogging.





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WING ANTI-ICE VALVE OPEN

RA-73314, RA-73315, RA-73319

L VALVE
OPEN

R VALVE
OPEN

Condition: A wing anti-ice L VALVE OPEN or R VALVE OPEN light stays illuminated bright blue. The wing anti-ice valve is not in the commanded position.

1 Choose one:

◆ WING ANTI-ICE switch is **ON**:

The wing anti-ice valve is failed closed.

WING ANTI-ICE switch OFF

Avoid icing conditions where wing anti-ice is needed.



◆ WING ANTI-ICE switch is **OFF**:

The wing anti-ice valve is failed open.

►► **Go to step 2**

2 ISOLATION VALVE switch. CLOSE

3 PACK switch (affected side) OFF

This causes the operating pack to regulate to high flow in flight with the flaps up.

▼ Continued on next page ▼

▼ WING ANTI-ICE VALVE OPEN continued ▼

4 Choose one:

◆ WING ANTI-ICE **is** needed:

▶▶ **Go to step 5**

◆ WING ANTI-ICE is **not** needed:

▶▶ **Go to step 8**

5 Engine BLEED air switch
(affected side) ON

6 WING ANTI-ICE switch ON

7 **When** wing anti-ice is no longer needed:

▶▶ **Go to step 8**

8 WING ANTI-ICE switch Verify OFF

9 Engine BLEED air switch (affected side) OFF

Wing anti-ice is not available on the affected side with the ISOLATION VALVE switch closed.

Note: Repeat steps 4 through 9 as needed.



WING ANTI-ICE VALVE OPEN

RA-73269 - RA-73313, RA-73316 - RA-73318, RA-73321

**L VALVE
OPEN**

**R VALVE
OPEN**

Condition: A wing anti-ice L VALVE OPEN or R VALVE OPEN light stays illuminated bright blue. The wing anti-ice valve is not in the commanded position.

1 Choose one:

◆ WING ANTI-ICE switch is **ON**:

The wing anti-ice valve is failed closed.

WING ANTI-ICE switch OFF

Avoid icing conditions where wing anti-ice is needed.

►►► **Go to step 10**

◆ WING ANTI-ICE switch is **OFF**:

The wing anti-ice valve is failed open.

►►► **Go to step 2**

2 ISOLATION VALVE switch. CLOSE

3 PACK switch (affected side) OFF

This causes the operating pack to regulate to high flow in flight with the flaps up.

▼ Continued on next page ▼

▼ WING ANTI-ICE VALVE OPEN continued ▼

4 Choose one:

◆ WING ANTI-ICE **is** needed:

▶▶ **Go to step 5**

◆ WING ANTI-ICE is **not** needed:

▶▶ **Go to step 8**

5 Engine BLEED air switch
(affected side) ON

6 WING ANTI-ICE switch ON

7 **When** wing anti-ice is no longer needed:

▶▶ **Go to step 8**

8 WING ANTI-ICE switch Verify OFF

9 Engine BLEED air switch (affected side) OFF

Wing anti-ice is not available on the affected side with the ISOLATION VALVE switch closed.

Note: Repeat steps 4 through 9 as needed.

▼ Continued on next page ▼



▼ WING ANTI-ICE VALVE OPEN continued ▼

10 Choose one:

- ◆ Temperature at landing airport is below 10°C **or** ice formations are observed on airplane surfaces:
▶▶ Go to step 11
- ◆ Temperature at landing airport is 10°C or above **and** ice formations are **not** observed on airplane surfaces:



11 Set VREF 30 + 10 knots or VREF 40 + 10 knots.

Note: The maximum wind additive should not exceed 5 knots..

12 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake ____
- Landing data **VREF 30 + 10 knots or**
VREF 40 + 10 knots ____
Minimums ____
- Approach briefing Completed

▼ Continued on next page ▼

▼ WING ANTI-ICE VALVE OPEN continued ▼

Approach Checklist

Altimeters _____

Landing Checklist

ENGINE START switches CONT

Speedbrake ARMED

Landing gear Down

Flaps _____, Green light



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Non-Normal Checklists

Chapter NNC

Automatic Flight

Section 4

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ROLL/YAW ASYMMETRY	4.3



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AUTOPILOT DISENGAGE

A/P
P/RST

Condition: All autopilots are disengaged. The red light flashes and the aural tone sounds.

- 1 Fly the airplane manually or re-engage an autopilot.



AUTOTHROTTLE DISENGAGE

A/T
P/RST

Condition: The autothrottle is disengaged. The red light flashes.

- 1 Control thrust manually or re-engage the autothrottle.



ROLL AUTHORITY**RA-73319**

Condition: The autopilot has reached its roll authority limit due to an airplane roll or yaw asymmetry.

1 Possible causes of the ROLL AUTHORITY alert include:

- Thrust asymmetry
- Incorrect airplane trim for the conditions
- Fuel imbalance
- Asymmetric wing icing
- A flap or slat asymmetry
- Jammed or restricted flight controls



ROLL/YAW ASYMMETRY**RA-73319**

Condition: The autopilot is approaching its roll authority limit due to an airplane roll or yaw asymmetry.

- 1 Possible causes of the ROLL/YAW ASYMMETRY alert include:
 - Thrust asymmetry
 - Incorrect airplane trim for the conditions
 - Fuel imbalance
 - Asymmetric wing icing
 - A flap or slat asymmetry
 - Jammed or restricted flight controls



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**Non-Normal Checklists
Communications**

**Chapter NNC
Section 5**

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ACARS Electrical Power Loss

RA-73269, RA-73313 - RA-73321

Condition: ACARS AC power is lost.

Note: The ACARS automatically reverts to VOX MODE. The DATA MODE is inoperative.

**ACARS MU Fail or DU Fail**

RA-73269, RA-73313 - RA-73321

Condition: The ACARS system is failed.

- 1 Use normal voice procedures for reporting.



Radio Transmit Continuous (Stuck Microphone Switch)

Condition: A radio transmits continuously without crew input.

- 1 MIC SELECTOR switches
(all audio selector panels)FLT INT

This deselects radios and stops radio transmissions.

Note: The microphone/interphone with the stuck switch continuously transmits on flight interphone.

- 2 The associated audio selector panel should stay on flight interphone. All other audio selector panels may be used normally.





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TRANSFER BUS OFF.....6.16



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**BAT
DISCHARGE****BATTERY DISCHARGE**

Condition: A battery discharge exceedance occurs.

(VP-BSA, VP-BSQ - VP-BZY) Do not use

Note: A fully charged battery supplies a minimum of 30 minutes of standby power.

Note: Fully charged batteries supply a minimum of 60 minutes of standby power.



DRIVE**DRIVE**

Condition: A generator drive malfunction occurs.

Action is irreversible.

- 1  Generator drive
DISCONNECT switch
(affected side) Confirm Hold in the
DISCONNECT
position momentarily

This prevents generator drive damage.

- 2 Choose one:

◆ APU is **available** for start:

APU START

When APU is running:APU GEN switch
(affected side) ON◆ APU is **not** available:Plan to land at the nearest suitable airport.
Only one main AC power source remains.

ELEC**ELEC**

Condition: A standby power or DC system fault occurs.

Note: The ELEC light illuminates on the ground only.

- 1 Do **not** takeoff.



LOSS OF BOTH ENGINE DRIVEN GENERATORS

GEN 1 & 2 GEN 1 & 2 GEN 1 & 2

**TRANSFER
BUS OFF**

**SOURCE
OFF**

**GEN OFF
BUS**

Condition: Both engine driven generators are off.

Note: At high altitude, thrust deterioration or engine flameout can occur.

1 Engine GEN switches (both) . . . ON, one at a time

2 Choose one:

◆ **A single SOURCE OFF light stays illuminated:**

▶▶ **Go to step 3**

◆ **Both SOURCE OFF lights stay illuminated:**

▶▶ **Go to step 5**

◆ **Both SOURCE OFF lights extinguish:**

YAW DAMPER switch ON

▶▶ **Go to step 17**

A single SOURCE OFF light stays illuminated

3 YAW DAMPER switch ON

▼ Continued on next page ▼

▼ **LOSS OF BOTH ENGINE DRIVEN GENERATORS** continued ▼

4 Choose one:

◆ APU is **available** for start:

Note: APU start attempts are not recommended above 25,000 feet.

APU START

When APU is running:

APU GEN switch
(affected side) ON

►► **Go to step 17**

◆ APU is **not** available:

Plan to land at the nearest suitable airport.
Only one main AC power source remains.

►► **Go to step 17**

▼ Continued on next page ▼

▼ **LOSS OF BOTH ENGINE DRIVEN GENERATORS** continued ▼

Both SOURCE OFF lights stay illuminated

5 Choose one:

◆ APU is **available** for start:

BUS TRANSFER switch OFF

ELEC HYD PUMP switches (both) OFF

Note: APU start attempts are not recommended above 25,000 feet. With both buses off, only one start attempt is recommended. Multiple start attempts reduce standby power capacity.

APU START

►► **Go to step 6**

◆ APU is **not** available:

►► **Go to step 14**

6 **When** APU is running:

APU GEN
switches (both) ON,
one at a time

7 Check for the REMOTE CONTROL Circuit Breaker (RCCB Remote P6-SPCU-A-4).

▼ **Continued on next page** ▼

▼ **LOSS OF BOTH ENGINE DRIVEN GENERATORS** continued ▼

8 Choose one:

◆ **RCCB REMOTE** circuit breaker **is** tripped:

Reset circuit breaker. Only one reset is allowed.

▶▶ **Go to step 9**

◆ **RCCB REMOTE** circuit breaker is **not** tripped:

▶▶ **Go to step 9**

9 Choose one:

◆ **A single or both** SOURCE OFF lights **extinguish**:

▶▶ **Go to step 10**

◆ **Both** SOURCE OFF lights **stay illuminated**:

▶▶ **Go to step 14**

10 BUS TRANSFER switch AUTO

This restores power to the remaining transfer bus if one SOURCE OFF light stays illuminated.

11 ELEC HYD PUMP

switches (both) ON, one at a time

12 YAW DAMPER switch ON

13 Plan to land at the nearest suitable airport. Only one main AC power source remains.

▶▶ **Go to step 17**

▼ **Continued on next page** ▼

▼ LOSS OF BOTH ENGINE DRIVEN GENERATORS continued ▼**Both SOURCE OFF lights stay illuminated**

14 Avoid icing conditions.

Note: Flight in icing conditions can result in erroneous flight instrument indications.

15 Plan to land at the nearest suitable airport.

(VP-BSA, VP-BSQ - VP-BZY) Do not use

Note: A fully charged battery supplies a minimum of 30 minutes of standby power.

Note: Fully charged batteries supply a minimum of 60 minutes of standby power.

16 The right IRS will operate on DC power for 5 minutes.

▼ Continued on next page ▼

▼ **LOSS OF BOTH ENGINE DRIVEN GENERATORS** continued ▼

17 Choose one:

◆ **Both** the captain's and first officer's primary attitude displays are **operative and** ATT flags are **not** shown:



◆ **Both** the captain's and first officer's primary attitude displays are **failed**:

►► **Go to step 19**

◆ **Only** the **first officer's** primary attitude display is **failed**:

IRS TRANSFER switch BOTH ON L

Do **not** engage either autopilot.

►► **Go to step 18**

▼ Continued on next page ▼

▼ **LOSS OF BOTH ENGINE DRIVEN GENERATORS** continued ▼

18 Choose one:

◆ **A single SOURCE OFF light is illuminated:**

Plan to land at the nearest suitable airport.
Only one main AC power source remains.



◆ **Both SOURCE OFF lights are extinguished:**



◆ **Both SOURCE OFF lights are illuminated:**

The left IRS will operate as long as battery power remains.

Plan to land at the nearest suitable airport.



Action is irreversible. Do this step only if **both** the captain's and first officer's primary attitude displays are **failed**.

19  IRS MODE selectors (both). ATT

Maintain straight and level, constant airspeed flight until attitude displays recover (approximately 30 seconds).

The primary attitude displays stay failed and the SET IRS HDG prompt on the POS INIT page is blank until the attitude mode alignment is complete.

▼ **Continued on next page** ▼

▼ LOSS OF BOTH ENGINE DRIVEN GENERATORS continued ▼

20 Enter magnetic heading on the POS INIT page or on the overhead IRS display unit by selecting HDG/STS.

Note: The MAP display is not available.

Note: Periodically enter updated heading on the POS INIT page or on the overhead IRS display unit by selecting HDG/STS.

21 Do **not** engage either autopilot.



**SOURCE
OFF****SOURCE OFF**

Condition: The transfer bus is not powered by the last selected source.

1 Choose one:

◆ **Both** SOURCE OFF lights are illuminated:

▶▶ **Go to the LOSS OF BOTH ENGINE
DRIVEN GENERATORS checklist on
page 6.4**



◆ **Only one** SOURCE OFF light is illuminated:

▶▶ **Go to step 2**

2 Engine GEN switch (affected side) ON

3 Choose one:

◆ SOURCE OFF light **extinguishes**:



◆ SOURCE OFF light **stays illuminated**:

▶▶ **Go to step 4**

▼ Continued on next page ▼

▼ SOURCE OFF continued ▼

4 Choose one:

◆ APU is **available** for start:

APU START

When APU is running:

APU GEN switch
(affected side) ON

►► **Go to step 5**

◆ APU is **not** available:

Plan to land at the nearest suitable airport.
Only one main AC power source remains.



5 Choose one:

◆ SOURCE OFF light **extinguishes**:



◆ SOURCE OFF light **stays illuminated**:

Plan to land at the nearest suitable airport.
Only one main AC power source remains.



**STANDBY
PWR OFF****STANDBY POWER OFF**

Condition: One or more of these busses are not energized:

- AC standby bus
- DC standby bus
- Battery bus.

1 STANDBY POWER switch BAT

**TR UNIT****TR UNIT**

Condition: One or more transformer rectifiers are failed.

RA-73319

- 1 Do not use the AFDS approach mode during an ILS or GLS approach.

RA-73269 - RA-73318, RA-73321

- 2 Do not use the AFDS approach mode during an ILS approach.

Note: Autoland is not available.

RA-73269, RA-73314, RA-73315, RA-73319, RA-73321

Note: The AFDS approach mode can be used during an instrument approach using IAN.





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**TRANSFER
BUS OFF****TRANSFER BUS OFF**

Condition: The transfer bus is not energized.

- 1 Engine GEN switch (affected side) ON
- 2 Choose one:

◆ TRANSFER BUS OFF light **extinguishes**:



◆ TRANSFER BUS OFF light **stays illuminated**:

►► **Go to step 3**

- 3 Choose one:

◆ APU is **available** for start:

APU START

When APU is running:

APU GEN switch
(affected side) ON

►► **Go to step 4**

◆ APU is **not** available:

Plan to land at the nearest suitable airport.
Only one main AC power source remains.



▼ Continued on next page ▼

▼TRANSFER BUS OFF continued▼

4 Choose one:

◆TRANSFER BUS OFF light **extinguishes:**



◆TRANSFER BUS OFF light **stays illuminated:**

Plan to land at the nearest suitable airport.
Only one main AC power source remains.



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Non-Normal Checklists

Chapter NNC

Engines, APU

Section 7

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Aborted Engine Start

Condition: On the ground, an aborted engine start is needed.

Objective: To shut down the engine and motor it.

1 Engine start lever (affected engine) CUTOFF

2 Choose one:

◆ ENGINE START switch is in **GRD**:

Motor the engine for 60 seconds.

ENGINE START switch
(affected engine) OFF



◆ ENGINE START switch is in **OFF**:

►► **Go to step 3**

3 **After** N2 decreases below 20%:

ENGINE START switch
(affected engine) GRD

Motor the engine for 60 seconds.

ENGINE START switch
(affected engine) OFF



Aborted Engine Start

(VP-BPI, VP-BPY) Do not use

Condition: On the ground, an aborted engine start is needed.

Objective: To shut down the engine and motor it.

1 Engine start lever (affected engine) CUTOFF

2 Choose one:

◆ ENGINE START switch is in **GRD**:

Motor the engine for 60 seconds.

ENGINE START switch
(affected engine) AUTO



◆ ENGINE START switch is in **AUTO**:

▶▶ **Go to step 3**

3 **After** N2 decreases below 20%:

ENGINE START switch
(affected engine) GRD

Motor the engine for 60 seconds.

ENGINE START switch
(affected engine) AUTO



Intentionally
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Engine Limit or Surge or Stall

Condition: One or more of these occur:

- Engine indications are abnormal
- Engine indications are rapidly approaching or exceeding limits
- Abnormal engine noises are heard, possibly with airframe vibration
- There is no response to thrust lever movement or the response is abnormal
- Flames in the engine inlet or exhaust are reported.

Objective: To attempt to recover normal engine operation or shut down the engine if recovery is not possible.

- 1 Autothrottle (if engaged) Disengage
- 2 Thrust lever
(affected engine) Confirm Retard until
engine indications
stay within limits or
the thrust lever is closed

3 Choose one:

◆ Engine indications are **stabilized and** EGT is **stabilized or decreasing**:

▶▶ **Go to step 4**

◆ Engine indications are **abnormal or** EGT continues to **increase**:

▶▶ **Go to step 7**

▼ Continued on next page ▼

▼ Engine Limit or Surge or Stall continued ▼

Check that RPM and EGT follow thrust lever movement.

4  Thrust lever (affected engine). Advance slowly

5 Run the engine normally or at a reduced thrust setting that is surge and stall free.

6 Choose one:

◆ Engine runs **normally**:



◆ Engine runs at **reduced** thrust:

Note: Do not use FMC performance predictions.

RA-73311, RA-73312, RA-73314, RA-73315, RA-73319

Transponder mode selector TA

This step prevents climb commands which can exceed reduced thrust performance capability.

RA-73269, RA-73313, RA-73316 - RA-73318, RA-73321

Transponder mode selector TA ONLY

This step prevents climb commands which can exceed reduced thrust performance capability.



7 Engine start lever (affected engine) Confirm CUTOFF

▼ Continued on next page ▼

▼ Engine Limit or Surge or Stall continued ▼

8 PACK switch (affected side) OFF

This causes the operating pack to regulate to high flow in flight with flaps up.

9 Choose one:

◆ APU is **available** for start:

APU START

When APU is running:

APU GEN switch
(affected side) ON

►► **Go to step 10**

◆ APU is **not** available:

►► **Go to step 10**

10 Balance fuel as needed.

RA-73311, RA-73312, RA-73314, RA-73315,
RA-73319

11 Transponder mode selector TA

This prevents climb commands which can exceed single engine performance capability.

RA-73269, RA-73313, RA-73316 - RA-73318,
RA-73321

12 Transponder mode selector TA ONLY

This prevents climb commands which can exceed single engine performance capability.

▼ Continued on next page ▼

▼ Engine Limit or Surge or Stall continued ▼

13 ISOLATION VALVE switch Verify AUTO

This ensures bleed air is available to both wings if wing anti-ice is needed.

14 A restart may be attempted if there is N1 rotation and no abnormal airframe vibration.

15 Choose one:

◆ Restart **will** be attempted:

▶▶ **Go to the Engine In-Flight Start checklist on page 7.27**



◆ Restart will **not** be attempted:

▶▶ **Go to step 16**

16 Plan to land at the nearest suitable airport.

Note: Do not use FMC performance predictions.

▶▶ **Go to the One Engine Inoperative Landing checklist on page 7.34**



Loss Of Thrust On Both Engines

Condition: Both of these occur:

- Both engines have a loss of thrust
- Both ENG FAIL alerts show.

Objective: To restart at least one engine.

- 1 ENGINE START switches (both) FLT
- 2 Engine start levers (both) CUTOFF
- 3 **When** EGT decreases:

Engine start levers (both) IDLE detent

- 4 **If** EGT reaches a redline or there is no increase in EGT within 30 seconds:

Engine start lever
(affected engine) Confirm CUTOFF,
then IDLE detent

If EGT again reaches a redline or there is no increase in EGT within 30 seconds, repeat as needed.

-
- 5 At or above FL270, set airspeed to 275 knots.
Below FL270, set airspeed to 300 knots.
 - 6 Engines can accelerate to idle very slowly, especially at high altitudes or in heavy precipitation. If N2 is steadily increasing and EGT stays within limits, do not interrupt the start.
 - 7 Do **not** wait for a successful engine start before starting the APU.

▼ Continued on next page ▼

▼ Loss Of Thrust On Both Engines continued ▼

8 Choose one:

◆ APU is **available** for start:

APU START

▶▶ **Go to step 9**

◆ APU is **not** available:

▶▶ **Go to step 10**

9 **When** APU is running:

APU GEN switches

(both). ON, one at a time

10 Choose one:

◆ **One or both** engines **start**:

▶▶ **Go to step 15**

◆ **Neither** engine starts:

▶▶ **Go to step 11**

11 Choose one:

◆ **N2** is **at or above 11%**:

Attempt a windmill start.

▶▶ **Go to step 13**

◆ **N2** is **below 11%**:

▶▶ **Go to step 12**

▼ Continued on next page ▼

▼ Loss Of Thrust On Both Engines continued ▼

12 Choose one:

◆ APU bleed air is **available**:

Attempt a starter assist start.

▶▶ **Go to step 16**

◆ APU bleed air is **not** available:

Attempt a windmill start.

▶▶ **Go to step 13**

13 Thrust levers (both) Close

14 Engine start
 lever (**either**) Confirm. CUTOFF,
 then IDLE detent

The engine can accelerate to idle very slowly especially at high altitudes or in heavy precipitation. If N2 is steadily increasing and EGT stays within limits, do not interrupt the start.

15 **When** engine parameters have stabilized:

ENGINE START switch
 (operating engine) As needed

Thrust lever
 (operating engine) Advance slowly

Engine GEN switch
 (operating engine side) ON

▶▶ **Go to step 25**

16 Thrust levers (both) Close

▼ Continued on next page ▼

▼ **Loss Of Thrust On Both Engines continued** ▼

- 17 WING ANTI-ICE switch OFF
- 18 PACK switches (both) OFF
- 19 APU BLEED air switch ON
- 20 Ignition select switch BOTH
- 21 Engine start
 lever (**either**) Confirm. CUTOFF
- 22 ENGINE START switch GRD
- 23 **When** N2 is at or above 11%:
- Engine start lever IDLE detent

The engine can accelerate to idle very slowly especially at high altitudes or in heavy precipitation. If N2 is steadily increasing and EGT stays within limits, do not interrupt the start.

24 **When** engine parameters have stabilized:

- APU BLEED air switch OFF
- ENGINE START switch
 (operating engine) As needed
- Thrust lever
 (operating engine) Advance slowly
- Engine GEN switch
 (operating engine side) ON
- PACK switch
 (operating engine side) AUTO

▼ **Continued on next page** ▼

▼ Loss Of Thrust On Both Engines continued ▼

25 Choose one:

◆ **Both** the captain's and first officer's primary attitude displays are **operative and** ATT flags are **not** shown:

▶▶ **Go to step 30**

◆ **Both** the captain's and first officer's primary attitude displays are **failed**:

▶▶ **Go to step 26**

◆ **Only** the **first officer's** primary attitude display is **failed**:

IRS TRANSFER switch. BOTH ON L

Do **not** engage either autopilot.

▶▶ **Go to step 30**

Action is irreversible. Do this step only if **both** the captain's and first officer's primary attitude displays are **failed**.

26  IRS MODE selectors (both). ATT

Maintain straight and level, constant airspeed flight until attitude displays recover (approximately 30 seconds).

The primary attitude displays stay failed and the SET IRS HDG prompt on the POS INIT page is blank until the attitude mode alignment is complete.

▼ Continued on next page ▼

▼ **Loss Of Thrust On Both Engines continued** ▼

27 Enter magnetic heading on the POS INIT page or on the overhead IRS display unit by selecting HDG/STS.

28 The MAP display is not available.

Note: Periodically enter updated heading on the POS INIT page or on the overhead IRS display unit by selecting HDG/STS.

29 Do **not** engage either autopilot.

30 Choose one:

◆ **Both** engines are **running**:

Run the APU as needed.



◆ **One** engine stays **failed**:

► ► **Go to the Engine In-Flight Start checklist on page 7.27**


FAULT
APU FAULT

Condition: An APU malfunction occurs.

Note: The APU shuts down automatically.

1 APU switch OFF

If the APU FAULT light extinguishes after 5 minutes, restarts may be attempted.



**LOW OIL
PRESSURE**

APU LOW OIL PRESSURE

Condition: The APU oil pressure is low.

Note: The APU shuts down automatically.

1 APU switch. OFF

The LOW OIL PRESSURE light extinguishes in 5 minutes.



**OVER
SPEED**

APU OVERSPEED

Condition: One of these occurs:

- An APU RPM limit exceedance causes automatic shutdown
- During a normal APU shutdown the overspeed shutdown protection logic fails a self-test.

1 APU switch. OFF

The APU OVERSPEED light extinguishes in 5 minutes.



ALTN
EEC ALTERNATE MODE

Condition: An EEC operates in the alternate control mode.

- 1 Autothrottle (if engaged) Disengage
- 2 Thrust levers (both) Retard to mid position

This prevents exceeding thrust limits when switching to the EEC alternate mode.

- 3 EEC mode switches (one at a time) ALTN

This ensures both engines operate in alternate mode.

- 4 Autothrottle (if needed) Engage

Note: Maximum thrust limiting is available with autothrottle engaged.

- 5 Do not exceed engine limits. Engine limit protection in alternate mode is not the same as in normal mode.

**RA-73269, RA-73314 - RA-73318, RA-73321
(SB Changes RA-73313)**

- 6 Choose one:

◆ DSPLY SOURCE annunciation is **shown**:

▶▶ **Go to step 8**

◆ DSPLY SOURCE annunciation is **not** shown:



▼ Continued on next page ▼

▼ EEC ALTERNATE MODE continued ▼

RA-73311, RA-73312, RA-73319
(SB Changes RA-73313)

7 Choose one:

◆ DSPLY SOURCE 1 **or** DSPLY SOURCE 2
annunciation is **shown**:

►► **Go to step 9**

◆ DSPLY SOURCE 1 **and** DSPLY SOURCE 2
annunciations are **not** shown:



RA-73269, RA-73314 - RA-73318, RA-73321
(SB Changes RA-73313)

8 Choose one:

◆ DISPLAY SOURCE checklist **has** been completed:



◆ DISPLAY SOURCE checklist has **not** been
completed:

►► **Go to the DISPLAY SOURCE
checklist on page 10.26**



▼ Continued on next page ▼

▼ EEC ALTERNATE MODE continued ▼

RA-73311, RA-73312, RA-73319
(SB Changes RA-73313)

9 Choose one:

◆ DISPLAY SOURCE checklist **has** been completed:



◆ DISPLAY SOURCE checklist has **not** been completed:

►► **Go to the DISPLAY SOURCE
checklist on page 10.28**



**ENGINE
CONTROL**

ENGINE CONTROL

Condition: An engine control system fault occurs.

Note: An ENGINE CONTROL light illuminates on the ground only.

1 Do **not** takeoff.



Engine Failure or Shutdown

Condition: One of these occurs:

- An engine failure
- An ENG FAIL alert shows
- An engine flameout
- Another checklist directs an engine shutdown.

1 Choose one:

◆ **Airframe vibrations with abnormal engine indications** exist:

▶▶ **Go to the ENGINE FIRE or Engine Severe Damage or Separation checklist on page 8.8**



◆ An engine has **separated**:

▶▶ **Go to the ENGINE FIRE or Engine Severe Damage or Separation checklist on page 8.8**



◆ Airframe vibrations with abnormal engine indications do **not** exist **and** an engine has **not** separated:

▶▶ **Go to step 2**

- 2 Do an engine shutdown only when flight conditions allow.
- 3 Autothrottle (if engaged) Disengage

▼ Continued on next page ▼

▼ Engine Failure or Shutdown continued ▼

4 Thrust lever
(affected engine) Confirm Close

5 **When** the affected engine is at idle thrust:

Engine start lever
(affected engine) Confirm CUTOFF

6 PACK switch (affected side) OFF

This causes the operating pack to regulate to high flow in flight with flaps up.

7 Choose one:

◆ APU is **available** for start:

APU START

When APU is running:

APU GEN switch
(affected side) ON

▶▶ **Go to step 8**

◆ APU is **not** available:

▶▶ **Go to step 8**

8 Balance fuel as needed.

**RA-73311, RA-73312, RA-73314, RA-73315,
RA-73319**

9 Transponder mode selector TA

This prevents climb commands which can exceed single engine performance capability.

▼ Continued on next page ▼

RA-73269, RA-73313, RA-73316 - RA-73318,
RA-73321

10 Transponder mode selector TA ONLY

This prevents climb commands which can exceed single engine performance capability.

11 ISOLATION VALVE switch. Verify AUTO

This ensures bleed air is available to both wings if wing anti-ice is needed.

12 A restart may be attempted if there is N1 rotation and no abnormal airframe vibration.

13 Choose one:

◆ A restart will be **attempted**:

▶▶ **Go to the Engine In-Flight Start checklist on page 7.27**



◆ A restart will **not** be attempted:

▶▶ **Go to step 14**

14 Plan to land at the nearest suitable airport.

Note: Do not use FMC performance predictions.

▶▶ **Go to the One Engine Inoperative Landing checklist on page 7.34**



Engine High Oil Temperature

Condition: The engine oil temperature is high.

1 Choose one:

◆ Temperature is **at or above** the **redline**:

▶▶ **Go to the Engine Failure or
Shutdown checklist on page 7.18**



◆ Temperature is in the **amber band**:

▶▶ **Go to step 2**

2 Autothrottle (if engaged) Disengage

3 Thrust lever
(affected engine) Confirm Retard
slowly until the
engine oil temperature
is within the normal operating range
or the thrust lever is closed

▼ Continued on next page ▼

▼ Engine High Oil Temperature continued ▼

RA-73311, RA-73312, RA-73314, RA-73315,
RA-73319

4 Choose one:

◆ Oil temperature is in the **amber** band for **45 minutes or less**:

Note: Run the affected engine at a thrust setting that keeps the engine oil temperature within the normal operating range.

Do not use FMC performance predictions.

Transponder mode selector TA

This step prevents climb commands which can exceed reduced thrust performance capability.



◆ Oil temperature is in the **amber** band for **more than 45 minutes**:

►► **Go to the Engine Failure or Shutdown checklist on page 7.18**



▼ Continued on next page ▼

▼ Engine High Oil Temperature continued ▼

RA-73269, RA-73313, RA-73316 - RA-73318, RA-73321

5 Choose one:

◆ Oil temperature is in the **amber** band for **45 minutes or less**:

Note: Run the affected engine at a thrust setting that keeps the engine oil temperature within the normal operating range.

Do not use FMC performance predictions.

Transponder mode selector TA ONLY

This step prevents climb commands which can exceed reduced thrust performance capability.


◆ Oil temperature is in the **amber** band for **more than 45 minutes**:

►► **Go to the Engine Failure or Shutdown checklist on page 7.18**



Engine High Vibration

Condition: The vibration level is 4.0 units or greater.
Airframe vibration may or may not be felt.

1 Choose one:

◆ In **icing** conditions:

▶▶ **Go to step 2**

◆ **Not** in icing conditions:

▶▶ **Go to step 8**

2 ENGINE START switches (both) FLT

3 Autothrottle (if engaged) Disengage

Do the next two steps on one engine at a time.

4  Thrust lever
(selected engine) Retard to 45% N1
for five seconds,
then slowly advance
to a minimum of 80% N1
for approximately 1 second

5 Thrust lever
(selected engine) Reduce as needed
for flight conditions

6 **Wait** 15 seconds. This allows the engine vibration
level to stabilize.

▼ Continued on next page ▼

▼ Engine High Vibration continued ▼

7 Choose one:

◆ Vibration level is **less than** 4 units:

Continue normal operation.


◆ Vibration level is 4 units **or greater**:

▶▶ **Go to step 9**

8 Autothrottle (if engaged) Disengage

9 Thrust lever
(affected engine) Confirm. Retard to
maintain vibration
levels below 4 units or
until the thrust lever is closed

Note: If the VIB indication does not decrease when the thrust lever is retarded, check other engine indications on the affected engine. If other engine indications are normal, run the engine at reduced thrust.

Note: Do not use FMC performance predictions.

**RA-73311, RA-73312, RA-73314, RA-73315,
RA-73319**

10 Transponder mode selector TA

This step prevents climb commands which can exceed reduced thrust performance capability.

▼ Continued on next page ▼

RA-73269, RA-73313, RA-73316 - RA-73318,

RA-73321

11 Transponder mode selector TA ONLY

This step prevents climb commands which can exceed reduced thrust performance capability.



Engine In-Flight Start

Condition: An engine start is needed and all of the following are true:

- There was **no** engine fire
- There is N1 rotation
- There is **no** abnormal airframe vibration.

Note: Oil quantity indication as low as zero is normal if windmilling N2 RPM is below approximately 8%.

- 1 Do this checklist **only** after completion of the Engine Failure or Shutdown checklist or as directed by the Engine Limit or Surge or Stall checklist or by the Loss of Thrust on Both Engines checklist.
- 2 Check the In-Flight Start Envelope. X-BLD or XB indication may not match the envelope. Starts are not assured outside of the In-flight Start Envelope.

Note: If the N2 is less than 8%, ENGINE START switch must be in CONT to display the EGT.

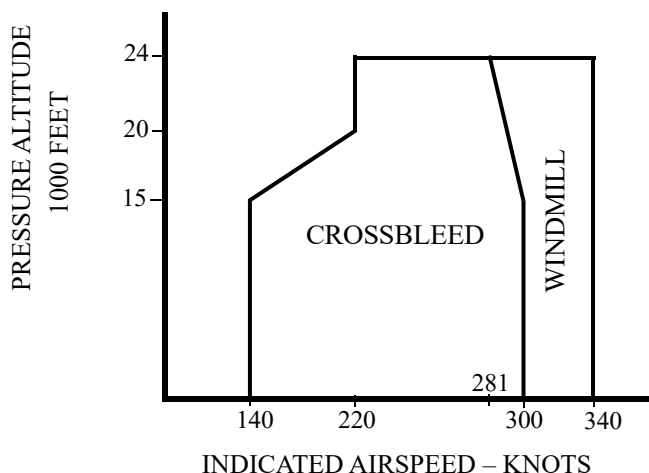
Note: For engines shut down one hour or more, or if EGT is less than 30°C, attempt a restart:

- At an altitude at or below 20,000 feet
- With airspeed at or above 220 knots
- Using a crossbleed start.

▼ Continued on next page ▼

▼ Engine In-Flight Start continued ▼

IN-FLIGHT START ENVELOPE



- 3 Thrust lever
(affected engine) Confirm Close
- 4 Engine start lever
(affected engine) Confirm CUTOFF
- 5 Engines can accelerate to idle very slowly, especially at high altitudes or in heavy precipitation. If N2 is steadily increasing and EGT stays within limits, do not interrupt the start.

▼ Continued on next page ▼

▼ Engine In-Flight Start continued ▼

6 Choose one:

◆ **Windmill** start:

ENGINE START switch
(affected engine) FLT

►► **Go to step 7**

◆ **Crossbleed** start:

PACK switch (affected side) OFF

DUCT PRESSURE Minimum 30 PSI

Advance the thrust lever to increase
duct pressure if needed.

ENGINE START switch
(affected engine) GRD

►► **Go to step 7**

7 **When** N2 is at or above 11%:

Engine start lever
(affected engine) IDLE detent

Monitor EGT to ensure it does not rise rapidly or
exceed the start limit during the start attempt.

▼ Continued on next page ▼

▼ Engine In-Flight Start continued ▼

8 Choose one:

◆ EGT **increases** within 30 seconds **and** a normal start occurs:

▶▶ **Go to step 10**

◆ EGT does **not** increase within 30 seconds **or** another abort start condition as listed in the Normal Procedures occurs:

Engine start lever
(affected engine) . . . Confirm . . . CUTOFF

ENGINE START switch
(affected engine) OFF

(VP-BPI, VP-BPY) **Do not use**

ENGINE START switch
(affected engine) AUTO

Note: If the engine has been shutdown for more than one hour, multiple start attempts can be needed.

▶▶ **Go to step 9**

9 Plan to land at the nearest suitable airport.

Note: Do not use FMC performance predictions.

▶▶ **Go to the One Engine Inoperative Landing checklist on page 7.34**



10 Engine GEN switch (affected side) ON

11 PACK switch (affected side) AUTO

▼ Continued on next page ▼

▼ Engine In-Flight Start continued ▼

- 12 ENGINE START switchAs needed
- 13 APUAs needed
- 14 Transponder mode selectorTA/RA



ENGINE LOW OIL PRESSURE

**LOW OIL
PRESSURE**

May or may not be shown

Condition: The engine oil pressure is low.

1 Choose one:

◆ Engine oil pressure is in the **amber band** with **takeoff thrust** set:

Do **not** takeoff.



◆ Engine oil pressure is **at or below** the **redline**:

▶▶ **Go to the Engine Failure or
Shutdown checklist on page 7.18**



**OIL FILTER
BYPASS**
ENGINE OIL FILTER BYPASS

Condition: The OIL FILTER BYPASS alert indicates oil filter contamination can cause oil to bypass the oil filter.

- 1 Autothrottle (if engaged) Disengage
- 2 Thrust lever
(affected engine) Confirm Retard
slowly until the
OIL FILTER BYPASS
alert extinguishes or
the thrust lever is closed
- 3 Choose one:
 - ◆ OIL FILTER BYPASS alert **extinguishes**:
 ▶▶ **Go to step 4**
 - ◆ OIL FILTER BYPASS alert **stays illuminated**:
 ▶▶ **Go to the Engine Failure or
Shutdown checklist on page 7.18**
- 4 Run the engine at reduced thrust to keep the alert extinguished.

Note: Do not use FMC performance predictions.

**RA-73311, RA-73312, RA-73314, RA-73315,
RA-73319**

- 5 Transponder mode selector TA
 This step prevents climb commands which can exceed reduced thrust performance capability.

▼ Continued on next page ▼

RA-73269, RA-73313, RA-73316 - RA-73318,

RA-73321

6 Transponder mode selector TA ONLY

This step prevents climb commands which can exceed reduced thrust performance capability.



One Engine Inoperative Landing

Condition: Landing must be made with one engine inoperative.

- 1 Plan a flaps 15 landing.
- 2 Set VREF 15 or VREF ICE.

Note: If any of the following conditions apply, set VREF ICE = VREF 15 + 10 knots:

- Engine anti-ice will be used during landing
- Wing anti-ice has been used any time during the flight
- Icing conditions were encountered during the flight and the landing temperature is below 10° C.

Note: When VREF ICE is needed, the wind additive should not exceed 5 knots.

- 3 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.
- 4 Maintain VREF 15 + wind additive or VREF ICE + wind additive on final approach to assure sufficient maneuver margin and speed for go-around. The minimum wind additive is 5 knots.
- 5 When engine anti-ice is needed, use on the operating engine only.
- 6 **Checklist Complete Except Deferred Items**

▼ Continued on next page ▼

▼ One Engine Inoperative Landing continued ▼

Deferred Items

Descent Checklist

Pressurization LAND ALT

Recall Checked

Autobrake

Landing data **VREF 15 or VREF ICE ____,**
Minimums

Approach briefing Completed

Additional Go-Around Thrust

Choose one:

◆ Additional go-around thrust is **needed**:

►► Go to No Engine Bleed Landing below

◆ Additional go-around thrust is **not** needed:

►► **Go to Go-Around Procedure Review below**

No Engine Bleed Landing

When below 10,000 feet:

WING ANTI-ICE switch OFF

ISOLATION VALVE switch CLOSE

BLEED 1 air switch OFF

▼ Continued on next page ▼



▼ One Engine Inoperative Landing continued ▼

Do not open the APU bleed air valve if the engine fire switch is illuminated.



APU BLEED air switch ON
Left PACK switch AUTO
BLEED 2 air switch OFF

Go-Around Procedure Review

Do the normal go-around procedure except:

- Use flaps 1.
- Maintain VREF 15 + 5 knots or VREF ICE + 5 knots until reaching flap retraction altitude.
- Limit bank angle to 15° when airspeed is less than VREF 15 + 15 knots or VREF ICE + 5 knots or the minimum maneuver speed, whichever is lower.
- Accelerate to flaps 1 maneuvering speed before flap retraction.

Approach Checklist

Altimeters _____

Additional Deferred Item

GROUND PROXIMITY
FLAP INHIBIT switch FLAP INHIBIT

▼ Continued on next page ▼

▼ One Engine Inoperative Landing continued ▼

Landing Checklist

ENGINE START switch
(operating engine) CONT
Speedbrake ARMED
Landing gear Down
Flaps. **15, Green light**



REVERSER

REVERSER

Condition: A fault occurs in the thrust reverser system.

- 1 Expect normal reverser operation after landing.



REVERSER UNLOCKED (IN FLIGHT)

Condition: The amber REV indication shows with uncommanded reverse thrust.

Note: Only multiple failures could allow the engine to go into reverse thrust.

Unstowed reverser sleeves produce buffet, yaw, roll and increased airplane drag.

- 1 Check movement of the forward thrust lever on the affected engine.

The EECs prevent power above idle if the related thrust reverser has moved from the stowed position.

Warning! Do not actuate the reverse thrust lever.

- 2 Choose one:

◆ Engine **responds** to forward thrust lever movement **and no** buffet or yaw exists:

Continue normal operation.



◆ Engine does **not** respond to forward thrust lever movement **or** buffet or yaw **exists**:

▶ ▶ **Go to the Engine Failure or Shutdown checklist on page 7.18**



Intentionally
Blank

**START VALVE
OPEN**
START VALVE OPEN

Condition: The START VALVE OPEN alert indicates the

- 1 ENGINE START switch OFF
(VP-BPI, VP-BPY) **Do not use**
- 2 ENGINE START switch AUTO
- 3 Choose one:
 - ◆ START VALVE OPEN alert **extinguishes:**

■ ■ ■ ■
 - ◆ START VALVE OPEN alert **stays illuminated:**

▶▶ **Go to step 4**
- 4 ISOLATION VALVE switch. CLOSE
- 5 PACK switch (affected side) OFF
 This causes the operating pack to regulate to high flow in flight with flaps up.
- 6 Engine BLEED air switch
(affected side) OFF

▼ Continued on next page ▼

▼ **START VALVE OPEN continued** ▼

7 Choose one:

◆ **START VALVE OPEN** alert stays illuminated for **engine 1**:

APU BLEED air switch OFF

▶▶ **Go to step 8**

◆ **START VALVE OPEN** alert stays illuminated for **engine 2**:

▶▶ **Go to step 8**

◆ **START VALVE OPEN** alert **extinguishes**:

▶▶ **Go to step 8**

8 Choose one:

◆ **In flight**:

WING ANTI-ICE switch OFF

This prevents possible asymmetrical ice buildup on the wings.

Avoid icing conditions where wing anti-ice is needed.



◆ **On the ground**:

Ground air source

(if in use) Disconnect

Engine start lever

(affected engine) CUTOFF





Volcanic Ash

Condition: Volcanic ash is suspected when one or more of these occur:

- A static discharge around the windshield
- A bright glow in the engine inlets
- Smoke or dust on the flight deck
- An acrid odor.

Objective: To exit the ash cloud and restart engines if needed.

Caution! Exit the volcanic ash as quickly as possible. Consider a 180° turn. Consider a descending turn.

- 1 Don oxygen masks and smoke goggles, as needed.
- 2 Establish crew communications, as needed.
- 3 Autothrottle (if engaged)Disengage

If conditions allow, run the engines at idle thrust.

- 4  Thrust levers (both) Close

This reduces possible engine damage or flameout, or both, by decreasing EGT.

- 5 ENGINE START switches (both) FLT
- 6 PACK switches HIGH
- 7 ENG ANTI-ICE switches (both).ON
- 8 WING ANTI-ICE switchON

▼ Continued on next page ▼

▼ Volcanic Ash continued ▼

- 9 APU switch
(if APU available) START

This supplies backup electrical and pneumatic sources, if needed.

Note: Volcanic ash can cause non-normal system reactions such as:

- Engine malfunctions, increasing EGT, engine stall or flameout
- Decrease in indicated airspeed or loss of airspeed indications
- Equipment cooling OFF light.

- 10 If failed, engines can accelerate to idle very slowly, especially at high altitudes. If N2 is steadily increasing and EGT stays within limits, the start is progressing normally.

- 11 Plan to land at the nearest suitable airport.

- 12 Choose one:

◆ Engines run **normally**:



◆ Engines do **not** run normally:

►► **Go to the Loss Of Thrust On Both Engines checklist on page 7.8**



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Blank



Non-Normal Checklists

Chapter NNC

Fire Protection

Section 8

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APU FIRE

Condition: Fire is detected in the APU.

1 APU fire switch . . . ConfirmPull, rotate to the
stop, and hold
for 1 second

2 APU switch OFF

3 Choose one:

◆ APU fire switch **extinguishes:**

■ ■ ■ ■

◆ APU fire switch **stays illuminated:**

►► **Go to step 4**

4 Plan to land at the nearest suitable airport.

■ ■ ■ ■



CARGO FIRE (MAIN)

(VQ-BUV) Do not use

MAIN

Condition: Fire is detected in the main deck cargo compartment.

Objective: To suppress the fire. To land at the nearest suitable airport.

- 1 Don oxygen masks and set regulators to 100%, as needed.
- 2 Don smoke goggles, as needed.
- 3 Close the flight deck door. This step prevents smoke or fumes contamination from other compartments.
- 4 Establish crew and cabin communications.

Warning! One pack must be operating to prevent excessive smoke accumulation on the flight deck.

- 5 L PACK switch Verify AUTO
- 6 R PACK switch OFF

This step causes the operating pack to regulate to high flow in flight with the flaps up.

▼ Continued on next page ▼

▼ **CARGO FIRE (MAIN) continued** ▼

Note: If the L PACK light illuminates any time during the rest of the flight:

- Set the R PACK switch to AUTO
- Set the L PACK switch to OFF.

7 MAIN CARGO FIRE

ARMED switch Confirm. Push,
Verify ARMED

8 CARGO FIRE

DEPR/DISCH switch Push and hold
for 1 second

This step ensures closure of the main deck air distribution shut-off valves.

Note: DEPR light can take several seconds to illuminate.

9 CAB/UTIL switch OFF

▼ **Continued on next page** ▼

▼ CARGO FIRE (MAIN) continued ▼

10 Choose one:

◆ On the **ground**:

Warning! Inform ground personnel NOT to open any cargo door until all supernumeraries and crew have exited the airplane and fire fighting equipment is nearby.

◆ In **flight**:▶▶ **Go to step 11**

11 Pressurization mode selector MAN

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

12  Outflow VALVE

switch Move to OPEN until the outflow valve indication is at the 1/4 open position

Note: The airplane depressurizes within 4 minutes.

The intermittent cabin altitude/configuration warning horn sounds and the CABIN ALTITUDE lights (if installed and operative) illuminate at a cabin altitude of approximately 10,000 feet.

▼ Continued on next page ▼

▼ **CARGO FIRE (MAIN) continued** ▼

- 13 Plan to land at the nearest suitable airport.
- 14 If the fire situation becomes uncontrollable, consider an immediate landing.
- 15 Distances in the following steps assume airport elevation at sea level and no wind. Adjust distances for actual conditions.

16 Choose one:

- ◆ The distance to the nearest suitable airport is **50 nm or less**:

Start an expedited uninterrupted descent to the lowest safe altitude or 3,000 feet AFE, whichever is higher.

►► **Go to step 17**

- ◆ The distance to the nearest suitable airport is **more than 50 nm**:

Expedite a climb or descent to 25,000 feet when conditions and terrain allow.

Plan to stay at 25,000 feet until approximately 50 nm from the runway.

At approximately 50 nm from the runway start an expedited, uninterrupted descent to the lowest safe altitude or 3,000 feet AFE, whichever is higher.

►► **Go to step 17**

- 17 Descend at VMO/MMO with the speedbrakes extended.

▼ **Continued on next page** ▼

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▼ CARGO FIRE (MAIN) continued ▼

Note: Maintain VMO/MMO until 15 nm from the runway.

18 Plan to level off 15 nm from the runway.

19 Do not delay the approach and landing.

20 Use autobrake MAX for landing.

21 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake **MAX**
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

Approach Checklist

- Altimeters ____

Warning! Inform ground personnel NOT to open any cargo door after landing until all supernumeraries and crew have exited the airplane and fire fighting equipment is nearby.

Approximately 15nm from the runway

▼ Continued on next page ▼

▼ CARGO FIRE (MAIN) continued ▼

Extend the speedbrakes, reduce speed, and configure for landing. Retract the speedbrakes, as needed.

Landing Checklist

Engine start switches CONT

Speedbrake ARMED

Landing gear Down

Flaps ____ , Green light



ENGINE FIRE or Engine Severe Damage or Separation

Condition: One or more of these occur:

- Engine fire warning
- Airframe vibrations with abnormal engine indications
- Engine separation.

- 1 Autothrottle (if engaged) Disengage
- 2 Thrust lever
(affected engine) Confirm Close
- 3 Engine start lever
(affected engine) Confirm CUTOFF
- 4 Engine fire switch
(affected engine) Confirm Pull

To manually unlock the engine fire switch, press
the override and pull.

- 5 **If** the engine fire switch or ENG OVERHEAT light is
illuminated:

Engine fire switch
(affected engine) Rotate to the stop
and hold for 1 second

▼ Continued on next page ▼

▼ **ENGINE FIRE or Engine Severe Damage or Separation continued** ▼

6 Choose one:

◆ On the **ground**:

▶▶ **Go to the Engine Fire on the Ground checklist on page Back Cover.2**



◆ In **flight**:

▶▶ **Go to step 7**

7 **If** after 30 seconds the engine fire switch or ENG OVERHEAT light stays illuminated:

Engine fire switch

(affected engine) Rotate to the
other stop and
hold for 1 second

▼ **Continued on next page** ▼

▼ **ENGINE FIRE or Engine Severe Damage or Separation continued** ▼

8 Choose one:

- ◆ High airframe vibration **occurs** and **continues** after the engine is shut down:

Without delay, reduce airspeed and descend to a safe altitude which results in an acceptable vibration level.

Note: If high vibration returns and further airspeed reduction and descent are not practical, increasing airspeed can reduce the vibration.

►► **Go to step 9**

- ◆ High airframe vibration does **not** occur or does **not** continue after the engine is shut down:

►► **Go to step 9**

9 ISOLATION VALVE switch. CLOSE

10 PACK switch (affected side) OFF

This step causes the operating pack to regulate to high flow in flight with the flaps up.

11 APU BLEED air switch OFF

▼ **Continued on next page** ▼

▼ **ENGINE FIRE or Engine Severe Damage or Separation continued ▼**

12 Choose one:

◆ APU is **available** for start:

APU START

When APU is running:

APU GEN switch
(affected side) ON

►► **Go to step 13**

◆ APU is **not** available:

►► **Go to step 13**

13 Balance fuel as needed.

**RA-73311, RA-73312, RA-73314, RA-73315,
RA-73319**

14 Transponder mode selector TA

This prevents climb commands which can
exceed single engine performance capability.

**RA-73269, RA-73313, RA-73316 - RA-73318,
RA-73321**

15 Transponder mode selector TA ONLY

This prevents climb commands which can
exceed single engine performance capability.

16 ISOLATION VALVE switch

(after the fire has been extinguished) AUTO

This step ensures bleed air is available to both
wings if wing anti-ice is needed.

▼ **Continued on next page ▼**

▼ **ENGINE FIRE or Engine Severe Damage or Separation continued** ▼

17 Plan to land at the nearest suitable airport.

Note: Do not use FMC performance predictions.

►► **Go to the One Engine Inoperative Landing checklist on page 7.34**



ENGINE OVERHEAT

**ENG 1
OVERHEAT**

**ENG 2
OVERHEAT**

Condition: An overheat is detected in the engine.

- 1 Autothrottle (if engaged) Disengage
- 2 Thrust lever
(affected engine) Confirm Close
- 3 **If the ENG OVERHEAT light **stays illuminated**:**

►► **Go to the ENGINE FIRE or Engine Severe Damage or Separation checklist on page 8.8**



-
- 4 **If the ENG OVERHEAT light **extinguishes**:**

Run the engine at reduced thrust to keep the light extinguished.

Note: Do not use FMC performance predictions.

▼ **Continued on next page** ▼

**RA-73311, RA-73312, RA-73314, RA-73315,
RA-73319**

- 5 Transponder mode selector TA

This step prevents climb commands which can exceed reduced thrust performance capability.

**RA-73269, RA-73313, RA-73316 - RA-73318,
RA-73321**

- 6 Transponder mode selector TA ONLY

This step prevents climb commands which can exceed reduced thrust performance capability.



Engine Tailpipe Fire

Condition: An engine tailpipe fire occurs on the ground with no engine fire warning.

- 1 Engine start lever
(affected engine) CUTOFF
- 2 Advise the cabin.
- 3 Choose one:
 - ◆ Bleed air is **available**:
 - ▶▶ **Go to step 4**
 - ◆ Bleed air is **not** available:
 - Advise ATC.
 - ■ ■ ■
- 4 PACK switches (both) OFF
- 5 ISOLATION VALVE switch. AUTO
- 6 Engine BLEED air switches (both) ON
- 7 Choose one:
 - ◆ APU is **running**:
 - APU BLEED air switch ON
 - ▶▶ **Go to step 8**
 - ◆ APU is **not** running:
 - ▶▶ **Go to step 8**

▼ Continued on next page ▼

▼ Engine Tailpipe Fire continued ▼

8 Choose one:

◆ Affected ENGINE START switch is in **GRD**:

►► **Go to step 9**

◆ Affected ENGINE START switch is **not** in GRD:

Allow the affected N2 to decrease below 20%.

ENGINE START switch
(affected engine) GRD

►► **Go to step 9**

9 Advise ATC.

10 **When** the tailpipe fire is extinguished:

ENGINE START switch
(affected engine) OFF

(VP-BPI, VP-BPY) **Do not use**

ENGINE START switch
(affected engine) AUTO





Smoke, Fire or Fumes

Condition: Smoke, fire or fumes occur.

- 1 Diversion may be needed.
- 2 Don oxygen masks and set regulators to 100%, as needed.
- 3 Don smoke goggles, as needed.
- 4 Establish crew and cabin communications.
- 5 BUS TRANSFER switch OFF

RA-73319

- 6 Advise the cabin crew that the cabin lighting will be extinguished, but passenger reading lights will continue to work.
- 7 CAB/UTIL switch. OFF
- 8 IFE/PASS SEAT switch. OFF
- 9 RECIRC FAN switches (both) OFF
- 10 APU BLEED air switch OFF
- 11 **Anytime** the smoke or fumes become the greatest threat:

►► Go to the Smoke or Fumes Removal checklist on page 8.30

▼ Continued on next page ▼

▼ Smoke, Fire or Fumes continued ▼

12 Choose one:

- ◆ Source of the smoke, fire or fumes is **obvious and** can be **extinguished quickly**:

Isolate and extinguish the source.

If possible, remove power from the affected equipment by switch or circuit breaker in the flight deck or cabin.

▶▶ **Go to step 13**

- ◆ Source of the smoke, fire or fumes is **not** obvious **or cannot** be extinguished quickly:

▶▶ **Go to step 14**

13 Choose one:

- ◆ Source is **visually confirmed** to be extinguished **and** the smoke or fumes are **decreasing**:

Continue the flight at the captain's discretion.

Restore unpowered items at the captain's discretion.

▶▶ **Go to the Smoke or Fumes Removal checklist on page 8.30, if needed**



- ◆ Source is **not** visually confirmed to be extinguished **or** smoke or fumes are **not** decreasing:

▶▶ **Go to step 14**

▼ Continued on next page ▼



▼ Smoke, Fire or Fumes continued ▼

- 14 EQUIP COOLING SUPPLY
and EXHAUST switches (both) ALTN

- 15 Instruct the cabin crew to:
 - Turn on cabin reading lights.
 - Turn on galley attendants work lights.
 - All except RA-73319**
Turn off cabin fluorescent light switches.

- 16 Divert to the nearest suitable airport while
continuing the checklist.

- 17 Consider an immediate landing if the smoke, fire or
fumes situation becomes uncontrollable.

- 18 Do **not** delay landing in an attempt to complete all
of the following steps.

- 19 ISOLATION VALVE switch. CLOSE

- 20 R PACK switch OFF

- 21 **Wait** 2 minutes unless the smoke or fumes are
increasing. This allows time for the smoke or fumes
to clear.

▼ Continued on next page ▼

▼ Smoke, Fire or Fumes continued ▼

22 Choose one:

◆ Smoke or fumes are **decreasing**:

▶▶ **Go to the Smoke or Fumes Removal checklist on page 8.30, if needed**

◆ Smoke or fumes **continue or are increasing**:

R PACK switch AUTO

L PACK switch OFF

▶▶ **Go to step 23**

23 **Wait** 2 minutes unless the smoke or fumes are increasing. This allows time for the smoke or fumes to clear.

24 Choose one:

◆ Smoke or fumes are **decreasing**:

▶▶ **Go to the Smoke or Fumes Removal checklist on page 8.30, if needed**

◆ Smoke or fumes **continue or are increasing**:

L PACK switch AUTO

Consider an immediate landing.

▶▶ **Go to the Smoke or Fumes Removal checklist on page 8.30, if needed**




APU DET
INOP

APU DETECTION
INOPERATIVE

Condition: APU fire detection is inoperative.

- 1 APU switch. OFF

Caution! Do not run the APU. An APU fire would not be detected and the APU would continue to run.



CARGO FIRE

FWD

AFT

Condition: Fire is detected in the related cargo compartment.

Objective: To suppress the cargo fire.

- 1 CARGO FIRE ARM switch
(affected compartment) . . . Confirm Push,
Verify ARMED
- 2 CARGO FIRE DISCH switch Push and hold
for 1 second

Note: DISCH light may need up to 30 seconds to illuminate.

Note: Halon or residual smoke can cause flight deck cargo fire indications to remain or occur again. More information is found in the Additional Information section.

▼ Continued on next page ▼



▼ CARGO FIRE continued ▼

3 Choose one:

◆ On the **ground**:

Warning! Inform ground personnel NOT to open any cargo door until all passengers and crew have exited the airplane and fire fighting equipment is nearby.



◆ In flight:

▶▶ Go to step 4

- 4 RECIRC FAN switches (both) OFF
- 5 PACK switches (both) HIGH
- 6 Plan to land at the nearest suitable airport.
- 7 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake ____
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

▼ Continued on next page ▼

▼ CARGO FIRE continued ▼

Approach Checklist

Altimeters _____

Warning! Inform ground personnel NOT to open any cargo door after landing until all passengers and crew have exited the airplane and fire fighting equipment is nearby.

Landing Checklist

ENGINE START switches CONT

Speedbrake ARMED

Landing gear Down

Flaps ____, Green light



Additional Information

The red FWD/AFT cargo fire warning light(s) can extinguish, remain illuminated, or re-illuminate over the remainder of the flight. If the FWD/AFT cargo fire warning light(s) re-illuminate, both master FIRE WARN lights also re-illuminate and the fire warning bell sounds. These flight deck indications alone of a cargo fire within the **same** compartment do not indicate the fire is uncontrolled.

Illumination of the cargo fire DISCH light indicates the fire suppression system has been fully activated.



CARGO FIRE (FWD/AFT)

(VQ-BUV) Do not use

FWD	AFT
------------	------------

Condition: Fire is detected in the related cargo compartment.

Objective: To suppress the cargo fire.

- 1 CARGO FIRE ARM
switch (FWD/AFT). Push,
Verify ARMED
- 2 CARGO FIRE DEPR/DISCH
switch Push and hold
for 1 second

Note: DISCH light may need up to 30 seconds to illuminate.

Note: Halon or residual smoke can cause flight deck cargo fire indications to remain or occur again. More information is found in the Additional Information section.

▼ Continued on next page ▼

▼ CARGO FIRE (FWD/AFT) continued ▼

3 Choose one:

◆ On the **ground**:

**Warning! Inform ground personnel
NOT to open any cargo door
until all passengers and crew
have exited the airplane and
fire fighting equipment is
nearby.**



◆ In flight:

▶▶ **Go to step 4**

4 PACK switches (both) HIGH |

5 Plan to land at the nearest suitable airport.

6 **Checklist Complete Except Deferred Items**

▼ Continued on next page ▼



▼ CARGO FIRE (FWD/AFT) continued ▼

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake ____
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

Approach Checklist

- Altimeters ____

Warning! Inform ground personnel NOT to open any cargo door after landing until all passengers and crew have exited the airplane and fire fighting equipment is nearby.

Landing Checklist

- ENGINE START switches. CONT
- Speedbrake ARMED
- Landing gear Down
- Flaps ____, Green light



▼ Continued on next page ▼

▼ **CARGO FIRE (FWD/AFT) continued ▼**

Additional Information

The red FWD/AFT cargo fire warning light(s) can extinguish, remain illuminated, or re-illuminate over the remainder of the flight. If the FWD/AFT cargo fire warning light(s) re-illuminate, both master FIRE WARN lights also re-illuminate and the fire warning bell sounds.

Continued indications of a cargo fire within the **same** compartment do not indicate the fire is uncontrolled.

Illumination of the cargo fire DISCH light indicates the fire suppression system has been fully activated.

DEPR

CARGO FIRE DEPR

(VQ-BUV) Do not use

Condition: The main deck suppression system is armed and the air distribution shutoff valves are closed.


**DETECTOR
FAULT**

CARGO FIRE DETECTOR FAULT

Condition: Fire detection is inoperative in one or both cargo compartments.



**DETECTOR
FAULT****CARGO FIRE DETECTOR FAULT****(VQ-BUV) Do not use**

Condition: Fire detection is inoperative in one or more cargo compartments.

**DISCH****CARGO FIRE DISCHARGE****(VQ-BUV) Do not use**

Condition: There is low pressure in one or more cargo fire extinguisher bottles.

- 1 One or more cargo fire extinguisher bottles have discharged.

**FAULT****ENGINE FIRE/OVERHEAT
DETECTOR FAULT**

Condition: Engine fire and overheat detection is inoperative in one or both engines.

**SMOKE****LAVATORY SMOKE****RA-73319**

Condition: Smoke is detected in one or more lavatories.

- 1 Verify that the lavatory fire is contained.





Intentionally
Blank

Smoke or Fumes Removal

Condition: Smoke or fumes removal is needed.

- 1 Do this checklist **only** when directed by the Smoke, Fire or Fumes checklist.
- 2 Do **not** delay landing in an attempt to complete the following steps.
- 3 Close the flight deck door.
- 4 Choose one:
 - ◆ **Both** PACK switches are **OFF**:
 - ▶▶ **Go to step 5**
 - ◆ **A single or both** PACK switch(es) are in **AUTO**:
 - ▶▶ **Go to step 6**

▼ Continued on next page ▼

▼ Smoke or Fumes Removal continued ▼

5 Choose one:

◆ Smoke or fumes source is confirmed to be **outside** the flight deck:



◆ Smoke or fumes source is confirmed to be **on** the flight deck:

Caution! Window should not be opened unless the source is confirmed to be on the flight deck.

Establish normal holding speed. High airspeed may prevent opening the window.

Open the first officer's sliding window.

▶▶ **Go to the Smoke, Fire or Fumes checklist on page 8.16 and do the remaining steps**



Do **not** turn on any PACK switch that was turned off by the Smoke, Fire or Fumes checklist.

- 6  Operating PACK switch(es) HIGH
7 LAND ALT indicator 10,000 feet

Note: The intermittent cabin altitude/configuration warning horn will sound and the CABIN ALTITUDE lights (if installed and operative) will illuminate at a cabin altitude of approximately 10,000 feet.

▼ Continued on next page ▼

▼ Smoke or Fumes Removal continued ▼

- 8 Engine BLEED air switches (both) Verify ON
- 9 Set thrust to maximum practical N1 (minimum 45%).
- 10 Open flight deck air conditioning and gasper outlets.

**Caution! Do not open any flight deck window.
Keep the flight deck door closed.**

11 Choose one:

◆ Smoke or fumes are **controllable**:

▶▶ **Go to the Smoke, Fire or Fumes
checklist on page 8.16 and do the
remaining steps**



◆ Smoke or fumes are **uncontrollable**:

▶▶ **Go to step 12**

12 Descend to the lowest safe altitude or 10,000 feet, whichever is higher.

13 **When** at 14,000 feet or below:

Pressurization mode selector MAN

Outflow VALVE switch Hold in OPEN
until the outflow VALVE
indication shows fully OPEN

This causes the cabin airflow to carry
smoke or fumes aft.

▼ Continued on next page ▼

▼ Smoke or Fumes Removal continued ▼

Note: The outflow valve can take up to 20 seconds to open.

►► **Go to the Smoke, Fire or Fumes checklist on page 8.16 and do the remaining steps**



**WHEEL
WELL**
WHEEL WELL FIRE

Condition: Fire is detected in the main wheel well.

270K/.82M maximum.

- 1  LANDING GEAR lever DN

This attempts to remove and extinguish the fire source.

- 2 Plan to land at the nearest suitable airport.

Note: Do **not** use FMC performance predictions with gear extended.

- 3 Choose one:

◆ Gear **must be retracted** for airplane performance:

▶▶ **Go to step 4**

◆ Gear does **not** need to be retracted for airplane performance:



- 4 **When** the WHEEL WELL light extinguishes:

Wait 20 minutes.

235K maximum.

- 5  LANDING GEAR lever UP

After landing gear retraction is complete:

LANDING GEAR lever OFF





Non-Normal Checklists

Chapter NNC

Flight Controls

Section 9

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Intentionally
Blank

Runaway Stabilizer

Condition: Uncommanded stabilizer trim movement occurs continuously or in a manner not appropriate for flight conditions.

- 1 Control column. Hold firmly
- 2 Autopilot (if engaged) Disengage
- 3 Autothrottle (if engaged) Disengage
- 4 Control column and
thrust levers Control airplane pitch
attitude and airspeed
- 5 Main Electric
Stabilizer trim Reduce control
column forces
- 6 **If** the runaway **stops** after the autopilot is
disengaged:

Do **not** re-engage the autopilot or autothrottle.

■ ■ ■ ■
- 7 **If** the runaway **continues** after the autopilot is
disengaged:

STAB TRIM cutout
switches (both) CUTOUT

If the runaway **continues**:

Stabilizer
trim wheel Grasp and hold

▼ Continued on next page ▼



▼ Runaway Stabilizer continued ▼

- 8 Stabilizer Trim manually
A two-pilot effort may be used to correct an out of trim condition.

Note: Reducing airspeed reduces airloads on the stabilizer which can reduce the effort needed to manually trim.

- 9 Anticipate trim requirements.
- 10 Do **not** re-engage the autopilot or autothrottle.
- 11 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake ____
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

Approach Checklist

- Altimeters ____

Landing Configuration and Trim

Establish landing configuration and in-trim condition early on final approach.

▼ Continued on next page ▼

▼ Runaway Stabilizer continued ▼

Landing Checklist

ENGINE START switches CONT
Speedbrake ARMED
Landing gear Down
Flaps ____, Green light



All Flaps Up Landing

Condition: The leading edge devices fail to extend and trailing edge flaps are less than 1.

Objective: To configure for a landing with leading edge devices retracted and trailing edge flaps less than 1.

- 1 Do this checklist **only** when directed by the Trailing Edge Flaps Up Landing checklist.
- 2 Consider burning off fuel to reduce touchdown speed.
- 3 Set VREF 40 + 55 knots.
- 4 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.
- 5 Maintain flaps up maneuvering speed until established on final approach.
- 6 Limit bank angle to 15° when airspeed is less than the flaps up maneuvering speed.
- 7 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

Pressurization LAND ALT

Recall Checked

Autobrake

Landing data **VREF 40 + 55 knots** ____,
Minimums

▼ Continued on next page ▼

▼ All Flaps Up Landing continued ▼

Approach briefing Completed

Go-Around Procedure Review

Do the normal go-around procedure except:

Limit bank angle to 15° when the airspeed is less than the flaps up maneuvering speed.

Accelerate to flaps up maneuvering speed.

Approach Checklist

Altimeters _____

Additional Deferred Items

FASTEN BELTS switch ON

(VP-BPI, VP-BPY) Do not use

ENGINE START switches CONT

GROUND PROXIMITY

FLAP INHIBIT switch FLAP INHIBIT

▼ Continued on next page ▼



▼ All Flaps Up Landing continued ▼

Landing Checklist

ENGINE START switches. CONT
Speedbrake ARMED

| RA-73313 - RA-73319

Note: The SPEED BRAKE lever will not move beyond the FLIGHT DETENT on landing and the spoilers will not fully deploy.

Landing gear Down
Flaps _____, **No lights**



AUTO SLAT
FAIL

AUTO SLAT FAIL

Condition: The auto slat system is failed.

- 1 Continue normal operation.



Elevator Tab Vibration

Condition: An elevator tab vibration occurs in flight. One or more of the following may be an indication of an elevator tab vibration:

- Vibration that originates, and is strongest, in the aft part of the airplane but can be felt throughout the airplane
- Vibration that is felt in the control wheel and rudder pedals
- Vibration that causes items attached to the airplane, such as sun visors, to move.

1 Passenger signs ON

Do **not** use speedbrakes or change aircraft configuration to reduce airspeed. Do **not** reduce airspeed below the minimum speed for the existing flap setting and gross weight.

2  Smoothly reduce airspeed until the vibration stops.

3 Consider landing at the nearest suitable airport.

4 Stay at or below the reduced airspeed at which the vibration stopped for the rest of the flight. Limit bank angle to 15° until below 20,000 feet.

5 Do **not** deploy speedbrakes in flight.

Note: Flaps and landing gear can be extended normally for the approach.

The speedbrakes can be armed for landing.



**FEEL
DIFF PRESS****FEEL DIFFERENTIAL
PRESSURE**

Condition: High differential pressure is measured by the elevator feel computer.

Note: Column forces can be significantly higher than normal, particularly during landing flare.

The autopilot can disengage when pitch changes are commanded.

Do not attempt an autoland.

- 1 Continue normal operation.

**FLAP LOAD
RELIEF****FLAP LOAD RELIEF**

RA-73313, RA-73316 - RA-73318

Condition: Flap load relief occurs.

- 1 Check flap position and maintain the appropriate airspeed.



**LOW
PRESSURE****FLIGHT CONTROL LOW
PRESSURE**

Condition: Hydraulic system pressure to the ailerons, elevators and rudder is low.

Objective: To activate the standby hydraulic system and standby rudder PCU.

- 1 FLT CONTROL switch
(affected side) Confirm. STBY RUD



Jammed or Restricted Flight Controls

Condition: A flight control is jammed or restricted in roll, pitch, or yaw.

- 1 Autopilot (if engaged) Disengage
- 2 Autothrottle (if engaged) Disengage
- 3 Verify that the thrust is symmetrical.
- 4 Overpower the jammed or restricted system. Use maximum force, including a combined effort of both pilots, if needed. A maximum two-pilot effort on the controls will not cause a cable or system failure.
- 5 Do **not** turn off any flight control switches.
- 6 Choose one:

◆ The failure could be **due** to freezing water **and** conditions **allow**:

Consider a descent to a warmer temperature and attempt to overpower the jammed or restricted system again.

►► **Go to step 7**

◆ The failure could **not** be due to freezing water **or** conditions do **not** allow:

►► **Go to step 7**

▼ Continued on next page ▼

▼ Jammed or Restricted Flight Controls continued ▼

7 Choose one:

◆ Controls are **normal**:



◆ Controls are **not** normal:

►► **Go to step 8**

- 8 Use stabilizer or rudder trim to offload control forces. If electric stabilizer trim is needed, move the Stabilizer Trim Override switch to OVERRIDE.
- 9 Do not make abrupt thrust changes. Extend or retract speedbrake slowly and smoothly.
- 10 Limit bank angle to 15°.
- 11 Plan to land at the nearest suitable airport.
- 12 Plan a flaps 15 landing.
- 13 Set VREF 15 or VREF ICE.

Note: If any of the following conditions apply, set VREF ICE = VREF 15 + 10 knots:

- Engine anti-ice will be used during landing
- Wing anti-ice has been used any time during the flight
- Icing conditions were encountered during the flight and the landing temperature is below 10° C.

Note: When VREF ICE is needed, the wind additive should not exceed 5 knots.

▼ Continued on next page ▼



▼ Jammed or Restricted Flight Controls continued ▼

14 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.

15 Checklist Complete Except Deferred Items

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake ____
- Landing data **VREF 15 or VREF ICE** ____,
Minimums ____
- Approach briefing Completed

Go-Around Procedure Review

- Do the normal go-around procedure.
- Advance thrust to go-around smoothly and slowly to avoid excessive pitch-up.

Approach Checklist

- Altimeters ____

Additional Deferred Item

- GROUND PROXIMITY
- FLAP INHIBIT switch FLAP INHIBIT

▼ Continued on next page ▼

▼ Jammed or Restricted Flight Controls continued ▼**Landing Checklist**

ENGINE START switches CONT
Speedbrake ARMED
Landing gear Down
Flaps. **15, Green light**



**LE FLAPS
TRANSIT**
**LEADING EDGE FLAPS
TRANSIT**

Condition: One or more of these occur:

- The leading edge devices are not in the commanded position
- A leading edge device asymmetry is detected
- A leading edge device skew is detected.

Note: Do not use FMC performance predictions with any flaps or slats extended.

1 Choose one:

◆ **Trailing** edge flaps are **extended** and the trailing edge flap position indication **disagrees** with the flap handle position:

▶▶ **Go to the Trailing Edge Flap Disagree checklist on page 9.36**



◆ **Trailing** edge flaps are **extended** and the trailing edge flap position indication **agrees** with the flap handle position:

▶▶ **Go to step 7**

◆ **Trailing** edge flaps are **up**:

Limit airspeed to 230 knots maximum.

▶▶ **Go to step 2**

▼ Continued on next page ▼

▼ LEADING EDGE FLAPS TRANSIT continued ▼

2 Choose one:

◆ Roll is **encountered**:

►► **Go to step 7**

◆ Roll is **not** encountered:

Note: Roll may be difficult to identify with the autopilot engaged.

►► **Go to step 3**

Maximum flap extension altitude 20,000 feet.

3  Flaps Extend to flaps 1, then retract to flaps up

4 Choose one:

◆ LE FLAPS TRANSIT light **extinguishes** after the flaps are up:

Continue normal operation.



◆ LE FLAPS TRANSIT light **stays illuminated** after the flaps are up:

►► **Go to step 5**

5 Check LE DEVICES annunciator panel.

▼ Continued on next page ▼

▼ LEADING EDGE FLAPS TRANSIT continued ▼

6 Choose one:

◆ Light(s) for **only one** leading edge device is illuminated:

Limit airspeed to 300 knots (280 knots for turbulent air penetration) or 0.65 Mach, whichever is lower.

►► **Go to step 7**

◆ Light(s) for **more than one** leading edge device is illuminated:

Limit airspeed to 230 knots maximum.

►► **Go to step 7**

7 Plan a flaps 15 landing.

8 Set VREF 15 + 15 knots.

9 Limit bank angle to 15° when airspeed is less than the flaps up maneuvering speed.

10 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.

11 Checklist Complete Except Deferred Items

Deferred Items

Descent Checklist

Pressurization LAND ALT ____

Recall Checked

Autobrake ____

▼ Continued on next page ▼

▼ LEADING EDGE FLAPS TRANSIT continued ▼

Landing data **VREF 15 + 15 knots** ____,
Minimums ____

Approach briefing Completed

Approach Checklist

Altimeters ____

Additional Deferred Item

GROUND PROXIMITY

FLAP INHIBIT switch FLAP INHIBIT

Note: The amber LE FLAPS TRANSIT light may be illuminated. Operation within the lower amber airspeed band for landing is normal for this condition.

(VP-BPI, VP-BPY) Do not use

Note: V/S and VNAV PTH modes can revert to LVL CHG mode.

Landing Checklist

ENGINE START switches CONT

Speedbrake ARMED

Landing gear Down

Flaps. **15, Green or amber light**

Note: The light may be green or amber depending on the cause of the failure.



**MACH TRIM
FAIL****MACH TRIM FAIL**

Condition: The mach trim system is failed.

- 1 Limit airspeed to 280 knots/0.82 Mach.



**SPEED BRAKE
DO NOT ARM**
SPEED BRAKE DO NOT ARM
(VP-BZU - VP-BZY) Do not use

Condition: An automatic speedbrake fault occurs.

Note: Speedbrakes may be used in flight.

1 Choose one:

- ◆ **SPEED BRAKE DO NOT ARM** light illuminates **before** the flaps are retracted:
Retract the flaps on schedule.
▶▶ Go to step 2
- ◆ **SPEED BRAKE DO NOT ARM** light illuminates with the flaps **up**:
Limit airspeed to 320 knots maximum.
▶▶ Go to step 3
- ◆ **SPEED BRAKE DO NOT ARM** light is illuminated **after** flap extension for landing:
▶▶ Go to step 3

▼ Continued on next page ▼



▼ SPEED BRAKE DO NOT ARM continued ▼

2 Choose one:

- ◆ SPEED BRAKE DO NOT ARM light **stays illuminated** after the flaps are retracted:
Limit airspeed to 320 knots maximum.
▶▶ **Go to step 3**
- ◆ SPEED BRAKE DO NOT ARM light **extinguishes** after the flaps are retracted:
▶▶ **Go to step 3**

3 Checklist Complete Except Deferred Items

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake ____
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

Approach Checklist

- Altimeters ____

▼ Continued on next page ▼

▼ **SPEED BRAKE DO NOT ARM continued** ▼

Additional Deferred Item

Choose one:

◆ **SPEED BRAKE DO NOT ARM light is extinguished** after flap extension for landing:



◆ **SPEED BRAKE DO NOT ARM light is illuminated** after flap extension for landing:

Do **not** arm the speedbrakes for landing. Manually deploy the speedbrakes immediately upon landing. Increased force may be needed to move the SPEED BRAKE lever to the UP position.

►► **Go to Landing Checklist below**

Landing Checklist

ENGINE START switches CONT

Speedbrake **DOWN detent**

Landing gear Down

Flaps ____, Green light





SPEED BRAKE
DO NOT ARM

SPEED BRAKE DO NOT ARM

Condition: An automatic speedbrake fault occurs.

Note: Speedbrakes may be used in flight.

- 1 Do **not** arm the speedbrake for landing. Manually deploy the speedbrakes immediately upon landing.
- 2 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

Pressurization LAND ALT ____
Recall Checked
Autobrake ____
Landing data VREF ____, Minimums ____
Approach briefing Completed

Approach Checklist

Altimeters ____

Landing Checklist

ENGINE START switches. CONT
Speedbrake **DOWN detent**
Landing gear Down

▼ Continued on next page ▼

▼ **SPEED BRAKE DO NOT ARM continued** ▼

Flaps ____, Green light



**SPEED TRIM
FAIL**

SPEED TRIM FAIL

Condition: Speed Trim function is inoperative.

Note: The Speed Trim System will not provide stabilizer trim inputs when deviating from a trimmed airspeed.

1 Continue normal operation.



**SPEEDBRAKES
EXTENDED**

SPEEDBRAKES EXTENDED

Condition: In flight, the speedbrakes are extended beyond the ARMED position and one or more of these occur:

- The radio altitude is below 800 feet
- The flap lever setting is more than flaps 10.

On the ground, the SPEED BRAKE lever is down and the speedbrakes are extended.

1 Choose one:

◆ **In flight:**

SPEED BRAKE lever ARMED



◆ **On the ground:**

SPEED BRAKE lever DOWN detent

Note: If the light stays illuminated, do **not** takeoff.





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STABILIZER OUT OF TRIM

STAB
OUT OF
TRIM

Condition: The autopilot does not set the stabilizer trim correctly.

Note: During large changes in trim requirements, the STAB OUT OF TRIM light can illuminate momentarily.

1 Choose one:

◆ Stabilizer is **trimming**:
Continue normal operation.
■ ■ ■ ■

◆ Stabilizer is **not** trimming:
▶▶ **Go to step 2**

- 2 Control column. Hold firmly
- 3 AutopilotDisengage
- 4 Autothrottle (if engaged)Disengage
- 5 Main Electric
Stabilizer trimAs needed

▼ Continued on next page ▼

▼ **STABILIZER OUT OF TRIM continued** ▼

6 Choose one:

◆ Stabilizer **responds** to electric trim inputs:

Do **not** re-engage the autopilot or autothrottle.



◆ Stabilizer does **not** respond to electric trim inputs:

►► **Go to the Stabilizer Trim Inoperative checklist on page 9.28**



Stabilizer Trim Inoperative

Condition: Loss of electric trim through the main electric stabilizer trim switches, or as directed by the STAB OUT OF TRIM checklist.

Objective: To land the airplane using manual trim or, if manual trim is not available, to land the airplane using elevator control only.

1 STAB TRIM cutout switches (both) CUTOUT
The autopilot is not available.

2 Stabilizer Trim manually
A two-pilot effort may be used and will not cause system damage.

Note: Reducing airspeed reduces airloads on the stabilizer which can reduce the effort needed to manually trim.

If the failure could be due to ice accumulation, descend to a warmer temperature and attempt again to trim manually.

3 Choose one:

◆ Stabilizer **can** be trimmed manually:

▶▶ **Go to step 4**

◆ Stabilizer **cannot** be trimmed manually:

▶▶ **Go to step 5**

▼ Continued on next page ▼

▼ Stabilizer Trim Inoperative continued ▼

4 Anticipate trim requirements.

►► **Go to step 7**

5 Expect higher than normal elevator forces during approach and landing.

6 The thrust reduction at flare will cause a nose down pitch.

7 Plan a flaps 15 landing.

8 Set VREF 15 or VREF ICE.

Note: If any of the following conditions apply, set VREF ICE = VREF 15 + 10 knots:

- Engine anti-ice will be used during landing
- Wing anti-ice has been used any time during the flight
- Icing conditions were encountered during the flight and the landing temperature is below 10° C.

Note: When VREF ICE is needed, the wind additive should not exceed 5 knots.

9 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.

10 Checklist Complete Except Deferred Items

Deferred Items

Descent Checklist

Pressurization. LAND ALT ____

▼ Continued on next page ▼



▼ Stabilizer Trim Inoperative continued ▼

- Recall Checked
- Autobrake ____
- Landing data **VREF 15 or VREF ICE** ____,
Minimums ____
- Approach briefing Completed

Go-Around Procedure Review

Do the normal go-around procedure.

Advance thrust to go-around smoothly and slowly to avoid excessive pitch-up.

Approach Checklist

- Altimeters ____

Landing Configuration

Establish landing configuration early on final approach.

Additional Deferred Item

- GROUND PROXIMITY
- FLAP INHIBIT switch FLAP INHIBIT

Landing Checklist

- ENGINE START switches. CONT

▼ Continued on next page ▼

▼ Stabilizer Trim Inoperative continued ▼

Speedbrake ARMED
 Landing gear Down
 Flaps. **15, Green light**


**STBY
RUD ON**
STANDBY RUDDER ON

Condition: The standby rudder hydraulic system is commanded on.

1 Choose one:

- ◆ STBY RUD ON light is illuminated with **no other flight deck indications:**

Avoid large or abrupt rudder pedal inputs.



- ◆ STBY RUD ON light is illuminated due to the **pilot moving** the FLT CONTROL A or B switch to STBY RUD:



- ◆ STBY RUD ON light is illuminated in response to a hydraulic system **non-normal** situation:



Trailing Edge Flap Asymmetry

Condition: One or more of these occur:

- An uncommanded roll occurs when the flaps change position
- The left and right flap indications disagree.

Objective: To configure the airplane for landing.

- 1 Set the flap lever to the nearest detent that is equal to or less than the smallest indicated flap position.

Caution! Do not attempt to move the trailing edge flaps with the ALTERNATE FLAPS switch because there is no asymmetry protection.

Note: Do not use FMC performance predictions with any flaps or slats extended.

▼ Continued on next page ▼

▼ Trailing Edge Flap Asymmetry continued ▼

2 Choose one:

◆ Flap **lever** is set to **30**:

Set VREF 30.

Note: VREF + wind additive must not exceed the flap placard speed for flaps 40.

▶▶ **Go to step 4**

◆ Flap **lever** is set to **15 or 25**:

▶▶ **Go to step 3**

◆ Flap **lever** is set to **1 or greater and less than 15**:

Consider burning off fuel to reduce touchdown speed.

Set VREF 40 + 30 knots.

▶▶ **Go to step 4**

◆ Flap **lever** is set to **UP**:

▶▶ **Go to the Trailing Edge Flaps Up Landing checklist on page 9.43**


▼ Continued on next page ▼



▼ **Trailing Edge Flap Asymmetry continued** ▼

3 Set VREF 15 or VREF ICE.

Note: If any of the following conditions apply, set VREF ICE = VREF 15 + 10 knots:

- Engine anti-ice will be used during landing
- Wing anti-ice has been used any time during the flight
- Icing conditions were encountered during the flight and the landing temperature is below 10° C.

Note: When VREF ICE is needed, the wind additive should not exceed 5 knots.

VREF 15 + wind additive, or VREF ICE + wind additive if needed, must not exceed the flap placard speed for the next larger flap setting.

4 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.

5 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake ____
- Landing data **VREF ____ as directed by checklist, Minimums ____**

▼ **Continued on next page** ▼

▼ Trailing Edge Flap Asymmetry continued ▼

Approach briefing Completed

Approach Checklist

Altimeters _____

Additional Deferred Item

Choose one:

◆ Flap **lever** is set to **30**:

▶▶ **Go to Landing Checklist below**

◆ Flap **lever** is set to **less than 30**:

GROUND PROXIMITY FLAP

INHIBIT switch FLAP INHIBIT

▶▶ **Go to Landing Checklist below**

Landing Checklist

ENGINE START switches CONT

Speedbrake ARMED

Landing gear Down

Flaps. _____, **Green or amber light**

Note: The light can be green or amber depending on the cause of the failure.



Trailing Edge Flap Disagree

Condition: The trailing edge flaps are not in the commanded position.

Objective: To configure the airplane for landing.

1 Choose one:

◆ Trailing edge flap asymmetry **exists**:

▶▶ **Go to the Trailing Edge Flap Asymmetry checklist on page 9.32**



◆ Trailing edge flap asymmetry does **not** exist:

▶▶ **Go to step 2**

2 Choose one:

◆ Indicated flap position is **30 or greater and less than 40**:

Land using existing flaps.

▶▶ **Go to step 3**

◆ Indicated flap position is **15 or greater and less than 30**:

Land using existing flaps.

▶▶ **Go to step 5**

◆ Indicated flap position is **less than 15**:

▶▶ **Go to step 4**

3 Set VREF 30.

▼ Continued on next page ▼

▼ Trailing Edge Flap Disagree continued ▼

Note: VREF 30 + wind additive must not exceed the flap placard speed for flaps 40.

▶▶ Go to step 6

- 4 Plan to extend flaps to 15 using alternate flap extension.

Note: Alternate flap extension time to flaps 15 is approximately 2 minutes.

The drag penalty with the leading edge devices extended can make it impossible to reach an alternate field.

- 5 Set VREF 15 or VREF ICE.

Note: If any of the following conditions apply, set VREF ICE = VREF 15 + 10 knots:

- Engine anti-ice will be used during landing
- Wing anti-ice has been used any time during the flight
- Icing conditions were encountered during the flight and the landing temperature is below 10° C.

Note: When VREF ICE is needed, the wind additive should not exceed 5 knots.

VREF 15 + wind additive, or VREF ICE + wind additive if needed, must not exceed the flap placard speed for the next larger flap setting.

▼ Continued on next page ▼



▼Trailing Edge Flap Disagree continued▼

- 6 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.
- 7 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

Pressurization LAND ALT ____

Recall Checked

Autobrake ____

Landing data **VREF ____ as directed
by checklist, Minimums ____**

Approach briefing Completed

Approach Checklist

Altimeters ____

▼ Continued on next page ▼

▼ Trailing Edge Flap Disagree continued ▼

Additional Deferred Item

Choose one:

- ◆ Indicated flap position is **30 or greater**:
 - ▶▶ **Go to Landing Checklist below**
- ◆ Indicated flap position is **15 or greater and less than 30**:
 - GROUND PROXIMITY FLAP
INHIBIT switch FLAP INHIBIT
 - ▶▶ **Go to Landing Checklist below**
- ◆ Indicated flap position is **less than 15**:
 - GROUND PROXIMITY FLAP
INHIBIT switch FLAP INHIBIT
 - ▶▶ **Go to Alternate Flap Extension below**

Alternate Flap Extension

During flap extension, set the flap lever to the desired flap position.

 **230K maximum during alternate flap extension.**
ALTERNATE FLAPS master switch **ARM**

Note: The landing gear configuration warning can sound if the flaps are between 10 and 15 and the landing gear are retracted.

▼ Continued on next page ▼

Note: The amber LE FLAPS TRANSIT light stays illuminated until the flaps approach the flaps 15 position.

Note: The amber LE FLAPS TRANSIT light stays illuminated until the flaps approach the flaps 10 position.

Note: Operation within the lower amber airspeed band can be needed until the LE FLAPS TRANSIT light extinguishes.

If flap asymmetry occurs, release the switch immediately. There is no asymmetry protection.



ALTERNATE FLAPS

position switch Hold DOWN
to extend flaps
to 15 on schedule

As flaps are extending, slow to respective maneuvering speed.

▼ Continued on next page ▼

▼ Trailing Edge Flap Disagree continued ▼

Choose one:

- ◆ Trailing edge flaps **asymmetry** occurs:
 - ▶▶ **Go to the Trailing Edge Flap Asymmetry checklist on page 9.32**
 - ■ ■ ■
- ◆ Trailing edge flaps extend to **15**:
 - ▶▶ **Go to Landing Checklist below**
- ◆ Indicated flap position is **less than 1** after attempting alternate flap extension:
 - ▶▶ **Go to the Trailing Edge Flaps Up Landing checklist on page 9.43**
 - ■ ■ ■
- ◆ Indicated flap position is **1 or greater and less than 15** after attempting alternate flap extension:

Land using existing flaps.

Consider burning off fuel to reduce touchdown speed.

Set VREF 40 + 30 knots.

Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.

▶▶ **Go to Landing Checklist below**

▼ Continued on next page ▼



▼Trailing Edge Flap Disagree continued▼

Landing Checklist

- ENGINE START switches. CONT
- Speedbrake ARMED
- Landing gear Down
- Flaps. ____, **Green or amber light**

Note: The light can be green or amber depending on the cause of the failure.



Trailing Edge Flaps Up Landing

Condition: The trailing edge flaps are less than 1.

Objective: To configure for a landing with trailing edge flaps less than 1.

1 Choose one:

◆ Trailing edge flap asymmetry **exists**:

▶▶ **Go to step 2**

◆ Trailing edge flap asymmetry does **not** exist:

Do this checklist **only** when directed by the Trailing Edge Flap Disagree checklist.

▶▶ **Go to step 4**

230K maximum.

2 ALTERNATE FLAPS master switch ARM

Note: This procedure extends the leading edge devices only.

3 ALTERNATE FLAPS position switch Momentary DOWN

Verify that the LE DEVICES annunciator indicates FULL EXT for all leading edge slats and flaps.

Note: The LE FLAPS TRANSIT light can stay illuminated after the LE devices are fully extended.

▼ Continued on next page ▼



▼ Trailing Edge Flaps Up Landing continued ▼

4 Choose one:

◆ LE DEVICES annunciator **shows** FULL EXT:

▶▶ **Go to step 5**

◆ LE DEVICES annunciator does **not** show FULL EXT:

▶▶ **Go to the All Flaps Up Landing checklist on page 9.4**



- 5 Consider burning off fuel to reduce touchdown speed.
- 6 Set VREF 40 + 40 knots.
- 7 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.
- 8 Maintain flaps up maneuvering speed until on final.
- 9 Limit bank angle to 15° when airspeed is less than the flaps up maneuvering speed.

10 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

Pressurization LAND ALT ____
Recall Checked
Autobrake ____

▼ Continued on next page ▼

▼ Trailing Edge Flaps Up Landing continued ▼

Landing data **VREF 40 + 40 knots** ____,
Minimums ____

Approach briefing Completed

Go-Around Procedure Review

Do the normal go-around procedure except:

Limit bank angle to 15° when the airspeed is less than the flaps up maneuvering speed.

Accelerate to flaps up maneuvering speed.

Do not exceed 230 knots with leading edge devices extended.

Approach Checklist

Altimeters ____

▼ Continued on next page ▼



▼ Trailing Edge Flaps Up Landing continued ▼

Additional Deferred Items

- FASTEN BELTS switch ON
- (VP-BPI, VP-BPY) Do not use
- ENGINE START switches. CONT
- GROUND PROXIMITY
- FLAP INHIBIT switch FLAP INHIBIT

Note: A nuisance stick shaker can occur when slowing to VREF 40 + 40 knots at high gross weights and/or bank angles greater than 15°.

Operation within the lower amber airspeed band for landing is normal for this condition.

Note: V/S and VNAV PTH modes can revert to LVL CHG mode.

Landing Checklist

- ENGINE START switches. CONT
- Speedbrake ARMED

| RA-73313 - RA-73319

Note: The SPEED BRAKE lever will not move beyond the FLIGHT DETENT on landing and the spoilers will not fully deploy.

- Landing gear Down
- Flaps _____, **Green or amber light**

▼ Continued on next page ▼

▼ Trailing Edge Flaps Up Landing continued ▼

Note: The light can be green or amber depending on the cause of the failure.


**YAW
DAMPER**
YAW DAMPER

Condition: The yaw damper is disengaged.

1 YAW DAMPER switch OFF then ON

2 Choose one:

◆ YAW DAMPER light **extinguishes:**



◆ YAW DAMPER light **stays illuminated:**

YAW DAMPER switch OFF

►► **Go to step 3**

3 Avoid areas of predicted moderate or severe turbulence. If turbulence is encountered and passenger comfort becomes affected, reduce airspeed and/or descend to a lower altitude.

(VQ-BUV) Do not use

4 Avoid areas of predicted moderate or severe turbulence. If turbulence is encountered and supernumerary comfort becomes affected, reduce airspeed and/or descend to a lower altitude.

5 Do not exceed flaps 30.



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Airspeed Unreliable

RA-73314, RA-73315, RA-73319

Condition: Airspeed or Mach indications are suspected to be unreliable. (Items which might indicate unreliable airspeed are listed in the Additional Information section.)

Objective: To identify a reliable airspeed indication, if possible, or to use the Flight With Unreliable Airspeed table in the Performance Inflight chapter for the remainder of the flight.

- 1 Autopilot (if engaged) Disengage
- 2 Autothrottle (if engaged) Disengage
- 3 F/D switches (both) OFF
- 4 Set the following gear up pitch attitude and thrust:
 - Flaps extended 10° and 80% N1
 - Flaps up 4° and 75% N1
-
- 5 PROBE HEAT switches Check ON
- 6 The following indications are **reliable**:
 - Attitude
 - N1
 - Ground speed
 - Radio altitude

▼ Continued on next page ▼

▼ **Airspeed Unreliable continued** ▼

Note: Stick shaker, overspeed warning and AIRSPEED LOW (as installed) alerts may sound erroneously or simultaneously.

RA-73319

Note: The Flight Path Vector and Pitch Limit Indicator may be unreliable.

RA-73314, RA-73315

Note: The Flight Path Vector and Pitch Limit Indicator may be unreliable on the PFD and HUD.

7 Choose one:

◆ Reliable airspeed indication **can** be determined:

Use the reliable airspeed indication for the remainder of the flight.

►► **Go to step 11**

◆ Reliable airspeed indication **cannot** be determined:

►► **Go to step 8**

8 Refer to the Flight With Unreliable Airspeed table in the Performance Inflight chapter and set the pitch attitude and thrust setting for the current airplane configuration and phase of flight.

▼ **Continued on next page** ▼

▼ **Airspeed Unreliable continued** ▼

9 **When** in trim and stabilized:

Compare the captain, first officer and standby airspeed indicators with the airspeed shown in the table.

An airspeed indication that differs by more than 20 knots or 0.03 Mach from the airspeed shown in the table should be considered **unreliable**.

10 Choose one:

- ◆ Reliable airspeed indication **can** be determined:

Use the reliable airspeed indication for the remainder of the flight.

►► **Go to step 11**

- ◆ Reliable airspeed indication **cannot** be determined:

►► **Go to step 12**

▼ **Continued on next page** ▼



▼ **Airspeed Unreliable continued** ▼

11 Choose one:

◆ **Captain’s or First Officer’s** airspeed indication is **reliable**:

Flight director switch
(reliable side)ON

▶▶ **Go to step 14**

◆ **Only** the **standby airspeed** indication is **reliable**:

Note: Do not use the autopilot,
autothrottle, or flight directors.

▶▶ **Go to step 16**

12 Set pitch attitude and thrust from the Flight With Unreliable Airspeed table in the Performance Inflight chapter for the airplane configuration and phase of flight, as needed.

13 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.

▼ **Continued on next page** ▼

▼ **Airspeed Unreliable continued** ▼

Note: Maintain visual conditions if possible.

Establish landing configuration early.

Radio altitude reference is available below 2,500 feet.

Use electronic and visual glideslope indicators, where available, for approach and landing.

Do not use the autopilot, autothrottle, or flight directors.

►► **Go to step 16**

14 Autopilot

(reliable side, if needed) Engage

Note: Autopilot may not engage or may disengage automatically.

15 Do not use the autothrottle.

16 Choose one:

◆ Captain's **or** First Officer's altitude indication **is reliable:**

Use the reliable altitude indication for the remainder of the flight.

►► **Go to step 17**

◆ Captain's **and** First Officer's altitude indications **are unreliable:**

Note: Airplane does not meet RVSM airspace requirements.

►► **Go to step 21**

▼ **Continued on next page** ▼



▼ **Airspeed Unreliable continued** ▼

17 Airplane may not meet RVSM airspace requirements.

18 Set transponder to provide reliable altitude reporting.

RA-73314, RA-73315, RA-73319

ALT selector (1 or 2) Select reliable side

(VP-BSA, VP-BSQ, VP-BSZ) **Do not use**

ALT SOURCE selector (1 or 2) Select reliable side

RA-73314, RA-73315, RA-73319

19 Transponder mode selector TA

(VP-BSA, VP-BSQ, VP-BSZ) **Do not use**

20 Transponder mode selector TA ONLY

►► **Go to step 22**

21 Set transponder altitude reporting off.

Transponder mode selector ALT OFF

22 Checklist Complete Except Deferred Items

Deferred Items

Review Before Descent

For approach, only set the BARO minimums on the **reliable** PFD. Remove the BARO minimums from the **unreliable** PFD.

▼ **Continued on next page** ▼

▼ Airspeed Unreliable continued ▼

Note: If BARO minimums are set only on the First Officer's PFD, DA/MDA aural callouts are not provided.

Go-Around Procedure Review

Refer to the Performance Inflight chapter to determine the go-around pitch attitude and thrust setting:

For **pitch attitude**, refer to the Flight with Unreliable Airspeed Go-Around table.

For **thrust setting**, refer to the Go-Around %N1 table.

▼ Continued on next page ▼

▼ **Airspeed Unreliable continued** ▼

Choose one:

◆ **Captain's or First Officer's** airspeed indication is **reliable**:

When TO/GA is pushed, the flight director pitch bar may automatically be removed. An AFDS pitch mode change, such as LVL CHG, restores the flight director pitch bar.

Note: Only use flight director guidance on the reliable PFD.



◆ **Only the standby airspeed** indication is **reliable**:

Do not use TO/GA.



◆ **Captain's, First Officer's, and standby airspeed** indications are **not reliable**:

Do not use TO/GA.

▼ **Continued on next page** ▼

▼ Airspeed Unreliable continued ▼**Additional Information**

One or more of the following may be associated with unreliable airspeed or Mach indications:

- Difference between captain and first officer airspeed indications
- Speed/altitude information not consistent with pitch attitude and thrust setting
- Blank or fluctuating airspeed indication
- Continuous or intermittent stick shaker
- IAS DISAGREE alert
- ALT DISAGREE alert
- AOA DISAGREE alert
- SPD failure flag
- SPD LIM failure flag
- Erroneous minimum speed bars
- Erroneous maximum speed bars
- **RA-73314, RA-73315, RA-73319**
- Airspeed low aural
- Airspeed low PFD indications on one side
- Overspeed warning
- Erroneous Flight Director (FD) pitch command bar
- Radome damage or loss

If the AOA DISAGREE alert is shown, one or more of the following flight deck effects can occur:

- Erroneous Pitch Limit Indicator (PLI)
- Windshear alerts
- Autoslat operation



Airspeed Unreliable
EASA-Europe or TCCA-Canada

| RA-73269, RA-73311 - RA-73313, RA-73316 - RA-73318, RA-73321

Condition: Airspeed or Mach indications are suspected to be unreliable. (Items which might indicate unreliable airspeed are listed in the Additional Information section.)

Objective: To identify a reliable airspeed indication, if possible, or to use the Flight With Unreliable Airspeed table in the Performance Inflight chapter for the remainder of the flight.

- 1 Autopilot (if engaged)Disengage
- 2 Autothrottle (if engaged)Disengage
- 3 F/D switches (both) OFF
- 4 Set the following gear up pitch attitude and thrust:
 Flaps extended. 10° and 80% N1
 Flaps up 4° and 75% N1

- 5 PROBE HEAT switches Check ON
- 6 The following indications are **reliable**:
 Attitude
 N1
 Ground speed
 Radio altitude

▼ Continued on next page ▼

▼ **Airspeed Unreliable EASA-Europe or TCCA-Canada continued** ▼

Note: Stick shaker, overspeed warning and AIRSPEED LOW (as installed) alerts may sound erroneously or simultaneously.

Note: The Flight Path Vector and Pitch Limit Indicator may be unreliable.

7 Choose one:

◆ Reliable airspeed indication **can** be determined:

Use the reliable airspeed indication for the remainder of the flight.

►► **Go to step 11**

◆ Reliable airspeed indication **cannot** be determined:

►► **Go to step 8**

8 Refer to the Flight With Unreliable Airspeed table in the Performance Inflight chapter and set the pitch attitude and thrust setting for the current airplane configuration and phase of flight.

9 **When** in trim and stabilized:

Compare the captain, first officer and standby airspeed indicators with the airspeed shown in the table.

An airspeed indication that differs by more than 20 knots or 0.03 Mach from the airspeed shown in the table should be considered **unreliable**.

▼ **Continued on next page** ▼



▼ **Airspeed Unreliable EASA-Europe or TCCA-Canada continued** ▼

10 Choose one:

- ◆ Reliable airspeed indication **can** be determined:

Use the reliable airspeed indication for the remainder of the flight.

▶▶ **Go to step 11**
- ◆ Reliable airspeed indication **cannot** be determined:

▶▶ **Go to step 12**

11 Choose one:

- ◆ **Captain’s or First Officer’s** airspeed indication is **reliable**:

Flight director switch
(reliable side) ON

▶▶ **Go to step 14**
- ◆ **Only** the **standby airspeed** indication is **reliable**:

Note: Do not use the autopilot, autothrottle, or flight directors.

▶▶ **Go to step 16**

12 Set pitch attitude and thrust from the Flight With Unreliable Airspeed table in the Performance Inflight chapter for the airplane configuration and phase of flight, as needed.

▼ **Continued on next page** ▼

▼ Airspeed Unreliable EASA-Europe or TCCA-Canada continued ▼

13 Check the Non-Normal Configuration Landing Distance table in the Performance Inflight-QRH chapter or other approved source.

Note: Maintain visual conditions if possible.

Establish landing configuration early.

Radio altitude reference is available below 2,500 feet.

Use electronic and visual glideslope indicators, where available, for approach and landing.

Do not use the autopilot, autothrottle, or flight directors.

►► Go to step 16

14 Autopilot
(reliable side, if needed) Engage

Note: Autopilot may not engage or may disengage automatically.

15 Do not use the autothrottle.

▼ Continued on next page ▼

▼ **Airspeed Unreliable EASA-Europe or TCCA-Canada continued** ▼

16 Choose one:

◆ Captain's **or** First Officer's altitude indication **is reliable**:

Use the reliable altitude indication for the remainder of the flight.

►► **Go to step 17**

◆ Captain's **and** First Officer's altitude indications **are unreliable**:

Note: Airplane does not meet RVSM airspace requirements.

►► **Go to step 21**

17 Airplane may not meet RVSM airspace requirements.

18 Set transponder to provide reliable altitude reporting.

RA-73311, RA-73312

ALT selector (1 or 2) Select reliable side

RA-73269, RA-73313, RA-73316 - RA-73318,

RA-73321

ALT SOURCE selector (1 or 2) Select reliable side

RA-73311, RA-73312

19 Transponder mode selector TA

RA-73269, RA-73313, RA-73316 - RA-73318,

RA-73321

20 Transponder mode selector TA ONLY

▼ **Continued on next page** ▼

▼ **Airspeed Unreliable EASA-Europe or TCCA-Canada continued** ▼

▶▶ **Go to step 22**

21 Set transponder altitude reporting off.

RA-73311 - RA-73313, RA-73316 - RA-73318

Transponder mode

selector. ALT OFF

RA-73269, RA-73321

Transponder mode

selector. ALT RPTG OFF

22 A nuisance stick shaker may be deactivated at pilot's discretion.

This improves recognition of a stall warning on the opposite side.

Note: Elevator Feel Shift may be active, resulting in increased control column forces.

23 Choose one:

◆ Deactivating stick shaker **is** needed:

Only the active stick shaker should be deactivated.

▶▶ **Go to step 24**

◆ Deactivating stick shaker **is not** needed:

▶▶ **Go to step 26**

24 **When** the circuit breaker is pulled, increased control column forces due to Elevator Feel Shift activation are removed.

Note: The stick shaker on the opposite side is not deactivated.

▼ **Continued on next page** ▼

▼ **Airspeed Unreliable EASA-Europe or TCCA-Canada continued** ▼

25 Choose one:

◆ Stick shaker **left** needs to be deactivated:

STICK SHAKER LEFT

circuit breaker (P18-2:E4) Pull

►► **Go to step 26**

◆ Stick shaker **right** needs to be deactivated:

STICK SHAKER RIGHT

circuit breaker (P6-1:B6) Pull

►► **Go to step 26**

26 Checklist Complete Except Deferred Items

Deferred Items

Review Before Descent

For approach, only set the BARO minimums on the **reliable** PFD. Remove the BARO minimums from the **unreliable** PFD.

Note: If BARO minimums are set only on the First Officer's PFD, DA/MDA aural callouts are not provided.

Go-Around Procedure Review

Refer to the Performance Inflight chapter to determine the go-around pitch attitude and thrust setting:

For **pitch attitude**, refer to the Flight with Unreliable Airspeed Go-Around table.

▼ **Continued on next page** ▼

▼ Airspeed Unreliable EASA-Europe or TCCA-Canada continued ▼

For **thrust setting**, refer to the Go-Around %N1 table.

Choose one:

◆ **Captain's or First Officer's** airspeed indication is **reliable**:

When TO/GA is pushed, the flight director pitch bar may automatically be removed. An AFDS pitch mode change, such as LVL CHG, restores the flight director pitch bar.

Note: Only use flight director guidance on the reliable PFD.



◆ **Only** the **standby airspeed** indication is **reliable**:

Do not use TO/GA.



◆ **Captain's, First Officer's, and standby** airspeed indications are **not reliable**:

Do not use TO/GA.



▼ Continued on next page ▼

▼ **Airspeed Unreliable EASA-Europe or TCCA-Canada continued** ▼

Additional Information

One or more of the following may be associated with unreliable airspeed or Mach indications:

- Difference between captain and first officer airspeed indications
- Speed/altitude information not consistent with pitch attitude and thrust setting
- Blank or fluctuating airspeed indication
- Continuous or intermittent stick shaker
- IAS DISAGREE alert
- ALT DISAGREE alert

| RA-73311 - RA-73313, RA-73316 - RA-73318
(SB Changes RA-73269. SB adds AOA DISAGREE alert.)

- AOA DISAGREE alert
- SPD failure flag
- SPD LIM failure flag
- Erroneous minimum speed bars
- Erroneous maximum speed bars

| RA-73311 - RA-73313, RA-73316 - RA-73318, RA-73321
(SB Changes RA-73269. Aural alert is added when SB is completed.)

- Airspeed low aural
- Airspeed low PFD indications on one side
- Overspeed warning
- Erroneous Flight Director (FD) pitch command bar
- Radome damage or loss

| RA-73311 - RA-73313, RA-73316 - RA-73318
(SB Changes RA-73269. SB adds AOA DISAGREE alert.)

▼ **Continued on next page** ▼

▼ Airspeed Unreliable EASA-Europe or TCCA-Canada continued ▼

If the AOA DISAGREE alert is shown, one or more of the following flight deck effects can occur:

- Erroneous Pitch Limit Indicator (PLI)
- Windshear alerts
- Autoslat operation

ALT DISAGREE

Condition: The ALT DISAGREE alert indicates the captain's and first officer's altitude indications disagree.

RA-73314, RA-73315, RA-73319

1 Choose one:

◆ IAS DISAGREE alert **is** shown:

▶▶ **Go to the Airspeed Unreliable checklist on page 10.1**



◆ IAS DISAGREE alert is **not** shown:

▶▶ **Go to step 3**

RA-73269 - RA-73313, RA-73316 - RA-73318, RA-73321

2 Choose one:

◆ IAS DISAGREE alert **is** shown:

▶▶ **Go to the Airspeed Unreliable EASA-Europe or TCCA-Canada checklist on page 10.10**



◆ IAS DISAGREE alert is **not** shown:

▶▶ **Go to step 3**

3 Verify all altimeters are set to correct barometric setting.

▼ Continued on next page ▼

▼ ALT DISAGREE continued ▼

4 Choose one:

◆ ALT DISAGREE alert **extinguishes**:

Continue normal operation.



◆ ALT DISAGREE alert **stays illuminated**:

Note: Airplane does not meet RVSM airspace requirements.

►► **Go to step 5**

5 Standby altimeter is available.

6 Do **not** use the flight path vector.

7 Maintain visual conditions if possible.

8 Choose one:

◆ A reliable altitude **can** be determined:

►► **Go to step 9**

◆ A reliable altitude **cannot** be determined:

►► **Go to step 10**

▼ Continued on next page ▼



▼ ALT DISAGREE continued ▼

9 Set transponder to provide reliable altitude reporting

- RA-73311, RA-73312, RA-73314, RA-73315, RA-73319
ALT selector (1 or 2) Select reliable side
- RA-73269, RA-73313, RA-73316 - RA-73318, RA-73321
ALT SOURCE selector (1 or 2) Select reliable side

▶▶ Go to step 11

10 Select transponder altitude reporting off.

- RA-73311 - RA-73319
Transponder mode selector ALT OFF
- RA-73269 - RA-73321
Transponder mode selector ALT RPTG OFF

11 Checklist Complete Except Deferred Items

Deferred Items

Review Before Descent:

For approach, only set the BARO minimums on the **reliable** PFD. Remove the BARO minimums from the **unreliable** PFD.

Note: If BARO minimums are set only on the First Officer’s PFD, DA/MDA aural callouts are not provided.

▼ Continued on next page ▼

▼ **ALT DISAGREE** continued ▼

Establish landing configuration early

Radio altitude reference is available below 2,500 feet

Use electronic and visual glideslope indicators, where available, for approach and landing.



AOA DISAGREE

RA-73311 - RA73319
(SB Changes RA-73269)

Condition: The AOA DISAGREE alert indicates the left and right angle of attack vanes disagree.

RA-73314, RA-73315, RA-73319

►► **Go to the Airspeed Unreliable checklist on page 10.1**

RA-73269 - RA-73313, RA-73316 - RA-73318

►► **Go to the Airspeed Unreliable EASA-Europe or TCCA-Canada checklist on page 10.10**



CDS FAULT

Condition: The CDS FAULT annunciation indicates a CDS fault occurs.

Note: CDS FAULT annunciates on the ground only, before the second engine start.

1 Do **not** takeoff.





Display Failure

Condition: A display in the common display system is failed.

1 Choose one:

◆ A single display is not usable and automatic switching **has** occurred:

Continue normal operation.



◆ A single display is not usable and automatic switching has **not** occurred:

▶▶ **Go to step 2**

2 MAIN PANEL DUs selectorAs needed

3 LOWER DU selector.As needed



DISPLAYS CONTROL PANEL

Condition: The DISPLAYS CONTROL PANEL annunciation indicates the EFIS control panel is failed.

Note: The altimeter blanks and an ALT flag illuminates on the side corresponding to the failed control panel.

- 1 CONTROL PANEL select
switch BOTH ON 1 or BOTH ON 2
Select the operating control panel.
- 2 Verify that the DISPLAYS CONTROL PANEL
annunciation and ALT flag extinguish.



DISPLAY SOURCE

**RA-73269, RA-73314 - RA-73318, RA-73321
(SB Changes RA-73313)**

Condition: The DSDLY SOURCE annunciation indicates only one DEU is supplying display information. Indications may include:

- No hydraulic pressure indication on the failed side
- Speed limit flag shown on the failed side
- Minimum maneuver speed and stick shaker band removed on the failed side
- Both EEC ALTN lights illuminated.

Note: Flight director indications may be removed and autoflight mode reversions may occur.

Dual autopilot approach is not available.

1 Choose one:

◆ DEU fails on the **same** side as the engaged autopilot:

Verify that the correct flight director indications and flight mode annunciations are shown on the operating side.

Select autopilot on the operating side.

►► **Go to step 2**

◆ DEU fails on the **opposite** side as the engaged autopilot:

►► **Go to step 2**

▼ **Continued on next page** ▼

▼ DISPLAY SOURCE continued ▼

2 Choose one:

◆ **EEC ALTN** lights are **illuminated**:

▶▶ **Go to step 3**

◆ **EEC ALTN** lights are **not** illuminated:



3 Choose one:

◆ **EEC ALTERNATE MODE** checklist has been **completed**:



◆ **EEC ALTERNATE MODE** checklist has **not** been completed:

▶▶ **Go to the EEC ALTERNATE MODE checklist on page 7.15**



DISPLAY SOURCE

**RA-73311, RA-73312, RA-73319
(SB Changes RA-73313)**

Condition: The DSDLY SOURCE 1 or DSDLY SOURCE 2 annunciation indicates only one DEU is supplying display information. Indications may include:

- No hydraulic pressure indication on the failed side
- Speed limit flag shown on the failed side
- Minimum maneuver speed and stick shaker band removed on the failed side
- Both EEC ALTN lights illuminated.

Note: Flight director indications may be removed and autoflight mode reversions may occur.

Dual autopilot approach is not available.

▼ Continued on next page ▼

▼ **DISPLAY SOURCE** continued ▼

1 Choose one:

◆ **DSPLY SOURCE 1** annunciation is shown:

Verify that the correct flight director indications and flight mode annunciations are shown on the first officer's side.

Select Autopilot B.

▶▶ **Go to step 2**

◆ **DSPLY SOURCE 2** annunciation is shown:

Verify that the correct flight director indications and flight mode annunciations are shown on the captain's side.

Select Autopilot A.

▶▶ **Go to step 2**

2 Choose one:

◆ **EEC ALTN** lights are **illuminated**:

▶▶ **Go to step 3**

◆ **EEC ALTN** lights are **not** illuminated:


▼ **Continued on next page** ▼

▼ DISPLAY SOURCE continued ▼

3 Choose one:

◆ **EEC ALTERNATE MODE** checklist has been completed:



◆ **EEC ALTERNATE MODE** checklist has **not** been completed:

▶▶ **Go to the EEC ALTERNATE MODE checklist on page 7.15**

**OFF****FLIGHT RECORDER OFF**

Condition: The flight recorder is off.

1 Continue normal operation.

**IAS DISAGREE**

Condition: The IAS DISAGREE alert indicates the captain's and first officer's airspeed indications disagree.

RA-73314, RA-73315, RA-73319

▶▶ **Go to the Airspeed Unreliable checklist on page 10.1**

RA-73269 - RA-73313, RA-73316 - RA-73318, RA-73321

▶▶ **Go to the Airspeed Unreliable EASA-Europe or TCCA-Canada checklist on page 10.10**





Non-Normal Checklists

Chapter NNC

Flight Management, Navigation

Section 11

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Intentionally
Blank

ADS-B Out Failure

Condition: One or more of these occur:

- ATC reports ADS-B Out is lost or degraded
- GPS-L or GPS-R INVALID message is in the FMC scratchpad and the transponder is selected to the related side.
- Transponder is failed

- 1 Transponder selector. Select opposite transponder



M
S
G

FMC DISAGREE

FMC
P/RST

Condition: Data needed for dual FMC operation disagree.

- 1 Choose one:

◆ Flying an approach with an **RNP alerting requirement**:

Go-around unless suitable visual references can be established and maintained.



◆ Flying an approach **without** an RNP alerting requirement:

Verify position.



**M
S
G****FMC DISAGREE - VERTICAL****FMC**
P/RST

Condition: One of the following occur:

- Left FMC and right FMC target airspeeds disagree during descent
- Left FMC and right FMC vertical paths disagree during descent.

- 1 Do **not** move the FMC source select switch.

The FMC will attempt to correct the difference without crew action.

- 2 Monitor crossing altitudes to ensure compliance.

**Additional Information**

The disagreement will most likely be corrected when crossing a waypoint with an altitude constraint.

If the disagreement remains when entering the approach phase, the FMC DISAGREE - VERTICAL message will be replaced with the FMC DISAGREE message. An approach with an RNP alerting requirement is not authorized when the FMC DISAGREE message is shown.

FMC
P/RST

FMC FAIL

Condition: Display of VTK followed by MAP failure flags on one or both navigation displays.

Objective: To restore dual FMC operation, configure for single FMC operation or resume conventional navigation.

Note: LNAV/VNAV may disengage.

- 1 Check both navigation display MAP modes and FMC CDUs.
- 2 Choose one:
 - ◆ No FMC data on the captain's and first officer's navigation display MAP modes **and** no FMC data on both CDUs:

Both FMCs are failed.

▶▶ **Go to step 7**
 - ◆ No FMC data on the captain's navigation display MAP mode **and** no FMC data on both CDUs:

Left FMC is failed.

▶▶ **Go to step 3**
 - ◆ No FMC data on the first officer's navigation display MAP mode, MSG light is illuminated **and** SINGLE FMC OPERATION scratchpad message is shown:

Right FMC is failed.

▶▶ **Go to step 3**

▼ Continued on next page ▼

▼ FMC FAIL continued ▼

- 3 FMC source
select switch BOTH ON L or BOTH ON R
Select the operating FMC.
- 4 **Wait** 1 minute.
- 5 Check for FMC-L and FMC-R bearing/distance
information on POS SHIFT page 3/3.
- 6 Choose one:
 - ◆ Bearing/distance information for **both** FMCs are
displayed:
FMC source select switch NORMAL
■ ■ ■ ■
 - ◆ Bearing/distance information for only a **single**
FMC is displayed:
Continue with single FMC operation.
■ ■ ■ ■
- 7 Resume conventional navigation. Without an
operating FMC, LNAV and VNAV are not available.
- 8 **RA-73311 - RA-73319**
Verify position relative to terrain using conventional
navigation.

Note: EGPWS may use inaccurate GPS position data or an inappropriate value of RNP. This could result in a VSD terrain display that is incorrectly positioned relative to the airplane track.

▼ Continued on next page ▼

▼ FMC FAIL continued ▼

9 **When** preparing for the approach:

Use the SPD REF selector to set the current gross weight.

Use the SPD REF selector to set the reference airspeed bugs.

Use the N1 SET selector to set the N1 bugs.

Note: For additional information on using the N1 SET and SPD REF selectors refer to:

- Setting N1 Bugs with No Operative FMC Supplementary Procedure (SP 7).
- Setting Airspeed Bugs with No Operative FMC Supplementary Procedure (SP 10).



MSG

FMC/CDU ALERTING MESSAGE

FMC
P/RST

Condition: An alert message is in the FMC scratchpad.

1 Take action as needed by the message.



GLS**GLS****RA-73319**

Condition: A GLS failure occurs.

- 1 Do not fly a GLS approach.
- 2 ILS and non-ILS approaches may be flown.

**GPS****GPS**

Condition: Both GPS receivers are failed.

Note: The FMC uses only IRS or radio inputs.

Look-ahead terrain alerting and display are unavailable due to position uncertainty.

ADS-B is inoperative.

- 1 Continue normal operation if ANP meets the requirements for the phase of flight.

**ILS****ILS****RA-73319**

Condition: An ILS failure occurs.

- 1 Do not fly an ILS approach.
- 2 GLS and non-ILS approaches may be flown.



DC FAIL
IRS DC FAIL

Condition: IRS backup DC power is failed.

1 Choose one:

- ◆ **One** IRS DC FAIL light is illuminated **and** all other IRS lights are **extinguished**:

Continue normal operation. The IRS is operating normally on AC power.



- ◆ **Both** IRS DC FAIL lights are illuminated:

The IRS is operating normally on AC power.

The battery is nearly discharged or the switched hot battery bus is not powered.
The following systems may be inoperative:

Engine and APU fire extinguishing
APU start.



FAULT**IRS FAULT****RA-73313, RA-73316 - RA-73319**

Condition: One or more of these occur:

- An IRS fault occurs
- On the ground, if the ALIGN light is also illuminated, the present position entry is possibly incorrect.

1 Choose one:

◆ On the **ground**:▶▶ **Go to step 2**◆ In **flight**:▶▶ **Go to step 6**

On the ground**▼ Continued on next page ▼**

▼ IRS FAULT continued ▼

2 Choose one:

◆ ALIGN light is **extinguished**:

Do not takeoff.



◆ ALIGN light is also **illuminated**:

IRS mode selector

(failed side) OFF

The FAULT light extinguishes immediately and the ALIGN light extinguishes after approximately 30 seconds.

▶▶ **Go to step 3**

3 **After** the ALIGN light extinguishes:

IRS mode selector

(failed side) NAV

Enter present position.

4 Choose one:

◆ ALIGN light is **flashing**:

Re-enter present position using the most accurate latitude and longitude.

▶▶ **Go to step 5**

ALIGN light is **not** flashing:

▶▶ **Go to step 5**

▼ Continued on next page ▼

▼ IRS FAULT continued ▼

5 Choose one:

◆ FAULT light **illuminates** again:

Do not takeoff.



◆ FAULT light does **not** illuminate again:

Verify IRS alignment is complete.



In flight

6 IRS transfer switch . . . BOTH ON L or BOTH ON R
Select the operating IRS.

Note: The selected IRS is the source of:

- Attitude and heading for both PFDs and NDs
- AFDS guidance
- Weather radar information
- GPWS and Windshear alerts

7 The autopilot on the side with the operational IRS can be used **except** during approach. Autopilot use during approach is not authorized.

Note: If the autopilot is engaged, the yaw damper disconnects after approximately 1 minute and cannot be reconnected until the autopilot is disengaged.

8 Checklist Complete Except Deferred Items

▼ Continued on next page ▼

▼ IRS FAULT continued ▼

Deferred Items

Descent Checklist

Pressurization LAND ALT ____
 Recall Checked
 Autobrake ____
 Landing data VREF ____, Minimums ____
 Approach briefing Completed

Approach Checklist

Altimeters ____

Prior to Start of the Approach

Autopilot Disengage
 Autothrottle (if engaged) Disengage
 YAW DAMPER switch ON

Landing Checklist

ENGINE START switches CONT
 Speedbrake ARMED
 Landing gear Down
 Flaps ____, Green light



FAULT**IRS FAULT**

RA-73269, RA-73311, RA-73312, RA-73314, RA-73315, RA-73319, RA-73321

Condition: One or more of these occur:

- An IRS fault occurs
- On the ground, if the ALIGN light is also illuminated, the present position entry is possibly incorrect.

1 Choose one:

◆ On the **ground**:

▶▶ **Go to step 2**

◆ In **flight**:

▶▶ **Go to step 6**

On the ground

▼ Continued on next page ▼

▼ IRS FAULT continued ▼

2 Choose one:

◆ ALIGN light is **extinguished**:

Do not takeoff.



◆ ALIGN light is also **illuminated**:

IRS mode selector

(failed side) OFF

The FAULT light extinguishes immediately and the ALIGN light extinguishes after approximately 30 seconds.

►► **Go to step 3**

3 **After** the ALIGN light extinguishes:

IRS mode selector

(failed side) NAV

Enter present position.

4 Choose one:

◆ ALIGN light is **flashing**:

Re-enter present position using the most accurate latitude and longitude.

►► **Go to step 5**

◆ ALIGN light is **not** flashing:

►► **Go to step 5**

▼ Continued on next page ▼



▼IRS FAULT continued▼

5 Choose one:

◆FAULT light **illuminates** again:

Do not takeoff.



◆FAULT light does **not** illuminate again:

Verify IRS alignment is complete



In flight

6 Autopilot (if engaged)Disengage

7 IRS transfer switchBOTH ON L or BOTH ON R

Select the operating IRS.

Note: The selected IRS is the source of:

- Attitude and heading for both PFDs and NDs
- AFDS guidance
- Weather radar information
- GPWS and Windshear alerts

Note: Autopilot(s) cannot be engaged.

8 Autothrottle (if engaged)Disengage



ON DC
IRS ON DC

Condition: IRS AC power is failed.

1 Choose one:

◆ **Left** IRS ON DC light is illuminated:

The left IRS continues to operate as long as DC power is available.


◆ **Right** IRS ON DC light is illuminated:

Power to the right IRS is removed after 5 minutes.



UNABLE REQD NAV PERF - RNP

Condition: UNABLE REQD NAV PERF-RNP is shown. The actual navigation performance is not sufficient.

1 Choose one:

◆ On a procedure or airway with an **RNP alerting requirement**:

▶▶ **Go to step 2**

◆ On a procedure or airway **without** an RNP alerting requirement:

▶▶ **Go to step 3**

2 Select an alternate procedure or airway. During an approach, go-around unless suitable visual references can be established and maintained.

Note: The FMC uses inertial inputs only, unless radio updating is enabled.

If radio updating is allowed, consider doing the following steps:

- Select the FMC NAV OPTIONS page

RA-73314, RA-73315, RA-73319

- Select DME UPDATE ON
- Select VOR UPDATE ON



▼ Continued on next page ▼

▼UNABLE REQD NAV PERF - RNP continued▼**3** Verify position.

Note: The FMC uses inertial inputs only, unless radio updating is enabled.

If radio updating is allowed, consider doing the following steps:

- Select the FMC NAV OPTIONS page

RA-73314, RA-73315, RA-73319

- Select DME UPDATE ON
- Select VOR UPDATE ON



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CONFIG**(VP-BPI, VP-BPY) Do not use**

Condition: All of these occur:

- Both center tank fuel pump switches are off
- There is more than 1600 lbs of fuel in the center tank
- An engine is running.

Condition: All of these occur:

- Both center tank fuel pump switches are off
- There is more than 726 kgs of fuel in the center tank
- An engine is running.

- 1 Do not accomplish this procedure until established in a level flight attitude.
- 2 CTR FUEL PUMP switches (both). ON
Verify that the LOW PRESSURE lights extinguish.
- 3 Resume normal fuel management.



**CROSSFEED SELECTOR
INOPERATIVE****VALVE
OPEN****May or may not be illuminated**

Condition: One of the following occurs:

- The crossfeed VALVE OPEN light stays illuminated bright blue
- The CROSSFEED selector is in the open position and the crossfeed VALVE OPEN light is extinguished
- When the CROSSFEED selector is rotated to a new position, the crossfeed VALVE OPEN light does not illuminate bright blue.

- 1 Check the FUEL CROSSFEED VALVE circuit breaker (P6-3:B7).

▼ Continued on next page ▼

▼ CROSSFEED SELECTOR INOPERATIVE continued ▼

2 Choose one:

◆ FUEL CROSSFEED VALVE circuit breaker is **tripped**:

It is not possible to determine the position of the fuel crossfeed valve.

Note: Verify that sufficient fuel is available to both engines to complete the flight.

Do **not** accomplish the following checklist:

IMBAL



◆ FUEL CROSSFEED VALVE circuit breaker is **not** tripped:

►► **Go to step 3**

▼ Continued on next page ▼

▼ CROSSFEED SELECTOR INOPERATIVE continued ▼

3 Choose one:

◆ CROSSFEED selector is in the **closed** position:

Crossfeed **valve** is failed open.

Note: Maintain fuel balance with selective use of fuel pumps.

◆ CROSSFEED selector is in the **open** position:

Crossfeed **valve** is failed closed.

Note: As conditions allow, vary engine thrust as needed to maintain fuel balance.

Verify that sufficient fuel is available to both engines to complete the flight.



**FILTER
BYPASS**
FUEL FILTER BYPASS

Condition: Fuel contamination can cause fuel to bypass the engine fuel filter.

1 Choose one:

- ◆ Only **one** FILTER BYPASS light, ENG 1 or ENG 2, has illuminated during the flight:

Note: Erratic engine operation and flameout may occur on the affected engine due to fuel contamination.



- ◆ FILTER BYPASS lights for **both** engines illuminate or have illuminated at any time during the flight (either separately or at the same time):

►► **Go to step 2**

2 Plan to land at the nearest suitable airport.

Note: Erratic engine operation and flameout may occur on either or both engines due to fuel contamination.





Fuel Leak Engine

Condition: An engine fuel leak is suspected for the reasons listed in the Additional Information section of this checklist.

Objective: To confirm there is an engine fuel leak and shut down the affected engine if needed. This checklist does not address the unlikely possibility of a tank leak.

- 1 A diversion may be needed.
 - 2 Main tank FUEL PUMPS switches (all) ON
 - 3 CROSSFEED selector Close
 - 4 CTR FUEL PUMPS switches (both) OFF
- The fuel CONFIG alert may show with fuel in the center tank.

The following steps check for an engine fuel leak

- 5 Record the main tank fuel quantities and the current time.
- 6 An engine fuel leak is confirmed if one or both of the following are true:
 - Fuel spray is observed from an engine or strut
 - (VP-BPI, VP-BPY) Do not use
 - A change in fuel imbalance of 500 lbs within 30 minutes or less

 - A change in fuel imbalance of 230 kgs within 30 minutes or less

▼ Fuel Leak Engine continued ▼

7 Choose one:

◆ Engine fuel leak is **confirmed**:

▶▶ **Go to step 11**

◆ Engine fuel leak is **not** confirmed:

▶▶ **Go to step 8**

8 Choose one:

◆ The center tank **contains** usable fuel:

▶▶ **Go to step 9**

◆ The center tank does **not** contain usable fuel:

▶▶ **Go to step 10**

9 CTR FUEL PUMPS switches (both) ON

10 Resume normal fuel management.



An engine fuel leak is confirmed

11 The following steps shut down the engine to stop an engine fuel leak.

12 The engine to be shut down is the engine on the side where the fuel quantity decreased faster.

13 Autothrottle Disengage

14 Thrust lever
(affected engine) Confirm Close

▼ Continued on next page ▼



▼Fuel Leak Engine continued▼

15 **When** the affected engine is at idle thrust:

Engine start lever
(affected engine) Confirm CUTOFF

This closes the spar valve and stops an engine fuel leak.

16 PACK switch (affected side) OFF

This causes the operating pack to regulate to high flow in flight with the flaps up.

RA-73311, RA-73412, RA-73314, RA-73315,
RA-73319

17 Choose one:

◆APU is **available** for start:

APU START

When APU is running:

APU GEN switch
(affected side)ON

►►Go to step 18

◆APU is **not** available:

►►Go to step 18

RA-73311, RA-73412, RA-73314, RA-73315,
RA-73319

18 Transponder mode selector TA

This prevents climb commands which can exceed single engine performance capability.

▼ Continued on next page ▼

▼ Fuel Leak Engine continued ▼

RA-73269, RA-73313, RA-73316 - RA-73318,
RA-73321

19 Choose one:

◆ APU is **available** for start:

APU START

When APU is running:

APU GEN switch
(affected side) ON

▶▶ **Go to step 20**

◆ APU is **not** available:

▶▶ **Go to step 20**

RA-73269, RA-73313, RA-73316 - RA-73318,
RA-73321

20 Transponder mode selector TA ONLY

This prevents climb commands which can
exceed single engine performance capability.

21 ISOLATION VALVE switch Verify AUTO

This ensures bleed air is available to both wings
if wing anti-ice is needed.

22 Choose one:

◆ Fuel LOW alert is **shown**:

▶▶ **Go to step 23**

◆ Fuel LOW alert is **not** shown:

▶▶ **Go to step 25**

▼ Continued on next page ▼



▼Fuel Leak Engine continued▼

23 CROSSFEED selector. Open

This ensures that all fuel is available to the running engine.

24 FUEL PUMPS switches (all). ON

This ensures that all fuel is available for use.

25 Plan to land at the nearest suitable airport.

Note: Balance fuel as needed. All remaining fuel can be used for the running engine.

Note: Do not use FMC performance predictions.

►► **Go to the One Engine Inoperative Landing checklist on page 7.34**



Additional Information

Reasons that an engine fuel leak should be suspected:

- A visual observation of fuel spray
- The total fuel quantity is decreasing at an abnormal rate
- An engine has excessive fuel flow
- The fuel IMBAL alert shows
- The fuel LOW alert shows
- The USING RSV FUEL message shows on the FMC CDU
- The INSUFFICIENT FUEL message shows on the FMC CDU
- The CHECK FMC FUEL QUANTITY message shows on the FMC CDU.



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**LOW
PRESSURE**
FUEL PUMP LOW PRESSURE

Condition: The fuel pump pressure is low.

Note: Fuel pump LOW PRESSURE lights may flicker when tank quantity is low and the airplane is in turbulent air or during climb or descent.

1 Choose one:

◆ **One main** tank fuel pump LOW PRESSURE light is illuminated:

Main tank FUEL PUMP switch
(affected pump). OFF

Sufficient fuel pressure is available
for normal operation.



◆ **Both main** tank fuel pump LOW PRESSURE lights are illuminated:

Note: At high altitude, thrust deterioration or engine flameout may occur.



◆ **One CTR** tank fuel pump LOW PRESSURE light is illuminated:

▶▶ **Go to step 2**

◆ **Both CTR** tank fuel pump LOW PRESSURE lights are illuminated:

▶▶ **Go to step 5**

▼ Continued on next page ▼

▼ **FUEL PUMP LOW PRESSURE** continued ▼

- 2 CROSSFEED selector. Open
This prevents fuel imbalance.
- 3 CTR FUEL PUMP switch (affected side) OFF
- 4 **When** the other CTR tank fuel pump LOW PRESSURE light illuminates:
CROSSFEED selector Close
Remaining CTR FUEL PUMP switch OFF



Both CTR tank fuel pump LOW PRESSURE lights are illuminated

- 5 CTR FUEL PUMP switches (both). OFF
- 6 Fuel CONFIG alert may show with fuel in the center tank.
- 7 Center tank fuel is unusable. Main tank fuel may not be sufficient for the planned flight.



Fuel Quantity Indication Inoperative

Condition: The fuel quantity indication is blank.

- 1 Enter and periodically update the manually calculated FUEL weight on the FMC PERF INIT page.



Fuel Temperature Low

Condition: Fuel temperature is near the minimum.

- 1 **When** fuel temperature is approaching the fuel temperature limit (3° C /5° F above the fuel freeze point or - 43° C /- 45° F whichever is higher):

Increase speed, change altitude and/or deviate to a warmer air mass to achieve a TAT equal to or higher than the fuel temperature limit.

TAT will increase approximately 0.5 to 0.7° C for each 0.01 Mach increase in speed. In extreme conditions, it may be necessary to descend as low as FL250.



IMBAL

Condition: There is a fuel imbalance between the main tanks.

Objective: To decide if a fuel leak is suspected. To balance fuel if a fuel leak is not suspected.

- 1 If an engine has low fuel flow and unusual engine indications, the IMBAL alert may show due to an engine malfunction instead of a fuel leak.
- 2 The IMBAL alert may be caused by a fuel leak, an inoperative crossfeed valve or a fuel imbalance.
- 3 A fuel leak should be suspected if one or more of the following are true:

The total fuel remaining is less than the planned fuel remaining

An engine has excessive fuel flow.

- 4 Choose one:

◆ A fuel leak is **suspected**:

▶▶ **Go to the Fuel Leak Engine checklist on page 12.6**



◆ A fuel leak is **not** suspected:

▶▶ **Go to step 5**

▼ Continued on next page ▼



▼IMBAL continued▼

5 CROSSFEED selector Open

Verify that the VALVE OPEN light illuminates bright, then dim. This indicates that the crossfeed valve is operating correctly.

6 Choose one:

◆ Crossfeed valve is operating **correctly**:

▶▶ **Go to step 7**

◆ Crossfeed valve is **not** operating correctly:

▶▶ **Go to the CROSSFEED SELECTOR
INOPERATIVE checklist on page
12.2**



▼ Continued on next page ▼

▼ **IMBAL continued ▼**

7 Choose one:

◆ **Main tank 1** quantity is low:

Main tank 1 FUEL PUMPS

switches (both) OFF

This allows fuel from the higher
quantity tank to feed both engines.

►► **Go to step 8**

◆ **Main tank 2** quantity is low:

Main tank 2 FUEL PUMPS

switches (both) OFF

This allows fuel from the higher
quantity tank to feed both engines.

►► **Go to step 8**

8 **When** fuel balancing is complete:

Main tank FUEL PUMPS switches (all) ON

CROSSFEED selector Close



LOW

Condition: The fuel quantity is low in a main tank.

Objective: To decide if a fuel leak is suspected. To ensure that all fuel is available for use.

Note: Avoid high nose up attitude. Make thrust changes slowly and smoothly. This reduces the possibility of uncovering fuel pumps.

- 1 The fuel LOW alert may be caused by a fuel leak or low fuel.
- 2 A fuel leak should be suspected if one or more of the following are true:

The total fuel remaining is less than the planned fuel remaining.

An engine has excessive fuel flow.

One main tank is abnormally low compared to the other main tank and to the expected fuel remaining in the tanks.

- 3 Choose one:

◆ A fuel leak is **suspected**:

▶▶ **Go to the Fuel Leak Engine checklist on page 12.6**



◆ A fuel leak is **not** suspected:

▶▶ **Go to step 4**

▼ Continued on next page ▼

▼LOW continued▼

- 4 CROSSFEED selector. Open
This ensures that fuel is available to both engines if the low tank empties.
- 5 FUEL PUMPS switches (all). ON
This ensures that all fuel is available for use.
- 6 Plan to land at the nearest suitable airport.



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Non-Normal Checklists

Chapter NNC

Hydraulics

Section 13

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**LOW
PRESSURE**
**HYDRAULIC PUMP LOW
PRESSURE**

Condition: The hydraulic pump pressure is low.

1 HYD PUMP switch (affected side) OFF

Note: Loss of an engine-driven hydraulic pump and a high demand on the system may result in an intermittent illumination of the LOW PRESSURE light for the remaining electric motor-driven hydraulic pump.



OVERHEAT

HYDRAULIC PUMP OVERHEAT

Condition: The hydraulic pump temperature is high.

1 ELEC HYD PUMP switch (affected side) OFF

Note: One pump supplies sufficient pressure for normal system operation.

2 MFD SYS switch Push

3 Choose one:

◆ Affected side hydraulic pressure is 3300 psi **or less:**

■ ■ ■ ■

◆ Affected side hydraulic pressure is **greater than** 3300 psi:

ENG HYD PUMP switch
(affected side) OFF

►► **Go to step 4**

4 Choose one:

◆ **System A** HYD PUMPS switches are OFF:

►► **Go to the LOSS OF SYSTEM A checklist on page 13.3**

■ ■ ■ ■

◆ **System B** HYD PUMPS switches are OFF:

►► **Go to the LOSS OF SYSTEM B checklist on page 13.6**

■ ■ ■ ■

LOSS OF SYSTEM A

FLT CONTROL
A HYD PUMPS
A
ENG 1
ELEC 2

**LOW
PRESSURE**

**LOW
PRESSURE**

**LOW
PRESSURE**

Condition: Hydraulic system A pressure is low.

- 1 System A
FLT CONTROL switch. . . . Confirm STBY RUD
- 2 System A
HYD PUMP switches (both) OFF
- 3 Check the Non-Normal Configuration Landing
Distance tables in the Performance Inflight-QRH
chapter or other approved source.
- 4 NOSE WHEEL STEERING switch ALT
- 5 Plan for manual gear extension.

Note: When the gear has been lowered manually, it cannot be retracted. The drag penalty with gear extended may make it impossible to reach an alternate field.

▼ Continued on next page ▼



▼ LOSS OF SYSTEM A continued ▼

Note: Inoperative Items

Autopilot A inop
Autopilot B is available.

Flight spoilers (two on each wing) inop
Roll rate and speedbrake effectiveness may be reduced in flight.

Normal landing gear extension and retraction inop
Manual gear extension is needed.

Ground spoilers inop
Landing distance will be increased.

Alternate brakes inop
Normal brakes are available.

Engine 1 thrust reverser normal hydraulic pressure inop
Thrust reverser will deploy and retract at a slower rate and some thrust asymmetry can be anticipated during thrust reverser deployment.

Normal nose wheel steering inop
Alternate nose wheel steering is available.

6 Checklist Complete Except Deferred Items

Deferred Items

Descent Checklist

Pressurization LAND ALT ____
Recall Checked
Autobrake ____
Landing data VREF ____, Minimums ____

▼ Continued on next page ▼

▼ LOSS OF SYSTEM A continued ▼

Approach briefing Completed

Approach Checklist

Altimeters _____

Manual Gear Extension

LANDING GEAR lever. OFF

Manual gear extension handles. Pull

The uplock is released when the handle is pulled to its limit.

The related red landing gear indicator light illuminates, indicating uplock release.

Wait 15 seconds after the last manual gear extension handle is pulled:

LANDING GEAR lever DN

Landing Checklist

ENGINE START switches CONT

Speedbrake ARMED

Landing gear Down

Flaps _____, Green light



LOSS OF SYSTEM B

FLT CONTROL**B**
**LOW
PRESSURE**
B HYD PUMPS**ELEC 1**
**LOW
PRESSURE**
ENG 2
**LOW
PRESSURE**

Condition: Hydraulic system B pressure is low.

- 1 System B
FLT CONTROL switch Confirm STBY RUD
- 2 System B
HYD PUMP switches (both). OFF
- 3 Plan a flaps 15 landing.
- 4 Set VREF 15 or VREF ICE.

Note: If any of the following conditions apply, set VREF ICE = VREF 15 + 10 knots:

- Engine anti-ice will be used during landing
- Wing anti-ice has been used any time during the flight
- Icing conditions were encountered during the flight and the landing temperature is below 10° C.

Note: When VREF ICE is needed, the wind additive should not exceed 5 knots.

- 5 Plan to extend flaps to 15 using alternate flap extension.

Note: The drag penalty with the leading edge devices extended may make it impossible to reach an alternate field.

▼ Continued on next page ▼

▼ LOSS OF SYSTEM B continued ▼

- 6 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.
- 7 Do **not** arm the autobrake for landing. Use manual braking.

▼ Continued on next page ▼

▼ LOSS OF SYSTEM B continued ▼

Note: Inoperative Items**Autopilot B inop**

Autopilot A is available.

Flight spoilers (two on each wing) inop

Roll rate and speedbrake effectiveness may be reduced in flight.

Yaw damper inop**Trailing edge flaps normal hydraulic system inop**

The trailing edge flaps can be operated with the alternate electrical system. Alternate flap extension time to flaps 15 is approximately 2 minutes.

Leading edge flaps and slats normal hydraulic system inop

The leading edge flaps and slats can be extended with standby pressure. Once extended, they can not be retracted.

Autobrake inop

Use manual braking.

Normal brakes inop

Alternate brakes are available.

Engine 2 thrust reverser normal hydraulic pressure inop

Thrust reverser will deploy and retract at a slower rate and some thrust asymmetry can be anticipated during thrust reverser deployment.

Alternate nose wheel steering inop

Normal nose wheel steering is available.

8 Checklist Complete Except Deferred Items

▼ Continued on next page ▼

▼ LOSS OF SYSTEM B continued ▼

Deferred Items

Descent Checklist

Pressurization LAND ALT _____

Recall Checked

Autobrake **OFF**

Landing data **VREF 15 or VREF ICE ____,**
Minimums

Approach briefing Completed

Approach Checklist

Altimeters

▼ Continued on next page ▼

▼ LOSS OF SYSTEM B continued ▼

Alternate Flap Extension

During flap extension, set the flap lever to the desired flap position.

 **230K maximum during alternate flap extension.**
ALTERNATE FLAPS master switch ARM

Note: The landing gear configuration warning may sound if the flaps are between 10 and 15 and the landing gear are retracted.

| RA-73313 - RA-73319

Note: The amber LE FLAPS TRANSIT light stays illuminated until the flaps approach the flaps 15 position.

| RA-73311, RA-73312, RA-73269, RA-73321

Note: The amber LE FLAPS TRANSIT light stays illuminated until the flaps approach the flaps 10 position.

Note: Operation within the lower amber airspeed band may be needed until the LE FLAPS TRANSIT light extinguishes.

▼ Continued on next page ▼

▼ **LOSS OF SYSTEM B continued** ▼

 If flap asymmetry occurs, release the switch immediately. There is no asymmetry protection.

ALTERNATE FLAPS
position switch Hold DOWN

to extend flaps
to 15 on schedule

As flaps are extending, slow to respective
maneuvering speed.

Additional Deferred Item

GROUND PROXIMITY FLAP

INHIBIT switch FLAP INHIBIT

Landing Checklist

ENGINE START switches CONT

Speedbrake ARMED

Landing gear Down

Flaps. **15, Green light**



MANUAL REVERSION or LOSS OF SYSTEM A AND SYSTEM B

FLT CONTROL**A****B****LOW
PRESSURE****LOW
PRESSURE****HYD PUMPS****ENG 1 ELEC 2****ELEC 1 ENG 2****LOW
PRESSURE****LOW
PRESSURE**

Condition: Hydraulic system A and B pressures are low.

- 1 System A and B FLT CONTROL switches (both) Confirm STBY RUD
- 2 YAW DAMPER switch ON
- 3 System A and B HYD PUMPS switches (all) OFF
- 4 Plan to land at the nearest suitable airport.
- 5 Plan a flaps 15 landing.
- 6 Set VREF 15 or VREF ICE.

Note: If any of the following conditions apply, set VREF ICE = VREF 15 + 10 knots:

- Engine anti-ice will be used during landing
- Wing anti-ice has been used any time during the flight
- Icing conditions were encountered during the flight and the landing temperature is below 10° C.

Note: When VREF ICE is needed, the wind additive should not exceed 5 knots.

▼ Continued on next page ▼

**▼ MANUAL REVERSION or LOSS OF SYSTEM A AND SYSTEM B
continued ▼**

- 7 Plan to extend flaps to 15 using alternate flap extension.

Note: The drag penalty with the leading edge devices extended may make it impossible to reach an alternate field.

- 8 Plan for manual gear extension.

Note: When the gear has been lowered manually, it cannot be retracted. The drag penalty with gear extended may make it impossible to reach an alternate field.

- 9 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.

Note: The crosswind capability of the airplane is greatly reduced.

- 10 Do not arm the autobrake for landing.
- 11 Do not arm the speedbrakes for landing.
- 12 On touchdown, apply steady brake pressure without modulating the brakes.
- 13 Do not attempt to taxi the airplane after stopping.

▼ Continued on next page ▼

**▼ MANUAL REVERSION or LOSS OF SYSTEM A AND SYSTEM B
continued ▼****Note: Inoperative Items****Autopilots A and B inop****All flight spoilers inop**

Roll rate will be reduced and speedbrakes will not be available in flight.

Trailing edge flaps normal hydraulic system inop

The trailing edge flaps can be operated with the alternate electrical system. Alternate flap extension time to flaps 15 is approximately 2 minutes.

Leading edge flaps and slats normal hydraulic system inop

The leading edge flaps and slats can be extended with standby hydraulic pressure. Once extended, they can not be retracted.

Normal landing gear extension and retraction inop

Manual gear extension is needed.

Autobrake inop**Ground spoilers inop**

Landing distance will be increased.

Normal and alternate brakes inop

Inboard and outboard brakes have accumulator pressure only. On landing, apply steady brake pressure without modulating the brakes.

Both thrust reversers normal pressure inop

Thrust reversers will deploy and retract at a slower rate.

Nose wheel steering inop

Do not attempt to taxi the airplane after stopping.

▼ Continued on next page ▼

**▼ MANUAL REVERSION or LOSS OF SYSTEM A AND SYSTEM B
continued ▼**

14 Checklist Complete Except Deferred Items

Deferred Items

Descent Checklist

Pressurization LAND ALT ____

Recall Checked

Autobrake **OFF**

Landing data **VREF 15 or VREF ICE** ____,
Minimums ____

Approach briefing Completed

Go-Around Procedure Review

Do the normal go-around procedure except:

- Advance thrust to go-around smoothly and slowly to avoid excessive pitch-up.
- Be prepared to trim.
- Limit bank angle to 15° when airspeed is less than the minimum maneuver speed.

Approach Checklist

Altimeters ____

▼ Continued on next page ▼

**▼ MANUAL REVERSION or LOSS OF SYSTEM A AND SYSTEM B
continued ▼****Alternate Flap Extension**

During flap extension, set the flap lever to the desired flap position.

 **230K maximum during alternate flap extension.**
! ALTERNATE FLAPS master switch ARM

Note: The landing gear configuration warning may sound if the flaps are between 10 and 15 and the landing gear are retracted.

| RA-73313 - RA-73319

Note: The amber LE FLAPS TRANSIT light stays illuminated until the flaps approach the flaps 15 position.

| RA-73311, RA-73312, RA-73269, RA-73321

Note: The amber LE FLAPS TRANSIT light stays illuminated until the flaps approach the flaps 10 position.

Note: Operation within the lower amber airspeed band may be needed until the LE FLAPS TRANSIT light extinguishes.

▼ Continued on next page ▼

▼ **MANUAL REVERSION or LOSS OF SYSTEM A AND SYSTEM B**
continued ▼

 If flap asymmetry occurs, release the switch immediately. There is no asymmetry protection.

ALTERNATE FLAPS

position switch Hold DOWN
to extend flaps
to 15 on schedule

As flaps are extending, slow to respective
maneuvering speed.

Manual Gear Extension

LANDING GEAR lever. OFF

Manual gear extension handles. Pull

The uplock is released when the handle is pulled
to its limit.

The related red landing gear indicator light
illuminates, indicating uplock release.

Wait 15 seconds after the last manual gear
extension handle is pulled:

LANDING GEAR lever DN

Additional Deferred Item

GROUND PROXIMITY FLAP

INHIBIT switch FLAP INHIBIT

▼ **Continued on next page** ▼



▼ **MANUAL REVERSION or LOSS OF SYSTEM A AND SYSTEM B**
continued ▼

Landing Checklist

- ENGINE START switches. CONT
- Speedbrake **DOWN detent**
- Landing gear Down
- Flaps **15, Green light**



LOW
PRESSURE

STANDBY HYDRAULIC LOW
PRESSURE

Condition: The standby hydraulic pump pressure is low.

Note: With a loss of hydraulic system A or B, the thrust reverser on the affected side is inoperative.

Note: With a loss of hydraulic system A and B, the rudder is inoperative.



LOW
QUANTITY

STANDBY HYDRAULIC LOW
QUANTITY

Condition: The standby hydraulic quantity is low.

- 1 Continue normal operation.





Non-Normal Checklists

Chapter NNC

Landing Gear

Section 14

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**ANTISKID
INOP**

ANTISKID INOPERATIVE

Condition: An antiskid system fault occurs.

Caution! Locked wheel protection is not available.

- 1 AUTO BRAKE select switch. OFF

The autobrake system is inoperative.

- 2 Do **not** arm the speedbrakes for landing. Manually deploy the speedbrakes immediately upon landing.

Automatic speedbrake extension may be inoperative.

- 3 Check the Non-Normal Configuration Landing Distance tables in the Performance Inflight-QRH chapter or other approved source.

- 4 **Checklist Complete Except Deferred Items**

Deferred Items

Landing Procedure Review

Use minimum braking consistent with runway length and conditions to reduce the possibility of a tire blowout.

Do **not** apply the brakes until the nose wheel is on the ground and the speedbrakes have been manually deployed.

Brake initially using light steady pedal pressure. Increase pressure as ground speed decreases. Do **not** pump the brakes.

▼ Continued on next page ▼



▼ ANTISKID INOPERATIVE continued ▼

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake **OFF**
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

Approach Checklist

- Altimeters ____

Landing Checklist

- ENGINE START switches. CONT
- Speedbrake **DOWN detent**
- Landing gear Down
- Flaps ____, Green light



**AUTO BRAKE
DISARM**
AUTO BRAKE DISARM

Condition: The autobrake system disarms after being set.

1 Choose one:

◆ On the **ground**:

AUTO BRAKE select switch OFF

▶▶ **Go to step 2**

◆ **In flight**:

▶▶ **Go to step 3**

2 Choose one:

◆ AUTO BRAKE DISARM light **extinguishes**:

◆ AUTO BRAKE DISARM light **stays illuminated**:

Do not takeoff.



3 AUTO BRAKE select switch. . . OFF, then reselect

▼ **Continued on next page ▼**

◆AUTO BRAKE DISARM light **stays extinguished**:



AUTO BRAKE select switch OFF

Use manual brakes for landing.

►► Go to step 5

5 Checklist Complete Except Deferred Items

Deferred Items

Descent Checklist

Pressurization LAND ALT _____

Recall Checked

Autobrake **OFF**

Landing data VREF , Minimums

Approach briefing Completed

Approach Checklist

Altimeters

Landing Checklist

ENGINE START switches. CONT

Speedbrake ARMED

▼ Continued on next page ▼

▼ AUTO BRAKE DISARM continued ▼

Landing gear Down

Flaps ____, Green light

**Brake Pressure Indicator Zero PSI**

Condition: The brake accumulator has no nitrogen precharge.

- 1 Accumulator braking is not available.

Note: If hydraulic systems indications are normal, brake operation is unaffected.





**BRAKE
TEMP**

BRAKE TEMPERATURE

(VP-BPI, VP-BPY) **Do not use**

Condition: One or more brake temperatures are high.

- 1 Choose one:

◆ On the **ground**:

▶▶ **Go to step 2**

◆ In **flight**:

▶▶ **Go to step 6**
- 2 Minimize taxiing.
- 3 Unless needed, do not set the parking brake.
- 4 Wheel fuse plugs can melt.
- 5 Check the Recommended Brake Cooling Schedule in the Advisory Information section of the Performance Inflight chapter for needed cooling time and precautions.

■ ■ ■ ■
- 6

270K/.82M maximum.

LANDING GEAR lever. DN
This allows cooling air to flow around the brakes.
- 7 **When** the BRAKE TEMP light is **extinguished**:

Wait 7 minutes. This ensures sufficient cooling time.

▼ Continued on next page ▼

▼ BRAKE TEMPERATURE continued ▼

235K maximum.

8  LANDING GEAR lever UP**After** landing gear retraction is complete:

LANDING GEAR lever OFF



GEAR DISAGREE**LEFT
GEAR****NOSE
GEAR****RIGHT
GEAR**

Condition: The landing gear position disagrees with the LANDING GEAR lever position.

Note: Do not exceed the gear EXTEND limit speed (270K/.82M).

Do not use FMC performance predictions with gear extended.

1 Choose one:

◆ LANDING GEAR lever is **UP**:

▶▶ **Go to step 5**

◆ LANDING GEAR lever is **OFF**:

▶▶ **Go to step 2**

◆ LANDING GEAR lever is **DN**:

▶▶ **Go to step 9**

▼ Continued on next page ▼

▼ GEAR DISAGREE continued ▼

2 Choose one:

◆ LANDING GEAR lever **moved** to the UP position after takeoff:

▶▶ **Go to step 3**

◆ LANDING GEAR lever did **not** move to the UP position after takeoff:

▶▶ **Go to the Landing Gear Lever Will Not Move Up After Takeoff checklist on page 14.16**



235K maximum

3  LANDING GEAR lever. UP

4 Choose one:

◆ **All red and green** landing gear indicator lights are **extinguished**:

The landing gear lever should be kept in the UP position to keep the landing gear retracted.



◆ **Any red** landing gear indicator light is **illuminated**:

▶▶ **Go to step 8**

▼ Continued on next page ▼

▼ **GEAR DISAGREE** continued ▼

5 Choose one:

◆ **All red and green** landing gear indicator lights are **illuminated**:

Open and close the manual gear extension access door. Verify the door is fully closed.

▶▶ **Go to step 6**

◆ **Any other combination** of landing gear indicator lights is **illuminated**:

▶▶ **Go to step 8**

235K maximum

6  **LANDING GEAR lever** DN, then UP

7 Choose one:

◆ **All** landing gear indicator lights **extinguish**:

LANDING GEAR lever OFF



◆ **Any red** landing gear indicator light is **illuminated**:

▶▶ **Go to step 8**

8 Flight with gear down increases fuel consumption and decreases climb performance. Refer to the Gear Down performance tables in the Performance Inflight section.



▼ **Continued on next page** ▼

▼ **GEAR DISAGREE** continued ▼

9 Check landing gear indicator lights.

Note: If a green landing gear indicator light is illuminated on either the center main panel or the overhead panel, the related landing gear is down and locked.

10 Choose one:

◆ **All** landing gear indicate **down and locked** and one or more **red** landing gear indicator lights are also **illuminated**:

▶▶ **Go to step 11**

◆ **Any** landing gear is **not** down and locked:

▶▶ **Go to the Manual Gear Extension checklist on page 14.20**



11 Verify landing gear lever is pushed in and fully in the DN detent.

12 Choose one:

◆ **All red** landing gear indicator lights **extinguish**:



◆ **One or more red** landing gear indicator lights stay **illuminated**:

GROUND PROXIMITY GEAR
INHIBIT switch GEAR INHIBIT
Land normally.



Landing Gear Lever Jammed in the Up Position

Condition: The LANDING GEAR lever will not move from the UP position.

Note: Start this checklist **only** when ready to extend the gear for landing.

Once the gear is extended, do **not** retract.

270K/.82M maximum.

- 1  LANDING GEAR override trigger Pull
- 2 LANDING GEAR lever DN
- 3 Choose one:

◆ LANDING GEAR **lever** moves to the **DN** position:

▶▶ **Go to step 4**

◆ LANDING GEAR **lever** does **not** move to the DN position:

▶▶ **Go to step 6**

- 4 Check landing gear indicator lights.

Note: If a green landing gear indicator light is illuminated on either the center main panel or the overhead panel, the related landing gear is down and locked.

▼ Continued on next page ▼

▼ Landing Gear Lever Jammed in the Up Position continued ▼

5 Choose one:

◆ **All** landing gear **indicate** down and locked:

Plan to land at the nearest suitable airport.



◆ **One or more** landing gear do **not** indicate down and locked:

►► **Go to the Manual Gear Extension checklist on page 14.20**



6 NOSE WHEEL

STEERING switch Verify NORM

Nose wheel steering is not available.

Warning! Do not use alternate nose wheel steering because the landing gear may retract on the ground.

7  270K/.82M maximum.
Manual gear extension handles (all) Pull
The uplock is released when the handle is pulled to its limit. The related red landing gear indicator light illuminates, indicating uplock released.

Note: With the LANDING GEAR lever in the UP or OFF position, the red landing gear indicator lights will stay illuminated.

▼ Continued on next page ▼



▼ Landing Gear Lever Jammed in the Up Position continued ▼

8 Check landing gear indicator lights.

Note: If a green landing gear indicator light is illuminated on either the center main panel or the overhead panel, the related landing gear is down and locked.

9 Choose one:

◆ **All** landing gear **indicate** down and locked:

▶▶ **Go to step 10**

◆ **One or more** landing gear do **not** indicate down and locked:

▶▶ **Go to the Partial or All Gear Up Landing checklist on page 14.22**



10 Checklist Complete Except Deferred Items

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake ____
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

▼ Continued on next page ▼

▼ Landing Gear Lever Jammed in the Up Position continued ▼

Approach Checklist

Altimeters _____

Additional Deferred Item

GROUND PROXIMITY GEAR

INHIBIT switch GEAR INHIBIT

Landing Checklist

ENGINE START switches CONT

Speedbrake ARMED

Landing gear **Down, Three green**

Flaps ____, Green light

Note: Nose wheel steering is not available.

Warning! Do not use alternate nose wheel steering because the landing gear may retract on the ground.





Landing Gear Lever Will Not Move Up After Takeoff

Condition: The LANDING GEAR lever cannot be moved to the UP position due to one of the following:

- Failure of the landing gear lever lock solenoid
- Failure of the air/ground system
- Failure of the ground spoiler interlock valve to close.

Note: Do not use FMC performance predictions.

- 1 LANDING GEAR leverDN
- 2 Retract the flaps on schedule.

▼ Continued on next page ▼

▼ Landing Gear Lever Will Not Move Up After Takeoff continued ▼

3 Choose one:

- ◆ Intermittent cabin altitude/configuration warning horn **stays silent** and the TAKEOFF CONFIG lights (if installed and operative) do **not illuminate** after the flaps are fully retracted and the thrust levers are advanced beyond the vertical position:

Note: This indicates a failure of the landing gear lever lock solenoid.

►► **Go to step 4**

- ◆ Intermittent cabin altitude/configuration warning horn **sounds** or the TAKEOFF CONFIG lights (if installed and operative) **illuminate** when the flaps are fully retracted:

Note: This indicates either a failure of the air/ground system or a failure of the ground spoiler interlock valve to close.

Do **not** retract the gear.

►► **Go to step 7**

235K maximum.

4  LANDING GEAR override trigger Pull

5 LANDING GEAR lever UP

After landing gear retraction is complete:

LANDING GEAR lever OFF

▼ Continued on next page ▼



▼ Landing Gear Lever Will Not Move Up After Takeoff continued ▼

6 Continue normal operation.



7 LANDING GEAR
TAKEOFF WARNING CUTOFF
circuit breaker (P6-3:C18) Pull

Note: The intermittent cabin altitude/configuration warning horn may still sound and the TAKEOFF CONFIG lights (if installed and operative) may still illuminate depending on thrust lever and flap position.

Caution! Do not use the speedbrakes in flight.

- 8 Plan to land at the nearest suitable airport.
- 9 Do **not** arm the autobrake for landing. Use manual braking.
- 10 Do **not** arm the speedbrakes for landing. Manually deploy the speedbrakes immediately upon landing.
- 11 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

- Pressurization LAND ALT ____
- Recall Checked
- Autobrake **OFF**
- Landing data VREF ____, Minimums ____
- Approach briefing Completed

▼ Continued on next page ▼

▼ Landing Gear Lever Will Not Move Up After Takeoff continued ▼

Approach Checklist

Altimeters _____

Gear Down Verification

LANDING GEAR lever. Verify DN

Landing Checklist

ENGINE START switches CONT

Speedbrake **DOWN detent**

Landing gear **Down (previously verified)**

Flaps _____, Green light

Note: Manually deploy the speedbrakes immediately upon touchdown. Use manual braking.



Manual Gear Extension

Condition: One of these occurs:

- Any landing gear is not down and locked when the LANDING GEAR lever is down
- The LANDING GEAR lever is jammed in the OFF position.

Note: If a green landing gear indicator light is illuminated on either the center main panel or the overhead panel, the related landing gear is down and locked.

1 LANDING GEAR lever OFF (if possible)

2  270K/.82M maximum.
Manual gear extension handles (affected gear) Pull
The uplock is released when the handle is pulled to its limit. The related red landing gear indicator light illuminates, indicating uplock released.

3 **Wait** 15 seconds after the last manual gear extension handle is pulled:

LANDING GEAR lever DN (if possible)

4 Check landing gear indicator lights.

Note: If the LANDING GEAR lever is in the OFF position, the red landing gear indicator lights will also be illuminated.

▼ Continued on next page ▼

▼ Manual Gear Extension continued ▼

5 Choose one:

◆ **All** landing gear **indicate** down and locked:

▶▶ **Go to step 6**

◆ **One or more** landing gear do **not** indicate down and locked:

▶▶ **Go to the Partial or All Gear Up Landing checklist on page 14.22**



6 Choose one:

◆ LANDING GEAR **lever** is in the **DN** position:

Land normally.



◆ LANDING GEAR **lever** is in the **OFF** position:

GROUND PROXIMITY GEAR

INHIBIT switch GEAR INHIBIT

Land normally.

Note: Nose wheel steering is not available.



Partial or All Gear Up Landing

Condition: All landing gear are not down and locked after attempting manual gear extension.

1 Choose one:

◆ Manual gear extension **has** been attempted:

▶▶ **Go to step 2**

◆ Manual gear extension has **not** been attempted:

▶▶ **Go to the Manual Gear Extension checklist on page 14.20**



2 Brief the crew and passengers on emergency landing and evacuation procedures.

(VQ-BUV) Do not use

3 Brief supernumeraries on emergency landing and evacuation procedures.

4 Consider burning off fuel to reduce touchdown speed.

5 Plan a flaps 40 landing.

6 Set VREF 40.

▼ Continued on next page ▼

▼ Partial or All Gear Up Landing continued ▼

7 LANDING GEAR

AURAL WARN circuit breaker (P6-3:D18). . . . Pull

This prevents the landing gear warning horn with gear retracted and landing flaps selected.

The flight deck chime for an incoming call from the cabin crew is unavailable.

(VQ-BUV) Do not use

8 LANDING GEAR

AURAL WARN circuit breaker (P6-3:D18). . . . Pull

This prevents the landing gear warning horn with gear retracted and landing flaps selected.

9 FLIGHT CONTROL

AUTO SPEED BRAKE

circuit breaker (P6-2:B9). Pull

This prevents inadvertent deployment of ground spoilers after landing.

10 Do **not** arm the autobrake for landing. Use manual braking.

11 Do **not** arm the speedbrakes for landing.

12 **Checklist Complete Except Deferred Items**

Deferred Items

Descent Checklist

Pressurization. LAND ALT ____

Recall Checked

Autobrake **OFF**

▼ Continued on next page ▼



▼ Partial or All Gear Up Landing continued ▼

Landing data **VREF 40**____, **Minimums** ____
Approach briefing Completed

Approach Checklist

Altimeters ____

Landing Procedure Review

Do not extend the speedbrakes unless stopping distance is critical. When stopping distance is critical, extend the speedbrakes after all landing gear, the nose or the engine nacelle have contacted the runway.

Do not use the thrust reversers unless stopping distance is critical.

Turn all fuel pump switches OFF just before the flare.

After stopping, do the Evacuation checklist, if needed.

Additional Deferred Items

APU switch OFF

GROUND PROXIMITY GEAR
INHIBIT switch GEAR INHIBIT

▼ Continued on next page ▼

▼Partial or All Gear Up Landing continued▼**When** on approach:

Engine BLEED air switches OFF

This ensures the airplane is depressurized
at touchdown.

Landing Checklist

ENGINE START switches CONT

Speedbrake **DOWN detent**Landing gear **Down**Flaps. **40, Green light**

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Non-Normal Checklists

Chapter NNC

Warning Systems

Section 15

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LANDING CONFIGURATION

Condition: In flight, the steady warning horn sounds.

- 1 Assure correct airplane landing configuration.



TAKEOFF CONFIGURATION

**TAKEOFF
CONFIG****(If installed and operative)**

Condition: On the ground, the intermittent cabin altitude/configuration warning horn sounds or a TAKEOFF CONFIG light (if installed and operative) illuminates when advancing the thrust levers to takeoff thrust.

- 1 Assure correct airplane takeoff configuration.



**WARNING HORN (INTERMITTENT)
or
WARNING LIGHT - CABIN ALTITUDE
OR TAKEOFF CONFIGURATION**

(If installed and operative)

Left Forward Panel



Right Forward Panel



Condition: One of these occurs:

- In flight, at an airplane flight altitude above 10,000 feet MSL, the intermittent warning horn sounds or a CABIN ALTITUDE light (if installed and operative) illuminates
- On the ground, the intermittent warning horn sounds or a TAKEOFF CONFIG light (if installed and operative) illuminates when advancing the thrust levers to takeoff thrust.

- 1 **If** the intermittent warning horn sounds or a CABIN ALTITUDE light (if installed and operative) illuminates **in flight** at an airplane flight altitude above 10,000 feet MSL:

Don the oxygen masks and set the regulators to 100%.

Establish crew communications.

▼ Continued on next page ▼

▼ **WARNING HORN (INTERMITTENT) or WARNING LIGHT - CABIN ALTITUDE OR TAKEOFF CONFIGURATION** continued ▼

►► **Go to the CABIN ALTITUDE WARNING or Rapid Depressurization checklist on page 2.1**



- 2 **If** the intermittent warning horn sounds or a TAKEOFF CONFIG light (if installed and operative) illuminates **on the ground** when advancing the thrust levers to takeoff thrust:

Assure correct airplane takeoff configuration.



INOP

**GROUND PROXIMITY
GND PROX INOPERATIVE**

(VP-BDU) Do not use

Condition: A ground proximity warning system fault occurs.

Note: Some or all GPWS alerts are not available. GPWS alerts which occur are valid.





INOP

GROUND PROXIMITY
GPWS INOPERATIVE

Condition: A ground proximity warning system (GPWS) fault occurs.

Note: Some or all GPWS alerts are not available. GPWS alerts which occur are valid.



INOP

GROUND PROXIMITY RUNWAY INOPERATIVE (RAAS)

(VP-BDU) Do not use

Condition: One of these occurs to the Runway Awareness and Advisory System (RAAS):

- it is failed
- it is unavailable because GPS position accuracy is inadequate
- it is unavailable because the airport is not in the GPWS database

1 Choose one:

◆ The RUNWAY INHIBIT switch is in the **INHIBIT position**:

Note: The RUNWAY INHIBIT switch is in the INHIBIT position and the airspeed has been 250 knots or greater for 60 seconds or more. The RUNWAY INOP light extinguishes when airspeed is below 250 knots.



◆ The RUNWAY INHIBIT switch is **not** in the INHIBIT position:

Some or all RAAS alerts are not available.
RAAS alerts which occur are valid.



Overspeed

Condition: Airspeed is more than Vmo/Mmo.

**PSEU****PSEU**

Condition: A proximity switch electronics unit fault occurs.

Note: The PSEU light illuminates on the ground only.

1 Choose one:

◆ PSEU light **stays illuminated** when the Master Caution system is reset:

►► **Go to step 2**

◆ PSEU light **extinguishes** when the Master Caution system is reset:



▼ Continued on next page ▼

▼ PSEU continued ▼

2 Choose one:

- ◆ PSEU light **stays illuminated** when the parking brake is set or when both engines are shut down:

Do **not** takeoff.



- ◆ PSEU light **extinguishes** when the parking brake is set or when both engines are shut down:



(VP-BZU - VP-BZY) Do not use

3 Choose one:

- ◆ PSEU light **stays illuminated** when the parking brake is set or when both engines are shut down:

Do **not** takeoff.



- ◆ PSEU light **extinguishes** when the parking brake is set or when both engines are shut down:





SPSEU

SPSEU

(VP-BZU - VP-BZY) Do not use

Condition: A supplemental proximity switch electronics unit fault occurs.

Note: The SPSEU light illuminates on the ground only.

- 1 Do **not** takeoff.



Tail Strike

Condition: A tail strike is suspected or confirmed.

Caution! Continued pressurization of the airplane can cause further structural damage.

- 1 Pressurization mode selectorMAN

Use momentary actuation of the outflow valve switch to avoid large and rapid pressurization changes.

- 2  Outflow VALVE switchMove to OPEN until the outflow VALVE indication shows fully open to depressurize the airplane

- 3 Plan to land at the nearest suitable airport.





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Introduction

Note: This section is reserved for operator-developed information.



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737 Flight Crew Operations Manual

Performance Inflight - QRH

Chapter PI-QRH

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General

The table below shows the airplanes that have been identified with the following performance package. Note, some airplanes may be identified with more than one performance package. This configuration table information reflects the Boeing delivered configuration updated for service bulletin incorporations in conformance with the policy stated in the introduction section of the FCOM. The performance data is prepared for the owner/operator named on the title page. The intent of this information is to assist flight crews and airlines in knowing which performance package is applicable to a given airplane. The performance package model identification information is based on Boeing's knowledge of the airline's fleet at a point in time approximately three months prior to the page date. Notice of Errata (NOE) will not be provided to airlines to identify airplanes that are moved between performance packages within this manual or airplanes added to the airline's fleet whose performance packages are already represented in this manual. These types of changes will be updated in the next block revision. Owners/operators are responsible for ensuring the operational documentation they are using is complete and matches the current configuration of their airplanes, and the accuracy and validity of all information furnished by the owner/operator or any other party. Owners/operators receiving active revision service are responsible to ensure that any modifications to the listed airplanes are properly reflected in this manual.

Serial and tabulation number are supplied by Boeing.

Registry Number	Old Registry Number	Serial Number	Tabulation Number
RA-73311	VP-BDP	40309	YF619
RA-73312	VP-BDQ	40311	YF621

Intentionally
Blank



Performance Inflight - QRH

Chapter PI-QRH

General

Section 10

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

CLIMB (280/.76)

Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT	4.0	4.0	4.0		
	V/S (FT/MIN)	1700	1100	600		
30000	PITCH ATT	4.0	4.0	3.5	4.0	4.0
	V/S (FT/MIN)	2500	1900	1500	1100	800
20000	PITCH ATT	7.0	6.5	6.0	6.0	6.0
	V/S (FT/MIN)	4200	3300	2600	2100	1700
10000	PITCH ATT	11.0	9.5	8.5	8.0	8.0
	V/S (FT/MIN)	5600	4400	3600	3000	2500
SEA LEVEL	PITCH ATT	14.5	12.5	11.0	10.0	9.5
	V/S (FT/MIN)	6700	5300	4400	3700	3100

CRUISE (.76/280)

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT	2.0	2.5	3.5		
	%N1	82.9	85.4	88.9		
35000	PITCH ATT	1.0	2.0	2.5	3.0	3.5
	%N1	81.2	82.6	84.4	86.8	90.4
30000	PITCH ATT	1.0	1.5	2.0	2.5	3.0
	%N1	80.7	81.5	82.7	84.2	86.1
25000	PITCH ATT	1.0	1.5	2.0	2.5	3.0
	%N1	77.2	77.9	79.0	80.5	82.3
20000	PITCH ATT	1.0	1.5	2.0	2.5	3.5
	%N1	73.6	74.2	75.3	76.6	78.2
15000	PITCH ATT	1.0	1.5	2.0	3.0	3.5
	%N1	69.8	70.6	71.6	72.9	74.4

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

DESCENT (.76/280)

Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT V/S (FT/MIN)	-1.5 -2700	-0.5 -2400	0.5 -2300	1.0 -2500	1.5 -2700
30000	PITCH ATT V/S (FT/MIN)	-3.5 -3100	-2.0 -2600	-1.0 -2300	-0.5 -2100	0.5 -2000
20000	PITCH ATT V/S (FT/MIN)	-3.5 -2800	-2.0 -2300	-1.0 -2000	0.0 -1900	0.5 -1700
10000	PITCH ATT V/S (FT/MIN)	-3.5 -2500	-2.0 -2100	-1.0 -1800	0.0 -1700	0.5 -1500
SEA LEVEL	PITCH ATT V/S (FT/MIN)	-3.5 -2300	-2.5 -1900	-1.0 -1700	-0.5 -1500	0.5 -1400

HOLDING (VREF40 + 70)

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
15000	PITCH ATT	5.0	5.0	5.0	5.0	5.0
	%N1	56.1	61.7	65.9	69.8	73.2
	KIAS	178	193	212	229	246
10000	PITCH ATT	5.0	5.0	5.0	5.0	5.0
	%N1	52.5	57.5	62.1	65.9	69.1
	KIAS	178	192	211	228	244
5000	PITCH ATT	5.0	5.5	5.0	5.0	5.0
	%N1	48.9	53.9	58.1	62.0	65.5
	KIAS	178	192	210	227	243

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = -2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 47.6	5.0 52.3	5.5 56.3	6.0 60.0	6.5 63.6
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 49.7	5.0 54.5	5.5 58.6	6.0 62.4	6.0 65.8
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 49.4	6.0 54.5	6.0 59.0	6.5 63.1	6.5 66.7
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 53.2	5.0 58.6	5.5 63.4	5.5 67.5	6.0 71.1
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 53.3	5.5 58.7	6.0 63.7	6.0 67.9	6.5 71.6
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 54.0	6.5 59.7	6.5 64.8	7.0 69.0	7.0 72.9
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 57.5	6.0 63.3	6.0 68.2	6.0 72.6	6.5 76.3
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = -1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 48.3	5.0 52.9	5.5 57.0	6.0 60.8	6.5 64.5
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 50.4	5.0 55.2	5.5 59.4	6.0 63.3	6.0 66.6
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 50.0	6.0 55.2	6.0 59.8	6.5 64.0	6.5 67.6
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 53.9	5.5 59.4	5.5 64.2	5.5 68.3	6.0 72.0
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 54.0	5.5 59.5	6.0 64.5	6.0 68.7	6.5 72.5
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 54.7	6.5 60.6	6.5 65.6	7.0 69.8	7.0 73.7
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 58.3	6.0 64.1	6.0 69.0	6.0 73.4	6.5 77.1
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = SEA LEVEL

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 49.0	5.0 53.6	5.5 57.7	6.0 61.6	6.5 65.2
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 51.1	5.0 55.9	5.5 60.2	6.0 64.1	6.0 67.3
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 50.7	6.0 56.0	6.0 60.6	6.5 64.7	6.5 68.3
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 54.7	5.5 60.2	5.5 65.0	5.5 69.1	6.0 72.9
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 54.7	5.5 60.4	6.0 65.3	6.0 69.5	6.5 73.4
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 55.5	6.5 61.4	6.5 66.3	7.0 70.6	7.0 74.6
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 59.1	6.0 65.0	6.0 69.9	6.0 74.3	6.5 77.9
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 49.8 178	5.0 54.3 192	5.5 58.4 204	6.0 62.5 215	6.5 66.0 225
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 51.8 158	5.0 56.7 172	5.5 61.0 184	6.0 64.8 195	6.0 68.1 205
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 51.4 138	6.0 56.7 152	6.0 61.4 164	6.5 65.5 175	6.5 69.1 185
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.0 55.4 138	5.5 61.0 152	5.5 65.8 164	5.5 69.9 175	6.0 73.7 185
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.5 55.4 128	5.5 61.2 142	6.0 66.1 154	6.0 70.3 165	6.5 74.2 175
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.5 56.3 118	6.5 62.2 132	6.5 67.1 144	7.0 71.5 155	7.0 75.4 165
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.5 59.9 128	6.0 65.8 142	6.0 70.7 154	6.0 75.1 165	6.5 78.8 175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 50.4	5.0 55.1	5.5 59.2	6.0 63.3	6.5 66.7
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 52.5	5.0 57.4	5.5 61.8	6.0 65.6	6.0 68.9
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 52.2	6.0 57.5	6.0 62.3	6.5 66.3	6.5 69.9
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 56.1	5.5 61.8	5.5 66.6	5.5 70.8	6.0 74.5
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 56.2	5.5 62.0	6.0 66.9	6.0 71.2	6.5 75.1
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 57.0	6.5 63.0	6.5 67.9	7.0 72.4	7.0 76.2
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 60.7	6.0 66.5	6.0 71.6	6.0 75.9	6.5 79.7
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 3000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 51.1	5.0 55.8	5.5 60.0	6.0 64.1	6.5 67.4
VREF40+70	KIAS	178	192	205	216	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 53.3	5.0 58.1	5.5 62.6	6.0 66.3	6.0 69.7
VREF40+50	KIAS	158	172	185	196	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 52.9	6.0 58.2	6.0 63.1	6.5 67.1	6.5 70.7
VREF40+30	KIAS	138	152	165	176	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 56.9	5.5 62.6	5.5 67.4	5.5 71.6	6.0 75.3
VREF40+30	KIAS	138	152	165	176	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 56.9	5.5 62.8	6.0 67.7	6.0 72.1	6.5 75.9
VREF40+20	KIAS	128	142	155	166	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 57.8	6.5 63.8	6.5 68.7	7.0 73.2	7.0 77.0
VREF40+10	KIAS	118	132	145	156	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 61.5	6.0 67.3	6.0 72.5	6.0 76.7	6.5 80.5
VREF40+20	KIAS	128	142	155	166	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 4000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 51.7	5.0 56.5	5.5 60.8	6.0 64.9	6.5 68.2
VREF40+70	KIAS	178	192	205	216	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 54.0	5.0 58.9	5.5 63.4	6.0 67.1	6.0 70.5
VREF40+50	KIAS	158	172	185	196	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 53.6	6.0 59.0	6.0 63.9	6.5 67.9	6.5 71.6
VREF40+30	KIAS	138	152	165	176	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 57.7	5.5 63.4	5.5 68.2	5.5 72.5	6.0 76.1
VREF40+30	KIAS	138	152	165	176	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 57.7	5.5 63.6	6.0 68.5	6.0 73.0	6.5 76.6
VREF40+20	KIAS	128	142	155	166	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 58.6	6.5 64.6	6.5 69.6	7.0 74.1	7.0 77.9
VREF40+10	KIAS	118	132	145	156	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 62.3	6.0 68.1	6.0 73.3	6.0 77.6	6.5 81.4
VREF40+20	KIAS	128	142	155	166	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 5000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 52.4 178	5.0 57.3 192	5.5 61.6 205	6.0 65.7 216	6.5 69.0 225
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 54.7 158	5.0 59.7 172	5.5 64.2 185	6.0 67.9 196	6.0 71.4 205
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 54.4 138	6.0 59.9 152	6.0 64.7 165	6.5 68.7 176	6.5 72.5 185
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.0 58.5 138	5.5 64.2 152	5.5 69.1 165	5.5 73.3 176	6.0 76.9 185
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.5 58.5 128	5.5 64.4 142	6.0 69.4 155	6.0 73.8 166	6.5 77.5 175
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.5 59.4 118	6.5 65.4 132	6.5 70.5 145	7.0 74.9 156	7.0 78.8 165
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.5 63.1 128	6.0 69.0 142	6.0 74.2 155	6.0 78.5 166	6.5 82.3 175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 6000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 53.1	5.0 58.0	5.5 62.5	6.0 66.4	6.5 69.8
VREF40+70	KIAS	178	192	205	216	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 55.5	5.0 60.5	5.5 64.9	6.0 68.7	6.0 72.3
VREF40+50	KIAS	158	172	185	196	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 55.1	6.0 60.7	6.0 65.5	6.5 69.5	6.5 73.3
VREF40+30	KIAS	138	152	165	176	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 59.3	5.5 65.1	5.5 69.9	5.5 74.1	6.0 77.7
VREF40+30	KIAS	138	152	165	176	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 59.4	5.5 65.2	6.0 70.3	6.0 74.7	6.5 78.3
VREF40+20	KIAS	128	142	155	166	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 60.3	6.5 66.2	6.5 71.4	7.0 75.7	7.0 79.6
VREF40+10	KIAS	118	132	145	156	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 64.0	6.0 69.9	6.0 75.0	6.0 79.3	6.5 83.2
VREF40+20	KIAS	128	142	155	166	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 7000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 53.9	5.0 58.8	5.5 63.4	6.0 67.1	6.5 70.6
VREF40+70	KIAS	178	192	205	216	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 56.3	5.0 61.4	5.5 65.7	6.0 69.5	6.0 73.1
VREF40+50	KIAS	158	172	185	196	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 55.9	6.0 61.6	6.0 66.3	6.5 70.4	6.5 74.1
VREF40+30	KIAS	138	152	165	176	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 60.1	5.5 65.9	5.5 70.8	5.5 75.0	6.0 78.6
VREF40+30	KIAS	138	152	165	176	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 60.2	5.5 66.0	6.0 71.2	6.0 75.4	6.5 79.2
VREF40+20	KIAS	128	142	155	166	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 61.1	6.5 67.0	6.5 72.2	7.0 76.6	7.0 80.5
VREF40+10	KIAS	118	132	145	156	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 64.8	6.0 70.8	6.0 75.8	6.0 80.2	6.5 84.1
VREF40+20	KIAS	128	142	155	166	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 8000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 54.6	5.0 59.6	5.5 64.1	6.0 67.9	6.5 71.4
VREF40+70	KIAS	178	192	205	216	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 57.0	5.0 62.2	5.5 66.5	6.0 70.4	6.0 74.0
VREF40+50	KIAS	158	172	185	196	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 56.7	6.0 62.4	6.0 67.0	6.5 71.2	6.5 74.9
VREF40+30	KIAS	138	152	165	176	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 61.0	5.5 66.7	5.5 71.7	5.5 75.8	6.0 79.6
VREF40+30	KIAS	138	152	165	176	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 61.0	5.5 66.9	6.0 72.1	6.0 76.2	6.0 80.1
VREF40+20	KIAS	128	142	155	166	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 61.9	6.5 67.9	6.5 73.1	7.0 77.5	7.0 81.4
VREF40+10	KIAS	118	132	145	156	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 65.6	6.0 71.6	6.0 76.7	6.0 81.1	6.5 85.0
VREF40+20	KIAS	128	142	155	166	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 9000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 55.4	5.5 60.5	5.5 64.9	6.0 68.6	6.5 72.3
VREF40+70	KIAS	178	192	205	216	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 57.7	5.0 63.0	5.5 67.3	6.0 71.3	6.0 74.8
VREF40+50	KIAS	158	172	185	196	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 57.5	6.0 63.2	6.0 67.9	6.5 72.1	6.5 75.8
VREF40+30	KIAS	138	152	165	176	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 61.8	5.5 67.5	5.5 72.5	5.5 76.6	6.0 80.4
VREF40+30	KIAS	138	152	165	176	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 61.8	5.5 67.7	6.0 72.9	6.0 77.1	6.0 81.0
VREF40+20	KIAS	128	142	155	166	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 62.8	6.5 68.7	6.5 73.9	7.0 78.4	7.0 82.3
VREF40+10	KIAS	118	132	145	156	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 66.4	6.0 72.5	6.0 77.6	6.0 82.0	6.5 85.9
VREF40+20	KIAS	128	142	155	166	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 10000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 56.1	5.5 61.3	5.5 65.6	6.0 69.5	6.5 73.0
VREF40+70	KIAS	178	192	205	216	227
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 58.5	5.0 63.8	5.5 68.1	6.0 72.2	6.0 75.6
VREF40+50	KIAS	158	172	185	196	207
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 58.3	6.0 64.0	6.0 68.7	6.0 72.9	6.5 76.6
VREF40+30	KIAS	138	152	165	176	187
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 62.6	5.5 68.4	5.5 73.3	5.5 77.5	6.0 81.4
VREF40+30	KIAS	138	152	165	176	187
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 62.7	5.5 68.6	6.0 73.7	6.0 78.0	6.0 81.9
VREF40+20	KIAS	128	142	155	166	177
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 63.6	6.5 69.7	6.5 74.8	7.0 79.3	7.0 83.2
VREF40+10	KIAS	118	132	145	156	167
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 67.2	6.0 73.4	6.0 78.5	6.0 82.9	6.5 86.9
VREF40+20	KIAS	128	142	155	166	177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 11000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 56.9	5.5 62.3	5.5 66.3	6.0 70.3	6.5 73.9
VREF40+70	KIAS	178	192	205	217	227
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 59.5	5.0 64.6	5.5 69.1	6.0 73.1	6.0 76.4
VREF40+50	KIAS	158	172	185	197	207
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 59.4	6.0 64.9	6.0 69.7	6.0 73.8	6.5 77.5
VREF40+30	KIAS	138	152	165	177	187
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 63.6	5.5 69.4	5.5 74.2	5.5 78.5	6.0 82.3
VREF40+30	KIAS	138	152	165	177	187
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 63.8	5.5 69.7	6.0 74.6	6.0 79.0	6.0 82.9
VREF40+20	KIAS	128	142	155	167	177
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 64.7	6.5 70.8	6.5 75.8	6.5 80.3	7.0 84.2
VREF40+10	KIAS	118	132	145	157	167
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 68.4	6.0 74.3	6.0 79.5	6.0 84.0	6.5 88.0
VREF40+20	KIAS	128	142	155	167	177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 12000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	5.0 57.7	5.5 63.0	5.5 67.1	6.0 71.2	6.5 74.6
VREF40+70	KIAS	178	192	205	217	228
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 60.6	5.5 65.4	5.5 70.0	6.0 73.8	6.0 77.3
VREF40+50	KIAS	158	172	185	197	208
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 60.5	6.0 65.8	6.0 70.7	6.0 74.7	6.5 78.5
VREF40+30	KIAS	138	152	165	177	188
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 64.6	5.5 70.4	5.5 75.1	5.5 79.5	6.0 83.3
VREF40+30	KIAS	138	152	165	177	188
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 64.7	5.5 70.8	6.0 75.6	6.0 80.0	6.0 83.9
VREF40+20	KIAS	128	142	155	167	178
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 65.8	6.5 71.8	6.5 76.9	6.5 81.3	6.5 85.2
VREF40+10	KIAS	118	132	145	157	168
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 69.6	6.0 75.3	6.0 80.5	6.0 85.0	6.0 89.1
VREF40+20	KIAS	128	142	155	167	178

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 13000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	5.0 58.5	5.5 63.7	5.5 67.9	6.0 72.0	6.5 75.5
VREF40+70	KIAS	178	192	205	217	228
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 61.4	5.5 66.3	5.5 70.9	6.0 74.6	6.0 78.2
VREF40+50	KIAS	158	172	185	197	208
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 61.5	6.0 66.8	6.0 71.7	6.0 75.6	6.5 79.4
VREF40+30	KIAS	138	152	165	177	188
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 65.6	5.5 71.4	5.5 76.0	5.5 80.4	6.0 84.2
VREF40+30	KIAS	138	152	165	177	188
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 65.8	5.5 71.7	6.0 76.5	6.0 81.0	6.0 84.8
VREF40+20	KIAS	128	142	155	167	178
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 66.9	6.5 72.8	6.5 77.9	6.5 82.3	6.5 86.2
VREF40+10	KIAS	118	132	145	157	168
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 70.6	6.0 76.4	6.0 81.5	6.0 86.0	6.0 90.2
VREF40+20	KIAS	128	142	155	167	178

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 14000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	5.0 59.4	5.5 64.4	5.5 68.7	6.0 72.8	6.5 76.3
VREF40+70	KIAS	179	192	205	217	228
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 62.3	5.5 67.1	5.5 71.7	6.0 75.4	6.0 79.0
VREF40+50	KIAS	159	172	185	197	208
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 62.3	6.0 67.8	6.0 72.5	6.0 76.5	6.5 80.3
VREF40+30	KIAS	139	152	165	177	188
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 66.5	5.5 72.3	5.5 77.0	5.5 81.3	6.0 85.1
VREF40+30	KIAS	139	152	165	177	188
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 66.9	5.5 72.6	6.0 77.6	6.0 81.9	6.0 85.8
VREF40+20	KIAS	129	142	155	167	178
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 68.1	6.5 73.8	6.5 78.9	6.5 83.3	6.5 87.3
VREF40+10	KIAS	119	132	145	157	168
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 71.6	6.0 77.5	6.0 82.5	6.0 87.1	6.0 91.5
VREF40+20	KIAS	129	142	155	167	178

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 14500 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	5.0 59.9	5.5 64.8	5.5 69.2	6.0 73.2	6.5 76.7
VREF40+70	KIAS	179	192	206	218	229
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 62.7	5.5 67.6	5.5 72.1	6.0 75.9	6.0 79.5
VREF40+50	KIAS	159	172	186	198	209
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 62.8	6.0 68.3	6.0 72.9	6.0 76.9	6.5 80.7
VREF40+30	KIAS	139	152	166	178	189
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 67.1	5.5 72.7	5.5 77.5	5.5 81.8	6.0 85.5
VREF40+30	KIAS	139	152	166	178	189
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 67.4	5.5 73.1	6.0 78.1	6.0 82.4	6.0 86.3
VREF40+20	KIAS	129	142	156	168	179
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 68.6	6.5 74.2	6.5 79.4	6.5 83.7	6.5 87.8
VREF40+10	KIAS	119	132	146	158	169
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 72.1	6.0 78.0	6.0 83.0	6.0 87.6	6.0 92.3
VREF40+20	KIAS	129	142	156	168	179

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = -2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	42.2	46.7	50.6	54.1	57.0
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	46.3	51.0	55.3	59.0	62.3
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	50.8	56.2	60.9	64.9	68.3
	KIAS	118	132	144	155	164

Flap placard speed exceeded in shaded area.

Airport Altitude = -1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	42.7	47.3	51.2	54.8	57.8
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	46.9	51.7	56.0	59.8	63.1
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	51.5	56.9	61.7	65.7	69.0
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = SEA LEVEL

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	43.3	48.0	51.9	55.5	58.4
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	47.5	52.5	56.7	60.5	63.9
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	52.2	57.6	62.4	66.5	69.8
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	43.9	48.6	52.6	56.2	59.2
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	48.1	53.2	57.5	61.3	64.6
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	52.9	58.4	63.2	67.3	70.6
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	44.5	49.2	53.3	56.9	59.9
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	48.8	53.8	58.2	62.1	65.4
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	53.6	59.2	64.0	68.0	71.5
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 3000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	45.1	49.8	54.1	57.6	60.7
	KIAS	131	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	49.4	54.5	59.0	63.0	66.2
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	54.4	59.9	64.9	68.8	72.3
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 4000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	45.8	50.5	54.7	58.3	61.5
	KIAS	131	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	50.1	55.3	59.8	63.7	67.0
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	55.1	60.7	65.6	69.6	73.2
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 5000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	46.5	51.2	55.4	59.0	62.3
	KIAS	131	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	50.8	56.0	60.6	64.5	67.7
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	55.8	61.5	66.4	70.4	74.0
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 6000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	47.1	51.9	56.1	59.8	63.0
	KIAS	131	145	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	51.6	56.7	61.4	65.2	68.5
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	56.6	62.3	67.1	71.3	74.8
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 7000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	47.7	52.6	56.8	60.6	63.8
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	52.3	57.4	62.2	66.0	69.3
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	57.3	63.1	67.9	72.2	75.6
	KIAS	118	132	144	156	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 8000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	48.4	53.3	57.6	61.4	64.5
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	52.9	58.2	62.9	66.8	70.1
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	58.1	64.0	68.7	73.0	76.4
	KIAS	118	132	145	156	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 9000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	49.0	54.0	58.3	62.2	65.2
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	53.7	59.0	63.7	67.6	70.9
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	58.9	64.7	69.6	73.8	77.2
	KIAS	118	132	145	156	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 10000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	49.7	54.7	59.1	62.9	65.9
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	54.4	59.8	64.5	68.3	71.7
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	59.8	65.5	70.5	74.6	78.1
	KIAS	118	132	145	156	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 11000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	50.4	55.5	59.9	63.6	66.6
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	55.1	60.7	65.3	69.1	72.5
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	60.5	66.3	71.3	75.4	79.0
	KIAS	118	132	145	156	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 12000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	51.1	56.2	60.7	64.3	67.4
	KIAS	131	146	159	172	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	55.9	61.5	66.1	70.0	73.3
	KIAS	125	139	152	164	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	61.3	67.1	72.2	76.2	79.8
	KIAS	118	132	145	156	166

Flap placard speed exceeded in shaded area.

Airport Altitude = 13000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	51.9	56.9	61.5	65.1	68.1
	KIAS	131	146	160	172	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	56.6	62.2	66.8	70.8	74.1
	KIAS	125	139	152	164	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	62.2	67.9	73.0	77.1	80.7
	KIAS	118	132	145	156	166

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 14000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	52.7	57.8	62.4	65.8	69.0
	KIAS	131	146	160	172	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	57.6	63.1	67.7	71.7	74.9
	KIAS	125	139	152	164	174
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	63.2	68.9	73.9	78.0	81.7
	KIAS	118	132	145	156	166

Flap placard speed exceeded in shaded area.

Airport Altitude = 14500 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	53.1	58.3	62.8	66.3	69.4
	KIAS	131	146	160	172	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	58.2	63.6	68.2	72.1	75.3
	KIAS	125	139	152	164	174
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	63.7	69.5	74.4	78.5	82.2
	KIAS	118	132	145	156	167

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

GO-AROUND

Flaps 15, Gear Up, Set Go-Around Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
14500	PITCH ATT	17.0	13.5	11.5	10.0	9.0
	V/S (FT/MIN)	3200	2500	1900	1400	1100
	KIAS	128	142	155	166	177
10000	PITCH ATT	19.5	16.0	13.5	11.5	10.5
	V/S (FT/MIN)	3800	3000	2400	1900	1500
	KIAS	128	142	155	166	175
5000	PITCH ATT	22.5	18.0	15.0	13.0	11.5
	V/S (FT/MIN)	4200	3400	2700	2200	1800
	KIAS	128	142	154	165	175
SEA LEVEL	PITCH ATT	25.0	20.0	16.5	14.5	13.0
	V/S (FT/MIN)	4500	3700	3000	2500	2000
	KIAS	128	142	154	165	175
-2000	PITCH ATT	25.5	20.5	17.0	14.5	13.0
	V/S (FT/MIN)	4500	3600	3000	2500	2000
	KIAS	128	142	154	165	174

Max Climb %N1**Based on engine bleed for packs on or off and anti-ice off**

TAT (°C)	PRESSURE ALTITUDE (FT)/SPEED (KIAS/MACH)									
	0	5000	10000	15000	20000	25000	30000	35000	37000	41000
	280	280	280	280	280	280	280	.78	.78	.78
60	89.4	89.7	89.7	89.8	89.6	91.4	93.0	94.4	94.5	92.8
55	90.2	90.5	90.5	90.7	90.0	90.8	92.4	93.7	93.8	92.1
50	90.9	91.2	91.3	91.5	91.0	90.8	91.7	93.0	93.1	91.4
45	91.6	91.9	92.1	92.3	91.9	91.7	91.7	92.3	92.4	90.7
40	92.4	92.6	92.9	93.1	92.7	92.5	92.5	91.6	91.7	90.0
35	92.9	93.3	93.6	93.8	93.6	93.3	93.3	92.4	91.7	90.1
30	92.2	94.1	94.3	94.6	94.4	94.1	94.0	93.2	92.6	91.1
25	91.5	94.1	95.0	95.2	95.2	94.8	94.7	94.0	93.4	92.1
20	90.7	93.3	95.8	96.0	95.9	95.6	95.4	94.7	94.2	93.0
15	90.0	92.5	95.2	96.8	96.7	96.3	96.1	95.5	95.0	94.0
10	89.2	91.8	94.4	97.1	97.6	97.0	96.7	96.2	95.8	94.9
5	88.4	91.0	93.6	96.3	98.5	97.9	97.4	97.0	96.6	95.8
0	87.7	90.2	92.8	95.5	97.9	99.0	98.4	97.8	97.5	96.7
-5	86.9	89.4	92.0	94.7	97.2	98.9	99.4	98.6	98.3	97.7
-10	86.1	88.6	91.2	93.9	96.4	98.1	99.7	99.5	99.2	98.7
-15	85.3	87.8	90.3	93.1	95.6	97.4	98.9	100.5	100.1	99.7
-20	84.5	87.0	89.5	92.3	94.8	96.6	98.1	100.2	100.7	100.3
-25	83.7	86.1	88.7	91.4	94.1	95.8	97.3	99.3	99.9	99.5
-30	82.9	85.3	87.8	90.6	93.3	95.0	96.5	98.5	99.0	98.7
-35	82.0	84.5	87.0	89.8	92.4	94.1	95.6	97.6	98.2	97.8
-40	81.2	83.6	86.1	88.9	91.6	93.3	94.8	96.8	97.3	96.9

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)					
	0	10	20	30	35	41
ENGINE ANTI-ICE	-0.6	-0.8	-0.9	-0.9	-0.8	-0.8
ENGINE & WING ANTI-ICE*	-1.8	-2.1	-2.5	-2.7	-3.0	-3.0

*Dual bleed sources

Go-around %N1

Based on engine bleed for packs on, engine and wing anti-ice on or off

AIRPORT OAT		TAT (°C)	AIRPORT PRESSURE ALTITUDE (FT)											
°C	°F		-2000	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
57	134	60	91.0	91.8	91.8									
52	125	55	91.7	92.6	92.6	92.5	92.5							
47	116	50	92.5	93.3	93.3	93.3	93.3	93.2	93.2					
42	108	45	93.3	94.1	94.1	94.1	94.0	94.0	93.9	93.9	93.8			
37	99	40	94.1	94.9	94.9	94.8	94.8	94.7	94.7	94.6	94.6	94.6	94.5	94.4
32	90	35	94.3	95.8	95.8	95.7	95.7	95.6	95.5	95.5	95.4	95.3	95.3	95.2
27	81	30	93.5	95.7	96.3	96.5	96.5	96.4	96.4	96.3	96.2	96.2	96.1	96.0
22	72	25	92.8	94.9	95.5	96.1	96.7	97.3	97.3	97.2	97.1	97.0	97.0	96.9
17	63	20	92.0	94.2	94.7	95.3	95.9	96.5	97.2	97.9	98.3	98.2	98.1	98.0
12	54	15	91.3	93.4	94.0	94.5	95.1	95.8	96.5	97.2	97.9	98.7	99.4	99.4
7	45	10	90.5	92.6	93.2	93.8	94.4	95.0	95.7	96.4	97.1	97.9	98.7	99.5
2	36	5	89.7	91.8	92.4	93.0	93.6	94.2	94.9	95.6	96.4	97.1	98.0	98.8
-3	27	0	89.0	91.0	91.6	92.2	92.8	93.4	94.1	94.8	95.6	96.4	97.2	98.1
-8	18	-5	88.2	90.2	90.8	91.4	92.0	92.6	93.3	94.0	94.8	95.6	96.4	97.3
-13	9	-10	87.4	89.4	90.0	90.6	91.1	91.8	92.5	93.2	94.0	94.8	95.7	96.5
-17	1	-15	86.6	88.6	89.2	89.7	90.3	90.9	91.7	92.4	93.2	94.0	94.9	95.8
-22	-8	-20	85.8	87.8	88.3	88.9	89.5	90.1	90.8	91.6	92.3	93.2	94.1	95.0
-27	-17	-25	84.9	86.9	87.5	88.1	88.6	89.3	90.0	90.7	91.5	92.3	93.3	94.2
-32	-26	-30	84.1	86.1	86.7	87.2	87.8	88.4	89.2	89.9	90.7	91.5	92.5	93.4
-37	-35	-35	83.3	85.2	85.8	86.3	86.9	87.6	88.3	89.0	89.8	90.7	91.6	92.6
-42	-44	-40	82.4	84.4	84.9	85.5	86.1	86.7	87.4	88.2	89.0	89.8	90.8	91.8
-47	-53	-45	81.6	83.5	84.1	84.6	85.2	85.8	86.6	87.3	88.1	89.0	90.0	90.9
-52	-62	-50	80.7	82.6	83.2	83.7	84.3	84.9	85.7	86.4	87.2	88.1	89.1	90.1

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (FT)											
	-2000	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
PACKS OFF	0.6	0.8	0.8	0.8	0.8	0.8	0.8	0.9	0.9	0.9	1.0	0.9
A/C HIGH	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1



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WEIGHT (1000 KG)	FLAPS		
	40	30	15
85	160	168	177
80	155	163	172
75	151	158	167
70	146	153	161
65	141	148	156
60	135	142	149
55	128	136	143
50	122	129	136
45	115	122	128
40	108	115	121

Intentionally
Blank



Performance Inflight - QRH

Advisory Information

Chapter PI-QRH

Section 11

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF30	ONE REV NO REV

Dry Runway

MAX MANUAL	1105	65/-65	25/35	-40/140	10/-10	25/-25	40	25	45
AUTOBRAKE MAX	1395	70/-75	35/40	-50/175	0/0	35/-35	65	0	5
AUTOBRAKE 3	1985	110/-125	50/70	-85/290	0/0	60/-60	110	0	0
AUTOBRAKE 2	2520	160/-175	75/105	-115/395	35/-45	75/-75	110	70	70
AUTOBRAKE 1	2775	190/-205	90/120	-140/465	75/-85	80/-80	100	225	335

Good Reported Braking Action

MAX MANUAL	1530	85/-90	40/50	-70/235	40/-35	40/-40	60	80	180
AUTOBRAKE MAX	1625	90/-100	45/60	-70/240	35/-30	40/-40	70	85	195
AUTOBRAKE 3	1985	110/-125	50/70	-85/290	5/0	60/-60	110	5	15
AUTOBRAKE 2	2520	160/-175	75/105	-115/395	35/-45	75/-75	110	70	70
AUTOBRAKE 1	2775	190/-205	90/120	-140/465	75/-85	80/-80	100	225	335

Medium Reported Braking Action

MAX MANUAL	2085	130/-140	65/85	-110/385	100/-75	60/-60	75	220	520
AUTOBRAKE MAX	2130	140/-145	65/85	-110/390	90/-70	60/-60	85	220	525
AUTOBRAKE 3	2215	140/-145	65/85	-110/395	75/-50	65/-65	110	160	470
AUTOBRAKE 2	2580	160/-180	80/105	-125/450	75/-70	75/-75	110	120	260
AUTOBRAKE 1	2795	190/-205	90/120	-140/485	100/-90	80/-85	100	240	405

Poor Reported Braking Action

MAX MANUAL	2720	190/-195	90/125	-160/610	235/-155	75/-80	85	460	1200
AUTOBRAKE MAX	2725	190/-195	90/125	-160/610	235/-155	75/-80	90	460	1210
AUTOBRAKE 3	2745	195/-195	90/125	-160/615	230/-145	75/-80	100	460	1215
AUTOBRAKE 2	2905	200/-205	100/130	-165/635	220/-145	80/-85	105	385	1065
AUTOBRAKE 1	3025	215/-220	100/140	-175/650	225/-155	85/-90	100	435	1070

Reference distance is based on sea level, standard day, no wind or slope, VREF30, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 70 m.

For autobrake and manual speedbrakes, increase reference landing distance by 65 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

All reference distances and adjustments shown have been increased by 15%.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 40

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF40	ONE REV NO REV

Dry Runway

MAX MANUAL	1050	65/-60	25/30	-40/130	10/-10	25/-25	40	15	40
AUTOBRAKE MAX	1305	65/-70	30/40	-45/160	0/0	30/-30	65	0	0
AUTOBRAKE 3	1830	100/-115	45/65	-80/270	0/0	50/-50	105	0	0
AUTOBRAKE 2	2335	145/-160	70/90	-110/380	25/-40	70/-70	110	40	40
AUTOBRAKE 1	2600	175/-190	85/110	-130/450	65/-75	75/-75	100	180	255

Good Reported Braking Action

MAX MANUAL	1460	80/-85	40/50	-65/230	40/-35	35/-35	60	75	160
AUTOBRAKE MAX	1555	85/-90	40/50	-70/235	35/-30	40/-40	70	80	175
AUTOBRAKE 3	1840	100/-115	45/65	-80/275	10/-5	50/-50	110	5	15
AUTOBRAKE 2	2335	145/-160	70/90	-110/380	25/-40	70/-70	110	40	40
AUTOBRAKE 1	2600	175/-190	85/110	-130/450	65/-75	75/-75	100	180	255

Medium Reported Braking Action

MAX MANUAL	1990	120/-130	60/80	-105/380	100/-75	50/-50	75	195	465
AUTOBRAKE MAX	2015	125/-140	65/80	-105/385	85/-70	50/-60	85	195	465
AUTOBRAKE 3	2070	125/-140	65/80	-110/390	80/-50	60/-60	105	175	450
AUTOBRAKE 2	2405	150/-165	70/100	-120/430	65/-65	70/-70	110	85	220
AUTOBRAKE 1	2615	175/-190	85/110	-130/465	90/-85	75/-75	100	195	315

Poor Reported Braking Action

MAX MANUAL	2580	180/-185	85/115	-160/600	230/-150	70/-75	85	415	1070
AUTOBRAKE MAX	2590	180/-185	85/120	-160/600	230/-150	70/-75	85	415	1070
AUTOBRAKE 3	2600	180/-190	85/120	-160/605	225/-145	70/-75	100	415	1075
AUTOBRAKE 2	2725	185/-190	85/120	-160/615	215/-140	75/-80	105	335	955
AUTOBRAKE 1	2840	195/-205	90/125	-165/635	220/-150	80/-85	100	385	935

Reference distance is based on sea level, standard day, no wind or slope, VREF40, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 65 m.

For autobrake and manual speedbrakes, increase reference landing distance by 50 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

All reference distances and adjustments shown have been increased by 15%.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1075	80/-65	25/35	-40/130	15/-10	25/-25	N/A	30	60
AUTOBRAKE MAX	1435	70/-80	35/45	-50/165	5/-5	35/-35	N/A	0	5
AUTOBRAKE 2	2550	165/-180	85/110	-110/370	50/-55	75/-75	N/A	165	180

Good Reported Braking Action

MAX MANUAL	1475	80/-85	40/55	-60/215	35/-30	40/-40	N/A	90	205
AUTOBRAKE MAX	1600	85/-95	45/60	-65/225	30/-25	40/-40	N/A	100	230
AUTOBRAKE 2	2550	165/-180	85/110	-110/370	55/-55	75/-75	N/A	165	180

Medium Reported Braking Action

MAX MANUAL	2025	130/-135	65/85	-100/350	90/-70	55/-55	N/A	240	595
AUTOBRAKE MAX	2080	130/-140	65/90	-100/355	85/-65	55/-60	N/A	245	605
AUTOBRAKE 3	2255	135/-145	70/90	-105/370	60/-45	65/-65	N/A	145	460

Poor Reported Braking Action

MAX MANUAL	2635	185/-190	90/125	-145/550	210/-140	75/-80	N/A	505	1385
AUTOBRAKE MAX	2635	185/-190	90/125	-145/550	210/-135	75/-80	N/A	500	1380
AUTOBRAKE 3	2685	185/-190	90/125	-150/555	195/-125	75/-80	N/A	485	1375

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)									
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ		
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV	

Dry Runway

MAX MANUAL	1025	60/-60	25/30	-35/125	15/-10	25/-25	N/A	25	55
AUTOBRAKE MAX	1335	60/-70	30/40	-45/155	5/-5	30/-35	N/A	0	5
AUTOBRAKE 2	2345	150/-160	75/95	-105/355	45/-50	70/-70	N/A	140	160

Good Reported Braking Action

MAX MANUAL	1415	75/-80	40/50	-60/210	35/-30	35/-35	N/A	80	185
AUTOBRAKE MAX	1525	80/-90	40/55	-65/220	30/-30	40/-40	N/A	90	205
AUTOBRAKE 2	2345	150/-160	75/95	-105/355	45/-50	70/-70	N/A	140	160

Medium Reported Braking Action

MAX MANUAL	1915	120/-125	60/80	-95/340	85/-70	50/-55	N/A	215	520
AUTOBRAKE MAX	1965	120/-130	60/80	-95/345	80/-65	55/-55	N/A	215	530
AUTOBRAKE 3	2100	120/-135	60/85	-100/360	60/-50	60/-60	N/A	135	420

Poor Reported Braking Action

MAX MANUAL	2460	170/-170	85/115	-140/535	200/-130	65/-70	N/A	435	1165
AUTOBRAKE MAX	2470	170/-175	85/115	-140/535	200/-125	70/-75	N/A	430	1160
AUTOBRAKE 3	2510	170/-175	85/115	-145/540	190/-125	70/-75	N/A	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	985	55/-55	20/30	-35/120	15/-10	20/-20	N/A	20	45
AUTOBRAKE MAX	1245	55/-65	30/35	-45/150	5/0	30/-30	N/A	0	0
AUTOBRAKE 2	2195	135/-150	70/90	-100/340	40/-45	65/-65	N/A	105	110

Good Reported Braking Action

MAX MANUAL	1360	70/-80	35/50	-60/205	35/-30	35/-35	N/A	75	170
AUTOBRAKE MAX	1455	75/-85	40/50	-60/215	30/-25	35/-35	N/A	85	185
AUTOBRAKE 2	2195	135/-150	70/90	-100/340	40/-45	65/-65	N/A	105	110

Medium Reported Braking Action

MAX MANUAL	1830	110/-120	55/75	-95/335	85/-65	50/-50	N/A	195	465
AUTOBRAKE MAX	1870	115/-125	60/80	-95/340	80/-60	50/-50	N/A	195	475
AUTOBRAKE 3	1965	115/-125	60/80	-100/350	60/-50	55/-55	N/A	135	405

Poor Reported Braking Action

MAX MANUAL	2345	160/-165	80/110	-140/525	195/-130	65/-70	N/A	395	1035
AUTOBRAKE MAX	2355	160/-165	80/110	-140/525	195/-125	65/-70	N/A	390	1035
AUTOBRAKE 3	2380	160/-165	80/110	-140/530	190/-125	65/-70	N/A	395	1045

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

All Flaps Up Landing

VREF40 + 55

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1330	185/-85	50/105	-45/205	20/-15	35/-35	45	45	95
AUTOBRAKE MAX	1855	85/-90	45/70	-60/195	5/-5	50/-50	75	5	20
AUTOBRAKE 2	3360	195/-225	115/150	-130/430	75/-85	105/-105	100	280	330

Good Reported Braking Action

MAX MANUAL	1755	85/-95	50/65	-65/230	40/-35	45/-50	45	110	255
AUTOBRAKE MAX	2000	90/-100	55/75	-75/245	30/-25	55/-55	70	85	225
AUTOBRAKE 2	3360	195/-225	115/150	-130/430	75/-85	105/-105	100	280	330

Medium Reported Braking Action

MAX MANUAL	2495	145/-155	80/110	-110/385	105/-85	70/-75	65	315	775
AUTOBRAKE MAX	2580	150/-160	85/115	-110/390	100/-80	75/-75	75	325	800
AUTOBRAKE 3	2950	145/-170	90/120	-120/420	65/-60	90/-90	110	165	510

Poor Reported Braking Action

MAX MANUAL	3320	220/-225	120/165	-165/605	250/-170	95/-100	80	690	1915
AUTOBRAKE MAX	3325	215/-225	120/165	-165/605	245/-160	100/-100	90	685	1905
AUTOBRAKE 3	3445	210/-225	120/165	-170/615	220/-150	100/-105	110	600	1840

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1795	105/-110	50/65	-80/290	55/-45	45/-45	60	145	345
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2015	125/-130	60/80	-100/350	85/-65	50/-55	70	215	530
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2585	180/-180	85/120	-145/545	200/-135	70/-75	80	460	1280
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3450	260/-260	120/175	-245/1005	625/-305	85/-105	95	1100	3915
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1695	95/-105	45/60	-80/280	55/-45	40/-45	60	125	300
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1895	115/-120	55/75	-95/340	80/-65	50/-50	65	185	455
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2415	165/-165	80/105	-140/530	190/-125	65/-70	80	395	1075
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3205	235/-235	110/155	-235/980	590/-285	75/-100	90	945	3215
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1615	90/-100	45/60	-80/275	55/-45	40/-40	60	115	265
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1805	105/-115	50/70	-95/335	80/-60	45/-45	65	170	405
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2290	150/-155	70/100	-140/520	185/-120	60/-65	80	355	950
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3040	220/-225	100/145	-230/960	575/-275	70/-95	85	860	2860
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Jammed or Restricted Flight Controls (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1005	70/-60	25/30	-35/125	10/-10	20/-25	35	25	50
AUTOBRAKE MAX	1310	65/-75	30/40	-45/155	5/-5	30/-30	65	0	5
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Good Reported Braking Action

MAX MANUAL	1380	75/-80	35/50	-60/205	35/-30	35/-35	50	80	185
AUTOBRAKE MAX	1485	85/-90	40/55	-60/215	30/-25	35/-35	55	90	205
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Medium Reported Braking Action

MAX MANUAL	1900	125/-130	60/80	-95/340	85/-65	50/-55	65	225	550
AUTOBRAKE MAX	1935	125/-130	60/80	-95/345	80/-65	50/-55	75	225	555
AUTOBRAKE 3	2055	125/-135	60/85	-100/355	60/-40	55/-60	100	145	465

Poor Reported Braking Action

MAX MANUAL	2480	180/-180	85/120	-145/540	205/-135	70/-75	75	475	1310
AUTOBRAKE MAX	2480	180/-180	85/120	-145/540	205/-135	70/-75	80	475	1305
AUTOBRAKE 3	2500	180/-180	85/120	-145/540	200/-120	70/-75	95	475	1310

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LEADING EDGE FLAPS TRANSIT (Flaps 15)

VREF15 + 15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1130	80/-70	25/35	-40/135	15/-15	25/-25	35	30	70
AUTOBRAKE MAX	1500	70/-80	35/45	-50/170	5/-5	40/-40	65	0	5
AUTOBRAKE 2	2725	175/-190	90/115	-115/385	50/-60	85/-85	100	155	160

Good Reported Braking Action

MAX MANUAL	1570	85/-95	45/60	-65/220	40/-35	40/-40	50	105	240
AUTOBRAKE MAX	1690	90/-100	45/60	-65/230	35/-30	45/-45	60	115	260
AUTOBRAKE 2	2730	175/-190	90/115	-115/385	50/-60	85/-85	100	160	160

Medium Reported Braking Action

MAX MANUAL	2170	140/-145	70/95	-100/365	95/-75	60/-60	70	275	695
AUTOBRAKE MAX	2210	140/-150	70/95	-105/365	90/-70	60/-65	75	280	700
AUTOBRAKE 3	2375	140/-150	75/100	-110/380	65/-45	70/-70	110	175	570

Poor Reported Braking Action

MAX MANUAL	2825	200/-205	100/140	-155/570	225/-150	80/-85	80	580	1620
AUTOBRAKE MAX	2825	200/-205	100/140	-155/570	230/-155	80/-85	85	575	1615
AUTOBRAKE 3	2855	200/-200	100/140	-155/570	215/-130	80/-85	100	565	1615

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1120	70/-65	25/35	-40/135	15/-15	25/-25	45	35	60
AUTOBRAKE MAX	1300	65/-75	30/40	-45/155	0/0	30/-30	60	0	10
AUTOBRAKE 2	2465	150/-175	75/95	-110/365	0/-10	75/-75	140	0	0

Good Reported Braking Action

MAX MANUAL	1620	95/-100	45/60	-70/235	50/-40	40/-45	70	135	275
AUTOBRAKE MAX	1630	95/-105	45/65	-70/235	40/-35	45/-45	75	135	275
AUTOBRAKE 2	2465	150/-175	75/95	-110/365	0/-10	75/-75	140	0	0

Medium Reported Braking Action

MAX MANUAL	2235	150/-155	75/100	-110/380	115/-90	60/-65	90	350	840
AUTOBRAKE MAX	2220	150/-155	75/100	-105/380	120/-95	60/-65	90	345	830
AUTOBRAKE 3	2220	150/-155	75/100	-105/380	120/-85	60/-65	90	345	830

Poor Reported Braking Action

MAX MANUAL	2905	215/-215	105/145	-160/590	265/-175	80/-85	105	710	2025
AUTOBRAKE MAX	2900	215/-215	105/150	-160/590	265/-180	80/-85	105	710	2025
AUTOBRAKE 3	2900	215/-215	105/150	-160/590	265/-180	80/-85	105	710	2025

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1060	65/-55	25/35	-40/130	15/-15	25/-25	45	30	50
AUTOBRAKE MAX	1215	60/-65	30/35	-45/145	0/0	30/-30	55	10	15
AUTOBRAKE 2	2260	135/-155	65/85	-105/350	0/-10	70/-70	135	0	0

Good Reported Braking Action

MAX MANUAL	1535	85/-95	45/60	-65/225	45/-40	40/-40	70	120	240
AUTOBRAKE MAX	1550	90/-95	45/60	-65/230	40/-35	40/-40	75	120	240
AUTOBRAKE 2	2260	135/-155	65/85	-105/350	0/-10	70/-70	135	0	0

Medium Reported Braking Action

MAX MANUAL	2090	135/-140	65/90	-105/370	110/-85	55/-60	85	305	710
AUTOBRAKE MAX	2085	135/-140	70/90	-105/370	115/-90	55/-60	90	300	705
AUTOBRAKE 3	2085	135/-140	70/90	-105/370	115/-80	55/-60	90	300	705

Poor Reported Braking Action

MAX MANUAL	2695	195/-195	95/130	-155/570	250/-165	75/-80	100	605	1650
AUTOBRAKE MAX	2695	195/-195	95/135	-155/570	250/-165	75/-80	100	605	1650
AUTOBRAKE 3	2695	195/-195	95/135	-155/570	250/-165	75/-80	100	605	1650

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1015	60/-55	25/30	-35/125	15/-15	25/-25	50	30	45
AUTOBRAKE MAX	1140	55/-60	25/35	-40/140	5/0	25/-25	55	10	20
AUTOBRAKE 2	2075	125/-140	60/80	-100/335	0/-5	60/-60	130	0	0

Good Reported Braking Action

MAX MANUAL	1460	80/-90	40/55	-65/225	45/-40	35/-40	70	105	210
AUTOBRAKE MAX	1470	85/-90	40/55	-65/225	40/-35	40/-40	75	105	210
AUTOBRAKE 2	2075	125/-140	60/80	-100/335	0/-5	60/-60	130	0	0

Medium Reported Braking Action

MAX MANUAL	1970	125/-135	60/85	-100/360	105/-85	55/-55	85	265	615
AUTOBRAKE MAX	1970	125/-135	60/85	-100/360	110/-85	55/-55	85	265	615
AUTOBRAKE 3	1970	125/-135	60/85	-100/360	110/-80	55/-55	90	265	615

Poor Reported Braking Action

MAX MANUAL	2525	180/-185	85/120	-150/560	240/-155	70/-75	95	525	1400
AUTOBRAKE MAX	2530	180/-185	90/125	-150/560	245/-160	70/-75	95	530	1405
AUTOBRAKE 3	2530	180/-185	90/125	-150/560	245/-160	70/-75	95	530	1405

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1570	80/-90	40/50	-60/195	35/-35	40/-40	75	-10	65
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2290	135/-145	65/90	-100/335	100/-80	60/-60	105	95	440
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	3035	200/-210	100/140	-150/525	215/-160	80/-85	120	365	1415
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3770	275/-275	135/190	-210/785	475/-270	100/-110	130	815	3380
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM B (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1140	55/-60	25/35	-45/145	20/-15	25/-25	40	40	70
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1630	95/-100	45/65	-75/255	50/-45	45/-45	60	140	285
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2215	150/-155	70/100	-115/410	125/-95	60/-65	75	340	815
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	2835	210/-210	100/140	-170/640	295/-180	75/-85	90	665	1870
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

MANUAL REVERSION (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1570	80/-90	40/50	-60/195	35/-35	40/-40	75	-10	65
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2290	135/-145	65/90	-100/335	100/-80	60/-60	105	95	440
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	3035	200/-210	100/140	-150/525	215/-160	80/-85	120	365	1415
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3770	275/-275	135/190	-210/785	475/-270	100/-110	130	815	3380
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

One Engine Inoperative Landing (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1020	75/-65	25/30	-35/130	15/-10	25/-25	35	0	25
AUTOBRAKE MAX	1300	70/-75	30/40	-45/155	0/0	30/-30	60	0	0
AUTOBRAKE 2	2450	150/-170	75/95	-110/365	10/-25	75/-75	120	0	0

Good Reported Braking Action

MAX MANUAL	1440	80/-85	40/50	-65/215	40/-35	40/-40	50	0	100
AUTOBRAKE MAX	1545	85/-95	40/55	-65/225	35/-30	40/-40	60	0	110
AUTOBRAKE 2	2450	150/-170	75/95	-110/365	10/-25	75/-75	120	0	0

Medium Reported Braking Action

MAX MANUAL	2075	135/-140	65/85	-105/370	110/-85	60/-60	70	0	310
AUTOBRAKE MAX	2115	135/-145	65/85	-105/375	105/-80	60/-60	80	0	315
AUTOBRAKE 3	2165	135/-150	65/85	-105/380	90/-65	60/-65	100	0	295

Poor Reported Braking Action

MAX MANUAL	2850	200/-210	95/130	-165/605	290/-185	85/-85	90	0	765
AUTOBRAKE MAX	2850	200/-210	95/130	-165/605	290/-185	85/-85	95	0	765
AUTOBRAKE 3	2875	205/-210	95/130	-165/610	280/-180	85/-85	100	0	775

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

One Engine Inoperative Landing (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	970	60/-55	20/30	-35/125	15/-10	20/-20	35	0	25
AUTOBRAKE MAX	1215	60/-65	30/35	-45/150	0/0	30/-30	55	0	0
AUTOBRAKE 2	2240	135/-150	65/85	-105/350	10/-25	65/-65	110	0	0

Good Reported Braking Action

MAX MANUAL	1370	75/-80	35/50	-60/210	35/-30	35/-35	50	0	90
AUTOBRAKE MAX	1465	80/-90	40/50	-65/220	35/-30	35/-40	60	0	100
AUTOBRAKE 2	2240	135/-150	65/85	-105/350	10/-25	65/-65	110	0	0

Medium Reported Braking Action

MAX MANUAL	1940	120/-130	60/80	-100/360	105/-80	55/-55	70	0	265
AUTOBRAKE MAX	1975	125/-135	60/80	-100/365	95/-75	55/-55	80	0	270
AUTOBRAKE 3	2015	125/-135	60/80	-105/365	90/-65	55/-60	90	0	260

Poor Reported Braking Action

MAX MANUAL	2625	180/-190	85/115	-155/585	265/-170	75/-80	85	0	635
AUTOBRAKE MAX	2625	180/-190	85/115	-155/585	270/-165	75/-80	90	0	635
AUTOBRAKE 3	2655	185/-190	90/120	-160/585	260/-170	75/-80	90	0	640

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Stabilizer Trim Inoperative (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1005	70/-60	25/30	-35/125	10/-10	20/-25	35	25	50
AUTOBRAKE MAX	1310	65/-75	30/40	-45/155	5/-5	30/-30	65	0	5
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Good Reported Braking Action

MAX MANUAL	1380	75/-80	35/50	-60/205	35/-30	35/-35	50	80	185
AUTOBRAKE MAX	1485	85/-90	40/55	-60/215	30/-25	35/-35	55	90	205
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Medium Reported Braking Action

MAX MANUAL	1900	125/-130	60/80	-95/340	85/-65	50/-55	65	225	550
AUTOBRAKE MAX	1935	125/-130	60/80	-95/345	80/-65	50/-55	75	225	555
AUTOBRAKE 3	2055	125/-135	60/85	-100/355	60/-40	55/-60	100	145	465

Poor Reported Braking Action

MAX MANUAL	2480	180/-180	85/120	-145/540	205/-135	70/-75	75	475	1310
AUTOBRAKE MAX	2480	180/-180	85/120	-145/540	205/-135	70/-75	80	475	1305
AUTOBRAKE 3	2500	180/-180	85/120	-145/540	200/-120	70/-75	95	475	1310

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Trailing Edge Flap Asymmetry ($1 \leq \text{Flap Lever} < 15$)

VREF40 + 30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1110	90/-65	25/40	-40/135	15/-15	25/-25	35	30	60
AUTOBRAKE MAX	1510	70/-75	35/45	-50/170	5/-5	40/-40	65	0	5
AUTOBRAKE 2	2730	165/-185	90/115	-115/385	55/-60	85/-85	100	165	175

Good Reported Braking Action

MAX MANUAL	1525	80/-85	40/55	-60/215	35/-30	40/-40	45	90	210
AUTOBRAKE MAX	1665	80/-90	45/60	-65/225	30/-25	45/-45	65	95	225
AUTOBRAKE 2	2735	160/-185	90/115	-115/385	55/-65	85/-85	95	165	175

Medium Reported Braking Action

MAX MANUAL	2125	125/-135	70/90	-100/360	90/-75	60/-60	65	255	625
AUTOBRAKE MAX	2180	130/-140	70/95	-100/360	85/-70	60/-60	75	260	640
AUTOBRAKE 3	2385	125/-140	70/95	-110/380	60/-45	70/-70	110	150	470

Poor Reported Braking Action

MAX MANUAL	2795	190/-195	100/135	-150/565	220/-150	80/-85	80	545	1510
AUTOBRAKE MAX	2790	185/-195	100/135	-150/565	220/-145	80/-85	85	540	1500
AUTOBRAKE 3	2845	185/-190	100/135	-155/570	205/-130	80/-85	105	525	1495

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Trailing Edge Flap Asymmetry (Flap Lever 15 or 25) VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1005	70/-60	25/30	-35/125	10/-10	20/-25	35	25	50
AUTOBRAKE MAX	1310	65/-75	30/40	-45/155	5/-5	30/-30	65	0	5
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Good Reported Braking Action

MAX MANUAL	1380	75/-80	35/50	-60/205	35/-30	35/-35	50	80	185
AUTOBRAKE MAX	1485	85/-90	40/55	-60/215	30/-25	35/-35	55	90	205
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Medium Reported Braking Action

MAX MANUAL	1900	125/-130	60/80	-95/340	85/-65	50/-55	65	225	550
AUTOBRAKE MAX	1935	125/-130	60/80	-95/345	80/-65	50/-55	75	225	555
AUTOBRAKE 3	2055	125/-135	60/85	-100/355	60/-40	55/-60	100	145	465

Poor Reported Braking Action

MAX MANUAL	2480	180/-180	85/120	-145/540	205/-135	70/-75	75	475	1310
AUTOBRAKE MAX	2480	180/-180	85/120	-145/540	205/-135	70/-75	80	475	1305
AUTOBRAKE 3	2500	180/-180	85/120	-145/540	200/-120	70/-75	95	475	1310

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Asymmetry (Flap Lever 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	960	55/-55	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1215	60/-65	30/35	-45/150	5/0	30/-30	60	0	5
AUTOBRAKE 2	2165	140/-150	65/90	-100/340	30/-45	65/-65	90	85	85

Good Reported Braking Action

MAX MANUAL	1315	70/-75	35/45	-60/200	30/-30	35/-35	50	75	165
AUTOBRAKE MAX	1410	75/-85	35/50	-60/210	30/-25	35/-35	60	80	185
AUTOBRAKE 2	2165	140/-150	65/90	-100/340	35/-45	65/-65	90	85	85

Medium Reported Braking Action

MAX MANUAL	1790	115/-120	55/75	-90/330	80/-65	50/-50	65	195	480
AUTOBRAKE MAX	1820	115/-120	55/75	-95/335	75/-60	50/-50	70	200	480
AUTOBRAKE 3	1910	115/-125	55/75	-95/345	60/-40	50/-55	95	140	425

Poor Reported Braking Action

MAX MANUAL	2315	160/-165	80/105	-140/525	195/-125	60/-65	75	410	1100
AUTOBRAKE MAX	2320	165/-165	80/110	-140/525	195/-130	65/-70	75	410	1100
AUTOBRAKE 3	2335	165/-165	80/110	-140/525	190/-115	65/-70	90	410	1110

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (1 ≤ Indicated Flaps <15)

VREF40 + 30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1110	90/-65	25/40	-40/135	15/-15	25/-25	35	30	60
AUTOBRAKE MAX	1510	70/-75	35/45	-50/170	5/-5	40/-40	65	0	5
AUTOBRAKE 2	2730	165/-185	90/115	-115/385	55/-60	85/-85	100	165	175

Good Reported Braking Action

MAX MANUAL	1525	80/-85	40/55	-60/215	35/-30	40/-40	45	90	210
AUTOBRAKE MAX	1665	80/-90	45/60	-65/225	30/-25	45/-45	65	95	225
AUTOBRAKE 2	2735	160/-185	90/115	-115/385	55/-65	85/-85	95	165	175

Medium Reported Braking Action

MAX MANUAL	2125	125/-135	70/90	-100/360	90/-75	60/-60	65	255	625
AUTOBRAKE MAX	2180	130/-140	70/95	-100/360	85/-70	60/-60	75	260	640
AUTOBRAKE 3	2385	125/-140	70/95	-110/380	60/-45	70/-70	110	150	470

Poor Reported Braking Action

MAX MANUAL	2795	190/-195	100/135	-150/565	220/-150	80/-85	80	545	1510
AUTOBRAKE MAX	2790	185/-195	100/135	-150/565	220/-145	80/-85	85	540	1500
AUTOBRAKE 3	2845	185/-190	100/135	-155/570	205/-130	80/-85	105	525	1495

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (15 ≤ Indicated Flaps <30)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1005	70/-60	25/30	-35/125	10/-10	20/-25	35	25	50
AUTOBRAKE MAX	1310	65/-75	30/40	-45/155	5/-5	30/-30	65	0	5
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Good Reported Braking Action

MAX MANUAL	1380	75/-80	35/50	-60/205	35/-30	35/-35	50	80	185
AUTOBRAKE MAX	1485	85/-90	40/55	-60/215	30/-25	35/-35	55	90	205
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Medium Reported Braking Action

MAX MANUAL	1900	125/-130	60/80	-95/340	85/-65	50/-55	65	225	550
AUTOBRAKE MAX	1935	125/-130	60/80	-95/345	80/-65	50/-55	75	225	555
AUTOBRAKE 3	2055	125/-135	60/85	-100/355	60/-40	55/-60	100	145	465

Poor Reported Braking Action

MAX MANUAL	2480	180/-180	85/120	-145/540	205/-135	70/-75	75	475	1310
AUTOBRAKE MAX	2480	180/-180	85/120	-145/540	205/-135	70/-75	80	475	1305
AUTOBRAKE 3	2500	180/-180	85/120	-145/540	200/-120	70/-75	95	475	1310

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (30 ≤ Indicated Flaps <40)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	960	55/-55	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1215	60/-65	30/35	-45/150	5/0	30/-30	60	0	5
AUTOBRAKE 2	2165	140/-150	65/90	-100/340	30/-45	65/-65	90	85	85

Good Reported Braking Action

MAX MANUAL	1315	70/-75	35/45	-60/200	30/-30	35/-35	50	75	165
AUTOBRAKE MAX	1410	75/-85	35/50	-60/210	30/-25	35/-35	60	80	185
AUTOBRAKE 2	2165	140/-150	65/90	-100/340	35/-45	65/-65	90	85	85

Medium Reported Braking Action

MAX MANUAL	1790	115/-120	55/75	-90/330	80/-65	50/-50	65	195	480
AUTOBRAKE MAX	1820	115/-120	55/75	-95/335	75/-60	50/-50	70	200	480
AUTOBRAKE 3	1910	115/-125	55/75	-95/345	60/-40	50/-55	95	140	425

Poor Reported Braking Action

MAX MANUAL	2315	160/-165	80/105	-140/525	195/-125	60/-65	75	410	1100
AUTOBRAKE MAX	2320	165/-165	80/110	-140/525	195/-130	65/-70	75	410	1100
AUTOBRAKE 3	2335	165/-165	80/110	-140/525	190/-115	65/-70	90	410	1110

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flaps Up Landing

VREF40 + 40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1185	110/-70	30/70	-40/140	15/-15	30/-30	45	30	70
AUTOBRAKE MAX	1645	75/-80	40/55	-55/180	5/-5	40/-45	70	5	10
AUTOBRAKE 2	2970	175/-195	100/130	-120/400	65/-70	90/-90	95	205	235

Good Reported Braking Action

MAX MANUAL	1600	80/-90	45/60	-65/220	35/-30	40/-45	45	90	205
AUTOBRAKE MAX	1795	85/-95	50/65	-70/235	25/-25	45/-50	65	80	200
AUTOBRAKE 2	2970	175/-195	100/130	-120/400	65/-70	90/-90	95	205	235

Medium Reported Braking Action

MAX MANUAL	2255	135/-140	70/95	-105/365	95/-75	65/-65	65	260	625
AUTOBRAKE MAX	2330	135/-145	75/100	-105/370	90/-75	65/-65	70	265	645
AUTOBRAKE 3	2605	135/-155	80/105	-115/395	60/-55	75/-80	105	145	435

Poor Reported Braking Action

MAX MANUAL	2990	200/-205	105/145	-155/580	230/-155	85/-90	80	565	1530
AUTOBRAKE MAX	2995	195/-205	105/145	-155/580	230/-150	85/-90	90	560	1520
AUTOBRAKE 3	3080	190/-205	105/145	-160/585	210/-140	90/-95	100	520	1495

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Recommended Brake Cooling Schedule

Reference Brake Energy Per Brake (Millions of Foot Pounds)

		WIND CORRECTED BRAKES ON SPEED (KIAS)*																	
		80			100			120			140			160			180		
WEIGHT (1000 KG)	OAT (°C)	PRESSURE ALTITUDE (1000 FT)																	
		0	5	10	0	5	10	0	5	10	0	5	10	0	5	10	0	5	10
80	0	15.1	17.0	19.3	22.4	25.3	28.9	30.9	35.0	40.2	40.4	45.9	53.0	50.8	57.9	67.3	60.8	69.6	81.2
	10	15.6	17.6	20.0	23.1	26.1	29.8	31.9	36.2	41.5	41.8	47.5	54.8	52.5	59.9	69.5	62.8	71.9	83.9
	15	15.8	17.8	20.2	23.5	26.5	30.3	32.4	36.7	42.1	42.4	48.2	55.6	53.3	60.7	70.5	63.7	72.9	85
	20	16.0	18.1	20.5	23.8	26.9	30.7	32.8	37.2	42.7	42.9	48.8	56.3	54.0	61.5	71.4	64.6	73.9	86.2
	30	16.4	18.5	21.1	24.4	27.6	31.5	33.7	38.2	43.8	44.0	50.0	57.7	55.3	63.1	73.2	66.2	75.7	88.4
	40	16.6	18.7	21.3	24.7	27.9	31.9	34.1	38.7	44.4	44.7	50.9	58.8	56.3	64.3	74.8	67.5	77.4	90.5
50	16.6	18.7	21.3	24.8	28.0	32.1	34.3	39.0	44.9	45.2	51.5	59.7	57.1	65.4	76.3	68.7	79.0	92.9	
70	0	13.7	15.4	17.5	20.2	22.8	26.0	27.7	31.3	35.9	36.1	41.0	47.2	45.3	51.6	59.7	54.9	62.7	72.9
	10	14.2	15.9	18.1	20.8	23.5	26.8	28.6	32.4	37.1	37.3	42.3	48.7	46.8	53.3	61.6	56.7	64.8	75.4
	15	14.4	16.2	18.4	21.1	23.9	27.2	29.0	32.8	37.6	37.8	43.0	49.4	47.5	54.0	62.5	57.5	65.7	76.4
	20	14.6	16.4	18.6	21.4	24.2	27.6	29.4	33.3	38.1	38.4	43.5	50.1	48.1	54.8	63.4	58.3	66.5	77.4
	30	14.9	16.8	19.1	22.0	24.8	28.3	30.2	34.1	39.1	39.3	44.6	51.4	49.3	56.1	64.9	59.8	68.2	79.4
	40	15.1	17.0	19.3	22.2	25.1	28.6	30.5	34.6	39.6	39.9	45.3	52.2	50.1	57.1	66.2	60.9	69.6	81.2
50	15.1	17.0	19.3	22.3	25.2	28.8	30.7	34.8	40.0	40.2	45.8	52.9	50.7	58.0	67.4	61.8	70.9	83.0	
60	0	12.3	13.9	15.7	18.0	20.3	23.1	24.4	27.6	31.6	31.7	35.9	41.2	39.6	45.0	51.8	48.1	54.8	63.5
	10	12.7	14.3	16.3	18.5	20.9	23.3	25.2	28.5	32.6	32.7	37.1	42.6	40.9	46.5	53.6	49.7	56.6	65.6
	15	12.9	14.6	16.5	18.8	21.2	24.2	25.6	29.0	33.1	33.2	37.6	43.2	41.5	47.1	54.4	50.4	57.4	66.5
	20	13.1	14.8	16.7	19.1	21.5	24.5	26.0	29.4	33.5	33.6	38.1	43.8	42.0	47.8	55.1	51.1	58.2	67.4
	30	13.4	15.1	17.2	19.6	22.1	25.1	26.6	30.1	34.4	34.5	39.1	44.9	43.1	49.0	56.5	52.3	59.6	69.1
	40	13.6	15.3	17.3	19.8	22.3	25.4	26.9	30.5	34.9	35.0	39.7	45.6	43.8	49.8	57.5	53.2	60.7	70.5
50	13.5	15.3	17.3	19.8	22.4	25.5	27.0	30.6	35.1	35.2	40.0	46.0	44.2	50.4	58.3	53.9	61.7	71.9	
50	0	11.0	12.3	14.0	15.7	17.7	20.2	21.2	23.9	27.3	27.2	30.8	35.3	33.8	38.3	44.1	40.9	46.4	53.6
	10	11.3	12.7	14.4	16.3	18.3	20.8	21.9	24.7	28.2	28.1	31.8	36.5	34.9	39.6	45.5	42.2	48.0	55.4
	15	11.5	12.9	14.7	16.5	18.6	21.1	22.2	25.1	28.6	28.6	32.3	37.0	35.4	40.2	46.2	42.8	48.7	56.2
	20	11.6	13.1	14.9	16.7	18.9	21.4	22.5	25.4	29.0	28.9	32.8	37.5	35.9	40.7	46.8	43.4	49.3	56.9
	30	11.9	13.4	15.2	17.2	19.3	22.0	23.1	26.1	29.7	29.7	33.6	38.4	36.8	41.4	48.0	44.5	50.6	58.4
	40	12.1	13.6	15.4	17.3	19.5	22.2	23.4	26.4	30.1	30.1	34.0	39.0	37.4	42.4	48.8	45.2	51.4	59.4
50	12.0	13.6	15.4	17.3	19.6	22.3	23.4	26.5	30.3	30.2	34.2	39.3	37.6	42.8	49.3	45.7	52.1	60.3	
40	0	9.6	10.8	12.3	13.5	15.2	17.3	17.9	20.2	23.0	22.8	25.8	29.4	28.1	31.8	36.4	33.7	38.2	43.9
	10	10.0	11.2	12.7	14.0	15.8	17.9	18.5	20.9	23.8	23.6	26.6	30.4	29.0	32.8	37.6	34.8	39.5	45.4
	15	10.1	11.4	12.9	14.2	16.0	18.1	18.8	21.2	24.1	23.9	27.0	30.8	29.4	33.3	38.2	35.3	40.0	46.0
	20	10.2	11.5	13.1	14.4	16.2	18.4	19.1	21.5	24.5	24.2	27.4	31.3	29.8	33.8	38.7	35.8	40.6	46.6
	30	10.5	11.8	13.4	14.8	16.6	18.9	19.6	22.1	25.1	24.9	28.1	32.1	30.6	34.6	39.7	36.7	41.6	47.8
	40	10.6	11.9	13.5	14.9	16.8	19.1	19.8	22.3	25.4	25.2	28.4	32.5	31.0	35.1	40.2	37.2	42.2	48.6
50	10.6	11.9	13.5	14.9	16.8	19.1	19.8	22.3	25.5	25.2	28.6	32.7	31.1	35.3	40.6	37.5	42.6	49.1	

*To correct for wind, enter table with the brakes on speed minus one half the headwind or plus 1.5 times the tailwind. If ground speed is used for brakes on speed, ignore wind and enter table with sea level, 15°C.

ADVISORY INFORMATION

Recommended Brake Cooling Schedule

Adjusted Brake Energy Per Brake (Millions of Foot Pounds)

No Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	7.8	16.3	25.3	34.7	44.7	55.0	65.7	76.6	87.9
	MAX AUTO	7.5	15.4	23.6	32.4	41.8	51.8	62.5	74.1	86.5
	AUTOBRAKE 3	7.3	14.7	22.3	30.2	38.6	47.6	57.4	68.1	80.0
	AUTOBRAKE 2	7.0	13.8	20.5	27.4	34.8	42.7	51.5	61.3	72.4
	AUTOBRAKE 1	6.7	13.1	19.2	25.3	31.8	38.8	46.6	55.4	65.5

Two Engine Detent Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	7.0	14.6	22.8	31.4	40.5	49.9	59.7	69.8	80.0
	MAX AUTO	5.8	12.3	19.5	27.2	35.6	44.5	53.9	63.7	74.1
	AUTOBRAKE 3	4.3	9.2	14.7	20.7	27.2	34.4	42.0	50.2	59.0
	AUTOBRAKE 2	2.5	5.6	9.1	13.1	17.8	23.0	28.8	35.2	42.3
	AUTOBRAKE 1	1.8	3.8	6.1	8.8	11.9	15.5	19.6	24.4	29.8

Cooling Time (Minutes) - Category C Steel Brakes

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)								
		16 & BELOW	17	20	23	25	28	32	33 TO 48	49 & ABOVE
		BRAKE TEMPERATURE MONITOR SYSTEM INDICATION ON CDS								
		UP TO 2.4	2.6	3.1	3.5	3.9	4.4	4.9	5.0 TO 7.5	7.5 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE	1	2	3	4	5	6	CAUTION	FUSE PLUG MELT ZONE	
GROUND	REQUIRED	10	20	30	40	50	60			

Cooling Time (Minutes) - Category N Carbon Brakes

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)								
		16 & BELOW	17	19	20.9	23.5	26.9	29.4	30 TO 41	41 & ABOVE
		BRAKE TEMPERATURE MONITOR SYSTEM INDICATION ON CDS								
		UP TO 2.5	2.6	3	3.3	3.8	4.5	4.9	5.0 TO 7.1	7.1 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE	1	4	5	6	7	7.6	CAUTION	FUSE PLUG MELT ZONE	
GROUND	REQUIRED	6.7	16.0	24.1	34.2	45.9	53.3			

Observe maximum quick turnaround limit.

Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.

Add 1.0 million foot pounds per brake for each taxi mile.

When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after one hour. If overheat occurs after takeoff, extend gear soon for at least 7 minutes.

When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not approach gear or attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

Brake temperature monitor system (BTMS) indication on CDS systems page may be used 10 to 15 minutes after airplane has come to a complete stop or inflight with gear retracted to determine recommended cooling schedule.

Intentionally
Blank

Performance Inflight - QRH

Engine Inoperative

Chapter PI-QRH

Section 12

ENGINE INOP

Initial Max Continuous %N1

Based on .79M, A/C high and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)								
	25	27	29	31	33	35	37	39	41
20	96.1	95.9	95.6	95.5	95.2	94.8	94.3	94.0	93.2
15	96.7	96.5	96.2	96.1	96.0	95.5	95.1	94.8	94.1
10	97.3	97.2	96.8	96.7	96.7	96.2	95.8	95.6	95.0
5	97.5	97.9	97.6	97.4	97.4	97.0	96.6	96.4	95.9
0	96.8	98.1	98.5	98.3	98.2	97.8	97.5	97.2	96.8
-5	96.0	97.3	98.5	99.2	99.1	98.6	98.3	98.1	97.8
-10	95.2	96.5	97.7	99.0	99.9	99.5	99.2	99.0	98.7
-15	94.4	95.8	96.9	98.2	99.5	100.4	100.1	99.9	99.7
-20	93.6	95.0	96.2	97.4	98.7	99.8	100.4	100.2	100.0
-25	92.8	94.2	95.4	96.6	97.9	99.0	99.6	99.4	99.2
-30	91.9	93.4	94.6	95.8	97.0	98.2	98.7	98.5	98.3
-35	91.1	92.6	93.7	94.9	96.2	97.3	97.9	97.7	97.5
-40	90.3	91.8	92.9	94.1	95.3	96.5	97.0	96.8	96.6

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)								
	25	27	29	31	33	35	37	39	41
ENGINE ANTI-ICE	-1.2	-1.1	-1.0	-0.9	-0.8	-0.8	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE	-4.2	-4.4	-4.5	-4.7	-5.0	-4.8	-4.8	-4.8	-4.8

ENGINE INOP

Max Continuous %N1

37000 FT to 29000 FT Pressure Altitudes

37000 FT PRESS ALT													TAT (°C)
CIAS	M	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0
160	.51	96.0	97.0	97.9	98.7	99.6	98.9	98.1	96.9	95.6	94.0	92.5	91.1
200	.63	95.4	96.3	97.2	98.1	98.9	99.8	99.5	98.7	97.8	96.8	95.6	94.5
240	.74	94.4	95.3	96.2	97.1	98.0	98.8	99.7	100.1	99.3	98.5	97.7	96.7
280	.86	93.7	94.6	95.5	96.4	97.2	98.1	98.9	99.7	100.5	100.2	99.3	98.5
35000 FT PRESS ALT													TAT (°C)
CIAS	M	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0
160	.49	95.9	96.8	97.7	98.6	99.5	99.2	98.4	97.3	96.1	94.7	93.3	92.0
200	.60	95.5	96.4	97.3	98.2	99.1	100.0	99.9	98.9	98.0	97.0	95.8	94.7
240	.71	94.4	95.3	96.2	97.1	98.0	98.8	99.6	100.2	99.5	98.9	98.0	97.0
280	.82	93.2	94.0	94.9	95.8	96.6	97.5	98.3	99.1	99.9	99.7	98.9	98.1
33000 FT PRESS ALT													TAT (°C)
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
160	.47	96.8	97.7	98.5	99.4	100.2	99.3	98.5	97.3	96.0	94.6	93.2	92.0
200	.58	96.4	97.3	98.2	99.1	100.0	100.8	99.9	99.0	98.0	96.8	95.6	94.5
240	.68	95.3	96.2	97.1	97.9	98.8	99.6	100.4	100.2	99.6	98.7	97.7	96.7
280	.79	93.6	94.5	95.4	96.2	97.1	97.9	98.7	99.5	99.9	99.1	98.2	97.4
320	.89	93.0	93.9	94.7	95.6	96.4	97.2	98.1	98.9	99.7	100.4	100.0	99.2
31000 FT PRESS ALT													TAT (°C)
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
160	.45	96.7	97.6	98.5	99.4	100.3	100.4	99.5	98.5	97.3	95.9	94.5	93.2
200	.55	96.5	97.3	98.2	99.1	100.0	100.8	101.0	100.1	99.1	98.0	96.7	95.5
240	.66	95.0	95.9	96.8	97.6	98.5	99.3	100.2	100.7	99.9	99.1	98.1	97.1
280	.76	93.2	94.0	94.9	95.7	96.6	97.4	98.2	99.0	99.8	99.1	98.2	97.3
320	.85	91.8	92.6	93.5	94.3	95.1	95.9	96.7	97.5	98.3	99.1	99.3	98.4
29000 FT PRESS ALT													TAT (°C)
CIAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10
160	.43	97.5	98.4	99.3	100.1	101.0	100.5	99.6	98.5	97.2	95.7	94.4	93.1
200	.53	96.9	97.8	98.7	99.5	100.4	101.2	100.7	99.7	98.7	97.5	96.2	95.1
240	.63	95.7	96.5	97.4	98.2	99.1	99.9	100.7	100.4	99.5	98.6	97.5	96.6
280	.73	93.6	94.4	95.2	96.1	96.9	97.7	98.5	99.3	99.4	98.5	97.5	96.8
320	.82	91.4	92.3	93.1	93.9	94.7	95.5	96.3	97.0	97.8	98.6	97.8	97.0
360	.91	91.4	92.3	93.1	93.9	94.7	95.5	96.3	97.0	97.8	98.6	99.3	99.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	29	31	33	35	37
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE ON	-4.1	-4.3	-4.5	-4.7	-4.7

ENGINE INOP

Max Continuous %N1

27000 FT to 20000 FT Pressure Altitudes

27000 FT PRESS ALT												TAT (°C)	
KIAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10
160	.41	97.3	98.2	99.1	100.0	100.8	101.5	100.6	99.6	98.4	97.0	95.7	94.4
200	.51	96.3	97.2	98.1	98.9	99.8	100.6	101.1	100.2	99.2	98.1	96.9	95.7
240	.60	95.0	95.9	96.7	97.6	98.4	99.2	100.1	100.7	99.7	98.7	97.7	96.8
280	.70	93.0	93.8	94.6	95.5	96.3	97.1	97.9	98.7	99.4	98.7	97.7	96.9
320	.79	90.9	91.7	92.6	93.4	94.2	95.0	95.7	96.5	97.3	98.0	97.9	97.2
360	.88	90.2	91.0	91.8	92.7	93.5	94.3	95.1	95.9	96.6	97.4	98.2	98.7
25000 FT PRESS ALT												TAT (°C)	
KIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
160	.39	98.2	99.0	99.9	100.7	101.6	101.7	100.7	99.6	98.4	97.0	95.8	94.5
200	.49	96.8	97.7	98.5	99.4	100.2	101.0	100.9	99.9	98.9	97.7	96.6	95.5
240	.58	95.1	95.9	96.8	97.6	98.4	99.2	100.0	99.8	98.9	97.9	96.9	96.0
280	.67	93.2	94.1	94.9	95.7	96.5	97.3	98.1	98.8	98.9	97.9	96.9	96.2
320	.76	90.9	91.8	92.6	93.4	94.2	95.0	95.8	96.6	97.3	97.9	97.2	96.5
360	.85	89.6	90.5	91.3	92.1	93.0	93.8	94.6	95.4	96.2	97.0	97.7	97.5
24000 FT PRESS ALT												TAT (°C)	
KIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
160	.38	97.7	98.5	99.4	100.3	101.1	101.9	100.8	99.7	98.5	97.2	96.0	94.7
200	.48	96.4	97.2	98.1	98.9	99.7	100.6	101.0	99.9	98.9	97.8	96.7	95.6
240	.57	94.7	95.6	96.4	97.2	98.0	98.8	99.6	99.9	99.0	97.9	97.0	96.1
280	.66	93.0	93.8	94.6	95.4	96.2	97.0	97.8	98.6	99.1	98.0	97.0	96.3
320	.75	90.6	91.4	92.3	93.1	93.9	94.7	95.5	96.3	97.1	97.8	97.2	96.5
360	.83	89.0	89.8	90.7	91.5	92.4	93.2	94.0	94.8	95.6	96.4	97.2	97.2
22000 FT PRESS ALT												TAT (°C)	
KIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
160	.37	97.5	98.4	99.2	100.1	100.9	101.0	99.9	98.7	97.5	96.3	95.2	94.0
200	.46	96.3	97.1	98.0	98.8	99.6	100.4	100.1	98.9	97.8	96.8	95.8	94.8
240	.55	94.8	95.6	96.4	97.2	98.0	98.8	99.6	99.1	98.1	97.1	96.2	95.4
280	.63	93.2	94.0	94.8	95.6	96.4	97.1	97.9	98.7	98.4	97.4	96.6	95.8
320	.72	90.9	91.8	92.6	93.4	94.2	95.0	95.8	96.6	97.4	97.5	96.8	96.1
360	.80	89.0	89.9	90.7	91.6	92.4	93.2	94.0	94.8	95.6	96.4	97.0	96.4
20000 FT PRESS ALT												TAT (°C)	
KIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
160	.35	96.5	97.4	98.2	99.0	99.8	100.6	100.2	98.9	97.7	96.6	95.5	94.4
200	.44	95.4	96.2	97.0	97.9	98.7	99.4	100.2	99.1	97.8	96.8	95.8	94.9
240	.53	93.9	94.7	95.5	96.3	97.1	97.9	98.7	99.3	98.2	97.1	96.2	95.4
280	.61	92.4	93.3	94.1	94.8	95.6	96.4	97.2	97.9	98.5	97.6	96.7	95.9
320	.69	90.3	91.1	92.0	92.8	93.6	94.4	95.2	96.0	96.8	97.6	96.9	96.2
360	.77	88.5	89.3	90.2	91.0	91.8	92.6	93.5	94.3	95.1	95.8	96.6	96.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	20	22	24	25	27
ENGINE ANTI-ICE ON	-0.9	-0.9	-1.0	-1.0	-1.0
ENGINE & WING ANTI-ICE ON	-3.6	-3.8	-3.8	-3.9	-4.0

ENGINE INOP

Max Continuous %N1
18000 FT to 12000 FT Pressure Altitudes

18000 FT PRESS ALT												TAT (°C)	
KLAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
160	.34	96.0	96.8	97.6	98.4	99.2	100.0	98.9	97.5	96.5	95.5	94.5	93.5
200	.42	95.1	95.9	96.7	97.5	98.2	99.0	99.3	98.0	96.7	95.9	95.0	94.1
240	.51	93.7	94.5	95.2	96.0	96.8	97.6	98.3	98.2	97.1	96.2	95.4	94.6
280	.59	92.0	92.9	93.7	94.5	95.3	96.1	96.8	97.6	97.5	96.6	95.8	95.1
320	.67	90.3	91.1	92.0	92.8	93.6	94.4	95.2	96.0	96.8	96.9	96.2	95.5
360	.75	88.7	89.5	90.4	91.2	92.0	92.8	93.6	94.4	95.2	96.0	96.4	95.8
16000 FT PRESS ALT												TAT (°C)	
KLAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
160	.33	95.0	95.8	96.6	97.4	98.2	99.0	99.4	98.2	97.0	96.1	95.2	94.2
200	.41	93.9	94.7	95.5	96.3	97.1	97.8	98.6	98.2	97.0	96.0	95.2	94.4
240	.49	92.5	93.3	94.1	94.9	95.7	96.5	97.2	98.0	97.3	96.3	95.5	94.7
280	.57	91.0	91.8	92.6	93.5	94.3	95.1	95.9	96.6	97.4	96.7	95.8	95.1
320	.64	89.4	90.3	91.1	91.9	92.8	93.6	94.4	95.2	95.9	96.7	96.1	95.5
360	.72	88.0	88.9	89.7	90.6	91.4	92.2	93.0	93.8	94.6	95.4	96.2	95.8
14000 FT PRESS ALT												TAT (°C)	
KLAS	M	-25	-20	-15	-10	-5	0	5	10	15	20	25	30
160	.31	94.9	95.7	96.5	97.3	98.0	98.8	99.2	98.2	97.3	96.4	95.5	94.6
200	.39	93.6	94.4	95.2	96.0	96.7	97.5	98.3	97.5	96.5	95.7	94.9	94.1
240	.47	92.1	92.9	93.8	94.6	95.4	96.2	96.9	97.4	96.5	95.6	94.8	94.1
280	.54	90.9	91.7	92.5	93.4	94.2	95.0	95.8	96.5	96.8	96.0	95.2	94.5
320	.62	89.6	90.4	91.2	92.1	92.9	93.7	94.5	95.3	96.1	96.2	95.5	94.8
360	.69	88.3	89.1	89.9	90.7	91.6	92.4	93.2	94.0	94.8	95.5	95.8	95.2
12000 FT PRESS ALT												TAT (°C)	
KLAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35
160	.30	94.8	95.6	96.4	97.1	97.9	98.6	97.9	96.8	95.9	95.2	94.4	93.5
200	.38	92.7	93.5	94.3	95.1	95.9	96.7	97.1	96.1	95.1	94.4	93.6	92.8
240	.45	91.6	92.5	93.3	94.1	94.9	95.7	96.4	96.4	95.5	94.7	94.0	93.2
280	.52	90.6	91.4	92.2	93.0	93.8	94.6	95.4	96.2	95.9	95.1	94.4	93.7
320	.60	89.5	90.3	91.2	92.0	92.8	93.6	94.4	95.2	96.0	95.5	94.8	94.1
360	.67	88.3	89.1	90.0	90.8	91.6	92.4	93.2	93.9	94.7	95.5	95.1	94.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)			
	12	14	16	18
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.9	-0.9
ENGINE & WING ANTI-ICE ON	-3.2	-3.4	-3.4	-3.5

ENGINE INOP

Max Continuous %N1

10000 FT to 1000 FT Pressure Altitudes

10000 FT PRESS ALT		TAT (°C)											
KIAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35
160	.29	92.7	93.5	94.4	95.2	95.9	96.7	97.5	96.5	95.6	94.9	94.2	93.4
200	.36	91.3	92.1	93.0	93.8	94.6	95.4	96.1	96.1	95.2	94.4	93.7	92.9
240	.43	90.3	91.1	92.0	92.8	93.6	94.4	95.2	95.9	95.4	94.6	93.8	93.1
280	.51	89.5	90.3	91.1	91.9	92.7	93.5	94.3	95.1	95.7	95.0	94.2	93.5
320	.58	88.6	89.4	90.2	91.0	91.8	92.6	93.4	94.2	95.0	95.4	94.7	93.9
360	.65	87.5	88.3	89.2	90.0	90.8	91.6	92.3	93.1	93.9	94.7	95.0	94.3
5000 FT PRESS ALT		TAT (°C)											
KIAS	M	-10	-5	0	5	10	15	20	25	30	35	40	45
160	.26	90.5	91.4	92.2	93.0	93.8	94.5	95.1	94.4	93.6	92.9	92.2	91.4
200	.33	90.0	90.8	91.6	92.4	93.2	93.9	94.7	94.4	93.7	93.0	92.3	91.5
240	.40	89.2	90.0	90.8	91.6	92.4	93.2	93.9	94.4	93.7	92.9	92.2	91.5
280	.46	88.5	89.3	90.1	90.9	91.7	92.5	93.3	94.0	94.0	93.2	92.5	91.8
320	.53	87.8	88.6	89.4	90.2	90.9	91.7	92.5	93.2	94.0	93.6	92.9	92.2
360	.57	86.9	87.7	88.5	89.3	90.1	90.8	91.6	92.3	93.1	93.8	93.3	92.6
3000 FT PRESS ALT		TAT (°C)											
KIAS	M	-5	0	5	10	15	20	25	30	35	40	45	50
160	.26	90.5	91.3	92.1	92.8	93.6	94.4	94.6	93.9	93.2	92.4	91.6	90.7
200	.32	89.9	90.7	91.5	92.3	93.1	93.8	94.6	94.0	93.3	92.5	91.8	91.0
240	.38	88.8	89.6	90.4	91.2	92.0	92.7	93.5	93.5	92.8	92.0	91.3	90.6
280	.45	88.3	89.1	89.9	90.6	91.4	92.2	92.9	93.7	93.1	92.4	91.7	91.0
320	.51	87.6	88.4	89.2	90.0	90.7	91.5	92.2	93.0	93.5	92.8	92.0	91.3
360	.57	86.8	87.6	88.4	89.1	89.9	90.6	91.4	92.1	92.8	93.1	92.4	91.7
1000 FT PRESS ALT		TAT (°C)											
KIAS	M	-5	0	5	10	15	20	25	30	35	40	45	50
160	.25	89.0	89.8	90.6	91.4	92.2	92.9	93.7	93.4	92.7	91.9	91.2	90.3
200	.31	88.7	89.5	90.3	91.0	91.8	92.6	93.3	93.7	93.0	92.2	91.5	90.7
240	.37	87.8	88.6	89.4	90.2	90.9	91.7	92.5	93.2	92.8	92.0	91.3	90.6
280	.43	87.3	88.1	88.8	89.6	90.4	91.1	91.9	92.6	93.1	92.3	91.6	90.9
320	.49	86.7	87.5	88.2	89.0	89.8	90.5	91.3	92.0	92.7	92.7	91.9	91.2
360	.55	85.9	86.7	87.5	88.2	89.0	89.7	90.5	91.2	91.9	92.6	92.3	91.6

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)			
	1	3	5	10
ENGINE ANTI-ICE ON	-0.6	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE ON	-2.9	-3.0	-2.7	-3.2

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	81	269	16300	14700	12300
80	76	262	18400	16800	14900
75	72	254	20400	19000	17300
70	67	246	22500	21200	19700
65	62	238	24700	23500	22100
60	57	229	26800	25800	24600
55	53	219	29100	28100	27000
50	48	209	31200	30400	29400
45	43	199	33300	32600	31700
40	38	187	35600	34900	34000

Includes APU fuel burn.

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown/LRC Cruise Range Capability

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
138	128	120	112	106	100	95	90	86	82	78
275	256	239	225	212	200	190	180	172	164	157
413	384	359	337	317	300	284	270	258	246	235
551	512	479	449	423	400	379	360	344	328	314
689	640	598	562	529	500	474	451	429	410	392
826	768	718	674	635	600	569	541	515	492	471
964	896	838	786	741	700	664	631	601	574	549
1102	1025	957	898	846	800	758	721	687	656	628
1240	1153	1077	1011	952	900	853	811	773	738	706
1377	1281	1197	1123	1058	1000	948	901	859	820	785
1515	1409	1317	1235	1164	1100	1043	991	945	902	863
1653	1537	1436	1348	1270	1200	1138	1081	1030	984	942
1792	1666	1556	1460	1375	1300	1232	1171	1116	1066	1020
1930	1794	1676	1573	1481	1400	1327	1261	1202	1148	1098
2068	1922	1796	1685	1587	1500	1422	1351	1288	1230	1177
2207	2051	1916	1798	1693	1600	1517	1441	1373	1312	1255
2345	2180	2036	1910	1799	1700	1611	1531	1459	1393	1333
2484	2309	2156	2023	1905	1800	1706	1621	1545	1475	1411

Driftdown/Cruise Fuel and Time

AIR DIST (NM)	FUEL REQUIRED (1000 KG)										TIME (HR:MIN)
	WEIGHT AT START OF DRIFTDOWN (1000 KG)										
	40	45	50	55	60	65	70	75	80	85	
100	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0:17
200	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.2	0:33
300	1.2	1.3	1.4	1.5	1.6	1.7	1.7	1.8	1.9	2.0	0:50
400	1.6	1.8	1.9	2.0	2.2	2.3	2.4	2.6	2.7	2.9	1:06
500	2.0	2.2	2.4	2.6	2.8	3.0	3.1	3.3	3.5	3.7	1:23
600	2.4	2.7	2.9	3.1	3.3	3.6	3.8	4.0	4.2	4.5	1:39
700	2.8	3.1	3.4	3.6	3.9	4.2	4.4	4.7	5.0	5.2	1:56
800	3.2	3.6	3.9	4.2	4.5	4.8	5.0	5.4	5.7	6.0	2:12
900	3.6	4.0	4.3	4.7	5.0	5.4	5.7	6.0	6.4	6.8	2:29
1000	4.0	4.4	4.8	5.2	5.6	6.0	6.3	6.7	7.1	7.6	2:46
1100	4.4	4.8	5.3	5.7	6.1	6.6	6.9	7.4	7.8	8.3	3:02
1200	4.8	5.3	5.7	6.2	6.7	7.1	7.6	8.1	8.6	9.1	3:19
1300	5.2	5.7	6.2	6.7	7.2	7.7	8.2	8.7	9.3	9.8	3:36
1400	5.5	6.1	6.6	7.2	7.7	8.3	8.8	9.4	10.0	10.6	3:52
1500	5.9	6.5	7.1	7.7	8.3	8.9	9.4	10.0	10.7	11.3	4:09
1600	6.3	6.9	7.5	8.2	8.8	9.4	10.0	10.7	11.3	12.1	4:26
1700	6.6	7.3	8.0	8.6	9.3	10.0	10.6	11.3	12.0	12.8	4:43
1800	7.0	7.7	8.4	9.1	9.8	10.5	11.2	12.0	12.7	13.5	4:59

Includes APU fuel burn.

Driftdown at optimum driftdown speed and cruise at long range cruise speed.

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Altitude Capability

100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	9500	6400	3400
80	12400	9500	6500
75	15300	12800	9800
70	18400	15800	13200
65	21100	18600	16300
60	23700	21800	19400
55	26300	24700	22400
50	29000	27700	25800
45	31400	30500	29200
40	33800	33000	31800

With engine anti-ice on, decrease altitude capability by 2000 ft.

With engine and wing anti-ice on, decrease altitude capability by 6500 ft.

ENGINE INOP

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		10	15	17	19	21	23	25	27	29	31
85	%N1	91.8									
	MACH	.561									
	KIAS	311									
	FF/ENG	3067									
80	%N1	90.1	94.0								
	MACH	.545	.590								
	KIAS	302	299								
	FF/ENG	2875	2870								
75	%N1	88.4	92.5	94.0							
	MACH	.528	.579	.593							
	KIAS	293	293	288							
	FF/ENG	2684	2709	2674							
70	%N1	86.5	90.7	92.3	94.0						
	MACH	.510	.562	.582	.595						
	KIAS	282	284	283	278						
	FF/ENG	2494	2518	2520	2481						
65	%N1	84.5	88.7	90.4	92.2	93.9					
	MACH	.491	.542	.563	.584	.596					
	KIAS	271	274	274	273	268					
	FF/ENG	2306	2327	2330	2330	2295					
60	%N1	82.3	86.5	88.3	90.0	91.9	93.7	96.4			
	MACH	.471	.521	.543	.564	.585	.597	.614			
	KIAS	261	263	263	263	263	258	254			
	FF/ENG	2124	2137	2139	2140	2143	2114	2146			
55	%N1	80.2	84.2	85.9	87.7	89.5	91.4	93.3	96.2		
	MACH	.453	.498	.520	.541	.563	.585	.597	.614		
	KIAS	250	251	252	252	253	252	247	244		
	FF/ENG	1954	1948	1950	1950	1953	1958	1938	1971		
50	%N1	77.8	81.6	83.4	85.2	87.0	88.7	90.7	92.7	95.7	
	MACH	.434	.475	.495	.516	.538	.561	.583	.596	.613	
	KIAS	240	239	239	240	241	241	241	236	233	
	FF/ENG	1791	1764	1762	1762	1764	1767	1777	1765	1793	
45	%N1	75.5	79.1	80.6	82.3	84.1	85.9	87.7	89.7	91.8	94.8
	MACH	.415	.452	.469	.489	.511	.533	.556	.578	.593	.610
	KIAS	229	227	227	227	228	229	229	229	225	222
	FF/ENG	1636	1594	1582	1575	1577	1580	1586	1600	1593	1613
40	%N1	73.0	76.2	77.8	79.4	81.0	82.8	84.6	86.4	88.3	90.7
	MACH	.395	.429	.445	.462	.480	.502	.525	.548	.571	.589
	KIAS	218	215	215	214	214	215	216	216	216	214
	FF/ENG	1485	1434	1416	1402	1392	1394	1400	1410	1421	1424

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time
Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
298	272	249	230	214	200	190	180	172	164	158
600	547	501	462	429	400	379	361	344	328	315
903	823	753	694	644	600	570	542	517	494	473
1209	1100	1005	926	859	800	759	721	687	657	630
1516	1379	1259	1159	1075	1000	949	902	859	820	786
1825	1659	1513	1393	1290	1200	1139	1082	1031	984	943
2137	1940	1768	1626	1506	1400	1328	1262	1202	1147	1099
2450	2222	2024	1860	1722	1600	1518	1442	1373	1311	1256
2766	2507	2281	2095	1938	1800	1707	1622	1544	1474	1412
3083	2792	2539	2331	2155	2000	1896	1801	1715	1637	1568

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		18		22		26	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.4	0:43	1.2	0:41	1.1	0:39	1.0	0:38	0.9	0:37
400	2.8	1:23	2.6	1:19	2.4	1:14	2.2	1:11	2.1	1:09
600	4.3	2:04	3.9	1:57	3.6	1:50	3.4	1:45	3.2	1:42
800	5.7	2:46	5.2	2:36	4.9	2:26	4.5	2:19	4.4	2:14
1000	7.1	3:28	6.6	3:15	6.1	3:03	5.7	2:53	5.5	2:47
1200	8.5	4:10	7.9	3:55	7.3	3:40	6.8	3:28	6.6	3:21
1400	9.8	4:53	9.1	4:36	8.5	4:18	8.0	4:02	7.7	3:54
1600	11.2	5:36	10.4	5:16	9.7	4:55	9.1	4:38	8.7	4:28
1800	12.5	6:20	11.7	5:58	10.9	5:34	10.2	5:13	9.8	5:02
2000	13.9	7:05	12.9	6:39	12.0	6:13	11.3	5:49	10.8	5:36

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)									
	40	45	50	55	60	65	70	75	80	
1	-0.1	-0.1	-0.1	0.0	0.0	0.1	0.1	0.2	0.3	
2	-0.3	-0.2	-0.1	-0.1	0.0	0.2	0.3	0.6	0.8	
3	-0.4	-0.3	-0.2	-0.1	0.0	0.3	0.5	0.9	1.2	
4	-0.6	-0.4	-0.3	-0.1	0.0	0.3	0.7	1.2	1.6	
5	-0.7	-0.5	-0.4	-0.2	0.0	0.4	0.9	1.4	2.0	
6	-0.8	-0.6	-0.4	-0.2	0.0	0.5	1.1	1.7	2.4	
7	-1.0	-0.8	-0.5	-0.3	0.0	0.6	1.2	2.0	2.8	
8	-1.1	-0.9	-0.6	-0.3	0.0	0.6	1.4	2.2	3.2	
9	-1.3	-1.0	-0.7	-0.3	0.0	0.7	1.5	2.4	3.5	
10	-1.4	-1.1	-0.7	-0.4	0.0	0.7	1.6	2.6	3.8	
11	-1.6	-1.2	-0.8	-0.4	0.0	0.8	1.7	2.8	4.1	
12	-1.7	-1.3	-0.9	-0.4	0.0	0.8	1.9	3.0	4.4	
13	-1.9	-1.4	-0.9	-0.5	0.0	0.9	2.0	3.2	4.7	
14	-2.0	-1.5	-1.0	-0.5	0.0	0.9	2.0	3.4	4.9	

Includes APU fuel burn.



ENGINE INOP
MAX CONTINUOUS THRUST

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)							
		1500	5000	10000	15000	20000	25000	30000	35000
85	%N1	81.1	84.1	88.3	92.8				
	KIAS	250	251	252	253				
	FF/ENG	2740	2730	2750	2800				
80	%N1	79.5	82.4	86.5	91.0				
	KIAS	242	243	244	245				
	FF/ENG	2580	2570	2570	2610				
75	%N1	77.8	80.5	84.7	89.1	95.0			
	KIAS	235	236	236	238	239			
	FF/ENG	2420	2400	2400	2420	2490			
70	%N1	76.0	78.6	82.8	87.1	92.1			
	KIAS	227	227	228	229	231			
	FF/ENG	2260	2240	2230	2250	2270			
65	%N1	74.0	76.7	80.8	85.0	89.7	97.7		
	KIAS	219	219	220	221	222	224		
	FF/ENG	2100	2090	2070	2070	2080	2230		
60	%N1	71.7	74.6	78.5	82.8	87.4	93.7		
	KIAS	210	210	211	212	213	214		
	FF/ENG	1950	1930	1910	1910	1910	1970		
55	%N1	69.4	72.3	76.3	80.5	84.9	90.0		
	KIAS	200	201	202	203	204	205		
	FF/ENG	1800	1770	1750	1740	1730	1760		
50	%N1	66.9	69.7	73.8	77.8	82.3	87.0	94.9	
	KIAS	192	192	192	193	194	195	196	
	FF/ENG	1650	1620	1600	1580	1570	1570	1680	
45	%N1	64.2	66.9	70.9	75.0	79.4	84.0	89.6	
	KIAS	185	185	185	185	185	185	186	
	FF/ENG	1500	1470	1440	1420	1400	1400	1450	
40	%N1	61.1	64.0	67.8	72.0	76.2	80.7	85.4	94.0
	KIAS	178	178	178	178	178	178	178	178
	FF/ENG	1350	1330	1300	1270	1250	1240	1260	1360

This table includes 5% additional fuel for holding in a racetrack pattern.

ENGINE INOP

ADVISORY INFORMATION

**Gear Down Landing Rate of Climb Available
Flaps 15**

TAT (°C)	RATE OF CLIMB (FT/MIN)					
	PRESSURE ALTITUDE (FT)					
	-2000	0	2000	4000	6000	8000
52	-220	-290				
50	-200	-270	-370			
48	-170	-240	-350			
46	-140	-210	-320	-430		
44	-120	-190	-300	-410		
42	-90	-160	-270	-380	-490	
40	-60	-130	-250	-360	-470	
38	-30	-110	-220	-330	-450	-560
36	-20	-80	-190	-310	-420	-540
34	-10	-50	-160	-280	-400	-520
32	-10	-30	-140	-260	-370	-490
30	-10	-30	-110	-230	-350	-470
20	0	-30	-100	-170	-240	-340
10	0	-20	-90	-170	-240	-320
0	10	-20	-90	-160	-240	-320
-20	10	-10	-90	-160	-250	-330
-40	10	-10	-90	-170	-250	-340

Rate of climb capability shown is valid for 60000 kg, gear down at VREF15+5.
Decrease rate of climb 130 ft/min per 5000 kg greater than 60000 kg.
Increase rate of climb 180 ft/min per 5000 kg less than 60000 kg.

Flaps 30

TAT (°C)	RATE OF CLIMB (FT/MIN)					
	PRESSURE ALTITUDE (FT)					
	-2000	0	2000	4000	6000	8000
52	-400	-470				
50	-370	-440	-550			
48	-350	-420	-530			
46	-320	-400	-500	-610		
44	-300	-370	-480	-590		
42	-270	-340	-460	-570	-680	
40	-250	-320	-430	-540	-660	
38	-220	-290	-410	-520	-640	-750
36	-200	-270	-380	-500	-610	-730
34	-200	-240	-360	-470	-590	-710
32	-200	-220	-330	-450	-570	-690
30	-200	-220	-300	-420	-540	-660
20	-200	-220	-290	-360	-440	-540
10	-190	-220	-290	-370	-440	-530
0	-190	-220	-290	-370	-450	-530
-20	-200	-220	-300	-380	-460	-550
-40	-200	-230	-310	-390	-480	-570

Rate of climb capability shown is valid for 60000 kg, gear down at VREF30+5.
Decrease rate of climb 130 ft/min per 5000 kg greater than 60000 kg.
Increase rate of climb 180 ft/min per 5000 kg less than 60000 kg.



Performance Inflight - QRH

Gear Down

Chapter PI-QRH

Section 13

GEAR DOWN

Long Range Cruise Altitude Capability

Max Cruise Thrust, 100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	15500	12500	9300
80	18400	15500	12600
75	21100	18500	15700
70	23700	21500	18600
65	26100	24400	21800
60	28600	27100	25300
55	30800	29600	28100
50	32900	31900	30700
45	35100	34100	33000
40	37500	36500	35400

GEAR DOWN

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)								
		10	21	23	25	27	29	31	33	35
85	%N1	85.9								
	MACH	.482								
	KIAS	267								
	FF/ENG	2421								
80	%N1	84.2								
	MACH	.468								
	KIAS	259								
	FF/ENG	2271								
75	%N1	82.5	91.7							
	MACH	.454	.554							
	KIAS	251	248							
	FF/ENG	2123	2101							
70	%N1	80.6	89.8	91.7						
	MACH	.440	.541	.557						
	KIAS	243	242	240						
	FF/ENG	1977	1960	1950						
65	%N1	78.6	87.9	89.5	91.6	94.5				
	MACH	.425	.524	.543	.560	.578				
	KIAS	235	234	233	231	229				
	FF/ENG	1835	1812	1806	1805	1836				
60	%N1	76.5	85.6	87.4	89.1	91.3	94.5			
	MACH	.409	.504	.525	.544	.562	.580			
	KIAS	226	225	225	224	222	220			
	FF/ENG	1696	1661	1661	1658	1664	1696			
55	%N1	74.4	83.3	85.0	86.8	88.5	90.9	94.1		
	MACH	.393	.484	.504	.525	.545	.562	.581		
	KIAS	217	216	216	216	215	213	211		
	FF/ENG	1559	1515	1512	1515	1517	1523	1555		
50	%N1	71.9	80.7	82.5	84.2	86.0	87.8	90.2	93.5	
	MACH	.376	.463	.482	.502	.523	.544	.561	.580	
	KIAS	207	206	206	206	206	205	203	201	
	FF/ENG	1424	1371	1367	1368	1374	1377	1381	1411	
45	%N1	69.1	78.0	79.7	81.4	83.1	85.0	86.8	89.1	92.5
	MACH	.358	.441	.458	.477	.498	.520	.541	.559	.578
	KIAS	197	196	196	196	196	196	195	193	191
	FF/ENG	1294	1231	1224	1224	1230	1235	1237	1239	1265
40	%N1	66.2	74.9	76.6	78.3	80.0	81.8	83.6	85.5	87.7
	MACH	.340	.417	.434	.452	.471	.491	.513	.535	.554
	KIAS	187	185	185	185	185	185	185	185	183
	FF/ENG	1170	1098	1085	1083	1089	1092	1094	1096	1097



GEAR DOWN

Long Range Cruise Enroute Fuel and Time Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
324	290	260	236	217	200	188	178	168	160	153
654	583	523	474	435	400	377	357	338	321	307
989	880	787	713	653	600	566	535	507	483	461
1329	1181	1054	953	871	800	754	713	676	643	614
1674	1484	1322	1194	1090	1000	943	891	844	803	766
2024	1791	1593	1436	1310	1200	1131	1069	1013	962	918
2381	2103	1865	1680	1530	1400	1320	1247	1181	1122	1070
2743	2417	2140	1924	1751	1600	1508	1424	1348	1280	1221
3113	2737	2418	2171	1972	1800	1695	1600	1514	1438	1371

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	2.4	0:49	2.2	0:47	1.9	0:44	1.7	0:42	1.6	0:41
400	4.9	1:36	4.5	1:31	4.0	1:25	3.7	1:20	3.5	1:17
600	7.4	2:25	6.8	2:17	6.1	2:06	5.7	1:59	5.4	1:54
800	9.8	3:14	9.1	3:03	8.1	2:48	7.6	2:38	7.2	2:31
1000	12.1	4:04	11.3	3:50	10.1	3:30	9.5	3:18	9.0	3:08
1200	14.4	4:56	13.5	4:39	12.1	4:14	11.3	3:58	10.7	3:46
1400	16.7	5:49	15.6	5:28	14.0	4:58	13.1	4:40	12.4	4:24
1600	18.9	6:43	17.7	6:18	15.9	5:44	14.9	5:22	14.1	5:03
1800	21.1	7:38	19.7	7:10	17.7	6:30	16.6	6:05	15.7	5:43

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	40	50	60	70	80
2	-0.3	-0.2	0.0	0.3	0.7
4	-0.7	-0.3	0.0	0.6	1.3
6	-1.0	-0.5	0.0	0.9	2.0
8	-1.3	-0.7	0.0	1.2	2.6
10	-1.7	-0.8	0.0	1.4	3.2
12	-2.0	-1.0	0.0	1.6	3.7
14	-2.4	-1.2	0.0	1.8	4.2
16	-2.7	-1.3	0.0	2.0	4.6
18	-3.0	-1.5	0.0	2.2	5.0
20	-3.4	-1.7	0.0	2.4	5.3
22	-3.7	-1.8	0.0	2.5	5.6

GEAR DOWN

Descent

VREF40 + 70 KIAS

PRESSURE ALTITUDE (FT)	TIME (MIN)	FUEL (KG)	DISTANCE (NM)
41000	21	280	91
39000	20	270	86
37000	19	270	81
35000	19	260	77
33000	18	260	72
31000	17	250	68
29000	17	250	64
27000	16	240	60
25000	15	230	56
23000	14	230	52
21000	13	220	48
19000	13	210	44
17000	12	200	40
15000	11	190	36
10000	8	170	26
5000	6	140	16
1500	4	110	9

Allowances for a straight-in approach are included.



GEAR DOWN

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)							
		1500	5000	10000	15000	20000	25000	30000	35000
85	%N1	75.8	78.5	82.7	87.0	92.0			
	KIAS	230	230	230	230	230			
	FF/ENG	2240	2230	2220	2240	2260			
80	%N1	74.2	77.0	81.1	85.4	90.0			
	KIAS	225	225	225	225	225			
	FF/ENG	2120	2110	2100	2100	2110			
75	%N1	72.5	75.4	79.4	83.7	88.3	94.8		
	KIAS	220	220	220	220	220	220		
	FF/ENG	2000	1990	1970	1970	1970	2050		
70	%N1	70.8	73.7	77.6	81.9	86.4	91.8		
	KIAS	216	216	216	216	216	216		
	FF/ENG	1890	1870	1850	1840	1840	1870		
65	%N1	69.0	71.9	75.9	80.1	84.5	89.3		
	KIAS	211	211	211	211	211	211		
	FF/ENG	1770	1750	1730	1720	1710	1730		
60	%N1	67.1	69.8	74.0	78.0	82.5	87.1	94.3	
	KIAS	204	204	204	204	204	204	204	
	FF/ENG	1660	1630	1610	1600	1580	1590	1670	
55	%N1	65.1	67.8	71.9	75.9	80.3	84.8	90.4	
	KIAS	198	198	198	198	198	198	198	
	FF/ENG	1540	1520	1490	1480	1460	1460	1500	
50	%N1	62.8	65.6	69.6	73.7	78.0	82.4	87.1	
	KIAS	192	192	192	192	192	192	192	
	FF/ENG	1430	1400	1380	1360	1330	1330	1350	
45	%N1	60.3	63.3	67.1	71.4	75.5	79.9	84.5	91.5
	KIAS	185	185	185	185	185	185	185	185
	FF/ENG	1310	1290	1270	1250	1220	1210	1220	1270
40	%N1	57.9	60.6	64.6	68.7	72.9	77.3	81.7	86.8
	KIAS	178	178	178	178	178	178	178	178
	FF/ENG	1200	1180	1160	1130	1110	1090	1100	1110

This table includes 5% additional fuel for holding in a racetrack pattern.

Intentionally
Blank



Performance Inflight - QRH

Chapter PI-QRH

Gear Down, Engine Inop

Section 14

GEAR DOWN

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
75	70	218	1500		
70	66	213	4700	2200	
65	61	208	8000	5900	3400
60	57	202	11100	9300	7100
55	52	196	14000	12900	10900
50	47	190	16900	16000	14800
45	43	183	19800	18700	17600
40	38	176	22800	21700	20600

Includes APU fuel burn.

Long Range Cruise Altitude Capability

100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
65	1600		
60	6300	3200	
55	10600	8200	5300
50	14400	13000	10400
45	18000	16900	15700
40	21700	20500	19300



GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		5	7	9	11	13	15	17	19	21	23
60	%N1	90.2	91.9								
	MACH	.364	.375								
	KIAS	220	219								
	FF/ENG	3192	3191								
55	%N1	87.8	89.3	91.0	92.8						
	MACH	.351	.362	.374	.387						
	KIAS	212	211	210	209						
	FF/ENG	2924	2909	2906	2913						
50	%N1	85.3	86.7	88.2	89.9	91.7	94.2				
	MACH	.338	.348	.359	.371	.384	.398				
	KIAS	204	203	202	201	200	199				
	FF/ENG	2672	2647	2630	2626	2633	2657				
45	%N1	82.7	84.0	85.4	86.9	88.6	90.4	92.7	96.6		
	MACH	.325	.334	.344	.355	.367	.380	.393	.408		
	KIAS	196	195	193	192	191	190	189	189		
	FF/ENG	2432	2400	2374	2356	2351	2352	2359	2417		
40	%N1	79.8	81.1	82.5	83.9	85.4	87.0	88.8	90.8	94.1	98.4
	MACH	.311	.320	.329	.339	.349	.361	.374	.387	.402	.418
	KIAS	188	186	184	183	182	181	180	179	179	178
	FF/ENG	2206	2166	2133	2107	2088	2076	2069	2065	2101	2201

GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time
Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
172	151	134	120	109	100	93	88	83	78	75
352	308	270	242	219	200	187	175	165	156	148
533	465	408	364	330	300	280	262	246	232	220
716	623	545	486	440	400	373	349	328	309	293
900	783	684	609	551	500	466	436	409	385	365
1086	943	823	733	661	600	559	523	490	462	438
1273	1105	964	856	772	700	652	610	572	538	510
1462	1267	1103	980	883	800	745	696	652	614	581
1653	1431	1245	1104	994	900	838	782	733	690	653
1845	1595	1386	1228	1105	1000	931	868	813	765	724

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)					
	6		10		14	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
100	1.3	0:27	1.1	0:26	1.0	0:26
200	2.6	0:53	2.4	0:50	2.3	0:48
300	3.9	1:18	3.7	1:15	3.6	1:11
400	5.2	1:44	4.9	1:39	4.8	1:35
500	6.5	2:10	6.1	2:04	6.0	1:58
600	7.8	2:37	7.3	2:29	7.1	2:22
700	9.1	3:03	8.5	2:55	8.3	2:46
800	10.3	3:30	9.7	3:20	9.4	3:10
900	11.6	3:58	10.9	3:46	10.5	3:35
1000	12.8	4:25	12.0	4:12	11.6	3:59

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	40	50	60	70	80
1	-0.2	-0.1	0.0	0.1	0.3
2	-0.3	-0.2	0.0	0.3	0.6
3	-0.5	-0.3	0.0	0.5	1.0
4	-0.6	-0.3	0.0	0.7	1.3
5	-0.8	-0.4	0.0	0.9	1.7
6	-1.0	-0.5	0.0	1.0	2.0
7	-1.1	-0.6	0.0	1.2	2.4
8	-1.3	-0.7	0.0	1.4	2.7
9	-1.5	-0.7	0.0	1.6	3.1
10	-1.6	-0.8	0.0	1.8	3.5
11	-1.8	-0.9	0.0	1.9	3.8
12	-1.9	-1.0	0.0	2.1	4.2
13	-2.1	-1.1	0.0	2.3	4.5
14	-2.3	-1.1	0.0	2.5	4.9

Includes APU fuel burn.



GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

**Holding
Flaps Up**

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)			
		1500	5000	10000	15000
70	%N1	89.4			
	KIAS	216			
	FF/ENG	3610			
65	%N1	87.4	90.5		
	KIAS	211	211		
	FF/ENG	3360	3380		
60	%N1	85.2	88.2	92.9	
	KIAS	204	204	204	
	FF/ENG	3110	3110	3150	
55	%N1	82.9	85.9	90.4	
	KIAS	198	198	198	
	FF/ENG	2860	2860	2880	
50	%N1	80.4	83.4	87.7	92.8
	KIAS	192	192	192	192
	FF/ENG	2630	2620	2620	2670
45	%N1	77.8	80.7	85.0	89.6
	KIAS	185	185	185	185
	FF/ENG	2400	2380	2380	2400
40	%N1	75.1	77.8	82.1	86.5
	KIAS	178	178	178	178
	FF/ENG	2180	2160	2140	2140

This table includes 5% additional fuel for holding in a racetrack pattern.

Performance Inflight - QRH**Chapter PI-QRH****Text****Section 15**

Introduction

This chapter contains information to supplement performance data from the Flight Management Computer (FMC). In addition, sufficient inflight data is provided to complete a flight with the FMC inoperative. In the event of conflict between data presented in this chapter and that contained in the approved Airplane Flight Manual, the Flight Manual shall always take precedence.

General**Flight with Unreliable Airspeed / Turbulent Air Penetration**

Pitch attitude and average %N1 information is provided for use in all phases of flight in the event of unreliable airspeed/Mach indications resulting from blocking or freezing of the pitot system. Loss of radome or turbulent air may also cause unreliable airspeed/Mach indications. The cruise table in this section may also be used for turbulent air penetration.

Pitch attitude is shown in bold type for emphasis since altitude and/or vertical speed indications may also be unreliable.

Max Climb %N1

This table shows Max Climb %N1 for a 280/.78 climb speed schedule, normal engine bleed for packs on or off and anti-ice off. Enter the table with airport pressure altitude and TAT and read %N1. %N1 adjustments are shown for anti-ice operation.

Go-around %N1

To find Max Go-around %N1 based on normal engine bleed for packs on (AUTO) and anti-ice on or off, enter the Go-around %N1 table with airport pressure altitude and reported OAT or TAT and read %N1. For packs OFF or HIGH operation, apply the %N1 adjustment shown below the table.

VREF

This table contains flaps 40, 30 and 15 reference speeds for a given weight.

Advisory Information

Normal Configuration Landing Distance

The normal configuration distance tables are provided as advisory information to help determine the actual landing distance performance of the airplane for different runway surface conditions and brake configurations.

If the surface is affected by water, snow or ice, and the braking action is reported as "good", conditions should not be expected to be as good as on clean, dry runways. The value "good" is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when landing. The performance level used to calculate the "good" data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate "poor" data reflects runways covered with wet ice.

Dry runway landing performance is shown for max manual braking configuration and autobrake settings max, 3, 2, and 1. The autobrake performance may be used to assist in the selection of the most desirable autobrake setting for a given field length. Selection of an autobrake setting results in a constant rate of deceleration. Maximum effort manual braking should achieve shorter landing distance than the max autobrake setting. The reference landing distance is a reference distance from 50 ft above the threshold to stop based on a reference landing weight and normal approach speed for the selected landing flap at sea level, zero wind, zero slope, and two engine detent reverse thrust. Subsequent columns provide adjustments for off-reference landing weight, altitude, wind, slope, temperature, speed, and reverse thrust. Each adjustment is independently added to the reference landing distance.

Flaps 30 and 40 landing distances and adjustments are provided for dry runways as well as runways with good, medium, and poor reported braking actions, which are commonly referred to as slippery runway conditions. All landing distances (reference distances plus adjustments) are 115% of the actual landing distance.

Non-normal Configuration Landing Distance

Advisory information is provided to support non-normal configurations that affect the landing performance of the airplane. Landing distances and adjustments are provided for dry runways and runways with good, medium, and poor reported braking action.



Enter the table with the applicable non-normal configuration and read the normal approach speed. The reference landing distance is a reference distance from 50 ft above the threshold to stop based on a reference landing weight and speed at sea level, zero wind, and zero slope. Subsequent columns provide adjustments for off-reference landing weight, altitude, wind, slope, and speed conditions. Each adjustment is independently added to the reference landing distance. Landing distance includes the effect of reverse thrust.

Recommended Brake Cooling Schedule

Advisory information is provided to assist in avoiding the problems associated with hot brakes. For normal operation, most landings are at weights below the AFM quick turnaround limit weight.

Use of the recommended cooling schedule will help avoid brake overheat and fuse plug problems that could result from repeated landings at short time intervals or a rejected takeoff.

Enter the Recommended Brake Cooling Schedule table with the airplane weight and brakes on speed, adjusted for wind at the appropriate temperature and altitude condition. Instructions for applying wind adjustments are included below the table. Linear interpolation may be used to obtain intermediate values. The resulting number is the reference brake energy per brake in millions of foot-pounds, and represents the amount of energy absorbed by each brake during a rejected takeoff. Notes providing adjustments for wind are included below the table.

To determine the energy per brake absorbed during landing, enter the appropriate Adjusted Brake Energy Per Brake table (No Reverse Thrust or 2 Engine Reverse) with the reference brake energy per brake and the type of braking used during landing (Max Manual, Max Auto, or Autobrake). The resulting number is the adjusted brake energy per brake and represents the energy absorbed in each brake during the landing.

The recommended cooling time is found in the appropriate (steel or carbon brakes) final table by entering with the adjusted brake energy per brake. Times are provided for ground cooling and inflight gear down cooling.

Brake Temperature Monitor System (BTMS) indications are also shown. If brake cooling is determined from the BTMS, use the hottest brake indication 10 to 15 minutes after the airplane has come to a complete stop, or inflight with gear retracted to determine recommended cooling schedule.

Engine Inoperative

Initial Max Continuous %N1

The Initial Max Continuous %N1 setting for use following an engine failure is shown. The table is based on the typical all engine cruise speed of .79M to provide a target %N1 setting at the start of driftdown. Once driftdown is established, the Max Continuous %N1 table should be used to determine %N1 for the given conditions.

Max Continuous %N1

Power setting is based on one engine operating with one A/C pack operating and all anti-ice bleeds off. Enter the table with pressure altitude, TAT, and IAS or Mach to read %N1.

It is desirable to maintain engine thrust level within the limits of the Max Cruise thrust rating. However, where thrust level in excess of Max Cruise rating is required, such as for meeting terrain clearance, ATC altitude assignments, or to attain maximum range capability, it is permissible to use the thrust needed up to the Max Continuous thrust rating. The Max Continuous thrust rating is intended primarily for emergency use at the discretion of the pilot and is the maximum thrust that may be used continuously.

Driftdown Speed/Level Off Altitude

The table shows optimum driftdown speed as a function of cruise weight at start of driftdown. Also shown are the approximate weight and pressure altitude at which the airplane will level off considering 100 ft/min residual rate of climb.

The level off altitude is dependent on air temperature (ISA deviation).

Driftdown/LRC Range Capability

This table shows the range capability from the start of driftdown. Driftdown is continued to level off altitude. As weight decreases due to fuel burn, the airplane is accelerated to Long Range Cruise speed. Cruise is continued at level off altitude and Long Range Cruise speed.

To determine fuel required, enter the Ground to Air Miles Conversion table with the desired ground distance and adjust for anticipated winds to obtain air distance to destination. Then enter the Driftdown/Cruise Fuel and Time table with air distance and weight at start of driftdown to determine fuel and time required. If altitudes other than the level off altitude is used, fuel and time required may be obtained by using the Engine Inoperative Long Range Cruise Enroute Fuel and Time table.

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Long Range Cruise Altitude Capability

The table shows the maximum altitude that can be maintained at a given weight and air temperature (ISA deviation), based on Long Range Cruise speed, Max Continuous thrust, and 100 ft/min residual rate of climb.

Long Range Cruise Control

The table provides target %N1, engine inoperative Long Range Cruise Mach number, IAS and fuel flow for the airplane weight and pressure altitude. The fuel flow values in this table reflect single engine fuel burn.

APU Operation During Flight

For APU operation during flight, increase fuel flow according to the following table. These increments include the APU fuel flow and the effect of increased drag from the APU door.

PRESSURE ALTITUDE (1000 FT)	APU FUEL FLOW (KG/HR)
39	45
35	45
31	50
25	60
20	65
15	75
10	85
5	95

Long Range Cruise Diversion Fuel and Time

Tables are provided for crews to determine the fuel and time required to proceed to an alternate airfield with one engine inoperative. The data is based on single engine Long Range Cruise speed and .78/280/250 descent. Enter with Air Distance as determined from the Ground to Air Miles Conversion table and read Fuel and Time required at the cruise pressure altitude. Adjust the fuel obtained for deviation from the reference weight at checkpoint as required by entering the off reference fuel adjustments table with the fuel required for the reference weight and the actual weight at checkpoint. Read fuel required and time for the actual weight.

Holding

Target %N1, indicated airspeed and fuel flow per engine information is tabulated for holding with flaps up based on the FMC optimum holding speed schedule. This is the higher of the maximum endurance speed and the maneuvering speed. Small variations in airspeed will not appreciably affect the overall endurance time. Enter the table with weight and pressure altitude to read %N1, IAS and fuel flow per engine.

Gear Down Landing Rate of Climb Available

Rate of climb data is provided as guidance information in the event an engine inoperative landing (manual or autoland) is planned. The tables show gear down rate of climb available for Flaps 15 and Flaps 30. Enter the table with TAT and pressure altitude to read rate of climb available. Apply adjustments shown to correct for weight.

Gear Down

This section contains performance for airplane operation with the landing gear extended. The data is based on engine bleeds for normal air conditioning.

Note: The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS may generate inappropriate enroute speed schedules, display non-conservative predictions of fuel burn, estimated time of arrival (ETA), maximum altitude, and compute overly shallow descent path. An accurate estimated time of arrival (ETA) is available if current speed or Mach is entered into the VNAV cruise page.

Tables for gear down performance in this section are identical in format and used in the same manner as tables for the gear up configuration previously described.

**Performance Inflight - QRH****Chapter PI-QRH****Table of Contents****Section 40****737-800W CFM56-7B26 C M KG JAA CATC/N****Pkg Model Identification PI-QRH.ModID.40.1****General PI-QRH.40.1**

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**Performance Inflight - QRH**
Pkg Model Identification**Chapter PI-QRH**
Section 40**General**

The table below shows the airplanes that have been identified with the following performance package. Note, some airplanes may be identified with more than one performance package. This configuration table information reflects the Boeing delivered configuration updated for service bulletin incorporations in conformance with the policy stated in the introduction section of the FCOM. The performance data is prepared for the owner/operator named on the title page. The intent of this information is to assist flight crews and airlines in knowing which performance package is applicable to a given airplane. The performance package model identification information is based on Boeing's knowledge of the airline's fleet at a point in time approximately three months prior to the page date. Notice of Errata (NOE) will not be provided to airlines to identify airplanes that are moved between performance packages within this manual or airplanes added to the airline's fleet whose performance packages are already represented in this manual. These types of changes will be updated in the next block revision. Owners/operators are responsible for ensuring the operational documentation they are using is complete and matches the current configuration of their airplanes, and the accuracy and validity of all information furnished by the owner/operator or any other party. Owners/operators receiving active revision service are responsible to ensure that any modifications to the listed airplanes are properly reflected in this manual.

Serial and tabulation number are supplied by Boeing.

Registry Number	Old Registry Number	Serial Number	Tabulation Number
RA-73269	VP-BOW	30040	YK136
RA-73321	VQ-BVY	32841	YK137

Intentionally
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Performance Inflight - QRH**Chapter PI-QRH****General****Section 40****Flight With Unreliable Airspeed/ Turbulent Air Penetration**

Altitude and/or vertical speed indications may also be unreliable.

CLIMB (280/.76)

Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT	4.0	4.0	4.0		
	V/S (FT/MIN)	1700	1100	600		
30000	PITCH ATT	4.0	4.0	3.5	4.0	4.0
	V/S (FT/MIN)	2500	1900	1500	1100	800
20000	PITCH ATT	7.0	6.5	6.0	6.0	6.0
	V/S (FT/MIN)	4200	3300	2600	2100	1700
10000	PITCH ATT	11.0	9.5	8.5	8.0	8.0
	V/S (FT/MIN)	5600	4400	3600	3000	2500
SEA LEVEL	PITCH ATT	14.5	12.5	11.0	10.0	9.5
	V/S (FT/MIN)	6700	5300	4400	3700	3100

CRUISE (.76/280)

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT	2.0	2.5	3.5		
	%N1	82.9	85.4	88.9		
35000	PITCH ATT	1.0	2.0	2.5	3.0	3.5
	%N1	81.2	82.6	84.4	86.8	90.4
30000	PITCH ATT	1.0	1.5	2.0	2.5	3.0
	%N1	80.7	81.5	82.7	84.2	86.1
25000	PITCH ATT	1.0	1.5	2.0	2.5	3.0
	%N1	77.2	77.9	79.0	80.5	82.3
20000	PITCH ATT	1.0	1.5	2.0	2.5	3.5
	%N1	73.6	74.2	75.3	76.6	78.2
15000	PITCH ATT	1.0	1.5	2.0	3.0	3.5
	%N1	69.8	70.6	71.6	72.9	74.4

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

DESCENT (.76/280)

Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT V/S (FT/MIN)	-1.5 -2700	-0.5 -2400	0.5 -2300	1.0 -2500	1.5 -2700
30000	PITCH ATT V/S (FT/MIN)	-3.5 -3100	-2.0 -2600	-1.0 -2300	-0.5 -2100	0.5 -2000
20000	PITCH ATT V/S (FT/MIN)	-3.5 -2800	-2.0 -2300	-1.0 -2000	0.0 -1900	0.5 -1700
10000	PITCH ATT V/S (FT/MIN)	-3.5 -2500	-2.0 -2100	-1.0 -1800	0.0 -1700	0.5 -1500
SEA LEVEL	PITCH ATT V/S (FT/MIN)	-3.5 -2300	-2.5 -1900	-1.0 -1700	-0.5 -1500	0.5 -1400

HOLDING (VREF40 + 70)

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
15000	PITCH ATT	5.0	5.0	5.0	5.0	5.0
	%N1	56.1	61.7	65.9	69.8	73.2
	KIAS	178	193	212	229	246
10000	PITCH ATT	5.0	5.0	5.0	5.0	5.0
	%N1	52.5	57.5	62.1	65.9	69.1
	KIAS	178	192	211	228	244
5000	PITCH ATT	5.0	5.5	5.0	5.0	5.0
	%N1	48.9	53.9	58.1	62.0	65.5
	KIAS	178	192	210	227	243

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = -2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 47.6	5.0 52.3	5.5 56.3	6.0 60.0	6.5 63.6
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 49.7	5.0 54.5	5.5 58.6	6.0 62.4	6.0 65.8
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 49.4	6.0 54.5	6.0 59.0	6.5 63.1	6.5 66.7
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 53.2	5.0 58.6	5.5 63.4	5.5 67.5	6.0 71.1
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 53.3	5.5 58.7	6.0 63.7	6.0 67.9	6.5 71.6
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 54.0	6.5 59.7	6.5 64.8	7.0 69.0	7.0 72.9
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 57.5	6.0 63.3	6.0 68.2	6.0 72.6	6.5 76.3
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = -1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 48.3	5.0 52.9	5.5 57.0	6.0 60.8	6.5 64.5
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 50.4	5.0 55.2	5.5 59.4	6.0 63.3	6.0 66.6
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 50.0	6.0 55.2	6.0 59.8	6.5 64.0	6.5 67.6
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 53.9	5.5 59.4	5.5 64.2	5.5 68.3	6.0 72.0
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 54.0	5.5 59.5	6.0 64.5	6.0 68.7	6.5 72.5
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 54.7	6.5 60.6	6.5 65.6	7.0 69.8	7.0 73.7
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 58.3	6.0 64.1	6.0 69.0	6.0 73.4	6.5 77.1
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = SEA LEVEL

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 49.0	5.0 53.6	5.5 57.7	6.0 61.6	6.5 65.2
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 51.1	5.0 55.9	5.5 60.2	6.0 64.1	6.0 67.3
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 50.7	6.0 56.0	6.0 60.6	6.5 64.7	6.5 68.3
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 54.7	5.5 60.2	5.5 65.0	5.5 69.1	6.0 72.9
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 54.7	5.5 60.4	6.0 65.3	6.0 69.5	6.5 73.4
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 55.5	6.5 61.4	6.5 66.3	7.0 70.6	7.0 74.6
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 59.1	6.0 65.0	6.0 69.9	6.0 74.3	6.5 77.9
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 49.8	5.0 54.3	5.5 58.4	6.0 62.5	6.5 66.0
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 51.8	5.0 56.7	5.5 61.0	6.0 64.8	6.0 68.1
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 51.4	6.0 56.7	6.0 61.4	6.5 65.5	6.5 69.1
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 55.4	5.5 61.0	5.5 65.8	5.5 69.9	6.0 73.7
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 55.4	5.5 61.2	6.0 66.1	6.0 70.3	6.5 74.2
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 56.3	6.5 62.2	6.5 67.1	7.0 71.5	7.0 75.4
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 59.9	6.0 65.8	6.0 70.7	6.0 75.1	6.5 78.8
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 50.4	5.0 55.1	5.5 59.2	6.0 63.3	6.5 66.7
VREF40+70	KIAS	178	192	204	215	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 52.5	5.0 57.4	5.5 61.8	6.0 65.6	6.0 68.9
VREF40+50	KIAS	158	172	184	195	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 52.2	6.0 57.5	6.0 62.3	6.5 66.3	6.5 69.9
VREF40+30	KIAS	138	152	164	175	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 56.1	5.5 61.8	5.5 66.6	5.5 70.8	6.0 74.5
VREF40+30	KIAS	138	152	164	175	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 56.2	5.5 62.0	6.0 66.9	6.0 71.2	6.5 75.1
VREF40+20	KIAS	128	142	154	165	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 57.0	6.5 63.0	6.5 67.9	7.0 72.4	7.0 76.2
VREF40+10	KIAS	118	132	144	155	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 60.7	6.0 66.5	6.0 71.6	6.0 75.9	6.5 79.7
VREF40+20	KIAS	128	142	154	165	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 3000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 51.1	5.0 55.8	5.5 60.0	6.0 64.1	6.5 67.4
VREF40+70	KIAS	178	192	205	216	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 53.3	5.0 58.1	5.5 62.6	6.0 66.3	6.0 69.7
VREF40+50	KIAS	158	172	185	196	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 52.9	6.0 58.2	6.0 63.1	6.5 67.1	6.5 70.7
VREF40+30	KIAS	138	152	165	176	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 56.9	5.5 62.6	5.5 67.4	5.5 71.6	6.0 75.3
VREF40+30	KIAS	138	152	165	176	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 56.9	5.5 62.8	6.0 67.7	6.0 72.1	6.5 75.9
VREF40+20	KIAS	128	142	155	166	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 57.8	6.5 63.8	6.5 68.7	7.0 73.2	7.0 77.0
VREF40+10	KIAS	118	132	145	156	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 61.5	6.0 67.3	6.0 72.5	6.0 76.7	6.5 80.5
VREF40+20	KIAS	128	142	155	166	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 4000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 51.7	5.0 56.5	5.5 60.8	6.0 64.9	6.5 68.2
VREF40+70	KIAS	178	192	205	216	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 54.0	5.0 58.9	5.5 63.4	6.0 67.1	6.0 70.5
VREF40+50	KIAS	158	172	185	196	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 53.6	6.0 59.0	6.0 63.9	6.5 67.9	6.5 71.6
VREF40+30	KIAS	138	152	165	176	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 57.7	5.5 63.4	5.5 68.2	5.5 72.5	6.0 76.1
VREF40+30	KIAS	138	152	165	176	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 57.7	5.5 63.6	6.0 68.5	6.0 73.0	6.5 76.6
VREF40+20	KIAS	128	142	155	166	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 58.6	6.5 64.6	6.5 69.6	7.0 74.1	7.0 77.9
VREF40+10	KIAS	118	132	145	156	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 62.3	6.0 68.1	6.0 73.3	6.0 77.6	6.5 81.4
VREF40+20	KIAS	128	142	155	166	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 5000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 52.4	5.0 57.3	5.5 61.6	6.0 65.7	6.5 69.0
VREF40+70	KIAS	178	192	205	216	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 54.7	5.0 59.7	5.5 64.2	6.0 67.9	6.0 71.4
VREF40+50	KIAS	158	172	185	196	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 54.4	6.0 59.9	6.0 64.7	6.5 68.7	6.5 72.5
VREF40+30	KIAS	138	152	165	176	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 58.5	5.5 64.2	5.5 69.1	5.5 73.3	6.0 76.9
VREF40+30	KIAS	138	152	165	176	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 58.5	5.5 64.4	6.0 69.4	6.0 73.8	6.5 77.5
VREF40+20	KIAS	128	142	155	166	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 59.4	6.5 65.4	6.5 70.5	7.0 74.9	7.0 78.8
VREF40+10	KIAS	118	132	145	156	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 63.1	6.0 69.0	6.0 74.2	6.0 78.5	6.5 82.3
VREF40+20	KIAS	128	142	155	166	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 6000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 53.1	5.0 58.0	5.5 62.5	6.0 66.4	6.5 69.8
VREF40+70	KIAS	178	192	205	216	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 55.5	5.0 60.5	5.5 64.9	6.0 68.7	6.0 72.3
VREF40+50	KIAS	158	172	185	196	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 55.1	6.0 60.7	6.0 65.5	6.5 69.5	6.5 73.3
VREF40+30	KIAS	138	152	165	176	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 59.3	5.5 65.1	5.5 69.9	5.5 74.1	6.0 77.7
VREF40+30	KIAS	138	152	165	176	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 59.4	5.5 65.2	6.0 70.3	6.0 74.7	6.5 78.3
VREF40+20	KIAS	128	142	155	166	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 60.3	6.5 66.2	6.5 71.4	7.0 75.7	7.0 79.6
VREF40+10	KIAS	118	132	145	156	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 64.0	6.0 69.9	6.0 75.0	6.0 79.3	6.5 83.2
VREF40+20	KIAS	128	142	155	166	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 7000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 53.9	5.0 58.8	5.5 63.4	6.0 67.1	6.5 70.6
VREF40+70	KIAS	178	192	205	216	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 56.3	5.0 61.4	5.5 65.7	6.0 69.5	6.0 73.1
VREF40+50	KIAS	158	172	185	196	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 55.9	6.0 61.6	6.0 66.3	6.5 70.4	6.5 74.1
VREF40+30	KIAS	138	152	165	176	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 60.1	5.5 65.9	5.5 70.8	5.5 75.0	6.0 78.6
VREF40+30	KIAS	138	152	165	176	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 60.2	5.5 66.0	6.0 71.2	6.0 75.4	6.5 79.2
VREF40+20	KIAS	128	142	155	166	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 61.1	6.5 67.0	6.5 72.2	7.0 76.6	7.0 80.5
VREF40+10	KIAS	118	132	145	156	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 64.8	6.0 70.8	6.0 75.8	6.0 80.2	6.5 84.1
VREF40+20	KIAS	128	142	155	166	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 8000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 54.6	5.0 59.6	5.5 64.1	6.0 67.9	6.5 71.4
VREF40+70	KIAS	178	192	205	216	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 57.0	5.0 62.2	5.5 66.5	6.0 70.4	6.0 74.0
VREF40+50	KIAS	158	172	185	196	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 56.7	6.0 62.4	6.0 67.0	6.5 71.2	6.5 74.9
VREF40+30	KIAS	138	152	165	176	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 61.0	5.5 66.7	5.5 71.7	5.5 75.8	6.0 79.6
VREF40+30	KIAS	138	152	165	176	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 61.0	5.5 66.9	6.0 72.1	6.0 76.2	6.0 80.1
VREF40+20	KIAS	128	142	155	166	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 61.9	6.5 67.9	6.5 73.1	7.0 77.5	7.0 81.4
VREF40+10	KIAS	118	132	145	156	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 65.6	6.0 71.6	6.0 76.7	6.0 81.1	6.5 85.0
VREF40+20	KIAS	128	142	155	166	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 9000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 55.4	5.5 60.5	5.5 64.9	6.0 68.6	6.5 72.3
VREF40+70	KIAS	178	192	205	216	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 57.7	5.0 63.0	5.5 67.3	6.0 71.3	6.0 74.8
VREF40+50	KIAS	158	172	185	196	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 57.5	6.0 63.2	6.0 67.9	6.5 72.1	6.5 75.8
VREF40+30	KIAS	138	152	165	176	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 61.8	5.5 67.5	5.5 72.5	5.5 76.6	6.0 80.4
VREF40+30	KIAS	138	152	165	176	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 61.8	5.5 67.7	6.0 72.9	6.0 77.1	6.0 81.0
VREF40+20	KIAS	128	142	155	166	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 62.8	6.5 68.7	6.5 73.9	7.0 78.4	7.0 82.3
VREF40+10	KIAS	118	132	145	156	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 66.4	6.0 72.5	6.0 77.6	6.0 82.0	6.5 85.9
VREF40+20	KIAS	128	142	155	166	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 10000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT	4.5	5.5	5.5	6.0	6.5
VREF40+70	%N1	56.1	61.3	65.6	69.5	73.0
	KIAS	178	192	205	216	227
FLAPS 1 (GEAR UP)	PITCH ATT	5.0	5.0	5.5	6.0	6.0
VREF40+50	%N1	58.5	63.8	68.1	72.2	75.6
	KIAS	158	172	185	196	207
FLAPS 5 (GEAR UP)	PITCH ATT	5.5	6.0	6.0	6.0	6.5
VREF40+30	%N1	58.3	64.0	68.7	72.9	76.6
	KIAS	138	152	165	176	187
FLAPS 10 (GEAR UP)	PITCH ATT	5.0	5.5	5.5	5.5	6.0
VREF40+30	%N1	62.6	68.4	73.3	77.5	81.4
	KIAS	138	152	165	176	187
FLAPS 15 (GEAR UP)	PITCH ATT	5.5	5.5	6.0	6.0	6.0
VREF40+20	%N1	62.7	68.6	73.7	78.0	81.9
	KIAS	128	142	155	166	177
FLAPS 25 (GEAR UP)	PITCH ATT	6.5	6.5	6.5	7.0	7.0
VREF40+10	%N1	63.6	69.7	74.8	79.3	83.2
	KIAS	118	132	145	156	167
FLAPS 15 (GEAR DOWN)	PITCH ATT	5.5	6.0	6.0	6.0	6.5
VREF40+20	%N1	67.2	73.4	78.5	82.9	86.9
	KIAS	128	142	155	166	177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 11000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 56.9	5.5 62.3	5.5 66.3	6.0 70.3	6.5 73.9
VREF40+70	KIAS	178	192	205	217	227
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 59.5	5.0 64.6	5.5 69.1	6.0 73.1	6.0 76.4
VREF40+50	KIAS	158	172	185	197	207
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 59.4	6.0 64.9	6.0 69.7	6.0 73.8	6.5 77.5
VREF40+30	KIAS	138	152	165	177	187
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 63.6	5.5 69.4	5.5 74.2	5.5 78.5	6.0 82.3
VREF40+30	KIAS	138	152	165	177	187
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 63.8	5.5 69.7	6.0 74.6	6.0 79.0	6.0 82.9
VREF40+20	KIAS	128	142	155	167	177
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 64.7	6.5 70.8	6.5 75.8	6.5 80.3	7.0 84.2
VREF40+10	KIAS	118	132	145	157	167
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 68.4	6.0 74.3	6.0 79.5	6.0 84.0	6.5 88.0
VREF40+20	KIAS	128	142	155	167	177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 12000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	5.0 57.7	5.5 63.0	5.5 67.1	6.0 71.2	6.5 74.6
VREF40+70	KIAS	178	192	205	217	228
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 60.6	5.5 65.4	5.5 70.0	6.0 73.8	6.0 77.3
VREF40+50	KIAS	158	172	185	197	208
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 60.5	6.0 65.8	6.0 70.7	6.0 74.7	6.5 78.5
VREF40+30	KIAS	138	152	165	177	188
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 64.6	5.5 70.4	5.5 75.1	5.5 79.5	6.0 83.3
VREF40+30	KIAS	138	152	165	177	188
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 64.7	5.5 70.8	6.0 75.6	6.0 80.0	6.0 83.9
VREF40+20	KIAS	128	142	155	167	178
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 65.8	6.5 71.8	6.5 76.9	6.5 81.3	6.5 85.2
VREF40+10	KIAS	118	132	145	157	168
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 69.6	6.0 75.3	6.0 80.5	6.0 85.0	6.0 89.1
VREF40+20	KIAS	128	142	155	167	178

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 13000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	5.0 58.5	5.5 63.7	5.5 67.9	6.0 72.0	6.5 75.5
VREF40+70	KIAS	178	192	205	217	228
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 61.4	5.5 66.3	5.5 70.9	6.0 74.6	6.0 78.2
VREF40+50	KIAS	158	172	185	197	208
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 61.5	6.0 66.8	6.0 71.7	6.0 75.6	6.5 79.4
VREF40+30	KIAS	138	152	165	177	188
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 65.6	5.5 71.4	5.5 76.0	5.5 80.4	6.0 84.2
VREF40+30	KIAS	138	152	165	177	188
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 65.8	5.5 71.7	6.0 76.5	6.0 81.0	6.0 84.8
VREF40+20	KIAS	128	142	155	167	178
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 66.9	6.5 72.8	6.5 77.9	6.5 82.3	6.5 86.2
VREF40+10	KIAS	118	132	145	157	168
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 70.6	6.0 76.4	6.0 81.5	6.0 86.0	6.0 90.2
VREF40+20	KIAS	128	142	155	167	178

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 14000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT	5.0	5.5	5.5	6.0	6.5
	%N1	59.4	64.4	68.7	72.8	76.3
VREF40+70	KIAS	179	192	205	217	228
FLAPS 1 (GEAR UP)	PITCH ATT	5.0	5.5	5.5	6.0	6.0
	%N1	62.3	67.1	71.7	75.4	79.0
VREF40+50	KIAS	159	172	185	197	208
FLAPS 5 (GEAR UP)	PITCH ATT	5.5	6.0	6.0	6.0	6.5
	%N1	62.3	67.8	72.5	76.5	80.3
VREF40+30	KIAS	139	152	165	177	188
FLAPS 10 (GEAR UP)	PITCH ATT	5.0	5.5	5.5	5.5	6.0
	%N1	66.5	72.3	77.0	81.3	85.1
VREF40+30	KIAS	139	152	165	177	188
FLAPS 15 (GEAR UP)	PITCH ATT	5.5	5.5	6.0	6.0	6.0
	%N1	66.9	72.6	77.6	81.9	85.8
VREF40+20	KIAS	129	142	155	167	178
FLAPS 25 (GEAR UP)	PITCH ATT	6.5	6.5	6.5	6.5	6.5
	%N1	68.1	73.8	78.9	83.3	87.3
VREF40+10	KIAS	119	132	145	157	168
FLAPS 15 (GEAR DOWN)	PITCH ATT	5.5	6.0	6.0	6.0	6.0
	%N1	71.6	77.5	82.5	87.1	91.5
VREF40+20	KIAS	129	142	155	167	178

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 14500 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	5.0 59.9	5.5 64.8	5.5 69.2	6.0 73.2	6.5 76.7
VREF40+70	KIAS	179	192	206	218	229
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 62.7	5.5 67.6	5.5 72.1	6.0 75.9	6.0 79.5
VREF40+50	KIAS	159	172	186	198	209
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 62.8	6.0 68.3	6.0 72.9	6.0 76.9	6.5 80.7
VREF40+30	KIAS	139	152	166	178	189
FLAPS 10 (GEAR UP)	PITCH ATT %N1	5.0 67.1	5.5 72.7	5.5 77.5	5.5 81.8	6.0 85.5
VREF40+30	KIAS	139	152	166	178	189
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.5 67.4	5.5 73.1	6.0 78.1	6.0 82.4	6.0 86.3
VREF40+20	KIAS	129	142	156	168	179
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.5 68.6	6.5 74.2	6.5 79.4	6.5 83.7	6.5 87.8
VREF40+10	KIAS	119	132	146	158	169
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.5 72.1	6.0 78.0	6.0 83.0	6.0 87.6	6.0 92.3
VREF40+20	KIAS	129	142	156	168	179

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = -2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	42.2	46.7	50.6	54.1	57.0
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	46.3	51.0	55.3	59.0	62.3
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	50.8	56.2	60.9	64.9	68.3
	KIAS	118	132	144	155	164

Flap placard speed exceeded in shaded area.

Airport Altitude = -1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	42.7	47.3	51.2	54.8	57.8
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	46.9	51.7	56.0	59.8	63.1
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	51.5	56.9	61.7	65.7	69.0
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = SEA LEVEL

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	43.3	48.0	51.9	55.5	58.4
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	47.5	52.5	56.7	60.5	63.9
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	52.2	57.6	62.4	66.5	69.8
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	43.9	48.6	52.6	56.2	59.2
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	48.1	53.2	57.5	61.3	64.6
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	52.9	58.4	63.2	67.3	70.6
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	44.5	49.2	53.3	56.9	59.9
	KIAS	130	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	48.8	53.8	58.2	62.1	65.4
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	53.6	59.2	64.0	68.0	71.5
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 3000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	45.1	49.8	54.1	57.6	60.7
	KIAS	131	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	49.4	54.5	59.0	63.0	66.2
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	54.4	59.9	64.9	68.8	72.3
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 4000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	45.8	50.5	54.7	58.3	61.5
	KIAS	131	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	50.1	55.3	59.8	63.7	67.0
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	55.1	60.7	65.6	69.6	73.2
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 5000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	46.5	51.2	55.4	59.0	62.3
	KIAS	131	145	159	171	181
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	50.8	56.0	60.6	64.5	67.7
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	55.8	61.5	66.4	70.4	74.0
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 6000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	47.1	51.9	56.1	59.8	63.0
	KIAS	131	145	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	51.6	56.7	61.4	65.2	68.5
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	56.6	62.3	67.1	71.3	74.8
	KIAS	118	132	144	155	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 7000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	47.7	52.6	56.8	60.6	63.8
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	52.3	57.4	62.2	66.0	69.3
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	57.3	63.1	67.9	72.2	75.6
	KIAS	118	132	144	156	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 8000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	48.4	53.3	57.6	61.4	64.5
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	52.9	58.2	62.9	66.8	70.1
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	58.1	64.0	68.7	73.0	76.4
	KIAS	118	132	145	156	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 9000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	49.0	54.0	58.3	62.2	65.2
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	53.7	59.0	63.7	67.6	70.9
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	58.9	64.7	69.6	73.8	77.2
	KIAS	118	132	145	156	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 10000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	49.7	54.7	59.1	62.9	65.9
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	54.4	59.8	64.5	68.3	71.7
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	0.0
	%N1	59.8	65.5	70.5	74.6	78.1
	KIAS	118	132	145	156	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 11000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	50.4	55.5	59.9	63.6	66.6
	KIAS	131	146	159	171	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	55.1	60.7	65.3	69.1	72.5
	KIAS	125	139	152	163	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	60.5	66.3	71.3	75.4	79.0
	KIAS	118	132	145	156	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 12000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	51.1	56.2	60.7	64.3	67.4
	KIAS	131	146	159	172	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	55.9	61.5	66.1	70.0	73.3
	KIAS	125	139	152	164	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	61.3	67.1	72.2	76.2	79.8
	KIAS	118	132	145	156	166

Flap placard speed exceeded in shaded area.

Airport Altitude = 13000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	51.9	56.9	61.5	65.1	68.1
	KIAS	131	146	160	172	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	56.6	62.2	66.8	70.8	74.1
	KIAS	125	139	152	164	173
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	62.2	67.9	73.0	77.1	80.7
	KIAS	118	132	145	156	166

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 14000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	52.7	57.8	62.4	65.8	69.0
	KIAS	131	146	160	172	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	57.6	63.1	67.7	71.7	74.9
	KIAS	125	139	152	164	174
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	63.2	68.9	73.9	78.0	81.7
	KIAS	118	132	145	156	166

Flap placard speed exceeded in shaded area.

Airport Altitude = 14500 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	53.1	58.3	62.8	66.3	69.4
	KIAS	131	146	160	172	182
FLAPS 30 (VREF30 + 10)	PITCH ATT	0.5	0.5	1.0	1.0	1.0
	%N1	58.2	63.6	68.2	72.1	75.3
	KIAS	125	139	152	164	174
FLAPS 40 (VREF40 + 10)	PITCH ATT	-0.5	-0.5	-0.5	-0.5	-0.5
	%N1	63.7	69.5	74.4	78.5	82.2
	KIAS	118	132	145	156	167

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

GO-AROUND

Flaps 15, Gear Up, Set Go-Around Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
14500	PITCH ATT	17.0	14.0	11.5	10.0	9.0
	V/S (FT/MIN)	3400	2600	2000	1600	1200
	KIAS	128	142	155	166	177
10000	PITCH ATT	20.0	16.0	13.5	11.5	10.5
	V/S (FT/MIN)	3900	3000	2400	1900	1500
	KIAS	128	142	155	166	175
5000	PITCH ATT	24.0	19.0	16.0	13.5	12.5
	V/S (FT/MIN)	4500	3600	3000	2400	2000
	KIAS	128	142	154	165	175
SEA LEVEL	PITCH ATT	28.0	22.0	18.5	16.0	14.0
	V/S (FT/MIN)	5100	4200	3500	2900	2400
	KIAS	128	142	154	165	175
-2000	PITCH ATT	28.0	22.0	18.5	16.0	14.0
	V/S (FT/MIN)	5000	4100	3400	2800	2400
	KIAS	128	142	154	165	174

Max Climb %N1**Based on engine bleed for packs on or off and anti-ice off**

TAT (°C)	PRESSURE ALTITUDE (FT)/SPEED (KIAS/MACH)									
	0	5000	10000	15000	20000	25000	30000	35000	37000	41000
	280	280	280	280	280	280	280	.78	.78	.78
60	90.2	90.5	90.4	90.6	90.4	92.1	93.8	95.1	95.2	93.5
55	91.0	91.2	91.3	91.4	90.8	91.5	93.1	94.4	94.5	92.8
50	91.7	92.0	92.1	92.2	91.7	91.5	92.4	93.7	93.8	92.1
45	92.4	92.6	92.8	93.0	92.6	92.4	92.4	93.0	93.1	91.4
40	93.1	93.3	93.6	93.8	93.4	93.2	93.2	92.3	92.4	90.7
35	93.6	94.0	94.3	94.5	94.3	94.0	94.0	93.0	92.4	90.8
30	92.9	94.8	95.0	95.2	95.1	94.8	94.7	93.9	93.3	91.8
25	92.2	94.8	95.7	95.9	95.9	95.5	95.4	94.7	94.1	92.8
20	91.4	94.0	96.5	96.7	96.6	96.2	96.1	95.4	94.9	93.7
15	90.6	93.2	95.9	97.5	97.4	96.9	96.7	96.2	95.7	94.6
10	89.9	92.5	95.1	97.8	98.3	97.7	97.4	96.9	96.5	95.6
5	89.1	91.7	94.3	97.0	99.2	98.6	98.1	97.7	97.3	96.5
0	88.3	90.9	93.5	96.2	98.6	99.6	99.1	98.5	98.2	97.5
-5	87.6	90.1	92.7	95.4	97.8	99.6	100.0	99.2	99.0	98.4
-10	86.8	89.3	91.9	94.6	97.1	98.8	100.3	100.2	99.8	99.4
-15	86.0	88.5	91.0	93.8	96.3	98.0	99.6	101.1	100.8	100.4
-20	85.2	87.6	90.2	93.0	95.5	97.2	98.7	100.8	101.3	101.0
-25	84.3	86.8	89.4	92.2	94.7	96.4	97.9	100.0	100.5	100.1
-30	83.5	86.0	88.5	91.3	93.9	95.6	97.1	99.1	99.6	99.3
-35	82.7	85.1	87.7	90.5	93.1	94.8	96.3	98.3	98.8	98.4
-40	81.8	84.3	86.8	89.6	92.3	93.9	95.4	97.4	97.9	97.6

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)					
	0	10	20	30	35	41
ENGINE ANTI-ICE	-0.6	-0.8	-0.9	-0.9	-0.8	-0.8
ENGINE & WING ANTI-ICE*	-1.8	-2.1	-2.5	-2.7	-3.0	-3.0

*Dual bleed sources

Go-around %N1

Based on engine bleed for packs on, engine and wing anti-ice on or off

AIRPORT OAT		TAT (°C)	AIRPORT PRESSURE ALTITUDE (FT)											
°C	°F		-2000	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
57	134	60	95.0	96.2	96.8									
52	125	55	95.9	96.7	96.6	96.8	97.5							
47	116	50	96.6	97.6	97.8	97.8	97.7	97.5	98.2	98.8				
42	108	45	97.4	98.4	98.5	98.6	98.7	98.8	98.7	98.5	99.0			
37	99	40	98.0	99.1	99.2	99.3	99.4	99.5	99.6	99.5	99.1	98.9	98.8	99.1
32	90	35	98.1	99.9	100.0	100.1	100.1	100.3	100.3	100.2	99.9	99.6	99.6	99.5
27	81	30	97.3	99.8	100.4	100.7	100.7	100.7	100.7	100.7	100.6	100.4	100.4	100.3
22	72	25	96.6	99.1	99.7	100.2	100.6	100.9	100.9	100.9	100.9	100.9	100.9	100.8
17	63	20	95.8	98.3	98.9	99.5	99.8	100.2	100.5	100.9	101.0	101.1	101.0	101.0
12	54	15	95.0	97.5	98.1	98.7	99.1	99.4	99.8	100.1	100.5	100.9	101.3	101.2
7	45	10	94.2	96.8	97.4	98.0	98.3	98.7	99.0	99.4	99.8	100.2	100.5	100.9
2	36	5	93.4	96.0	96.6	97.2	97.6	97.9	98.3	98.7	99.0	99.4	99.8	100.2
-3	27	0	92.6	95.2	95.8	96.4	96.8	97.2	97.5	97.9	98.3	98.7	99.0	99.4
-8	18	-5	91.8	94.4	95.0	95.6	96.0	96.4	96.8	97.2	97.5	97.9	98.3	98.6
-13	9	-10	91.0	93.6	94.2	94.8	95.2	95.6	96.0	96.4	96.8	97.1	97.5	97.9
-17	1	-15	90.2	92.8	93.4	94.0	94.4	94.8	95.2	95.6	96.0	96.4	96.7	97.1
-22	-8	-20	89.3	92.0	92.6	93.2	93.6	94.0	94.4	94.8	95.2	95.6	95.9	96.3
-27	-17	-25	88.5	91.1	91.8	92.4	92.8	93.2	93.6	94.0	94.4	94.8	95.1	95.5
-32	-26	-30	87.6	90.3	90.9	91.6	92.0	92.4	92.8	93.3	93.6	94.0	94.3	94.7
-37	-35	-35	86.8	89.4	90.1	90.7	91.1	91.6	92.0	92.4	92.8	93.2	93.5	93.9
-42	-44	-40	85.9	88.6	89.2	89.9	90.3	90.7	91.2	91.6	92.0	92.4	92.7	93.0
-47	-53	-45	85.0	87.7	88.4	89.0	89.4	89.9	90.3	90.8	91.2	91.5	91.9	92.2
-52	-62	-50	84.1	86.8	87.5	88.2	88.6	89.0	89.5	90.0	90.3	90.7	91.0	91.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (FT)											
	-2000	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
PACKS OFF	0.7	0.8	0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9
A/C HIGH	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1



VREF

WEIGHT (1000 KG)	FLAPS		
	40	30	15
85	160	168	177
80	155	163	172
75	151	158	167
70	146	153	161
65	141	148	156
60	135	142	149
55	128	136	143
50	122	129	136
45	115	122	128
40	108	115	121

Intentionally
Blank



Performance Inflight - QRH

Advisory Information

Chapter PI-QRH

Section 41

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF30	ONE REV	NO REV

Dry Runway

MAX MANUAL	1105	65/-65	25/35	-40/140	10/-10	25/-25	40	25	45
AUTOBRAKE MAX	1395	70/-75	35/40	-50/175	0/0	35/-35	65	0	5
AUTOBRAKE 3	1985	110/-125	50/70	-85/290	0/0	60/-60	110	0	0
AUTOBRAKE 2	2520	160/-175	75/105	-115/395	35/-45	75/-75	110	70	70
AUTOBRAKE 1	2775	190/-205	90/120	-140/465	75/-85	80/-80	100	225	335

Good Reported Braking Action

MAX MANUAL	1530	85/-90	40/50	-70/235	40/-35	40/-40	60	80	180
AUTOBRAKE MAX	1625	90/-100	45/60	-70/240	35/-30	40/-40	70	85	195
AUTOBRAKE 3	1985	110/-125	50/70	-85/290	5/0	60/-60	110	5	15
AUTOBRAKE 2	2520	160/-175	75/105	-115/395	35/-45	75/-75	110	70	70
AUTOBRAKE 1	2775	190/-205	90/120	-140/465	75/-85	80/-80	100	225	335

Medium Reported Braking Action

MAX MANUAL	2085	130/-140	65/85	-110/385	100/-75	60/-60	75	220	520
AUTOBRAKE MAX	2130	140/-145	65/85	-110/390	90/-70	60/-60	85	220	525
AUTOBRAKE 3	2215	140/-145	65/85	-110/395	75/-50	65/-65	110	160	470
AUTOBRAKE 2	2580	160/-180	80/105	-125/450	75/-70	75/-75	110	120	260
AUTOBRAKE 1	2795	190/-205	90/120	-140/485	100/-90	80/-85	100	240	405

Poor Reported Braking Action

MAX MANUAL	2720	190/-195	90/125	-160/610	235/-155	75/-80	85	460	1200
AUTOBRAKE MAX	2725	190/-195	90/125	-160/610	235/-155	75/-80	90	460	1210
AUTOBRAKE 3	2745	195/-195	90/125	-160/615	230/-145	75/-80	100	460	1215
AUTOBRAKE 2	2905	200/-205	100/130	-165/635	220/-145	80/-85	105	385	1065
AUTOBRAKE 1	3025	215/-220	100/140	-175/650	225/-155	85/-90	100	435	1070

Reference distance is based on sea level, standard day, no wind or slope, VREF30, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 70 m.

For autobrake and manual speedbrakes, increase reference landing distance by 65 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

All reference distances and adjustments shown have been increased by 15%.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 40

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF40	ONE REV NO REV

Dry Runway

MAX MANUAL	1050	65/-60	25/30	-40/130	10/-10	25/-25	40	15	40
AUTOBRAKE MAX	1305	65/-70	30/40	-45/160	0/0	30/-30	65	0	0
AUTOBRAKE 3	1830	100/-115	45/65	-80/270	0/0	50/-50	105	0	0
AUTOBRAKE 2	2335	145/-160	70/90	-110/380	25/-40	70/-70	110	40	40
AUTOBRAKE 1	2600	175/-190	85/110	-130/450	65/-75	75/-75	100	180	255

Good Reported Braking Action

MAX MANUAL	1460	80/-85	40/50	-65/230	40/-35	35/-35	60	75	160
AUTOBRAKE MAX	1555	85/-90	40/50	-70/235	35/-30	40/-40	70	80	175
AUTOBRAKE 3	1840	100/-115	45/65	-80/275	10/-5	50/-50	110	5	15
AUTOBRAKE 2	2335	145/-160	70/90	-110/380	25/-40	70/-70	110	40	40
AUTOBRAKE 1	2600	175/-190	85/110	-130/450	65/-75	75/-75	100	180	255

Medium Reported Braking Action

MAX MANUAL	1990	120/-130	60/80	-105/380	100/-75	50/-50	75	195	465
AUTOBRAKE MAX	2015	125/-140	65/80	-105/385	85/-70	50/-60	85	195	465
AUTOBRAKE 3	2070	125/-140	65/80	-110/390	80/-50	60/-60	105	175	450
AUTOBRAKE 2	2405	150/-165	70/100	-120/430	65/-65	70/-70	110	85	220
AUTOBRAKE 1	2615	175/-190	85/110	-130/465	90/-85	75/-75	100	195	315

Poor Reported Braking Action

MAX MANUAL	2580	180/-185	85/115	-160/600	230/-150	70/-75	85	415	1070
AUTOBRAKE MAX	2590	180/-185	85/120	-160/600	230/-150	70/-75	85	415	1070
AUTOBRAKE 3	2600	180/-190	85/120	-160/605	225/-145	70/-75	100	415	1075
AUTOBRAKE 2	2725	185/-190	85/120	-160/615	215/-140	75/-80	105	335	955
AUTOBRAKE 1	2840	195/-205	90/125	-165/635	220/-150	80/-85	100	385	935

Reference distance is based on sea level, standard day, no wind or slope, VREF40, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 65 m.

For autobrake and manual speedbrakes, increase reference landing distance by 50 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

All reference distances and adjustments shown have been increased by 15%.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1075	80/-65	25/35	-40/130	15/-10	25/-25	N/A	30	60
AUTOBRAKE MAX	1435	70/-80	35/45	-50/165	5/-5	35/-35	N/A	0	5
AUTOBRAKE 2	2550	165/-180	85/110	-110/370	50/-55	75/-75	N/A	165	180

Good Reported Braking Action

MAX MANUAL	1475	80/-85	40/55	-60/215	35/-30	40/-40	N/A	90	205
AUTOBRAKE MAX	1600	85/-95	45/60	-65/225	30/-25	40/-40	N/A	100	230
AUTOBRAKE 2	2550	165/-180	85/110	-110/370	55/-55	75/-75	N/A	165	180

Medium Reported Braking Action

MAX MANUAL	2025	130/-135	65/85	-100/350	90/-70	55/-55	N/A	240	595
AUTOBRAKE MAX	2080	130/-140	65/90	-100/355	85/-65	55/-60	N/A	245	605
AUTOBRAKE 3	2255	135/-145	70/90	-105/370	60/-45	65/-65	N/A	145	460

Poor Reported Braking Action

MAX MANUAL	2635	185/-190	90/125	-145/550	210/-140	75/-80	N/A	505	1385
AUTOBRAKE MAX	2635	185/-190	90/125	-145/550	210/-135	75/-80	N/A	500	1380
AUTOBRAKE 3	2685	185/-190	90/125	-150/555	195/-125	75/-80	N/A	485	1375

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)									
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ		
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV	

Dry Runway

MAX MANUAL	1025	60/-60	25/30	-35/125	15/-10	25/-25	N/A	25	55
AUTOBRAKE MAX	1335	60/-70	30/40	-45/155	5/-5	30/-35	N/A	0	5
AUTOBRAKE 2	2345	150/-160	75/95	-105/355	45/-50	70/-70	N/A	140	160

Good Reported Braking Action

MAX MANUAL	1415	75/-80	40/50	-60/210	35/-30	35/-35	N/A	80	185
AUTOBRAKE MAX	1525	80/-90	40/55	-65/220	30/-30	40/-40	N/A	90	205
AUTOBRAKE 2	2345	150/-160	75/95	-105/355	45/-50	70/-70	N/A	140	160

Medium Reported Braking Action

MAX MANUAL	1915	120/-125	60/80	-95/340	85/-70	50/-55	N/A	215	520
AUTOBRAKE MAX	1965	120/-130	60/80	-95/345	80/-65	55/-55	N/A	215	530
AUTOBRAKE 3	2100	120/-135	60/85	-100/360	60/-50	60/-60	N/A	135	420

Poor Reported Braking Action

MAX MANUAL	2460	170/-170	85/115	-140/535	200/-130	65/-70	N/A	435	1165
AUTOBRAKE MAX	2470	170/-175	85/115	-140/535	200/-125	70/-75	N/A	430	1160
AUTOBRAKE 3	2510	170/-175	85/115	-145/540	190/-125	70/-75	N/A	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	985	55/-55	20/30	-35/120	15/-10	20/-20	N/A	20	45
AUTOBRAKE MAX	1245	55/-65	30/35	-45/150	5/0	30/-30	N/A	0	0
AUTOBRAKE 2	2195	135/-150	70/90	-100/340	40/-45	65/-65	N/A	105	110

Good Reported Braking Action

MAX MANUAL	1360	70/-80	35/50	-60/205	35/-30	35/-35	N/A	75	170
AUTOBRAKE MAX	1455	75/-85	40/50	-60/215	30/-25	35/-35	N/A	85	185
AUTOBRAKE 2	2195	135/-150	70/90	-100/340	40/-45	65/-65	N/A	105	110

Medium Reported Braking Action

MAX MANUAL	1830	110/-120	55/75	-95/335	85/-65	50/-50	N/A	195	465
AUTOBRAKE MAX	1870	115/-125	60/80	-95/340	80/-60	50/-50	N/A	195	475
AUTOBRAKE 3	1965	115/-125	60/80	-100/350	60/-50	55/-55	N/A	135	405

Poor Reported Braking Action

MAX MANUAL	2345	160/-165	80/110	-140/525	195/-130	65/-70	N/A	395	1035
AUTOBRAKE MAX	2355	160/-165	80/110	-140/525	195/-125	65/-70	N/A	390	1035
AUTOBRAKE 3	2380	160/-165	80/110	-140/530	190/-125	65/-70	N/A	395	1045

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

All Flaps Up Landing

VREF40 + 55

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1330	185/-85	50/105	-45/205	20/-15	35/-35	45	45	95
AUTOBRAKE MAX	1855	85/-90	45/70	-60/195	5/-5	50/-50	75	5	20
AUTOBRAKE 2	3360	195/-225	115/150	-130/430	75/-85	105/-105	100	280	330

Good Reported Braking Action

MAX MANUAL	1755	85/-95	50/65	-65/230	40/-35	45/-50	45	110	255
AUTOBRAKE MAX	2000	90/-100	55/75	-75/245	30/-25	55/-55	70	85	225
AUTOBRAKE 2	3360	195/-225	115/150	-130/430	75/-85	105/-105	100	280	330

Medium Reported Braking Action

MAX MANUAL	2495	145/-155	80/110	-110/385	105/-85	70/-75	65	315	775
AUTOBRAKE MAX	2580	150/-160	85/115	-110/390	100/-80	75/-75	75	325	800
AUTOBRAKE 3	2950	145/-170	90/120	-120/420	65/-60	90/-90	110	165	510

Poor Reported Braking Action

MAX MANUAL	3320	220/-225	120/165	-165/605	250/-170	95/-100	80	690	1915
AUTOBRAKE MAX	3325	215/-225	120/165	-165/605	245/-160	100/-100	90	685	1905
AUTOBRAKE 3	3445	210/-225	120/165	-170/615	220/-150	100/-105	110	600	1840

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1795	105/-110	50/65	-80/290	55/-45	45/-45	60	145	345
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2015	125/-130	60/80	-100/350	85/-65	50/-55	70	215	530
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2585	180/-180	85/120	-145/545	200/-135	70/-75	80	460	1280
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3450	260/-260	120/175	-245/1005	625/-305	85/-105	95	1100	3915
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1695	95/-105	45/60	-80/280	55/-45	40/-45	60	125	300
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1895	115/-120	55/75	-95/340	80/-65	50/-50	65	185	455
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2415	165/-165	80/105	-140/530	190/-125	65/-70	80	395	1075
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3205	235/-235	110/155	-235/980	590/-285	75/-100	90	945	3215
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1615	90/-100	45/60	-80/275	55/-45	40/-40	60	115	265
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1805	105/-115	50/70	-95/335	80/-60	45/-45	65	170	405
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2290	150/-155	70/100	-140/520	185/-120	60/-65	80	355	950
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3040	220/-225	100/145	-230/960	575/-275	70/-95	85	860	2860
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Jammed or Restricted Flight Controls (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1005	70/-60	25/30	-35/125	10/-10	20/-25	35	25	50
AUTOBRAKE MAX	1310	65/-75	30/40	-45/155	5/-5	30/-30	65	0	5
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Good Reported Braking Action

MAX MANUAL	1380	75/-80	35/50	-60/205	35/-30	35/-35	50	80	185
AUTOBRAKE MAX	1485	85/-90	40/55	-60/215	30/-25	35/-35	55	90	205
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Medium Reported Braking Action

MAX MANUAL	1900	125/-130	60/80	-95/340	85/-65	50/-55	65	225	550
AUTOBRAKE MAX	1935	125/-130	60/80	-95/345	80/-65	50/-55	75	225	555
AUTOBRAKE 3	2055	125/-135	60/85	-100/355	60/-40	55/-60	100	145	465

Poor Reported Braking Action

MAX MANUAL	2480	180/-180	85/120	-145/540	205/-135	70/-75	75	475	1310
AUTOBRAKE MAX	2480	180/-180	85/120	-145/540	205/-135	70/-75	80	475	1305
AUTOBRAKE 3	2500	180/-180	85/120	-145/540	200/-120	70/-75	95	475	1310

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LEADING EDGE FLAPS TRANSIT (Flaps 15)

VREF15 + 15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1130	80/-70	25/35	-40/135	15/-15	25/-25	35	30	70
AUTOBRAKE MAX	1500	70/-80	35/45	-50/170	5/-5	40/-40	65	0	5
AUTOBRAKE 2	2725	175/-190	90/115	-115/385	50/-60	85/-85	100	155	160

Good Reported Braking Action

MAX MANUAL	1570	85/-95	45/60	-65/220	40/-35	40/-40	50	105	240
AUTOBRAKE MAX	1690	90/-100	45/60	-65/230	35/-30	45/-45	60	115	260
AUTOBRAKE 2	2730	175/-190	90/115	-115/385	50/-60	85/-85	100	160	160

Medium Reported Braking Action

MAX MANUAL	2170	140/-145	70/95	-100/365	95/-75	60/-60	70	275	695
AUTOBRAKE MAX	2210	140/-150	70/95	-105/365	90/-70	60/-65	75	280	700
AUTOBRAKE 3	2375	140/-150	75/100	-110/380	65/-45	70/-70	110	175	570

Poor Reported Braking Action

MAX MANUAL	2825	200/-205	100/140	-155/570	225/-150	80/-85	80	580	1620
AUTOBRAKE MAX	2825	200/-205	100/140	-155/570	230/-155	80/-85	85	575	1615
AUTOBRAKE 3	2855	200/-200	100/140	-155/570	215/-130	80/-85	100	565	1615

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1120	70/-65	25/35	-40/135	15/-15	25/-25	45	35	60
AUTOBRAKE MAX	1300	65/-75	30/40	-45/155	0/0	30/-30	60	0	10
AUTOBRAKE 2	2465	150/-175	75/95	-110/365	0/-10	75/-75	140	0	0

Good Reported Braking Action

MAX MANUAL	1620	95/-100	45/60	-70/235	50/-40	40/-45	70	135	275
AUTOBRAKE MAX	1630	95/-105	45/65	-70/235	40/-35	45/-45	75	135	275
AUTOBRAKE 2	2465	150/-175	75/95	-110/365	0/-10	75/-75	140	0	0

Medium Reported Braking Action

MAX MANUAL	2235	150/-155	75/100	-110/380	115/-90	60/-65	90	350	840
AUTOBRAKE MAX	2220	150/-155	75/100	-105/380	120/-95	60/-65	90	345	830
AUTOBRAKE 3	2220	150/-155	75/100	-105/380	120/-85	60/-65	90	345	830

Poor Reported Braking Action

MAX MANUAL	2905	215/-215	105/145	-160/590	265/-175	80/-85	105	710	2025
AUTOBRAKE MAX	2900	215/-215	105/150	-160/590	265/-180	80/-85	105	710	2025
AUTOBRAKE 3	2900	215/-215	105/150	-160/590	265/-180	80/-85	105	710	2025

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1060	65/-55	25/35	-40/130	15/-15	25/-25	45	30	50
AUTOBRAKE MAX	1215	60/-65	30/35	-45/145	0/0	30/-30	55	10	15
AUTOBRAKE 2	2260	135/-155	65/85	-105/350	0/-10	70/-70	135	0	0

Good Reported Braking Action

MAX MANUAL	1535	85/-95	45/60	-65/225	45/-40	40/-40	70	120	240
AUTOBRAKE MAX	1550	90/-95	45/60	-65/230	40/-35	40/-40	75	120	240
AUTOBRAKE 2	2260	135/-155	65/85	-105/350	0/-10	70/-70	135	0	0

Medium Reported Braking Action

MAX MANUAL	2090	135/-140	65/90	-105/370	110/-85	55/-60	85	305	710
AUTOBRAKE MAX	2085	135/-140	70/90	-105/370	115/-90	55/-60	90	300	705
AUTOBRAKE 3	2085	135/-140	70/90	-105/370	115/-80	55/-60	90	300	705

Poor Reported Braking Action

MAX MANUAL	2695	195/-195	95/130	-155/570	250/-165	75/-80	100	605	1650
AUTOBRAKE MAX	2695	195/-195	95/135	-155/570	250/-165	75/-80	100	605	1650
AUTOBRAKE 3	2695	195/-195	95/135	-155/570	250/-165	75/-80	100	605	1650

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1015	60/-55	25/30	-35/125	15/-15	25/-25	50	30	45
AUTOBRAKE MAX	1140	55/-60	25/35	-40/140	5/0	25/-25	55	10	20
AUTOBRAKE 2	2075	125/-140	60/80	-100/335	0/-5	60/-60	130	0	0

Good Reported Braking Action

MAX MANUAL	1460	80/-90	40/55	-65/225	45/-40	35/-40	70	105	210
AUTOBRAKE MAX	1470	85/-90	40/55	-65/225	40/-35	40/-40	75	105	210
AUTOBRAKE 2	2075	125/-140	60/80	-100/335	0/-5	60/-60	130	0	0

Medium Reported Braking Action

MAX MANUAL	1970	125/-135	60/85	-100/360	105/-85	55/-55	85	265	615
AUTOBRAKE MAX	1970	125/-135	60/85	-100/360	110/-85	55/-55	85	265	615
AUTOBRAKE 3	1970	125/-135	60/85	-100/360	110/-80	55/-55	90	265	615

Poor Reported Braking Action

MAX MANUAL	2525	180/-185	85/120	-150/560	240/-155	70/-75	95	525	1400
AUTOBRAKE MAX	2530	180/-185	90/125	-150/560	245/-160	70/-75	95	530	1405
AUTOBRAKE 3	2530	180/-185	90/125	-150/560	245/-160	70/-75	95	530	1405

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1570	80/-90	40/50	-60/195	35/-35	40/-40	75	-10	65
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2290	135/-145	65/90	-100/335	100/-80	60/-60	105	95	440
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	3035	200/-210	100/140	-150/525	215/-160	80/-85	120	365	1415
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3770	275/-275	135/190	-210/785	475/-270	100/-110	130	815	3380
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM B (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1140	55/-60	25/35	-45/145	20/-15	25/-25	40	40	70
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1630	95/-100	45/65	-75/255	50/-45	45/-45	60	140	285
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2215	150/-155	70/100	-115/410	125/-95	60/-65	75	340	815
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	2835	210/-210	100/140	-170/640	295/-180	75/-85	90	665	1870
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

MANUAL REVERSION (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1570	80/-90	40/50	-60/195	35/-35	40/-40	75	-10	65
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2290	135/-145	65/90	-100/335	100/-80	60/-60	105	95	440
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	3035	200/-210	100/140	-150/525	215/-160	80/-85	120	365	1415
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3770	275/-275	135/190	-210/785	475/-270	100/-110	130	815	3380
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

One Engine Inoperative Landing (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1020	75/-65	25/30	-35/130	15/-10	25/-25	35	0	25
AUTOBRAKE MAX	1300	70/-75	30/40	-45/155	0/0	30/-30	60	0	0
AUTOBRAKE 2	2450	150/-170	75/95	-110/365	10/-25	75/-75	120	0	0

Good Reported Braking Action

MAX MANUAL	1440	80/-85	40/50	-65/215	40/-35	40/-40	50	0	100
AUTOBRAKE MAX	1545	85/-95	40/55	-65/225	35/-30	40/-40	60	0	110
AUTOBRAKE 2	2450	150/-170	75/95	-110/365	10/-25	75/-75	120	0	0

Medium Reported Braking Action

MAX MANUAL	2075	135/-140	65/85	-105/370	110/-85	60/-60	70	0	310
AUTOBRAKE MAX	2115	135/-145	65/85	-105/375	105/-80	60/-60	80	0	315
AUTOBRAKE 3	2165	135/-150	65/85	-105/380	90/-65	60/-65	100	0	295

Poor Reported Braking Action

MAX MANUAL	2850	200/-210	95/130	-165/605	290/-185	85/-85	90	0	765
AUTOBRAKE MAX	2850	200/-210	95/130	-165/605	290/-185	85/-85	95	0	765
AUTOBRAKE 3	2875	205/-210	95/130	-165/610	280/-180	85/-85	100	0	775

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

One Engine Inoperative Landing (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	970	60/-55	20/30	-35/125	15/-10	20/-20	35	0	25
AUTOBRAKE MAX	1215	60/-65	30/35	-45/150	0/0	30/-30	55	0	0
AUTOBRAKE 2	2240	135/-150	65/85	-105/350	10/-25	65/-65	110	0	0

Good Reported Braking Action

MAX MANUAL	1370	75/-80	35/50	-60/210	35/-30	35/-35	50	0	90
AUTOBRAKE MAX	1465	80/-90	40/50	-65/220	35/-30	35/-40	60	0	100
AUTOBRAKE 2	2240	135/-150	65/85	-105/350	10/-25	65/-65	110	0	0

Medium Reported Braking Action

MAX MANUAL	1940	120/-130	60/80	-100/360	105/-80	55/-55	70	0	265
AUTOBRAKE MAX	1975	125/-135	60/80	-100/365	95/-75	55/-55	80	0	270
AUTOBRAKE 3	2015	125/-135	60/80	-105/365	90/-65	55/-60	90	0	260

Poor Reported Braking Action

MAX MANUAL	2625	180/-190	85/115	-155/585	265/-170	75/-80	85	0	635
AUTOBRAKE MAX	2625	180/-190	85/115	-155/585	270/-165	75/-80	90	0	635
AUTOBRAKE 3	2655	185/-190	90/120	-160/585	260/-170	75/-80	90	0	640

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Stabilizer Trim Inoperative (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1005	70/-60	25/30	-35/125	10/-10	20/-25	35	25	50
AUTOBRAKE MAX	1310	65/-75	30/40	-45/155	5/-5	30/-30	65	0	5
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Good Reported Braking Action

MAX MANUAL	1380	75/-80	35/50	-60/205	35/-30	35/-35	50	80	185
AUTOBRAKE MAX	1485	85/-90	40/55	-60/215	30/-25	35/-35	55	90	205
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Medium Reported Braking Action

MAX MANUAL	1900	125/-130	60/80	-95/340	85/-65	50/-55	65	225	550
AUTOBRAKE MAX	1935	125/-130	60/80	-95/345	80/-65	50/-55	75	225	555
AUTOBRAKE 3	2055	125/-135	60/85	-100/355	60/-40	55/-60	100	145	465

Poor Reported Braking Action

MAX MANUAL	2480	180/-180	85/120	-145/540	205/-135	70/-75	75	475	1310
AUTOBRAKE MAX	2480	180/-180	85/120	-145/540	205/-135	70/-75	80	475	1305
AUTOBRAKE 3	2500	180/-180	85/120	-145/540	200/-120	70/-75	95	475	1310

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Trailing Edge Flap Asymmetry (1 ≤ Flap Lever <15) VREF40 + 30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1110	90/-65	25/40	-40/135	15/-15	25/-25	35	30	60
AUTOBRAKE MAX	1510	70/-75	35/45	-50/170	5/-5	40/-40	65	0	5
AUTOBRAKE 2	2730	165/-185	90/115	-115/385	55/-60	85/-85	100	165	175

Good Reported Braking Action

MAX MANUAL	1525	80/-85	40/55	-60/215	35/-30	40/-40	45	90	210
AUTOBRAKE MAX	1665	80/-90	45/60	-65/225	30/-25	45/-45	65	95	225
AUTOBRAKE 2	2735	160/-185	90/115	-115/385	55/-65	85/-85	95	165	175

Medium Reported Braking Action

MAX MANUAL	2125	125/-135	70/90	-100/360	90/-75	60/-60	65	255	625
AUTOBRAKE MAX	2180	130/-140	70/95	-100/360	85/-70	60/-60	75	260	640
AUTOBRAKE 3	2385	125/-140	70/95	-110/380	60/-45	70/-70	110	150	470

Poor Reported Braking Action

MAX MANUAL	2795	190/-195	100/135	-150/565	220/-150	80/-85	80	545	1510
AUTOBRAKE MAX	2790	185/-195	100/135	-150/565	220/-145	80/-85	85	540	1500
AUTOBRAKE 3	2845	185/-190	100/135	-155/570	205/-130	80/-85	105	525	1495

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Trailing Edge Flap Asymmetry (Flap Lever 15 or 25) VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1005	70/-60	25/30	-35/125	10/-10	20/-25	35	25	50
AUTOBRAKE MAX	1310	65/-75	30/40	-45/155	5/-5	30/-30	65	0	5
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Good Reported Braking Action

MAX MANUAL	1380	75/-80	35/50	-60/205	35/-30	35/-35	50	80	185
AUTOBRAKE MAX	1485	85/-90	40/55	-60/215	30/-25	35/-35	55	90	205
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Medium Reported Braking Action

MAX MANUAL	1900	125/-130	60/80	-95/340	85/-65	50/-55	65	225	550
AUTOBRAKE MAX	1935	125/-130	60/80	-95/345	80/-65	50/-55	75	225	555
AUTOBRAKE 3	2055	125/-135	60/85	-100/355	60/-40	55/-60	100	145	465

Poor Reported Braking Action

MAX MANUAL	2480	180/-180	85/120	-145/540	205/-135	70/-75	75	475	1310
AUTOBRAKE MAX	2480	180/-180	85/120	-145/540	205/-135	70/-75	80	475	1305
AUTOBRAKE 3	2500	180/-180	85/120	-145/540	200/-120	70/-75	95	475	1310

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Asymmetry (Flap Lever 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	960	55/-55	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1215	60/-65	30/35	-45/150	5/0	30/-30	60	0	5
AUTOBRAKE 2	2165	140/-150	65/90	-100/340	30/-45	65/-65	90	85	85

Good Reported Braking Action

MAX MANUAL	1315	70/-75	35/45	-60/200	30/-30	35/-35	50	75	165
AUTOBRAKE MAX	1410	75/-85	35/50	-60/210	30/-25	35/-35	60	80	185
AUTOBRAKE 2	2165	140/-150	65/90	-100/340	35/-45	65/-65	90	85	85

Medium Reported Braking Action

MAX MANUAL	1790	115/-120	55/75	-90/330	80/-65	50/-50	65	195	480
AUTOBRAKE MAX	1820	115/-120	55/75	-95/335	75/-60	50/-50	70	200	480
AUTOBRAKE 3	1910	115/-125	55/75	-95/345	60/-40	50/-55	95	140	425

Poor Reported Braking Action

MAX MANUAL	2315	160/-165	80/105	-140/525	195/-125	60/-65	75	410	1100
AUTOBRAKE MAX	2320	165/-165	80/110	-140/525	195/-130	65/-70	75	410	1100
AUTOBRAKE 3	2335	165/-165	80/110	-140/525	190/-115	65/-70	90	410	1110

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (1 ≤ Indicated Flaps <15)

VREF40 + 30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1110	90/-65	25/40	-40/135	15/-15	25/-25	35	30	60
AUTOBRAKE MAX	1510	70/-75	35/45	-50/170	5/-5	40/-40	65	0	5
AUTOBRAKE 2	2730	165/-185	90/115	-115/385	55/-60	85/-85	100	165	175

Good Reported Braking Action

MAX MANUAL	1525	80/-85	40/55	-60/215	35/-30	40/-40	45	90	210
AUTOBRAKE MAX	1665	80/-90	45/60	-65/225	30/-25	45/-45	65	95	225
AUTOBRAKE 2	2735	160/-185	90/115	-115/385	55/-65	85/-85	95	165	175

Medium Reported Braking Action

MAX MANUAL	2125	125/-135	70/90	-100/360	90/-75	60/-60	65	255	625
AUTOBRAKE MAX	2180	130/-140	70/95	-100/360	85/-70	60/-60	75	260	640
AUTOBRAKE 3	2385	125/-140	70/95	-110/380	60/-45	70/-70	110	150	470

Poor Reported Braking Action

MAX MANUAL	2795	190/-195	100/135	-150/565	220/-150	80/-85	80	545	1510
AUTOBRAKE MAX	2790	185/-195	100/135	-150/565	220/-145	80/-85	85	540	1500
AUTOBRAKE 3	2845	185/-190	100/135	-155/570	205/-130	80/-85	105	525	1495

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (15 ≤ Indicated Flaps <30)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1005	70/-60	25/30	-35/125	10/-10	20/-25	35	25	50
AUTOBRAKE MAX	1310	65/-75	30/40	-45/155	5/-5	30/-30	65	0	5
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Good Reported Braking Action

MAX MANUAL	1380	75/-80	35/50	-60/205	35/-30	35/-35	50	80	185
AUTOBRAKE MAX	1485	85/-90	40/55	-60/215	30/-25	35/-35	55	90	205
AUTOBRAKE 2	2360	155/-170	75/100	-105/355	35/-50	70/-70	95	100	100

Medium Reported Braking Action

MAX MANUAL	1900	125/-130	60/80	-95/340	85/-65	50/-55	65	225	550
AUTOBRAKE MAX	1935	125/-130	60/80	-95/345	80/-65	50/-55	75	225	555
AUTOBRAKE 3	2055	125/-135	60/85	-100/355	60/-40	55/-60	100	145	465

Poor Reported Braking Action

MAX MANUAL	2480	180/-180	85/120	-145/540	205/-135	70/-75	75	475	1310
AUTOBRAKE MAX	2480	180/-180	85/120	-145/540	205/-135	70/-75	80	475	1305
AUTOBRAKE 3	2500	180/-180	85/120	-145/540	200/-120	70/-75	95	475	1310

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (30 ≤ Indicated Flaps <40)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	960	55/-55	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1215	60/-65	30/35	-45/150	5/0	30/-30	60	0	5
AUTOBRAKE 2	2165	140/-150	65/90	-100/340	30/-45	65/-65	90	85	85

Good Reported Braking Action

MAX MANUAL	1315	70/-75	35/45	-60/200	30/-30	35/-35	50	75	165
AUTOBRAKE MAX	1410	75/-85	35/50	-60/210	30/-25	35/-35	60	80	185
AUTOBRAKE 2	2165	140/-150	65/90	-100/340	35/-45	65/-65	90	85	85

Medium Reported Braking Action

MAX MANUAL	1790	115/-120	55/75	-90/330	80/-65	50/-50	65	195	480
AUTOBRAKE MAX	1820	115/-120	55/75	-95/335	75/-60	50/-50	70	200	480
AUTOBRAKE 3	1910	115/-125	55/75	-95/345	60/-40	50/-55	95	140	425

Poor Reported Braking Action

MAX MANUAL	2315	160/-165	80/105	-140/525	195/-125	60/-65	75	410	1100
AUTOBRAKE MAX	2320	165/-165	80/110	-140/525	195/-130	65/-70	75	410	1100
AUTOBRAKE 3	2335	165/-165	80/110	-140/525	190/-115	65/-70	90	410	1110

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flaps Up Landing

VREF40 + 40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1185	110/-70	30/70	-40/140	15/-15	30/-30	45	30	70
AUTOBRAKE MAX	1645	75/-80	40/55	-55/180	5/-5	40/-45	70	5	10
AUTOBRAKE 2	2970	175/-195	100/130	-120/400	65/-70	90/-90	95	205	235

Good Reported Braking Action

MAX MANUAL	1600	80/-90	45/60	-65/220	35/-30	40/-45	45	90	205
AUTOBRAKE MAX	1795	85/-95	50/65	-70/235	25/-25	45/-50	65	80	200
AUTOBRAKE 2	2970	175/-195	100/130	-120/400	65/-70	90/-90	95	205	235

Medium Reported Braking Action

MAX MANUAL	2255	135/-140	70/95	-105/365	95/-75	65/-65	65	260	625
AUTOBRAKE MAX	2330	135/-145	75/100	-105/370	90/-75	65/-65	70	265	645
AUTOBRAKE 3	2605	135/-155	80/105	-115/395	60/-55	75/-80	105	145	435

Poor Reported Braking Action

MAX MANUAL	2990	200/-205	105/145	-155/580	230/-155	85/-90	80	565	1530
AUTOBRAKE MAX	2995	195/-205	105/145	-155/580	230/-150	85/-90	90	560	1520
AUTOBRAKE 3	3080	190/-205	105/145	-160/585	210/-140	90/-95	100	520	1495

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Recommended Brake Cooling Schedule

Reference Brake Energy Per Brake (Millions of Foot Pounds)

		WIND CORRECTED BRAKES ON SPEED (KIAS)*																	
		80			100			120			140			160			180		
WEIGHT (1000 KG)	OAT (°C)	PRESSURE ALTITUDE (1000 FT)																	
		0	5	10	0	5	10	0	5	10	0	5	10	0	5	10	0	5	10
80	0	15.1	17.0	19.3	22.4	25.3	28.9	30.9	35.0	40.2	40.4	45.9	53.0	50.8	57.9	67.3	60.8	69.6	81.2
	10	15.6	17.6	20.0	23.1	26.1	29.8	31.9	36.2	41.5	41.8	47.5	54.8	52.5	59.9	69.5	62.8	71.9	83.9
	15	15.8	17.8	20.2	23.5	26.5	30.3	32.4	36.7	42.1	42.4	48.2	55.6	53.3	60.7	70.5	63.7	72.9	85.0
	20	16.0	18.1	20.5	23.8	26.9	30.7	32.8	37.2	42.7	42.9	48.8	56.3	54.0	61.5	71.4	64.6	73.9	86.2
	30	16.4	18.5	21.1	24.4	27.6	31.5	33.7	38.2	43.8	44.0	50.0	57.7	55.3	63.1	73.2	66.2	75.7	88.4
	40	16.6	18.7	21.3	24.7	27.9	31.9	34.1	38.7	44.4	44.7	50.9	58.8	56.3	64.3	74.8	67.5	77.4	90.5
70	50	16.6	18.7	21.3	24.8	28.0	32.1	34.3	39.0	44.9	45.2	51.5	59.7	57.1	65.4	76.3	68.7	79.0	92.9
	0	13.7	15.4	17.5	20.2	22.8	26.0	27.7	31.3	35.9	36.1	41.0	47.2	45.3	51.6	59.7	54.9	62.7	72.9
	10	14.2	15.9	18.1	20.8	23.5	26.8	28.6	32.4	37.1	37.3	42.3	48.7	46.8	53.3	61.6	56.7	64.8	75.4
	15	14.4	16.2	18.4	21.1	23.9	27.2	29.0	32.8	37.6	37.8	43.0	49.4	47.5	54.0	62.5	57.5	65.7	76.4
	20	14.6	16.4	18.6	21.4	24.2	27.6	29.4	33.3	38.1	38.4	43.5	50.1	48.1	54.8	63.4	58.3	66.5	77.4
	30	14.9	16.8	19.1	22.0	24.8	28.3	30.2	34.1	39.1	39.3	44.6	51.4	49.3	56.1	64.9	59.8	68.2	79.4
60	40	15.1	17.0	19.3	22.2	25.1	28.6	30.5	34.6	39.6	39.9	45.3	52.2	50.1	57.1	66.2	60.9	69.6	81.2
	50	15.1	17.0	19.3	22.3	25.2	28.8	30.7	34.8	40.0	40.2	45.8	52.9	50.7	58.0	67.4	61.8	70.9	83.0
	0	12.3	13.9	15.7	18.0	20.3	23.1	24.4	27.6	31.6	31.7	35.9	41.2	39.6	45.0	51.8	48.1	54.8	63.5
	10	12.7	14.3	16.3	18.5	20.9	23.3	25.2	28.5	32.6	32.7	37.1	42.6	40.9	46.5	53.6	49.7	56.6	65.6
	15	12.9	14.6	16.5	18.8	21.2	24.2	25.6	29.0	33.1	33.2	37.6	43.2	41.5	47.1	54.4	50.4	57.4	66.5
	20	13.1	14.8	16.7	19.1	21.5	24.5	26.0	29.4	33.5	33.6	38.1	43.8	42.0	47.8	55.1	51.1	58.2	67.4
50	30	13.4	15.1	17.2	19.6	22.1	25.1	26.6	30.1	34.4	34.5	39.1	44.9	43.1	49.0	56.5	52.3	59.6	69.1
	40	13.6	15.3	17.3	19.8	22.3	25.4	26.9	30.5	34.9	35.0	39.7	45.6	43.8	49.8	57.5	53.2	60.7	70.5
	50	13.5	15.3	17.3	19.8	22.4	25.5	27.0	30.6	35.1	35.2	40.0	46.0	44.2	50.4	58.3	53.9	61.7	71.9
	0	11.0	12.3	14.0	15.7	17.7	20.2	21.2	23.9	27.3	27.2	30.8	35.3	33.8	38.3	44.1	40.9	46.4	53.6
	10	11.2	12.7	14.4	16.3	18.3	20.8	21.9	24.7	28.2	28.1	31.8	36.5	34.9	39.6	45.5	42.2	48.0	55.4
	15	11.5	12.9	14.7	16.5	18.6	21.1	22.2	25.1	28.6	28.6	32.3	37.0	35.4	40.2	46.2	42.8	48.7	56.2
40	20	11.6	13.1	14.9	16.7	18.9	21.4	22.5	25.4	29.0	28.9	32.8	37.5	35.9	40.7	46.8	43.4	49.3	56.9
	30	11.9	13.4	15.2	17.2	19.3	22.0	23.1	26.1	29.7	29.7	33.6	38.4	36.8	41.4	48.0	44.5	50.6	58.4
	40	12.1	13.6	15.4	17.3	19.5	22.2	23.4	26.4	30.1	30.1	34.0	39.0	37.4	42.4	48.8	45.2	51.4	59.4
	50	12.0	13.6	15.4	17.3	19.6	22.3	23.4	26.5	30.3	30.2	34.2	39.3	37.6	42.8	49.3	45.7	52.1	60.3
	0	9.6	10.8	12.3	13.5	15.2	17.3	17.9	20.2	23.0	22.8	25.8	29.4	28.1	31.8	36.4	33.7	38.2	43.9
	10	10.0	11.2	12.7	14.0	15.8	17.9	18.5	20.9	23.8	23.6	26.6	30.4	29.0	32.8	37.6	34.8	39.5	45.4
30	15	10.1	11.4	12.9	14.2	16.0	18.1	18.8	21.2	24.1	23.9	27.0	30.8	29.4	33.3	38.2	35.3	40.0	46.0
	20	10.2	11.5	13.1	14.4	16.2	18.4	19.1	21.5	24.5	24.2	27.4	31.3	29.8	33.8	38.7	35.8	40.6	46.6
	30	10.5	11.8	13.4	14.8	16.6	18.9	19.6	22.1	25.1	24.9	28.1	32.1	30.6	34.6	39.7	36.7	41.6	47.8
	40	10.6	11.9	13.5	14.9	16.8	19.1	19.8	22.3	25.4	25.2	28.4	32.5	31.0	35.1	40.2	37.2	42.2	48.6
	50	10.6	11.9	13.5	14.9	16.8	19.1	19.8	22.3	25.5	25.2	28.6	32.7	31.1	35.3	40.6	37.5	42.6	49.1

*To correct for wind, enter table with the brakes on speed minus one half the headwind or plus 1.5 times the tailwind. If ground speed is used for brakes on speed, ignore wind and enter table with sea level, 15°C.

ADVISORY INFORMATION

Recommended Brake Cooling Schedule

Adjusted Brake Energy Per Brake (Millions of Foot Pounds)

No Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	7.8	16.3	25.3	34.7	44.7	55.0	65.7	76.6	87.9
	MAX AUTO	7.5	15.4	23.6	32.4	41.8	51.8	62.5	74.1	86.5
	AUTOBRAKE 3	7.3	14.7	22.3	30.2	38.6	47.6	57.4	68.1	80.0
	AUTOBRAKE 2	7.0	13.8	20.5	27.4	34.8	42.7	51.5	61.3	72.4
	AUTOBRAKE 1	6.7	13.1	19.2	25.3	31.8	38.8	46.6	55.4	65.5

Two Engine Detent Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	7.0	14.6	22.8	31.4	40.5	49.9	59.7	69.8	80.0
	MAX AUTO	5.8	12.3	19.5	27.2	35.6	44.5	53.9	63.7	74.1
	AUTOBRAKE 3	4.3	9.2	14.7	20.7	27.2	34.4	42.0	50.2	59.0
	AUTOBRAKE 2	2.5	5.6	9.1	13.1	17.8	23.0	28.8	35.2	42.3
	AUTOBRAKE 1	1.8	3.8	6.1	8.8	11.9	15.5	19.6	24.4	29.8

Cooling Time (Minutes) - Category C Steel Brakes

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)								
		16 & BELOW	17	20	23	25	28	32	33 TO 48	49 & ABOVE
		BRAKE TEMPERATURE MONITOR SYSTEM INDICATION ON CDS								
		UP TO 2.4	2.6	3.1	3.5	3.9	4.4	4.9	5.0 TO 7.5	7.5 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE	1	2	3	4	5	6	CAUTION	FUSE PLUG MELT ZONE	
GROUND	REQUIRED	10	20	30	40	50	60			

Cooling Time (Minutes) - Category N Carbon Brakes

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)								
		16 & BELOW	17	19	20.9	23.5	26.9	29.4	30 TO 41	41 & ABOVE
		BRAKE TEMPERATURE MONITOR SYSTEM INDICATION ON CDS								
		UP TO 2.5	2.6	3	3.3	3.8	4.5	4.9	5.0 TO 7.1	7.1 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE	1	4	5	6	7	7.6	CAUTION	FUSE PLUG MELT ZONE	
GROUND	REQUIRED	6.7	16.0	24.1	34.2	45.9	53.3			

Observe maximum quick turnaround limit.

Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.

Add 1.0 million foot pounds per brake for each taxi mile.

When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after one hour. If overheat occurs after takeoff, extend gear soon for at least 7 minutes.

When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not approach gear or attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

Brake temperature monitor system (BTMS) indication on CDS systems page may be used 10 to 15 minutes after airplane has come to a complete stop or inflight with gear retracted to determine recommended cooling schedule.

Intentionally
Blank

Performance Inflight - QRH

Chapter PI-QRH

Engine Inoperative

Section 42

ENGINE INOP
Initial Max Continuous %N1

Based on .79M, A/C high and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)								
	25	27	29	31	33	35	37	39	41
20	96.8	96.6	96.3	96.1	95.9	95.4	95.0	94.7	93.9
15	97.4	97.2	96.9	96.8	96.6	96.2	95.7	95.5	94.8
10	98.0	97.8	97.5	97.4	97.4	96.9	96.5	96.3	95.7
5	98.3	98.6	98.3	98.1	98.1	97.7	97.3	97.1	96.6
0	97.5	98.7	99.2	99.0	98.9	98.5	98.2	98.0	97.5
-5	96.7	98.0	99.1	99.8	99.7	99.3	98.9	98.7	98.4
-10	96.0	97.2	98.4	99.6	100.5	100.2	99.8	99.6	99.4
-15	95.2	96.4	97.6	98.8	100.1	101.0	100.8	100.6	100.3
-20	94.4	95.6	96.8	98.0	99.3	100.5	101.1	100.8	100.6
-25	93.6	94.9	96.0	97.2	98.5	99.7	100.2	100.0	99.8
-30	92.8	94.1	95.2	96.4	97.7	98.8	99.4	99.2	99.0
-35	92.0	93.2	94.4	95.6	96.8	98.0	98.5	98.3	98.1
-40	91.2	92.4	93.5	94.7	96.0	97.1	97.6	97.4	97.2

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)								
	25	27	29	31	33	35	37	39	41
ENGINE ANTI-ICE	-1.2	-1.1	-1.0	-0.9	-0.8	-0.8	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE	-4.2	-4.4	-4.5	-4.7	-5.0	-4.8	-4.8	-4.8	-4.8

ENGINE INOP

**Max Continuous %N1
37000 FT to 29000 FT Pressure Altitudes**

37000 FT PRESS ALT													TAT (°C)	
KLAS	M	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	
160	.51	96.6	97.6	98.5	99.4	100.2	99.6	98.8	97.6	96.3	94.7	93.2	91.8	
200	.63	96.0	96.9	97.8	98.7	99.6	100.4	100.1	99.3	98.4	97.5	96.3	95.2	
240	.74	95.1	96.0	96.8	97.7	98.6	99.4	100.3	100.7	100.0	99.2	98.4	97.5	
280	.86	94.3	95.2	96.1	97.0	97.8	98.7	99.5	100.4	101.2	100.9	100.0	99.1	
35000 FT PRESS ALT													TAT (°C)	
KLAS	M	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	
160	.49	96.5	97.4	98.3	99.2	100.1	99.8	99.0	98.0	96.8	95.4	94.0	92.7	
200	.60	96.1	97.0	97.9	98.8	99.7	100.6	100.5	99.6	98.6	97.6	96.5	95.4	
240	.71	95.0	95.9	96.8	97.7	98.6	99.4	100.3	100.8	100.2	99.5	98.6	97.7	
280	.82	93.8	94.6	95.5	96.4	97.3	98.1	98.9	99.8	100.6	100.3	99.5	98.8	
33000 FT PRESS ALT													TAT (°C)	
KLAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	
160	.47	97.4	98.3	99.2	100.0	100.8	100.0	99.1	97.9	96.7	95.3	93.9	92.6	
200	.58	97.0	97.9	98.8	99.7	100.6	101.4	100.6	99.6	98.6	97.5	96.3	95.1	
240	.68	95.9	96.8	97.7	98.5	99.4	100.2	101.1	100.9	100.2	99.4	98.4	97.4	
280	.79	94.3	95.1	96.0	96.8	97.7	98.5	99.3	100.2	100.5	99.7	98.9	98.1	
320	.89	93.6	94.5	95.4	96.2	97.1	97.9	98.7	99.5	100.3	101.1	100.7	99.8	
31000 FT PRESS ALT													TAT (°C)	
KLAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	
160	.45	97.3	98.2	99.1	100.0	100.9	101.1	100.2	99.2	98.0	96.6	95.2	93.9	
200	.55	97.1	98.0	98.9	99.7	100.6	101.5	101.6	100.7	99.7	98.6	97.4	96.2	
240	.66	95.6	96.5	97.4	98.3	99.1	100.0	100.8	101.3	100.5	99.8	98.8	97.8	
280	.76	93.8	94.7	95.5	96.4	97.2	98.0	98.8	99.7	100.5	99.8	98.9	98.0	
320	.85	92.4	93.2	94.1	94.9	95.7	96.5	97.4	98.2	98.9	99.7	99.9	99.1	
29000 FT PRESS ALT													TAT (°C)	
KLAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	
160	.43	98.1	99.0	99.9	100.8	101.6	101.2	100.2	99.1	97.9	96.4	95.1	93.8	
200	.53	97.5	98.4	99.3	100.2	101.0	101.9	101.3	100.4	99.3	98.2	96.9	95.8	
240	.63	96.3	97.1	98.0	98.9	99.7	100.5	101.4	101.1	100.2	99.2	98.3	97.2	
280	.73	94.2	95.0	95.9	96.7	97.5	98.3	99.1	99.9	100.1	99.1	98.2	97.5	
320	.82	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.5	99.2	98.5	97.6	
360	.91	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.5	99.2	100.0	100.1	

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	29	31	33	35	37
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE ON	-4.1	-4.3	-4.5	-4.7	-4.7

ENGINE INOP

Max Continuous %N1

27000 FT to 20000 FT Pressure Altitudes

27000 FT PRESS ALT												TAT (°C)	
KIAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10
160	.41	98.0	98.8	99.7	100.6	101.4	102.2	101.2	100.2	99.0	97.8	96.4	95.1
200	.51	96.9	97.8	98.7	99.6	100.4	101.2	101.8	100.8	99.9	98.8	97.6	96.4
240	.60	95.6	96.5	97.4	98.2	99.1	99.9	100.7	101.3	100.4	99.4	98.5	97.5
280	.70	93.6	94.4	95.3	96.1	96.9	97.7	98.5	99.3	100.1	99.4	98.4	97.6
320	.79	91.6	92.4	93.2	94.0	94.8	95.6	96.4	97.2	98.0	98.7	98.6	97.8
360	.88	91.0	91.8	92.6	93.4	94.2	95.0	95.8	96.6	97.3	98.1	98.8	99.4
25000 FT PRESS ALT												TAT (°C)	
KIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
160	.39	98.8	99.7	100.5	101.4	102.2	102.4	101.4	100.3	99.1	97.7	96.5	95.2
200	.49	97.5	98.3	99.2	100.0	100.9	101.7	101.5	100.6	99.5	98.4	97.3	96.2
240	.58	95.7	96.5	97.4	98.2	99.0	99.9	100.7	100.5	99.5	98.6	97.6	96.7
280	.67	93.9	94.7	95.5	96.3	97.1	97.9	98.7	99.5	99.5	98.6	97.6	96.9
320	.76	91.7	92.6	93.4	94.2	95.0	95.8	96.5	97.3	98.0	98.6	97.8	97.2
360	.85	90.4	91.2	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.6	98.4	98.2
24000 FT PRESS ALT												TAT (°C)	
KIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
160	.38	98.6	99.5	100.4	101.2	102.1	102.9	101.9	100.8	99.6	98.4	97.1	95.8
200	.48	97.5	98.4	99.2	100.1	100.9	101.8	102.2	101.1	100.1	99.0	97.8	96.7
240	.57	95.9	96.8	97.6	98.5	99.3	100.1	100.9	101.2	100.2	99.2	98.2	97.3
280	.66	94.2	95.1	95.9	96.7	97.5	98.3	99.1	99.9	100.4	99.4	98.3	97.5
320	.75	92.1	93.0	93.8	94.6	95.4	96.2	96.9	97.7	98.5	99.2	98.6	97.8
360	.83	90.6	91.4	92.2	93.1	93.9	94.7	95.5	96.2	97.0	97.8	98.5	98.6
22000 FT PRESS ALT												TAT (°C)	
KIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
160	.37	99.1	100.0	100.9	101.7	102.5	102.8	101.8	100.7	99.5	98.2	97.0	95.8
200	.46	98.4	99.3	100.1	101.0	101.8	102.6	102.3	101.2	100.0	98.9	97.8	96.8
240	.55	97.2	98.1	98.9	99.7	100.5	101.3	102.1	101.6	100.5	99.4	98.5	97.5
280	.63	95.7	96.5	97.4	98.2	99.0	99.8	100.6	101.3	101.0	99.8	98.9	98.1
320	.72	93.9	94.7	95.5	96.3	97.1	97.9	98.6	99.4	100.1	100.2	99.3	98.6
360	.80	92.2	93.0	93.8	94.6	95.4	96.1	96.9	97.7	98.4	99.2	99.7	99.1
20000 FT PRESS ALT												TAT (°C)	
KIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
160	.35	98.7	99.5	100.4	101.2	102.0	102.8	102.5	101.5	100.4	99.2	98.0	96.8
200	.44	98.3	99.2	100.0	100.9	101.7	102.5	103.3	102.3	101.1	100.0	98.9	97.8
240	.53	97.5	98.4	99.2	100.0	100.8	101.7	102.5	103.1	101.8	100.5	99.5	98.6
280	.61	96.2	97.0	97.8	98.7	99.5	100.3	101.1	101.8	102.5	101.3	100.1	99.3
320	.69	94.7	95.5	96.3	97.1	97.9	98.7	99.5	100.2	101.0	101.7	100.9	99.9
360	.77	93.0	93.8	94.6	95.4	96.2	97.0	97.7	98.5	99.2	100.0	100.7	100.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	20	22	24	25	27
ENGINE ANTI-ICE ON	-0.9	-0.9	-1.0	-1.0	-1.0
ENGINE & WING ANTI-ICE ON	-3.6	-3.8	-3.8	-3.9	-4.0

ENGINE INOP

**Max Continuous %N1
18000 FT to 12000 FT Pressure Altitudes**

18000 FT PRESS ALT												TAT (°C)	
KLAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
160	.34	98.5	99.3	100.2	101.0	101.8	102.6	101.6	100.3	99.2	98.1	97.0	95.9
200	.42	98.7	99.6	100.4	101.2	102.0	102.8	103.1	101.7	100.4	99.3	98.3	97.3
240	.51	97.8	98.7	99.5	100.3	101.1	101.9	102.7	102.5	101.1	99.9	99.0	98.1
280	.59	96.3	97.1	97.9	98.7	99.5	100.3	101.0	101.8	101.6	100.5	99.6	98.8
320	.67	94.8	95.6	96.4	97.2	97.9	98.7	99.5	100.2	101.0	100.9	100.0	99.2
360	.75	93.0	93.8	94.6	95.3	96.1	96.9	97.6	98.4	99.1	99.9	100.2	99.6
16000 FT PRESS ALT												TAT (°C)	
KLAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
160	.33	97.1	98.0	98.8	99.6	100.4	101.2	101.6	100.3	99.1	98.1	97.1	96.1
200	.41	98.0	98.8	99.6	100.4	101.2	102.0	102.8	102.5	101.3	100.2	99.3	98.3
240	.49	97.1	97.9	98.7	99.5	100.3	101.1	101.9	102.7	101.8	100.5	99.6	98.7
280	.57	95.6	96.4	97.2	98.0	98.8	99.6	100.3	101.1	101.8	100.9	99.8	99.0
320	.64	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.4	100.2	100.9	100.2	99.4
360	.72	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.4	99.2	99.9	99.6
14000 FT PRESS ALT												TAT (°C)	
KLAS	M	-25	-20	-15	-10	-5	0	5	10	15	20	25	30
160	.31	96.6	97.4	98.2	99.0	99.8	100.6	100.4	99.1	98.0	97.1	96.2	95.3
200	.39	97.1	97.9	98.7	99.5	100.3	101.1	101.8	101.5	101.0	100.1	99.3	98.4
240	.47	96.6	97.4	98.2	99.0	99.8	100.6	101.3	101.8	101.1	100.3	99.5	98.7
280	.54	95.5	96.3	97.1	97.8	98.6	99.4	100.1	100.9	101.0	100.1	99.2	98.5
320	.62	94.1	94.9	95.7	96.5	97.2	98.0	98.7	99.5	100.2	100.3	99.5	98.8
360	.69	92.2	93.1	93.9	94.7	95.5	96.3	97.0	97.8	98.6	99.3	99.6	99.0
12000 FT PRESS ALT												TAT (°C)	
KLAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35
160	.30	96.3	97.0	97.8	98.6	99.4	100.1	99.3	98.1	97.1	96.3	95.4	94.5
200	.38	97.1	97.9	98.7	99.5	100.3	101.0	101.5	100.8	99.8	99.0	98.2	97.3
240	.45	96.5	97.3	98.0	98.8	99.6	100.3	101.1	101.0	100.1	99.4	98.6	97.9
280	.52	95.5	96.3	97.0	97.8	98.6	99.3	100.0	100.8	100.3	99.4	98.6	98.0
320	.60	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.4	100.2	99.7	98.9	98.2
360	.67	92.3	93.2	94.0	94.8	95.6	96.4	97.1	97.9	98.7	99.4	99.1	98.5

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)			
	12	14	16	18
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.9	-0.9
ENGINE & WING ANTI-ICE ON	-3.2	-3.4	-3.4	-3.5

ENGINE INOP

Max Continuous %N1

10000 FT to 1000 FT Pressure Altitudes

10000 FT PRESS ALT		TAT (°C)											
KIAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35
160	.29	95.2	96.0	96.8	97.6	98.3	99.1	99.8	98.6	97.4	96.6	95.8	94.9
200	.36	96.0	96.7	97.5	98.3	99.0	99.8	100.5	100.5	99.4	98.5	97.8	97.0
240	.43	95.6	96.4	97.2	97.9	98.7	99.4	100.2	100.9	100.1	99.2	98.4	97.7
280	.51	94.5	95.3	96.1	96.9	97.6	98.4	99.1	99.9	100.4	99.5	98.7	98.0
320	.58	93.0	93.9	94.7	95.5	96.2	97.0	97.8	98.6	99.3	99.7	99.0	98.2
360	.65	91.6	92.4	93.2	94.0	94.8	95.6	96.4	97.2	98.0	98.7	99.1	98.5
5000 FT PRESS ALT		TAT (°C)											
KIAS	M	-10	-5	0	5	10	15	20	25	30	35	40	45
160	.26	94.9	95.7	96.4	97.2	98.0	98.8	99.2	98.3	97.4	96.6	95.9	95.1
200	.33	94.7	95.5	96.3	97.1	97.8	98.6	99.4	98.9	98.0	97.3	96.6	95.8
240	.40	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.5	98.7	97.9	97.2	96.5
280	.46	93.3	94.1	94.9	95.7	96.5	97.3	98.1	98.8	98.9	98.2	97.5	96.8
320	.53	92.5	93.3	94.1	94.9	95.7	96.5	97.2	98.0	98.7	98.4	97.7	97.1
360	.59	91.5	92.3	93.1	93.9	94.7	95.5	96.2	97.0	97.8	98.5	98.0	97.3
3000 FT PRESS ALT		TAT (°C)											
KIAS	M	-5	0	5	10	15	20	25	30	35	40	45	50
160	.26	94.8	95.6	96.4	97.2	98.0	98.7	98.8	97.9	97.1	96.4	95.6	94.8
200	.32	94.5	95.3	96.1	96.9	97.6	98.4	99.2	98.3	97.5	96.8	96.1	95.3
240	.38	94.1	94.9	95.6	96.4	97.2	98.0	98.7	98.8	98.0	97.2	96.6	95.9
280	.45	93.2	94.0	94.8	95.6	96.4	97.2	97.9	98.7	98.3	97.5	96.9	96.2
320	.51	92.5	93.3	94.1	94.9	95.7	96.4	97.2	98.0	98.5	97.8	97.1	96.5
360	.57	91.6	92.4	93.2	94.0	94.7	95.5	96.3	97.1	97.8	98.1	97.4	96.8
1000 FT PRESS ALT		TAT (°C)											
KIAS	M	-5	0	5	10	15	20	25	30	35	40	45	50
160	.25	93.9	94.7	95.4	96.2	97.0	97.8	98.5	98.2	97.4	96.7	96.0	95.2
200	.31	93.5	94.3	95.1	95.9	96.7	97.4	98.2	98.5	97.8	97.0	96.3	95.6
240	.37	93.0	93.8	94.6	95.4	96.1	96.9	97.7	98.4	98.1	97.3	96.6	95.9
280	.43	92.3	93.2	93.9	94.7	95.5	96.3	97.1	97.8	98.3	97.6	96.9	96.2
320	.49	91.6	92.4	93.2	94.0	94.8	95.6	96.3	97.1	97.9	97.9	97.2	96.5
360	.55	90.7	91.5	92.3	93.1	93.9	94.7	95.4	96.2	96.9	97.7	97.3	96.6

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)			
	1	3	5	10
ENGINE ANTI-ICE ON	-0.6	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE ON	-2.9	-3.0	-2.7	-3.2

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	82	271	18500	17300	15900
80	77	263	20200	19000	17700
75	72	255	21600	20600	19400
70	67	247	23100	22200	21100
65	62	238	24700	23800	22800
60	57	229	26800	25800	24700
55	53	219	29100	28100	27000
50	48	209	31200	30400	29400
45	43	199	33300	32600	31700
40	38	187	35600	34900	34000

Includes APU fuel burn.

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown/LRC Cruise Range Capability

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
138	128	120	112	106	100	95	90	86	82	78
275	256	239	225	212	200	190	180	172	164	157
413	384	359	337	317	300	284	270	258	246	235
551	512	479	449	423	400	379	360	344	328	314
689	640	598	562	529	500	474	451	429	410	392
826	768	718	674	635	600	569	541	515	492	471
964	896	838	786	741	700	664	631	601	574	549
1102	1025	957	898	846	800	758	721	687	656	628
1240	1153	1077	1011	952	900	853	811	773	738	706
1377	1281	1197	1123	1058	1000	948	901	859	820	785
1515	1409	1317	1235	1164	1100	1043	991	945	902	863
1653	1537	1436	1348	1270	1200	1138	1081	1030	984	942
1792	1666	1556	1460	1375	1300	1232	1171	1116	1066	1020
1930	1794	1676	1573	1481	1400	1327	1261	1202	1148	1098
2068	1922	1796	1685	1587	1500	1422	1351	1288	1230	1177
2207	2051	1916	1798	1693	1600	1517	1441	1373	1312	1255
2345	2180	2036	1910	1799	1700	1611	1531	1459	1393	1333
2484	2309	2156	2023	1905	1800	1706	1621	1545	1475	1411

Driftdown/Cruise Fuel and Time

AIR DIST (NM)	FUEL REQUIRED (1000 KG)										TIME (HR:MIN)
	WEIGHT AT START OF DRIFTDOWN (1000 KG)										
	40	45	50	55	60	65	70	75	80	85	
100	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0:16
200	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.3	0:33
300	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	0:49
400	1.6	1.8	1.9	2.0	2.2	2.3	2.5	2.6	2.8	2.9	1:06
500	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.3	3.5	3.7	1:22
600	2.4	2.7	2.9	3.1	3.3	3.6	3.8	4.0	4.3	4.5	1:39
700	2.8	3.1	3.4	3.6	3.9	4.2	4.5	4.7	5.0	5.3	1:55
800	3.2	3.6	3.9	4.2	4.5	4.8	5.1	5.4	5.7	6.1	2:11
900	3.6	4.0	4.3	4.7	5.0	5.4	5.7	6.1	6.4	6.8	2:28
1000	4.0	4.4	4.8	5.2	5.6	6.0	6.4	6.7	7.1	7.6	2:44
1100	4.4	4.8	5.3	5.7	6.1	6.6	7.0	7.4	7.9	8.3	3:01
1200	4.8	5.3	5.7	6.2	6.7	7.1	7.6	8.1	8.6	9.0	3:17
1300	5.2	5.7	6.2	6.7	7.2	7.7	8.2	8.7	9.2	9.8	3:34
1400	5.5	6.1	6.6	7.2	7.7	8.3	8.8	9.4	9.9	10.5	3:51
1500	5.9	6.5	7.1	7.7	8.3	8.9	9.4	10.0	10.6	11.2	4:07
1600	6.3	6.9	7.5	8.2	8.8	9.4	10.0	10.7	11.3	12.0	4:24
1700	6.6	7.3	8.0	8.6	9.3	10.0	10.6	11.3	12.0	12.7	4:41
1800	7.0	7.7	8.4	9.1	9.8	10.5	11.2	11.9	12.6	13.4	4:57

Includes APU fuel burn.

Driftdown at optimum driftdown speed and cruise at long range cruise speed.

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Altitude Capability

100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	15200	12600	9900
80	17200	15300	12500
75	19200	17400	15000
70	20900	19700	17300
65	22500	21300	19800
60	24100	23000	21600
55	26300	24800	23500
50	29000	27700	25800
45	31400	30500	29200
40	33800	33000	31800

With engine anti-ice on, decrease altitude capability by 1200 ft.

With engine and wing anti-ice on, decrease altitude capability by 5500 ft.

ENGINE INOP

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		10	15	17	19	21	23	25	27	29	31
85	%N1	91.8	95.5	97.9							
	MACH	.561	.600	.616							
	KIAS	311	303	300							
	FF/ENG	3067	3033	3052							
80	%N1	90.1	94.0	95.9	98.5						
	MACH	.545	.590	.603	.621						
	KIAS	302	299	294	291						
	FF/ENG	2875	2870	2846	2886						
75	%N1	88.4	92.5	94.0	96.1						
	MACH	.528	.579	.593	.607						
	KIAS	293	293	288	284						
	FF/ENG	2684	2709	2674	2662						
70	%N1	86.5	90.7	92.3	94.0	96.2					
	MACH	.510	.562	.582	.595	.610					
	KIAS	282	284	283	278	274					
	FF/ENG	2494	2518	2520	2481	2487					
65	%N1	84.5	88.7	90.4	92.2	93.9	96.4				
	MACH	.491	.542	.563	.584	.596	.612				
	KIAS	271	274	274	273	268	265				
	FF/ENG	2306	2327	2330	2330	2295	2317				
60	%N1	82.3	86.5	88.3	90.0	91.9	93.7	96.4			
	MACH	.471	.521	.543	.564	.585	.597	.614			
	KIAS	261	263	263	263	263	258	254			
	FF/ENG	2124	2137	2139	2140	2143	2114	2146			
55	%N1	80.2	84.2	85.9	87.7	89.5	91.4	93.3	96.2		
	MACH	.453	.498	.520	.541	.563	.585	.597	.614		
	KIAS	250	251	252	252	253	252	247	244		
	FF/ENG	1954	1948	1950	1950	1953	1958	1938	1971		
50	%N1	77.8	81.6	83.4	85.2	87.0	88.7	90.7	92.7	95.7	
	MACH	.434	.475	.495	.516	.538	.561	.583	.596	.613	
	KIAS	240	239	239	240	241	241	241	236	233	
	FF/ENG	1791	1764	1762	1762	1764	1767	1777	1765	1793	
45	%N1	75.5	79.1	80.6	82.3	84.1	85.9	87.7	89.7	91.8	94.8
	MACH	.415	.452	.469	.489	.511	.533	.556	.578	.593	.610
	KIAS	229	227	227	227	228	229	229	229	225	222
	FF/ENG	1636	1594	1582	1575	1577	1580	1586	1600	1593	1613
40	%N1	73.0	76.2	77.8	79.4	81.0	82.8	84.6	86.4	88.3	90.7
	MACH	.395	.429	.445	.462	.480	.502	.525	.548	.571	.589
	KIAS	218	215	215	214	214	215	216	216	216	214
	FF/ENG	1485	1434	1416	1402	1392	1394	1400	1410	1421	1424

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
298	272	249	230	214	200	190	180	172	164	158
600	547	501	462	429	400	379	361	344	328	315
903	823	753	694	644	600	570	542	517	494	473
1209	1100	1005	926	859	800	759	721	687	657	630
1516	1379	1259	1159	1075	1000	949	902	859	820	786
1825	1659	1513	1393	1290	1200	1139	1082	1031	984	943
2137	1940	1768	1626	1506	1400	1328	1262	1202	1147	1099
2450	2222	2024	1860	1722	1600	1518	1442	1373	1311	1256
2766	2507	2281	2095	1938	1800	1707	1622	1544	1474	1412
3083	2792	2539	2331	2155	2000	1896	1801	1715	1637	1568

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		18		22		26	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.4	0:43	1.2	0:41	1.1	0:39	1.0	0:38	0.9	0:37
400	2.8	1:23	2.6	1:19	2.4	1:14	2.2	1:11	2.1	1:09
600	4.3	2:04	3.9	1:57	3.6	1:50	3.4	1:45	3.2	1:42
800	5.7	2:46	5.2	2:36	4.9	2:26	4.5	2:19	4.4	2:14
1000	7.1	3:28	6.6	3:15	6.1	3:03	5.7	2:53	5.5	2:47
1200	8.5	4:10	7.9	3:55	7.3	3:40	6.8	3:28	6.6	3:21
1400	9.8	4:53	9.1	4:36	8.5	4:18	8.0	4:02	7.7	3:54
1600	11.2	5:36	10.4	5:16	9.7	4:55	9.1	4:38	8.7	4:28
1800	12.5	6:20	11.7	5:58	10.9	5:34	10.2	5:13	9.8	5:02
2000	13.9	7:05	12.9	6:39	12.0	6:13	11.3	5:49	10.8	5:36

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)									
	40	45	50	55	60	65	70	75	80	
1	-0.1	-0.1	-0.1	0.0	0.0	0.1	0.1	0.2	0.3	
2	-0.3	-0.2	-0.1	-0.1	0.0	0.2	0.3	0.6	0.8	
3	-0.4	-0.3	-0.2	-0.1	0.0	0.3	0.5	0.9	1.2	
4	-0.6	-0.4	-0.3	-0.1	0.0	0.3	0.7	1.2	1.6	
5	-0.7	-0.5	-0.4	-0.2	0.0	0.4	0.9	1.4	2.0	
6	-0.8	-0.6	-0.4	-0.2	0.0	0.5	1.1	1.7	2.4	
7	-1.0	-0.8	-0.5	-0.3	0.0	0.6	1.2	2.0	2.8	
8	-1.1	-0.9	-0.6	-0.3	0.0	0.6	1.4	2.2	3.2	
9	-1.3	-1.0	-0.7	-0.3	0.0	0.7	1.5	2.4	3.5	
10	-1.4	-1.1	-0.7	-0.4	0.0	0.7	1.6	2.6	3.8	
11	-1.6	-1.2	-0.8	-0.4	0.0	0.8	1.7	2.8	4.1	
12	-1.7	-1.3	-0.9	-0.4	0.0	0.8	1.9	3.0	4.4	
13	-1.9	-1.4	-0.9	-0.5	0.0	0.9	2.0	3.2	4.7	
14	-2.0	-1.5	-1.0	-0.5	0.0	0.9	2.0	3.4	4.9	

Includes APU fuel burn.



ENGINE INOP
MAX CONTINUOUS THRUST

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)							
		1500	5000	10000	15000	20000	25000	30000	35000
85	%N1	81.1	84.1	88.3	92.8				
	KIAS	250	251	252	253				
	FF/ENG	2740	2730	2750	2800				
80	%N1	79.5	82.4	86.5	91.0	98.3			
	KIAS	242	243	244	245	247			
	FF/ENG	2580	2570	2570	2610	2740			
75	%N1	77.8	80.5	84.7	89.1	95.0			
	KIAS	235	236	236	238	239			
	FF/ENG	2420	2400	2400	2420	2490			
70	%N1	76.0	78.6	82.8	87.1	92.1			
	KIAS	227	227	228	229	231			
	FF/ENG	2260	2240	2230	2250	2270			
65	%N1	74.0	76.7	80.8	85.0	89.7	97.7		
	KIAS	219	219	220	221	222	224		
	FF/ENG	2100	2090	2070	2070	2080	2230		
60	%N1	71.7	74.6	78.5	82.8	87.4	93.7		
	KIAS	210	210	211	212	213	214		
	FF/ENG	1950	1930	1910	1910	1910	1970		
55	%N1	69.4	72.3	76.3	80.5	84.9	90.0		
	KIAS	200	201	202	203	204	205		
	FF/ENG	1800	1770	1750	1740	1730	1760		
50	%N1	66.9	69.7	73.8	77.8	82.3	87.0	94.9	
	KIAS	192	192	192	193	194	195	196	
	FF/ENG	1650	1620	1600	1580	1570	1570	1680	
45	%N1	64.2	66.9	70.9	75.0	79.4	84.0	89.6	
	KIAS	185	185	185	185	185	185	186	
	FF/ENG	1500	1470	1440	1420	1400	1400	1450	
40	%N1	61.1	64.0	67.8	72.0	76.2	80.7	85.4	94.0
	KIAS	178	178	178	178	178	178	178	178
	FF/ENG	1350	1330	1300	1270	1250	1240	1260	1360

This table includes 5% additional fuel for holding in a racetrack pattern.

ENGINE INOP

ADVISORY INFORMATION

Gear Down Landing Rate of Climb Available Flaps 15

TAT (°C)	RATE OF CLIMB (FT/MIN)					
	PRESSURE ALTITUDE (FT)					
	-2000	0	2000	4000	6000	8000
52	-80	-140				
50	-50	-110	-220			
48	-20	-90	-190			
46	10	-60	-160	-270		
44	40	-30	-140	-250		
42	70	0	-110	-220	-340	
40	100	30	-80	-190	-310	
38	120	60	-50	-160	-290	-430
36	140	90	-20	-140	-260	-400
34	140	120	0	-120	-240	-380
32	140	130	20	-100	-220	-360
30	140	130	40	-80	-210	-340
20	160	140	60	-50	-160	-280
10	170	150	60	-50	-160	-280
0	170	160	70	-50	-160	-280
-20	190	170	80	-40	-160	-280
-40	200	180	80	-40	-170	-290

Rate of climb capability shown is valid for 60000 kg, gear down at VREF15+5.
Decrease rate of climb 130 ft/min per 5000 kg greater than 60000 kg.
Increase rate of climb 160 ft/min per 5000 kg less than 60000 kg.

Flaps 30

TAT (°C)	RATE OF CLIMB (FT/MIN)					
	PRESSURE ALTITUDE (FT)					
	-2000	0	2000	4000	6000	8000
52	-260	-320				
50	-230	-300	-400			
48	-200	-270	-380			
46	-180	-250	-350	-460		
44	-150	-220	-330	-430		
42	-120	-190	-300	-410	-530	
40	-100	-170	-280	-390	-500	
38	-70	-140	-250	-360	-480	-620
36	-60	-110	-220	-340	-460	-600
34	-50	-80	-200	-320	-440	-580
32	-50	-70	-180	-300	-420	-560
30	-50	-60	-160	-280	-410	-540
20	-40	-60	-150	-260	-370	-490
10	-40	-50	-140	-260	-370	-480
0	-30	-50	-140	-260	-370	-490
-20	-30	-40	-140	-260	-380	-500
-40	-20	-40	-140	-270	-390	-520

Rate of climb capability shown is valid for 60000 kg, gear down at VREF30+5.
Decrease rate of climb 130 ft/min per 5000 kg greater than 60000 kg.
Increase rate of climb 170 ft/min per 5000 kg less than 60000 kg.



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Chapter PI-QRH

Gear Down

Section 43

GEAR DOWN

Long Range Cruise Altitude Capability
Max Cruise Thrust, 100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	15600	12500	9400
80	18400	15500	12600
75	21100	18500	15700
70	23600	21400	18600
65	26100	24400	21800
60	28600	27100	25300
55	30800	29600	28100
50	32900	31900	30700
45	35100	34100	33000
40	37500	36500	35400

GEAR DOWN

Long Range Cruise Enroute Fuel and Time

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
324	290	260	236	217	200	188	178	168	160	153
654	583	523	474	435	400	377	357	338	321	307
989	880	787	713	653	600	566	535	507	483	461
1329	1181	1054	953	871	800	754	713	676	643	614
1674	1484	1322	1194	1090	1000	943	891	844	803	766
2024	1791	1593	1436	1310	1200	1131	1069	1013	962	918
2381	2103	1865	1680	1530	1400	1320	1247	1181	1122	1070
2743	2417	2140	1924	1751	1600	1508	1424	1348	1280	1221
3113	2737	2418	2171	1972	1800	1695	1600	1514	1438	1371

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	2.4	0:49	2.2	0:47	1.9	0:44	1.7	0:42	1.6	0:41
400	4.9	1:36	4.5	1:31	4.0	1:25	3.7	1:20	3.5	1:17
600	7.4	2:25	6.8	2:17	6.1	2:06	5.7	1:59	5.4	1:54
800	9.8	3:14	9.1	3:03	8.1	2:48	7.6	2:38	7.2	2:31
1000	12.1	4:04	11.3	3:50	10.1	3:30	9.5	3:18	9.0	3:08
1200	14.4	4:56	13.5	4:39	12.1	4:14	11.3	3:58	10.7	3:46
1400	16.7	5:49	15.6	5:28	14.0	4:58	13.1	4:40	12.4	4:24
1600	18.9	6:43	17.7	6:18	15.9	5:44	14.9	5:22	14.1	5:03
1800	21.1	7:38	19.7	7:10	17.7	6:30	16.6	6:05	15.7	5:43

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	40	50	60	70	80
2	-0.3	-0.2	0.0	0.3	0.7
4	-0.7	-0.3	0.0	0.6	1.3
6	-1.0	-0.5	0.0	0.9	2.0
8	-1.3	-0.7	0.0	1.2	2.6
10	-1.7	-0.8	0.0	1.4	3.2
12	-2.0	-1.0	0.0	1.6	3.7
14	-2.4	-1.2	0.0	1.8	4.2
16	-2.7	-1.3	0.0	2.0	4.6
18	-3.0	-1.5	0.0	2.2	5.0
20	-3.4	-1.7	0.0	2.4	5.3
22	-3.7	-1.8	0.0	2.5	5.6

GEAR DOWN

Descent

VREF40 + 70 KIAS

PRESSURE ALTITUDE (FT)	TIME (MIN)	FUEL (KG)	DISTANCE (NM)
41000	21	280	91
39000	20	270	86
37000	19	270	81
35000	19	260	77
33000	18	260	72
31000	17	250	68
29000	17	250	64
27000	16	240	60
25000	15	230	56
23000	14	230	52
21000	13	220	48
19000	13	210	44
17000	12	200	40
15000	11	190	36
10000	8	170	26
5000	6	140	16
1500	4	110	9

Allowances for a straight-in approach are included.



GEAR DOWN

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)							
		1500	5000	10000	15000	20000	25000	30000	35000
85	%N1	75.8	78.5	82.7	87.0	92.0			
	KIAS	230	230	230	230	230			
	FF/ENG	2240	2230	2220	2240	2260			
80	%N1	74.2	77.0	81.1	85.4	90.0			
	KIAS	225	225	225	225	225			
	FF/ENG	2120	2110	2100	2100	2110			
75	%N1	72.5	75.4	79.4	83.7	88.3	94.8		
	KIAS	220	220	220	220	220	220		
	FF/ENG	2000	1990	1970	1970	1970	2050		
70	%N1	70.8	73.7	77.6	81.9	86.4	91.8		
	KIAS	216	216	216	216	216	216		
	FF/ENG	1890	1870	1850	1840	1840	1870		
65	%N1	69.0	71.9	75.9	80.1	84.5	89.3		
	KIAS	211	211	211	211	211	211		
	FF/ENG	1770	1750	1730	1720	1710	1730		
60	%N1	67.1	69.8	74.0	78.0	82.5	87.1	94.3	
	KIAS	204	204	204	204	204	204	204	
	FF/ENG	1660	1630	1610	1600	1580	1590	1670	
55	%N1	65.1	67.8	71.9	75.9	80.3	84.8	90.4	
	KIAS	198	198	198	198	198	198	198	
	FF/ENG	1540	1520	1490	1480	1460	1460	1500	
50	%N1	62.8	65.6	69.6	73.7	78.0	82.4	87.1	
	KIAS	192	192	192	192	192	192	192	
	FF/ENG	1430	1400	1380	1360	1330	1330	1350	
45	%N1	60.3	63.3	67.1	71.4	75.5	79.9	84.5	91.5
	KIAS	185	185	185	185	185	185	185	185
	FF/ENG	1310	1290	1270	1250	1220	1210	1220	1270
40	%N1	57.9	60.6	64.6	68.7	72.9	77.3	81.7	86.8
	KIAS	178	178	178	178	178	178	178	178
	FF/ENG	1200	1180	1160	1130	1110	1090	1100	1110

This table includes 5% additional fuel for holding in a racetrack pattern.

Intentionally
Blank



Performance Inflight - QRH

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Gear Down, Engine Inop

Section 44

GEAR DOWN

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	80	227	1700		
80	76	223	4000	2300	200
75	71	218	6300	4900	2800
70	66	213	8600	7300	5300
65	62	208	10900	9800	8000
60	57	202	13200	12300	10900
55	52	196	15600	14800	13900
50	47	190	18100	17300	16500
45	43	183	20600	19800	18900
40	38	176	23100	22300	21400

Includes APU fuel burn.

Long Range Cruise Altitude Capability

100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
75	1500		
70	4500	2500	
65	7500	5900	3400
60	10600	9200	6900
55	13300	12300	10600
50	16200	15400	14500
45	19300	18300	17500
40	22200	21400	20500

GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		5	7	9	11	13	15	17	19	21	23
70	%N1	94.8									
	MACH	.389									
	KIAS	235									
	FF/ENG	3774									
65	%N1	92.6	94.3	96.9							
	MACH	.376	.389	.402							
	KIAS	228	227	226							
	FF/ENG	3477	3485	3527							
60	%N1	90.2	91.9	93.7	96.3						
	MACH	.364	.375	.388	.402						
	KIAS	220	219	218	218						
	FF/ENG	3192	3191	3198	3240						
55	%N1	87.8	89.3	91.0	92.8	95.4					
	MACH	.351	.362	.374	.387	.400					
	KIAS	212	211	210	209	209					
	FF/ENG	2924	2909	2906	2913	2951					
50	%N1	85.3	86.7	88.2	89.9	91.7	94.2	98.2			
	MACH	.338	.348	.359	.371	.384	.398	.412			
	KIAS	204	203	202	201	200	199	198			
	FF/ENG	2672	2647	2630	2626	2633	2657	2737			
45	%N1	82.7	84.0	85.4	86.9	88.6	90.4	92.7	96.6		
	MACH	.325	.334	.344	.355	.367	.380	.393	.408		
	KIAS	196	195	193	192	191	190	189	189		
	FF/ENG	2432	2400	2374	2356	2351	2352	2359	2417		
40	%N1	79.8	81.1	82.5	83.9	85.4	87.0	88.8	90.8	94.1	98.4
	MACH	.311	.320	.329	.339	.349	.361	.374	.387	.402	.418
	KIAS	188	186	184	183	182	181	180	179	179	178
	FF/ENG	2206	2166	2133	2107	2088	2076	2069	2065	2101	2201



GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time
Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
172	151	134	120	109	100	93	88	83	78	75
352	308	270	242	219	200	187	175	165	156	148
533	465	408	364	330	300	280	262	246	232	220
716	623	545	486	440	400	373	349	328	309	293
900	783	684	609	551	500	466	436	409	385	365
1086	943	823	733	661	600	559	523	490	462	438
1273	1105	964	856	772	700	652	610	572	538	510
1462	1267	1103	980	883	800	745	696	652	614	581
1653	1431	1245	1104	994	900	838	782	733	690	653
1845	1595	1386	1228	1105	1000	931	868	813	765	724

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)					
	6		10		14	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
100	1.3	0:27	1.1	0:26	1.0	0:26
200	2.6	0:53	2.4	0:50	2.3	0:48
300	3.9	1:18	3.7	1:15	3.6	1:11
400	5.2	1:44	4.9	1:39	4.8	1:35
500	6.5	2:10	6.1	2:04	6.0	1:58
600	7.8	2:37	7.3	2:29	7.1	2:22
700	9.1	3:03	8.5	2:55	8.3	2:46
800	10.3	3:30	9.7	3:20	9.4	3:10
900	11.6	3:58	10.9	3:46	10.5	3:35
1000	12.8	4:25	12.0	4:12	11.6	3:59

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	40	50	60	70	80
1	-0.2	-0.1	0.0	0.1	0.3
2	-0.3	-0.2	0.0	0.3	0.6
3	-0.5	-0.3	0.0	0.5	1.0
4	-0.6	-0.3	0.0	0.7	1.3
5	-0.8	-0.4	0.0	0.9	1.7
6	-1.0	-0.5	0.0	1.0	2.0
7	-1.1	-0.6	0.0	1.2	2.4
8	-1.3	-0.7	0.0	1.4	2.7
9	-1.5	-0.7	0.0	1.6	3.1
10	-1.6	-0.8	0.0	1.8	3.5
11	-1.8	-0.9	0.0	1.9	3.8
12	-1.9	-1.0	0.0	2.1	4.2
13	-2.1	-1.1	0.0	2.3	4.5
14	-2.3	-1.1	0.0	2.5	4.9

Includes APU fuel burn.

GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

**Holding
Flaps Up**

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)			
		1500	5000	10000	15000
80	%N1	93.4			
	KIAS	225			
	FF/ENG	4140			
75	%N1	91.4	94.7		
	KIAS	220	220		
	FF/ENG	3870	3910		
70	%N1	89.4	92.6		
	KIAS	216	216		
	FF/ENG	3610	3640		
65	%N1	87.4	90.5	95.9	
	KIAS	211	211	211	
	FF/ENG	3360	3380	3460	
60	%N1	85.2	88.2	92.9	
	KIAS	204	204	204	
	FF/ENG	3110	3110	3150	
55	%N1	82.9	85.9	90.4	97.2
	KIAS	198	198	198	198
	FF/ENG	2860	2860	2880	3010
50	%N1	80.4	83.4	87.7	92.8
	KIAS	192	192	192	192
	FF/ENG	2630	2620	2620	2670
45	%N1	77.8	80.7	85.0	89.6
	KIAS	185	185	185	185
	FF/ENG	2400	2380	2380	2400
40	%N1	75.1	77.8	82.1	86.5
	KIAS	178	178	178	178
	FF/ENG	2180	2160	2140	2140

This table includes 5% additional fuel for holding in a racetrack pattern.

Performance Inflight - QRH

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Text

Section 45

Introduction

This chapter contains information to supplement performance data from the Flight Management Computer (FMC). In addition, sufficient inflight data is provided to complete a flight with the FMC inoperative. In the event of conflict between data presented in this chapter and that contained in the approved Airplane Flight Manual, the Flight Manual shall always take precedence.

General

Flight with Unreliable Airspeed / Turbulent Air Penetration

Pitch attitude and average %N1 information is provided for use in all phases of flight in the event of unreliable airspeed/Mach indications resulting from blocking or freezing of the pitot system. Loss of radome or turbulent air may also cause unreliable airspeed/Mach indications. The cruise table in this section may also be used for turbulent air penetration.

Pitch attitude is shown in bold type for emphasis since altitude and/or vertical speed indications may also be unreliable.

Max Climb %N1

This table shows Max Climb %N1 for a 280/.78 climb speed schedule, normal engine bleed for packs on or off and anti-ice off. Enter the table with airport pressure altitude and TAT and read %N1. %N1 adjustments are shown for anti-ice operation.

Go-around %N1

To find Max Go-around %N1 based on normal engine bleed for packs on (AUTO) and anti-ice on or off, enter the Go-around %N1 table with airport pressure altitude and reported OAT or TAT and read %N1. For packs OFF or HIGH operation, apply the %N1 adjustment shown below the table.

VREF

This table contains flaps 40, 30 and 15 reference speeds for a given weight.

Advisory Information

Normal Configuration Landing Distance

The normal configuration distance tables are provided as advisory information to help determine the actual landing distance performance of the airplane for different runway surface conditions and brake configurations.

If the surface is affected by water, snow or ice, and the braking action is reported as "good", conditions should not be expected to be as good as on clean, dry runways. The value "good" is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when landing. The performance level used to calculate the "good" data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate "poor" data reflects runways covered with wet ice.

Dry runway landing performance is shown for max manual braking configuration and autobrake settings max, 3, 2, and 1. The autobrake performance may be used to assist in the selection of the most desirable autobrake setting for a given field length. Selection of an autobrake setting results in a constant rate of deceleration. Maximum effort manual braking should achieve shorter landing distance than the max autobrake setting. The reference landing distance is a reference distance from 50 ft above the threshold to stop based on a reference landing weight and normal approach speed for the selected landing flap at sea level, zero wind, zero slope, and two engine detent reverse thrust. Subsequent columns provide adjustments for off-reference landing weight, altitude, wind, slope, temperature, speed, and reverse thrust. Each adjustment is independently added to the reference landing distance.

Flaps 30 and 40 landing distances and adjustments are provided for dry runways as well as runways with good, medium, and poor reported braking actions, which are commonly referred to as slippery runway conditions. All landing distances (reference distances plus adjustments) are 115% of the actual landing distance.

Non-normal Configuration Landing Distance

Advisory information is provided to support non-normal configurations that affect the landing performance of the airplane. Landing distances and adjustments are provided for dry runways and runways with good, medium, and poor reported braking action.

Enter the table with the applicable non-normal configuration and read the normal approach speed. The reference landing distance is a reference distance from 50 ft above the threshold to stop based on a reference landing weight and speed at sea level, zero wind, and zero slope. Subsequent columns provide adjustments for off-reference landing weight, altitude, wind, slope, and speed conditions. Each adjustment is independently added to the reference landing distance. Landing distance includes the effect of reverse thrust.

Recommended Brake Cooling Schedule

Advisory information is provided to assist in avoiding the problems associated with hot brakes. For normal operation, most landings are at weights below the AFM quick turnaround limit weight.

Use of the recommended cooling schedule will help avoid brake overheat and fuse plug problems that could result from repeated landings at short time intervals or a rejected takeoff.

Enter the Recommended Brake Cooling Schedule table with the airplane weight and brakes on speed, adjusted for wind at the appropriate temperature and altitude condition. Instructions for applying wind adjustments are included below the table. Linear interpolation may be used to obtain intermediate values. The resulting number is the reference brake energy per brake in millions of foot-pounds, and represents the amount of energy absorbed by each brake during a rejected takeoff. Notes providing adjustments for wind are included below the table.

To determine the energy per brake absorbed during landing, enter the appropriate Adjusted Brake Energy Per Brake table (No Reverse Thrust or 2 Engine Reverse) with the reference brake energy per brake and the type of braking used during landing (Max Manual, Max Auto, or Autobrake). The resulting number is the adjusted brake energy per brake and represents the energy absorbed in each brake during the landing.

The recommended cooling time is found in the appropriate (steel or carbon brakes) final table by entering with the adjusted brake energy per brake. Times are provided for ground cooling and inflight gear down cooling.

Brake Temperature Monitor System (BTMS) indications are also shown. If brake cooling is determined from the BTMS, use the hottest brake indication 10 to 15 minutes after the airplane has come to a complete stop, or inflight with gear retracted to determine recommended cooling schedule.

Engine Inoperative

Initial Max Continuous %N1

The Initial Max Continuous %N1 setting for use following an engine failure is shown. The table is based on the typical all engine cruise speed of .79M to provide a target %N1 setting at the start of driftdown. Once driftdown is established, the Max Continuous %N1 table should be used to determine %N1 for the given conditions.

Max Continuous %N1

Power setting is based on one engine operating with one A/C pack operating and all anti-ice bleeds off. Enter the table with pressure altitude, TAT, and IAS or Mach to read %N1.

It is desirable to maintain engine thrust level within the limits of the Max Cruise thrust rating. However, where thrust level in excess of Max Cruise rating is required, such as for meeting terrain clearance, ATC altitude assignments, or to attain maximum range capability, it is permissible to use the thrust needed up to the Max Continuous thrust rating. The Max Continuous thrust rating is intended primarily for emergency use at the discretion of the pilot and is the maximum thrust that may be used continuously.

Driftdown Speed/Level Off Altitude

The table shows optimum driftdown speed as a function of cruise weight at start of driftdown. Also shown are the approximate weight and pressure altitude at which the airplane will level off considering 100 ft/min residual rate of climb.

The level off altitude is dependent on air temperature (ISA deviation).

Driftdown/LRC Range Capability

This table shows the range capability from the start of driftdown. Driftdown is continued to level off altitude. As weight decreases due to fuel burn, the airplane is accelerated to Long Range Cruise speed. Cruise is continued at level off altitude and Long Range Cruise speed.

To determine fuel required, enter the Ground to Air Miles Conversion table with the desired ground distance and adjust for anticipated winds to obtain air distance to destination. Then enter the Driftdown/Cruise Fuel and Time table with air distance and weight at start of driftdown to determine fuel and time required. If altitudes other than the level off altitude is used, fuel and time required may be obtained by using the Engine Inoperative Long Range Cruise Enroute Fuel and Time table.

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Long Range Cruise Altitude Capability

The table shows the maximum altitude that can be maintained at a given weight and air temperature (ISA deviation), based on Long Range Cruise speed, Max Continuous thrust, and 100 ft/min residual rate of climb.

Long Range Cruise Control

The table provides target %N1, engine inoperative Long Range Cruise Mach number, IAS and fuel flow for the airplane weight and pressure altitude. The fuel flow values in this table reflect single engine fuel burn.

APU Operation During Flight

For APU operation during flight, increase fuel flow according to the following table. These increments include the APU fuel flow and the effect of increased drag from the APU door.

PRESSURE ALTITUDE (1000 FT)	APU FUEL FLOW (KG/HR)
39	45
35	45
31	50
25	60
20	65
15	75
10	85
5	95

Long Range Cruise Diversion Fuel and Time

Tables are provided for crews to determine the fuel and time required to proceed to an alternate airfield with one engine inoperative. The data is based on single engine Long Range Cruise speed and .78/280/250 descent. Enter with Air Distance as determined from the Ground to Air Miles Conversion table and read Fuel and Time required at the cruise pressure altitude. Adjust the fuel obtained for deviation from the reference weight at checkpoint as required by entering the off reference fuel adjustments table with the fuel required for the reference weight and the actual weight at checkpoint. Read fuel required and time for the actual weight.

Holding

Target %N1, indicated airspeed and fuel flow per engine information is tabulated for holding with flaps up based on the FMC optimum holding speed schedule. This is the higher of the maximum endurance speed and the maneuvering speed. Small variations in airspeed will not appreciably affect the overall endurance time. Enter the table with weight and pressure altitude to read %N1, IAS and fuel flow per engine.

Gear Down Landing Rate of Climb Available

Rate of climb data is provided as guidance information in the event an engine inoperative landing (manual or autoland) is planned. The tables show gear down rate of climb available for Flaps 15 and Flaps 30. Enter the table with TAT and pressure altitude to read rate of climb available. Apply adjustments shown to correct for weight.

Gear Down

This section contains performance for airplane operation with the landing gear extended. The data is based on engine bleeds for normal air conditioning.

Note: The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS may generate inappropriate enroute speed schedules, display non-conservative predictions of fuel burn, estimated time of arrival (ETA), maximum altitude, and compute overly shallow descent path. An accurate estimated time of arrival (ETA) is available if current speed or Mach is entered into the VNAV cruise page.

Tables for gear down performance in this section are identical in format and used in the same manner as tables for the gear up configuration previously described.

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Flight With Unreliable Airspeed/

Turbulent Air Penetration PI-QRH.60.1

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Airport Altitude = 8000 FT PI-QRH.60.13

Airport Altitude = 9000 FT PI-QRH.60.14

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General

The table below shows the airplanes that have been identified with the following performance package. Note, some airplanes may be identified with more than one performance package. This configuration table information reflects the Boeing delivered configuration updated for service bulletin incorporations in conformance with the policy stated in the introduction section of the FCOM. The performance data is prepared for the owner/operator named on the title page. The intent of this information is to assist flight crews and airlines in knowing which performance package is applicable to a given airplane. The performance package model identification information is based on Boeing's knowledge of the airline's fleet at a point in time approximately three months prior to the page date. Notice of Errata (NOE) will not be provided to airlines to identify airplanes that are moved between performance packages within this manual or airplanes added to the airline's fleet whose performance packages are already represented in this manual. These types of changes will be updated in the next block revision.

Owners/operators are responsible for ensuring the operational documentation they are using is complete and matches the current configuration of their airplanes, and the accuracy and validity of all information furnished by the owner/operator or any other party. Owners/operators receiving active revision service are responsible to ensure that any modifications to the listed airplanes are properly reflected in this manual.

Serial and tabulation number are supplied by Boeing.

Registry Number	Old Registry Number	Serial Number	Tabulation Number
RA-73314	VP-BSC	40233	YF701
RA-73315	VP-BSE	40236	YF704
RA-73319	VQ-BDC	42059	YV256



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Performance Inflight - QRH

Chapter PI-QRH

General

Section 60

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

CLIMB (280/.76)

Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT	4.0	4.0	4.0		
	V/S (FT/MIN)	1700	1100	600		
30000	PITCH ATT	4.0	4.0	3.5	4.0	4.0
	V/S (FT/MIN)	2500	1900	1500	1100	800
20000	PITCH ATT	7.0	6.5	6.0	6.0	6.0
	V/S (FT/MIN)	4200	3300	2600	2100	1700
10000	PITCH ATT	11.0	9.5	8.5	8.0	8.0
	V/S (FT/MIN)	5600	4400	3600	3000	2500
SEA LEVEL	PITCH ATT	14.5	12.5	11.0	10.0	9.5
	V/S (FT/MIN)	6700	5300	4400	3700	3100

CRUISE (.76/280)

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT	2.0	2.5	3.5		
	%N1	82.9	85.4	88.9		
35000	PITCH ATT	1.0	2.0	2.5	3.0	3.5
	%N1	81.2	82.6	84.4	86.8	90.4
30000	PITCH ATT	1.0	1.5	2.0	2.5	3.0
	%N1	80.7	81.5	82.7	84.2	86.1
25000	PITCH ATT	1.0	1.5	2.0	2.5	3.0
	%N1	77.2	77.9	79.0	80.5	82.3
20000	PITCH ATT	1.0	1.5	2.0	2.5	3.5
	%N1	73.6	74.2	75.3	76.6	78.2
15000	PITCH ATT	1.0	1.5	2.0	3.0	3.5
	%N1	69.8	70.6	71.6	72.9	74.4

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

DESCENT (.76/280)

Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT V/S (FT/MIN)	-1.5 -2700	-0.5 -2400	0.5 -2300	1.0 -2500	1.5 -2700
30000	PITCH ATT V/S (FT/MIN)	-3.5 -3100	-2.0 -2600	-1.0 -2300	-0.5 -2100	0.5 -2000
20000	PITCH ATT V/S (FT/MIN)	-3.5 -2800	-2.0 -2300	-1.0 -2000	0.0 -1900	0.5 -1700
10000	PITCH ATT V/S (FT/MIN)	-3.5 -2500	-2.0 -2100	-1.0 -1800	0.0 -1700	0.5 -1500
SEA LEVEL	PITCH ATT V/S (FT/MIN)	-3.5 -2300	-2.5 -1900	-1.0 -1700	-0.5 -1500	0.5 -1400

HOLDING (VREF40 + 70)

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
15000	PITCH ATT	5.0	5.0	5.0	5.0	5.0
	%N1	56	62	66	70	73
	KIAS	177	193	212	229	246
10000	PITCH ATT	5.0	5.0	5.0	5.0	5.0
	%N1	52	58	62	66	69
	KIAS	177	192	211	228	244
5000	PITCH ATT	5.0	5.5	5.0	5.0	5.0
	%N1	49	54	58	62	66
	KIAS	177	191	210	227	243

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = -2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT	4.5	5.0	5.5	6.0	6.0
	%N1	47.5	52.2	56.3	60.0	63.6
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT	5.0	5.0	5.5	6.0	6.0
	%N1	49.6	54.3	58.5	62.3	65.8
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT	5.5	5.5	6.0	6.5	6.5
	%N1	49.3	54.4	58.9	63.1	66.7
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT	4.5	4.5	5.0	5.5	5.5
	%N1	50.4	55.6	60.2	64.4	68.0
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT	5.0	5.0	5.5	5.5	6.0
	%N1	50.9	56.3	61.1	65.3	69.0
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT	6.0	6.0	6.0	6.5	6.5
	%N1	52.1	57.6	62.5	66.8	70.6
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT	5.0	5.5	5.5	6.0	6.0
	%N1	55.5	61.2	66.1	70.2	74.1
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = -1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 48.2 177	5.0 52.9 191	5.5 57.0 203	6.0 60.8 213	6.5 64.4 223
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 50.2 157	5.0 55.1 171	5.5 59.3 183	6.0 63.1 193	6.0 66.5 203
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 49.9 137	5.5 55.2 151	6.0 59.7 163	6.5 63.9 173	6.5 67.5 183
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 51.1 137	4.5 56.3 151	5.0 61.0 163	5.5 65.2 173	5.5 68.8 183
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 51.6 127	5.0 57.0 141	5.5 61.9 153	5.5 66.1 163	6.0 69.8 173
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 52.8 117	6.0 58.4 131	6.0 63.3 143	6.5 67.6 153	6.5 71.5 163
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 56.2 127	5.5 62.0 141	5.5 66.9 153	6.0 71.0 163	6.0 74.9 173

**Flight With Unreliable Airspeed/ Turbulent Air Penetration**

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = SEA LEVEL

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 48.9 177	5.0 53.5 191	5.5 57.7 203	6.0 61.6 213	6.5 65.2 223
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 50.9 157	5.0 55.8 171	5.5 60.0 183	6.0 63.9 193	6.0 67.3 203
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 50.6 137	5.5 55.9 151	6.0 60.5 163	6.5 64.7 173	6.5 68.3 183
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 51.8 137	4.5 57.1 151	5.0 61.9 163	5.5 66.0 173	5.5 69.6 183
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 52.3 127	5.5 57.8 141	5.5 62.7 153	5.5 66.9 163	6.0 70.6 173
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 53.5 117	6.0 59.2 131	6.0 64.2 143	6.5 68.4 153	6.5 72.4 163
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 57.0 127	5.5 62.8 141	5.5 67.7 153	6.0 71.9 163	6.0 75.8 173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 49.7	5.0 54.3	5.5 58.4	6.0 62.4	6.5 66.0
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 51.6	5.0 56.5	5.5 60.8	6.0 64.7	6.0 68.0
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 51.4	5.5 56.6	6.0 61.3	6.5 65.5	6.5 69.1
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 52.6	4.5 57.9	5.0 62.7	5.5 66.8	5.5 70.4
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 53.0	5.5 58.6	5.5 63.5	5.5 67.7	6.0 71.5
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 54.3	6.0 60.0	6.0 65.0	6.5 69.2	6.5 73.2
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 57.7	5.5 63.6	5.5 68.5	6.0 72.8	6.0 76.6
VREF40+20	KIAS	127	141	153	163	173

**Flight With Unreliable Airspeed/ Turbulent Air Penetration**

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 50.4	5.0 55.0	5.5 59.2	6.0 63.3	6.5 66.7
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 52.3	5.0 57.3	5.5 61.7	6.0 65.4	6.0 68.8
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 52.1	5.5 57.4	6.0 62.2	6.5 66.3	6.5 69.9
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 53.3	4.5 58.6	5.0 63.5	5.5 67.6	5.5 71.3
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 53.7	5.5 59.4	5.5 64.3	5.5 68.5	6.0 72.4
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 55.0	6.0 60.8	6.0 65.8	6.5 70.1	6.5 74.1
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 58.5	5.5 64.4	5.5 69.3	6.0 73.6	6.0 77.4
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 3000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 51.0	5.0 55.8	5.5 60.0	6.0 64.1	6.5 67.4
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 53.1	5.0 58.0	5.5 62.5	6.0 66.2	6.0 69.6
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 52.9	5.5 58.2	6.0 63.0	6.5 67.1	6.5 70.7
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 54.0	4.5 59.5	5.0 64.3	5.5 68.4	5.5 72.1
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 54.5	5.5 60.2	5.5 65.2	5.5 69.4	6.0 73.3
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 55.8	6.0 61.6	6.0 66.5	6.5 71.0	6.5 74.9
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 59.3	5.5 65.2	5.5 70.2	6.0 74.5	6.0 78.2
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 4000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 51.7	5.0 56.4	5.5 60.8	6.0 64.9	6.5 68.2
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 53.9	5.0 58.8	5.5 63.3	6.0 66.9	6.0 70.5
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 53.6	5.5 59.0	6.0 63.8	6.5 67.9	6.5 71.6
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 54.7	4.5 60.3	5.0 65.1	5.5 69.2	5.5 73.0
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 55.2	5.5 61.1	5.5 66.0	5.5 70.2	6.0 74.1
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 56.5	6.0 62.4	6.0 67.4	6.5 71.9	6.5 75.7
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 60.1	5.5 66.0	5.5 71.1	6.0 75.3	6.0 79.1
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 5000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 52.4	5.0 57.2	5.5 61.6	6.0 65.6	6.5 69.0
VREF40+70	KIAS	177	191	203	213	224
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 54.6	5.0 59.6	5.5 64.1	6.0 67.7	6.0 71.4
VREF40+50	KIAS	157	171	183	193	204
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 54.3	5.5 59.8	6.0 64.6	6.5 68.7	6.5 72.5
VREF40+30	KIAS	137	151	163	173	184
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 55.5	4.5 61.1	5.0 65.9	5.5 70.0	5.5 73.8
VREF40+30	KIAS	137	151	163	173	184
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 56.0	5.5 61.9	5.5 66.7	5.5 71.1	6.0 74.9
VREF40+20	KIAS	127	141	153	163	174
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 57.3	6.0 63.3	6.0 68.2	6.5 72.7	6.5 76.5
VREF40+10	KIAS	117	131	143	153	164
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 61.0	5.5 66.8	5.5 71.9	6.0 76.1	6.0 80.0
VREF40+20	KIAS	127	141	153	163	174

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 6000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 53.1	5.0 57.9	5.5 62.5	6.0 66.3	6.5 69.8
VREF40+70	KIAS	177	191	203	213	224
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 55.3	5.0 60.4	5.5 64.8	6.0 68.5	6.0 72.2
VREF40+50	KIAS	157	171	183	193	204
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 55.1	5.5 60.7	6.0 65.4	6.5 69.5	6.5 73.3
VREF40+30	KIAS	137	151	163	173	184
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 56.2	4.5 62.0	5.0 66.7	5.5 70.9	5.5 74.6
VREF40+30	KIAS	137	151	163	173	184
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 56.7	5.5 62.7	5.5 67.6	5.5 72.0	6.0 75.7
VREF40+20	KIAS	127	141	153	163	174
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 58.1	6.0 64.1	6.0 69.0	6.5 73.6	6.5 77.3
VREF40+10	KIAS	117	131	143	153	164
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 61.8	5.5 67.6	5.5 72.8	6.0 77.0	6.0 80.9
VREF40+20	KIAS	127	141	153	163	174

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 7000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 53.8	5.0 58.7	5.5 63.3	6.0 67.1	6.5 70.6
VREF40+70	KIAS	177	191	203	213	224
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 56.1	5.0 61.3	5.5 65.6	6.0 69.4	6.0 73.1
VREF40+50	KIAS	157	171	183	193	204
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 55.8	5.5 61.5	6.0 66.2	6.5 70.4	6.5 74.1
VREF40+30	KIAS	137	151	163	173	184
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 57.0	4.5 62.8	5.0 67.5	5.5 71.8	5.5 75.4
VREF40+30	KIAS	137	151	163	173	184
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 57.6	5.5 63.5	5.5 68.4	5.5 72.8	6.0 76.5
VREF40+20	KIAS	127	141	153	163	174
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 58.9	6.0 64.9	6.0 69.9	6.5 74.4	6.5 78.2
VREF40+10	KIAS	117	131	143	153	164
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 62.6	5.5 68.5	5.5 73.7	6.0 77.8	6.0 81.8
VREF40+20	KIAS	127	141	153	163	174

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 8000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 54.5	5.0 59.5	5.5 64.1	6.0 67.8	6.0 71.4
VREF40+70	KIAS	177	191	203	214	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 56.8	5.0 62.1	5.5 66.4	6.0 70.3	6.0 73.9
VREF40+50	KIAS	157	171	183	194	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 56.6	5.5 62.4	6.0 67.0	6.5 71.2	6.5 74.9
VREF40+30	KIAS	137	151	163	174	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 57.8	4.5 63.6	5.0 68.3	5.5 72.6	5.5 76.2
VREF40+30	KIAS	137	151	163	174	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 58.4	5.5 64.3	5.5 69.3	5.5 73.7	6.0 77.3
VREF40+20	KIAS	127	141	153	164	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 59.8	6.0 65.7	6.0 70.8	6.5 75.2	6.5 79.1
VREF40+10	KIAS	117	131	143	154	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 63.5	5.5 69.4	5.5 74.5	6.0 78.7	6.0 82.7
VREF40+20	KIAS	127	141	153	164	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 9000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 55.3	5.0 60.4	5.5 64.9	6.0 68.6	6.0 72.2
VREF40+70	KIAS	177	191	204	214	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 57.6	5.0 62.9	5.5 67.2	6.0 71.1	6.0 74.7
VREF40+50	KIAS	157	171	184	194	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 57.4	5.5 63.1	6.0 67.8	6.5 72.1	6.5 75.7
VREF40+30	KIAS	137	151	164	174	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 58.7	4.5 64.4	5.0 69.2	5.5 73.4	5.5 77.1
VREF40+30	KIAS	137	151	164	174	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 59.2	5.5 65.1	5.5 70.2	5.5 74.5	5.5 78.2
VREF40+20	KIAS	127	141	154	164	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 60.6	6.0 66.5	6.0 71.7	6.5 76.1	6.5 80.0
VREF40+10	KIAS	117	131	144	154	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 64.3	5.5 70.3	5.5 75.3	6.0 79.6	6.0 83.6
VREF40+20	KIAS	127	141	154	164	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 10000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT	4.5	5.0	5.5	6.0	6.0
VREF40+70	%N1	56.0	61.2	65.6	69.4	73.0
	KIAS	177	191	204	214	225
FLAPS 1 (GEAR UP)	PITCH ATT	5.0	5.0	5.5	6.0	6.0
VREF40+50	%N1	58.4	63.7	68.0	72.0	75.5
	KIAS	157	171	184	194	205
FLAPS 5 (GEAR UP)	PITCH ATT	5.5	5.5	6.0	6.0	6.5
VREF40+30	%N1	58.2	64.0	68.7	72.9	76.6
	KIAS	137	151	164	174	185
FLAPS 10 (GEAR UP)	PITCH ATT	4.5	4.5	5.0	5.0	5.5
VREF40+30	%N1	59.5	65.3	70.1	74.3	78.0
	KIAS	137	151	164	174	185
FLAPS 15 (GEAR UP)	PITCH ATT	5.0	5.0	5.5	5.5	5.5
VREF40+20	%N1	60.1	65.9	71.1	75.3	79.1
	KIAS	127	141	154	164	175
FLAPS 25 (GEAR UP)	PITCH ATT	6.0	6.0	6.0	6.5	6.0
VREF40+10	%N1	61.4	67.4	72.6	76.9	80.8
	KIAS	117	131	144	154	165
FLAPS 15 (GEAR DOWN)	PITCH ATT	5.0	5.5	5.5	6.0	6.0
VREF40+20	%N1	65.0	71.2	76.2	80.6	84.5
	KIAS	127	141	154	164	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 11000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 56.8 177	5.0 62.2 191	5.5 66.3 204	6.0 70.3 215	6.0 73.8 226
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 59.4 157	5.0 64.5 171	5.5 69.0 184	6.0 73.0 195	6.0 76.3 206
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 59.4 137	5.5 64.9 151	6.0 69.7 164	6.0 73.8 175	6.5 77.5 186
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 60.7 137	4.5 66.2 151	5.0 71.1 164	5.0 75.1 175	5.5 79.0 186
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 61.2 127	5.0 67.0 141	5.5 72.1 154	5.5 76.2 165	5.5 80.1 176
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 62.6 117	6.0 68.5 131	6.0 73.6 144	6.5 78.0 155	6.0 81.8 166
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 66.1 127	5.5 72.2 141	5.5 77.2 154	6.0 81.6 165	6.0 85.5 176

**Flight With Unreliable Airspeed/ Turbulent Air Penetration**

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 12000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 57.6	5.0 63.0	5.5 67.1	6.0 71.2	6.0 74.6
VREF40+70	KIAS	178	191	204	215	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 60.4	5.0 65.3	5.5 69.9	6.0 73.7	6.0 77.2
VREF40+50	KIAS	158	171	184	195	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 60.4	5.5 65.8	6.0 70.6	6.0 74.7	6.5 78.5
VREF40+30	KIAS	138	151	164	175	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 61.7	4.5 67.1	5.0 72.0	5.0 76.0	5.5 79.9
VREF40+30	KIAS	138	151	164	175	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 62.2	5.0 68.0	5.5 73.0	5.5 77.2	5.5 81.1
VREF40+20	KIAS	128	141	154	165	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 63.6	6.0 69.7	6.0 74.5	6.0 78.9	6.0 82.8
VREF40+10	KIAS	118	131	144	155	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 67.3	5.5 73.2	5.5 78.3	5.5 82.6	5.5 86.6
VREF40+20	KIAS	128	141	154	165	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 13000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 58.5	5.0 63.7	5.5 67.9	6.0 72.0	6.0 75.4
VREF40+70	KIAS	178	192	205	215	227
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 61.3	5.0 66.2	5.5 70.8	6.0 74.5	6.0 78.1
VREF40+50	KIAS	158	172	185	195	207
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 61.4	5.5 66.7	6.0 71.6	6.0 75.6	6.0 79.4
VREF40+30	KIAS	138	152	165	175	187
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 62.6	4.5 68.1	5.0 72.9	5.0 77.0	5.0 80.8
VREF40+30	KIAS	138	152	165	175	187
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 63.2	5.0 69.1	5.5 73.9	5.5 78.2	5.5 82.0
VREF40+20	KIAS	128	142	155	165	177
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 64.7	6.0 70.7	6.0 75.4	6.0 79.9	6.0 83.7
VREF40+10	KIAS	118	132	145	155	167
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 68.4	5.5 74.2	5.5 79.3	5.5 83.6	5.5 87.6
VREF40+20	KIAS	128	142	155	165	177

**Flight With Unreliable Airspeed/ Turbulent Air Penetration**

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 14000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 59.4	5.0 64.4	5.5 68.7	6.0 72.8	6.0 76.3
VREF40+70	KIAS	178	192	205	215	227
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 62.2	5.0 67.1	5.5 71.6	6.0 75.3	6.0 79.0
VREF40+50	KIAS	158	172	185	195	207
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 62.3	5.5 67.7	6.0 72.4	6.0 76.5	6.0 80.3
VREF40+30	KIAS	138	152	165	175	187
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 63.5	4.5 69.1	5.0 73.7	5.0 77.9	5.5 81.7
VREF40+30	KIAS	138	152	165	175	187
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 64.1	5.0 70.1	5.5 74.8	5.5 79.1	5.5 82.9
VREF40+20	KIAS	128	142	155	165	177
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 65.8	6.0 71.7	6.0 76.5	6.0 80.9	6.0 84.7
VREF40+10	KIAS	118	132	145	155	167
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 69.5	5.5 75.1	5.5 80.2	5.5 84.5	5.5 88.6
VREF40+20	KIAS	128	142	155	165	177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 14500 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 59.8 178	5.0 64.8 192	5.5 69.2 205	6.0 73.2 216	6.0 76.7 227
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 62.6 158	5.0 67.5 172	5.5 72.1 185	6.0 75.8 196	6.0 79.4 207
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 62.7 138	5.5 68.2 152	6.0 72.9 165	6.0 76.9 176	6.0 80.7 187
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 63.9 138	4.5 69.6 152	5.0 74.2 165	5.0 78.4 176	5.0 82.1 187
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 64.6 128	5.0 70.6 142	5.5 75.2 155	5.5 79.6 166	5.5 83.4 177
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 66.3 118	6.0 72.1 132	6.0 77.0 145	6.0 81.4 156	6.0 85.2 167
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 70.1 128	5.5 75.6 142	5.5 80.7 155	5.5 85.0 166	5.5 89.2 177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = -2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	38.9	43.0	46.7	49.9	52.8
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	46.1	50.8	55.1	58.6	62.0
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	50.5	55.9	60.6	64.4	67.9
	KIAS	117	130	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = -1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	39.5	43.6	47.3	50.5	53.5
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	46.7	51.5	55.8	59.3	62.8
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	51.2	56.6	61.3	65.2	68.7
	KIAS	117	130	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = SEA LEVEL

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	40.2	44.1	48.0	51.1	54.2
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	47.3	52.2	56.5	60.1	63.6
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	51.9	57.3	62.1	66.1	69.5
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	40.7	44.7	48.7	51.8	54.9
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	47.9	52.9	57.2	60.9	64.4
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	52.6	58.1	62.9	66.8	70.3
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	41.2	45.4	49.3	52.5	55.6
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	48.6	53.6	58.0	61.7	65.1
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	53.3	58.9	63.7	67.5	71.1
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 3000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	41.7	46.0	49.9	53.2	56.3
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	49.2	54.3	58.7	62.5	65.9
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	54.1	59.6	64.5	68.3	72.0
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 4000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	42.2	46.7	50.5	53.9	57.0
	KIAS	128	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	49.9	55.0	59.5	63.3	66.7
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	54.8	60.4	65.4	69.1	72.8
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 5000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	42.8	47.3	51.2	54.6	57.7
	KIAS	129	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	50.6	55.7	60.3	64.0	67.5
	KIAS	124	138	151	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	55.6	61.2	66.1	69.9	73.6
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

**Flight With Unreliable Airspeed/ Turbulent Air Penetration**

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 6000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	43.4	48.0	51.9	55.3	58.4
	KIAS	129	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	51.3	56.4	61.1	64.8	68.2
	KIAS	124	138	151	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	56.3	62.0	66.8	70.8	74.5
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 7000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	44.0	48.6	52.6	56.0	59.1
	KIAS	129	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	52.0	57.2	61.9	65.6	69.0
	KIAS	124	138	151	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	57.0	62.8	67.6	71.6	75.3
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 8000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	44.6	49.2	53.3	56.7	59.8
	KIAS	129	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	52.7	58.0	62.7	66.4	69.8
	KIAS	124	138	151	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	57.8	63.6	68.4	72.5	76.1
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 9000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	45.3	49.9	54.1	57.4	60.6
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	53.4	58.8	63.4	67.1	70.6
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	58.6	64.4	69.3	73.3	76.9
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 10000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	45.9	50.6	54.7	58.1	61.4
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	54.2	59.6	64.2	67.9	71.4
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	59.4	65.2	70.1	74.1	77.8
	KIAS	117	131	143	153	164

Flap placard speed exceeded in shaded area.

Airport Altitude = 11000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	46.7	51.3	55.5	58.8	62.2
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	54.9	60.4	65.1	68.7	72.3
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	60.2	66.0	71.0	74.9	78.7
	KIAS	117	131	143	153	164

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 12000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	47.3	52.0	56.2	59.6	62.9
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	55.6	61.2	65.8	69.5	73.0
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	61.0	66.8	71.8	75.7	79.6
	KIAS	117	131	143	153	164

Flap placard speed exceeded in shaded area.

Airport Altitude = 13000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	47.9	52.8	56.8	60.4	63.6
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	56.4	62.0	66.6	70.4	73.8
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	61.9	67.6	72.7	76.6	80.5
	KIAS	117	131	143	154	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 14000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	48.7	53.5	57.6	61.3	64.4
	KIAS	129	143	157	168	180
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	57.4	62.9	67.5	71.3	74.6
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	62.9	68.6	73.6	77.6	81.4
	KIAS	117	131	144	154	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 14500 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	49.1	53.9	58.0	61.7	64.7
	KIAS	129	144	157	168	180
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	58.0	63.4	68.0	71.7	75.0
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.0	0.5	0.5
	%N1	63.4	69.2	74.1	78.1	81.9
	KIAS	117	131	144	154	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

GO-AROUND

Flaps 15, Gear Up, Set Go-Around Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
14500	PITCH ATT	17.5	14.0	11.5	10.5	9.0
	V/S (FT/MIN)	3500	2700	2200	1700	1400
	KIAS	127	141	154	164	175
10000	PITCH ATT	20.0	16.0	13.5	12.0	10.5
	V/S (FT/MIN)	4000	3200	2600	2100	1700
	KIAS	127	141	153	163	174
5000	PITCH ATT	24.0	19.0	16.0	14.0	12.5
	V/S (FT/MIN)	4600	3800	3100	2600	2200
	KIAS	127	141	153	163	173
SEA LEVEL	PITCH ATT	28.0	22.0	18.5	16.0	14.0
	V/S (FT/MIN)	5200	4300	3600	3000	2600
	KIAS	127	141	153	163	173
-2000	PITCH ATT	28.0	22.5	18.5	16.0	14.5
	V/S (FT/MIN)	5100	4200	3500	3000	2500
	KIAS	127	140	153	163	173

Max Climb %N1

Based on engine bleed for packs on or off and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (FT)/SPEED (KIAS/MACH)									
	0	5000	10000	15000	20000	25000	30000	35000	37000	41000
	280	280	280	280	280	280	280	.78	.78	.78
60	90.2	90.5	90.4	90.6	90.4	92.1	93.8	95.1	95.2	93.5
55	91.0	91.2	91.3	91.4	90.8	91.5	93.1	94.4	94.5	92.8
50	91.7	92.0	92.1	92.2	91.7	91.5	92.4	93.7	93.8	92.1
45	92.4	92.6	92.8	93.0	92.6	92.4	92.4	93.0	93.1	91.4
40	93.1	93.3	93.6	93.8	93.4	93.2	93.2	92.3	92.4	90.7
35	93.6	94.0	94.3	94.5	94.3	94.0	94.0	93.0	92.4	90.8
30	92.9	94.8	95.0	95.2	95.1	94.8	94.7	93.9	93.3	91.8
25	92.2	94.8	95.7	95.9	95.9	95.5	95.4	94.7	94.1	92.8
20	91.4	94.0	96.5	96.7	96.6	96.2	96.1	95.4	94.9	93.7
15	90.6	93.2	95.9	97.5	97.4	96.9	96.7	96.2	95.7	94.6
10	89.9	92.5	95.1	97.8	98.3	97.7	97.4	96.9	96.5	95.6
5	89.1	91.7	94.3	97.0	99.2	98.6	98.1	97.7	97.3	96.5
0	88.3	90.9	93.5	96.2	98.6	99.6	99.1	98.5	98.2	97.5
-5	87.6	90.1	92.7	95.4	97.8	99.6	100.0	99.2	99.0	98.4
-10	86.8	89.3	91.9	94.6	97.1	98.8	100.3	100.2	99.8	99.4
-15	86.0	88.5	91.0	93.8	96.3	98.0	99.6	101.1	100.8	100.4
-20	85.2	87.6	90.2	93.0	95.5	97.2	98.7	100.8	101.3	101.0
-25	84.3	86.8	89.4	92.2	94.7	96.4	97.9	100.0	100.5	100.1
-30	83.5	86.0	88.5	91.3	93.9	95.6	97.1	99.1	99.6	99.3
-35	82.7	85.1	87.7	90.5	93.1	94.8	96.3	98.3	98.8	98.4
-40	81.8	84.3	86.8	89.6	92.3	93.9	95.4	97.4	97.9	97.6

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)					
	0	10	20	30	35	41
ENGINE ANTI-ICE	-0.6	-0.8	-0.9	-0.9	-0.8	-0.8
ENGINE & WING ANTI-ICE*	-1.8	-2.1	-2.5	-2.7	-3.0	-3.0

*Dual bleed sources

Go-around %N1

Based on engine bleed for packs on, engine and wing anti-ice on or off

AIRPORT OAT		TAT (°C)	AIRPORT PRESSURE ALTITUDE (FT)											
°C	°F		-2000	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
57	134	60	95.0	96.2	96.8									
52	125	55	95.9	96.7	96.6	96.8	97.5							
47	116	50	96.6	97.6	97.8	97.8	97.7	97.5	98.2	98.8				
42	108	45	97.4	98.4	98.5	98.6	98.7	98.8	98.7	98.5	99.0			
37	99	40	98.0	99.1	99.2	99.3	99.4	99.5	99.6	99.5	99.1	98.9	98.8	99.1
32	90	35	98.1	99.9	100.0	100.1	100.1	100.3	100.3	100.2	99.9	99.6	99.6	99.5
27	81	30	97.3	99.8	100.4	100.7	100.7	100.7	100.7	100.7	100.6	100.4	100.4	100.3
22	72	25	96.6	99.1	99.7	100.2	100.6	100.9	100.9	100.9	100.9	100.9	100.9	100.8
17	63	20	95.8	98.3	98.9	99.5	99.8	100.2	100.5	100.9	101.0	101.1	101.0	101.0
12	54	15	95.0	97.5	98.1	98.7	99.1	99.4	99.8	100.1	100.5	100.9	101.3	101.2
7	45	10	94.2	96.8	97.4	98.0	98.3	98.7	99.0	99.4	99.8	100.2	100.5	100.9
2	36	5	93.4	96.0	96.6	97.2	97.6	97.9	98.3	98.7	99.0	99.4	99.8	100.2
-3	27	0	92.6	95.2	95.8	96.4	96.8	97.2	97.5	97.9	98.3	98.7	99.0	99.4
-8	18	-5	91.8	94.4	95.0	95.6	96.0	96.4	96.8	97.2	97.5	97.9	98.3	98.6
-13	9	-10	91.0	93.6	94.2	94.8	95.2	95.6	96.0	96.4	96.8	97.1	97.5	97.9
-17	1	-15	90.2	92.8	93.4	94.0	94.4	94.8	95.2	95.6	96.0	96.4	96.7	97.1
-22	-8	-20	89.3	92.0	92.6	93.2	93.6	94.0	94.4	94.8	95.2	95.6	95.9	96.3
-27	-17	-25	88.5	91.1	91.8	92.4	92.8	93.2	93.6	94.0	94.4	94.8	95.1	95.5
-32	-26	-30	87.6	90.3	90.9	91.6	92.0	92.4	92.8	93.3	93.6	94.0	94.3	94.7
-37	-35	-35	86.8	89.4	90.1	90.7	91.1	91.6	92.0	92.4	92.8	93.2	93.5	93.9
-42	-44	-40	85.9	88.6	89.2	89.9	90.3	90.7	91.2	91.6	92.0	92.4	92.7	93.0
-47	-53	-45	85.0	87.7	88.4	89.0	89.4	89.9	90.3	90.8	91.2	91.5	91.9	92.2
-52	-62	-50	84.1	86.8	87.5	88.2	88.6	89.0	89.5	90.0	90.3	90.7	91.0	91.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (FT)											
	-2000	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
PACKS OFF	0.7	0.8	0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9
A/C HIGH	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1



737 Flight Crew Operations Manual

VREF

WEIGHT (1000 KG)	FLAPS		
	40	30	15
85	159	167	174
80	154	162	169
75	148	156	163
70	143	151	157
65	139	147	153
60	133	141	147
55	127	134	140
50	121	128	133
45	114	121	126
40	107	114	119

Intentionally
Blank



Performance Inflight - QRH

Advisory Information

Chapter PI-QRH

Section 61

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF15	ONE REV NO REV

Dry Runway

MAX MANUAL	975	75/-60	20/30	-35/120	10/-10	20/-20	35	20	40
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 3	1815	105/-115	50/65	-75/255	0/0	50/-50	100	0	0
AUTOBRAKE 2	2300	150/-160	70/95	-105/350	30/-45	70/-70	95	80	80
AUTOBRAKE 1	2530	180/-190	85/110	-120/410	70/-80	75/-75	90	220	335

Good Reported Braking Action

MAX MANUAL	1330	75/-80	35/45	-60/200	30/-25	35/-35	45	65	145
AUTOBRAKE MAX	1430	80/-85	40/50	-60/210	30/-25	35/-35	55	75	165
AUTOBRAKE 3	1820	105/-115	50/65	-75/260	5/-5	50/-50	100	5	10
AUTOBRAKE 2	2300	150/-160	70/95	-105/350	30/-45	70/-70	95	80	80
AUTOBRAKE 1	2530	180/-190	85/110	-120/410	70/-80	75/-75	90	220	335

Medium Reported Braking Action

MAX MANUAL	1845	120/-120	60/75	-95/335	80/-65	50/-50	60	190	455
AUTOBRAKE MAX	1885	125/-125	60/80	-95/340	75/-60	50/-55	70	190	460
AUTOBRAKE 3	2005	125/-130	60/80	-100/350	60/-40	55/-60	100	130	375
AUTOBRAKE 2	2350	155/-165	75/95	-115/395	65/-60	70/-70	95	115	230
AUTOBRAKE 1	2540	180/-190	85/110	-125/430	90/-85	75/-75	90	235	390

Poor Reported Braking Action

MAX MANUAL	2430	175/-175	85/115	-140/535	205/-135	70/-75	75	415	1105
AUTOBRAKE MAX	2430	175/-175	85/115	-140/535	205/-130	70/-75	80	415	1105
AUTOBRAKE 3	2460	175/-175	85/115	-145/535	195/-125	70/-75	90	420	1115
AUTOBRAKE 2	2625	185/-190	90/125	-150/555	190/-125	75/-80	95	350	960
AUTOBRAKE 1	2740	195/-200	95/130	-155/570	195/-140	80/-85	90	400	990

Reference distance is based on sea level, standard day, no wind or slope, VREF15, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 60 m.

For autobrake and manual speedbrakes, increase reference landing distance by 50 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF30	ONE REV	NO REV

Dry Runway

MAX MANUAL	935	60/-55	20/25	-35/120	10/-10	20/-20	35	15	35
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	5
AUTOBRAKE 3	1700	100/-105	45/60	-75/245	0/-5	45/-45	85	0	0
AUTOBRAKE 2	2120	140/-145	65/85	-100/335	30/-40	60/-60	85	80	80
AUTOBRAKE 1	2325	160/-170	75/100	-115/395	65/-70	70/-70	80	185	300

Good Reported Braking Action

MAX MANUAL	1275	70/-75	35/45	-55/195	30/-25	30/-30	45	60	130
AUTOBRAKE MAX	1375	75/-80	35/45	-60/205	30/-25	35/-35	55	65	145
AUTOBRAKE 3	1705	100/-105	45/60	-75/250	5/-10	45/-45	85	5	10
AUTOBRAKE 2	2120	140/-145	65/85	-100/335	30/-40	60/-60	85	80	80
AUTOBRAKE 1	2325	160/-170	75/100	-115/395	65/-70	70/-70	80	185	300

Medium Reported Braking Action

MAX MANUAL	1740	110/-115	55/70	-90/325	80/-60	45/-50	60	165	390
AUTOBRAKE MAX	1790	115/-120	55/75	-90/330	75/-60	50/-50	70	165	400
AUTOBRAKE 3	1885	115/-120	55/75	-95/340	55/-45	55/-55	85	115	330
AUTOBRAKE 2	2175	140/-150	65/90	-110/380	65/-60	65/-65	85	115	215
AUTOBRAKE 1	2340	160/-170	75/100	-120/410	90/-75	70/-70	80	195	350

Poor Reported Braking Action

MAX MANUAL	2265	160/-160	75/105	-135/520	190/-125	65/-70	70	355	920
AUTOBRAKE MAX	2275	160/-160	80/105	-135/520	190/-120	65/-70	80	350	920
AUTOBRAKE 3	2305	160/-160	80/105	-140/520	185/-120	65/-70	80	360	930
AUTOBRAKE 2	2435	170/-170	80/110	-145/535	180/-120	70/-75	85	315	810
AUTOBRAKE 1	2530	175/-180	85/115	-145/550	190/-130	75/-80	80	340	845

Reference distance is based on sea level, standard day, no wind or slope, VREF30, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 60 m.

For autobrake and manual speedbrakes, increase reference landing distance by 50 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF40	ONE REV	NO REV

Dry Runway

MAX MANUAL	890	50/-50	20/25	-35/115	10/-10	20/-20	35	15	30
AUTOBRAKE MAX	1120	55/-60	25/30	-40/140	0/0	25/-25	55	0	0
AUTOBRAKE 3	1565	90/-100	40/55	-70/235	0/-5	40/-40	85	0	0
AUTOBRAKE 2	1980	125/-135	60/75	-95/325	25/-35	55/-55	90	40	40
AUTOBRAKE 1	2185	150/-155	70/90	-110/380	55/-65	65/-65	85	145	225

Good Reported Braking Action

MAX MANUAL	1220	65/-70	30/40	-55/195	30/-25	30/-30	45	55	120
AUTOBRAKE MAX	1310	70/-75	35/45	-60/200	25/-25	30/-30	55	60	130
AUTOBRAKE 3	1570	90/-100	40/55	-70/240	10/-5	40/-45	90	5	10
AUTOBRAKE 2	1980	125/-135	60/75	-95/325	25/-35	55/-55	90	40	40
AUTOBRAKE 1	2185	150/-155	70/90	-110/380	55/-65	65/-65	85	145	225

Medium Reported Braking Action

MAX MANUAL	1660	105/-105	50/65	-90/320	75/-60	45/-45	60	150	350
AUTOBRAKE MAX	1695	110/-110	50/70	-90/325	70/-55	45/-45	70	150	355
AUTOBRAKE 3	1760	110/-115	50/70	-90/330	60/-45	50/-50	85	115	325
AUTOBRAKE 2	2035	130/-140	60/80	-105/370	60/-55	60/-60	90	80	175
AUTOBRAKE 1	2195	150/-155	70/95	-115/400	80/-70	65/-65	85	160	275

Poor Reported Braking Action

MAX MANUAL	2160	150/-150	70/100	-135/510	190/-120	60/-65	70	325	830
AUTOBRAKE MAX	2165	150/-150	75/100	-135/510	190/-120	60/-65	75	325	830
AUTOBRAKE 3	2185	155/-155	75/100	-135/510	185/-120	60/-65	80	330	840
AUTOBRAKE 2	2300	160/-160	75/105	-140/525	175/-115	65/-70	85	275	730
AUTOBRAKE 1	2390	165/-170	80/110	-145/540	180/-125	70/-75	80	305	745

Reference distance is based on sea level, standard day, no wind or slope, VREF40, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 55 m.

For autobrake and manual speedbrakes, increase reference landing distance by 45 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown. Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1035	85/-60	25/30	-35/125	10/-10	25/-25	N/A	25	50
AUTOBRAKE MAX	1390	70/-75	35/45	-50/160	0/0	35/-35	N/A	0	5
AUTOBRAKE 2	2450	160/-170	80/105	-110/360	50/-55	75/-75	N/A	160	185

Good Reported Braking Action

MAX MANUAL	1410	80/-80	40/50	-60/205	30/-30	35/-35	N/A	75	175
AUTOBRAKE MAX	1530	85/-90	40/55	-60/215	30/-20	40/-40	N/A	85	195
AUTOBRAKE 2	2450	160/-170	80/105	-110/360	50/-55	75/-75	N/A	160	185

Medium Reported Braking Action

MAX MANUAL	1935	125/-125	60/80	-95/340	80/-65	55/-55	N/A	215	520
AUTOBRAKE MAX	1995	125/-130	65/85	-95/345	75/-60	55/-55	N/A	220	535
AUTOBRAKE 3	2185	130/-140	65/90	-100/365	55/-50	65/-65	N/A	125	385

Poor Reported Braking Action

MAX MANUAL	2515	175/-175	90/120	-145/535	195/-130	75/-80	N/A	455	1230
AUTOBRAKE MAX	2530	180/-180	90/125	-145/535	195/-125	75/-80	N/A	450	1225
AUTOBRAKE 3	2590	175/-180	90/125	-145/540	185/-120	75/-80	N/A	430	1215

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	995	65/-55	20/30	-35/125	10/-10	20/-20	N/A	20	45
AUTOBRAKE MAX	1315	65/-70	30/40	-45/155	0/0	30/-30	N/A	0	5
AUTOBRAKE 2	2265	145/-155	70/95	-105/345	45/-50	65/-65	N/A	135	185

Good Reported Braking Action

MAX MANUAL	1355	75/-75	35/50	-60/200	30/-25	35/-35	N/A	70	155
AUTOBRAKE MAX	1465	80/-85	40/50	-60/210	30/-20	35/-35	N/A	80	175
AUTOBRAKE 2	2265	145/-155	70/95	-105/345	45/-50	65/-65	N/A	135	185

Medium Reported Braking Action

MAX MANUAL	1830	115/-115	55/75	-90/330	80/-60	50/-50	N/A	185	450
AUTOBRAKE MAX	1895	120/-120	60/80	-95/335	75/-60	50/-55	N/A	190	465
AUTOBRAKE 3	2040	120/-125	60/80	-100/350	60/-50	60/-60	N/A	115	340

Poor Reported Braking Action

MAX MANUAL	2360	160/-160	80/110	-140/520	185/-125	65/-70	N/A	390	1025
AUTOBRAKE MAX	2385	165/-165	85/110	-140/525	180/-120	70/-75	N/A	390	1025
AUTOBRAKE 3	2420	165/-165	80/110	-140/525	180/-115	70/-75	N/A	365	1010

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	955	55/-50	20/30	-35/120	10/-10	20/-20	N/A	20	40
AUTOBRAKE MAX	1225	60/-65	30/35	-45/150	0/0	30/-30	N/A	0	0
AUTOBRAKE 2	2130	135/-145	65/85	-100/335	40/-45	60/-60	N/A	105	125

Good Reported Braking Action

MAX MANUAL	1305	70/-75	35/45	-55/200	30/-25	35/-35	N/A	65	145
AUTOBRAKE MAX	1400	75/-80	35/50	-60/210	30/-25	35/-35	N/A	70	160
AUTOBRAKE 2	2130	135/-145	65/85	-100/335	40/-45	60/-60	N/A	105	125

Medium Reported Braking Action

MAX MANUAL	1755	110/-110	55/70	-90/325	75/-60	50/-50	N/A	170	410
AUTOBRAKE MAX	1810	110/-115	55/75	-90/330	75/-60	50/-50	N/A	175	420
AUTOBRAKE 3	1910	110/-120	55/75	-95/345	60/-45	55/-55	N/A	115	335

Poor Reported Braking Action

MAX MANUAL	2255	155/-155	75/105	-135/515	185/-120	65/-70	N/A	355	925
AUTOBRAKE MAX	2275	155/-155	80/105	-135/515	185/-115	65/-70	N/A	355	925
AUTOBRAKE 3	2300	155/-155	75/105	-135/520	180/-115	65/-70	N/A	355	935

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

All Flaps Up Landing

VREF40 + 55

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1320	190/-80	50/105	-45/200	20/-15	35/-35	45	40	85
AUTOBRAKE MAX	1815	90/-85	45/75	-60/190	5/0	50/-50	75	5	15
AUTOBRAKE 2	3435	195/-225	115/150	-130/435	60/-75	105/-105	115	215	225

Good Reported Braking Action

MAX MANUAL	1800	90/-100	50/70	-70/235	40/-35	50/-50	50	110	255
AUTOBRAKE MAX	1985	90/-100	55/75	-75/245	30/-25	55/-55	75	95	250
AUTOBRAKE 2	3435	195/-225	115/150	-130/435	60/-75	105/-105	115	215	225

Medium Reported Braking Action

MAX MANUAL	2570	155/-160	85/115	-110/390	110/-90	75/-75	70	320	795
AUTOBRAKE MAX	2625	155/-160	90/115	-115/395	105/-85	75/-80	80	325	805
AUTOBRAKE 3	2930	150/-165	90/120	-120/420	65/-45	90/-90	125	185	570

Poor Reported Braking Action

MAX MANUAL	3440	230/-235	130/175	-170/620	270/-180	105/-110	90	715	1990
AUTOBRAKE MAX	3430	230/-235	130/175	-170/615	270/-175	105/-110	95	710	1975
AUTOBRAKE 3	3510	225/-225	130/175	-170/625	245/-155	110/-115	120	675	1955

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1710	100/-105	50/65	-80/280	50/-45	45/-45	60	125	295
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1920	120/-120	55/75	-95/340	75/-60	50/-50	65	185	460
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2465	170/-170	85/115	-140/535	190/-125	70/-75	75	410	1125
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3305	250/-245	120/175	-235/985	600/-290	90/-110	90	1005	3480
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1635	95/-100	45/60	-75/275	50/-40	40/-40	55	110	260
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1825	110/-115	55/70	-90/335	75/-60	45/-50	65	165	395
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2320	155/-155	75/105	-135/520	180/-120	65/-70	75	355	940
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3085	225/-225	110/160	-230/960	560/-270	85/-105	85	860	2845
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1565	90/-95	40/55	-75/270	50/-40	40/-40	60	100	235
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1750	105/-110	50/65	-90/330	75/-60	45/-45	65	150	360
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2220	150/-150	70/100	-135/510	180/-115	60/-65	75	325	850
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	2955	215/-215	105/150	-225/945	550/-265	80/-100	85	795	2590
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Jammed or Restricted Flight Controls (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	970	75/-60	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Good Reported Braking Action

MAX MANUAL	1320	75/-75	35/45	-55/200	30/-25	35/-35	45	70	160
AUTOBRAKE MAX	1420	80/-85	40/50	-60/210	30/-20	35/-35	55	80	175
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Medium Reported Braking Action

MAX MANUAL	1815	120/-120	55/75	-90/330	80/-65	50/-50	60	195	485
AUTOBRAKE MAX	1860	120/-125	60/80	-95/335	75/-60	50/-55	70	200	490
AUTOBRAKE 3	1990	125/-130	60/80	-95/350	55/-40	55/-60	100	125	390

Poor Reported Braking Action

MAX MANUAL	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1165
AUTOBRAKE MAX	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1160
AUTOBRAKE 3	2405	170/-170	85/115	-140/530	185/-115	70/-75	90	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LEADING EDGE FLAPS TRANSIT (Flaps 15)

VREF15 + 15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1080	80/-65	25/35	-40/130	15/-10	25/-25	35	25	55
AUTOBRAKE MAX	1450	75/-75	35/45	-50/165	0/0	35/-35	65	0	5
AUTOBRAKE 2	2575	170/-180	85/110	-110/370	50/-60	75/-75	90	175	210

Good Reported Braking Action

MAX MANUAL	1480	80/-85	40/55	-60/210	35/-30	40/-40	45	85	195
AUTOBRAKE MAX	1605	85/-90	45/60	-65/220	30/-20	40/-40	65	95	215
AUTOBRAKE 2	2575	170/-180	85/110	-110/370	50/-60	75/-75	90	175	210

Medium Reported Braking Action

MAX MANUAL	2030	130/-130	65/90	-100/350	85/-70	55/-60	60	235	575
AUTOBRAKE MAX	2095	135/-135	70/90	-100/355	80/-65	60/-60	70	240	590
AUTOBRAKE 3	2295	135/-145	70/95	-105/370	60/-50	65/-70	95	140	430

Poor Reported Braking Action

MAX MANUAL	2635	185/-185	95/130	-145/545	205/-140	75/-80	75	490	1340
AUTOBRAKE MAX	2655	185/-185	95/130	-145/550	205/-130	80/-85	85	490	1340
AUTOBRAKE 3	2715	185/-185	95/130	-150/555	190/-125	80/-85	90	465	1325

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1085	70/-60	25/35	-40/130	15/-15	25/-25	45	30	50
AUTOBRAKE MAX	1275	65/-70	30/40	-45/150	5/0	30/-30	60	0	5
AUTOBRAKE 2	2395	150/-165	70/90	-105/360	0/-10	70/-70	140	0	0

Good Reported Braking Action

MAX MANUAL	1570	90/-95	45/60	-65/230	45/-40	40/-40	70	120	245
AUTOBRAKE MAX	1605	95/-100	45/60	-70/230	40/-35	40/-45	75	125	255
AUTOBRAKE 2	2395	150/-165	70/90	-105/360	5/-10	70/-70	140	0	0

Medium Reported Braking Action

MAX MANUAL	2165	145/-150	70/95	-105/375	110/-90	60/-65	90	325	775
AUTOBRAKE MAX	2165	150/-150	70/100	-105/375	115/-90	60/-65	90	325	775
AUTOBRAKE 3	2165	150/-150	70/100	-105/375	115/-80	60/-65	90	325	775

Poor Reported Braking Action

MAX MANUAL	2820	210/-210	105/145	-160/580	255/-170	80/-90	105	675	1905
AUTOBRAKE MAX	2820	215/-210	105/145	-160/580	260/-175	85/-90	105	675	1910
AUTOBRAKE 3	2820	215/-210	105/145	-160/580	260/-175	85/-90	105	675	1910

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1035	65/-55	25/30	-40/125	15/-15	25/-25	45	25	45
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	5
AUTOBRAKE 2	2230	140/-150	65/85	-105/345	0/-10	65/-65	130	0	0

Good Reported Braking Action

MAX MANUAL	1490	85/-90	40/55	-65/220	45/-40	40/-40	65	105	210
AUTOBRAKE MAX	1515	90/-95	40/55	-65/225	40/-35	40/-40	70	105	215
AUTOBRAKE 2	2230	140/-150	65/85	-105/345	0/-10	65/-65	130	0	0

Medium Reported Braking Action

MAX MANUAL	2035	135/-135	65/90	-100/360	105/-85	55/-60	85	275	645
AUTOBRAKE MAX	2025	135/-135	65/90	-100/360	110/-85	55/-60	85	275	640
AUTOBRAKE 3	2030	135/-135	65/90	-100/360	105/-75	55/-60	90	275	645

Poor Reported Braking Action

MAX MANUAL	2625	190/-190	95/130	-150/565	240/-155	75/-80	95	565	1530
AUTOBRAKE MAX	2625	195/-190	95/130	-150/565	245/-160	75/-80	95	565	1530
AUTOBRAKE 3	2625	195/-190	95/130	-150/565	245/-160	75/-80	95	565	1530

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	990	60/-50	20/30	-35/125	15/-15	20/-20	45	25	35
AUTOBRAKE MAX	1120	55/-60	25/30	-40/140	0/0	25/-25	55	5	10
AUTOBRAKE 2	2035	125/-140	60/75	-95/330	0/-5	60/-60	125	0	0

Good Reported Braking Action

MAX MANUAL	1420	80/-85	40/50	-65/220	45/-35	35/-35	70	95	190
AUTOBRAKE MAX	1420	80/-90	40/50	-65/220	35/-30	35/-35	70	95	185
AUTOBRAKE 2	2035	125/-140	60/75	-95/330	0/-5	60/-60	125	0	0

Medium Reported Braking Action

MAX MANUAL	1920	125/-130	60/85	-100/355	100/-80	55/-55	85	245	560
AUTOBRAKE MAX	1905	125/-130	60/85	-100/355	105/-80	55/-55	85	240	555
AUTOBRAKE 3	1910	125/-130	60/85	-100/355	105/-75	55/-55	90	240	555

Poor Reported Braking Action

MAX MANUAL	2470	180/-180	85/120	-150/550	230/-150	70/-75	95	495	1305
AUTOBRAKE MAX	2465	180/-180	90/120	-145/550	235/-155	70/-75	95	495	1305
AUTOBRAKE 3	2465	180/-180	90/120	-145/550	235/-155	70/-75	95	495	1305

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1535	80/-85	40/50	-60/195	35/-30	40/-40	75	-5	60
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2240	135/-140	65/90	-100/335	100/-80	60/-60	105	90	425
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2980	200/-205	100/135	-145/520	215/-160	85/-85	125	355	1385
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3710	275/-270	140/195	-210/780	475/-270	105/-115	135	805	3355
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM B (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1095	55/-60	25/35	-40/145	15/-15	25/-25	40	35	55
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1565	95/-95	45/60	-70/245	45/-40	40/-40	55	120	250
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2125	145/-145	70/95	-110/400	115/-90	60/-60	75	305	730
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	2725	200/-200	100/140	-165/625	285/-170	80/-85	85	610	1695
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

MANUAL REVERSION (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1535	80/-85	40/50	-60/195	35/-30	40/-40	75	-5	60
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2240	135/-140	65/90	-100/335	100/-80	60/-60	105	90	425
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2980	200/-205	100/135	-145/520	215/-160	85/-85	125	355	1385
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3710	275/-270	140/195	-210/780	475/-270	105/-115	135	805	3355
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

One Engine Inoperative Landing (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	980	80/-60	20/30	-35/125	10/-10	20/-20	35	0	20
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	0
AUTOBRAKE 2	2370	150/-160	70/90	-105/355	10/-30	70/-70	115	0	5

Good Reported Braking Action

MAX MANUAL	1370	75/-80	35/50	-60/210	35/-30	35/-35	50	0	85
AUTOBRAKE MAX	1485	85/-90	40/50	-65/220	35/-25	40/-40	60	0	95
AUTOBRAKE 2	2370	150/-160	70/90	-105/355	10/-30	70/-70	115	0	5

Medium Reported Braking Action

MAX MANUAL	1965	125/-130	60/80	-100/360	100/-80	55/-55	70	0	270
AUTOBRAKE MAX	2025	130/-135	60/80	-100/365	95/-75	60/-60	80	0	275
AUTOBRAKE 3	2085	135/-140	65/85	-105/370	80/-60	60/-60	95	0	245

Poor Reported Braking Action

MAX MANUAL	2700	190/-195	90/120	-160/590	270/-170	80/-85	85	0	675
AUTOBRAKE MAX	2705	195/-195	90/125	-160/590	270/-165	80/-85	95	0	680
AUTOBRAKE 3	2740	195/-200	95/125	-160/590	265/-170	80/-85	90	0	685

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

One Engine Inoperative Landing (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	940	60/-55	20/25	-35/120	10/-10	20/-20	35	0	20
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	0
AUTOBRAKE 2	2185	140/-145	65/85	-100/340	15/-35	65/-65	100	0	10

Good Reported Braking Action

MAX MANUAL	1310	75/-75	35/45	-60/205	35/-30	35/-35	50	0	75
AUTOBRAKE MAX	1420	80/-85	35/50	-60/215	30/-25	35/-35	60	0	85
AUTOBRAKE 2	2185	140/-145	65/85	-100/340	15/-35	65/-65	100	0	10

Medium Reported Braking Action

MAX MANUAL	1850	115/-120	55/75	-95/350	95/-75	50/-55	65	0	230
AUTOBRAKE MAX	1905	120/-125	55/75	-100/355	90/-70	55/-55	75	0	235
AUTOBRAKE 3	1955	120/-130	60/75	-100/360	80/-65	55/-55	85	0	210

Poor Reported Braking Action

MAX MANUAL	2500	175/-175	85/110	-150/565	245/-155	75/-75	80	0	555
AUTOBRAKE MAX	2505	175/-180	85/110	-150/565	245/-150	75/-80	90	0	555
AUTOBRAKE 3	2535	175/-180	85/110	-155/570	245/-160	75/-80	80	0	565

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Stabilizer Trim Inoperative (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	970	75/-60	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Good Reported Braking Action

MAX MANUAL	1320	75/-75	35/45	-55/200	30/-25	35/-35	45	70	160
AUTOBRAKE MAX	1420	80/-85	40/50	-60/210	30/-20	35/-35	55	80	175
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Medium Reported Braking Action

MAX MANUAL	1815	120/-120	55/75	-90/330	80/-65	50/-50	60	195	485
AUTOBRAKE MAX	1860	120/-125	60/80	-95/335	75/-60	50/-55	70	200	490
AUTOBRAKE 3	1990	125/-130	60/80	-95/350	55/-40	55/-60	100	125	390

Poor Reported Braking Action

MAX MANUAL	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1165
AUTOBRAKE MAX	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1160
AUTOBRAKE 3	2405	170/-170	85/115	-140/530	185/-115	70/-75	90	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Trailing Edge Flap Asymmetry ($1 \leq \text{Flap Lever} < 15$) VREF40 + 30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1075	95/-60	25/40	-40/130	15/-10	25/-25	40	25	55
AUTOBRAKE MAX	1475	70/-75	35/45	-50/170	0/0	35/-35	65	0	5
AUTOBRAKE 2	2615	160/-175	85/110	-110/375	55/-60	80/-80	90	180	220

Good Reported Braking Action

MAX MANUAL	1450	75/-80	40/55	-60/210	30/-30	35/-40	45	80	175
AUTOBRAKE MAX	1610	80/-85	45/55	-65/220	25/-20	40/-40	65	70	175
AUTOBRAKE 2	2615	160/-175	85/110	-110/375	55/-60	80/-80	90	180	220

Medium Reported Braking Action

MAX MANUAL	2015	120/-125	65/85	-95/345	85/-65	55/-60	60	220	535
AUTOBRAKE MAX	2090	125/-130	65/90	-100/350	80/-65	60/-60	70	225	550
AUTOBRAKE 3	2325	125/-135	70/95	-105/375	55/-50	70/-70	95	125	375

Poor Reported Braking Action

MAX MANUAL	2650	180/-180	95/130	-145/545	205/-135	80/-85	75	480	1290
AUTOBRAKE MAX	2670	180/-180	95/130	-145/550	200/-130	80/-85	80	475	1290
AUTOBRAKE 3	2740	175/-180	95/130	-150/555	190/-125	80/-85	95	435	1260

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Asymmetry (Flap Lever 15 or 25)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	970	75/-60	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Good Reported Braking Action

MAX MANUAL	1320	75/-75	35/45	-55/200	30/-25	35/-35	45	70	160
AUTOBRAKE MAX	1420	80/-85	40/50	-60/210	30/-20	35/-35	55	80	175
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Medium Reported Braking Action

MAX MANUAL	1815	120/-120	55/75	-90/330	80/-65	50/-50	60	195	485
AUTOBRAKE MAX	1860	120/-125	60/80	-95/335	75/-60	50/-55	70	200	490
AUTOBRAKE 3	1990	125/-130	60/80	-95/350	55/-40	55/-60	100	125	390

Poor Reported Braking Action

MAX MANUAL	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1165
AUTOBRAKE MAX	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1160
AUTOBRAKE 3	2405	170/-170	85/115	-140/530	185/-115	70/-75	90	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Asymmetry (Flap Lever 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	930	60/-55	20/25	-35/120	10/-10	20/-20	35	20	35
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	5
AUTOBRAKE 2	2100	140/-145	65/85	-100/335	35/-45	60/-60	85	95	100

Good Reported Braking Action

MAX MANUAL	1265	70/-70	35/45	-55/195	30/-25	30/-30	45	65	140
AUTOBRAKE MAX	1355	75/-80	35/45	-60/205	25/-20	35/-35	55	70	155
AUTOBRAKE 2	2100	140/-145	65/85	-100/335	35/-45	60/-60	85	95	100

Medium Reported Braking Action

MAX MANUAL	1715	110/-110	55/70	-90/325	75/-60	45/-50	60	170	415
AUTOBRAKE MAX	1760	115/-115	55/70	-90/325	70/-55	50/-50	70	175	425
AUTOBRAKE 3	1870	115/-120	55/75	-95/340	55/-45	50/-55	85	115	340

Poor Reported Braking Action

MAX MANUAL	2220	155/-155	75/105	-135/510	180/-120	65/-70	70	365	970
AUTOBRAKE MAX	2230	155/-155	75/105	-135/510	185/-115	65/-70	80	365	965
AUTOBRAKE 3	2260	160/-160	75/105	-135/515	175/-115	65/-70	80	365	975

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree ($1 \leq$ Indicated Flaps <15)

VREF40 + 30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1075	95/-60	25/40	-40/130	15/-10	25/-25	40	25	55
AUTOBRAKE MAX	1475	70/-75	35/45	-50/170	0/0	35/-35	65	0	5
AUTOBRAKE 2	2615	160/-175	85/110	-110/375	55/-60	80/-80	90	180	220

Good Reported Braking Action

MAX MANUAL	1450	75/-80	40/55	-60/210	30/-30	35/-40	45	80	175
AUTOBRAKE MAX	1610	80/-85	45/55	-65/220	25/-20	40/-40	65	70	175
AUTOBRAKE 2	2615	160/-175	85/110	-110/375	55/-60	80/-80	90	180	220

Medium Reported Braking Action

MAX MANUAL	2015	120/-125	65/85	-95/345	85/-65	55/-60	60	220	535
AUTOBRAKE MAX	2090	125/-130	65/90	-100/350	80/-65	60/-60	70	225	550
AUTOBRAKE 3	2325	125/-135	70/95	-105/375	55/-50	70/-70	95	125	375

Poor Reported Braking Action

MAX MANUAL	2650	180/-180	95/130	-145/545	205/-135	80/-85	75	480	1290
AUTOBRAKE MAX	2670	180/-180	95/130	-145/550	200/-130	80/-85	80	475	1290
AUTOBRAKE 3	2740	175/-180	95/130	-150/555	190/-125	80/-85	95	435	1260

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (15 ≤ Indicated Flaps <30)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	970	75/-60	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Good Reported Braking Action

MAX MANUAL	1320	75/-75	35/45	-55/200	30/-25	35/-35	45	70	160
AUTOBRAKE MAX	1420	80/-85	40/50	-60/210	30/-20	35/-35	55	80	175
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Medium Reported Braking Action

MAX MANUAL	1815	120/-120	55/75	-90/330	80/-65	50/-50	60	195	485
AUTOBRAKE MAX	1860	120/-125	60/80	-95/335	75/-60	50/-55	70	200	490
AUTOBRAKE 3	1990	125/-130	60/80	-95/350	55/-40	55/-60	100	125	390

Poor Reported Braking Action

MAX MANUAL	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1165
AUTOBRAKE MAX	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1160
AUTOBRAKE 3	2405	170/-170	85/115	-140/530	185/-115	70/-75	90	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (30 ≤ Indicated Flaps <40)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	930	60/-55	20/25	-35/120	10/-10	20/-20	35	20	35
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	5
AUTOBRAKE 2	2100	140/-145	65/85	-100/335	35/-45	60/-60	85	95	100

Good Reported Braking Action

MAX MANUAL	1265	70/-70	35/45	-55/195	30/-25	30/-30	45	65	140
AUTOBRAKE MAX	1355	75/-80	35/45	-60/205	25/-20	35/-35	55	70	155
AUTOBRAKE 2	2100	140/-145	65/85	-100/335	35/-45	60/-60	85	95	100

Medium Reported Braking Action

MAX MANUAL	1715	110/-110	55/70	-90/325	75/-60	45/-50	60	170	415
AUTOBRAKE MAX	1760	115/-115	55/70	-90/325	70/-55	50/-50	70	175	425
AUTOBRAKE 3	1870	115/-120	55/75	-95/340	55/-45	50/-55	85	115	340

Poor Reported Braking Action

MAX MANUAL	2220	155/-155	75/105	-135/510	180/-120	65/-70	70	365	970
AUTOBRAKE MAX	2230	155/-155	75/105	-135/510	185/-115	65/-70	80	365	965
AUTOBRAKE 3	2260	160/-160	75/105	-135/515	175/-115	65/-70	80	365	975

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flaps Up Landing

VREF40 + 40

	LANDING DISTANCE AND ADJUSTMENTS (M)									
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ		
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV	

Dry Runway

MAX MANUAL	1175	110/-70	30/70	-40/140	15/-15	30/-30	40	30	65
AUTOBRAKE MAX	1605	75/-75	40/50	-55/175	0/0	40/-40	70	0	10
AUTOBRAKE 2	2980	175/-195	100/125	-120/400	55/-65	90/-90	105	170	180

Good Reported Braking Action

MAX MANUAL	1620	85/-90	45/60	-65/225	40/-35	45/-45	50	95	210
AUTOBRAKE MAX	1770	85/-95	50/65	-70/235	30/-20	45/-50	70	90	220
AUTOBRAKE 2	2980	175/-195	100/125	-120/400	55/-65	90/-90	105	170	180

Medium Reported Braking Action

MAX MANUAL	2285	135/-145	75/100	-105/370	100/-80	65/-70	65	270	655
AUTOBRAKE MAX	2335	140/-145	75/100	-105/375	95/-75	65/-70	75	270	665
AUTOBRAKE 3	2565	135/-150	80/105	-115/395	60/-45	75/-80	115	160	485

Poor Reported Braking Action

MAX MANUAL	3035	205/-210	110/150	-160/585	240/-160	90/-95	85	590	1610
AUTOBRAKE MAX	3025	205/-210	110/150	-160/585	240/-155	90/-95	90	585	1600
AUTOBRAKE 3	3090	200/-205	110/150	-160/590	225/-140	95/-100	110	565	1595

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.



ADVISORY INFORMATION

Recommended Brake Cooling Schedule
Reference Brake Energy Per Brake (Millions of Foot Pounds)

		WIND CORRECTED BRAKES ON SPEED (KIAS)*																	
		80			100			120			140			160			180		
WEIGHT (1000 KG)	OAT (°C)	PRESSURE ALTITUDE (1000 FT)																	
		0	5	10	0	5	10	0	5	10	0	5	10	0	5	10	0	5	10
80	0	15.1	17.0	19.3	22.4	25.3	28.9	30.9	35.0	40.2	40.4	45.9	53.0	50.8	57.9	67.3	60.8	69.6	81.2
	10	15.6	17.6	20.0	23.1	26.1	29.8	31.9	36.2	41.5	41.8	47.5	54.8	52.5	59.9	69.5	62.8	71.9	83.9
	15	15.8	17.8	20.2	23.5	26.5	30.3	32.4	36.7	42.1	42.4	48.2	55.6	53.3	60.7	70.5	63.7	72.9	85.1
	20	16.0	18.1	20.5	23.8	26.9	30.7	32.8	37.2	42.7	42.9	48.8	56.3	54.0	61.5	71.4	64.6	73.9	86.2
	30	16.4	18.5	21.1	24.4	27.6	31.5	33.7	38.2	43.8	44.0	50.0	57.7	55.3	63.1	73.2	66.2	75.7	88.4
	40	16.6	18.7	21.3	24.7	27.9	31.9	34.1	38.7	44.4	44.7	50.9	58.8	56.3	64.3	74.8	67.5	77.4	90.5
50	16.6	18.7	21.3	24.8	28.0	32.1	34.3	39.0	44.9	45.2	51.5	59.7	57.1	65.4	76.3	68.7	79.0	92.9	
70	0	13.7	15.4	17.5	20.2	22.8	26.0	27.7	31.3	35.9	36.1	41.0	47.2	45.3	51.6	59.7	54.9	62.7	72.9
	10	14.2	15.9	18.1	20.8	23.5	26.8	28.6	32.4	37.1	37.3	42.3	48.7	46.8	53.3	61.6	56.7	64.8	75.4
	15	14.4	16.2	18.4	21.1	23.9	27.2	29.0	32.8	37.6	37.8	43.0	49.4	47.5	54.0	62.5	57.5	65.7	76.4
	20	14.6	16.4	18.6	21.4	24.2	27.6	29.4	33.3	38.1	38.4	43.5	50.1	48.1	54.8	63.4	58.3	66.5	77.4
	30	14.9	16.8	19.1	22.0	24.8	28.3	30.2	34.1	39.1	39.3	44.6	51.4	49.3	56.1	64.9	59.8	68.2	79.4
	40	15.1	17.0	19.3	22.2	25.1	28.6	30.5	34.6	39.6	39.9	45.3	52.2	50.1	57.1	66.2	60.9	69.6	81.2
50	15.1	17.0	19.3	22.3	25.2	28.8	30.7	34.8	40.0	40.2	45.8	52.9	50.7	58.0	67.4	61.8	70.9	83.0	
60	0	12.3	13.9	15.7	18.0	20.3	23.1	24.4	27.6	31.6	31.7	35.9	41.2	39.6	45.0	51.8	48.1	54.8	63.5
	10	12.7	14.3	16.3	18.5	20.9	23.8	25.2	28.5	32.6	32.7	37.1	42.6	40.9	46.5	53.6	49.7	56.6	65.6
	15	12.9	14.6	16.5	18.8	21.2	24.2	25.6	29.0	33.1	33.2	37.6	43.2	41.5	47.1	54.4	50.4	57.4	66.5
	20	13.1	14.8	16.7	19.1	21.5	24.5	26.0	29.4	33.5	33.6	38.1	43.8	42.0	47.8	55.1	51.1	58.2	67.4
	30	13.4	15.1	17.2	19.6	22.1	25.1	26.6	30.1	34.4	34.5	39.1	44.9	43.1	49.0	56.5	52.3	59.6	69.1
	40	13.6	15.3	17.3	19.8	22.3	25.4	26.9	30.5	34.9	35.0	39.7	45.6	43.8	49.8	57.5	53.2	60.7	70.5
50	13.5	15.3	17.3	19.8	22.4	25.5	27.0	30.6	35.1	35.2	40.0	46.0	44.2	50.4	58.3	53.9	61.7	71.9	
50	0	11.0	12.3	14.0	15.7	17.7	20.2	21.2	23.9	27.3	27.2	30.8	35.3	33.8	38.3	44.1	40.9	46.4	53.6
	10	11.3	12.7	14.4	16.3	18.3	20.8	21.9	24.7	28.2	28.1	31.8	36.5	34.9	39.6	45.5	42.2	48.0	55.4
	15	11.5	12.9	14.7	16.5	18.6	21.1	22.2	25.1	28.6	28.6	32.3	37.0	35.4	40.2	46.2	42.8	48.7	56.2
	20	11.6	13.1	14.9	16.7	18.9	21.4	22.5	25.4	29.0	28.9	32.8	37.5	35.9	40.7	46.8	43.4	49.3	56.9
	30	11.9	13.4	15.2	17.2	19.3	22.0	23.1	26.1	29.7	29.7	33.6	38.4	36.8	41.8	48.0	44.5	50.6	58.4
	40	12.1	13.6	15.4	17.3	19.5	22.2	23.4	26.4	30.1	30.1	34.0	39.0	37.4	42.4	48.8	45.2	51.4	59.4
50	12.0	13.6	15.4	17.3	19.6	22.3	23.4	26.5	30.3	30.2	34.2	39.3	37.6	42.8	49.3	45.7	52.1	60.3	
40	0	9.6	10.8	12.3	13.5	15.2	17.3	17.9	20.2	23.0	22.8	25.8	29.4	28.1	31.8	36.4	33.7	38.2	43.9
	10	10.0	11.2	12.7	14.0	15.8	17.9	18.5	20.9	23.8	23.6	26.6	30.4	29.0	32.8	37.6	34.8	39.5	45.4
	15	10.1	11.4	12.9	14.2	16.0	18.1	18.8	21.2	24.1	23.9	27.0	30.8	29.4	33.3	38.2	35.3	40.0	46.0
	20	10.2	11.5	13.1	14.4	16.2	18.4	19.1	21.5	24.5	24.2	27.4	31.3	29.8	33.8	38.7	35.8	40.6	46.6
	30	10.5	11.8	13.4	14.8	16.6	18.9	19.6	22.1	25.1	24.9	28.1	32.1	30.6	34.6	39.7	36.7	41.6	47.8
	40	10.6	11.9	13.5	14.9	16.8	19.1	19.8	22.3	25.4	25.2	28.4	32.5	31.0	35.1	40.2	37.2	42.2	48.6
50	10.6	11.9	13.5	14.9	16.8	19.1	19.8	22.3	25.5	25.2	28.6	32.7	31.1	35.3	40.6	37.5	42.6	49.1	

*To correct for wind, enter table with the brakes on speed minus one half the headwind or plus 1.5 times the tailwind. If ground speed is used for brakes on speed, ignore wind and enter table with sea level, 15°C.

ADVISORY INFORMATION

Recommended Brake Cooling Schedule

Adjusted Brake Energy Per Brake (Millions of Foot Pounds)

No Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	7.8	16.3	25.3	34.7	44.7	55.0	65.7	76.6	87.9
	MAX AUTO	7.5	15.4	23.6	32.4	41.8	51.8	62.5	74.1	86.5
	AUTOBRAKE 3	7.3	14.7	22.3	30.2	38.6	47.6	57.4	68.1	80.0
	AUTOBRAKE 2	7.0	13.8	20.5	27.4	34.8	42.7	51.5	61.3	72.4
	AUTOBRAKE 1	6.7	13.1	19.2	25.3	31.8	38.8	46.6	55.4	65.5

Two Engine Detent Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	7.0	14.6	22.8	31.4	40.5	49.9	59.7	69.8	80.0
	MAX AUTO	5.8	12.3	19.5	27.2	35.6	44.5	53.9	63.7	74.1
	AUTOBRAKE 3	4.3	9.2	14.7	20.7	27.2	34.4	42.0	50.2	59.0
	AUTOBRAKE 2	2.5	5.6	9.1	13.1	17.8	23.0	28.8	35.2	42.3
	AUTOBRAKE 1	1.8	3.8	6.1	8.8	11.9	15.5	19.6	24.4	29.8

Cooling Time (Minutes) - Category C Steel Brakes

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)								
		16 & BELOW	17	20	23	25	28	32	33 TO 48	49 & ABOVE
		BRAKE TEMPERATURE MONITOR SYSTEM INDICATION ON CDS								
		UP TO 2.4	2.6	3.1	3.5	3.9	4.4	4.9	5.0 TO 7.5	7.5 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE	1	2	3	4	5	6	CAUTION	FUSE PLUG MELT ZONE	
GROUND	REQUIRED	10	20	30	40	50	60			

Cooling Time (Minutes) - Category N Carbon Brakes

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)								
		16 & BELOW	17	19	20.9	23.5	26.9	29.4	30 TO 41	41 & ABOVE
		BRAKE TEMPERATURE MONITOR SYSTEM INDICATION ON CDS								
		UP TO 2.5	2.6	3	3.3	3.8	4.5	4.9	5.0 TO 7.1	7.1 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE	1	4	5	6	7	7.6	CAUTION	FUSE PLUG MELT ZONE	
GROUND	REQUIRED	6.7	16.0	24.1	34.2	45.9	53.3			

Observe maximum quick turnaround limit.

Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.

Add 1.0 million foot pounds per brake for each taxi mile.

When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after one hour. If overheat occurs after takeoff, extend gear soon for at least 7 minutes.

When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not approach gear or attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

Brake temperature monitor system (BTMS) indication on CDS systems page may be used 10 to 15 minutes after airplane has come to a complete stop or inflight with gear retracted to determine recommended cooling schedule.



Performance Inflight - QRH

Chapter PI-QRH

Engine Inoperative

Section 62

ENGINE INOP

Initial Max Continuous %N1

Based on .79M, A/C high and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)								
	25	27	29	31	33	35	37	39	41
20	96.8	96.6	96.3	96.1	95.9	95.4	95.0	94.7	93.9
15	97.4	97.2	96.9	96.8	96.6	96.2	95.7	95.5	94.8
10	98.0	97.8	97.5	97.4	97.4	96.9	96.5	96.3	95.7
5	98.3	98.6	98.3	98.1	98.1	97.7	97.3	97.1	96.6
0	97.5	98.7	99.2	99.0	98.9	98.5	98.2	98.0	97.5
-5	96.7	98.0	99.1	99.8	99.7	99.3	98.9	98.7	98.4
-10	96.0	97.2	98.4	99.6	100.5	100.2	99.8	99.6	99.4
-15	95.2	96.4	97.6	98.8	100.1	101.0	100.8	100.6	100.3
-20	94.4	95.6	96.8	98.0	99.3	100.5	101.1	100.8	100.6
-25	93.6	94.9	96.0	97.2	98.5	99.7	100.2	100.0	99.8
-30	92.8	94.1	95.2	96.4	97.7	98.8	99.4	99.2	99.0
-35	92.0	93.2	94.4	95.6	96.8	98.0	98.5	98.3	98.1
-40	91.2	92.4	93.5	94.7	96.0	97.1	97.6	97.4	97.2

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)								
	25	27	29	31	33	35	37	39	41
ENGINE ANTI-ICE	-1.2	-1.1	-1.0	-0.9	-0.8	-0.8	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE	-4.2	-4.4	-4.5	-4.7	-5.0	-4.8	-4.8	-4.8	-4.8

ENGINE INOP

**Max Continuous %N1
37000 FT to 29000 FT Pressure Altitudes**

37000 FT PRESS ALT													TAT (°C)
CIAS	M	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0
160	.51	96.6	97.6	98.5	99.4	100.2	99.6	98.8	97.6	96.3	94.7	93.2	91.8
200	.63	96.0	96.9	97.8	98.7	99.6	100.4	100.1	99.3	98.4	97.5	96.3	95.2
240	.74	95.1	96.0	96.8	97.7	98.6	99.4	100.3	100.7	100.0	99.2	98.4	97.5
280	.86	94.3	95.2	96.1	97.0	97.8	98.7	99.5	100.4	101.2	100.9	100.0	99.1
35000 FT PRESS ALT													TAT (°C)
CIAS	M	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0
160	.49	96.5	97.4	98.3	99.2	100.1	99.8	99.0	98.0	96.8	95.4	94.0	92.7
200	.60	96.1	97.0	97.9	98.8	99.7	100.6	100.5	99.6	98.6	97.6	96.5	95.4
240	.71	95.0	95.9	96.8	97.7	98.6	99.4	100.3	100.8	100.2	99.5	98.6	97.7
280	.82	93.8	94.6	95.5	96.4	97.3	98.1	98.9	99.8	100.6	100.3	99.5	98.8
33000 FT PRESS ALT													TAT (°C)
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
160	.47	97.4	98.3	99.2	100.0	100.8	100.0	99.1	97.9	96.7	95.3	93.9	92.6
200	.58	97.0	97.9	98.8	99.7	100.6	101.4	100.6	99.6	98.6	97.5	96.3	95.1
240	.68	95.9	96.8	97.7	98.5	99.4	100.2	101.1	100.9	100.2	99.4	98.4	97.4
280	.79	94.3	95.1	96.0	96.8	97.7	98.5	99.3	100.2	100.5	99.7	98.9	98.1
320	.89	93.6	94.5	95.4	96.2	97.1	97.9	98.7	99.5	100.3	101.1	100.7	99.8
31000 FT PRESS ALT													TAT (°C)
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
160	.45	97.3	98.2	99.1	100.0	100.9	101.1	100.2	99.2	98.0	96.6	95.2	93.9
200	.55	97.1	98.0	98.9	99.7	100.6	101.5	101.6	100.7	99.7	98.6	97.4	96.2
240	.66	95.6	96.5	97.4	98.3	99.1	100.0	100.8	101.3	100.5	99.8	98.8	97.8
280	.76	93.8	94.7	95.5	96.4	97.2	98.0	98.8	99.7	100.5	99.8	98.9	98.0
320	.85	92.4	93.2	94.1	94.9	95.7	96.5	97.4	98.2	98.9	99.7	99.9	99.1
29000 FT PRESS ALT													TAT (°C)
CIAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10
160	.43	98.1	99.0	99.9	100.8	101.6	101.2	100.2	99.1	97.9	96.4	95.1	93.8
200	.53	97.5	98.4	99.3	100.2	101.0	101.9	101.3	100.4	99.3	98.2	96.9	95.8
240	.63	96.3	97.1	98.0	98.9	99.7	100.5	101.4	101.1	100.2	99.2	98.3	97.2
280	.73	94.2	95.0	95.9	96.7	97.5	98.3	99.1	99.9	100.1	99.1	98.2	97.5
320	.82	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.5	99.2	98.5	97.6
360	.91	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.5	99.2	100.0	100.1

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	29	31	33	35	37
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE ON	-4.1	-4.3	-4.5	-4.7	-4.7



ENGINE INOP

Max Continuous %N1

27000 FT to 20000 FT Pressure Altitudes

27000 FT PRESS ALT												TAT (°C)	
KIAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10
160	.41	98.0	98.8	99.7	100.6	101.4	102.2	101.2	100.2	99.0	97.8	96.4	95.1
200	.51	96.9	97.8	98.7	99.6	100.4	101.2	101.8	100.8	99.9	98.8	97.6	96.4
240	.60	95.6	96.5	97.4	98.2	99.1	99.9	100.7	101.3	100.4	99.4	98.5	97.5
280	.70	93.6	94.4	95.3	96.1	96.9	97.7	98.5	99.3	100.1	99.4	98.4	97.6
320	.79	91.6	92.4	93.2	94.0	94.8	95.6	96.4	97.2	98.0	98.7	98.6	97.8
360	.88	91.0	91.8	92.6	93.4	94.2	95.0	95.8	96.6	97.3	98.1	98.8	99.4
25000 FT PRESS ALT												TAT (°C)	
KIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
160	.39	98.8	99.7	100.5	101.4	102.2	102.4	101.4	100.3	99.1	97.7	96.5	95.2
200	.49	97.5	98.3	99.2	100.0	100.9	101.7	101.5	100.6	99.5	98.4	97.3	96.2
240	.58	95.7	96.5	97.4	98.2	99.0	99.9	100.7	100.5	99.5	98.6	97.6	96.7
280	.67	93.9	94.7	95.5	96.3	97.1	97.9	98.7	99.5	99.5	98.6	97.6	96.9
320	.76	91.7	92.6	93.4	94.2	95.0	95.8	96.5	97.3	98.0	98.6	97.8	97.2
360	.85	90.4	91.2	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.6	98.4	98.2
24000 FT PRESS ALT												TAT (°C)	
KIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
160	.38	98.6	99.5	100.4	101.2	102.1	102.9	101.9	100.8	99.6	98.4	97.1	95.8
200	.48	97.5	98.4	99.2	100.1	100.9	101.8	102.2	101.1	100.1	99.0	97.8	96.7
240	.57	95.9	96.8	97.6	98.5	99.3	100.1	100.9	101.2	100.2	99.2	98.2	97.3
280	.66	94.2	95.1	95.9	96.7	97.5	98.3	99.1	99.9	100.4	99.4	98.3	97.5
320	.75	92.1	93.0	93.8	94.6	95.4	96.2	96.9	97.7	98.5	99.2	98.6	97.8
360	.83	90.6	91.4	92.2	93.1	93.9	94.7	95.5	96.2	97.0	97.8	98.5	98.6
22000 FT PRESS ALT												TAT (°C)	
KIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
160	.37	99.1	100.0	100.9	101.7	102.5	102.8	101.8	100.7	99.5	98.2	97.0	95.8
200	.46	98.4	99.3	100.1	101.0	101.8	102.6	102.3	101.2	100.0	98.9	97.8	96.8
240	.55	97.2	98.1	98.9	99.7	100.5	101.3	102.1	101.6	100.5	99.4	98.5	97.5
280	.63	95.7	96.5	97.4	98.2	99.0	99.8	100.6	101.3	101.0	99.8	98.9	98.1
320	.72	93.9	94.7	95.5	96.3	97.1	97.9	98.6	99.4	100.1	100.2	99.3	98.6
360	.80	92.2	93.0	93.8	94.6	95.4	96.1	96.9	97.7	98.4	99.2	99.7	99.1
20000 FT PRESS ALT												TAT (°C)	
KIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
160	.35	98.7	99.5	100.4	101.2	102.0	102.8	102.5	101.5	100.4	99.2	98.0	96.8
200	.44	98.3	99.2	100.0	100.9	101.7	102.5	103.3	102.3	101.1	100.0	98.9	97.8
240	.53	97.5	98.4	99.2	100.0	100.8	101.7	102.5	103.1	101.8	100.5	99.5	98.6
280	.61	96.2	97.0	97.8	98.7	99.5	100.3	101.1	101.8	102.5	101.3	100.1	99.3
320	.69	94.7	95.5	96.3	97.1	97.9	98.7	99.5	100.2	101.0	101.7	100.9	99.9
360	.77	93.0	93.8	94.6	95.4	96.2	97.0	97.7	98.5	99.2	100.0	100.7	100.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	20	22	24	25	27
ENGINE ANTI-ICE ON	-0.9	-0.9	-1.0	-1.0	-1.0
ENGINE & WING ANTI-ICE ON	-3.6	-3.8	-3.8	-3.9	-4.0

ENGINE INOP

Max Continuous %N1
18000 FT to 12000 FT Pressure Altitudes

18000 FT PRESS ALT												TAT (°C)	
KLAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
160	.34	98.5	99.3	100.2	101.0	101.8	102.6	101.6	100.3	99.2	98.1	97.0	95.9
200	.42	98.7	99.6	100.4	101.2	102.0	102.8	103.1	101.7	100.4	99.3	98.3	97.3
240	.51	97.8	98.7	99.5	100.3	101.1	101.9	102.7	102.5	101.1	99.9	99.0	98.1
280	.59	96.3	97.1	97.9	98.7	99.5	100.3	101.0	101.8	101.6	100.5	99.6	98.8
320	.67	94.8	95.6	96.4	97.2	97.9	98.7	99.5	100.2	101.0	100.9	100.0	99.2
360	.75	93.0	93.8	94.6	95.3	96.1	96.9	97.6	98.4	99.1	99.9	100.2	99.6
16000 FT PRESS ALT												TAT (°C)	
KLAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
160	.33	97.1	98.0	98.8	99.6	100.4	101.2	101.6	100.3	99.1	98.1	97.1	96.1
200	.41	98.0	98.8	99.6	100.4	101.2	102.0	102.8	102.5	101.3	100.2	99.3	98.3
240	.49	97.1	97.9	98.7	99.5	100.3	101.1	101.9	102.7	101.8	100.5	99.6	98.7
280	.57	95.6	96.4	97.2	98.0	98.8	99.6	100.3	101.1	101.8	100.9	99.8	99.0
320	.64	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.4	100.2	100.9	100.2	99.4
360	.72	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.4	99.2	99.9	99.6
14000 FT PRESS ALT												TAT (°C)	
KLAS	M	-25	-20	-15	-10	-5	0	5	10	15	20	25	30
160	.31	96.6	97.4	98.2	99.0	99.8	100.6	100.4	99.1	98.0	97.1	96.2	95.3
200	.39	97.1	97.9	98.7	99.5	100.3	101.1	101.8	101.5	101.0	100.1	99.3	98.4
240	.47	96.6	97.4	98.2	99.0	99.8	100.6	101.3	101.8	101.1	100.3	99.5	98.7
280	.54	95.5	96.3	97.1	97.8	98.6	99.4	100.1	100.9	101.0	100.1	99.2	98.5
320	.62	94.1	94.9	95.7	96.5	97.2	98.0	98.7	99.5	100.2	100.3	99.5	98.8
360	.69	92.2	93.1	93.9	94.7	95.5	96.3	97.0	97.8	98.6	99.3	99.6	99.0
12000 FT PRESS ALT												TAT (°C)	
KLAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35
160	.30	96.3	97.0	97.8	98.6	99.4	100.1	99.3	98.1	97.1	96.3	95.4	94.5
200	.38	97.1	97.9	98.7	99.5	100.3	101.0	101.5	100.8	99.8	99.0	98.2	97.3
240	.45	96.5	97.3	98.0	98.8	99.6	100.3	101.1	101.0	100.1	99.4	98.6	97.9
280	.52	95.5	96.3	97.0	97.8	98.6	99.3	100.0	100.8	100.3	99.4	98.6	98.0
320	.60	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.4	100.2	99.7	98.9	98.2
360	.67	92.3	93.2	94.0	94.8	95.6	96.4	97.1	97.9	98.7	99.4	99.1	98.5

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)			
	12	14	16	18
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.9	-0.9
ENGINE & WING ANTI-ICE ON	-3.2	-3.4	-3.4	-3.5

ENGINE INOP

Max Continuous %N1

10000 FT to 1000 FT Pressure Altitudes

10000 FT PRESS ALT		TAT (°C)											
KIAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35
160	.29	95.2	96.0	96.8	97.6	98.3	99.1	99.8	98.6	97.4	96.6	95.8	94.9
200	.36	96.0	96.7	97.5	98.3	99.0	99.8	100.5	100.5	99.4	98.5	97.8	97.0
240	.43	95.6	96.4	97.2	97.9	98.7	99.4	100.2	100.9	100.1	99.2	98.4	97.7
280	.51	94.5	95.3	96.1	96.9	97.6	98.4	99.1	99.9	100.4	99.5	98.7	98.0
320	.58	93.0	93.9	94.7	95.5	96.2	97.0	97.8	98.6	99.3	99.7	99.0	98.2
360	.65	91.6	92.4	93.2	94.0	94.8	95.6	96.4	97.2	98.0	98.7	99.1	98.5
5000 FT PRESS ALT		TAT (°C)											
KIAS	M	-10	-5	0	5	10	15	20	25	30	35	40	45
160	.26	94.9	95.7	96.4	97.2	98.0	98.8	99.2	98.3	97.4	96.6	95.9	95.1
200	.33	94.7	95.5	96.3	97.1	97.8	98.6	99.4	98.9	98.0	97.3	96.6	95.8
240	.40	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.5	98.7	97.9	97.2	96.5
280	.46	93.3	94.1	94.9	95.7	96.5	97.3	98.1	98.8	98.9	98.2	97.5	96.8
320	.53	92.5	93.3	94.1	94.9	95.7	96.5	97.2	98.0	98.7	98.4	97.7	97.1
360	.59	91.5	92.3	93.1	93.9	94.7	95.5	96.2	97.0	97.8	98.5	98.0	97.3
3000 FT PRESS ALT		TAT (°C)											
KIAS	M	-5	0	5	10	15	20	25	30	35	40	45	50
160	.26	94.8	95.6	96.4	97.2	98.0	98.7	98.8	97.9	97.1	96.4	95.6	94.8
200	.32	94.5	95.3	96.1	96.9	97.6	98.4	99.2	98.3	97.5	96.8	96.1	95.3
240	.38	94.1	94.9	95.6	96.4	97.2	98.0	98.7	98.8	98.0	97.2	96.6	95.9
280	.45	93.2	94.0	94.8	95.6	96.4	97.2	97.9	98.7	98.3	97.5	96.9	96.2
320	.51	92.5	93.3	94.1	94.9	95.7	96.4	97.2	98.0	98.5	97.8	97.1	96.5
360	.57	91.6	92.4	93.2	94.0	94.7	95.5	96.3	97.1	97.8	98.1	97.4	96.8
1000 FT PRESS ALT		TAT (°C)											
KIAS	M	-5	0	5	10	15	20	25	30	35	40	45	50
160	.25	93.9	94.7	95.4	96.2	97.0	97.8	98.5	98.2	97.4	96.7	96.0	95.2
200	.31	93.5	94.3	95.1	95.9	96.7	97.4	98.2	98.5	97.8	97.0	96.3	95.6
240	.37	93.0	93.8	94.6	95.4	96.1	96.9	97.7	98.4	98.1	97.3	96.6	95.9
280	.43	92.3	93.2	93.9	94.7	95.5	96.3	97.1	97.8	98.3	97.6	96.9	96.2
320	.49	91.6	92.4	93.2	94.0	94.8	95.6	96.3	97.1	97.9	97.9	97.2	96.5
360	.55	90.7	91.5	92.3	93.1	93.9	94.7	95.4	96.2	96.9	97.7	97.3	96.6

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)			
	1	3	5	10
ENGINE ANTI-ICE ON	-0.6	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE ON	-2.9	-3.0	-2.7	-3.2

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	82	271	18500	17300	15900
80	77	263	20200	19000	17700
75	72	255	21600	20600	19400
70	67	247	23100	22200	21100
65	62	238	24700	23800	22800
60	57	229	26800	25800	24700
55	53	219	29100	28100	27000
50	48	209	31200	30400	29400
45	43	199	33300	32600	31700
40	38	187	35600	34900	34000

Includes APU fuel burn.



ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown/LRC Cruise Range Capability Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
138	128	120	112	106	100	95	90	86	82	78
275	256	239	225	212	200	190	180	172	164	157
413	384	359	337	317	300	284	270	258	246	235
551	512	479	449	423	400	379	360	344	328	314
689	640	598	562	529	500	474	451	429	410	392
826	768	718	674	635	600	569	541	515	492	471
964	896	838	786	741	700	664	631	601	574	549
1102	1025	957	898	846	800	758	721	687	656	628
1240	1153	1077	1011	952	900	853	811	773	738	706
1377	1281	1197	1123	1058	1000	948	901	859	820	785
1515	1409	1317	1235	1164	1100	1043	991	945	902	863
1653	1537	1436	1348	1270	1200	1138	1081	1030	984	942
1792	1666	1556	1460	1375	1300	1232	1171	1116	1066	1020
1930	1794	1676	1573	1481	1400	1327	1261	1202	1148	1098
2068	1922	1796	1685	1587	1500	1422	1351	1288	1230	1177
2207	2051	1916	1798	1693	1600	1517	1441	1373	1312	1255
2345	2180	2036	1910	1799	1700	1611	1531	1459	1393	1333
2484	2309	2156	2023	1905	1800	1706	1621	1545	1475	1411

Driftdown/Cruise Fuel and Time

AIR DIST (NM)	FUEL REQUIRED (1000 KG)										TIME (HR:MIN)
	WEIGHT AT START OF DRIFTDOWN (1000 KG)										
	40	45	50	55	60	65	70	75	80	85	
100	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0:16
200	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.3	0:33
300	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	0:49
400	1.6	1.8	1.9	2.0	2.2	2.3	2.5	2.6	2.8	2.9	1:06
500	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.3	3.5	3.7	1:22
600	2.4	2.7	2.9	3.1	3.3	3.6	3.8	4.0	4.3	4.5	1:39
700	2.8	3.1	3.4	3.6	3.9	4.2	4.5	4.7	5.0	5.3	1:55
800	3.2	3.6	3.9	4.2	4.5	4.8	5.1	5.4	5.7	6.1	2:11
900	3.6	4.0	4.3	4.7	5.0	5.4	5.7	6.1	6.4	6.8	2:28
1000	4.0	4.4	4.8	5.2	5.6	6.0	6.4	6.7	7.1	7.6	2:44
1100	4.4	4.8	5.3	5.7	6.1	6.6	7.0	7.4	7.9	8.3	3:01
1200	4.8	5.3	5.7	6.2	6.7	7.1	7.6	8.1	8.6	9.0	3:17
1300	5.2	5.7	6.2	6.7	7.2	7.7	8.2	8.7	9.2	9.8	3:34
1400	5.5	6.1	6.6	7.2	7.7	8.3	8.8	9.4	9.9	10.5	3:51
1500	5.9	6.5	7.1	7.7	8.3	8.9	9.4	10.0	10.6	11.2	4:07
1600	6.3	6.9	7.5	8.2	8.8	9.4	10.0	10.7	11.3	12.0	4:24
1700	6.6	7.3	8.0	8.6	9.3	10.0	10.6	11.3	12.0	12.7	4:41
1800	7.0	7.7	8.4	9.1	9.8	10.5	11.2	11.9	12.6	13.4	4:57

Includes APU fuel burn.

Driftdown at optimum driftdown speed and cruise at long range cruise speed.

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Altitude Capability

100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	15200	12600	9900
80	17200	15300	12500
75	19200	17400	15000
70	20900	19700	17300
65	22500	21300	19800
60	24100	23000	21600
55	26300	24800	23500
50	29000	27700	25800
45	31400	30500	29200
40	33800	33000	31800

With engine anti-ice on, decrease altitude capability by 1200 ft.

With engine and wing anti-ice on, decrease altitude capability by 5500 ft.



ENGINE INOP

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		10	15	17	19	21	23	25	27	29	31
85	%N1	91.8	95.5	97.9							
	MACH	.561	.600	.616							
	KIAS	311	303	300							
	FF/ENG	3067	3033	3052							
80	%N1	90.1	94.0	95.9	98.5						
	MACH	.545	.590	.603	.621						
	KIAS	302	299	294	291						
	FF/ENG	2875	2870	2846	2886						
75	%N1	88.4	92.5	94.0	96.1						
	MACH	.528	.579	.593	.607						
	KIAS	293	293	288	284						
	FF/ENG	2684	2709	2674	2662						
70	%N1	86.5	90.7	92.3	94.0	96.2					
	MACH	.510	.562	.582	.595	.610					
	KIAS	282	284	283	278	274					
	FF/ENG	2494	2518	2520	2481	2487					
65	%N1	84.5	88.7	90.4	92.2	93.9	96.4				
	MACH	.491	.542	.563	.584	.596	.612				
	KIAS	271	274	274	273	268	265				
	FF/ENG	2306	2327	2330	2330	2295	2317				
60	%N1	82.3	86.5	88.3	90.0	91.9	93.7	96.4			
	MACH	.471	.521	.543	.564	.585	.597	.614			
	KIAS	261	263	263	263	263	258	254			
	FF/ENG	2124	2137	2139	2140	2143	2114	2146			
55	%N1	80.2	84.2	85.9	87.7	89.5	91.4	93.3	96.2		
	MACH	.453	.498	.520	.541	.563	.585	.597	.614		
	KIAS	250	251	252	252	253	252	247	244		
	FF/ENG	1954	1948	1950	1950	1953	1958	1938	1971		
50	%N1	77.8	81.6	83.4	85.2	87.0	88.7	90.7	92.7	95.7	
	MACH	.434	.475	.495	.516	.538	.561	.583	.596	.613	
	KIAS	240	239	239	240	241	241	241	236	233	
	FF/ENG	1791	1764	1762	1762	1764	1767	1777	1765	1793	
45	%N1	75.5	79.1	80.6	82.3	84.1	85.9	87.7	89.7	91.8	94.8
	MACH	.415	.452	.469	.489	.511	.533	.556	.578	.593	.610
	KIAS	229	227	227	227	228	229	229	229	225	222
	FF/ENG	1636	1594	1582	1575	1577	1580	1586	1600	1593	1613
40	%N1	73.0	76.2	77.8	79.4	81.0	82.8	84.6	86.4	88.3	90.7
	MACH	.395	.429	.445	.462	.480	.502	.525	.548	.571	.589
	KIAS	218	215	215	214	214	215	216	216	216	214
	FF/ENG	1485	1434	1416	1402	1392	1394	1400	1410	1421	1424

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time
Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
298	272	249	230	214	200	190	180	172	164	158
600	547	501	462	429	400	379	361	344	328	315
903	823	753	694	644	600	570	542	517	494	473
1209	1100	1005	926	859	800	759	721	687	657	630
1516	1379	1259	1159	1075	1000	949	902	859	820	786
1825	1659	1513	1393	1290	1200	1139	1082	1031	984	943
2137	1940	1768	1626	1506	1400	1328	1262	1202	1147	1099
2450	2222	2024	1860	1722	1600	1518	1442	1373	1311	1256
2766	2507	2281	2095	1938	1800	1707	1622	1544	1474	1412
3083	2792	2539	2331	2155	2000	1896	1801	1715	1637	1568

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		18		22		26	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.4	0:43	1.2	0:41	1.1	0:39	1.0	0:38	0.9	0:37
400	2.8	1:23	2.6	1:19	2.4	1:14	2.2	1:11	2.1	1:09
600	4.3	2:04	3.9	1:57	3.6	1:50	3.4	1:45	3.2	1:42
800	5.7	2:46	5.2	2:36	4.9	2:26	4.5	2:19	4.4	2:14
1000	7.1	3:28	6.6	3:15	6.1	3:03	5.7	2:53	5.5	2:47
1200	8.5	4:10	7.9	3:55	7.3	3:40	6.8	3:28	6.6	3:21
1400	9.8	4:53	9.1	4:36	8.5	4:18	8.0	4:02	7.7	3:54
1600	11.2	5:36	10.4	5:16	9.7	4:55	9.1	4:38	8.7	4:28
1800	12.5	6:20	11.7	5:58	10.9	5:34	10.2	5:13	9.8	5:02
2000	13.9	7:05	12.9	6:39	12.0	6:13	11.3	5:49	10.8	5:36

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)									
	40	45	50	55	60	65	70	75	80	
1	-0.1	-0.1	-0.1	0.0	0.0	0.1	0.1	0.2	0.3	
2	-0.3	-0.2	-0.1	-0.1	0.0	0.2	0.3	0.6	0.8	
3	-0.4	-0.3	-0.2	-0.1	0.0	0.3	0.5	0.9	1.2	
4	-0.6	-0.4	-0.3	-0.1	0.0	0.3	0.7	1.2	1.6	
5	-0.7	-0.5	-0.4	-0.2	0.0	0.4	0.9	1.4	2.0	
6	-0.8	-0.6	-0.4	-0.2	0.0	0.5	1.1	1.7	2.4	
7	-1.0	-0.8	-0.5	-0.3	0.0	0.6	1.2	2.0	2.8	
8	-1.1	-0.9	-0.6	-0.3	0.0	0.6	1.4	2.2	3.2	
9	-1.3	-1.0	-0.7	-0.3	0.0	0.7	1.5	2.4	3.5	
10	-1.4	-1.1	-0.7	-0.4	0.0	0.7	1.6	2.6	3.8	
11	-1.6	-1.2	-0.8	-0.4	0.0	0.8	1.7	2.8	4.1	
12	-1.7	-1.3	-0.9	-0.4	0.0	0.8	1.9	3.0	4.4	
13	-1.9	-1.4	-0.9	-0.5	0.0	0.9	2.0	3.2	4.7	
14	-2.0	-1.5	-1.0	-0.5	0.0	0.9	2.0	3.4	4.9	

Includes APU fuel burn.



ENGINE INOP
MAX CONTINUOUS THRUST

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)							
		1500	5000	10000	15000	20000	25000	30000	35000
85	%N1	81.1	84.1	88.3	92.8				
	KIAS	250	251	252	253				
	FF/ENG	2740	2730	2750	2800				
80	%N1	79.5	82.4	86.5	91.0	98.3			
	KIAS	242	243	244	245	247			
	FF/ENG	2580	2570	2570	2610	2740			
75	%N1	77.8	80.5	84.7	89.1	95.0			
	KIAS	235	236	236	238	239			
	FF/ENG	2420	2400	2400	2420	2490			
70	%N1	76.0	78.6	82.8	87.1	92.1			
	KIAS	227	227	228	229	231			
	FF/ENG	2260	2240	2230	2250	2270			
65	%N1	74.0	76.7	80.8	85.0	89.7	97.7		
	KIAS	219	219	220	221	222	224		
	FF/ENG	2100	2090	2070	2070	2080	2230		
60	%N1	71.7	74.6	78.5	82.8	87.4	93.7		
	KIAS	210	210	211	212	213	214		
	FF/ENG	1950	1930	1910	1910	1910	1970		
55	%N1	69.4	72.3	76.3	80.5	84.9	90.0		
	KIAS	200	201	202	203	204	205		
	FF/ENG	1800	1770	1750	1740	1730	1760		
50	%N1	67.0	69.7	73.8	77.8	82.3	87.0	94.9	
	KIAS	191	191	192	193	194	195	196	
	FF/ENG	1650	1620	1600	1580	1570	1570	1680	
45	%N1	64.3	66.9	71.0	75.0	79.4	84.0	89.6	
	KIAS	184	184	184	184	184	185	186	
	FF/ENG	1500	1470	1440	1430	1400	1400	1450	
40	%N1	61.1	64.0	67.8	72.0	76.2	80.7	85.4	94.1
	KIAS	177	177	177	177	177	177	177	177
	FF/ENG	1350	1330	1300	1270	1250	1240	1260	1360

This table includes 5% additional fuel for holding in a racetrack pattern.

ENGINE INOP

ADVISORY INFORMATION

**Gear Down Landing Rate of Climb Available
Flaps 15**

TAT (°C)	RATE OF CLIMB (FT/MIN)					
	PRESSURE ALTITUDE (FT)					
	-2000	0	2000	4000	6000	8000
52	50	-10	-120			
50	80	20	-80	-200		
48	110	50	-50	-160		
46	140	70	-30	-130	-250	
44	170	100	0	-100	-220	
42	200	130	30	-80	-190	-320
40	220	160	50	-50	-160	-300
38	250	190	80	-20	-140	-280
36	270	220	110	0	-120	-250
34	270	250	140	20	-100	-230
32	270	270	160	40	-70	-210
30	280	270	180	60	-60	-190
20	290	280	200	90	-20	-130
10	300	290	210	100	-10	-120
0	310	300	210	100	-10	-120
-20	330	320	230	110	0	-120
-40	350	340	240	120	0	-120

Rate of climb capability shown is valid for 60000 kg, gear down at VREF15+5.
Decrease rate of climb 120 ft/min per 5000 kg greater than 60000 kg.
Increase rate of climb 160 ft/min per 5000 kg less than 60000 kg.

Flaps 30

TAT (°C)	RATE OF CLIMB (FT/MIN)					
	PRESSURE ALTITUDE (FT)					
	-2000	0	2000	4000	6000	8000
52	-250	-310	-420			
50	-220	-290	-390	-510		
48	-190	-260	-360	-480		
46	-170	-240	-340	-450	-570	
44	-140	-210	-320	-420	-540	
42	-110	-180	-290	-400	-510	-650
40	-90	-160	-260	-370	-490	-630
38	-60	-130	-240	-350	-470	-610
36	-40	-100	-210	-320	-450	-580
34	-40	-70	-180	-300	-430	-560
32	-40	-50	-160	-290	-410	-540
30	-40	-50	-150	-270	-390	-520
20	-30	-50	-130	-240	-350	-470
10	-20	-40	-130	-240	-360	-470
0	-20	-40	-130	-240	-360	-470
-20	-10	-30	-130	-250	-370	-490
-40	-10	-30	-130	-250	-380	-500

Rate of climb capability shown is valid for 60000 kg, gear down at VREF30+5.
Decrease rate of climb 120 ft/min per 5000 kg greater than 60000 kg.
Increase rate of climb 170 ft/min per 5000 kg less than 60000 kg.



Performance Inflight - QRH

Chapter PI-QRH

Gear Down

Section 63

GEAR DOWN

Long Range Cruise Altitude Capability
Max Cruise Thrust, 100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	15600	12500	9400
80	18400	15500	12600
75	21100	18500	15700
70	23600	21400	18600
65	26100	24400	21800
60	28600	27100	25300
55	30800	29600	28100
50	32900	31900	30700
45	35100	34100	33000
40	37500	36500	35400

GEAR DOWN

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)								
		10	21	23	25	27	29	31	33	35
85	%N1	85.9								
	MACH	.482								
	KIAS	267								
	FF/ENG	2421								
80	%N1	84.2								
	MACH	.468								
	KIAS	259								
	FF/ENG	2271								
75	%N1	82.5	91.7							
	MACH	.454	.554							
	KIAS	251	248							
	FF/ENG	2123	2101							
70	%N1	80.6	89.8	91.7						
	MACH	.440	.541	.557						
	KIAS	243	242	240						
	FF/ENG	1977	1960	1950						
65	%N1	78.6	87.9	89.5	91.6	94.5				
	MACH	.425	.524	.543	.560	.578				
	KIAS	235	234	233	231	229				
	FF/ENG	1835	1812	1806	1805	1836				
60	%N1	76.5	85.6	87.4	89.1	91.3	94.5			
	MACH	.409	.504	.525	.544	.562	.580			
	KIAS	226	225	225	224	222	220			
	FF/ENG	1696	1661	1661	1658	1664	1696			
55	%N1	74.4	83.3	85.0	86.8	88.5	90.9	94.1		
	MACH	.393	.484	.504	.525	.545	.562	.581		
	KIAS	217	216	216	216	215	213	211		
	FF/ENG	1559	1515	1512	1515	1517	1523	1555		
50	%N1	71.9	80.7	82.5	84.2	86.0	87.8	90.2	93.5	
	MACH	.376	.463	.482	.502	.523	.544	.561	.580	
	KIAS	207	206	206	206	206	205	203	201	
	FF/ENG	1424	1371	1367	1368	1374	1377	1381	1411	
45	%N1	69.1	78.0	79.7	81.4	83.1	85.0	86.8	89.1	92.5
	MACH	.358	.441	.458	.477	.498	.520	.541	.559	.578
	KIAS	197	196	196	196	196	196	195	193	191
	FF/ENG	1294	1231	1224	1224	1230	1235	1237	1239	1265
40	%N1	66.2	74.9	76.6	78.3	80.0	81.8	83.6	85.5	87.7
	MACH	.340	.417	.434	.452	.471	.491	.513	.535	.554
	KIAS	187	185	185	185	185	185	185	185	183
	FF/ENG	1170	1098	1085	1083	1089	1092	1094	1096	1097



GEAR DOWN

Long Range Cruise Enroute Fuel and Time
Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
324	290	260	236	217	200	188	178	168	160	153
654	583	523	474	435	400	377	357	338	321	307
989	880	787	713	653	600	566	535	507	483	461
1329	1181	1054	953	871	800	754	713	676	643	614
1674	1484	1322	1194	1090	1000	943	891	844	803	766
2024	1791	1593	1436	1310	1200	1131	1069	1013	962	918
2381	2103	1865	1680	1530	1400	1320	1247	1181	1122	1070
2743	2417	2140	1924	1751	1600	1508	1424	1348	1280	1221
3113	2737	2418	2171	1972	1800	1695	1600	1514	1438	1371

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	2.4	0:49	2.2	0:47	1.9	0:44	1.7	0:42	1.6	0:41
400	4.9	1:36	4.5	1:31	4.0	1:25	3.7	1:20	3.5	1:17
600	7.4	2:25	6.8	2:17	6.1	2:06	5.7	1:59	5.4	1:54
800	9.8	3:14	9.1	3:03	8.1	2:48	7.6	2:38	7.2	2:31
1000	12.1	4:04	11.3	3:50	10.1	3:30	9.5	3:18	9.0	3:08
1200	14.4	4:56	13.5	4:39	12.1	4:14	11.3	3:58	10.7	3:46
1400	16.7	5:49	15.6	5:28	14.0	4:58	13.1	4:40	12.4	4:24
1600	18.9	6:43	17.7	6:18	15.9	5:44	14.9	5:22	14.1	5:03
1800	21.1	7:38	19.7	7:10	17.7	6:30	16.6	6:05	15.7	5:43

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	40	50	60	70	80
2	-0.3	-0.2	0.0	0.3	0.7
4	-0.7	-0.3	0.0	0.6	1.3
6	-1.0	-0.5	0.0	0.9	2.0
8	-1.3	-0.7	0.0	1.2	2.6
10	-1.7	-0.8	0.0	1.4	3.2
12	-2.0	-1.0	0.0	1.6	3.7
14	-2.4	-1.2	0.0	1.8	4.2
16	-2.7	-1.3	0.0	2.0	4.6
18	-3.0	-1.5	0.0	2.2	5.0
20	-3.4	-1.7	0.0	2.4	5.3
22	-3.7	-1.8	0.0	2.5	5.6

GEAR DOWN

Descent
VREF40 + 70 KIAS

PRESSURE ALTITUDE (FT)	TIME (MIN)	FUEL (KG)	DISTANCE (NM)
41000	21	280	91
39000	20	270	86
37000	19	270	81
35000	19	260	77
33000	18	260	72
31000	17	250	68
29000	17	250	64
27000	16	240	60
25000	15	230	56
23000	14	230	52
21000	13	220	48
19000	13	210	44
17000	12	200	40
15000	11	190	36
10000	8	170	26
5000	6	140	16
1500	4	110	9

Allowances for a straight-in approach are included.



GEAR DOWN

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)							
		1500	5000	10000	15000	20000	25000	30000	35000
85	%N1	75.7	78.4	82.7	86.9	91.9			
	KIAS	229	229	229	229	229			
	FF/ENG	2240	2220	2220	2230	2250			
80	%N1	74.1	76.9	81.0	85.3	89.9			
	KIAS	224	224	224	224	224			
	FF/ENG	2110	2100	2090	2090	2100			
75	%N1	72.3	75.3	79.2	83.6	88.1			
	KIAS	218	218	218	218	218			
	FF/ENG	1990	1970	1960	1960	1960			
70	%N1	70.6	73.5	77.5	81.8	86.2	91.7		
	KIAS	213	213	213	213	213	213		
	FF/ENG	1870	1850	1840	1830	1830	1860		
65	%N1	68.8	71.7	75.8	80.0	84.4	89.1		
	KIAS	209	209	209	209	209	209		
	FF/ENG	1760	1740	1720	1710	1700	1720		
60	%N1	66.9	69.7	73.9	77.9	82.3	86.9	94.1	
	KIAS	203	203	203	203	203	203	203	
	FF/ENG	1650	1620	1600	1590	1580	1580	1660	
55	%N1	65.0	67.6	71.8	75.8	80.2	84.7	90.2	
	KIAS	197	197	197	197	197	197	197	
	FF/ENG	1530	1510	1490	1470	1450	1450	1490	
50	%N1	62.7	65.5	69.4	73.6	77.8	82.3	87.0	
	KIAS	191	191	191	191	191	191	191	
	FF/ENG	1420	1400	1370	1350	1330	1320	1350	
45	%N1	60.2	63.1	67.0	71.2	75.3	79.8	84.4	91.3
	KIAS	184	184	184	184	184	184	184	184
	FF/ENG	1310	1290	1260	1240	1210	1200	1220	1260
40	%N1	57.7	60.4	64.5	68.5	72.8	77.1	81.5	86.6
	KIAS	177	177	177	177	177	177	177	177
	FF/ENG	1200	1170	1150	1130	1100	1080	1090	1110

This table includes 5% additional fuel for holding in a racetrack pattern.

Intentionally
Blank



Performance Inflight - QRH

Chapter PI-QRH

Gear Down, Engine Inop

Section 64

GEAR DOWN

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	80	227	1700		
80	76	223	4000	2300	200
75	71	218	6300	4900	2800
70	66	213	8600	7300	5300
65	62	208	10900	9800	8000
60	57	202	13200	12300	10900
55	52	196	15600	14800	13900
50	47	190	18100	17300	16500
45	43	183	20600	19800	18900
40	38	176	23100	22300	21400

Includes APU fuel burn.

Long Range Cruise Altitude Capability

100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
75	1500		
70	4500	2500	
65	7500	5900	3400
60	10600	9200	6900
55	13300	12300	10600
50	16200	15400	14500
45	19300	18300	17500
40	22200	21400	20500

GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		5	7	9	11	13	15	17	19	21	23
70	%N1	94.8									
	MACH	.389									
	KIAS	235									
	FF/ENG	3774									
65	%N1	92.6	94.3	96.9							
	MACH	.376	.389	.402							
	KIAS	228	227	226							
	FF/ENG	3477	3485	3527							
60	%N1	90.2	91.9	93.7	96.3						
	MACH	.364	.375	.388	.402						
	KIAS	220	219	218	218						
	FF/ENG	3192	3191	3198	3240						
55	%N1	87.8	89.3	91.0	92.8	95.4					
	MACH	.351	.362	.374	.387	.400					
	KIAS	212	211	210	209	209					
	FF/ENG	2924	2909	2906	2913	2951					
50	%N1	85.3	86.7	88.2	89.9	91.7	94.2	98.2			
	MACH	.338	.348	.359	.371	.384	.398	.412			
	KIAS	204	203	202	201	200	199	198			
	FF/ENG	2672	2647	2630	2626	2633	2657	2737			
45	%N1	82.7	84.0	85.4	86.9	88.6	90.4	92.7	96.6		
	MACH	.325	.334	.344	.355	.367	.380	.393	.408		
	KIAS	196	195	193	192	191	190	189	189		
	FF/ENG	2432	2400	2374	2356	2351	2352	2359	2417		
40	%N1	79.8	81.1	82.5	83.9	85.4	87.0	88.8	90.8	94.1	98.4
	MACH	.311	.320	.329	.339	.349	.361	.374	.387	.402	.418
	KIAS	188	186	184	183	182	181	180	179	179	178
	FF/ENG	2206	2166	2133	2107	2088	2076	2069	2065	2101	2201



GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time
Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
172	151	134	120	109	100	93	88	83	78	75
352	308	270	242	219	200	187	175	165	156	148
533	465	408	364	330	300	280	262	246	232	220
716	623	545	486	440	400	373	349	328	309	293
900	783	684	609	551	500	466	436	409	385	365
1086	943	823	733	661	600	559	523	490	462	438
1273	1105	964	856	772	700	652	610	572	538	510
1462	1267	1103	980	883	800	745	696	652	614	581
1653	1431	1245	1104	994	900	838	782	733	690	653
1845	1595	1386	1228	1105	1000	931	868	813	765	724

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)					
	6		10		14	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
100	1.3	0:27	1.1	0:26	1.0	0:26
200	2.6	0:53	2.4	0:50	2.3	0:48
300	3.9	1:18	3.7	1:15	3.6	1:11
400	5.2	1:44	4.9	1:39	4.8	1:35
500	6.5	2:10	6.1	2:04	6.0	1:58
600	7.8	2:37	7.3	2:29	7.1	2:22
700	9.1	3:03	8.5	2:55	8.3	2:46
800	10.3	3:30	9.7	3:20	9.4	3:10
900	11.6	3:58	10.9	3:46	10.5	3:35
1000	12.8	4:25	12.0	4:12	11.6	3:59

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	40	50	60	70	80
1	-0.2	-0.1	0.0	0.1	0.3
2	-0.3	-0.2	0.0	0.3	0.6
3	-0.5	-0.3	0.0	0.5	1.0
4	-0.6	-0.3	0.0	0.7	1.3
5	-0.8	-0.4	0.0	0.9	1.7
6	-1.0	-0.5	0.0	1.0	2.0
7	-1.1	-0.6	0.0	1.2	2.4
8	-1.3	-0.7	0.0	1.4	2.7
9	-1.5	-0.7	0.0	1.6	3.1
10	-1.6	-0.8	0.0	1.8	3.5
11	-1.8	-0.9	0.0	1.9	3.8
12	-1.9	-1.0	0.0	2.1	4.2
13	-2.1	-1.1	0.0	2.3	4.5
14	-2.3	-1.1	0.0	2.5	4.9

Includes APU fuel burn.

GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

**Holding
Flaps Up**

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)			
		1500	5000	10000	15000
80	%N1	93.2			
	KIAS	224			
	FF/ENG	4120			
75	%N1	91.2	94.5		
	KIAS	218	218		
	FF/ENG	3840	3890		
70	%N1	89.2	92.4		
	KIAS	213	213		
	FF/ENG	3580	3610		
65	%N1	87.3	90.3	95.7	
	KIAS	209	209	209	
	FF/ENG	3340	3360	3430	
60	%N1	85.1	88.1	92.7	
	KIAS	203	203	203	
	FF/ENG	3090	3090	3130	
55	%N1	82.8	85.7	90.2	97.0
	KIAS	197	197	197	197
	FF/ENG	2850	2840	2860	2990
50	%N1	80.2	83.2	87.6	92.6
	KIAS	191	191	191	191
	FF/ENG	2610	2600	2610	2650
45	%N1	77.7	80.5	84.9	89.5
	KIAS	184	184	184	184
	FF/ENG	2390	2370	2360	2380
40	%N1	75.0	77.7	82.0	86.4
	KIAS	177	177	177	177
	FF/ENG	2170	2140	2120	2130

This table includes 5% additional fuel for holding in a racetrack pattern.

**Performance Inflight - QRH****Chapter PI-QRH****Text****Section 65**

Introduction

This chapter contains information to supplement performance data from the Flight Management Computer (FMC). In addition, sufficient inflight data is provided to complete a flight with the FMC inoperative. In the event of conflict between data presented in this chapter and that contained in the approved Airplane Flight Manual, the Flight Manual shall always take precedence.

General**Flight with Unreliable Airspeed / Turbulent Air Penetration**

Pitch attitude and average %N1 information is provided for use in all phases of flight in the event of unreliable airspeed/Mach indications resulting from blocking or freezing of the pitot system. Loss of radome or turbulent air may also cause unreliable airspeed/Mach indications. The cruise table in this section may also be used for turbulent air penetration.

Pitch attitude is shown in bold type for emphasis since altitude and/or vertical speed indications may also be unreliable.

Max Climb %N1

This table shows Max Climb %N1 for a 280/.78 climb speed schedule, normal engine bleed for packs on or off and anti-ice off. Enter the table with airport pressure altitude and TAT and read %N1. %N1 adjustments are shown for anti-ice operation.

Go-around %N1

To find Max Go-around %N1 based on normal engine bleed for packs on (AUTO) and anti-ice on or off, enter the Go-around %N1 table with airport pressure altitude and reported OAT or TAT and read %N1. For packs OFF or HIGH operation, apply the %N1 adjustment shown below the table.

VREF

This table contains flaps 40, 30 and 15 reference speeds for a given weight.

Advisory Information

Normal Configuration Landing Distance

The normal configuration distance tables are provided as advisory information to help determine the actual landing distance performance of the airplane for different runway surface conditions and brake configurations.

Flaps 15, 30, and 40 landing distances and adjustments are provided for dry runways as well as runways with good, medium, and poor reported braking action, which are commonly referred to as slippery runway conditions.

If the surface is affected by water, snow or ice, and the braking action is reported as "good", conditions should not be expected to be as good as on clean, dry runways. The value "good" is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when landing. The performance level used to calculate the "good" data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate "poor" data reflects runways covered with wet ice.

Dry runway landing performance is shown for max manual braking configuration and autobrake settings max, 3, 2, and 1. The autobrake performance may be used to assist in the selection of the most desirable autobrake setting for a given field length. Selection of an autobrake setting results in a constant rate of deceleration. Maximum effort manual braking should achieve shorter landing distance than the max autobrake setting. The reference landing distance is a reference distance from 50 ft above the threshold to stop based on a reference landing weight and normal approach speed for the selected landing flap at sea level, zero wind, zero slope, and two engine detent reverse thrust. Subsequent columns provide adjustments for off-reference landing weight, altitude, wind, slope, temperature, speed, and reverse thrust. Each adjustment is independently added to the reference landing distance.

Non-normal Configuration Landing Distance

Advisory information is provided to support non-normal configurations that affect the landing performance of the airplane. Landing distances and adjustments are provided for dry runways and runways with good, medium, and poor reported braking action.

Enter the table with the applicable non-normal configuration and read the normal approach speed. The reference landing distance is a reference distance from 50 ft above the threshold to stop based on a reference landing weight and speed at sea level, zero wind, and zero slope. Subsequent columns provide adjustments for off-reference landing weight, altitude, wind, slope, and speed conditions. Each adjustment is independently added to the reference landing distance. Landing distance includes the effect of reverse thrust.

Recommended Brake Cooling Schedule

Advisory information is provided to assist in avoiding the problems associated with hot brakes. For normal operation, most landings are at weights below the AFM quick turnaround limit weight.

Use of the recommended cooling schedule will help avoid brake overheat and fuse plug problems that could result from repeated landings at short time intervals or a rejected takeoff.

Enter the Recommended Brake Cooling Schedule table with the airplane weight and brakes on speed, adjusted for wind at the appropriate temperature and altitude condition. Instructions for applying wind adjustments are included below the table. Linear interpolation may be used to obtain intermediate values. The resulting number is the reference brake energy per brake in millions of foot-pounds, and represents the amount of energy absorbed by each brake during a rejected takeoff. Notes providing adjustments for wind are included below the table.

To determine the energy per brake absorbed during landing, enter the appropriate Adjusted Brake Energy Per Brake table (No Reverse Thrust or 2 Engine Reverse) with the reference brake energy per brake and the type of braking used during landing (Max Manual, Max Auto, or Autobrake). The resulting number is the adjusted brake energy per brake and represents the energy absorbed in each brake during the landing.

The recommended cooling time is found in the appropriate (steel or carbon brakes) final table by entering with the adjusted brake energy per brake. Times are provided for ground cooling and inflight gear down cooling.

Brake Temperature Monitor System (BTMS) indications are also shown. If brake cooling is determined from the BTMS, use the hottest brake indication 10 to 15 minutes after the airplane has come to a complete stop, or inflight with gear retracted to determine recommended cooling schedule.

Engine Inoperative

Initial Max Continuous %N1

The Initial Max Continuous %N1 setting for use following an engine failure is shown. The table is based on the typical all engine cruise speed of .79M to provide a target %N1 setting at the start of driftdown. Once driftdown is established, the Max Continuous %N1 table should be used to determine %N1 for the given conditions.

Max Continuous %N1

Power setting is based on one engine operating with one A/C pack operating and all anti-ice bleeds off. Enter the table with pressure altitude, TAT, and IAS or Mach to read %N1.

It is desirable to maintain engine thrust level within the limits of the Max Cruise thrust rating. However, where thrust level in excess of Max Cruise rating is required, such as for meeting terrain clearance, ATC altitude assignments, or to attain maximum range capability, it is permissible to use the thrust needed up to the Max Continuous thrust rating. The Max Continuous thrust rating is intended primarily for emergency use at the discretion of the pilot and is the maximum thrust that may be used continuously.

Driftdown Speed/Level Off Altitude

The table shows optimum driftdown speed as a function of cruise weight at start of driftdown. Also shown are the approximate weight and pressure altitude at which the airplane will level off considering 100 ft/min residual rate of climb.

The level off altitude is dependent on air temperature (ISA deviation).

Driftdown/LRC Range Capability

This table shows the range capability from the start of driftdown. Driftdown is continued to level off altitude. As weight decreases due to fuel burn, the airplane is accelerated to Long Range Cruise speed. Cruise is continued at level off altitude and Long Range Cruise speed.

To determine fuel required, enter the Ground to Air Miles Conversion table with the desired ground distance and adjust for anticipated winds to obtain air distance to destination. Then enter the Driftdown/Cruise Fuel and Time table with air distance and weight at start of driftdown to determine fuel and time required. If altitudes other than the level off altitude is used, fuel and time required may be obtained by using the Engine Inoperative Long Range Cruise Enroute Fuel and Time table.

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Long Range Cruise Altitude Capability

The table shows the maximum altitude that can be maintained at a given weight and air temperature (ISA deviation), based on Long Range Cruise speed, Max Continuous thrust, and 100 ft/min residual rate of climb.

Long Range Cruise Control

The table provides target %N1, engine inoperative Long Range Cruise Mach number, IAS and fuel flow for the airplane weight and pressure altitude. The fuel flow values in this table reflect single engine fuel burn.

APU Operation During Flight

For APU operation during flight, increase fuel flow according to the following table. These increments include the APU fuel flow and the effect of increased drag from the APU door.

PRESSURE ALTITUDE (1000 FT)	APU FUEL FLOW (KG/HR)
39	45
35	45
31	50
25	60
20	65
15	75
10	85
5	95

Long Range Cruise Diversion Fuel and Time

Tables are provided for crews to determine the fuel and time required to proceed to an alternate airfield with one engine inoperative. The data is based on single engine Long Range Cruise speed and .78/280/250 descent. Enter with Air Distance as determined from the Ground to Air Miles Conversion table and read Fuel and Time required at the cruise pressure altitude. Adjust the fuel obtained for deviation from the reference weight at checkpoint as required by entering the off reference fuel adjustments table with the fuel required for the reference weight and the actual weight at checkpoint. Read fuel required and time for the actual weight.

Holding

Target %N1, indicated airspeed and fuel flow per engine information is tabulated for holding with flaps up based on the FMC optimum holding speed schedule. This is the higher of the maximum endurance speed and the maneuvering speed. Small variations in airspeed will not appreciably affect the overall endurance time. Enter the table with weight and pressure altitude to read %N1, IAS and fuel flow per engine.

Gear Down Landing Rate of Climb Available

Rate of climb data is provided as guidance information in the event an engine inoperative landing (manual or autoland) is planned. The tables show gear down rate of climb available for Flaps 15 and Flaps 30. Enter the table with TAT and pressure altitude to read rate of climb available. Apply adjustments shown to correct for weight.

Gear Down

This section contains performance for airplane operation with the landing gear extended. The data is based on engine bleeds for normal air conditioning.

Note: The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS may generate inappropriate enroute speed schedules, display non-conservative predictions of fuel burn, estimated time of arrival (ETA), maximum altitude, and compute overly shallow descent path. An accurate estimated time of arrival (ETA) is available if current speed or Mach is entered into the VNAV cruise page.

Tables for gear down performance in this section are identical in format and used in the same manner as tables for the gear up configuration previously described.

**Performance Inflight - QRH****Chapter PI-QRH****Table of Contents****Section 70****737-800WSFP1 CFM56-7B26 C M KG JAA CATC/N (FMC Model 737-800W.1)****Pkg Model Identification PI-QRH.ModID.70.1****General PI-QRH.70.1**

Flight With Unreliable Airspeed/

Turbulent Air Penetration PI-QRH.70.1

CLIMB (280/.76) PI-QRH.70.1

CRUISE (.76/280) PI-QRH.70.1

DESCENT (.76/280) PI-QRH.70.2

HOLDING (VREF40 + 70) PI-QRH.70.2

TERMINAL AREA (5000 FT) PI-QRH.70.3

Airport Altitude = -2000 FT PI-QRH.70.3

Airport Altitude = -1000 FT PI-QRH.70.4

Airport Altitude = SEA LEVEL PI-QRH.70.5

Airport Altitude = 1000 FT PI-QRH.70.6

Airport Altitude = 2000 FT PI-QRH.70.7

Airport Altitude = 3000 FT PI-QRH.70.8

Airport Altitude = 4000 FT PI-QRH.70.9

Airport Altitude = 5000 FT PI-QRH.70.10

Airport Altitude = 6000 FT PI-QRH.70.11

Airport Altitude = 7000 FT PI-QRH.70.12

Airport Altitude = 8000 FT PI-QRH.70.13

Airport Altitude = 9000 FT PI-QRH.70.14

Airport Altitude = 10000 FT PI-QRH.70.15

Airport Altitude = 11000 FT PI-QRH.70.16

Airport Altitude = 12000 FT PI-QRH.70.17

Airport Altitude = 13000 FT PI-QRH.70.18

Airport Altitude = 14000 FT PI-QRH.70.19

Airport Altitude = 14500 FT PI-QRH.70.20

FINAL APPROACH (1500 FT) PI-QRH.70.21

Airport Altitude = -2000 FT PI-QRH.70.21



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Airport Altitude = -1000 FT	PI-QRH.70.21
Airport Altitude = SEA LEVEL	PI-QRH.70.22
Airport Altitude = 1000 FT	PI-QRH.70.22
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Airport Altitude = 3000 FT	PI-QRH.70.23
Airport Altitude = 4000 FT	PI-QRH.70.24
Airport Altitude = 5000 FT	PI-QRH.70.24
Airport Altitude = 6000 FT	PI-QRH.70.25
Airport Altitude = 7000 FT	PI-QRH.70.25
Airport Altitude = 8000 FT	PI-QRH.70.26
Airport Altitude = 9000 FT	PI-QRH.70.26
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Trailing Edge Flap Disagree (1 ≤ Indicated Flaps <15)	PI-QRH.71.24
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General

The table below shows the airplanes that have been identified with the following performance package. Note, some airplanes may be identified with more than one performance package. This configuration table information reflects the Boeing delivered configuration updated for service bulletin incorporations in conformance with the policy stated in the introduction section of the FCOM. The performance data is prepared for the owner/operator named on the title page. The intent of this information is to assist flight crews and airlines in knowing which performance package is applicable to a given airplane. The performance package model identification information is based on Boeing's knowledge of the airline's fleet at a point in time approximately three months prior to the page date. Notice of Errata (NOE) will not be provided to airlines to identify airplanes that are moved between performance packages within this manual or airplanes added to the airline's fleet whose performance packages are already represented in this manual. These types of changes will be updated in the next block revision. Owners/operators are responsible for ensuring the operational documentation they are using is complete and matches the current configuration of their airplanes, and the accuracy and validity of all information furnished by the owner/operator or any other party. Owners/operators receiving active revision service are responsible to ensure that any modifications to the listed airplanes are properly reflected in this manual.

Serial and tabulation number are supplied by Boeing.

Registry Number	Old Registry Number	Serial Number	Tabulation Number
RA-73313	VP-BDW	35700	YL931
RA-73316	VPBSK	35984	YL935
RA-73317	VP-BSO	40874	YL938
RA-73318	VP-BSP	40872	YL961



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Performance Inflight - QRH

Chapter PI-QRH

General

Section 70

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

CLIMB (280/.76)

Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT V/S (FT/MIN)	4.0 1700	4.0 1100	4.0 600		
30000	PITCH ATT V/S (FT/MIN)	4.0 2500	4.0 1900	3.5 1500	4.0 1100	4.0 800
20000	PITCH ATT V/S (FT/MIN)	7.0 4200	6.5 3300	6.0 2600	6.0 2100	6.0 1700
10000	PITCH ATT V/S (FT/MIN)	11.0 5600	9.5 4400	8.5 3600	8.0 3000	8.0 2500
SEA LEVEL	PITCH ATT V/S (FT/MIN)	14.5 6700	12.5 5300	11.0 4400	10.0 3700	9.5 3100

CRUISE (.76/280)

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT %N1	2.0 82.9	2.5 85.4	3.5 88.9		
35000	PITCH ATT %N1	1.0 81.2	2.0 82.6	2.5 84.4	3.0 86.8	3.5 90.4
30000	PITCH ATT %N1	1.0 80.7	1.5 81.5	2.0 82.7	2.5 84.2	3.0 86.1
25000	PITCH ATT %N1	1.0 77.2	1.5 77.9	2.0 79.0	2.5 80.5	3.0 82.3
20000	PITCH ATT %N1	1.0 73.6	1.5 74.2	2.0 75.3	2.5 76.6	3.5 78.2
15000	PITCH ATT %N1	1.0 69.8	1.5 70.6	2.0 71.6	3.0 72.9	3.5 74.4

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

DESCENT (.76/280)

Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
40000	PITCH ATT V/S (FT/MIN)	-1.5 -2700	-0.5 -2400	0.5 -2300	1.0 -2500	1.5 -2700
30000	PITCH ATT V/S (FT/MIN)	-3.5 -3100	-2.0 -2600	-1.0 -2300	-0.5 -2100	0.5 -2000
20000	PITCH ATT V/S (FT/MIN)	-3.5 -2800	-2.0 -2300	-1.0 -2000	0.0 -1900	0.5 -1700
10000	PITCH ATT V/S (FT/MIN)	-3.5 -2500	-2.0 -2100	-1.0 -1800	0.0 -1700	0.5 -1500
SEA LEVEL	PITCH ATT V/S (FT/MIN)	-3.5 -2300	-2.5 -1900	-1.0 -1700	-0.5 -1500	0.5 -1400

HOLDING (VREF40 + 70)

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
15000	PITCH ATT	5.0	5.0	5.0	5.0	5.0
	%N1	56	62	66	70	73
	KIAS	177	193	212	229	246
10000	PITCH ATT	5.0	5.0	5.0	5.0	5.0
	%N1	52	58	62	66	69
	KIAS	177	192	211	228	244
5000	PITCH ATT	5.0	5.5	5.0	5.0	5.0
	%N1	49	54	58	62	66
	KIAS	177	191	210	227	243

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = -2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT	4.5	5.0	5.5	6.0	6.0
	%N1	47.5	52.2	56.3	60.0	63.6
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT	5.0	5.0	5.5	6.0	6.0
	%N1	49.6	54.3	58.5	62.3	65.8
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT	5.5	5.5	6.0	6.5	6.5
	%N1	49.3	54.4	58.9	63.1	66.7
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT	4.5	4.5	5.0	5.5	5.5
	%N1	50.4	55.6	60.2	64.4	68.0
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT	5.0	5.0	5.5	5.5	6.0
	%N1	50.9	56.3	61.1	65.3	69.0
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT	6.0	6.0	6.0	6.5	6.5
	%N1	52.1	57.6	62.5	66.8	70.6
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT	5.0	5.5	5.5	6.0	6.0
	%N1	55.5	61.2	66.1	70.2	74.1
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = -1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 48.2	5.0 52.9	5.5 57.0	6.0 60.8	6.5 64.4
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 50.2	5.0 55.1	5.5 59.3	6.0 63.1	6.0 66.5
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 49.9	5.5 55.2	6.0 59.7	6.5 63.9	6.5 67.5
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 51.1	4.5 56.3	5.0 61.0	5.5 65.2	5.5 68.8
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 51.6	5.0 57.0	5.5 61.9	5.5 66.1	6.0 69.8
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 52.8	6.0 58.4	6.0 63.3	6.5 67.6	6.5 71.5
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 56.2	5.5 62.0	5.5 66.9	6.0 71.0	6.0 74.9
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = SEA LEVEL

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 48.9 177	5.0 53.5 191	5.5 57.7 203	6.0 61.6 213	6.5 65.2 223
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 50.9 157	5.0 55.8 171	5.5 60.0 183	6.0 63.9 193	6.0 67.3 203
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 50.6 137	5.5 55.9 151	6.0 60.5 163	6.5 64.7 173	6.5 68.3 183
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 51.8 137	4.5 57.1 151	5.0 61.9 163	5.5 66.0 173	5.5 69.6 183
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 52.3 127	5.5 57.8 141	5.5 62.7 153	5.5 66.9 163	6.0 70.6 173
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 53.5 117	6.0 59.2 131	6.0 64.2 143	6.5 68.4 153	6.5 72.4 163
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 57.0 127	5.5 62.8 141	5.5 67.7 153	6.0 71.9 163	6.0 75.8 173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 49.7 177	5.0 54.3 191	5.5 58.4 203	6.0 62.4 213	6.5 66.0 223
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 51.6 157	5.0 56.5 171	5.5 60.8 183	6.0 64.7 193	6.0 68.0 203
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 51.4 137	5.5 56.6 151	6.0 61.3 163	6.5 65.5 173	6.5 69.1 183
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 52.6 137	4.5 57.9 151	5.0 62.7 163	5.5 66.8 173	5.5 70.4 183
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 53.0 127	5.5 58.6 141	5.5 63.5 153	5.5 67.7 163	6.0 71.5 173
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 54.3 117	6.0 60.0 131	6.0 65.0 143	6.5 69.2 153	6.5 73.2 163
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 57.7 127	5.5 63.6 141	5.5 68.5 153	6.0 72.8 163	6.0 76.6 173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT	4.5	5.0	5.5	6.0	6.5
	%N1	50.4	55.0	59.2	63.3	66.7
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT	5.0	5.0	5.5	6.0	6.0
	%N1	52.3	57.3	61.7	65.4	68.8
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT	5.5	5.5	6.0	6.5	6.5
	%N1	52.1	57.4	62.2	66.3	69.9
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT	4.5	4.5	5.0	5.5	5.5
	%N1	53.3	58.6	63.5	67.6	71.3
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT	5.0	5.5	5.5	5.5	6.0
	%N1	53.7	59.4	64.3	68.5	72.4
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT	6.0	6.0	6.0	6.5	6.5
	%N1	55.0	60.8	65.8	70.1	74.1
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT	5.0	5.5	5.5	6.0	6.0
	%N1	58.5	64.4	69.3	73.6	77.4
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 3000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 51.0	5.0 55.8	5.5 60.0	6.0 64.1	6.5 67.4
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 53.1	5.0 58.0	5.5 62.5	6.0 66.2	6.0 69.6
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 52.9	5.5 58.2	6.0 63.0	6.5 67.1	6.5 70.7
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 54.0	4.5 59.5	5.0 64.3	5.5 68.4	5.5 72.1
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 54.5	5.5 60.2	5.5 65.2	5.5 69.4	6.0 73.3
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 55.8	6.0 61.6	6.0 66.5	6.5 71.0	6.5 74.9
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 59.3	5.5 65.2	5.5 70.2	6.0 74.5	6.0 78.2
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 4000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 51.7	5.0 56.4	5.5 60.8	6.0 64.9	6.5 68.2
VREF40+70	KIAS	177	191	203	213	223
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 53.9	5.0 58.8	5.5 63.3	6.0 66.9	6.0 70.5
VREF40+50	KIAS	157	171	183	193	203
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 53.6	5.5 59.0	6.0 63.8	6.5 67.9	6.5 71.6
VREF40+30	KIAS	137	151	163	173	183
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 54.7	4.5 60.3	5.0 65.1	5.5 69.2	5.5 73.0
VREF40+30	KIAS	137	151	163	173	183
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 55.2	5.5 61.1	5.5 66.0	5.5 70.2	6.0 74.1
VREF40+20	KIAS	127	141	153	163	173
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 56.5	6.0 62.4	6.0 67.4	6.5 71.9	6.5 75.7
VREF40+10	KIAS	117	131	143	153	163
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 60.1	5.5 66.0	5.5 71.1	6.0 75.3	6.0 79.1
VREF40+20	KIAS	127	141	153	163	173

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 5000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 52.4 177	5.0 57.2 191	5.5 61.6 203	6.0 65.6 213	6.5 69.0 224
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 54.6 157	5.0 59.6 171	5.5 64.1 183	6.0 67.7 193	6.0 71.4 204
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 54.3 137	5.5 59.8 151	6.0 64.6 163	6.5 68.7 173	6.5 72.5 184
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 55.5 137	4.5 61.1 151	5.0 65.9 163	5.5 70.0 173	5.5 73.8 184
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 56.0 127	5.5 61.9 141	5.5 66.7 153	5.5 71.1 163	6.0 74.9 174
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 57.3 117	6.0 63.3 131	6.0 68.2 143	6.5 72.7 153	6.5 76.5 164
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 61.0 127	5.5 66.8 141	5.5 71.9 153	6.0 76.1 163	6.0 80.0 174

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 6000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 53.1	5.0 57.9	5.5 62.5	6.0 66.3	6.5 69.8
VREF40+70	KIAS	177	191	203	213	224
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 55.3	5.0 60.4	5.5 64.8	6.0 68.5	6.0 72.2
VREF40+50	KIAS	157	171	183	193	204
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 55.1	5.5 60.7	6.0 65.4	6.5 69.5	6.5 73.3
VREF40+30	KIAS	137	151	163	173	184
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 56.2	4.5 62.0	5.0 66.7	5.5 70.9	5.5 74.6
VREF40+30	KIAS	137	151	163	173	184
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 56.7	5.5 62.7	5.5 67.6	5.5 72.0	6.0 75.7
VREF40+20	KIAS	127	141	153	163	174
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 58.1	6.0 64.1	6.0 69.0	6.5 73.6	6.5 77.3
VREF40+10	KIAS	117	131	143	153	164
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 61.8	5.5 67.6	5.5 72.8	6.0 77.0	6.0 80.9
VREF40+20	KIAS	127	141	153	163	174

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 7000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 53.8 177	5.0 58.7 191	5.5 63.3 203	6.0 67.1 213	6.5 70.6 224
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 56.1 157	5.0 61.3 171	5.5 65.6 183	6.0 69.4 193	6.0 73.1 204
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 55.8 137	5.5 61.5 151	6.0 66.2 163	6.5 70.4 173	6.5 74.1 184
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 57.0 137	4.5 62.8 151	5.0 67.5 163	5.5 71.8 173	5.5 75.4 184
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 57.6 127	5.5 63.5 141	5.5 68.4 153	5.5 72.8 163	6.0 76.5 174
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 58.9 117	6.0 64.9 131	6.0 69.9 143	6.5 74.4 153	6.5 78.2 164
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 62.6 127	5.5 68.5 141	5.5 73.7 153	6.0 77.8 163	6.0 81.8 174

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 8000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 54.5	5.0 59.5	5.5 64.1	6.0 67.8	6.0 71.4
VREF40+70	KIAS	177	191	203	214	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 56.8	5.0 62.1	5.5 66.4	6.0 70.3	6.0 73.9
VREF40+50	KIAS	157	171	183	194	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 56.6	5.5 62.4	6.0 67.0	6.5 71.2	6.5 74.9
VREF40+30	KIAS	137	151	163	174	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 57.8	4.5 63.6	5.0 68.3	5.5 72.6	5.5 76.2
VREF40+30	KIAS	137	151	163	174	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 58.4	5.5 64.3	5.5 69.3	5.5 73.7	6.0 77.3
VREF40+20	KIAS	127	141	153	164	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 59.8	6.0 65.7	6.0 70.8	6.5 75.2	6.5 79.1
VREF40+10	KIAS	117	131	143	154	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 63.5	5.5 69.4	5.5 74.5	6.0 78.7	6.0 82.7
VREF40+20	KIAS	127	141	153	164	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 9000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 55.3	5.0 60.4	5.5 64.9	6.0 68.6	6.0 72.2
VREF40+70	KIAS	177	191	204	214	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 57.6	5.0 62.9	5.5 67.2	6.0 71.1	6.0 74.7
VREF40+50	KIAS	157	171	184	194	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 57.4	5.5 63.1	6.0 67.8	6.5 72.1	6.5 75.7
VREF40+30	KIAS	137	151	164	174	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 58.7	4.5 64.4	5.0 69.2	5.5 73.4	5.5 77.1
VREF40+30	KIAS	137	151	164	174	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 59.2	5.5 65.1	5.5 70.2	5.5 74.5	5.5 78.2
VREF40+20	KIAS	127	141	154	164	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 60.6	6.0 66.5	6.0 71.7	6.5 76.1	6.5 80.0
VREF40+10	KIAS	117	131	144	154	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 64.3	5.5 70.3	5.5 75.3	6.0 79.6	6.0 83.6
VREF40+20	KIAS	127	141	154	164	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 10000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 56.0	5.0 61.2	5.5 65.6	6.0 69.4	6.0 73.0
VREF40+70	KIAS	177	191	204	214	225
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 58.4	5.0 63.7	5.5 68.0	6.0 72.0	6.0 75.5
VREF40+50	KIAS	157	171	184	194	205
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 58.2	5.5 64.0	6.0 68.7	6.0 72.9	6.5 76.6
VREF40+30	KIAS	137	151	164	174	185
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 59.5	4.5 65.3	5.0 70.1	5.0 74.3	5.5 78.0
VREF40+30	KIAS	137	151	164	174	185
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 60.1	5.0 65.9	5.5 71.1	5.5 75.3	5.5 79.1
VREF40+20	KIAS	127	141	154	164	175
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 61.4	6.0 67.4	6.0 72.6	6.5 76.9	6.0 80.8
VREF40+10	KIAS	117	131	144	154	165
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 65.0	5.5 71.2	5.5 76.2	6.0 80.6	6.0 84.5
VREF40+20	KIAS	127	141	154	164	175

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 11000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 56.8	5.0 62.2	5.5 66.3	6.0 70.3	6.0 73.8
VREF40+70	KIAS	177	191	204	215	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 59.4	5.0 64.5	5.5 69.0	6.0 73.0	6.0 76.3
VREF40+50	KIAS	157	171	184	195	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 59.4	5.5 64.9	6.0 69.7	6.0 73.8	6.5 77.5
VREF40+30	KIAS	137	151	164	175	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 60.7	4.5 66.2	5.0 71.1	5.0 75.1	5.5 79.0
VREF40+30	KIAS	137	151	164	175	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 61.2	5.0 67.0	5.5 72.1	5.5 76.2	5.5 80.1
VREF40+20	KIAS	127	141	154	165	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 62.6	6.0 68.5	6.0 73.6	6.5 78.0	6.0 81.8
VREF40+10	KIAS	117	131	144	155	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 66.1	5.5 72.2	5.5 77.2	6.0 81.6	6.0 85.5
VREF40+20	KIAS	127	141	154	165	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 12000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 57.6	5.0 63.0	5.5 67.1	6.0 71.2	6.0 74.6
VREF40+70	KIAS	178	191	204	215	226
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 60.4	5.0 65.3	5.5 69.9	6.0 73.7	6.0 77.2
VREF40+50	KIAS	158	171	184	195	206
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 60.4	5.5 65.8	6.0 70.6	6.0 74.7	6.5 78.5
VREF40+30	KIAS	138	151	164	175	186
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 61.7	4.5 67.1	5.0 72.0	5.0 76.0	5.5 79.9
VREF40+30	KIAS	138	151	164	175	186
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 62.2	5.0 68.0	5.5 73.0	5.5 77.2	5.5 81.1
VREF40+20	KIAS	128	141	154	165	176
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 63.6	6.0 69.7	6.0 74.5	6.0 78.9	6.0 82.8
VREF40+10	KIAS	118	131	144	155	166
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 67.3	5.5 73.2	5.5 78.3	5.5 82.6	5.5 86.6
VREF40+20	KIAS	128	141	154	165	176

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 13000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 58.5	5.0 63.7	5.5 67.9	6.0 72.0	6.0 75.4
VREF40+70	KIAS	178	192	205	215	227
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 61.3	5.0 66.2	5.5 70.8	6.0 74.5	6.0 78.1
VREF40+50	KIAS	158	172	185	195	207
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 61.4	5.5 66.7	6.0 71.6	6.0 75.6	6.0 79.4
VREF40+30	KIAS	138	152	165	175	187
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 62.6	4.5 68.1	5.0 72.9	5.0 77.0	5.0 80.8
VREF40+30	KIAS	138	152	165	175	187
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 63.2	5.0 69.1	5.5 73.9	5.5 78.2	5.5 82.0
VREF40+20	KIAS	128	142	155	165	177
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 64.7	6.0 70.7	6.0 75.4	6.0 79.9	6.0 83.7
VREF40+10	KIAS	118	132	145	155	167
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 68.4	5.5 74.2	5.5 79.3	5.5 83.6	5.5 87.6
VREF40+20	KIAS	128	142	155	165	177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 14000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP)	PITCH ATT %N1	4.5 59.4	5.0 64.4	5.5 68.7	6.0 72.8	6.0 76.3
VREF40+70	KIAS	178	192	205	215	227
FLAPS 1 (GEAR UP)	PITCH ATT %N1	5.0 62.2	5.0 67.1	5.5 71.6	6.0 75.3	6.0 79.0
VREF40+50	KIAS	158	172	185	195	207
FLAPS 5 (GEAR UP)	PITCH ATT %N1	5.5 62.3	5.5 67.7	6.0 72.4	6.0 76.5	6.0 80.3
VREF40+30	KIAS	138	152	165	175	187
FLAPS 10 (GEAR UP)	PITCH ATT %N1	4.5 63.5	4.5 69.1	5.0 73.7	5.0 77.9	5.5 81.7
VREF40+30	KIAS	138	152	165	175	187
FLAPS 15 (GEAR UP)	PITCH ATT %N1	5.0 64.1	5.0 70.1	5.5 74.8	5.5 79.1	5.5 82.9
VREF40+20	KIAS	128	142	155	165	177
FLAPS 25 (GEAR UP)	PITCH ATT %N1	6.0 65.8	6.0 71.7	6.0 76.5	6.0 80.9	6.0 84.7
VREF40+10	KIAS	118	132	145	155	167
FLAPS 15 (GEAR DOWN)	PITCH ATT %N1	5.0 69.5	5.5 75.1	5.5 80.2	5.5 84.5	5.5 88.6
VREF40+20	KIAS	128	142	155	165	177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

TERMINAL AREA (5000 FT)

%N1 for Level Flight

Airport Altitude = 14500 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS UP (GEAR UP) VREF40+70	PITCH ATT %N1 KIAS	4.5 59.8 178	5.0 64.8 192	5.5 69.2 205	6.0 73.2 216	6.0 76.7 227
FLAPS 1 (GEAR UP) VREF40+50	PITCH ATT %N1 KIAS	5.0 62.6 158	5.0 67.5 172	5.5 72.1 185	6.0 75.8 196	6.0 79.4 207
FLAPS 5 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	5.5 62.7 138	5.5 68.2 152	6.0 72.9 165	6.0 76.9 176	6.0 80.7 187
FLAPS 10 (GEAR UP) VREF40+30	PITCH ATT %N1 KIAS	4.5 63.9 138	4.5 69.6 152	5.0 74.2 165	5.0 78.4 176	5.0 82.1 187
FLAPS 15 (GEAR UP) VREF40+20	PITCH ATT %N1 KIAS	5.0 64.6 128	5.0 70.6 142	5.5 75.2 155	5.5 79.6 166	5.5 83.4 177
FLAPS 25 (GEAR UP) VREF40+10	PITCH ATT %N1 KIAS	6.0 66.3 118	6.0 72.1 132	6.0 77.0 145	6.0 81.4 156	6.0 85.2 167
FLAPS 15 (GEAR DOWN) VREF40+20	PITCH ATT %N1 KIAS	5.0 70.1 128	5.5 75.6 142	5.5 80.7 155	5.5 85.0 166	5.5 89.2 177

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = -2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	38.9	43.0	46.7	49.9	52.8
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	46.1	50.8	55.1	58.6	62.0
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	50.5	55.9	60.6	64.4	67.9
	KIAS	117	130	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = -1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	39.5	43.6	47.3	50.5	53.5
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	46.7	51.5	55.8	59.3	62.8
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	51.2	56.6	61.3	65.2	68.7
	KIAS	117	130	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = SEA LEVEL

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	40.2	44.1	48.0	51.1	54.2
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	47.3	52.2	56.5	60.1	63.6
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	51.9	57.3	62.1	66.1	69.5
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 1000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	40.7	44.7	48.7	51.8	54.9
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	47.9	52.9	57.2	60.9	64.4
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	52.6	58.1	62.9	66.8	70.3
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 2000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	41.2	45.4	49.3	52.5	55.6
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	48.6	53.6	58.0	61.7	65.1
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	53.3	58.9	63.7	67.5	71.1
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 3000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	41.7	46.0	49.9	53.2	56.3
	KIAS	128	143	156	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	49.2	54.3	58.7	62.5	65.9
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	54.1	59.6	64.5	68.3	72.0
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 4000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	42.2	46.7	50.5	53.9	57.0
	KIAS	128	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	49.9	55.0	59.5	63.3	66.7
	KIAS	124	138	150	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	54.8	60.4	65.4	69.1	72.8
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 5000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	42.8	47.3	51.2	54.6	57.7
	KIAS	129	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	50.6	55.7	60.3	64.0	67.5
	KIAS	124	138	151	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	55.6	61.2	66.1	69.9	73.6
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 6000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	43.4	48.0	51.9	55.3	58.4
	KIAS	129	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	51.3	56.4	61.1	64.8	68.2
	KIAS	124	138	151	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	56.3	62.0	66.8	70.8	74.5
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 7000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	44.0	48.6	52.6	56.0	59.1
	KIAS	129	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	52.0	57.2	61.9	65.6	69.0
	KIAS	124	138	151	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	57.0	62.8	67.6	71.6	75.3
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 8000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	44.6	49.2	53.3	56.7	59.8
	KIAS	129	143	157	167	178
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.0	1.5	1.5
	%N1	52.7	58.0	62.7	66.4	69.8
	KIAS	124	138	151	161	171
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	57.8	63.6	68.4	72.5	76.1
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Airport Altitude = 9000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	45.3	49.9	54.1	57.4	60.6
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	53.4	58.8	63.4	67.1	70.6
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	58.6	64.4	69.3	73.3	76.9
	KIAS	117	131	143	153	163

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 10000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	45.9	50.6	54.7	58.1	61.4
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	54.2	59.6	64.2	67.9	71.4
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	59.4	65.2	70.1	74.1	77.8
	KIAS	117	131	143	153	164

Flap placard speed exceeded in shaded area.

Airport Altitude = 11000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	46.7	51.3	55.5	58.8	62.2
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	54.9	60.4	65.1	68.7	72.3
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	60.2	66.0	71.0	74.9	78.7
	KIAS	117	131	143	153	164

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 12000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	47.3	52.0	56.2	59.6	62.9
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	55.6	61.2	65.8	69.5	73.0
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	61.0	66.8	71.8	75.7	79.6
	KIAS	117	131	143	153	164

Flap placard speed exceeded in shaded area.

Airport Altitude = 13000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	47.9	52.8	56.8	60.4	63.6
	KIAS	129	143	157	168	179
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	56.4	62.0	66.6	70.4	73.8
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.5	0.5	0.5	0.5
	%N1	61.9	67.6	72.7	76.6	80.5
	KIAS	117	131	143	154	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

FINAL APPROACH (1500 FT)

Gear Down, %N1 for 3° Glideslope

Airport Altitude = 14000 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	48.7	53.5	57.6	61.3	64.4
	KIAS	129	143	157	168	180
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	57.4	62.9	67.5	71.3	74.6
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.5	0.5	0.5
	%N1	62.9	68.6	73.6	77.6	81.4
	KIAS	117	131	144	154	165

Flap placard speed exceeded in shaded area.

Airport Altitude = 14500 FT

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 KG)				
		40	50	60	70	80
FLAPS 15 (VREF15 + 10)	PITCH ATT	2.0	2.5	2.5	2.5	2.5
	%N1	49.1	53.9	58.0	61.7	64.7
	KIAS	129	144	157	168	180
FLAPS 30 (VREF30 + 10)	PITCH ATT	1.0	1.0	1.5	1.5	1.5
	%N1	58.0	63.4	68.0	71.7	75.0
	KIAS	124	138	151	161	172
FLAPS 40 (VREF40 + 10)	PITCH ATT	0.0	0.0	0.0	0.5	0.5
	%N1	63.4	69.2	74.1	78.1	81.9
	KIAS	117	131	144	154	165

Flap placard speed exceeded in shaded area.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

GO-AROUND

Flaps 15, Gear Up, Set Go-Around Thrust

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		40	50	60	70	80
14500	PITCH ATT	17.5	14.0	11.5	10.5	9.0
	V/S (FT/MIN)	3500	2700	2200	1700	1400
	KIAS	127	141	154	164	175
10000	PITCH ATT	20.0	16.0	13.5	12.0	10.5
	V/S (FT/MIN)	4000	3200	2600	2100	1700
	KIAS	127	141	153	163	174
5000	PITCH ATT	24.0	19.0	16.0	14.0	12.5
	V/S (FT/MIN)	4600	3800	3100	2600	2200
	KIAS	127	141	153	163	173
SEA LEVEL	PITCH ATT	28.0	22.0	18.5	16.0	14.0
	V/S (FT/MIN)	5200	4300	3600	3000	2600
	KIAS	127	141	153	163	173
-2000	PITCH ATT	28.0	22.5	18.5	16.0	14.5
	V/S (FT/MIN)	5100	4200	3500	3000	2500
	KIAS	127	140	153	163	173

Max Climb %N1**Based on engine bleed for packs on or off and anti-ice off**

TAT (°C)	PRESSURE ALTITUDE (FT)/SPEED (KIAS/MACH)									
	0	5000	10000	15000	20000	25000	30000	35000	37000	41000
	280	280	280	280	280	280	280	.78	.78	.78
60	90.2	90.5	90.4	90.6	90.4	92.1	93.8	95.1	95.2	93.5
55	91.0	91.2	91.3	91.4	90.8	91.5	93.1	94.4	94.5	92.8
50	91.7	92.0	92.1	92.2	91.7	91.5	92.4	93.7	93.8	92.1
45	92.4	92.6	92.8	93.0	92.6	92.4	92.4	93.0	93.1	91.4
40	93.1	93.3	93.6	93.8	93.4	93.2	93.2	92.3	92.4	90.7
35	93.6	94.0	94.3	94.5	94.3	94.0	94.0	93.0	92.4	90.8
30	92.9	94.8	95.0	95.2	95.1	94.8	94.7	93.9	93.3	91.8
25	92.2	94.8	95.7	95.9	95.9	95.5	95.4	94.7	94.1	92.8
20	91.4	94.0	96.5	96.7	96.6	96.2	96.1	95.4	94.9	93.7
15	90.6	93.2	95.9	97.5	97.4	96.9	96.7	96.2	95.7	94.6
10	89.9	92.5	95.1	97.8	98.3	97.7	97.4	96.9	96.5	95.6
5	89.1	91.7	94.3	97.0	99.2	98.6	98.1	97.7	97.3	96.5
0	88.3	90.9	93.5	96.2	98.6	99.6	99.1	98.5	98.2	97.5
-5	87.6	90.1	92.7	95.4	97.8	99.6	100.0	99.2	99.0	98.4
-10	86.8	89.3	91.9	94.6	97.1	98.8	100.3	100.2	99.8	99.4
-15	86.0	88.5	91.0	93.8	96.3	98.0	99.6	101.1	100.8	100.4
-20	85.2	87.6	90.2	93.0	95.5	97.2	98.7	100.8	101.3	101.0
-25	84.3	86.8	89.4	92.2	94.7	96.4	97.9	100.0	100.5	100.1
-30	83.5	86.0	88.5	91.3	93.9	95.6	97.1	99.1	99.6	99.3
-35	82.7	85.1	87.7	90.5	93.1	94.8	96.3	98.3	98.8	98.4
-40	81.8	84.3	86.8	89.6	92.3	93.9	95.4	97.4	97.9	97.6

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)					
	0	10	20	30	35	41
ENGINE ANTI-ICE	-0.6	-0.8	-0.9	-0.9	-0.8	-0.8
ENGINE & WING ANTI-ICE*	-1.8	-2.1	-2.5	-2.7	-3.0	-3.0

*Dual bleed sources

Go-around %N1

Based on engine bleed for packs on, engine and wing anti-ice on or off

AIRPORT OAT		TAT (°C)	AIRPORT PRESSURE ALTITUDE (FT)											
°C	°F		-2000	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
57	134	60	95.0	96.2	96.8									
52	125	55	95.9	96.7	96.6	96.8	97.5							
47	116	50	96.6	97.6	97.8	97.8	97.7	97.5	98.2	98.8				
42	108	45	97.4	98.4	98.5	98.6	98.7	98.8	98.7	98.5	98.5	99.0		
37	99	40	98.0	99.1	99.2	99.3	99.4	99.5	99.6	99.5	99.1	98.9	98.8	99.1
32	90	35	98.1	99.9	100.0	100.1	100.1	100.3	100.3	100.2	99.9	99.6	99.6	99.5
27	81	30	97.3	99.8	100.4	100.7	100.7	100.7	100.7	100.7	100.6	100.4	100.4	100.3
22	72	25	96.6	99.1	99.7	100.2	100.6	100.9	100.9	100.9	100.9	100.9	100.9	100.8
17	63	20	95.8	98.3	98.9	99.5	99.8	100.2	100.5	100.9	101.0	101.1	101.0	101.0
12	54	15	95.0	97.5	98.1	98.7	99.1	99.4	99.8	100.1	100.5	100.9	101.3	101.2
7	45	10	94.2	96.8	97.4	98.0	98.3	98.7	99.0	99.4	99.8	100.2	100.5	100.9
2	36	5	93.4	96.0	96.6	97.2	97.6	97.9	98.3	98.7	99.0	99.4	99.8	100.2
-3	27	0	92.6	95.2	95.8	96.4	96.8	97.2	97.5	97.9	98.3	98.7	99.0	99.4
-8	18	-5	91.8	94.4	95.0	95.6	96.0	96.4	96.8	97.2	97.5	97.9	98.3	98.6
-13	9	-10	91.0	93.6	94.2	94.8	95.2	95.6	96.0	96.4	96.8	97.1	97.5	97.9
-17	1	-15	90.2	92.8	93.4	94.0	94.4	94.8	95.2	95.6	96.0	96.4	96.7	97.1
-22	-8	-20	89.3	92.0	92.6	93.2	93.6	94.0	94.4	94.8	95.2	95.6	95.9	96.3
-27	-17	-25	88.5	91.1	91.8	92.4	92.8	93.2	93.6	94.0	94.4	94.8	95.1	95.5
-32	-26	-30	87.6	90.3	90.9	91.6	92.0	92.4	92.8	93.3	93.6	94.0	94.3	94.7
-37	-35	-35	86.8	89.4	90.1	90.7	91.1	91.6	92.0	92.4	92.8	93.2	93.5	93.9
-42	-44	-40	85.9	88.6	89.2	89.9	90.3	90.7	91.2	91.6	92.0	92.4	92.7	93.0
-47	-53	-45	85.0	87.7	88.4	89.0	89.4	89.9	90.3	90.8	91.2	91.5	91.9	92.2
-52	-62	-50	84.1	86.8	87.5	88.2	88.6	89.0	89.5	90.0	90.3	90.7	91.0	91.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (FT)											
	-2000	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
PACKS OFF	0.7	0.8	0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9
A/C HIGH	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1



737 Flight Crew Operations Manual

VREF

WEIGHT (1000 KG)	FLAPS		
	40	30	15
85	159	167	174
80	154	162	169
75	148	156	163
70	143	151	157
65	139	147	153
60	133	141	147
55	127	134	140
50	121	128	133
45	114	121	126
40	107	114	119

Intentionally
Blank



Performance Inflight - QRH

Advisory Information

Chapter PI-QRH

Section 71

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF30	ONE REV	NO REV

Dry Runway

MAX MANUAL	1075	70/-65	25/30	-40/140	10/-10	25/-25	40	15	40
AUTOBRAKE MAX	1380	70/-75	30/40	-50/165	0/0	35/-35	65	0	5
AUTOBRAKE 3	1955	115/-120	50/70	-85/280	0/-5	50/-50	100	0	0
AUTOBRAKE 2	2440	160/-165	75/100	-115/385	35/-45	70/-70	100	90	90
AUTOBRAKE 1	2675	185/-195	85/115	-130/455	75/-80	80/-80	90	215	345

Good Reported Braking Action

MAX MANUAL	1465	80/-85	40/50	-65/225	35/-30	35/-35	50	70	150
AUTOBRAKE MAX	1580	85/-90	40/50	-70/235	35/-30	40/-40	65	75	165
AUTOBRAKE 3	1960	115/-120	50/70	-85/290	5/-10	50/-50	100	5	10
AUTOBRAKE 2	2440	160/-165	75/100	-115/385	35/-45	70/-70	100	90	90
AUTOBRAKE 1	2675	185/-195	85/115	-130/455	75/-80	80/-80	90	215	345

Medium Reported Braking Action

MAX MANUAL	2000	125/-130	65/80	-105/375	90/-70	50/-60	70	190	450
AUTOBRAKE MAX	2060	130/-140	65/85	-105/380	85/-70	60/-60	80	190	460
AUTOBRAKE 3	2170	130/-140	65/85	-110/390	65/-50	65/-65	100	130	380
AUTOBRAKE 2	2500	160/-175	75/105	-125/435	75/-70	75/-75	100	130	245
AUTOBRAKE 1	2690	185/-195	85/115	-140/470	105/-85	80/-80	90	225	405

Poor Reported Braking Action

MAX MANUAL	2605	185/-185	85/120	-155/600	220/-145	75/-80	80	410	1060
AUTOBRAKE MAX	2615	185/-185	90/120	-155/600	220/-140	75/-80	90	405	1060
AUTOBRAKE 3	2650	185/-185	90/120	-160/600	215/-140	75/-80	90	415	1070
AUTOBRAKE 2	2800	195/-195	90/125	-165/615	205/-140	80/-85	100	360	930
AUTOBRAKE 1	2910	200/-205	100/130	-165/635	220/-150	85/-90	90	390	970

Reference distance is based on sea level, standard day, no wind or slope, VREF30, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 70 m.

For autobrake and manual speedbrakes, increase reference landing distance by 60 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

All reference distances and adjustments shown have been increased by 15%.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 40

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF40	ONE REV NO REV

Dry Runway

MAX MANUAL	1025	60/-60	25/30	-40/130	10/-10	25/-25	40	15	35
AUTOBRAKE MAX	1290	65/-70	30/35	-45/160	0/0	30/-30	65	0	0
AUTOBRAKE 3	1800	105/-115	45/65	-80/270	0/-5	45/-45	100	0	0
AUTOBRAKE 2	2275	145/-155	70/85	-110/375	30/-40	65/-65	105	45	45
AUTOBRAKE 1	2515	175/-180	80/105	-125/435	65/-75	75/-75	100	165	260

Good Reported Braking Action

MAX MANUAL	1405	75/-80	35/45	-65/225	35/-30	35/-35	50	65	140
AUTOBRAKE MAX	1505	80/-85	40/50	-70/230	30/-30	35/-35	65	70	150
AUTOBRAKE 3	1805	105/-115	45/65	-80/275	10/-5	45/-50	105	5	10
AUTOBRAKE 2	2275	145/-155	70/85	-110/375	30/-40	65/-65	105	45	45
AUTOBRAKE 1	2515	175/-180	80/105	-125/435	65/-75	75/-75	100	165	260

Medium Reported Braking Action

MAX MANUAL	1910	120/-120	60/75	-105/370	85/-70	50/-50	70	175	405
AUTOBRAKE MAX	1950	125/-125	60/80	-105/375	80/-65	50/-50	80	175	410
AUTOBRAKE 3	2025	125/-130	60/80	-105/380	70/-50	60/-60	100	130	375
AUTOBRAKE 2	2340	150/-160	70/90	-120/425	70/-65	70/-70	105	90	200
AUTOBRAKE 1	2525	175/-180	80/110	-130/460	90/-80	75/-75	100	185	315

Poor Reported Braking Action

MAX MANUAL	2485	175/-175	80/115	-155/585	220/-140	70/-75	80	375	955
AUTOBRAKE MAX	2490	175/-175	85/115	-155/585	220/-140	70/-75	85	375	955
AUTOBRAKE 3	2515	180/-180	85/115	-155/585	215/-140	70/-75	90	380	965
AUTOBRAKE 2	2645	185/-185	85/120	-160/605	200/-130	75/-80	100	315	840
AUTOBRAKE 1	2750	190/-195	90/125	-165/620	205/-145	80/-85	90	350	855

Reference distance is based on sea level, standard day, no wind or slope, VREF40, two-engine detent No. 2 reverse thrust, and auto speedbrakes.

For max manual braking and manual speedbrakes, increase reference landing distance by 65 m.

For autobrake and manual speedbrakes, increase reference landing distance by 50 m.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

All reference distances and adjustments shown have been increased by 15%.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1035	85/-60	25/30	-35/125	10/-10	25/-25	N/A	25	50
AUTOBRAKE MAX	1390	70/-75	35/45	-50/160	0/0	35/-35	N/A	0	5
AUTOBRAKE 2	2450	160/-170	80/105	-110/360	50/-55	75/-75	N/A	160	185

Good Reported Braking Action

MAX MANUAL	1410	80/-80	40/50	-60/205	30/-30	35/-35	N/A	75	175
AUTOBRAKE MAX	1530	85/-90	40/55	-60/215	30/-20	40/-40	N/A	85	195
AUTOBRAKE 2	2450	160/-170	80/105	-110/360	50/-55	75/-75	N/A	160	185

Medium Reported Braking Action

MAX MANUAL	1935	125/-125	60/80	-95/340	80/-65	55/-55	N/A	215	520
AUTOBRAKE MAX	1995	125/-130	65/85	-95/345	75/-60	55/-55	N/A	220	535
AUTOBRAKE 3	2185	130/-140	65/90	-100/365	55/-50	65/-65	N/A	125	385

Poor Reported Braking Action

MAX MANUAL	2515	175/-175	90/120	-145/535	195/-130	75/-80	N/A	455	1230
AUTOBRAKE MAX	2530	180/-180	90/125	-145/535	195/-125	75/-80	N/A	450	1225
AUTOBRAKE 3	2590	175/-180	90/125	-145/540	185/-120	75/-80	N/A	430	1215

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	995	65/-55	20/30	-35/125	10/-10	20/-20	N/A	20	45
AUTOBRAKE MAX	1315	65/-70	30/40	-45/155	0/0	30/-30	N/A	0	5
AUTOBRAKE 2	2265	145/-155	70/95	-105/345	45/-50	65/-65	N/A	135	185

Good Reported Braking Action

MAX MANUAL	1355	75/-75	35/50	-60/200	30/-25	35/-35	N/A	70	155
AUTOBRAKE MAX	1465	80/-85	40/50	-60/210	30/-20	35/-35	N/A	80	175
AUTOBRAKE 2	2265	145/-155	70/95	-105/345	45/-50	65/-65	N/A	135	185

Medium Reported Braking Action

MAX MANUAL	1830	115/-115	55/75	-90/330	80/-60	50/-50	N/A	185	450
AUTOBRAKE MAX	1895	120/-120	60/80	-95/335	75/-60	50/-55	N/A	190	465
AUTOBRAKE 3	2040	120/-125	60/80	-100/350	60/-50	60/-60	N/A	115	340

Poor Reported Braking Action

MAX MANUAL	2360	160/-160	80/110	-140/520	185/-125	65/-70	N/A	390	1025
AUTOBRAKE MAX	2385	165/-165	85/110	-140/525	180/-120	70/-75	N/A	390	1025
AUTOBRAKE 3	2420	165/-165	80/110	-140/525	180/-115	70/-75	N/A	365	1010

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Airspeed Unreliable (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	955	55/-50	20/30	-35/120	10/-10	20/-20	N/A	20	40
AUTOBRAKE MAX	1225	60/-65	30/35	-45/150	0/0	30/-30	N/A	0	0
AUTOBRAKE 2	2130	135/-145	65/85	-100/335	40/-45	60/-60	N/A	105	125

Good Reported Braking Action

MAX MANUAL	1305	70/-75	35/45	-55/200	30/-25	35/-35	N/A	65	145
AUTOBRAKE MAX	1400	75/-80	35/50	-60/210	30/-25	35/-35	N/A	70	160
AUTOBRAKE 2	2130	135/-145	65/85	-100/335	40/-45	60/-60	N/A	105	125

Medium Reported Braking Action

MAX MANUAL	1755	110/-110	55/70	-90/325	75/-60	50/-50	N/A	170	410
AUTOBRAKE MAX	1810	110/-115	55/75	-90/330	75/-60	50/-50	N/A	175	420
AUTOBRAKE 3	1910	110/-120	55/75	-95/345	60/-45	55/-55	N/A	115	335

Poor Reported Braking Action

MAX MANUAL	2255	155/-155	75/105	-135/515	185/-120	65/-70	N/A	355	925
AUTOBRAKE MAX	2275	155/-155	80/105	-135/515	185/-115	65/-70	N/A	355	925
AUTOBRAKE 3	2300	155/-155	75/105	-135/520	180/-115	65/-70	N/A	355	935

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

All Flaps Up Landing

VREF40 + 55

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1320	190/-80	50/105	-45/200	20/-15	35/-35	45	40	85
AUTOBRAKE MAX	1815	90/-85	45/75	-60/190	5/0	50/-50	75	5	15
AUTOBRAKE 2	3435	195/-225	115/150	-130/435	60/-75	105/-105	115	215	225

Good Reported Braking Action

MAX MANUAL	1800	90/-100	50/70	-70/235	40/-35	50/-50	50	110	255
AUTOBRAKE MAX	1985	90/-100	55/75	-75/245	30/-25	55/-55	75	95	250
AUTOBRAKE 2	3435	195/-225	115/150	-130/435	60/-75	105/-105	115	215	225

Medium Reported Braking Action

MAX MANUAL	2570	155/-160	85/115	-110/390	110/-90	75/-75	70	320	795
AUTOBRAKE MAX	2625	155/-160	90/115	-115/395	105/-85	75/-80	80	325	805
AUTOBRAKE 3	2930	150/-165	90/120	-120/420	65/-45	90/-90	125	185	570

Poor Reported Braking Action

MAX MANUAL	3440	230/-235	130/175	-170/620	270/-180	105/-110	90	715	1990
AUTOBRAKE MAX	3430	230/-235	130/175	-170/615	270/-175	105/-110	95	710	1975
AUTOBRAKE 3	3510	225/-225	130/175	-170/625	245/-155	110/-115	120	675	1955

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1710	100/-105	50/65	-80/280	50/-45	45/-45	60	125	295
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1920	120/-120	55/75	-95/340	75/-60	50/-50	65	185	460
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2465	170/-170	85/115	-140/535	190/-125	70/-75	75	410	1125
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3305	250/-245	120/175	-235/985	600/-290	90/-110	90	1005	3480
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1635	95/-100	45/60	-75/275	50/-40	40/-40	55	110	260
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1825	110/-115	55/70	-90/335	75/-60	45/-50	65	165	395
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2320	155/-155	75/105	-135/520	180/-120	65/-70	75	355	940
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3085	225/-225	110/160	-230/960	560/-270	85/-105	85	860	2845
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

ANTISKID INOPERATIVE (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1565	90/-95	40/55	-75/270	50/-40	40/-40	60	100	235
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1750	105/-110	50/65	-90/330	75/-60	45/-45	65	150	360
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2220	150/-150	70/100	-135/510	180/-115	60/-65	75	325	850
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	2955	215/-215	105/150	-225/945	550/-265	80/-100	85	795	2590
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Jammed or Restricted Flight Controls (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	970	75/-60	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Good Reported Braking Action

MAX MANUAL	1320	75/-75	35/45	-55/200	30/-25	35/-35	45	70	160
AUTOBRAKE MAX	1420	80/-85	40/50	-60/210	30/-20	35/-35	55	80	175
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Medium Reported Braking Action

MAX MANUAL	1815	120/-120	55/75	-90/330	80/-65	50/-50	60	195	485
AUTOBRAKE MAX	1860	120/-125	60/80	-95/335	75/-60	50/-55	70	200	490
AUTOBRAKE 3	1990	125/-130	60/80	-95/350	55/-40	55/-60	100	125	390

Poor Reported Braking Action

MAX MANUAL	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1165
AUTOBRAKE MAX	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1160
AUTOBRAKE 3	2405	170/-170	85/115	-140/530	185/-115	70/-75	90	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LEADING EDGE FLAPS TRANSIT (Flaps 15)

VREF15 + 15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1080	80/-65	25/35	-40/130	15/-10	25/-25	35	25	55
AUTOBRAKE MAX	1450	75/-75	35/45	-50/165	0/0	35/-35	65	0	5
AUTOBRAKE 2	2575	170/-180	85/110	-110/370	50/-60	75/-75	90	175	210

Good Reported Braking Action

MAX MANUAL	1480	80/-85	40/55	-60/210	35/-30	40/-40	45	85	195
AUTOBRAKE MAX	1605	85/-90	45/60	-65/220	30/-20	40/-40	65	95	215
AUTOBRAKE 2	2575	170/-180	85/110	-110/370	50/-60	75/-75	90	175	210

Medium Reported Braking Action

MAX MANUAL	2030	130/-130	65/90	-100/350	85/-70	55/-60	60	235	575
AUTOBRAKE MAX	2095	135/-135	70/90	-100/355	80/-65	60/-60	70	240	590
AUTOBRAKE 3	2295	135/-145	70/95	-105/370	60/-50	65/-70	95	140	430

Poor Reported Braking Action

MAX MANUAL	2635	185/-185	95/130	-145/545	205/-140	75/-80	75	490	1340
AUTOBRAKE MAX	2655	185/-185	95/130	-145/550	205/-130	80/-85	85	490	1340
AUTOBRAKE 3	2715	185/-185	95/130	-150/555	190/-125	80/-85	90	465	1325

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1085	70/-60	25/35	-40/130	15/-15	25/-25	45	30	50
AUTOBRAKE MAX	1275	65/-70	30/40	-45/150	5/0	30/-30	60	0	5
AUTOBRAKE 2	2395	150/-165	70/90	-105/360	0/-10	70/-70	140	0	0

Good Reported Braking Action

MAX MANUAL	1570	90/-95	45/60	-65/230	45/-40	40/-40	70	120	245
AUTOBRAKE MAX	1605	95/-100	45/60	-70/230	40/-35	40/-45	75	125	255
AUTOBRAKE 2	2395	150/-165	70/90	-105/360	5/-10	70/-70	140	0	0

Medium Reported Braking Action

MAX MANUAL	2165	145/-150	70/95	-105/375	110/-90	60/-65	90	325	775
AUTOBRAKE MAX	2165	150/-150	70/100	-105/375	115/-90	60/-65	90	325	775
AUTOBRAKE 3	2165	150/-150	70/100	-105/375	115/-80	60/-65	90	325	775

Poor Reported Braking Action

MAX MANUAL	2820	210/-210	105/145	-160/580	255/-170	80/-90	105	675	1905
AUTOBRAKE MAX	2820	215/-210	105/145	-160/580	260/-175	85/-90	105	675	1910
AUTOBRAKE 3	2820	215/-210	105/145	-160/580	260/-175	85/-90	105	675	1910

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1035	65/-55	25/30	-40/125	15/-15	25/-25	45	25	45
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	5
AUTOBRAKE 2	2230	140/-150	65/85	-105/345	0/-10	65/-65	130	0	0

Good Reported Braking Action

MAX MANUAL	1490	85/-90	40/55	-65/220	45/-40	40/-40	65	105	210
AUTOBRAKE MAX	1515	90/-95	40/55	-65/225	40/-35	40/-40	70	105	215
AUTOBRAKE 2	2230	140/-150	65/85	-105/345	0/-10	65/-65	130	0	0

Medium Reported Braking Action

MAX MANUAL	2035	135/-135	65/90	-100/360	105/-85	55/-60	85	275	645
AUTOBRAKE MAX	2025	135/-135	65/90	-100/360	110/-85	55/-60	85	275	640
AUTOBRAKE 3	2030	135/-135	65/90	-100/360	105/-75	55/-60	90	275	645

Poor Reported Braking Action

MAX MANUAL	2625	190/-190	95/130	-150/565	240/-155	75/-80	95	565	1530
AUTOBRAKE MAX	2625	195/-190	95/130	-150/565	245/-160	75/-80	95	565	1530
AUTOBRAKE 3	2625	195/-190	95/130	-150/565	245/-160	75/-80	95	565	1530

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A (Flaps 40)

VREF40

	LANDING DISTANCE AND ADJUSTMENTS (M)									
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ		
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV	

Dry Runway

MAX MANUAL	990	60/-50	20/30	-35/125	15/-15	20/-20	45	25	35
AUTOBRAKE MAX	1120	55/-60	25/30	-40/140	0/0	25/-25	55	5	10
AUTOBRAKE 2	2035	125/-140	60/75	-95/330	0/-5	60/-60	125	0	0

Good Reported Braking Action

MAX MANUAL	1420	80/-85	40/50	-65/220	45/-35	35/-35	70	95	190
AUTOBRAKE MAX	1420	80/-90	40/50	-65/220	35/-30	35/-35	70	95	185
AUTOBRAKE 2	2035	125/-140	60/75	-95/330	0/-5	60/-60	125	0	0

Medium Reported Braking Action

MAX MANUAL	1920	125/-130	60/85	-100/355	100/-80	55/-55	85	245	560
AUTOBRAKE MAX	1905	125/-130	60/85	-100/355	105/-80	55/-55	85	240	555
AUTOBRAKE 3	1910	125/-130	60/85	-100/355	105/-75	55/-55	90	240	555

Poor Reported Braking Action

MAX MANUAL	2470	180/-180	85/120	-150/550	230/-150	70/-75	95	495	1305
AUTOBRAKE MAX	2465	180/-180	90/120	-145/550	235/-155	70/-75	95	495	1305
AUTOBRAKE 3	2465	180/-180	90/120	-145/550	235/-155	70/-75	95	495	1305

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM A AND SYSTEM B (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1535	80/-85	40/50	-60/195	35/-30	40/-40	75	-5	60
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2240	135/-140	65/90	-100/335	100/-80	60/-60	105	90	425
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2980	200/-205	100/135	-145/520	215/-160	85/-85	125	355	1385
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3710	275/-270	140/195	-210/780	475/-270	105/-115	135	805	3355
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

LOSS OF SYSTEM B (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1095	55/-60	25/35	-40/145	15/-15	25/-25	40	35	55
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	1565	95/-95	45/60	-70/245	45/-40	40/-40	55	120	250
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2125	145/-145	70/95	-110/400	115/-90	60/-60	75	305	730
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	2725	200/-200	100/140	-165/625	285/-170	80/-85	85	610	1695
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

MANUAL REVERSION (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1535	80/-85	40/50	-60/195	35/-30	40/-40	75	-5	60
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Good Reported Braking Action

MAX MANUAL	2240	135/-140	65/90	-100/335	100/-80	60/-60	105	90	425
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 2	Autobrake Inoperative								

Medium Reported Braking Action

MAX MANUAL	2980	200/-205	100/135	-145/520	215/-160	85/-85	125	355	1385
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Poor Reported Braking Action

MAX MANUAL	3710	275/-270	140/195	-210/780	475/-270	105/-115	135	805	3355
AUTOBRAKE MAX	Autobrake Inoperative								
AUTOBRAKE 3	Autobrake Inoperative								

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

One Engine Inoperative Landing (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	980	80/-60	20/30	-35/125	10/-10	20/-20	35	0	20
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	0
AUTOBRAKE 2	2370	150/-160	70/90	-105/355	10/-30	70/-70	115	0	5

Good Reported Braking Action

MAX MANUAL	1370	75/-80	35/50	-60/210	35/-30	35/-35	50	0	85
AUTOBRAKE MAX	1485	85/-90	40/50	-65/220	35/-25	40/-40	60	0	95
AUTOBRAKE 2	2370	150/-160	70/90	-105/355	10/-30	70/-70	115	0	5

Medium Reported Braking Action

MAX MANUAL	1965	125/-130	60/80	-100/360	100/-80	55/-55	70	0	270
AUTOBRAKE MAX	2025	130/-135	60/80	-100/365	95/-75	60/-60	80	0	275
AUTOBRAKE 3	2085	135/-140	65/85	-105/370	80/-60	60/-60	95	0	245

Poor Reported Braking Action

MAX MANUAL	2700	190/-195	90/120	-160/590	270/-170	80/-85	85	0	675
AUTOBRAKE MAX	2705	195/-195	90/125	-160/590	270/-165	80/-85	95	0	680
AUTOBRAKE 3	2740	195/-200	95/125	-160/590	265/-170	80/-85	90	0	685

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

One Engine Inoperative Landing (Flaps 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	940	60/-55	20/25	-35/120	10/-10	20/-20	35	0	20
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	0
AUTOBRAKE 2	2185	140/-145	65/85	-100/340	15/-35	65/-65	100	0	10

Good Reported Braking Action

MAX MANUAL	1310	75/-75	35/45	-60/205	35/-30	35/-35	50	0	75
AUTOBRAKE MAX	1420	80/-85	35/50	-60/215	30/-25	35/-35	60	0	85
AUTOBRAKE 2	2185	140/-145	65/85	-100/340	15/-35	65/-65	100	0	10

Medium Reported Braking Action

MAX MANUAL	1850	115/-120	55/75	-95/350	95/-75	50/-55	65	0	230
AUTOBRAKE MAX	1905	120/-125	55/75	-100/355	90/-70	55/-55	75	0	235
AUTOBRAKE 3	1955	120/-130	60/75	-100/360	80/-65	55/-55	85	0	210

Poor Reported Braking Action

MAX MANUAL	2500	175/-175	85/110	-150/565	245/-155	75/-75	80	0	555
AUTOBRAKE MAX	2505	175/-180	85/110	-150/565	245/-150	75/-80	90	0	555
AUTOBRAKE 3	2535	175/-180	85/110	-155/570	245/-160	75/-80	80	0	565

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Stabilizer Trim Inoperative (Flaps 15)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	970	75/-60	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Good Reported Braking Action

MAX MANUAL	1320	75/-75	35/45	-55/200	30/-25	35/-35	45	70	160
AUTOBRAKE MAX	1420	80/-85	40/50	-60/210	30/-20	35/-35	55	80	175
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Medium Reported Braking Action

MAX MANUAL	1815	120/-120	55/75	-90/330	80/-65	50/-50	60	195	485
AUTOBRAKE MAX	1860	120/-125	60/80	-95/335	75/-60	50/-55	70	200	490
AUTOBRAKE 3	1990	125/-130	60/80	-95/350	55/-40	55/-60	100	125	390

Poor Reported Braking Action

MAX MANUAL	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1165
AUTOBRAKE MAX	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1160
AUTOBRAKE 3	2405	170/-170	85/115	-140/530	185/-115	70/-75	90	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Trailing Edge Flap Asymmetry ($1 \leq \text{Flap Lever} < 15$) VREF40 + 30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1075	95/-60	25/40	-40/130	15/-10	25/-25	40	25	55
AUTOBRAKE MAX	1475	70/-75	35/45	-50/170	0/0	35/-35	65	0	5
AUTOBRAKE 2	2615	160/-175	85/110	-110/375	55/-60	80/-80	90	180	220

Good Reported Braking Action

MAX MANUAL	1450	75/-80	40/55	-60/210	30/-30	35/-40	45	80	175
AUTOBRAKE MAX	1610	80/-85	45/55	-65/220	25/-20	40/-40	65	70	175
AUTOBRAKE 2	2615	160/-175	85/110	-110/375	55/-60	80/-80	90	180	220

Medium Reported Braking Action

MAX MANUAL	2015	120/-125	65/85	-95/345	85/-65	55/-60	60	220	535
AUTOBRAKE MAX	2090	125/-130	65/90	-100/350	80/-65	60/-60	70	225	550
AUTOBRAKE 3	2325	125/-135	70/95	-105/375	55/-50	70/-70	95	125	375

Poor Reported Braking Action

MAX MANUAL	2650	180/-180	95/130	-145/545	205/-135	80/-85	75	480	1290
AUTOBRAKE MAX	2670	180/-180	95/130	-145/550	200/-130	80/-85	80	475	1290
AUTOBRAKE 3	2740	175/-180	95/130	-150/555	190/-125	80/-85	95	435	1260

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance Trailing Edge Flap Asymmetry (Flap Lever 15 or 25) VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	970	75/-60	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Good Reported Braking Action

MAX MANUAL	1320	75/-75	35/45	-55/200	30/-25	35/-35	45	70	160
AUTOBRAKE MAX	1420	80/-85	40/50	-60/210	30/-20	35/-35	55	80	175
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Medium Reported Braking Action

MAX MANUAL	1815	120/-120	55/75	-90/330	80/-65	50/-50	60	195	485
AUTOBRAKE MAX	1860	120/-125	60/80	-95/335	75/-60	50/-55	70	200	490
AUTOBRAKE 3	1990	125/-130	60/80	-95/350	55/-40	55/-60	100	125	390

Poor Reported Braking Action

MAX MANUAL	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1165
AUTOBRAKE MAX	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1160
AUTOBRAKE 3	2405	170/-170	85/115	-140/530	185/-115	70/-75	90	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Asymmetry (Flap Lever 30)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	930	60/-55	20/25	-35/120	10/-10	20/-20	35	20	35
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	5
AUTOBRAKE 2	2100	140/-145	65/85	-100/335	35/-45	60/-60	85	95	100

Good Reported Braking Action

MAX MANUAL	1265	70/-70	35/45	-55/195	30/-25	30/-30	45	65	140
AUTOBRAKE MAX	1355	75/-80	35/45	-60/205	25/-20	35/-35	55	70	155
AUTOBRAKE 2	2100	140/-145	65/85	-100/335	35/-45	60/-60	85	95	100

Medium Reported Braking Action

MAX MANUAL	1715	110/-110	55/70	-90/325	75/-60	45/-50	60	170	415
AUTOBRAKE MAX	1760	115/-115	55/70	-90/325	70/-55	50/-50	70	175	425
AUTOBRAKE 3	1870	115/-120	55/75	-95/340	55/-45	50/-55	85	115	340

Poor Reported Braking Action

MAX MANUAL	2220	155/-155	75/105	-135/510	180/-120	65/-70	70	365	970
AUTOBRAKE MAX	2230	155/-155	75/105	-135/510	185/-115	65/-70	80	365	965
AUTOBRAKE 3	2260	160/-160	75/105	-135/515	175/-115	65/-70	80	365	975

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (1 ≤ Indicated Flaps <15)

VREF40 + 30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	1075	95/-60	25/40	-40/130	15/-10	25/-25	40	25	55
AUTOBRAKE MAX	1475	70/-75	35/45	-50/170	0/0	35/-35	65	0	5
AUTOBRAKE 2	2615	160/-175	85/110	-110/375	55/-60	80/-80	90	180	220

Good Reported Braking Action

MAX MANUAL	1450	75/-80	40/55	-60/210	30/-30	35/-40	45	80	175
AUTOBRAKE MAX	1610	80/-85	45/55	-65/220	25/-20	40/-40	65	70	175
AUTOBRAKE 2	2615	160/-175	85/110	-110/375	55/-60	80/-80	90	180	220

Medium Reported Braking Action

MAX MANUAL	2015	120/-125	65/85	-95/345	85/-65	55/-60	60	220	535
AUTOBRAKE MAX	2090	125/-130	65/90	-100/350	80/-65	60/-60	70	225	550
AUTOBRAKE 3	2325	125/-135	70/95	-105/375	55/-50	70/-70	95	125	375

Poor Reported Braking Action

MAX MANUAL	2650	180/-180	95/130	-145/545	205/-135	80/-85	75	480	1290
AUTOBRAKE MAX	2670	180/-180	95/130	-145/550	200/-130	80/-85	80	475	1290
AUTOBRAKE 3	2740	175/-180	95/130	-150/555	190/-125	80/-85	95	435	1260

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (15 ≤ Indicated Flaps <30)

VREF15

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	970	75/-60	20/30	-35/120	10/-10	20/-20	35	20	45
AUTOBRAKE MAX	1270	70/-70	30/40	-45/155	0/0	30/-30	60	0	5
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Good Reported Braking Action

MAX MANUAL	1320	75/-75	35/45	-55/200	30/-25	35/-35	45	70	160
AUTOBRAKE MAX	1420	80/-85	40/50	-60/210	30/-20	35/-35	55	80	175
AUTOBRAKE 2	2270	150/-160	70/95	-105/350	35/-45	65/-65	90	105	105

Medium Reported Braking Action

MAX MANUAL	1815	120/-120	55/75	-90/330	80/-65	50/-50	60	195	485
AUTOBRAKE MAX	1860	120/-125	60/80	-95/335	75/-60	50/-55	70	200	490
AUTOBRAKE 3	1990	125/-130	60/80	-95/350	55/-40	55/-60	100	125	390

Poor Reported Braking Action

MAX MANUAL	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1165
AUTOBRAKE MAX	2375	170/-170	85/115	-140/525	195/-125	70/-75	75	425	1160
AUTOBRAKE 3	2405	170/-170	85/115	-140/530	185/-115	70/-75	90	425	1165

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flap Disagree (30 ≤ Indicated Flaps <40)

VREF30

	LANDING DISTANCE AND ADJUSTMENTS (M)							
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV NO REV

Dry Runway

MAX MANUAL	930	60/-55	20/25	-35/120	10/-10	20/-20	35	20	35
AUTOBRAKE MAX	1200	60/-65	25/35	-45/145	0/0	30/-30	55	0	5
AUTOBRAKE 2	2100	140/-145	65/85	-100/335	35/-45	60/-60	85	95	100

Good Reported Braking Action

MAX MANUAL	1265	70/-70	35/45	-55/195	30/-25	30/-30	45	65	140
AUTOBRAKE MAX	1355	75/-80	35/45	-60/205	25/-20	35/-35	55	70	155
AUTOBRAKE 2	2100	140/-145	65/85	-100/335	35/-45	60/-60	85	95	100

Medium Reported Braking Action

MAX MANUAL	1715	110/-110	55/70	-90/325	75/-60	45/-50	60	170	415
AUTOBRAKE MAX	1760	115/-115	55/70	-90/325	70/-55	50/-50	70	175	425
AUTOBRAKE 3	1870	115/-120	55/75	-95/340	55/-45	50/-55	85	115	340

Poor Reported Braking Action

MAX MANUAL	2220	155/-155	75/105	-135/510	180/-120	65/-70	70	365	970
AUTOBRAKE MAX	2230	155/-155	75/105	-135/510	185/-115	65/-70	80	365	965
AUTOBRAKE 3	2260	160/-160	75/105	-135/515	175/-115	65/-70	80	365	975

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Trailing Edge Flaps Up Landing

VREF40 + 40

	LANDING DISTANCE AND ADJUSTMENTS (M)								
	REF DIST	WT ADJ	ALT ADJ	WIND ADJ	SLOPE ADJ	TEMP ADJ	APP SPD ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	65000 KG LANDING WEIGHT	PER 5000 KG ABV/BLW 65000 KG	PER 1000 FT STD/HIGH*	PER 10 KTS HEAD/ TAIL WIND	PER 1% DOWN/ UP HILL	PER 10°C ABV/ BLW ISA	PER 5 KTS ABOVE VREF	ONE REV	NO REV

Dry Runway

MAX MANUAL	1175	110/-70	30/70	-40/140	15/-15	30/-30	40	30	65
AUTOBRAKE MAX	1605	75/-75	40/50	-55/175	0/0	40/-40	70	0	10
AUTOBRAKE 2	2980	175/-195	100/125	-120/400	55/-65	90/-90	105	170	180

Good Reported Braking Action

MAX MANUAL	1620	85/-90	45/60	-65/225	40/-35	45/-45	50	95	210
AUTOBRAKE MAX	1770	85/-95	50/65	-70/235	30/-20	45/-50	70	90	220
AUTOBRAKE 2	2980	175/-195	100/125	-120/400	55/-65	90/-90	105	170	180

Medium Reported Braking Action

MAX MANUAL	2285	135/-145	75/100	-105/370	100/-80	65/-70	65	270	655
AUTOBRAKE MAX	2335	140/-145	75/100	-105/375	95/-75	65/-70	75	270	665
AUTOBRAKE 3	2565	135/-150	80/105	-115/395	60/-45	75/-80	115	160	485

Poor Reported Braking Action

MAX MANUAL	3035	205/-210	110/150	-160/585	240/-160	90/-95	85	590	1610
AUTOBRAKE MAX	3025	205/-210	110/150	-160/585	240/-155	90/-95	90	585	1600
AUTOBRAKE 3	3090	200/-205	110/150	-160/590	225/-140	95/-100	110	565	1595

Reference distance is based on sea level, standard day, no wind or slope, and maximum available reverse thrust.

MAX MANUAL assumes maximum achievable manual braking.

Reference Distance includes an air distance allowance of 305 m from threshold to touchdown.

Actual (unfactored) distances are shown.

*For landing distance at or below 8000 ft pressure altitude, apply the STD adjustment. For altitudes higher than 8000 ft, first apply the STD adjustment to derive a new reference landing distance for 8000 ft then apply the HIGH adjustment to this new reference distance.

ADVISORY INFORMATION

Recommended Brake Cooling Schedule
Reference Brake Energy Per Brake (Millions of Foot Pounds)

		WIND CORRECTED BRAKES ON SPEED (KIAS)*																	
		80			100			120			140			160			180		
WEIGHT (1000 KG)	OAT (°C)	PRESSURE ALTITUDE (1000 FT)																	
		0	5	10	0	5	10	0	5	10	0	5	10	0	5	10	0	5	10
80	0	15.1	17.0	19.3	22.4	25.3	28.9	30.9	35.0	40.2	40.4	45.9	53.0	50.8	57.9	67.3	60.8	69.6	81.2
	10	15.6	17.6	20.0	23.1	26.1	29.8	31.9	36.2	41.5	41.8	47.5	54.8	52.5	59.9	69.5	62.8	71.9	83.9
	15	15.8	17.8	20.2	23.5	26.5	30.3	32.4	36.7	42.1	42.4	48.2	55.6	53.3	60.7	70.5	63.7	72.9	85.1
	20	16.0	18.1	20.5	23.8	26.9	30.7	32.8	37.2	42.7	42.9	48.8	56.3	54.0	61.5	71.4	64.6	73.9	86.2
	30	16.4	18.5	21.1	24.4	27.6	31.5	33.7	38.2	43.8	44.0	50.0	57.7	55.3	63.1	73.2	66.2	75.7	88.4
	40	16.6	18.7	21.3	24.7	27.9	31.9	34.1	38.7	44.4	44.7	50.9	58.8	56.3	64.3	74.8	67.5	77.4	90.5
50	16.6	18.7	21.3	24.8	28.0	32.1	34.3	39.0	44.9	45.2	51.5	59.7	57.1	65.4	76.3	68.7	79.0	92.9	
70	0	13.7	15.4	17.5	20.2	22.8	26.0	27.7	31.3	35.9	36.1	41.0	47.2	45.3	51.6	59.7	54.9	62.7	72.9
	10	14.2	15.9	18.1	20.8	23.5	26.8	28.6	32.4	37.1	37.3	42.3	48.7	46.8	53.3	61.6	56.7	64.8	75.4
	15	14.4	16.2	18.4	21.1	23.9	27.2	29.0	32.8	37.6	37.8	43.0	49.4	47.5	54.0	62.5	57.5	65.7	76.4
	20	14.6	16.4	18.6	21.4	24.2	27.6	29.4	33.3	38.1	38.4	43.5	50.1	48.1	54.8	63.4	58.3	66.5	77.4
	30	14.9	16.8	19.1	22.0	24.8	28.3	30.2	34.1	39.1	39.3	44.6	51.4	49.3	56.1	64.9	59.8	68.2	79.4
	40	15.1	17.0	19.3	22.2	25.1	28.6	30.5	34.6	39.6	39.9	45.3	52.2	50.1	57.1	66.2	60.9	69.6	81.2
50	15.1	17.0	19.3	22.3	25.2	28.8	30.7	34.8	40.0	40.2	45.8	52.9	50.7	58.0	67.4	61.8	70.9	83.0	
60	0	12.3	13.9	15.7	18.0	20.3	23.1	24.4	27.6	31.6	31.7	35.9	41.2	39.6	45.0	51.8	48.1	54.8	63.5
	10	12.7	14.3	16.3	18.5	20.9	23.8	25.2	28.5	32.6	32.7	37.1	42.6	40.9	46.5	53.6	49.7	56.6	65.6
	15	12.9	14.6	16.5	18.8	21.2	24.2	25.6	29.0	33.1	33.2	37.6	43.2	41.5	47.1	54.4	50.4	57.4	66.5
	20	13.1	14.8	16.7	19.1	21.5	24.5	26.0	29.4	33.5	33.6	38.1	43.8	42.0	47.8	55.1	51.1	58.2	67.4
	30	13.4	15.1	17.2	19.6	22.1	25.1	26.6	30.1	34.4	34.5	39.1	44.9	43.1	49.0	56.5	52.3	59.6	69.1
	40	13.6	15.3	17.3	19.8	22.3	25.4	26.9	30.5	34.9	35.0	39.7	45.6	43.8	49.8	57.5	53.2	60.7	70.5
50	13.5	15.3	17.3	19.8	22.4	25.5	27.0	30.6	35.1	35.2	40.0	46.0	44.2	50.4	58.3	53.9	61.7	71.9	
50	0	11.0	12.3	14.0	15.7	17.7	20.2	21.2	23.9	27.3	27.2	30.8	35.3	33.8	38.3	44.1	40.9	46.4	53.6
	10	11.3	12.7	14.4	16.3	18.3	20.8	21.9	24.7	28.2	28.1	31.8	36.5	34.9	39.6	45.5	42.2	48.0	55.4
	15	11.5	12.9	14.7	16.5	18.6	21.1	22.2	25.1	28.6	28.6	32.3	37.0	35.4	40.2	46.2	42.8	48.7	56.2
	20	11.6	13.1	14.9	16.7	18.9	21.4	22.5	25.4	29.0	28.9	32.8	37.5	35.9	40.7	46.8	43.4	49.3	56.9
	30	11.9	13.4	15.2	17.2	19.3	22.0	23.1	26.1	29.7	29.7	33.6	38.4	36.8	41.8	48.0	44.5	50.6	58.4
	40	12.1	13.6	15.4	17.3	19.5	22.2	23.4	26.4	30.1	30.1	34.0	39.0	37.4	42.4	48.8	45.2	51.4	59.4
50	12.0	13.6	15.4	17.3	19.6	22.3	23.4	26.5	30.3	30.2	34.2	39.3	37.6	42.8	49.3	45.7	52.1	60.3	
40	0	9.6	10.8	12.3	13.5	15.2	17.3	17.9	20.2	23.0	22.8	25.8	29.4	28.1	31.8	36.4	33.7	38.2	43.9
	10	10.0	11.2	12.7	14.0	15.8	17.9	18.5	20.9	23.8	23.6	26.6	30.4	29.0	32.8	37.6	34.8	39.5	45.4
	15	10.1	11.4	12.9	14.2	16.0	18.1	18.8	21.2	24.1	23.9	27.0	30.8	29.4	33.3	38.2	35.3	40.0	46.0
	20	10.2	11.5	13.1	14.4	16.2	18.4	19.1	21.5	24.5	24.2	27.4	31.3	29.8	33.8	38.7	35.8	40.6	46.6
	30	10.5	11.8	13.4	14.8	16.6	18.9	19.6	22.1	25.1	24.9	28.1	32.1	30.6	34.6	39.7	36.7	41.6	47.8
	40	10.6	11.9	13.5	14.9	16.8	19.1	19.8	22.3	25.4	25.2	28.4	32.5	31.0	35.1	40.2	37.2	42.2	48.6
50	10.6	11.9	13.5	14.9	16.8	19.1	19.8	22.3	25.5	25.2	28.6	32.7	31.1	35.3	40.6	37.5	42.6	49.1	

*To correct for wind, enter table with the brakes on speed minus one half the headwind or plus 1.5 times the tailwind. If ground speed is used for brakes on speed, ignore wind and enter table with sea level, 15°C.

ADVISORY INFORMATION

Recommended Brake Cooling Schedule

Adjusted Brake Energy Per Brake (Millions of Foot Pounds)

No Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	7.8	16.3	25.3	34.7	44.7	55.0	65.7	76.6	87.9
	MAX AUTO	7.5	15.4	23.6	32.4	41.8	51.8	62.5	74.1	86.5
	AUTOBRAKE 3	7.3	14.7	22.3	30.2	38.6	47.6	57.4	68.1	80.0
	AUTOBRAKE 2	7.0	13.8	20.5	27.4	34.8	42.7	51.5	61.3	72.4
	AUTOBRAKE 1	6.7	13.1	19.2	25.3	31.8	38.8	46.6	55.4	65.5

Two Engine Detent Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	7.0	14.6	22.8	31.4	40.5	49.9	59.7	69.8	80.0
	MAX AUTO	5.8	12.3	19.5	27.2	35.6	44.5	53.9	63.7	74.1
	AUTOBRAKE 3	4.3	9.2	14.7	20.7	27.2	34.4	42.0	50.2	59.0
	AUTOBRAKE 2	2.5	5.6	9.1	13.1	17.8	23.0	28.8	35.2	42.3
	AUTOBRAKE 1	1.8	3.8	6.1	8.8	11.9	15.5	19.6	24.4	29.8

Cooling Time (Minutes) - Category C Steel Brakes

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)								
		16 & BELOW	17	20	23	25	28	32	33 TO 48	49 & ABOVE
		BRAKE TEMPERATURE MONITOR SYSTEM INDICATION ON CDS								
		UP TO 2.4	2.6	3.1	3.5	3.9	4.4	4.9	5.0 TO 7.5	7.5 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE	1	2	3	4	5	6	CAUTION	FUSE PLUG MELT ZONE	
GROUND	REQUIRED	10	20	30	40	50	60			

Cooling Time (Minutes) - Category N Carbon Brakes

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)								
		16 & BELOW	17	19	20.9	23.5	26.9	29.4	30 TO 41	41 & ABOVE
		BRAKE TEMPERATURE MONITOR SYSTEM INDICATION ON CDS								
		UP TO 2.5	2.6	3	3.3	3.8	4.5	4.9	5.0 TO 7.1	7.1 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE	1	4	5	6	7	7.6	CAUTION	FUSE PLUG MELT ZONE	
GROUND	REQUIRED	6.7	16.0	24.1	34.2	45.9	53.3			

Observe maximum quick turnaround limit.

Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.

Add 1.0 million foot pounds per brake for each taxi mile.

When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after one hour. If overheat occurs after takeoff, extend gear soon for at least 7 minutes.

When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not approach gear or attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

Brake temperature monitor system (BTMS) indication on CDS systems page may be used 10 to 15 minutes after airplane has come to a complete stop or inflight with gear retracted to determine recommended cooling schedule.

Intentionally
Blank



Performance Inflight - QRH

Chapter PI-QRH

Engine Inoperative

Section 72

ENGINE INOP

Initial Max Continuous %N1

Based on .79M, A/C high and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)								
	25	27	29	31	33	35	37	39	41
20	96.8	96.6	96.3	96.1	95.9	95.4	95.0	94.7	93.9
15	97.4	97.2	96.9	96.8	96.6	96.2	95.7	95.5	94.8
10	98.0	97.8	97.5	97.4	97.4	96.9	96.5	96.3	95.7
5	98.3	98.6	98.3	98.1	98.1	97.7	97.3	97.1	96.6
0	97.5	98.7	99.2	99.0	98.9	98.5	98.2	98.0	97.5
-5	96.7	98.0	99.1	99.8	99.7	99.3	98.9	98.7	98.4
-10	96.0	97.2	98.4	99.6	100.5	100.2	99.8	99.6	99.4
-15	95.2	96.4	97.6	98.8	100.1	101.0	100.8	100.6	100.3
-20	94.4	95.6	96.8	98.0	99.3	100.5	101.1	100.8	100.6
-25	93.6	94.9	96.0	97.2	98.5	99.7	100.2	100.0	99.8
-30	92.8	94.1	95.2	96.4	97.7	98.8	99.4	99.2	99.0
-35	92.0	93.2	94.4	95.6	96.8	98.0	98.5	98.3	98.1
-40	91.2	92.4	93.5	94.7	96.0	97.1	97.6	97.4	97.2

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)								
	25	27	29	31	33	35	37	39	41
ENGINE ANTI-ICE	-1.2	-1.1	-1.0	-0.9	-0.8	-0.8	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE	-4.2	-4.4	-4.5	-4.7	-5.0	-4.8	-4.8	-4.8	-4.8

ENGINE INOP

**Max Continuous %N1
37000 FT to 29000 FT Pressure Altitudes**

37000 FT PRESS ALT													TAT (°C)
CIAS	M	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0
160	.51	96.6	97.6	98.5	99.4	100.2	99.6	98.8	97.6	96.3	94.7	93.2	91.8
200	.63	96.0	96.9	97.8	98.7	99.6	100.4	100.1	99.3	98.4	97.5	96.3	95.2
240	.74	95.1	96.0	96.8	97.7	98.6	99.4	100.3	100.7	100.0	99.2	98.4	97.5
280	.86	94.3	95.2	96.1	97.0	97.8	98.7	99.5	100.4	101.2	100.9	100.0	99.1
35000 FT PRESS ALT													TAT (°C)
CIAS	M	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0
160	.49	96.5	97.4	98.3	99.2	100.1	99.8	99.0	98.0	96.8	95.4	94.0	92.7
200	.60	96.1	97.0	97.9	98.8	99.7	100.6	100.5	99.6	98.6	97.6	96.5	95.4
240	.71	95.0	95.9	96.8	97.7	98.6	99.4	100.3	100.8	100.2	99.5	98.6	97.7
280	.82	93.8	94.6	95.5	96.4	97.3	98.1	98.9	99.8	100.6	100.3	99.5	98.8
33000 FT PRESS ALT													TAT (°C)
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
160	.47	97.4	98.3	99.2	100.0	100.8	100.0	99.1	97.9	96.7	95.3	93.9	92.6
200	.58	97.0	97.9	98.8	99.7	100.6	101.4	100.6	99.6	98.6	97.5	96.3	95.1
240	.68	95.9	96.8	97.7	98.5	99.4	100.2	101.1	100.9	100.2	99.4	98.4	97.4
280	.79	94.3	95.1	96.0	96.8	97.7	98.5	99.3	100.2	100.5	99.7	98.9	98.1
320	.89	93.6	94.5	95.4	96.2	97.1	97.9	98.7	99.5	100.3	101.1	100.7	99.8
31000 FT PRESS ALT													TAT (°C)
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
160	.45	97.3	98.2	99.1	100.0	100.9	101.1	100.2	99.2	98.0	96.6	95.2	93.9
200	.55	97.1	98.0	98.9	99.7	100.6	101.5	101.6	100.7	99.7	98.6	97.4	96.2
240	.66	95.6	96.5	97.4	98.3	99.1	100.0	100.8	101.3	100.5	99.8	98.8	97.8
280	.76	93.8	94.7	95.5	96.4	97.2	98.0	98.8	99.7	100.5	99.8	98.9	98.0
320	.85	92.4	93.2	94.1	94.9	95.7	96.5	97.4	98.2	98.9	99.7	99.9	99.1
29000 FT PRESS ALT													TAT (°C)
CIAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10
160	.43	98.1	99.0	99.9	100.8	101.6	101.2	100.2	99.1	97.9	96.4	95.1	93.8
200	.53	97.5	98.4	99.3	100.2	101.0	101.9	101.3	100.4	99.3	98.2	96.9	95.8
240	.63	96.3	97.1	98.0	98.9	99.7	100.5	101.4	101.1	100.2	99.2	98.3	97.2
280	.73	94.2	95.0	95.9	96.7	97.5	98.3	99.1	99.9	100.1	99.1	98.2	97.5
320	.82	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.5	99.2	98.5	97.6
360	.91	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.5	99.2	100.0	100.1

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	29	31	33	35	37
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE ON	-4.1	-4.3	-4.5	-4.7	-4.7

ENGINE INOP

Max Continuous %N1

27000 FT to 20000 FT Pressure Altitudes

27000 FT PRESS ALT												TAT (°C)	
KIAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10
160	.41	98.0	98.8	99.7	100.6	101.4	102.2	101.2	100.2	99.0	97.8	96.4	95.1
200	.51	96.9	97.8	98.7	99.6	100.4	101.2	101.8	100.8	99.9	98.8	97.6	96.4
240	.60	95.6	96.5	97.4	98.2	99.1	99.9	100.7	101.3	100.4	99.4	98.5	97.5
280	.70	93.6	94.4	95.3	96.1	96.9	97.7	98.5	99.3	100.1	99.4	98.4	97.6
320	.79	91.6	92.4	93.2	94.0	94.8	95.6	96.4	97.2	98.0	98.7	98.6	97.8
360	.88	91.0	91.8	92.6	93.4	94.2	95.0	95.8	96.6	97.3	98.1	98.8	99.4
25000 FT PRESS ALT												TAT (°C)	
KIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
160	.39	98.8	99.7	100.5	101.4	102.2	102.4	101.4	100.3	99.1	97.7	96.5	95.2
200	.49	97.5	98.3	99.2	100.0	100.9	101.7	101.5	100.6	99.5	98.4	97.3	96.2
240	.58	95.7	96.5	97.4	98.2	99.0	99.9	100.7	100.5	99.5	98.6	97.6	96.7
280	.67	93.9	94.7	95.5	96.3	97.1	97.9	98.7	99.5	99.5	98.6	97.6	96.9
320	.76	91.7	92.6	93.4	94.2	95.0	95.8	96.5	97.3	98.0	98.6	97.8	97.2
360	.85	90.4	91.2	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.6	98.4	98.2
24000 FT PRESS ALT												TAT (°C)	
KIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
160	.38	98.6	99.5	100.4	101.2	102.1	102.9	101.9	100.8	99.6	98.4	97.1	95.8
200	.48	97.5	98.4	99.2	100.1	100.9	101.8	102.2	101.1	100.1	99.0	97.8	96.7
240	.57	95.9	96.8	97.6	98.5	99.3	100.1	100.9	101.2	100.2	99.2	98.2	97.3
280	.66	94.2	95.1	95.9	96.7	97.5	98.3	99.1	99.9	100.4	99.4	98.3	97.5
320	.75	92.1	93.0	93.8	94.6	95.4	96.2	96.9	97.7	98.5	99.2	98.6	97.8
360	.83	90.6	91.4	92.2	93.1	93.9	94.7	95.5	96.2	97.0	97.8	98.5	98.6
22000 FT PRESS ALT												TAT (°C)	
KIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
160	.37	99.1	100.0	100.9	101.7	102.5	102.8	101.8	100.7	99.5	98.2	97.0	95.8
200	.46	98.4	99.3	100.1	101.0	101.8	102.6	102.3	101.2	100.0	98.9	97.8	96.8
240	.55	97.2	98.1	98.9	99.7	100.5	101.3	102.1	101.6	100.5	99.4	98.5	97.5
280	.63	95.7	96.5	97.4	98.2	99.0	99.8	100.6	101.3	101.0	99.8	98.9	98.1
320	.72	93.9	94.7	95.5	96.3	97.1	97.9	98.6	99.4	100.1	100.2	99.3	98.6
360	.80	92.2	93.0	93.8	94.6	95.4	96.1	96.9	97.7	98.4	99.2	99.7	99.1
20000 FT PRESS ALT												TAT (°C)	
KIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
160	.35	98.7	99.5	100.4	101.2	102.0	102.8	102.5	101.5	100.4	99.2	98.0	96.8
200	.44	98.3	99.2	100.0	100.9	101.7	102.5	103.3	102.3	101.1	100.0	98.9	97.8
240	.53	97.5	98.4	99.2	100.0	100.8	101.7	102.5	103.1	101.8	100.5	99.5	98.6
280	.61	96.2	97.0	97.8	98.7	99.5	100.3	101.1	101.8	102.5	101.3	100.1	99.3
320	.69	94.7	95.5	96.3	97.1	97.9	98.7	99.5	100.2	101.0	101.7	100.9	99.9
360	.77	93.0	93.8	94.6	95.4	96.2	97.0	97.7	98.5	99.2	100.0	100.7	100.4

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	20	22	24	25	27
ENGINE ANTI-ICE ON	-0.9	-0.9	-1.0	-1.0	-1.0
ENGINE & WING ANTI-ICE ON	-3.6	-3.8	-3.8	-3.9	-4.0

ENGINE INOP

Max Continuous %N1
18000 FT to 12000 FT Pressure Altitudes

18000 FT PRESS ALT												TAT (°C)	
KLAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
160	.34	98.5	99.3	100.2	101.0	101.8	102.6	101.6	100.3	99.2	98.1	97.0	95.9
200	.42	98.7	99.6	100.4	101.2	102.0	102.8	103.1	101.7	100.4	99.3	98.3	97.3
240	.51	97.8	98.7	99.5	100.3	101.1	101.9	102.7	102.5	101.1	99.9	99.0	98.1
280	.59	96.3	97.1	97.9	98.7	99.5	100.3	101.0	101.8	101.6	100.5	99.6	98.8
320	.67	94.8	95.6	96.4	97.2	97.9	98.7	99.5	100.2	101.0	100.9	100.0	99.2
360	.75	93.0	93.8	94.6	95.3	96.1	96.9	97.6	98.4	99.1	99.9	100.2	99.6
16000 FT PRESS ALT												TAT (°C)	
KLAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
160	.33	97.1	98.0	98.8	99.6	100.4	101.2	101.6	100.3	99.1	98.1	97.1	96.1
200	.41	98.0	98.8	99.6	100.4	101.2	102.0	102.8	102.5	101.3	100.2	99.3	98.3
240	.49	97.1	97.9	98.7	99.5	100.3	101.1	101.9	102.7	101.8	100.5	99.6	98.7
280	.57	95.6	96.4	97.2	98.0	98.8	99.6	100.3	101.1	101.8	100.9	99.8	99.0
320	.64	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.4	100.2	100.9	100.2	99.4
360	.72	92.1	92.9	93.7	94.5	95.3	96.1	96.9	97.7	98.4	99.2	99.9	99.6
14000 FT PRESS ALT												TAT (°C)	
KLAS	M	-25	-20	-15	-10	-5	0	5	10	15	20	25	30
160	.31	96.6	97.4	98.2	99.0	99.8	100.6	100.4	99.1	98.0	97.1	96.2	95.3
200	.39	97.1	97.9	98.7	99.5	100.3	101.1	101.8	101.5	101.0	100.1	99.3	98.4
240	.47	96.6	97.4	98.2	99.0	99.8	100.6	101.3	101.8	101.1	100.3	99.5	98.7
280	.54	95.5	96.3	97.1	97.8	98.6	99.4	100.1	100.9	101.0	100.1	99.2	98.5
320	.62	94.1	94.9	95.7	96.5	97.2	98.0	98.7	99.5	100.2	100.3	99.5	98.8
360	.69	92.2	93.1	93.9	94.7	95.5	96.3	97.0	97.8	98.6	99.3	99.6	99.0
12000 FT PRESS ALT												TAT (°C)	
KLAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35
160	.30	96.3	97.0	97.8	98.6	99.4	100.1	99.3	98.1	97.1	96.3	95.4	94.5
200	.38	97.1	97.9	98.7	99.5	100.3	101.0	101.5	100.8	99.8	99.0	98.2	97.3
240	.45	96.5	97.3	98.0	98.8	99.6	100.3	101.1	101.0	100.1	99.4	98.6	97.9
280	.52	95.5	96.3	97.0	97.8	98.6	99.3	100.0	100.8	100.3	99.4	98.6	98.0
320	.60	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.4	100.2	99.7	98.9	98.2
360	.67	92.3	93.2	94.0	94.8	95.6	96.4	97.1	97.9	98.7	99.4	99.1	98.5

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)			
	12	14	16	18
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.9	-0.9
ENGINE & WING ANTI-ICE ON	-3.2	-3.4	-3.4	-3.5

ENGINE INOP

Max Continuous %N1

10000 FT to 1000 FT Pressure Altitudes

10000 FT PRESS ALT		TAT (°C)											
KIAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35
160	.29	95.2	96.0	96.8	97.6	98.3	99.1	99.8	98.6	97.4	96.6	95.8	94.9
200	.36	96.0	96.7	97.5	98.3	99.0	99.8	100.5	100.5	99.4	98.5	97.8	97.0
240	.43	95.6	96.4	97.2	97.9	98.7	99.4	100.2	100.9	100.1	99.2	98.4	97.7
280	.51	94.5	95.3	96.1	96.9	97.6	98.4	99.1	99.9	100.4	99.5	98.7	98.0
320	.58	93.0	93.9	94.7	95.5	96.2	97.0	97.8	98.6	99.3	99.7	99.0	98.2
360	.65	91.6	92.4	93.2	94.0	94.8	95.6	96.4	97.2	98.0	98.7	99.1	98.5
5000 FT PRESS ALT		TAT (°C)											
KIAS	M	-10	-5	0	5	10	15	20	25	30	35	40	45
160	.26	94.9	95.7	96.4	97.2	98.0	98.8	99.2	98.3	97.4	96.6	95.9	95.1
200	.33	94.7	95.5	96.3	97.1	97.8	98.6	99.4	98.9	98.0	97.3	96.6	95.8
240	.40	94.0	94.8	95.6	96.4	97.2	97.9	98.7	99.5	98.7	97.9	97.2	96.5
280	.46	93.3	94.1	94.9	95.7	96.5	97.3	98.1	98.8	98.9	98.2	97.5	96.8
320	.53	92.5	93.3	94.1	94.9	95.7	96.5	97.2	98.0	98.7	98.4	97.7	97.1
360	.59	91.5	92.3	93.1	93.9	94.7	95.5	96.2	97.0	97.8	98.5	98.0	97.3
3000 FT PRESS ALT		TAT (°C)											
KIAS	M	-5	0	5	10	15	20	25	30	35	40	45	50
160	.26	94.8	95.6	96.4	97.2	98.0	98.7	98.8	97.9	97.1	96.4	95.6	94.8
200	.32	94.5	95.3	96.1	96.9	97.6	98.4	99.2	98.3	97.5	96.8	96.1	95.3
240	.38	94.1	94.9	95.6	96.4	97.2	98.0	98.7	98.8	98.0	97.2	96.6	95.9
280	.45	93.2	94.0	94.8	95.6	96.4	97.2	97.9	98.7	98.3	97.5	96.9	96.2
320	.51	92.5	93.3	94.1	94.9	95.7	96.4	97.2	98.0	98.5	97.8	97.1	96.5
360	.57	91.6	92.4	93.2	94.0	94.7	95.5	96.3	97.1	97.8	98.1	97.4	96.8
1000 FT PRESS ALT		TAT (°C)											
KIAS	M	-5	0	5	10	15	20	25	30	35	40	45	50
160	.25	93.9	94.7	95.4	96.2	97.0	97.8	98.5	98.2	97.4	96.7	96.0	95.2
200	.31	93.5	94.3	95.1	95.9	96.7	97.4	98.2	98.5	97.8	97.0	96.3	95.6
240	.37	93.0	93.8	94.6	95.4	96.1	96.9	97.7	98.4	98.1	97.3	96.6	95.9
280	.43	92.3	93.2	93.9	94.7	95.5	96.3	97.1	97.8	98.3	97.6	96.9	96.2
320	.49	91.6	92.4	93.2	94.0	94.8	95.6	96.3	97.1	97.9	97.9	97.2	96.5
360	.55	90.7	91.5	92.3	93.1	93.9	94.7	95.4	96.2	96.9	97.7	97.3	96.6

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)			
	1	3	5	10
ENGINE ANTI-ICE ON	-0.6	-0.8	-0.8	-0.8
ENGINE & WING ANTI-ICE ON	-2.9	-3.0	-2.7	-3.2



ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	82	271	18500	17300	15900
80	77	263	20200	19000	17700
75	72	255	21600	20600	19400
70	67	247	23100	22200	21100
65	62	238	24700	23800	22800
60	57	229	26800	25800	24700
55	53	219	29100	28100	27000
50	48	209	31200	30400	29400
45	43	199	33300	32600	31700
40	38	187	35600	34900	34000

Includes APU fuel burn.

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown/LRC Cruise Range Capability

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
138	128	120	112	106	100	95	90	86	82	78
275	256	239	225	212	200	190	180	172	164	157
413	384	359	337	317	300	284	270	258	246	235
551	512	479	449	423	400	379	360	344	328	314
689	640	598	562	529	500	474	451	429	410	392
826	768	718	674	635	600	569	541	515	492	471
964	896	838	786	741	700	664	631	601	574	549
1102	1025	957	898	846	800	758	721	687	656	628
1240	1153	1077	1011	952	900	853	811	773	738	706
1377	1281	1197	1123	1058	1000	948	901	859	820	785
1515	1409	1317	1235	1164	1100	1043	991	945	902	863
1653	1537	1436	1348	1270	1200	1138	1081	1030	984	942
1792	1666	1556	1460	1375	1300	1232	1171	1116	1066	1020
1930	1794	1676	1573	1481	1400	1327	1261	1202	1148	1098
2068	1922	1796	1685	1587	1500	1422	1351	1288	1230	1177
2207	2051	1916	1798	1693	1600	1517	1441	1373	1312	1255
2345	2180	2036	1910	1799	1700	1611	1531	1459	1393	1333
2484	2309	2156	2023	1905	1800	1706	1621	1545	1475	1411

Driftdown/Cruise Fuel and Time

AIR DIST (NM)	FUEL REQUIRED (1000 KG)										TIME (HR:MIN)
	WEIGHT AT START OF DRIFTDOWN (1000 KG)										
	40	45	50	55	60	65	70	75	80	85	
100	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0:16
200	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.3	0:33
300	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	0:49
400	1.6	1.8	1.9	2.0	2.2	2.3	2.5	2.6	2.8	2.9	1:06
500	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.3	3.5	3.7	1:22
600	2.4	2.7	2.9	3.1	3.3	3.6	3.8	4.0	4.3	4.5	1:39
700	2.8	3.1	3.4	3.6	3.9	4.2	4.5	4.7	5.0	5.3	1:55
800	3.2	3.6	3.9	4.2	4.5	4.8	5.1	5.4	5.7	6.1	2:11
900	3.6	4.0	4.3	4.7	5.0	5.4	5.7	6.1	6.4	6.8	2:28
1000	4.0	4.4	4.8	5.2	5.6	6.0	6.4	6.7	7.1	7.6	2:44
1100	4.4	4.8	5.3	5.7	6.1	6.6	7.0	7.4	7.9	8.3	3:01
1200	4.8	5.3	5.7	6.2	6.7	7.1	7.6	8.1	8.6	9.0	3:17
1300	5.2	5.7	6.2	6.7	7.2	7.7	8.2	8.7	9.2	9.8	3:34
1400	5.5	6.1	6.6	7.2	7.7	8.3	8.8	9.4	9.9	10.5	3:51
1500	5.9	6.5	7.1	7.7	8.3	8.9	9.4	10.0	10.6	11.2	4:07
1600	6.3	6.9	7.5	8.2	8.8	9.4	10.0	10.7	11.3	12.0	4:24
1700	6.6	7.3	8.0	8.6	9.3	10.0	10.6	11.3	12.0	12.7	4:41
1800	7.0	7.7	8.4	9.1	9.8	10.5	11.2	11.9	12.6	13.4	4:57

Includes APU fuel burn.

Driftdown at optimum driftdown speed and cruise at long range cruise speed.

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Altitude Capability

100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	15200	12600	9900
80	17200	15300	12500
75	19200	17400	15000
70	20900	19700	17300
65	22500	21300	19800
60	24100	23000	21600
55	26300	24800	23500
50	29000	27700	25800
45	31400	30500	29200
40	33800	33000	31800

With engine anti-ice on, decrease altitude capability by 1200 ft.

With engine and wing anti-ice on, decrease altitude capability by 5500 ft.

ENGINE INOP

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		10	15	17	19	21	23	25	27	29	31
85	%N1	91.8	95.5	97.9							
	MACH	.561	.600	.616							
	KIAS	311	303	300							
	FF/ENG	3067	3033	3052							
80	%N1	90.1	94.0	95.9	98.5						
	MACH	.545	.590	.603	.621						
	KIAS	302	299	294	291						
	FF/ENG	2875	2870	2846	2886						
75	%N1	88.4	92.5	94.0	96.1						
	MACH	.528	.579	.593	.607						
	KIAS	293	293	288	284						
	FF/ENG	2684	2709	2674	2662						
70	%N1	86.5	90.7	92.3	94.0	96.2					
	MACH	.510	.562	.582	.595	.610					
	KIAS	282	284	283	278	274					
	FF/ENG	2494	2518	2520	2481	2487					
65	%N1	84.5	88.7	90.4	92.2	93.9	96.4				
	MACH	.491	.542	.563	.584	.596	.612				
	KIAS	271	274	274	273	268	265				
	FF/ENG	2306	2327	2330	2330	2295	2317				
60	%N1	82.3	86.5	88.3	90.0	91.9	93.7	96.4			
	MACH	.471	.521	.543	.564	.585	.597	.614			
	KIAS	261	263	263	263	263	258	254			
	FF/ENG	2124	2137	2139	2140	2143	2114	2146			
55	%N1	80.2	84.2	85.9	87.7	89.5	91.4	93.3	96.2		
	MACH	.453	.498	.520	.541	.563	.585	.597	.614		
	KIAS	250	251	252	252	253	252	247	244		
	FF/ENG	1954	1948	1950	1950	1953	1958	1938	1971		
50	%N1	77.8	81.6	83.4	85.2	87.0	88.7	90.7	92.7	95.7	
	MACH	.434	.475	.495	.516	.538	.561	.583	.596	.613	
	KIAS	240	239	239	240	241	241	241	236	233	
	FF/ENG	1791	1764	1762	1762	1764	1767	1777	1765	1793	
45	%N1	75.5	79.1	80.6	82.3	84.1	85.9	87.7	89.7	91.8	94.8
	MACH	.415	.452	.469	.489	.511	.533	.556	.578	.593	.610
	KIAS	229	227	227	227	228	229	229	229	225	222
	FF/ENG	1636	1594	1582	1575	1577	1580	1586	1600	1593	1613
40	%N1	73.0	76.2	77.8	79.4	81.0	82.8	84.6	86.4	88.3	90.7
	MACH	.395	.429	.445	.462	.480	.502	.525	.548	.571	.589
	KIAS	218	215	215	214	214	215	216	216	216	214
	FF/ENG	1485	1434	1416	1402	1392	1394	1400	1410	1421	1424

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time
Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
298	272	249	230	214	200	190	180	172	164	158
600	547	501	462	429	400	379	361	344	328	315
903	823	753	694	644	600	570	542	517	494	473
1209	1100	1005	926	859	800	759	721	687	657	630
1516	1379	1259	1159	1075	1000	949	902	859	820	786
1825	1659	1513	1393	1290	1200	1139	1082	1031	984	943
2137	1940	1768	1626	1506	1400	1328	1262	1202	1147	1099
2450	2222	2024	1860	1722	1600	1518	1442	1373	1311	1256
2766	2507	2281	2095	1938	1800	1707	1622	1544	1474	1412
3083	2792	2539	2331	2155	2000	1896	1801	1715	1637	1568

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		18		22		26	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.4	0:43	1.2	0:41	1.1	0:39	1.0	0:38	0.9	0:37
400	2.8	1:23	2.6	1:19	2.4	1:14	2.2	1:11	2.1	1:09
600	4.3	2:04	3.9	1:57	3.6	1:50	3.4	1:45	3.2	1:42
800	5.7	2:46	5.2	2:36	4.9	2:26	4.5	2:19	4.4	2:14
1000	7.1	3:28	6.6	3:15	6.1	3:03	5.7	2:53	5.5	2:47
1200	8.5	4:10	7.9	3:55	7.3	3:40	6.8	3:28	6.6	3:21
1400	9.8	4:53	9.1	4:36	8.5	4:18	8.0	4:02	7.7	3:54
1600	11.2	5:36	10.4	5:16	9.7	4:55	9.1	4:38	8.7	4:28
1800	12.5	6:20	11.7	5:58	10.9	5:34	10.2	5:13	9.8	5:02
2000	13.9	7:05	12.9	6:39	12.0	6:13	11.3	5:49	10.8	5:36

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)									
	40	45	50	55	60	65	70	75	80	
1	-0.1	-0.1	-0.1	0.0	0.0	0.1	0.1	0.2	0.3	
2	-0.3	-0.2	-0.1	-0.1	0.0	0.2	0.3	0.6	0.8	
3	-0.4	-0.3	-0.2	-0.1	0.0	0.3	0.5	0.9	1.2	
4	-0.6	-0.4	-0.3	-0.1	0.0	0.3	0.7	1.2	1.6	
5	-0.7	-0.5	-0.4	-0.2	0.0	0.4	0.9	1.4	2.0	
6	-0.8	-0.6	-0.4	-0.2	0.0	0.5	1.1	1.7	2.4	
7	-1.0	-0.8	-0.5	-0.3	0.0	0.6	1.2	2.0	2.8	
8	-1.1	-0.9	-0.6	-0.3	0.0	0.6	1.4	2.2	3.2	
9	-1.3	-1.0	-0.7	-0.3	0.0	0.7	1.5	2.4	3.5	
10	-1.4	-1.1	-0.7	-0.4	0.0	0.7	1.6	2.6	3.8	
11	-1.6	-1.2	-0.8	-0.4	0.0	0.8	1.7	2.8	4.1	
12	-1.7	-1.3	-0.9	-0.4	0.0	0.8	1.9	3.0	4.4	
13	-1.9	-1.4	-0.9	-0.5	0.0	0.9	2.0	3.2	4.7	
14	-2.0	-1.5	-1.0	-0.5	0.0	0.9	2.0	3.4	4.9	

Includes APU fuel burn.



ENGINE INOP
MAX CONTINUOUS THRUST

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)							
		1500	5000	10000	15000	20000	25000	30000	35000
85	%N1	81.1	84.1	88.3	92.8				
	KIAS	250	251	252	253				
	FF/ENG	2740	2730	2750	2800				
80	%N1	79.5	82.4	86.5	91.0	98.3			
	KIAS	242	243	244	245	247			
	FF/ENG	2580	2570	2570	2610	2740			
75	%N1	77.8	80.5	84.7	89.1	95.0			
	KIAS	235	236	236	238	239			
	FF/ENG	2420	2400	2400	2420	2490			
70	%N1	76.0	78.6	82.8	87.1	92.1			
	KIAS	227	227	228	229	231			
	FF/ENG	2260	2240	2230	2250	2270			
65	%N1	74.0	76.7	80.8	85.0	89.7	97.7		
	KIAS	219	219	220	221	222	224		
	FF/ENG	2100	2090	2070	2070	2080	2230		
60	%N1	71.7	74.6	78.5	82.8	87.4	93.7		
	KIAS	210	210	211	212	213	214		
	FF/ENG	1950	1930	1910	1910	1910	1970		
55	%N1	69.4	72.3	76.3	80.5	84.9	90.0		
	KIAS	200	201	202	203	204	205		
	FF/ENG	1800	1770	1750	1740	1730	1760		
50	%N1	67.0	69.7	73.8	77.8	82.3	87.0	94.9	
	KIAS	191	191	192	193	194	195	196	
	FF/ENG	1650	1620	1600	1580	1570	1570	1680	
45	%N1	64.3	66.9	71.0	75.0	79.4	84.0	89.6	
	KIAS	184	184	184	184	184	185	186	
	FF/ENG	1500	1470	1440	1430	1400	1400	1450	
40	%N1	61.1	64.0	67.8	72.0	76.2	80.7	85.4	94.1
	KIAS	177	177	177	177	177	177	177	177
	FF/ENG	1350	1330	1300	1270	1250	1240	1260	1360

This table includes 5% additional fuel for holding in a racetrack pattern.

ENGINE INOP

ADVISORY INFORMATION

**Gear Down Landing Rate of Climb Available
Flaps 15**

TAT (°C)	RATE OF CLIMB (FT/MIN)					
	PRESSURE ALTITUDE (FT)					
	-2000	0	2000	4000	6000	8000
52	50	-10	-120			
50	80	20	-80	-200		
48	110	50	-50	-160		
46	140	70	-30	-130	-250	
44	170	100	0	-100	-220	
42	200	130	30	-80	-190	-320
40	220	160	50	-50	-160	-300
38	250	190	80	-20	-140	-280
36	270	220	110	0	-120	-250
34	270	250	140	20	-100	-230
32	270	270	160	40	-70	-210
30	280	270	180	60	-60	-190
20	290	280	200	90	-20	-130
10	300	290	210	100	-10	-120
0	310	300	210	100	-10	-120
-20	330	320	230	110	0	-120
-40	350	340	240	120	0	-120

Rate of climb capability shown is valid for 60000 kg, gear down at VREF15+5.
Decrease rate of climb 120 ft/min per 5000 kg greater than 60000 kg.
Increase rate of climb 160 ft/min per 5000 kg less than 60000 kg.

Flaps 30

TAT (°C)	RATE OF CLIMB (FT/MIN)					
	PRESSURE ALTITUDE (FT)					
	-2000	0	2000	4000	6000	8000
52	-250	-310	-420			
50	-220	-290	-390	-510		
48	-190	-260	-360	-480		
46	-170	-240	-340	-450	-570	
44	-140	-210	-320	-420	-540	
42	-110	-180	-290	-400	-510	-650
40	-90	-160	-260	-370	-490	-630
38	-60	-130	-240	-350	-470	-610
36	-40	-100	-210	-320	-450	-580
34	-40	-70	-180	-300	-430	-560
32	-40	-50	-160	-290	-410	-540
30	-40	-50	-150	-270	-390	-520
20	-30	-50	-130	-240	-350	-470
10	-20	-40	-130	-240	-360	-470
0	-20	-40	-130	-240	-360	-470
-20	-10	-30	-130	-250	-370	-490
-40	-10	-30	-130	-250	-380	-500

Rate of climb capability shown is valid for 60000 kg, gear down at VREF30+5.
Decrease rate of climb 120 ft/min per 5000 kg greater than 60000 kg.
Increase rate of climb 170 ft/min per 5000 kg less than 60000 kg.



Performance Inflight - QRH

Chapter PI-QRH

Gear Down

Section 73

GEAR DOWN

Long Range Cruise Altitude Capability

Max Cruise Thrust, 100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	15600	12500	9400
80	18400	15500	12600
75	21100	18500	15700
70	23600	21400	18600
65	26100	24400	21800
60	28600	27100	25300
55	30800	29600	28100
50	32900	31900	30700
45	35100	34100	33000
40	37500	36500	35400

GEAR DOWN

**Long Range Cruise Enroute Fuel and Time
Ground to Air Miles Conversion**

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
324	290	260	236	217	200	188	178	168	160	153
654	583	523	474	435	400	377	357	338	321	307
989	880	787	713	653	600	566	535	507	483	461
1329	1181	1054	953	871	800	754	713	676	643	614
1674	1484	1322	1194	1090	1000	943	891	844	803	766
2024	1791	1593	1436	1310	1200	1131	1069	1013	962	918
2381	2103	1865	1680	1530	1400	1320	1247	1181	1122	1070
2743	2417	2140	1924	1751	1600	1508	1424	1348	1280	1221
3113	2737	2418	2171	1972	1800	1695	1600	1514	1438	1371

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	2.4	0:49	2.2	0:47	1.9	0:44	1.7	0:42	1.6	0:41
400	4.9	1:36	4.5	1:31	4.0	1:25	3.7	1:20	3.5	1:17
600	7.4	2:25	6.8	2:17	6.1	2:06	5.7	1:59	5.4	1:54
800	9.8	3:14	9.1	3:03	8.1	2:48	7.6	2:38	7.2	2:31
1000	12.1	4:04	11.3	3:50	10.1	3:30	9.5	3:18	9.0	3:08
1200	14.4	4:56	13.5	4:39	12.1	4:14	11.3	3:58	10.7	3:46
1400	16.7	5:49	15.6	5:28	14.0	4:58	13.1	4:40	12.4	4:24
1600	18.9	6:43	17.7	6:18	15.9	5:44	14.9	5:22	14.1	5:03
1800	21.1	7:38	19.7	7:10	17.7	6:30	16.6	6:05	15.7	5:43

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	40	50	60	70	80
2	-0.3	-0.2	0.0	0.3	0.7
4	-0.7	-0.3	0.0	0.6	1.3
6	-1.0	-0.5	0.0	0.9	2.0
8	-1.3	-0.7	0.0	1.2	2.6
10	-1.7	-0.8	0.0	1.4	3.2
12	-2.0	-1.0	0.0	1.6	3.7
14	-2.4	-1.2	0.0	1.8	4.2
16	-2.7	-1.3	0.0	2.0	4.6
18	-3.0	-1.5	0.0	2.2	5.0
20	-3.4	-1.7	0.0	2.4	5.3
22	-3.7	-1.8	0.0	2.5	5.6

GEAR DOWN

Descent

VREF40 + 70 KIAS

PRESSURE ALTITUDE (FT)	TIME (MIN)	FUEL (KG)	DISTANCE (NM)
41000	21	280	91
39000	20	270	86
37000	19	270	81
35000	19	260	77
33000	18	260	72
31000	17	250	68
29000	17	250	64
27000	16	240	60
25000	15	230	56
23000	14	230	52
21000	13	220	48
19000	13	210	44
17000	12	200	40
15000	11	190	36
10000	8	170	26
5000	6	140	16
1500	4	110	9

Allowances for a straight-in approach are included.



GEAR DOWN

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)							
		1500	5000	10000	15000	20000	25000	30000	35000
85	%N1	75.7	78.4	82.7	86.9	91.9			
	KIAS	229	229	229	229	229			
	FF/ENG	2240	2220	2220	2230	2250			
80	%N1	74.1	76.9	81.0	85.3	89.9			
	KIAS	224	224	224	224	224			
	FF/ENG	2110	2100	2090	2090	2100			
75	%N1	72.3	75.3	79.2	83.6	88.1			
	KIAS	218	218	218	218	218			
	FF/ENG	1990	1970	1960	1960	1960			
70	%N1	70.6	73.5	77.5	81.8	86.2	91.7		
	KIAS	213	213	213	213	213	213		
	FF/ENG	1870	1850	1840	1830	1830	1860		
65	%N1	68.8	71.7	75.8	80.0	84.4	89.1		
	KIAS	209	209	209	209	209	209		
	FF/ENG	1760	1740	1720	1710	1700	1720		
60	%N1	66.9	69.7	73.9	77.9	82.3	86.9	94.1	
	KIAS	203	203	203	203	203	203	203	
	FF/ENG	1650	1620	1600	1590	1580	1580	1660	
55	%N1	65.0	67.6	71.8	75.8	80.2	84.7	90.2	
	KIAS	197	197	197	197	197	197	197	
	FF/ENG	1530	1510	1490	1470	1450	1450	1490	
50	%N1	62.7	65.5	69.4	73.6	77.8	82.3	87.0	
	KIAS	191	191	191	191	191	191	191	
	FF/ENG	1420	1400	1370	1350	1330	1320	1350	
45	%N1	60.2	63.1	67.0	71.2	75.3	79.8	84.4	91.3
	KIAS	184	184	184	184	184	184	184	184
	FF/ENG	1310	1290	1260	1240	1210	1200	1220	1260
40	%N1	57.7	60.4	64.5	68.5	72.8	77.1	81.5	86.6
	KIAS	177	177	177	177	177	177	177	177
	FF/ENG	1200	1170	1150	1130	1100	1080	1090	1110

This table includes 5% additional fuel for holding in a racetrack pattern.

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Performance Inflight - QRH

Chapter PI-QRH

Gear Down, Engine Inop

Section 74

GEAR DOWN

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
85	80	227	1700		
80	76	223	4000	2300	200
75	71	218	6300	4900	2800
70	66	213	8600	7300	5300
65	62	208	10900	9800	8000
60	57	202	13200	12300	10900
55	52	196	15600	14800	13900
50	47	190	18100	17300	16500
45	43	183	20600	19800	18900
40	38	176	23100	22300	21400

Includes APU fuel burn.

Long Range Cruise Altitude Capability

100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
75	1500		
70	4500	2500	
65	7500	5900	3400
60	10600	9200	6900
55	13300	12300	10600
50	16200	15400	14500
45	19300	18300	17500
40	22200	21400	20500

GEAR DOWN

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		5	7	9	11	13	15	17	19	21	23
70	%N1	94.8									
	MACH	.389									
	KIAS	235									
	FF/ENG	3774									
65	%N1	92.6	94.3	96.9							
	MACH	.376	.389	.402							
	KIAS	228	227	226							
	FF/ENG	3477	3485	3527							
60	%N1	90.2	91.9	93.7	96.3						
	MACH	.364	.375	.388	.402						
	KIAS	220	219	218	218						
	FF/ENG	3192	3191	3198	3240						
55	%N1	87.8	89.3	91.0	92.8	95.4					
	MACH	.351	.362	.374	.387	.400					
	KIAS	212	211	210	209	209					
	FF/ENG	2924	2909	2906	2913	2951					
50	%N1	85.3	86.7	88.2	89.9	91.7	94.2	98.2			
	MACH	.338	.348	.359	.371	.384	.398	.412			
	KIAS	204	203	202	201	200	199	198			
	FF/ENG	2672	2647	2630	2626	2633	2657	2737			
45	%N1	82.7	84.0	85.4	86.9	88.6	90.4	92.7	96.6		
	MACH	.325	.334	.344	.355	.367	.380	.393	.408		
	KIAS	196	195	193	192	191	190	189	189		
	FF/ENG	2432	2400	2374	2356	2351	2352	2359	2417		
40	%N1	79.8	81.1	82.5	83.9	85.4	87.0	88.8	90.8	94.1	98.4
	MACH	.311	.320	.329	.339	.349	.361	.374	.387	.402	.418
	KIAS	188	186	184	183	182	181	180	179	179	178
	FF/ENG	2206	2166	2133	2107	2088	2076	2069	2065	2101	2201



GEAR DOWN
ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time
Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
172	151	134	120	109	100	93	88	83	78	75
352	308	270	242	219	200	187	175	165	156	148
533	465	408	364	330	300	280	262	246	232	220
716	623	545	486	440	400	373	349	328	309	293
900	783	684	609	551	500	466	436	409	385	365
1086	943	823	733	661	600	559	523	490	462	438
1273	1105	964	856	772	700	652	610	572	538	510
1462	1267	1103	980	883	800	745	696	652	614	581
1653	1431	1245	1104	994	900	838	782	733	690	653
1845	1595	1386	1228	1105	1000	931	868	813	765	724

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)					
	6		10		14	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
100	1.3	0:27	1.1	0:26	1.0	0:26
200	2.6	0:53	2.4	0:50	2.3	0:48
300	3.9	1:18	3.7	1:15	3.6	1:11
400	5.2	1:44	4.9	1:39	4.8	1:35
500	6.5	2:10	6.1	2:04	6.0	1:58
600	7.8	2:37	7.3	2:29	7.1	2:22
700	9.1	3:03	8.5	2:55	8.3	2:46
800	10.3	3:30	9.7	3:20	9.4	3:10
900	11.6	3:58	10.9	3:46	10.5	3:35
1000	12.8	4:25	12.0	4:12	11.6	3:59

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	40	50	60	70	80
1	-0.2	-0.1	0.0	0.1	0.3
2	-0.3	-0.2	0.0	0.3	0.6
3	-0.5	-0.3	0.0	0.5	1.0
4	-0.6	-0.3	0.0	0.7	1.3
5	-0.8	-0.4	0.0	0.9	1.7
6	-1.0	-0.5	0.0	1.0	2.0
7	-1.1	-0.6	0.0	1.2	2.4
8	-1.3	-0.7	0.0	1.4	2.7
9	-1.5	-0.7	0.0	1.6	3.1
10	-1.6	-0.8	0.0	1.8	3.5
11	-1.8	-0.9	0.0	1.9	3.8
12	-1.9	-1.0	0.0	2.1	4.2
13	-2.1	-1.1	0.0	2.3	4.5
14	-2.3	-1.1	0.0	2.5	4.9

Includes APU fuel burn.

GEAR DOWN

ENGINE INOP

MAX CONTINUOUS THRUST

**Holding
Flaps Up**

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)			
		1500	5000	10000	15000
80	%N1	93.2			
	KIAS	224			
	FF/ENG	4120			
75	%N1	91.2	94.5		
	KIAS	218	218		
	FF/ENG	3840	3890		
70	%N1	89.2	92.4		
	KIAS	213	213		
	FF/ENG	3580	3610		
65	%N1	87.3	90.3	95.7	
	KIAS	209	209	209	
	FF/ENG	3340	3360	3430	
60	%N1	85.1	88.1	92.7	
	KIAS	203	203	203	
	FF/ENG	3090	3090	3130	
55	%N1	82.8	85.7	90.2	97.0
	KIAS	197	197	197	197
	FF/ENG	2850	2840	2860	2990
50	%N1	80.2	83.2	87.6	92.6
	KIAS	191	191	191	191
	FF/ENG	2610	2600	2610	2650
45	%N1	77.7	80.5	84.9	89.5
	KIAS	184	184	184	184
	FF/ENG	2390	2370	2360	2380
40	%N1	75.0	77.7	82.0	86.4
	KIAS	177	177	177	177
	FF/ENG	2170	2140	2120	2130

This table includes 5% additional fuel for holding in a racetrack pattern.

Performance Inflight - QRH**Chapter PI-QRH****Text****Section 75**

Introduction

This chapter contains information to supplement performance data from the Flight Management Computer (FMC). In addition, sufficient inflight data is provided to complete a flight with the FMC inoperative. In the event of conflict between data presented in this chapter and that contained in the approved Airplane Flight Manual, the Flight Manual shall always take precedence.

General**Flight with Unreliable Airspeed / Turbulent Air Penetration**

Pitch attitude and average %N1 information is provided for use in all phases of flight in the event of unreliable airspeed/Mach indications resulting from blocking or freezing of the pitot system. Loss of radome or turbulent air may also cause unreliable airspeed/Mach indications. The cruise table in this section may also be used for turbulent air penetration.

Pitch attitude is shown in bold type for emphasis since altitude and/or vertical speed indications may also be unreliable.

Max Climb %N1

This table shows Max Climb %N1 for a 280/.78 climb speed schedule, normal engine bleed for packs on or off and anti-ice off. Enter the table with airport pressure altitude and TAT and read %N1. %N1 adjustments are shown for anti-ice operation.

Go-around %N1

To find Max Go-around %N1 based on normal engine bleed for packs on (AUTO) and anti-ice on or off, enter the Go-around %N1 table with airport pressure altitude and reported OAT or TAT and read %N1. For packs OFF or HIGH operation, apply the %N1 adjustment shown below the table.

VREF

This table contains flaps 40, 30 and 15 reference speeds for a given weight.

Advisory Information

Normal Configuration Landing Distance

The normal configuration distance tables are provided as advisory information to help determine the actual landing distance performance of the airplane for different runway surface conditions and brake configurations.

If the surface is affected by water, snow or ice, and the braking action is reported as "good", conditions should not be expected to be as good as on clean, dry runways. The value "good" is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when landing. The performance level used to calculate the "good" data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate "poor" data reflects runways covered with wet ice.

Dry runway landing performance is shown for max manual braking configuration and autobrake settings max, 3, 2, and 1. The autobrake performance may be used to assist in the selection of the most desirable autobrake setting for a given field length. Selection of an autobrake setting results in a constant rate of deceleration. Maximum effort manual braking should achieve shorter landing distance than the max autobrake setting. The reference landing distance is a reference distance from 50 ft above the threshold to stop based on a reference landing weight and normal approach speed for the selected landing flap at sea level, zero wind, zero slope, and two engine detent reverse thrust. Subsequent columns provide adjustments for off-reference landing weight, altitude, wind, slope, temperature, speed, and reverse thrust. Each adjustment is independently added to the reference landing distance.

Flaps 30 and 40 landing distances and adjustments are provided for dry runways as well as runways with good, medium, and poor reported braking actions, which are commonly referred to as slippery runway conditions. All landing distances (reference distances plus adjustments) are 115% of the actual landing distance.

Non-normal Configuration Landing Distance

Advisory information is provided to support non-normal configurations that affect the landing performance of the airplane. Landing distances and adjustments are provided for dry runways and runways with good, medium, and poor reported braking action.

Enter the table with the applicable non-normal configuration and read the normal approach speed. The reference landing distance is a reference distance from 50 ft above the threshold to stop based on a reference landing weight and speed at sea level, zero wind, and zero slope. Subsequent columns provide adjustments for off-reference landing weight, altitude, wind, slope, and speed conditions. Each adjustment is independently added to the reference landing distance. Landing distance includes the effect of reverse thrust.

Recommended Brake Cooling Schedule

Advisory information is provided to assist in avoiding the problems associated with hot brakes. For normal operation, most landings are at weights below the AFM quick turnaround limit weight.

Use of the recommended cooling schedule will help avoid brake overheat and fuse plug problems that could result from repeated landings at short time intervals or a rejected takeoff.

Enter the Recommended Brake Cooling Schedule table with the airplane weight and brakes on speed, adjusted for wind at the appropriate temperature and altitude condition. Instructions for applying wind adjustments are included below the table. Linear interpolation may be used to obtain intermediate values. The resulting number is the reference brake energy per brake in millions of foot-pounds, and represents the amount of energy absorbed by each brake during a rejected takeoff. Notes providing adjustments for wind are included below the table.

To determine the energy per brake absorbed during landing, enter the appropriate Adjusted Brake Energy Per Brake table (No Reverse Thrust or 2 Engine Reverse) with the reference brake energy per brake and the type of braking used during landing (Max Manual, Max Auto, or Autobrake). The resulting number is the adjusted brake energy per brake and represents the energy absorbed in each brake during the landing.

The recommended cooling time is found in the appropriate (steel or carbon brakes) final table by entering with the adjusted brake energy per brake. Times are provided for ground cooling and inflight gear down cooling.

Brake Temperature Monitor System (BTMS) indications are also shown. If brake cooling is determined from the BTMS, use the hottest brake indication 10 to 15 minutes after the airplane has come to a complete stop, or inflight with gear retracted to determine recommended cooling schedule.

Engine Inoperative

Initial Max Continuous %N1

The Initial Max Continuous %N1 setting for use following an engine failure is shown. The table is based on the typical all engine cruise speed of .79M to provide a target %N1 setting at the start of driftdown. Once driftdown is established, the Max Continuous %N1 table should be used to determine %N1 for the given conditions.

Max Continuous %N1

Power setting is based on one engine operating with one A/C pack operating and all anti-ice bleeds off. Enter the table with pressure altitude, TAT, and IAS or Mach to read %N1.

It is desirable to maintain engine thrust level within the limits of the Max Cruise thrust rating. However, where thrust level in excess of Max Cruise rating is required, such as for meeting terrain clearance, ATC altitude assignments, or to attain maximum range capability, it is permissible to use the thrust needed up to the Max Continuous thrust rating. The Max Continuous thrust rating is intended primarily for emergency use at the discretion of the pilot and is the maximum thrust that may be used continuously.

Driftdown Speed/Level Off Altitude

The table shows optimum driftdown speed as a function of cruise weight at start of driftdown. Also shown are the approximate weight and pressure altitude at which the airplane will level off considering 100 ft/min residual rate of climb.

The level off altitude is dependent on air temperature (ISA deviation).

Driftdown/LRC Range Capability

This table shows the range capability from the start of driftdown. Driftdown is continued to level off altitude. As weight decreases due to fuel burn, the airplane is accelerated to Long Range Cruise speed. Cruise is continued at level off altitude and Long Range Cruise speed.

To determine fuel required, enter the Ground to Air Miles Conversion table with the desired ground distance and adjust for anticipated winds to obtain air distance to destination. Then enter the Driftdown/Cruise Fuel and Time table with air distance and weight at start of driftdown to determine fuel and time required. If altitudes other than the level off altitude is used, fuel and time required may be obtained by using the Engine Inoperative Long Range Cruise Enroute Fuel and Time table.

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Long Range Cruise Altitude Capability

The table shows the maximum altitude that can be maintained at a given weight and air temperature (ISA deviation), based on Long Range Cruise speed, Max Continuous thrust, and 100 ft/min residual rate of climb.

Long Range Cruise Control

The table provides target %N1, engine inoperative Long Range Cruise Mach number, IAS and fuel flow for the airplane weight and pressure altitude. The fuel flow values in this table reflect single engine fuel burn.

APU Operation During Flight

For APU operation during flight, increase fuel flow according to the following table. These increments include the APU fuel flow and the effect of increased drag from the APU door.

PRESSURE ALTITUDE (1000 FT)	APU FUEL FLOW (KG/HR)
39	45
35	45
31	50
25	60
20	65
15	75
10	85
5	95

Long Range Cruise Diversion Fuel and Time

Tables are provided for crews to determine the fuel and time required to proceed to an alternate airfield with one engine inoperative. The data is based on single engine Long Range Cruise speed and .78/280/250 descent. Enter with Air Distance as determined from the Ground to Air Miles Conversion table and read Fuel and Time required at the cruise pressure altitude. Adjust the fuel obtained for deviation from the reference weight at checkpoint as required by entering the off reference fuel adjustments table with the fuel required for the reference weight and the actual weight at checkpoint. Read fuel required and time for the actual weight.

Holding

Target %N1, indicated airspeed and fuel flow per engine information is tabulated for holding with flaps up based on the FMC optimum holding speed schedule. This is the higher of the maximum endurance speed and the maneuvering speed. Small variations in airspeed will not appreciably affect the overall endurance time. Enter the table with weight and pressure altitude to read %N1, IAS and fuel flow per engine.

Gear Down Landing Rate of Climb Available

Rate of climb data is provided as guidance information in the event an engine inoperative landing (manual or autoland) is planned. The tables show gear down rate of climb available for Flaps 15 and Flaps 30. Enter the table with TAT and pressure altitude to read rate of climb available. Apply adjustments shown to correct for weight.

Gear Down

This section contains performance for airplane operation with the landing gear extended. The data is based on engine bleeds for normal air conditioning.

Note: The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS may generate inappropriate enroute speed schedules, display non-conservative predictions of fuel burn, estimated time of arrival (ETA), maximum altitude, and compute overly shallow descent path. An accurate estimated time of arrival (ETA) is available if current speed or Mach is entered into the VNAV cruise page.

Tables for gear down performance in this section are identical in format and used in the same manner as tables for the gear up configuration previously described.

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Maneuvers

Introduction

Chapter MAN

Section 05

General

Non-Normal Maneuvers and Flight Patterns are included for training and review purposes.

Non-Normal Maneuvers

Flight crews are expected to do non-normal maneuvers from memory.

Flight Patterns

Flight patterns show procedures for some all-engine and engine-inoperative situations.

Flight patterns do not include all procedural items but show required/recommended:

- configuration changes
- thrust changes
- Mode Control Panel (MCP) changes
- pitch mode and roll mode changes
- checklist calls.



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**Maneuvers****Chapter MAN****Non-Normal Maneuvers****Section 1****Approach to Stall or Stall Recovery**

Do all recoveries from approach to stall as if an actual stall has occurred. Immediately do the following at the first indication of stall (buffet or stick shaker).

Note: Do not use flight director commands during the recovery.

Pilot Flying	Pilot Monitoring
- Initiate the recovery: - Hold the control column firmly. - Disengage autopilot and autothrottle. - Smoothly apply nose down elevator to reduce the angle of attack until buffet or stick shaker stops. Nose down stabilizer trim may be needed.*	- Monitor altitude and airspeed. - Verify all needed actions have been done and call out any omissions. - Call out any trend toward terrain contact.
- Continue the recovery: - Roll in the shortest direction to wings level if needed.** - Advance thrust levers as needed. - Retract the speedbrakes. - Do not change gear or flap configuration, except - During liftoff, if flaps are up, call for flaps 1.	- Monitor altitude and airspeed. - Verify all needed actions have been done and call out any omissions. - Call out any trend toward terrain contact. - Set the FLAP lever as directed.
- Complete the recovery: - Check airspeed and adjust thrust as needed. - Establish pitch attitude. - Return to the desired flight path. - Re-engage the autopilot and autothrottle if desired.	- Monitor altitude and airspeed. - Verify all needed actions have been done and call out any omissions. - Call out any trend toward terrain contact.

WARNING: *If the control column does not provide the needed response, stabilizer trim may be needed. Excessive use of pitch trim can aggravate the condition, or can result in loss of control or in high structural loads.

WARNING: ** Excessive use of pitch trim or rudder can aggravate the condition, or can result in loss of control or in high structural loads.

Rejected Takeoff

The captain has the sole responsibility for the decision to reject the takeoff. Make the decision in time to start the rejected takeoff maneuver by V1. If the decision is to reject the takeoff, the captain clearly announces “REJECT,” immediately starts the rejected takeoff maneuver and assume control of the airplane. If the first officer is making the takeoff, the first officer maintains control of the airplane until the captain makes a positive input to the controls.

Before 80 knots, reject the takeoff for any of the following:

- activation of the master caution system
- system failure(s)
- unusual noise or vibration
- tire failure
- abnormally slow acceleration
- takeoff configuration warning
- fire or fire warning
- engine failure
- predictive windshear warning
- if a side window opens
- if the airplane is unsafe or unable to fly.

Above 80 knots and before V1, reject the takeoff for any of the following:

- fire or fire warning
- engine failure
- predictive windshear warning
- if the airplane is unsafe or unable to fly.

During the takeoff, the crewmember observing the non-normal situation immediately calls it out as clearly as possible.



737 Flight Crew Operations Manual

Captain	First Officer
Without delay: Simultaneously close the thrust levers, disengage the autothrottles and apply maximum manual wheel brakes or verify operation of RTO autobrake. If RTO autobrake is selected, monitor system performance and apply manual wheel brakes if the AUTO BRAKE DISARM light illuminates or deceleration is not adequate. Raise SPEED BRAKE lever. Apply reverse thrust up to the maximum amount consistent with conditions. Continue maximum braking until certain the airplane can stop on the runway.	Verify actions as follows: Thrust levers closed. Autothrottles disengaged. Maximum brakes applied. Verify SPEED BRAKE lever UP and call "SPEEDBRAKES UP." If SPEED BRAKE lever is not UP, call "SPEEDBRAKES NOT UP." Reverse thrust applied. When both REV indications are green, call "REVERSERS NORMAL." If there is no REV indication(s) or the indication(s) stays amber, call "NO REVERSER ENGINE NUMBER 1", or "NO REVERSER ENGINE NUMBER 2", or "NO REVERSERS". Call out omitted action items.

Captain	First Officer
When stopping is assured: Start movement of the reverse thrust levers to reach the reverse idle detent before taxi speed. After the engines are at reverse idle, move the reverse thrust levers to full down.	Call out 60 knots. Communicate the reject decision to the control tower and cabin as soon as practical.
When the airplane is stopped, perform procedures as needed. Review Brake Cooling Schedule for brake cooling time and precautions (refer to Performance Inflight Chapter.) Consider the following: • The possibility of wheel fuse plugs melting • The need to clear the runway • The requirement for remote parking • Wind direction in case of fire • Alerting fire equipment • Not setting the parking brake unless passenger evacuation is needed • Advising the ground crew of the hot brake hazard • Advising passengers of the need to remain seated or evacuate • Completion of Non-Normal checklist (if appropriate) for conditions which caused the RTO.	

Runway Awareness and Advisory System (RAAS)

(VP-BDU) Do not use

Accomplish the following if a RAAS callout or alert differs from the flight crew's expectation:

Pilot Flying	Pilot Monitoring
Verify position. Contact ATC for assistance, if needed.	

Ground Proximity Warning System (GPWS) Response

GPWS Caution

Accomplish the following maneuver for any of these aural alerts:

- SINK RATE
- TERRAIN
- DON'T SINK
- TOO LOW FLAPS
- TOO LOW GEAR
- TOO LOW TERRAIN
- GLIDESLOPE
- BANK ANGLE
- AIRSPEED LOW (airplanes with AIRSPEED LOW aural)
- CAUTION TERRAIN

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- CAUTION OBSTACLE

Pilot Flying	Pilot Monitoring
Correct the flight path, airplane configuration, or airspeed.	

The below glideslope deviation alert can be cancelled or inhibited for:

- localizer or backcourse approach
- circling approach from an ILS
- when conditions require a deliberate approach below glideslope
- unreliable glideslope signal.

Note: If a terrain caution occurs when flying under daylight VMC, and positive visual verification is made that no obstacle or terrain hazard exists, the alert may be regarded as cautionary and the approach may be continued.

Note: Some aural alerts repeat.

GPWS Warning

Accomplish the following maneuver for any of these conditions:

- Activation of “PULL UP” or “TERRAIN TERRAIN PULL UP” warning.

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- Activation of the “PULL UP” or “OBSTACLE OBSTACLE PULL UP” warning.
- Other situations resulting in unacceptable flight toward terrain.

Pilot Flying	Pilot Monitoring
• Disengage autopilot. • Disengage autothrottle. • Aggressively apply maximum* thrust. • Simultaneously roll wings level and rotate to an initial pitch attitude of 20°. • Retract speedbrakes. • If terrain remains a threat, continue rotation up to the pitch limit indicator (if available) or stick shaker or initial buffet.	• Assure maximum* thrust. • Verify all needed actions have been completed and call out any omissions.
• Do not change gear or flap configuration until terrain separation is assured. • Monitor radio altimeter for sustained or increasing terrain separation. • When clear of terrain, slowly decrease pitch attitude and accelerate.	• Monitor vertical speed and altitude (radio altitude for terrain clearance and barometric altitude for a minimum safe altitude.) • Call out any trend toward terrain contact.

Note: Aft control column force increases as the airspeed decreases. In all cases, the pitch attitude that results in intermittent stick shaker or initial buffet is the upper pitch attitude limit. Flight at intermittent stick shaker may be needed to obtain a positive terrain separation. Use smooth, steady control to avoid a pitch attitude overshoot and stall.

Note: Do not use flight director commands.

Note: *Maximum thrust can be obtained by advancing the thrust levers full forward if the EECs are in the normal mode. If terrain contact is imminent, advance thrust levers full forward.

Note: If positive visual verification is made that no obstacle or terrain hazard exists when flying under daylight VMC conditions before a terrain or obstacle warning, the alert may be regarded as cautionary and the approach may be continued.

Traffic Avoidance

Immediately accomplish the following by recall whenever a TCAS traffic advisory (TA) or resolution advisory (RA) occurs.

WARNING: Comply with the RA if there is a conflict between the RA and air traffic control.

WARNING: Once an RA has been issued, safe separation could be compromised if current vertical speed is changed, except as needed to comply with the RA. This is because TCAS II-to-TCAS II coordination can be in progress with the intruder aircraft, and any change in vertical speed that does not comply with the RA can negate the effectiveness of the other aircraft's compliance with the RA.

Note: If stick shaker or initial buffet occurs during the maneuver, immediately accomplish the APPROACH TO STALL RECOVERY procedure.

Note: If high speed buffet occurs during the maneuver, relax pitch force as needed to reduce buffet, but continue the maneuver.

(VQ-BUV) Do not use

Note: Do not use flight director commands until clear of conflict.

Note: Do not use flight director pitch commands until clear of conflict.

For TA:

Pilot Flying	Pilot Monitoring
Look for traffic using traffic display as a guide. Call out any conflicting traffic.	
If traffic is sighted, maneuver if needed.	

Note: Maneuvers based solely on a TA can result in reduced separation and are not recommended.

For RA, except a climb in landing configuration:

WARNING: Do not follow a DESCEND (fly down) RA issued below 1000 feet AGL.

Pilot Flying	Pilot Monitoring
If maneuvering is needed, disengage the autopilot and disengage the autothrottle. Smoothly adjust pitch and thrust to satisfy the RA command. Follow the planned lateral flight path unless visual contact with the conflicting traffic requires other action.	
Attempt to establish visual contact. Call out any conflicting traffic.	

For a climb RA in landing configuration:

Pilot Flying	Pilot Monitoring
Disengage the autopilot and disengage the autothrottle. Advance thrust levers forward to ensure maximum thrust is attained and call for FLAPS 15. Smoothly adjust pitch to satisfy the RA command. Follow the planned lateral flight path unless visual contact with the conflicting traffic requires other action.	Verify maximum thrust set. Position flap lever to 15 detent.
Verify a positive rate of climb on the altimeter and call “GEAR UP.”	Verify a positive rate of climb on the altimeter and call “POSITIVE RATE.” Set the landing gear lever to UP.
Attempt to establish visual contact. Call out any conflicting traffic.	

Upset Recovery

Historically, an upset was defined as unintentionally exceeding one or more of the following conditions:

- Pitch attitude greater than 25 degrees nose up
- Pitch attitude greater than 10 degrees nose down
- Bank angle greater than 45 degrees
- Less than above parameters but flying at an airspeed inappropriate for the conditions.

An upset condition is now considered any time an airplane is diverting from the intended airplane state. An airplane upset can involve pitch or roll angle deviations as well as inappropriate airspeeds for the conditions.

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The following actions represent a logical progression for recovering the airplane. The sequence of actions is for guidance only and represents a series of options to be considered and used dependent on the situation. Not all actions can be needed once recovery is under way. If needed, use minimal pitch trim during initial recovery. Careful use of rudder to aid roll control should be considered only if roll control is ineffective and the airplane is not stalled.

These actions assume that the airplane is not stalled. A stalled condition can exist at any attitude and can be recognized by one or more of the following:

- Stick shaker
- Buffet that can be heavy at times
- Lack of pitch authority
- Lack of roll control
- Inability to stop a descent.

If the airplane is stalled, first recover from the stall by applying and maintaining nose down elevator until stall recovery is complete and stick shaker stops.

Nose High Recovery

Pilot Flying	Pilot Monitoring
Recognize and confirm the developing situation	
Disengage autopilot Disengage autothrottle Recover: • Apply nose-down elevator. Apply as much elevator as needed to obtain a nose down pitch rate • Apply appropriate nose down stabilizer trim* • Reduce thrust • Roll (adjust bank angle) to obtain a nose down pitch rate* Complete the recovery: • When approaching the horizon, roll to wings level • Check airspeed and adjust thrust • Establish pitch attitude.	Call out attitude, airspeed and altitude throughout the recovery. Verify all needed actions have been done and call out any continued deviation.

WARNING: * Excessive use of pitch trim or rudder can aggravate an upset, result in loss of control, or result in high structural loads.

Nose Low Recovery

Pilot Flying	Pilot Monitoring
Recognize and confirm the developing situation	
Disengage autopilot Disengage autothrottle Recover: • Recover from stall, if needed • Roll in the shortest direction to wings level. If bank angle is more than 90 degrees, unload and roll.* Complete the recovery: • Apply nose up elevator • Apply nose up trim, if needed* • Adjust thrust and drag, if needed.	Call out attitude, airspeed and altitude throughout the recovery. Verify all needed actions have been done and call out any continued deviation.

WARNING: * Excessive use of pitch trim or rudder can aggravate an upset, result in loss of control or result in high structural loads.

Windshear

Windshear Caution

For predictive windshear caution alert: (“MONITOR RADAR DISPLAY” aural).

Pilot Flying	Pilot Monitoring
Maneuver as needed to avoid the windshear.	

Windshear Warning

Predictive windshear warning during takeoff roll: (“WINDSHEAR AHEAD, WINDSHEAR AHEAD” aural)

- before V₁, reject takeoff
- after V₁, perform the Windshear Escape Maneuver.

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Windshear encountered during takeoff roll:

- If windshear is encountered before V₁, there may not be sufficient runway remaining to stop if an RTO is initiated at V₁. At VR, rotate at a normal rate toward a 15 degree pitch attitude. Once airborne, perform the Windshear Escape Maneuver.
- If windshear is encountered near the normal rotation speed and airspeed suddenly decreases, there may not be sufficient runway left to accelerate back to normal takeoff speed. If there is insufficient runway left to stop, initiate a normal rotation at least 2,000 feet before the end of the runway, even if airspeed is low. Higher than normal attitudes may be needed to lift off in the remaining runway. Ensure maximum thrust is set.

Predictive windshear warning during approach: (“GO-AROUND, WINDSHEAR AHEAD” aural)

- perform the Windshear Escape Maneuver, or, at pilot’s discretion, perform a normal go-around.

Windshear encountered in flight:

- perform the Windshear Escape Maneuver.

Note: The following are indications the airplane is in windshear:

- windshear warning (two-tone siren followed by “WINDSHEAR, WINDSHEAR, WINDSHEAR”) or
- unacceptable flight path deviations.

Note: Unacceptable flight path deviations are recognized as uncontrolled changes from normal steady state flight conditions below 1000 feet AGL, in excess of any of the following:

- 15 knots indicated airspeed
- 500 fpm vertical speed
- 5° pitch attitude
- 1 dot displacement from the glideslope
- unusual thrust lever position for a significant period of time.

Windshear Escape Maneuver

Pilot Flying	Pilot Monitoring
MANUAL FLIGHT - Disengage autopilot. - Push either TO/GA switch. - Aggressively apply maximum thrust* - Disengage autothrottle. - Simultaneously roll wings level and rotate toward an initial pitch attitude of 15°. - Retract speedbrakes. - Follow flight director TO/GA guidance (if available) **	- Verify maximum* thrust. - Verify all needed actions have been completed and call out any omissions.
AUTOMATIC FLIGHT - Push either TO/GA switch*** - Verify TO/GA mode annunciation. - Verify GA thrust. - Retract speedbrakes. - Monitor system performance****	- Verify GA* thrust. - Verify all needed actions have been completed and call out any omissions.
MANUAL OR AUTOMATIC FLIGHT - Do not change flap or gear configuration until windshear is no longer a factor. - Monitor vertical speed and altitude. - Do not attempt to regain lost airspeed until windshear is no longer a factor.	- Monitor vertical speed and altitude. - Call out any trend toward terrain contact, descending flight path, or significant airspeed changes.

Note: Aft control column force increases as the airspeed decreases. In all cases, the pitch attitude that results in intermittent stick shaker or initial buffet is the upper pitch attitude limit. Flight at intermittent stick shaker may be needed to obtain a positive terrain separation. Use smooth, steady controls to avoid a pitch attitude overshoot and stall.

Note: *Maximum thrust can be obtained by advancing the thrust levers full forward if the EECs are in the normal mode. If terrain contact is imminent, advance thrust levers full forward.

Note: **Do not exceed the Pitch Limit Indication.

Note: *** If TO/GA is not available, disengage autopilot and autothrottle and fly manually.



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WARNING: ** Severe windshear can exceed the performance of the AFDS. Be prepared to disengage the autopilot and autothrottle and fly manually.**

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Maneuvers

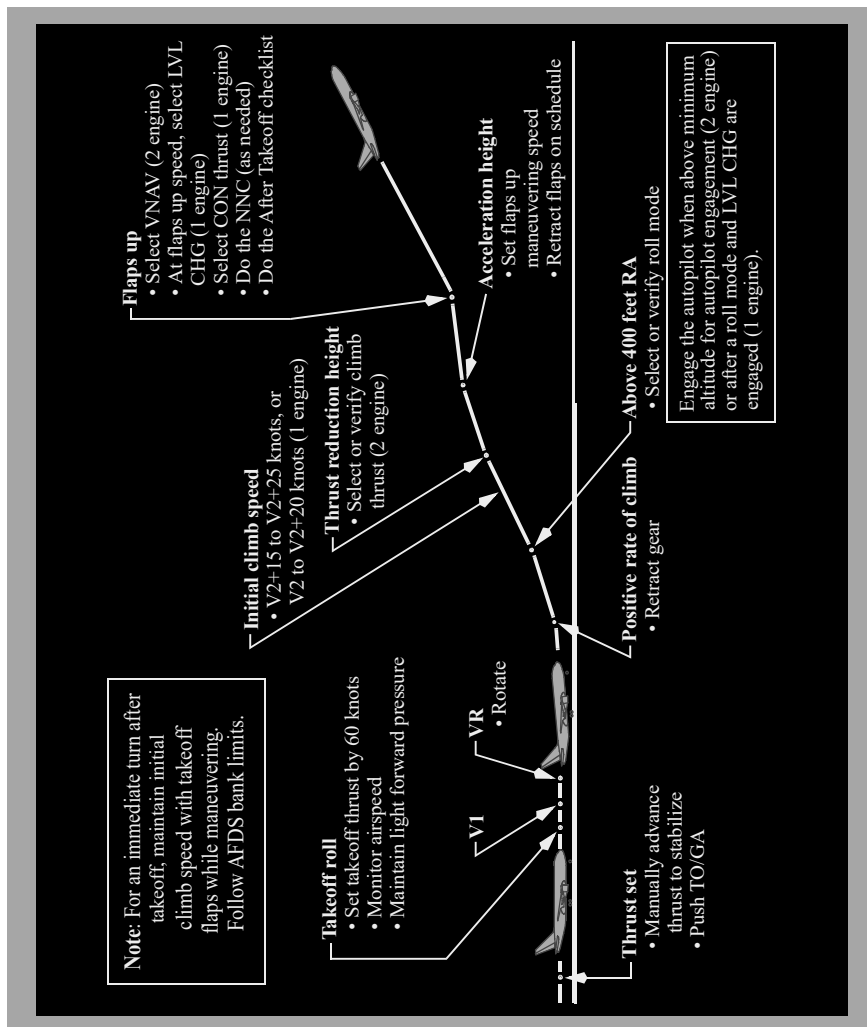
Flight Patterns

Chapter MAN

Section 2

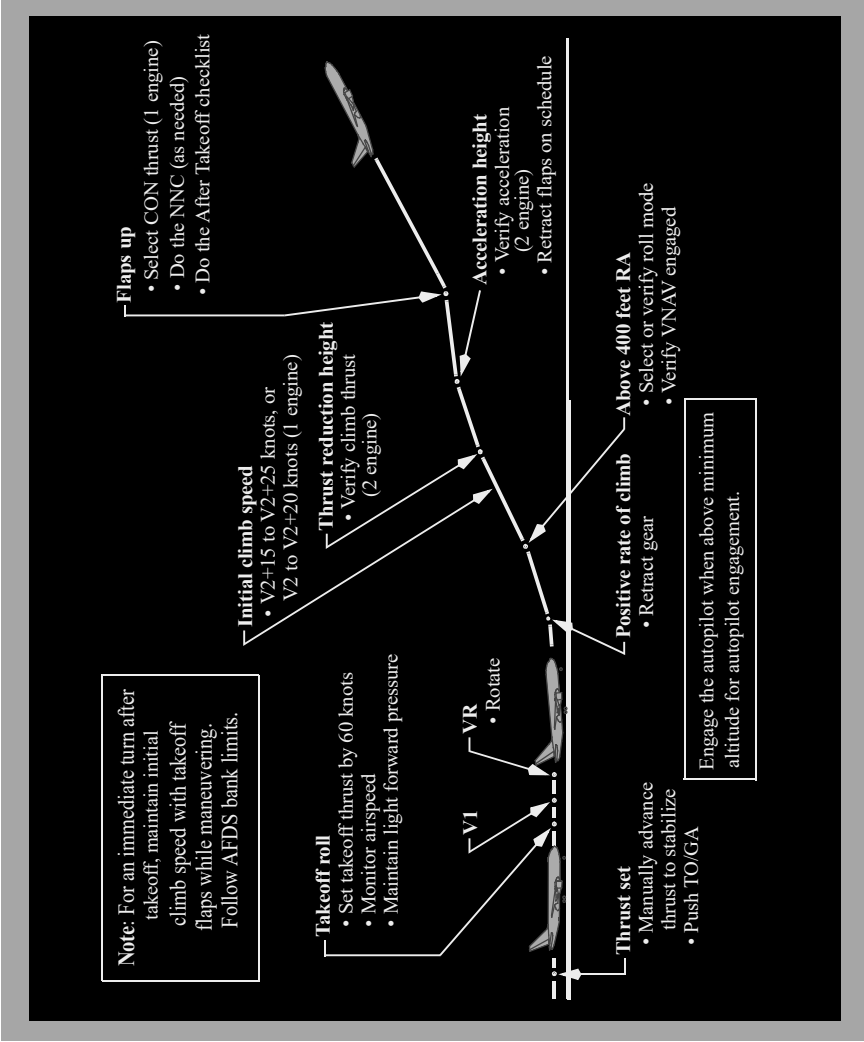
Takeoff

RA-73269 - RA-73321



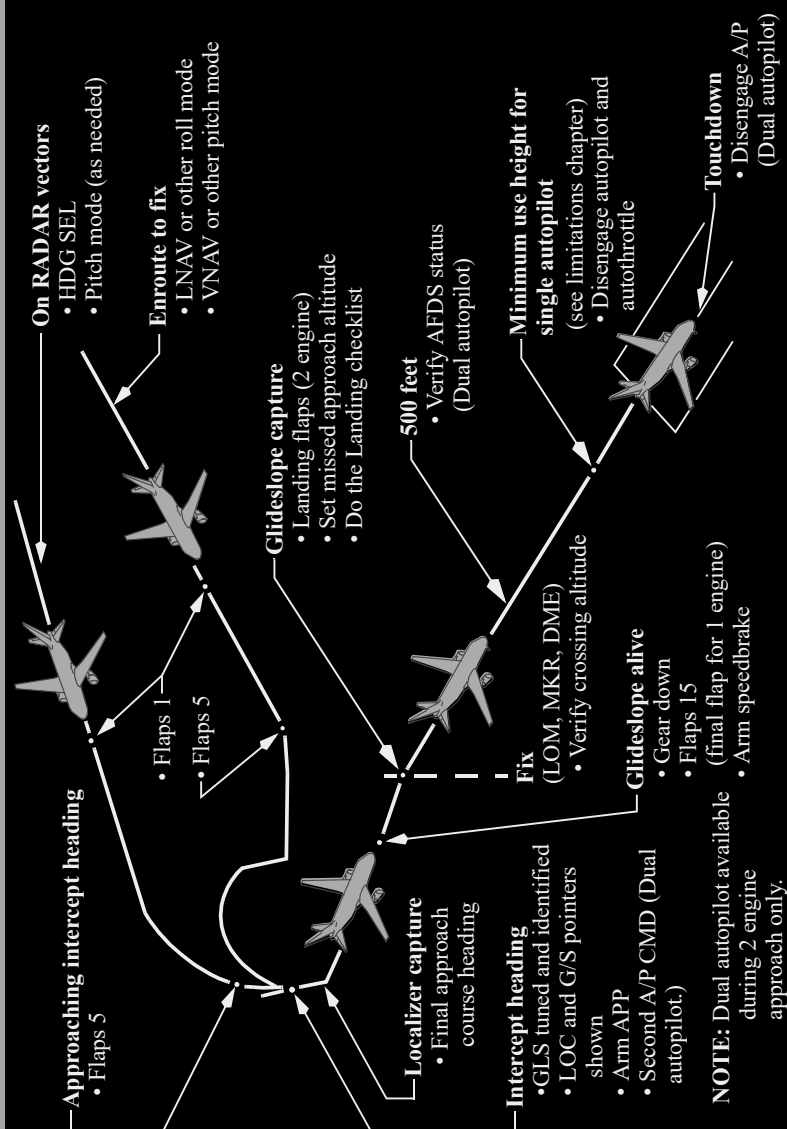
Takeoff

RA-73311 - RA-73319

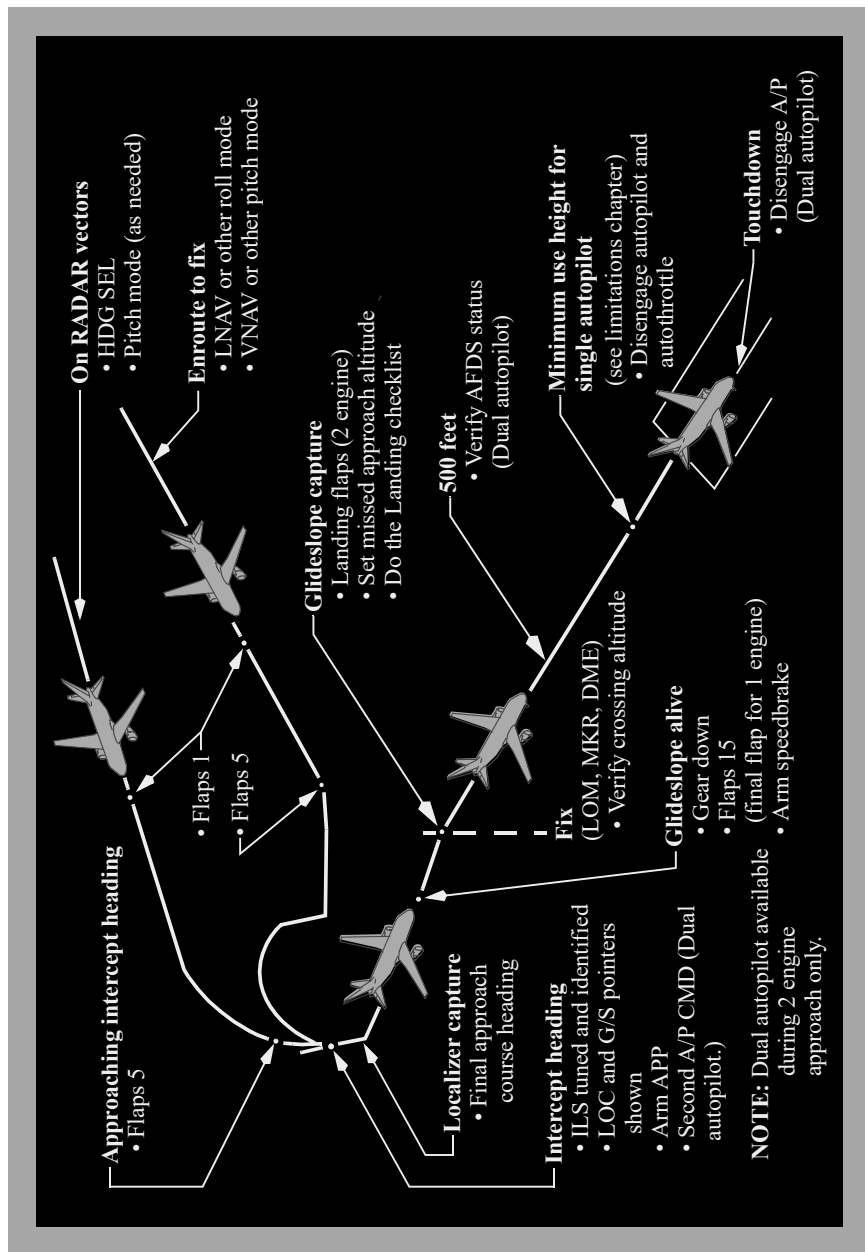


GLS Approach - Fail Passive

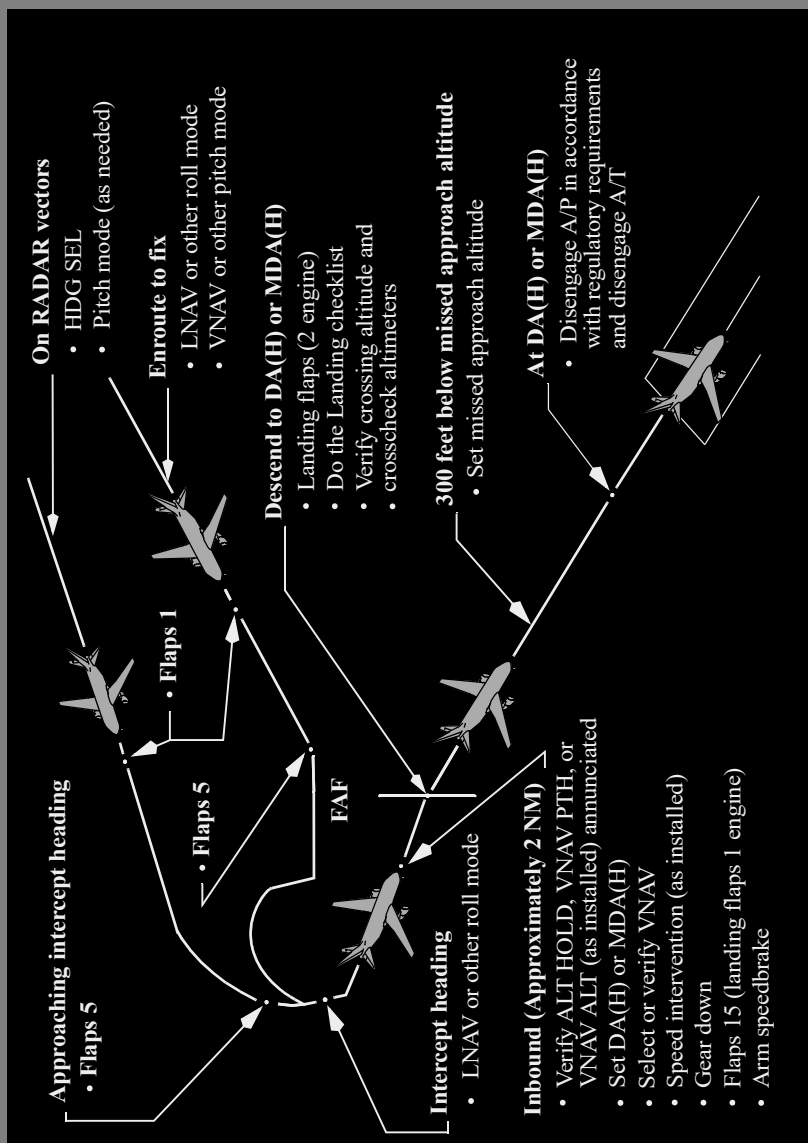
VQ-BDC



ILS Approach - Fail Passive

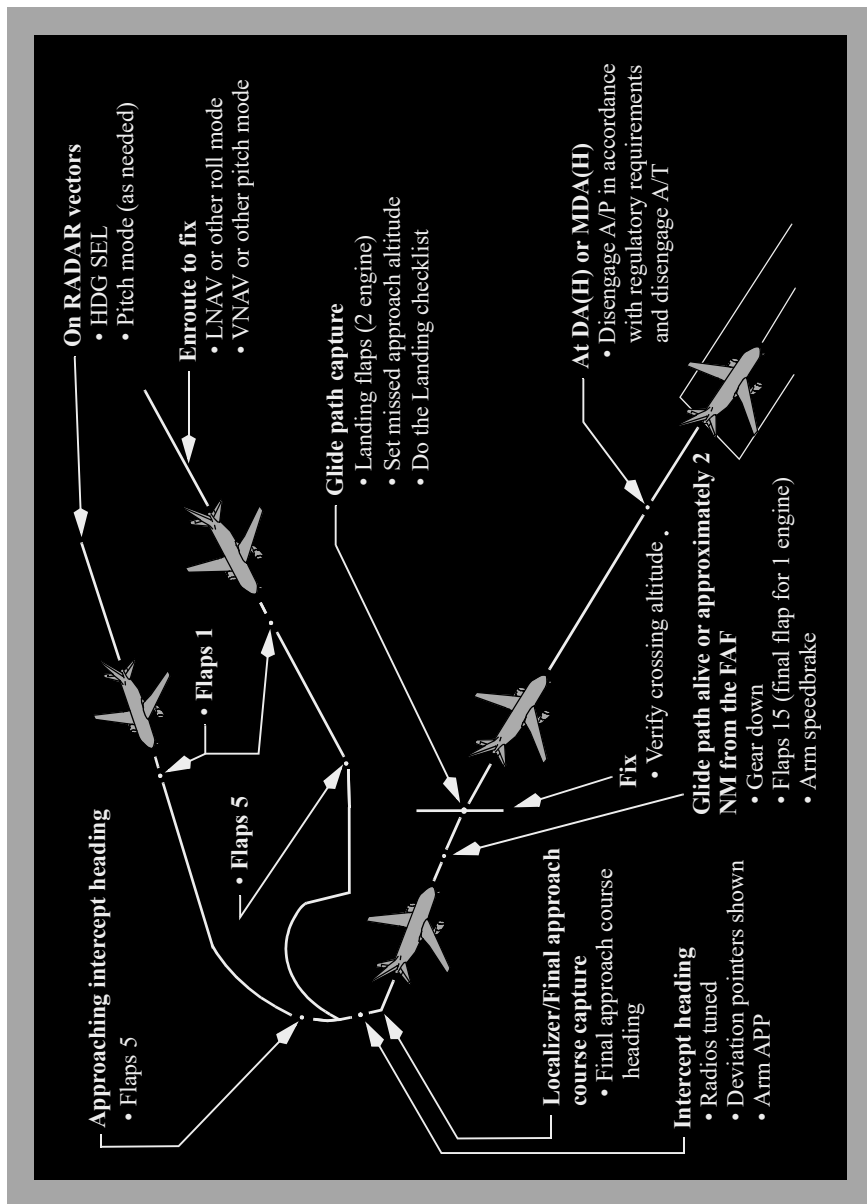


Instrument Approach Using VNAV

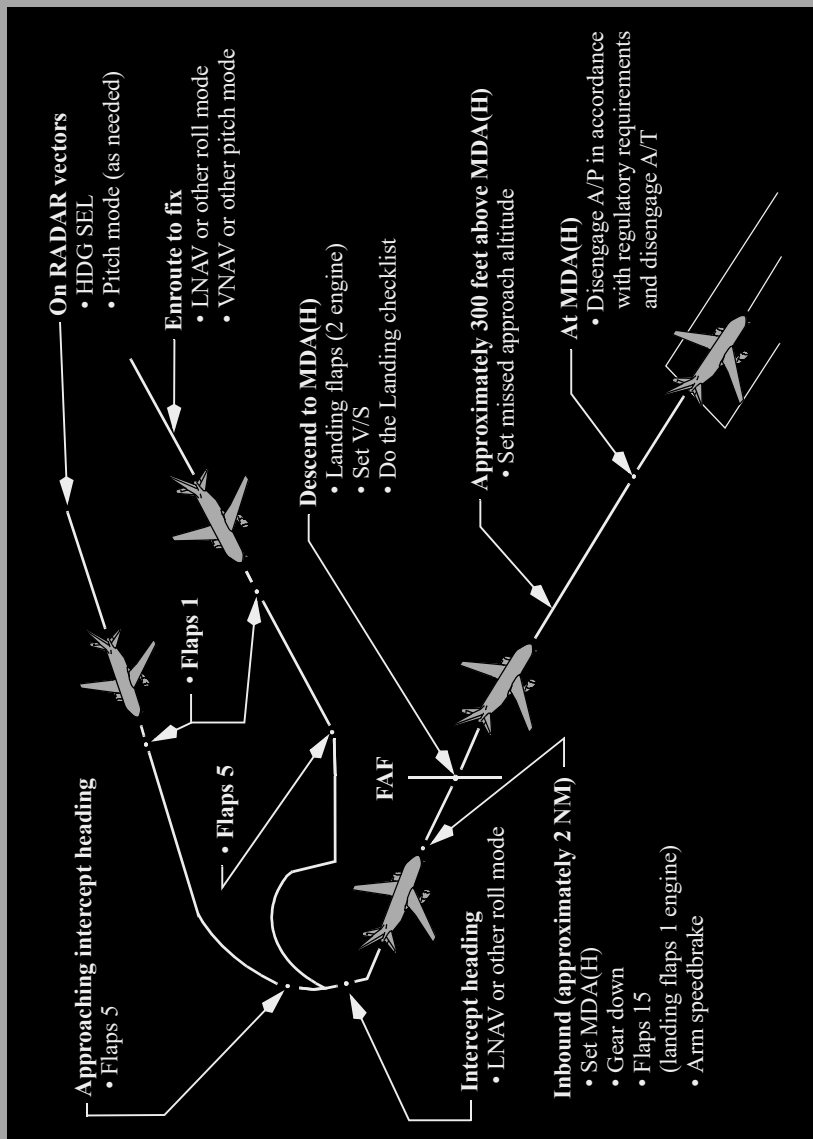


Instrument Approach Using IAN

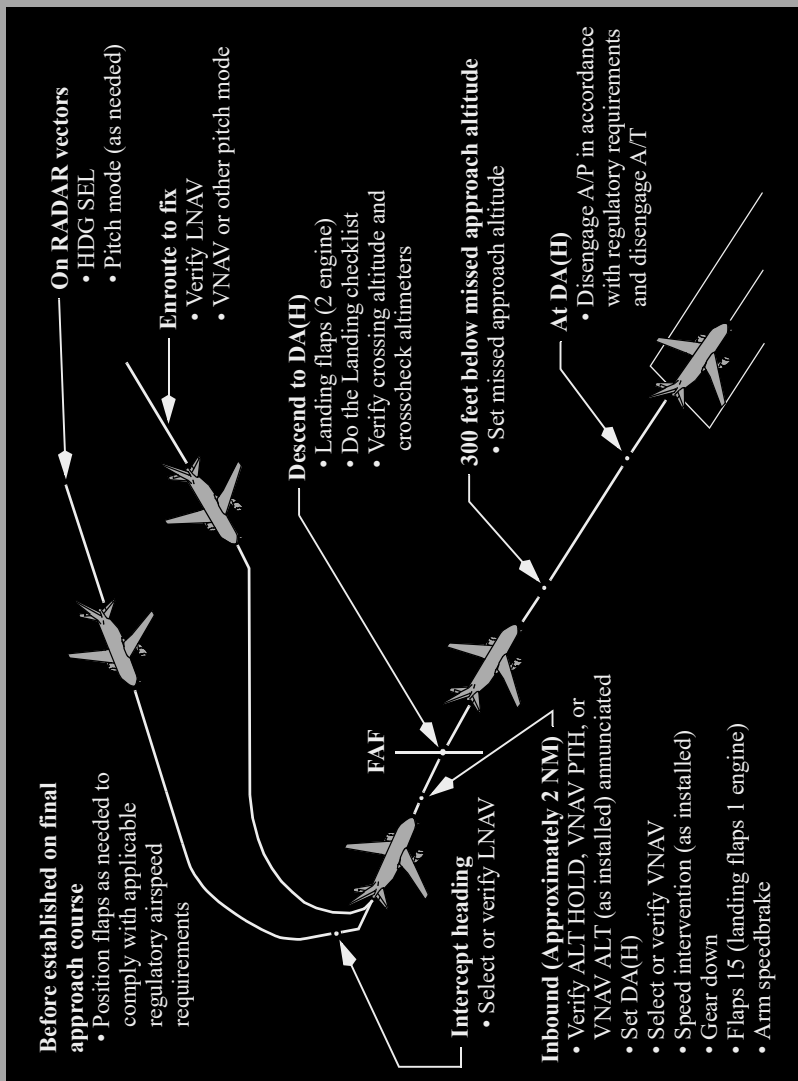
RA-73269, RA-73314, RA-73315, RA-73319, RA-73321



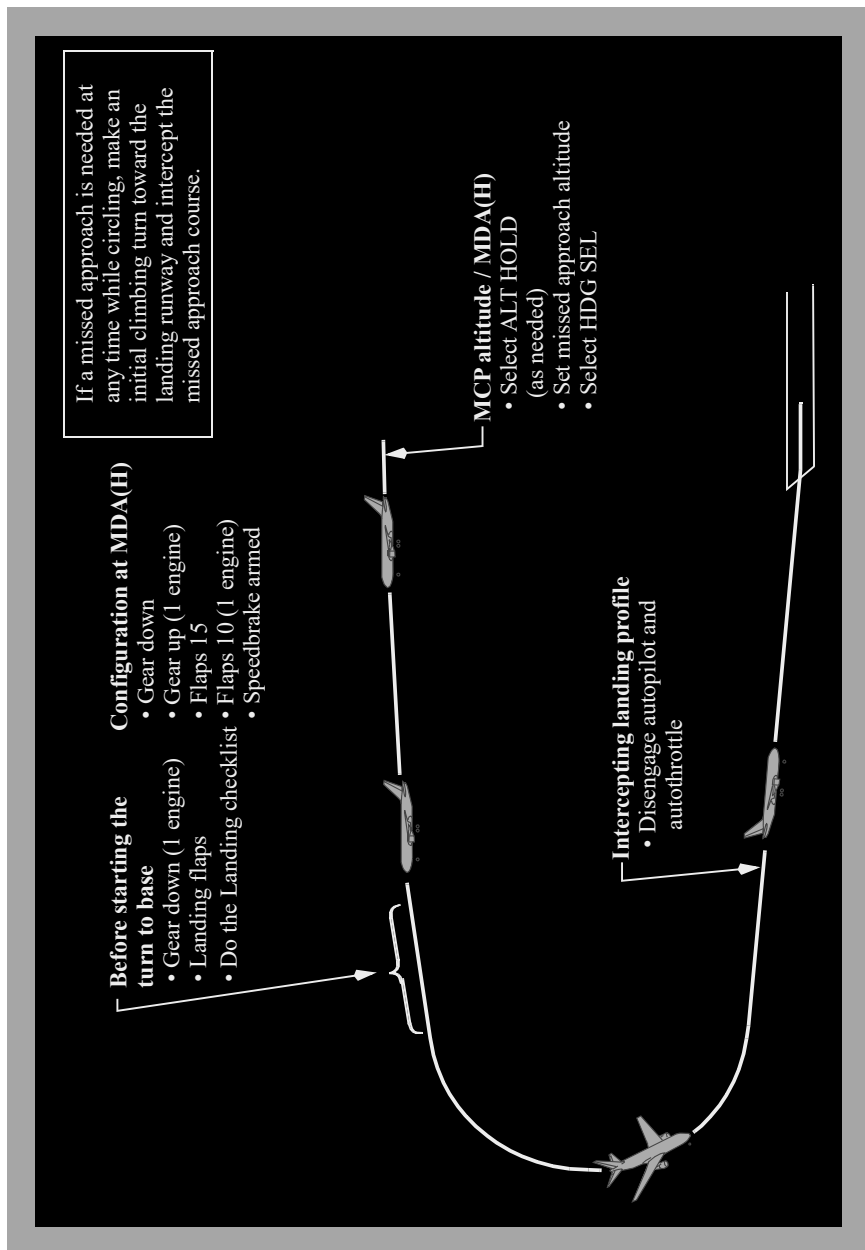
Instrument Approach Using V/S



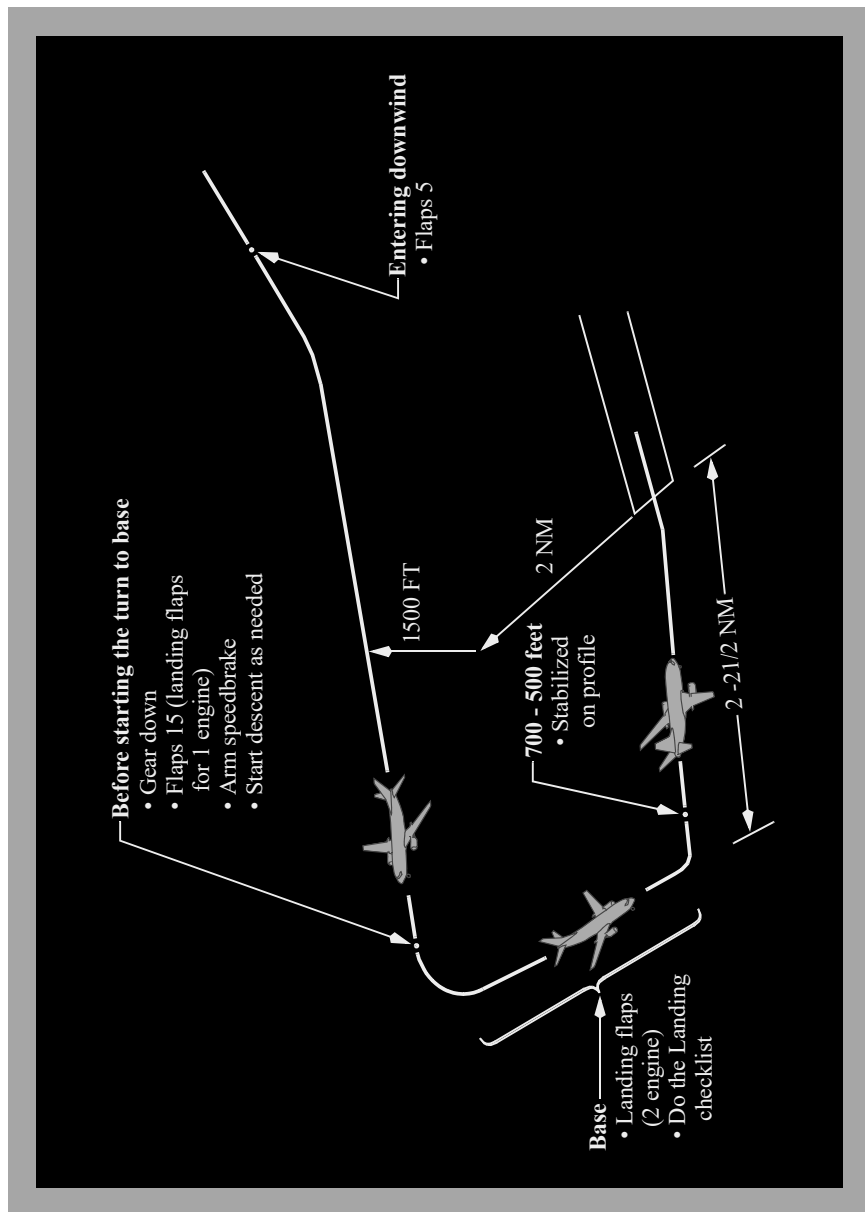
Instrument Approach - RNAV (RNP) AR



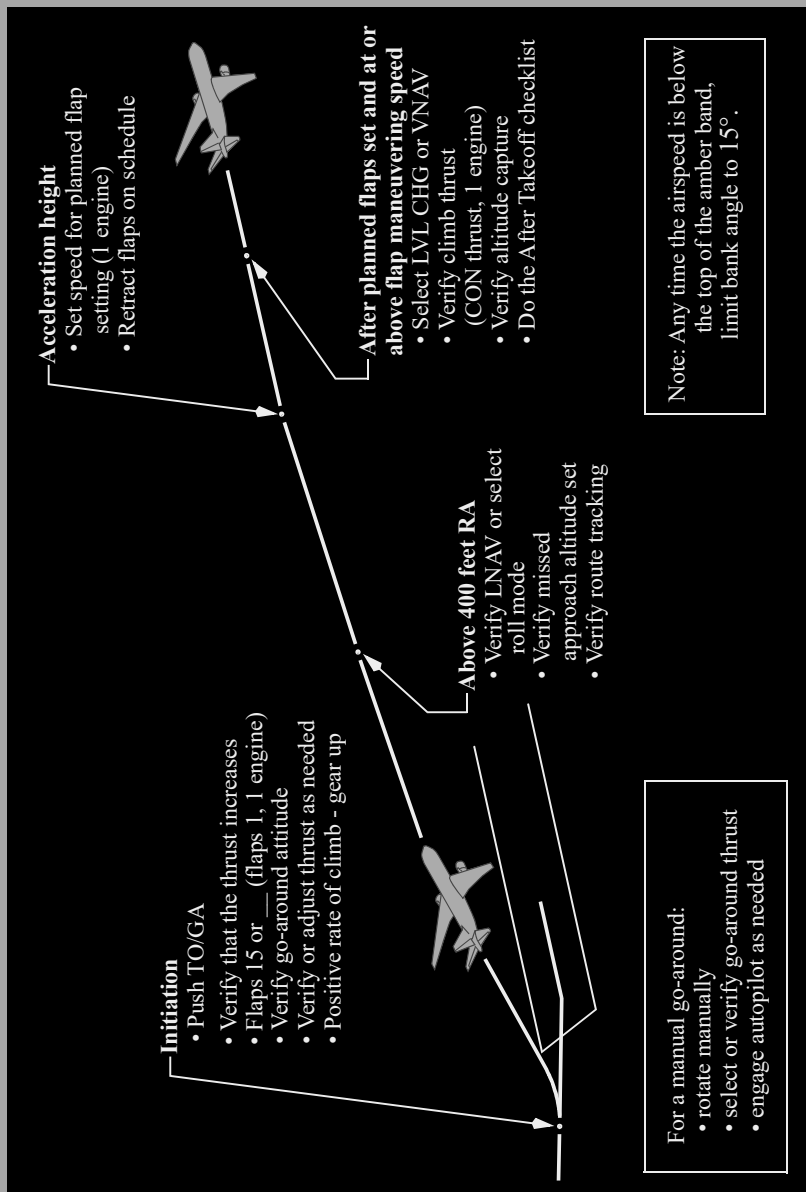
Circling Approach



Visual Traffic Pattern

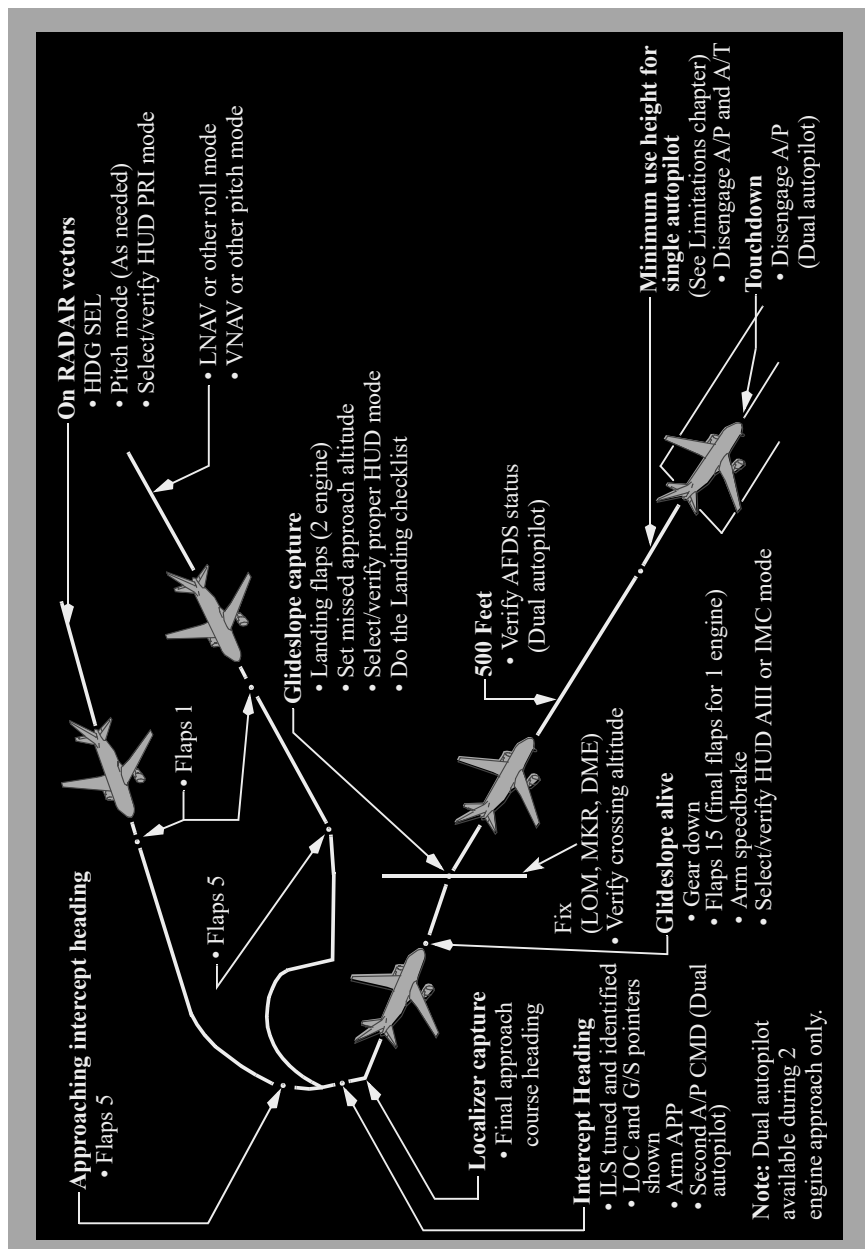


Go-Around and Missed Approach



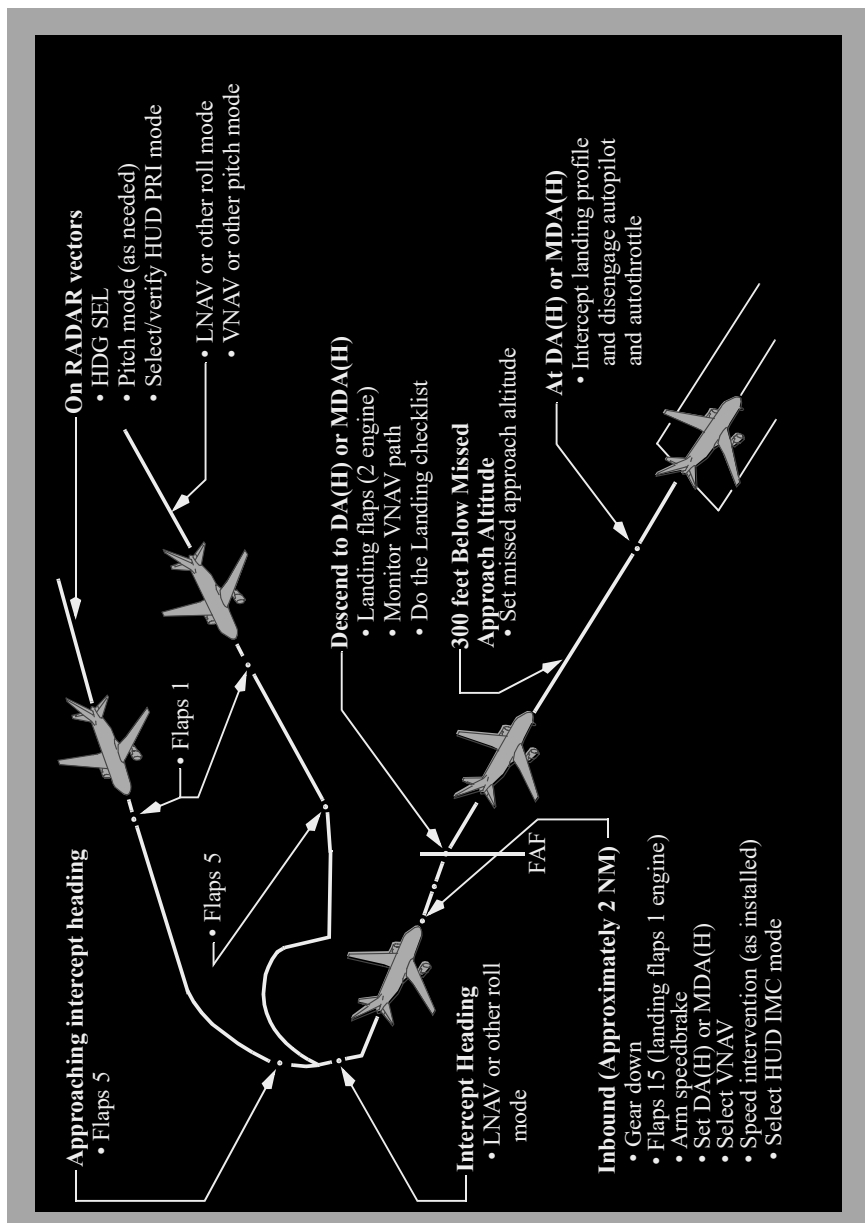
ILS Approach Using HUD

RA-73314, RA-73315



Instrument Approach Using VNAV and HUD

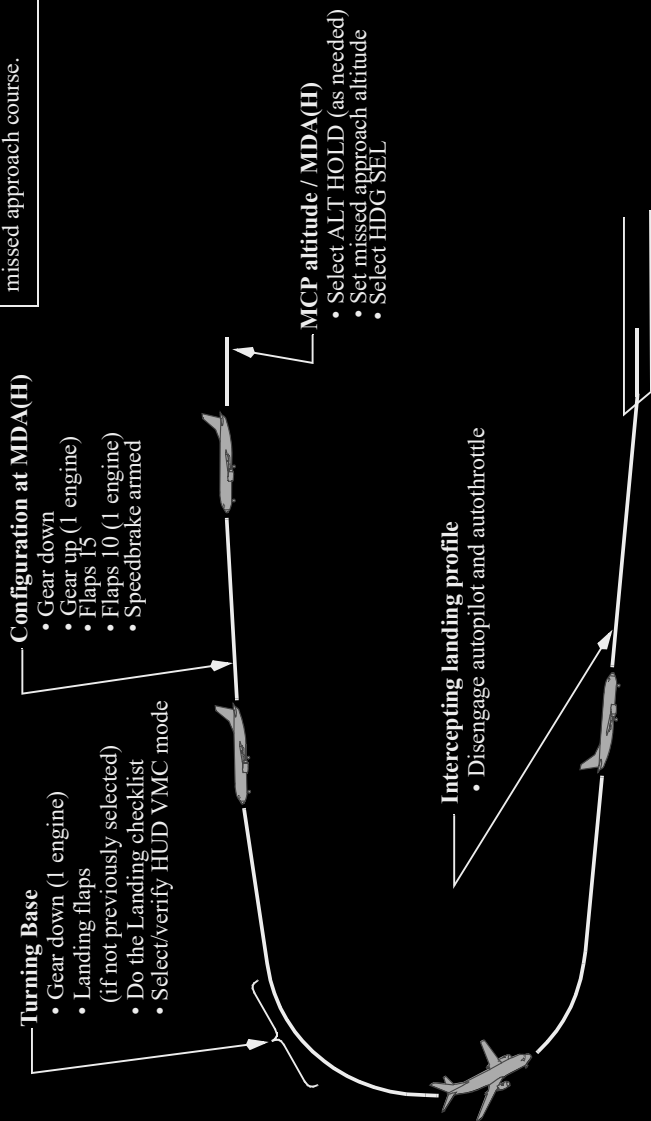
RA-73314, RA-73315



Circling Approach Using HUD

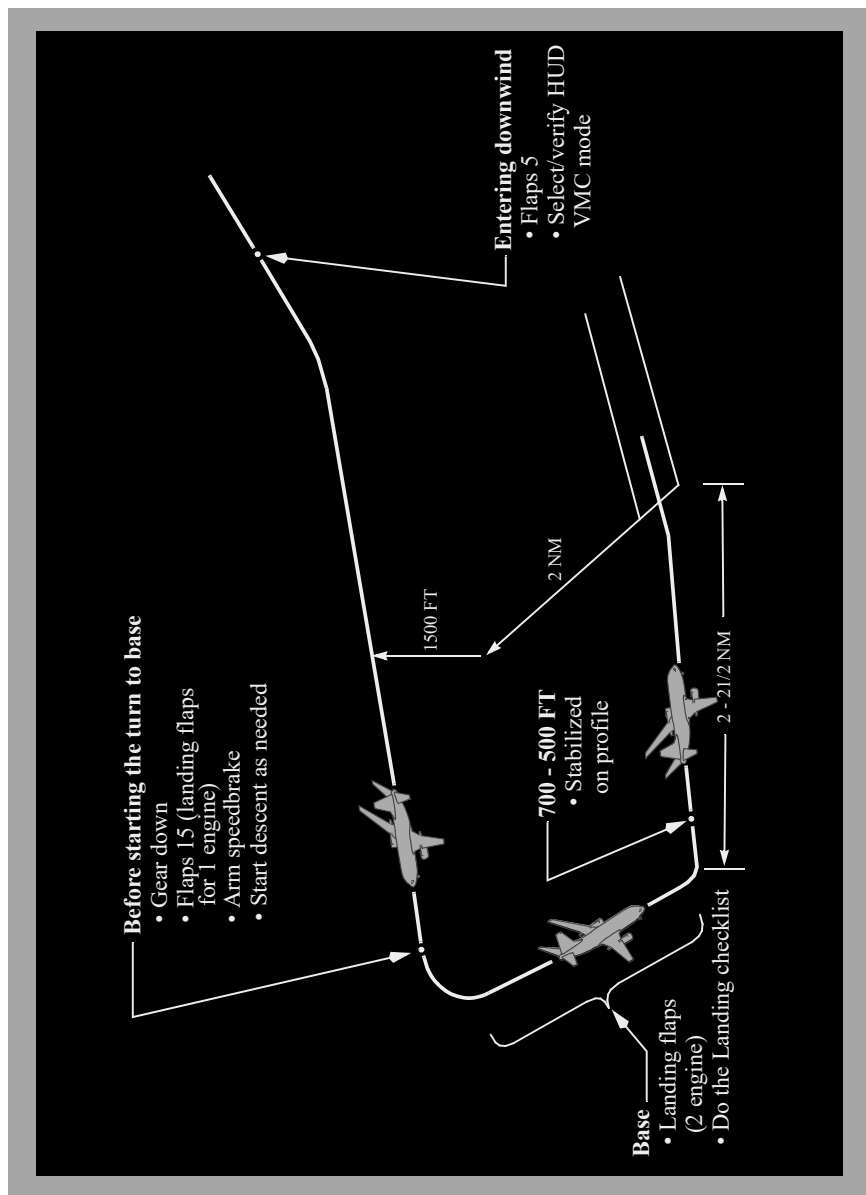
RA-73314, RA-73315

If a missed approach is needed at any time while circling, make an initial climbing turn toward the landing runway and intercept the missed approach course.



Visual Traffic Pattern Using HUD

RA-73314, RA-73315



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Checklist Instructions

Chapter CI

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**Checklist Instructions**
Model Identification**Chapter CI**
Section ModID**General**

The airplanes listed in the table below are covered in this Quick Reference Handbook (QRH). The numbers are used to distinguish data peculiar to one or more, but not all of the airplanes. Where data applies to all airplanes listed, no reference is made to individual airplane numbers. Configuration data reflects the airplane as delivered configuration and is updated for service bulletin incorporations in conformance with the policy stated in the introduction section of this chapter.

Registry number is supplied by the operator as provided by the national regulatory agency. Serial and tabulation numbers are supplied by Boeing.

Registry number(s) reflect the most current information supplied by the operator to the Boeing Company through the SR process and 60 days prior to the subject revision date. Registry numbers received after that date will be incorporated at the next scheduled revision. If a registry number is not provided the QRH will default to serial number.

Registry Number	Old Registry Number	Serial Number	Tabulation Number
RA-73269	VP-BOW	30040	YK136
RA-73311	VP-BDP	40309	YF619
RA-73312	VP-BDQ	40311	YF621
RA-73313	VP-BDW	35700	YL931
RA-73314	VP-BSC	40233	YF701
RA-73315	VP-BSE	40236	YF704
RA-73316	VP-BSK	35984	YL935
RA-73317	VP-BSO	40874	YL938
RA-73318	VP-BSP	40872	YL961
RA-73319	VQ-BDC	42059	YV256
RA-73321	VQ-BVY	32841	YK137



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**Checklist Instructions**
Revision Record**Chapter CI**
Section RR**Revision Transmittal Letter**

To: All holders of Nordwind Airlines 737 Flight Crew Operations Manual (FCOM), Boeing Document Number D6-27370-83N-NRW.

Subject: Flight Crew Operations Manual Revision.

This revision reflects the most current information available to The Boeing Company 60 days before the subject revision date. The following revision highlights explain changes in this revision. General information below explains the use of revision bars to identify new or revised information.

Revision Record

No.	Revision Date	Date Filed
12	September 14, 2017	
14	May 17, 2018	
16	October 18, 2018	
18	April 18, 2019	
20	March 19, 2020	
22	March 18, 2021	
23.1	April 12, 2023	
23.3	October 31, 2025	

No.	Revision Date	Date Filed
13	March 15, 2018	
15	September 20, 2018	
17	March 21, 2019	
19	September 19, 2019	
21	September 17, 2020	
23	September 2, 2021	
13	March 15, 2018	
23.2	September 15, 2025	
23.4	November 13, 2025	

General

The Boeing Company issues FCOM revisions to provide new or revised procedures and information. Formal revisions also incorporate appropriate information from previously issued FCOM bulletins.

The revision date is the approximate date the manual is approved for printing. The revision is mailed a few weeks after this date. This manual is effective upon receipt and supersedes any manual (with the same document number) with a previous revision number.

Formal revisions include a Transmittal Letter, a new Revision Record, Revision Highlights, and a current List of Effective Pages. Use the information on the new Revision Record and List of Effective Pages to verify the FCOM content.

Pages containing revised technical material have revision bars associated with the changed text or illustration. Editorial revisions (for example, spelling corrections) may have revision bars with no associated highlight.

The Revision Record should be completed by the person incorporating the revision into the manual.

Filing Instructions

Consult the List of Effective Pages (CI.LEP). Pages identified with an asterisk (*) are either replacement pages or new (original) issue pages. Remove corresponding old pages and replace or add new pages. Remove pages that are marked DELETED; there are no replacement pages for deleted pages.

Revision Highlights

Be careful when inserting changes not to throw away pages from the manual that are not replaced. Using the List of Effective Pages (CI.LEP) can help determine the correct content of the manual.

Throughout the manual, airplane effectivity may be updated to reflect coverage as listed on the Preface - Model Identification page, or to show service bulletin airplane effectivity. Highlights are not supplied.

This manual is published from a database; the text and illustrations are marked with configuration information. Occasionally, because the editors rearrange the database markers, or mark items with configuration information due to the addition of new database content, some customers may receive revision bars on content that appears to be unchanged. Pages may also be republished without revision bars due to slight changes in the flow of the document.



Checklist Instructions

Revision Highlights

Chapter CI

Section RR

Chapter NNC - Non-Normal Checklists

Section 0 - Miscellaneous

Ditching

NNC.0.3 - Added step to alert supernumeraries to prepare for ditching.

NNC.0.4 - Added step to pull the landing gear circuit breaker for freighter airplanes.

Section 2 - Air Systems

Cabin Temperature Hot

NNC.2.16 - Deleted "IFE/PASS SEAT switch . . . OFF".

Section 3 - Anti-Ice, Rain

ICE DETECTOR

NNC.3.5 - Added checklist for airplanes equipped with an ice detection system.

ICING

NNC.3.6 - Added checklist for airplanes equipped with an ice detection system.

WINDOW HEAT OFF

NNC.3.8 - On airplanes with the amber window heat OFF light, the OFF light can illuminate when the window has reached the normal upper temperature limit. Updated the WINDOW HEAT OFF Non-Normal Checklist to allow for continued normal operation when only the amber window heat OFF light(s) illuminate.

NNC.3.8 - On airplanes with the amber window heat OFF light, the OFF light can illuminate when the window has reached the normal upper temperature limit. Updated the WINDOW HEAT OFF Non-Normal Checklist to allow for continued normal operation when only the amber window heat OFF light(s) illuminate.

NNC.3.9 - On airplanes with the amber window heat OFF light, the OFF light can illuminate when the window has reached the normal upper temperature limit. Updated the WINDOW HEAT OFF Non-Normal Checklist to allow for continued normal operation when only the amber window heat OFF light(s) illuminate.

Section 6 - Electrical

TRANSFER BUS OFF

NNC.6.16 - Modified procedure removing end symbol and replacing with GOTO new step 4 as directed.

NNC.6.17 - Updated NNC by adding new step 4 as directed.

Section 8 - Fire Protection

ENGINE FIRE or Engine Severe Damage or Separation

NNC.8.9 - Added End Symbol to choose one.

CARGO FIRE (FWD/AFT)

NNC.8.25 - Revised step to "PACK switches (both) . . . HIGH".

Section 9 - Flight Controls

YAW DAMPER

NNC.9.47 - Added step to avoid areas of predicted moderate or severe turbulence for freighter airplanes.

Section 10 - Flight Instruments, Displays

Airspeed Unreliable

NNC.10.4 - Updated step to reflect Performance Inflight-QRH chapter or other approved source.

NNC.10.5 - Removed Transponder mode steps. They have been relocated further down in the procedure.

NNC.10.5 - Removed Transponder mode steps. They have been relocated further down in the procedure.

NNC.10.6 - Added transponder selection step to ensure reliable altitude reporting for those transponder panels that have an ALT selector switch.

NNC.10.6 - Added transponder selection step to ensure reliable altitude reporting for those transponder panels that have an ALT SOURCE selector switch.

NNC.10.6 - Added navigation to Checklist Complete step.

NNC.10.6 - Added transponder setting step to provide traffic advisories only for those transponder panels that have an ALT OFF switch.

NNC.10.8 - Added option to the step to account for the Captain's, First Officer's and standby airspeed indications to all be unreliable.

Airspeed Unreliable**EASA-Europe or TCCA-Canada**

NNC.10.13 - Updated step to reflect Performance Inflight-QRH chapter or other approved source.

NNC.10.14 - Removed Transponder mode steps. They have been relocated further down in the procedure.

NNC.10.14 - Added transponder selection step to ensure reliable altitude reporting for those transponder panels that have an ALT selector switch.

NNC.10.14 - Added transponder selection step to ensure reliable altitude reporting for those transponder panels that have an ALT SOURCE selector switch.

NNC.10.14 - Added step "Transponder mode selector . . . TA" for aircraft which have a transponder mode selection of "TA".

NNC.10.15 - Added navigation to nuisance stick shaker step.

NNC.10.15 - Added transponder setting step to provide traffic advisories only for those transponder panels that have an ALT OFF switch.

NNC.10.15 - Added transponder setting step to provide traffic advisories only for those transponder panels that have an ALT RPTG OFF switch.

NNC.10.17 - Added option to the step to account for the Captain's, First Officer's and standby airspeed indications to all be unreliable.

ALT DISAGREE

NNC.10.20 - Standardizing step to verify all altimeters are set to correct barometric setting across all Boeing models. Use of "verify" is meant to have the flight crew check the barometric setting, and correct if needed.

NNC.10.21 - Changed step to remove Transponder mode steps. They have been relocated further down in the procedure.

NNC.10.21 - Removed Transponder mode steps. They have been relocated further down in the procedure.

NNC.10.21 - Removed Transponder mode steps. They have been relocated further down in the procedure.

NNC.10.22 - Added transponder selection step to ensure reliable altitude reporting for those transponder panels that have an ALT selector switch.

NNC.10.22 - Added transponder selection step to ensure reliable altitude reporting for those transponder panels that have an ALT SOURCE selector switch.

NNC.10.22 - Added transponder setting step to provide traffic advisories only for those transponder panels that have an ALT OFF switch.

NNC.10.22 - Added transponder setting step to provide traffic advisories only for those transponder panels that have an ALT RPTG OFF switch.

Section 11 - Flight Management, Navigation

ADS-B Out Failure

NNC.11.1 - Added "Transponder is failed" to the condition statement as a failure of the transponder indicates ADS-B out failure.

Section 13 - Hydraulics

STANDBY HYDRAULIC LOW PRESSURE

NNC.13.18 - Added note to the STANDBY HYDRAULIC LOW PRESSURE NNC to inform the flight crew that the affected thrust reverser is inoperative. Allows for more accurate landing performance calculations.

Section 14 - Landing Gear

Partial or All Gear Up Landing

NNC.14.22 - Added step to brief supernumeraries on emergency landing and evacuation procedures.

NNC.14.23 - Added step to pull the landing gear circuit breaker for freighter airplanes.

Section 15 - Warning Systems

GROUND PROXIMITY GND PROX INOPERATIVE

NNC.15.3 - Added nnc "Ground Proximity Gnd Prox Inoperative" for aircraft with RAAS or ORW.

GROUND PROXIMITY RUNWAY INOPERATIVE (RAAS)

NNC.15.5 - Added GROUND PROXIMITY RUNWAY INOPERATIVE NNC for airplanes equipped with RAAS.

Performance Package 10

737-800W CFM56-7B24 KG JAA CATC/N

Section 10 - Pkg Model Identification

737-800W CFM56-7B24 KG JAA CATC/N was added as Section 10.

Section 10 -

737-800W CFM56-7B24 KG JAA CATC/N was added as Section 10.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

PI-QRH.10.3 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.4 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.5 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.6 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.7 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.8 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.9 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.10 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.11 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.12 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.13 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.10.15 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.16 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.17 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.18 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.19 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.20 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.10.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

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PI-QRH.10.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

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PI-QRH.10.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.10.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

Section 11 - Advisory Information

Added section "11".

Section 12 - Engine Inoperative

Added section "12".

Section 13 - Gear Down

Added section "13".

Section 14 - Gear Down, Engine Inop

Added section "14".

Section 15 - Text

Added section "15".

Performance Package 20

737-800W CFM56-7B26 LB FAA CATC/N

Section 20 - Pkg Model Identification

737-800W CFM56-7B26 LB FAA CATC/N moved from Section 10 to 20.

Section 20 -

737-800W CFM56-7B26 LB FAA CATC/N moved from Section 10 to 20.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

PI-QRH.20.3 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.4 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.5 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.6 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.7 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.8 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.9 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.10 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.11 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.12 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.13 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.14 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.15 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.16 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.17 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.18 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.19 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.20.20 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.20.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.20.30 - Expanded altitude to include entire operational envelope. Reformatted for improved readability. Other minor data changes for consistency between phases of flight.

Section 21 - Advisory Information

Section "11" moved to "21".

Section 22 - Engine Inoperative

Section "12" moved to "22".

Section 23 - Gear Down

Section "13" moved to "23".

Section 24 - Gear Down, Engine Inop

Section "14" moved to "24".

Section 25 - Text

Section "15" moved to "25".

Performance Package 30

737-800W CFM56-7B26 C M KG FAA CATC/N

Section 30 - Pkg Model Identification

737-800W CFM56-7B26 C M KG FAA CATC/N moved from Section 20 to 30.

Section 30 -

737-800W CFM56-7B26 C M KG FAA CATC/N moved from Section 20 to 30.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

PI-QRH.30.3 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.4 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.5 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.6 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.7 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.8 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.9 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.10 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.11 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.12 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.30.15 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.16 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.30.17 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.18 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.19 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.20 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.30.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

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PI-QRH.30.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

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PI-QRH.30.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.30.30 - Expanded altitude to include entire operational envelope. Reformatted for improved readability. Other minor data changes for consistency between phases of flight.

Section 31 - Advisory Information

Section "21" moved to "31".

Section 32 - Engine Inoperative

Section "22" moved to "32".

Section 33 - Gear Down

Section "23" moved to "33".

Section 34 - Gear Down, Engine Inop

Section "24" moved to "34".

Section 35 - Text

Section "25" moved to "35".

Performance Package 40**737-800W CFM56-7B26 C M KG JAA CATC/N****Section 40 - Pkg Model Identification**

737-800W CFM56-7B26 C M KG JAA CATC/N moved from Section 30 to 40.

General

PI-QRH.ModID.40.1 - Revised to add/change an airplane entry.

PI-QRH.ModID.40.1 - Revised to add/change an airplane entry.

Section 40 -

737-800W CFM56-7B26 C M KG JAA CATC/N moved from Section 30 to 40.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

PI-QRH.40.3 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.40.4 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.40.7 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.40.9 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.40.10 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.40.11 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.40.12 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.40.13 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.40.14 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.40.15 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.40.19 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.40.20 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.40.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

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PI-QRH.40.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.40.30 - Expanded altitude to include entire operational envelope. Reformatted for improved readability. Other minor data changes for consistency between phases of flight.

Section 41 - Advisory Information

Section "31" moved to "41".

Section 42 - Engine Inoperative

Section "32" moved to "42".

Section 43 - Gear Down

Section "33" moved to "43".

Section 44 - Gear Down, Engine Inop

Section "34" moved to "44".

Section 45 - Text

Section "35" moved to "45".

Performance Package 50

737-800W CFM56-7B26 C M KG JAA CATC/N HIGH_ALT

Section 50 - Pkg Model Identification

737-800W CFM56-7B26 C M KG JAA CATC/N HIGH_ALT moved from Section 40 to 50.

Section 50 -

737-800W CFM56-7B26 C M KG JAA CATC/N HIGH_ALT moved from Section 40 to 50.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

PI-QRH.50.3 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.4 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.5 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.6 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.7 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.50.8 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.9 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.10 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.11 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.12 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.13 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.14 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.15 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.16 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.17 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.18 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.19 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.20 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.50.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

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PI-QRH.50.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.50.30 - Expanded altitude to include entire operational envelope. Reformatted for improved readability. Other minor data changes for consistency between phases of flight.

Section 51 - Advisory Information

Section "41" moved to "51".

Section 52 - Engine Inoperative

Section "42" moved to "52".

Section 53 - Gear Down

Section "43" moved to "53".

Section 54 - Gear Down, Engine Inop

Section "44" moved to "54".

Section 55 - Text

Section "45" moved to "55".

Performance Package 60

737-800WSFP1 CFM56-7B26 C M KG FAA CATC/N (FMC Model 737-800W.1)

Section 60 - Pkg Model Identification

737-800WSFP1 CFM56-7B26 C M KG FAA CATC/N (FMC Model 737-800W.1) moved from Section 50 to 60.

Section 60 -

737-800WSFP1 CFM56-7B26 C M KG FAA CATC/N (FMC Model 737-800W.1) moved from Section 50 to 60.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

PI-QRH.60.3 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

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PI-QRH.60.4 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.5 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.6 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.7 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.8 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.9 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.10 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.11 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.12 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.13 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.14 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.15 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

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PI-QRH.60.16 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.17 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.18 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.19 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.20 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.60.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

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PI-QRH.60.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.60.30 - Expanded altitude to include entire operational envelope. Reformatted for improved readability. Other minor data changes for consistency between phases of flight.

Section 61 - Advisory Information

Section "51" moved to "61".

Section 62 - Engine Inoperative

Section "52" moved to "62".

Section 63 - Gear Down

Section "53" moved to "63".

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Section 64 - Gear Down, Engine Inop

Section "54" moved to "64".

Section 65 - Text

Section "55" moved to "65".

Performance Package 70

737-800WSFP1 CFM56-7B26 C M KG JAA CATC/N (FMC Model 737-800W.1)

Section 70 - Pkg Model Identification

737-800WSFP1 CFM56-7B26 C M KG JAA CATC/N (FMC Model 737-800W.1) moved from Section 60 to 70.

General

PI-QRH.ModID.70.1 - Revised to add/change an airplane entry.

PI-QRH.ModID.70.1 - Revised to add/change an airplane entry.

Section 70 -

737-800WSFP1 CFM56-7B26 C M KG JAA CATC/N (FMC Model 737-800W.1) moved from Section 60 to 70.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

PI-QRH.70.3 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.4 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.5 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.6 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.7 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.8 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.9 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.10 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.11 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

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PI-QRH.70.12 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.13 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.14 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.15 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.16 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.17 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.18 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.19 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.20 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Other minor data changes for consistency between phases of flight.

PI-QRH.70.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

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PI-QRH.70.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.27 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

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PI-QRH.70.28 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.29 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Power settings %N1 values are updated to the nearest tenth. Added KIAS speed data. Added shading and footnote for flap placard speed exceedance. Other minor data changes for consistency between phases of flight.

PI-QRH.70.30 - Expanded altitude to include entire operational envelope. Reformatted for improved readability. Other minor data changes for consistency between phases of flight.

Section 71 - Advisory Information

Section "61" moved to "71".

Section 72 - Engine Inoperative

Section "62" moved to "72".

Section 73 - Gear Down

Section "63" moved to "73".

Section 74 - Gear Down, Engine Inop

Section "64" moved to "74".

Section 75 - Text

Section "65" moved to "75".

Performance Package 80**737-900ERW CFM56-7B26 C KG M FAA CATH/P****Section 80 - Pkg Model Identification**

737-900ERW CFM56-7B26 C KG M FAA CATH/P moved from Section 70 to 80.

Section 80 -

737-900ERW CFM56-7B26 C KG M FAA CATH/P moved from Section 70 to 80.

Flight With Unreliable Airspeed/ Turbulent Air Penetration

PI-QRH.80.3 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.4 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.5 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.6 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.7 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.8 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.9 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.10 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

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PI-QRH.80.11 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.12 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.13 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.14 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.15 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.16 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.17 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.18 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.19 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.20 - Added altitude functionality. Reformatted for improved readability. Added configurations of Flaps 10, 15, and 25 with gear up. Corrected maneuver speeds to be based on airplane altitude instead of airport altitude. Other minor data changes for consistency between phases of flight.

PI-QRH.80.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.21 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.22 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.23 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

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PI-QRH.80.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.24 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.25 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.26 - Added altitude functionality. Reformatted for improved readability. Updated Flaps 40 drag polar, thrust lift curves, and calculations using KIAS approach speed increment. Other minor data changes for consistency between phases of flight.

PI-QRH.80.27 - Expanded altitude to include entire operational envelope. Reformatted for improved readability. Other minor data changes for consistency between phases of flight.

Section 81 - Advisory Information

Section "71" moved to "81".

Section 82 - Engine Inoperative

Section "72" moved to "82".

Section 83 - Gear Down

Section "73" moved to "83".

Section 84 - Gear Down, Engine Inop

Section "74" moved to "84".

Section 85 - Text

Section "75" moved to "85".

Chapter Man - Maneuvers

Section 1 - Non-Normal Maneuvers

Runway Awareness and Advisory System (RAAS)

MAN.1.4 - Added/revised Runway Awareness and Advisory System (RAAS).

Section 2 - Flight Patterns

Go-Around and Missed Approach

MAN.2.11 - Updated the maneuver diagram to match changes to the Go Around and Missed Approach procedure.

Chapter CI - Checklist Instructions

Section ModID - Model Identification

General

CI.ModID.1 - Revised to add/change an airplane entry.

CI.ModID.1 - Revised to add/change an airplane entry.

CI.ModID.1 - Revised to add/change an airplane entry.

CI.ModID.1 - Revised to add/change an airplane entry.

Section 1 - Normal Checklists

Normal Checklist Operation

CI.1.1 - Changed “entire procedure was not done” to “procedure was not completed” for accuracy and readability.

CI.1.1 - Changed wording for readability.

CI.1.2 - Changed wording for cross-model standardization.

Checklist Construction

CI.1.2 - Changed the punctuation.

CI.1.2 - Changed the punctuation.

Section 2 - Non-Normal Checklists

Introduction

CI.2.1 - Changed wording for cross-model standardization.

Non-Normal Checklist Operation

CI.2.2 - Changed the punctuation.

CI.2.3 - Changed wording for cross-model standardization.

CI.2.3 - Added abbreviation for Dispatch Deviations Guide.

CI.2.3 - Added abbreviation for Minimum Equipment List.

Non-Normal Checklist Use

CI.2.5 - Changed wording for cross-model standardization.

CI.2.5 - Changed wording for cross-model standardization.

CI.2.5 - Changed wording for cross-model standardization.

CI.2.6 - Changed wording for cross-model standardization.

CI.2.6 - Changed the punctuation.

CI.2.6 - Changed wording for cross-model standardization.

CI.2.7 - Changed wording for cross-model standardization.

CI.2.7 - Moved Checklist Complete symbol into the legend for readability.

Non-Normal Checklist Legend

CI.2.9 - Deleted "Choose one:" text in the symbol section for standardization.

CI.2.9 - Moved Checklist Complete symbol into the legend for readability.

Chapter NNC - NNC - Engine Fire on the Ground Checklist

Section Back Cover - Engine Fire on the Ground Checklist

Engine Fire on the Ground

Back Cover.2 - Deleted "Choose one" step to go to Engine Tailpipe Fire checklist.

Back Cover.2 - Changed step to include End Symbol and bolding.

Section Back Cover - Evacuation Checklist

Evacuation

Back Cover.4 - Changed "Advise the tower" to "Advise ATC" for cross-model standardization.

Back Cover.4 - Revised to account for an observed fire in addition to an indicated fire warning.



Intentionally
Blank



Checklist Instructions

QRH List of Effective Pages

Chapter CI

Section LEP

Quick Reference Handbook

Quick Action Index

* QA.Index.1-2 September 2, 2021

Lights (tab)

* Lights.Index.1-8 September 2, 2021

Unannunciated (tab)

* Unann.Index.1-4 September 2, 2021

Alphabetical Index

* Alpha.Index.1-14 September 2, 2021

Normal Checklists

* NC.1 September 2, 2021
 * NC.2 September 2, 2021
 * NC.3 September 2, 2021
 NC.4 September 14, 2017

0 Miscellaneous (tab)

* 0.TOC.1-2 September 2, 2021
 0.1 October 18, 2018
 * 0.2 September 2, 2021
 * 0.3 September 2, 2021
 * 0.4 September 2, 2021
 * 0.5 September 2, 2021
 * 0.6 September 2, 2021
 * 0.7 September 2, 2021
 0.8 March 26, 2015

1 Airplane Gen., Emer. Equip., Doors, Windows (tab)

* 1.TOC.1-2 September 2, 2021
 * 1.1 September 2, 2021
 1.2 June 12, 2014
 * 1.3 September 2, 2021
 1.4 June 12, 2014
 1.5 June 12, 2014
 1.6 June 12, 2014
 1.7 June 12, 2014
 * 1.8 September 2, 2021
 * 1.9 September 2, 2021

1.10 March 18, 2021
 1.11 March 18, 2021
 1.12 March 18, 2021
 * 1.13 September 2, 2021
 * 1.14 September 2, 2021
 * 1.15 September 2, 2021
 * 1.16 September 2, 2021
 * 1.17 September 2, 2021
 * 1.18 September 2, 2021
 * 1.19 September 2, 2021
 * 1.20 September 2, 2021
 * 1.21 September 2, 2021
 * 1.22 September 2, 2021
 1.23 March 18, 2021
 1.24 March 18, 2021
 1.25 March 18, 2021
 * 1.26 September 2, 2021
 1.27 March 18, 2021
 1.28 March 18, 2021
 1.29 March 18, 2021
 * 1.30 September 2, 2021
 1.31 March 18, 2021
 1.32 March 18, 2021
 1.33 March 18, 2021
 1.34 March 18, 2021
 1.35 March 18, 2021
 1.36 March 18, 2021
 1.37 March 18, 2021
 1.38 March 18, 2021

2 Air Systems (tab)

2.TOC.1-2 March 18, 2021
 2.1 September 25, 2014
 2.2 September 15, 2016
 * 2.3 September 2, 2021
 2.4 September 15, 2016
 2.5 September 15, 2016
 * 2.6 September 2, 2021
 2.7 September 15, 2016
 * 2.8 September 2, 2021
 * 2.9 September 2, 2021
 * 2.10 September 2, 2021
 * 2.11 September 2, 2021
 2.12 March 18, 2021

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2.13	March 18, 2021		
* 2.14	September 2, 2021	4 Automatic Flight (tab)	
* 2.15	September 2, 2021	4.TOC.1-2	May 18, 2017
* 2.16	September 2, 2021	4.1	October 29, 2013
* 2.17	September 2, 2021	4.2	October 18, 2018
2.18	March 18, 2021	4.3	October 18, 2018
2.19	March 18, 2021	4.4	May 18, 2017
* 2.20	September 2, 2021		
2.21	March 18, 2021	5 Communications (tab)	
2.22	March 18, 2021	5.TOC.1-2	October 18, 2018
* 2.23	September 2, 2021	* 5.1	September 2, 2021
* 2.24	September 2, 2021	5.2	October 18, 2018
2.25	March 18, 2021		
2.26	March 18, 2021	6 Electrical (tab)	
2.27	March 18, 2021	* 6.TOC.1-2	September 2, 2021
2.28	March 18, 2021	* 6.1	September 2, 2021
2.29	March 18, 2021	6.2	October 18, 2018
2.30	March 18, 2021	6.3	October 18, 2018
2.31	March 18, 2021	6.4	October 18, 2018
2.32	March 18, 2021	6.5	October 18, 2018
2.33	March 18, 2021	* 6.6	September 2, 2021
2.34	March 18, 2021	* 6.7	September 2, 2021
2.35	March 18, 2021	* 6.8	September 2, 2021
* 2.36	September 2, 2021	6.9	October 18, 2018
* 2.37	September 2, 2021	6.10	October 18, 2018
2.38	March 18, 2021	6.11	October 18, 2018
		6.12	October 18, 2018
		6.13	October 18, 2018
		* 6.14	September 2, 2021
		* 6.15	September 2, 2021
		* 6.16	September 2, 2021
		* 6.17	September 2, 2021
		* 6.18	September 2, 2021
		7 Engines, APU (tab)	
		7.TOC.1-2	March 18, 2021
		* 7.1	September 2, 2021
		7.2	April 18, 2019
		7.3	September 15, 2016
		7.4	September 15, 2016
		* 7.5	September 2, 2021
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		7.7	March 18, 2021
		7.8	March 16, 2017
		7.9	September 15, 2016
		7.10	April 18, 2019
		7.11	April 18, 2019
		7.12	September 15, 2016

3 Anti-Ice, Rain (tab)

* 3.TOC.1-2	September 2, 2021
* 3.1	September 2, 2021
3.2	March 27, 2014
* 3.3	September 2, 2021
3.4	September 19, 2019
* 3.5	September 2, 2021
* 3.6	September 2, 2021
3.7	September 19, 2019
* 3.8	September 2, 2021
* 3.9	September 2, 2021
3.10	September 19, 2019
3.11	September 19, 2019
* 3.12	September 2, 2021
3.13	September 19, 2019
* 3.14	September 2, 2021
3.15	September 19, 2019
3.16	September 19, 2019
3.17	September 19, 2019
3.18	September 19, 2019

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7.13	March 18, 2021	* 8.13	September 2, 2021
7.14	September 15, 2016	8.14	March 18, 2021
* 7.15	September 2, 2021	* 8.15	September 2, 2021
* 7.16	September 2, 2021	* 8.16	September 2, 2021
* 7.17	September 2, 2021	8.17	March 18, 2021
7.18	March 18, 2021	* 8.18	September 2, 2021
* 7.19	September 2, 2021	8.19	March 18, 2021
* 7.20	September 2, 2021	8.20	March 18, 2021
7.21	September 14, 2017	* 8.21	September 2, 2021
* 7.22	September 2, 2021	8.22	March 18, 2021
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* 7.25	September 2, 2021	* 8.25	September 2, 2021
* 7.26	September 2, 2021	8.26	March 18, 2021
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7.29	March 18, 2021	8.29	March 18, 2021
* 7.30	September 2, 2021	8.30	March 18, 2021
7.31	March 18, 2021	8.31	March 18, 2021
* 7.32	September 2, 2021	8.32	March 18, 2021
* 7.33	September 2, 2021	8.33	March 18, 2021
7.34	May 17, 2018	8.34	March 18, 2021
7.35	September 14, 2017	9 Flight Controls (tab)	
7.36	September 14, 2017	9.TOC.1-2	March 18, 2021
* 7.37	September 2, 2021	9.1	March 18, 2021
7.38	September 14, 2017	9.2	March 18, 2021
7.39	September 14, 2017	* 9.3	September 2, 2021
* 7.40	September 2, 2021	9.4	March 18, 2021
7.41	March 21, 2019	9.5	March 18, 2021
7.42	September 14, 2017	* 9.6	September 2, 2021
7.43	September 14, 2017	9.7	March 18, 2021
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8 Fire Protection (tab)		9.9	March 18, 2021
8.TOC.1-2	March 18, 2021	9.10	March 18, 2021
8.1	October 29, 2013	9.11	March 18, 2021
* 8.2	September 2, 2021	9.12	March 18, 2021
8.3	March 18, 2021	* 9.13	September 2, 2021
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8.5	March 18, 2021	9.15	March 18, 2021
8.6	March 18, 2021	9.16	March 18, 2021
* 8.7	September 2, 2021	* 9.17	September 2, 2021
8.8	March 18, 2021	9.18	March 18, 2021
* 8.9	September 2, 2021	* 9.19	September 2, 2021
8.10	March 18, 2021	9.20	March 18, 2021
* 8.11	September 2, 2021	9.21	March 18, 2021
* 8.12	September 2, 2021	* 9.22	September 2, 2021

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9.23	March 18, 2021	10.19	March 18, 2021
9.24	March 18, 2021	* 10.20	September 2, 2021
9.25	March 18, 2021	* 10.21	September 2, 2021
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9.27	March 18, 2021	* 10.23	September 2, 2021
* 9.28	September 2, 2021	10.24	March 18, 2021
* 9.29	September 2, 2021	10.25	March 18, 2021
* 9.30	September 2, 2021	* 10.26	September 2, 2021
9.31	March 18, 2021	10.27	March 18, 2021
9.32	March 18, 2021	* 10.28	September 2, 2021
9.33	March 18, 2021	10.29	March 18, 2021
9.34	March 18, 2021	* 10.30	September 2, 2021
* 9.35	September 2, 2021	11 Flight Management, Navigation (tab)	
9.36	March 18, 2021	11.TOC.1-2	March 19, 2020
9.37	March 18, 2021	* 11.1	September 2, 2021
9.38	March 18, 2021	11.2	September 20, 2018
9.39	March 18, 2021	11.3	May 17, 2018
* 9.40	September 2, 2021	* 11.4	September 2, 2021
9.41	March 18, 2021	11.5	September 20, 2018
* 9.42	September 2, 2021	11.6	September 20, 2018
9.43	March 18, 2021	11.7	May 17, 2018
9.44	March 18, 2021	* 11.8	September 2, 2021
9.45	March 18, 2021	11.9	March 19, 2020
* 9.46	September 2, 2021	11.10	March 19, 2020
* 9.47	September 2, 2021	11.11	March 19, 2020
9.48	March 18, 2021	* 11.12	September 2, 2021
10 Flight Instruments, Displays (tab)		11.13	March 19, 2020
10.TOC.1-2	March 18, 2021	11.14	March 19, 2020
* 10.1	September 2, 2021	11.15	March 19, 2020
* 10.2	September 2, 2021	11.16	March 19, 2020
10.3	March 18, 2021	11.17	March 19, 2020
* 10.4	September 2, 2021	11.18	March 19, 2020
* 10.5	September 2, 2021	12 Fuel (tab)	
* 10.6	September 2, 2021	12.TOC.1-2	September 15, 2016
* 10.7	September 2, 2021	* 12.1	September 2, 2021
* 10.8	September 2, 2021	12.2	September 15, 2016
10.9	March 18, 2021	12.3	September 15, 2016
* 10.10	September 2, 2021	12.4	September 15, 2016
10.11	March 18, 2021	12.5	September 15, 2016
10.12	March 18, 2021	* 12.6	September 2, 2021
* 10.13	September 2, 2021	12.7	September 15, 2016
* 10.14	September 2, 2021	* 12.8	September 2, 2021
* 10.15	September 2, 2021	* 12.9	September 2, 2021
* 10.16	September 2, 2021	12.10	May 17, 2018
* 10.17	September 2, 2021	12.11	September 15, 2016
* 10.18	September 2, 2021		

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12.12	October 28, 2014	* 14.15	September 2, 2021
12.13	October 28, 2014	14.16	May 17, 2018
12.14	October 28, 2014	14.17	September 20, 2018
12.15	September 15, 2016	14.18	May 17, 2018
12.16	September 15, 2016	* 14.19	September 2, 2021
12.17	October 28, 2014	14.20	October 29, 2013
12.18	September 15, 2016	14.21	October 29, 2013
12.19	October 28, 2014	* 14.22	September 2, 2021
12.20	October 28, 2014	* 14.23	September 2, 2021
13 Hydraulics (tab)		* 14.24	September 2, 2021
13.TOC.1-2	September 15, 2016	* 14.25	September 2, 2021
13.1	September 15, 2016	14.26	October 29, 2013
13.2	September 15, 2016	15 Warning Systems (tab)	
13.3	September 15, 2016	* 15.TOC.1-2	September 2, 2021
13.4	September 15, 2016	15.1	October 29, 2013
* 13.5	September 2, 2021	15.2	October 29, 2013
13.6	May 17, 2018	* 15.3	September 2, 2021
13.7	September 15, 2016	* 15.4	September 2, 2021
13.8	September 15, 2016	* 15.5	September 2, 2021
13.9	September 15, 2016	* 15.6	September 2, 2021
* 13.10	September 2, 2021	* 15.7	September 2, 2021
* 13.11	September 2, 2021	* 15.8	September 2, 2021
13.12	May 17, 2018	Operational Information (tab)	
13.13	September 15, 2016	OI.TOC.1-2	October 29, 2013
13.14	September 15, 2016	OI.1.1	September 25, 2014
13.15	September 15, 2016	OI.1.2	October 29, 2013
* 13.16	September 2, 2021	Performance Inflight - QRH (tab)	
13.17	May 18, 2017	* PI-QRH.TOC.1-2	September 2, 2021
* 13.18	September 2, 2021	* PI-QRH.TOC.10.1-4	Deleted
14 Landing Gear (tab)		* PI-QRH.ModID.10.1-2	Deleted
14.TOC.1-2	March 16, 2017	* PI-QRH.10.1-4	Deleted
14.1	September 15, 2016	* PI-QRH.10.5-6	Deleted
* 14.2	September 2, 2021	* PI-QRH.10.7-8	Deleted
14.3	March 27, 2014	* PI-QRH.11.1-3	Deleted
* 14.4	September 2, 2021	* PI-QRH.11.4-28	Deleted
14.5	June 12, 2014	* PI-QRH.11.29-30	Deleted
14.6	September 19, 2019	* PI-QRH.12.1-12	Deleted
14.7	September 19, 2019	* PI-QRH.13.1-6	Deleted
14.8	May 17, 2018	* PI-QRH.14.1-4	Deleted
14.9	March 16, 2017	* PI-QRH.15.1-6	Deleted
14.10	September 15, 2016	* PI-QRH.TOC.20.1-4	Deleted
14.11	March 16, 2017	* PI-QRH.ModID.20.1-2	Deleted
14.12	October 29, 2013	* PI-QRH.20.1-4	Deleted
14.13	March 27, 2014	* PI-QRH.20.5-6	Deleted
14.14	March 27, 2014	* PI-QRH.20.7-8	Deleted

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* PI-QRH.21.1-3	Deleted	* PI-QRH.10.25	September 2, 2021
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* PI-QRH.21.29-30	Deleted	* PI-QRH.10.27	September 2, 2021
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* PI-QRH.23.1-6	Deleted	* PI-QRH.10.29	September 2, 2021
* PI-QRH.24.1-4	Deleted	* PI-QRH.10.30	September 2, 2021
* PI-QRH.25.1-6	Deleted	* PI-QRH.10.31	September 2, 2021
* PI-QRH.TOC.30.1-4	Deleted	* PI-QRH.10.32	September 2, 2021
* PI-QRH.ModID.30.1-2	Deleted	* PI-QRH.10.33	September 2, 2021
* PI-QRH.30.1-4	Deleted	* PI-QRH.10.34	September 2, 2021
* PI-QRH.30.5-6	Deleted	* PI-QRH.11.1	September 2, 2021
* PI-QRH.30.7-8	Deleted	* PI-QRH.11.2	September 2, 2021
* PI-QRH.31.1-2	Deleted	* PI-QRH.11.3	September 2, 2021
* PI-QRH.31.3-27	Deleted	* PI-QRH.11.4	September 2, 2021
* PI-QRH.31.28-30	Deleted	* PI-QRH.11.5	September 2, 2021
* PI-QRH.32.1-12	Deleted	* PI-QRH.11.6	September 2, 2021
* PI-QRH.33.1-6	Deleted	* PI-QRH.11.7	September 2, 2021
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* PI-QRH.35.1-6	Deleted	* PI-QRH.11.9	September 2, 2021
737-800W CFM56-7B24 KG JAA CATC/N			
* PI-QRH.TOC.10.1-4	September 2, 2021	* PI-QRH.11.10	September 2, 2021
* PI-QRH.ModID.10.1-2	September 2, 2021	* PI-QRH.11.11	September 2, 2021
* PI-QRH.10.1	September 2, 2021	* PI-QRH.11.12	September 2, 2021
* PI-QRH.10.2	September 2, 2021	* PI-QRH.11.13	September 2, 2021
* PI-QRH.10.3	September 2, 2021	* PI-QRH.11.14	September 2, 2021
* PI-QRH.10.4	September 2, 2021	* PI-QRH.11.15	September 2, 2021
* PI-QRH.10.5	September 2, 2021	* PI-QRH.11.16	September 2, 2021
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* PI-QRH.10.9	September 2, 2021	* PI-QRH.11.20	September 2, 2021
* PI-QRH.10.10	September 2, 2021	* PI-QRH.11.21	September 2, 2021
* PI-QRH.10.11	September 2, 2021	* PI-QRH.11.22	September 2, 2021
* PI-QRH.10.12	September 2, 2021	* PI-QRH.11.23	September 2, 2021
* PI-QRH.10.13	September 2, 2021	* PI-QRH.11.24	September 2, 2021
* PI-QRH.10.14	September 2, 2021	* PI-QRH.11.25	September 2, 2021
* PI-QRH.10.15	September 2, 2021	* PI-QRH.11.26	September 2, 2021
* PI-QRH.10.16	September 2, 2021	* PI-QRH.11.27	September 2, 2021
* PI-QRH.10.17	September 2, 2021	* PI-QRH.11.28	September 2, 2021
* PI-QRH.10.18	September 2, 2021	* PI-QRH.11.29	September 2, 2021
* PI-QRH.10.19	September 2, 2021	* PI-QRH.11.30	September 2, 2021
* PI-QRH.10.20	September 2, 2021	* PI-QRH.12.1	September 2, 2021
* PI-QRH.10.21	September 2, 2021	* PI-QRH.12.2	September 2, 2021
* PI-QRH.10.22	September 2, 2021	* PI-QRH.12.3	September 2, 2021
* PI-QRH.10.23	September 2, 2021	* PI-QRH.12.4	September 2, 2021
* PI-QRH.10.24	September 2, 2021	* PI-QRH.12.5	September 2, 2021
		* PI-QRH.12.6	September 2, 2021
		* PI-QRH.12.7	September 2, 2021

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* PI-QRH.12.8	September 2, 2021	* PI-QRH.20.23	September 2, 2021
* PI-QRH.12.9	September 2, 2021	* PI-QRH.20.24	September 2, 2021
* PI-QRH.12.10	September 2, 2021	* PI-QRH.20.25	September 2, 2021
* PI-QRH.12.11	September 2, 2021	* PI-QRH.20.26	September 2, 2021
* PI-QRH.12.12	September 2, 2021	* PI-QRH.20.27	September 2, 2021
* PI-QRH.13.1	September 2, 2021	* PI-QRH.20.28	September 2, 2021
* PI-QRH.13.2	September 2, 2021	* PI-QRH.20.29	September 2, 2021
* PI-QRH.13.3	September 2, 2021	* PI-QRH.20.30	September 2, 2021
* PI-QRH.13.4	September 2, 2021	* PI-QRH.20.31	September 2, 2021
* PI-QRH.13.5	September 2, 2021	* PI-QRH.20.32	September 2, 2021
* PI-QRH.13.6	September 2, 2021	* PI-QRH.20.33	September 2, 2021
* PI-QRH.14.1	September 2, 2021	* PI-QRH.20.34	September 2, 2021
* PI-QRH.14.2	September 2, 2021	* PI-QRH.21.1	September 2, 2021
* PI-QRH.14.3	September 2, 2021	* PI-QRH.21.2	September 2, 2021
* PI-QRH.14.4	September 2, 2021	* PI-QRH.21.3	September 2, 2021
* PI-QRH.15.1	September 2, 2021	* PI-QRH.21.4	September 2, 2021
* PI-QRH.15.2	September 2, 2021	* PI-QRH.21.5	September 2, 2021
* PI-QRH.15.3	September 2, 2021	* PI-QRH.21.6	September 2, 2021
* PI-QRH.15.4	September 2, 2021	* PI-QRH.21.7	September 2, 2021
* PI-QRH.15.5	September 2, 2021	* PI-QRH.21.8	September 2, 2021
* PI-QRH.15.6	September 2, 2021	* PI-QRH.21.9	September 2, 2021
737-800W CFM56-7B26 LB FAA CATC/N		* PI-QRH.21.10	September 2, 2021
* PI-QRH.TOC.20.1-4	September 2, 2021	* PI-QRH.21.11	September 2, 2021
* PI-QRH.ModID.20.1-2	September 2, 2021	* PI-QRH.21.12	September 2, 2021
* PI-QRH.20.1	September 2, 2021	* PI-QRH.21.13	September 2, 2021
* PI-QRH.20.2	September 2, 2021	* PI-QRH.21.14	September 2, 2021
* PI-QRH.20.3	September 2, 2021	* PI-QRH.21.15	September 2, 2021
* PI-QRH.20.4	September 2, 2021	* PI-QRH.21.16	September 2, 2021
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* PI-QRH.20.8	September 2, 2021	* PI-QRH.21.20	September 2, 2021
* PI-QRH.20.9	September 2, 2021	* PI-QRH.21.21	September 2, 2021
* PI-QRH.20.10	September 2, 2021	* PI-QRH.21.22	September 2, 2021
* PI-QRH.20.11	September 2, 2021	* PI-QRH.21.23	September 2, 2021
* PI-QRH.20.12	September 2, 2021	* PI-QRH.21.24	September 2, 2021
* PI-QRH.20.13	September 2, 2021	* PI-QRH.21.25	September 2, 2021
* PI-QRH.20.14	September 2, 2021	* PI-QRH.21.26	September 2, 2021
* PI-QRH.20.15	September 2, 2021	* PI-QRH.21.27	September 2, 2021
* PI-QRH.20.16	September 2, 2021	* PI-QRH.21.28	September 2, 2021
* PI-QRH.20.17	September 2, 2021	* PI-QRH.21.29	September 2, 2021
* PI-QRH.20.18	September 2, 2021	* PI-QRH.21.30	September 2, 2021
* PI-QRH.20.19	September 2, 2021	* PI-QRH.22.1	September 2, 2021
* PI-QRH.20.20	September 2, 2021	* PI-QRH.22.2	September 2, 2021
* PI-QRH.20.21	September 2, 2021	* PI-QRH.22.3	September 2, 2021
* PI-QRH.20.22	September 2, 2021	* PI-QRH.22.4	September 2, 2021
		* PI-QRH.22.5	September 2, 2021

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* PI-QRH.22.6	September 2, 2021	* PI-QRH.30.21	September 2, 2021
* PI-QRH.22.7	September 2, 2021	* PI-QRH.30.22	September 2, 2021
* PI-QRH.22.8	September 2, 2021	* PI-QRH.30.23	September 2, 2021
* PI-QRH.22.9	September 2, 2021	* PI-QRH.30.24	September 2, 2021
* PI-QRH.22.10	September 2, 2021	* PI-QRH.30.25	September 2, 2021
* PI-QRH.22.11	September 2, 2021	* PI-QRH.30.26	September 2, 2021
* PI-QRH.22.12	September 2, 2021	* PI-QRH.30.27	September 2, 2021
* PI-QRH.23.1	September 2, 2021	* PI-QRH.30.28	September 2, 2021
* PI-QRH.23.2	September 2, 2021	* PI-QRH.30.29	September 2, 2021
* PI-QRH.23.3	September 2, 2021	* PI-QRH.30.30	September 2, 2021
* PI-QRH.23.4	September 2, 2021	* PI-QRH.30.31	September 2, 2021
* PI-QRH.23.5	September 2, 2021	* PI-QRH.30.32	September 2, 2021
* PI-QRH.23.6	September 2, 2021	* PI-QRH.30.33	September 2, 2021
* PI-QRH.24.1	September 2, 2021	* PI-QRH.30.34	September 2, 2021
* PI-QRH.24.2	September 2, 2021	* PI-QRH.31.1	September 2, 2021
* PI-QRH.24.3	September 2, 2021	* PI-QRH.31.2	September 2, 2021
* PI-QRH.24.4	September 2, 2021	* PI-QRH.31.3	September 2, 2021
* PI-QRH.25.1	September 2, 2021	* PI-QRH.31.4	September 2, 2021
* PI-QRH.25.2	September 2, 2021	* PI-QRH.31.5	September 2, 2021
* PI-QRH.25.3	September 2, 2021	* PI-QRH.31.6	September 2, 2021
* PI-QRH.25.4	September 2, 2021	* PI-QRH.31.7	September 2, 2021
* PI-QRH.25.5	September 2, 2021	* PI-QRH.31.8	September 2, 2021
* PI-QRH.25.6	September 2, 2021	* PI-QRH.31.9	September 2, 2021
737-800W CFM56-7B26 C M KG FAA CATC/N		* PI-QRH.31.10	September 2, 2021
* PI-QRH.TOC.30.1-4	September 2, 2021	* PI-QRH.31.11	September 2, 2021
* PI-QRH.ModID.30.1-2	September 2, 2021	* PI-QRH.31.12	September 2, 2021
* PI-QRH.30.1	September 2, 2021	* PI-QRH.31.13	September 2, 2021
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* PI-QRH.30.20	September 2, 2021	* PI-QRH.32.2	September 2, 2021
		* PI-QRH.32.3	September 2, 2021

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* PI-QRH.32.4	September 2, 2021	* PI-QRH.40.19	September 2, 2021
* PI-QRH.32.5	September 2, 2021	* PI-QRH.40.20	September 2, 2021
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* PI-QRH.33.6	September 2, 2021	* PI-QRH.40.33	September 2, 2021
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* PI-QRH.34.3	September 2, 2021	PI-QRH.41.2	September 17, 2020
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* PI-QRH.35.6	September 2, 2021	PI-QRH.41.9	March 19, 2020
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* PI-QRH.40.18	September 2, 2021	* PI-QRH.41.30	September 2, 2021
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* PI-QRH.42.1	September 2, 2021	* PI-QRH.61.3-27	Deleted
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* PI-QRH.44.1	September 2, 2021	737-800W CFM56-7B26 C M KG JAA CATC/N HIGH_ALT	
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PI-QRH.44.3	March 19, 2020	* PI-QRH.ModID.50.1-2	September 2, 2021
PI-QRH.44.4	March 19, 2020	* PI-QRH.50.1	September 2, 2021
* PI-QRH.45.1	September 2, 2021	* PI-QRH.50.2	September 2, 2021
PI-QRH.45.2	September 17, 2020	* PI-QRH.50.3	September 2, 2021
PI-QRH.45.3	March 19, 2020	* PI-QRH.50.4	September 2, 2021
PI-QRH.45.4	March 19, 2020	* PI-QRH.50.5	September 2, 2021
PI-QRH.45.5	March 19, 2020	* PI-QRH.50.6	September 2, 2021
PI-QRH.45.6	March 19, 2020	* PI-QRH.50.7	September 2, 2021
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* PI-QRH.51.17	September 2, 2021	CATC/N (FMC Model 737-800W.1)	
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* PI-QRH.61.13	September 2, 2021	CATC/N (FMC Model 737-800W.1)	
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* PI-QRH.62.2	September 2, 2021	* PI-QRH.70.17	September 2, 2021

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* PI-QRH.71.5	September 2, 2021	* PI-QRH.74.4	September 2, 2021
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* PI-QRH.71.13	September 2, 2021	* PI-QRH.TOC.80.1-4	September 2, 2021
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* PI-QRH.80.16	September 2, 2021	* PI-QRH.82.3	September 2, 2021
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* PI-QRH.81.11	September 2, 2021		
* PI-QRH.81.12	September 2, 2021	Maneuvers (tab)	
* PI-QRH.81.13	September 2, 2021	* Man.TOC.1-2	September 2, 2021
* PI-QRH.81.14	September 2, 2021	MAN.05.1	October 29, 2013
* PI-QRH.81.15	September 2, 2021	MAN.05.2	October 29, 2013
* PI-QRH.81.16	September 2, 2021	MAN.1.1	September 19, 2019
* PI-QRH.81.17	September 2, 2021	MAN.1.2	September 14, 2017
* PI-QRH.81.18	September 2, 2021	MAN.1.3	September 14, 2017
* PI-QRH.81.19	September 2, 2021	* MAN.1.4	September 2, 2021
* PI-QRH.81.20	September 2, 2021	* MAN.1.5	September 2, 2021
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* PI-QRH.81.22	September 2, 2021	* MAN.1.7	September 2, 2021
* PI-QRH.81.23	September 2, 2021	MAN.1.8	September 14, 2017
* PI-QRH.81.24	September 2, 2021	MAN.1.9	March 19, 2020
* PI-QRH.81.25	September 2, 2021	MAN.1.10	September 14, 2017
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* PI-QRH.81.27	September 2, 2021	MAN.1.12	September 19, 2019
* PI-QRH.81.28	September 2, 2021	MAN.1.13	September 19, 2019
* PI-QRH.81.29	September 2, 2021	MAN.1.14	September 19, 2019
* PI-QRH.81.30	September 2, 2021	MAN.2.1	April 18, 2019
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MAN.2.7	March 18, 2021	* CI.RR.29	September 2, 2021
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MAN.2.12	March 19, 2020	* CI.RR.34	September 2, 2021
MAN.2.13	March 19, 2020	* CI.RR.35	September 2, 2021
MAN.2.14	March 19, 2020	* CI.RR.36	September 2, 2021
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* CI.TOC.1-2	September 2, 2021	* CI.RR.40	September 2, 2021
Model Identification		* CI.RR.41	September 2, 2021
* CI.ModID.1-2	September 2, 2021	* CI.RR.42	September 2, 2021
Revision Record		* CI.RR.43	September 2, 2021
* CI.RR.1	September 2, 2021	* CI.RR.44	September 2, 2021
* CI.RR.2	September 2, 2021	* CI.RR.45	September 2, 2021
Revision Highlights		* CI.RR.46	September 2, 2021
* CI.RR.3	September 2, 2021	* CI.RR.47	September 2, 2021
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* CI.RR.5	September 2, 2021	List of Effective Pages	
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* CI.RR.8	September 2, 2021	* CI.1.1	September 2, 2021
* CI.RR.9	September 2, 2021	* CI.1.2	September 2, 2021
* CI.RR.10	September 2, 2021	Checklist Instructions - Non-Normal Checklists	
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* CI.RR.12	September 2, 2021	* CI.2.2	September 2, 2021
* CI.RR.13	September 2, 2021	* CI.2.3	September 2, 2021
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* CI.RR.15	September 2, 2021	* CI.2.5	September 2, 2021
* CI.RR.16	September 2, 2021	* CI.2.6	September 2, 2021
* CI.RR.17	September 2, 2021	* CI.2.7	September 2, 2021
* CI.RR.18	September 2, 2021	CI.2.8	March 18, 2021
* CI.RR.19	September 2, 2021	* CI.2.9	September 2, 2021
* CI.RR.20	September 2, 2021	CI.2.10	March 18, 2021
* CI.RR.21	September 2, 2021		
* CI.RR.22	September 2, 2021		
* CI.RR.23	September 2, 2021		
* CI.RR.24	September 2, 2021		

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Engine Fire on the Ground

Back Cover.1	March 18, 2021
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Evacuation

* Back Cover.5-6	Deleted
* Back Cover.3	September 2, 2021
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Checklist Instructions

Normal Checklists

Chapter CI

Section 1

Introduction

This introduction gives guidelines for use of the Normal Checklist (NC).

The NC is organized by phase of flight.

The NC is used to verify that critical items have been done.

Normal Checklist Operation

Normal checklists are used after doing all respective procedural items.

The following table shows which pilot calls for the checklist and which pilot reads the checklist. Both pilots visually verify that each item is in the needed configuration or that the step is done. The far right column shows which pilot gives the response. This is different than the normal procedures where the far right column can show which pilot does the step.

Checklist	Call	Read	Verify	Respond
PREFLIGHT	Captain	First officer	Both	Area of responsibility
BEFORE START	Captain	First officer	Both	Area of responsibility
BEFORE TAXI	Captain	First officer	Both	Area of responsibility
BEFORE TAKEOFF	Pilot flying	Pilot monitoring	Both	Pilot flying
AFTER TAKEOFF	Pilot flying	Pilot monitoring	Both	Pilot monitoring
DESCENT	Pilot flying	Pilot monitoring	Both	Area of responsibility
APPROACH	Pilot flying	Pilot monitoring	Both	Area of responsibility
LANDING	Pilot flying	Pilot monitoring	Both	Pilot flying
SHUTDOWN	Captain	First officer	Both	Area of responsibility
SECURE	Captain	First officer	Both	Area of responsibility

If the airplane configuration does not agree with the needed configuration:

- stop the checklist
- complete the respective procedure steps
- continue the checklist

If it becomes apparent that a procedure was not completed:

- stop the checklist
- complete the procedure
- do the checklist from the start

Try to do checklists before or after high work load times. The flight crew may need to stop a checklist for a short time to do other tasks. If the interruption is short, continue the checklist with the next step. If a pilot is not sure where the checklist was stopped, do the checklist from the start. If the checklist is stopped for a long time, also do the checklist from the start.

After completion of each checklist, the pilot reading the checklist calls, "____ CHECKLIST COMPLETE."

Checklist Content

The checklist has the minimum items needed to operate the airplane safely.

Normal checklists have items that meet any of the following criteria:

- items essential to safety of flight that are not monitored by an alerting system, or
- items essential to safety of flight that are monitored by an alerting system but if not done, would likely result in a catastrophic event if the alerting system fails, or
- items needed to meet regulatory requirements, or
- items needed to maintain fleet commonality between the 737, 747, 757, 767, 777, and 787, or
- items that enhance safety of flight and are not monitored by an alerting system (for example the autobrake), or
- during shutdown and secure, items that could result in injury to personnel or damage to equipment if not done.

Checklist Construction

When a checklist challenge does not end with "switch" or "lever", then the challenge refers to system status. For example, "Landing Gear...Down" refers to the status of the landing gear, not just the position of the lever.

When a checklist challenge ends with "switch" or "lever", then the challenge refers to the position of the switch or lever. For example, "Engine start levers...CUTOFF" refers to the position of the levers.

Because normal checklists are done routinely, some checklist items are simplified to be more conversational such as "Autobrake.....RTO" instead of "AUTOBRAKE select switch.....RTO".



Checklist Instructions

Non-Normal Checklists

Chapter CI

Section 2

Introduction

The non-normal checklists chapter contains checklists used by the flight crew to manage non-normal situations. The checklists are grouped in sections which match the system description chapters in Volume 2.

Most checklists correspond to a light, alert or other indication. In most cases, the MASTER CAUTION and system annunciator lights also illuminate to indicate the non-normal condition. These lights, alerts and other indications are the cues to select and do the associated checklist.

Checklists without a light, alert or other indication (such as Ditching) are called unannunciated checklists. Most unannunciated checklists are in the associated system section. For example, Fuel Leak Engine is in section 12, Fuel. Unannunciated checklists with no associated system are in section 0, Miscellaneous.

All checklists have condition statements. The condition statement briefly describes the situation that caused the light, alert or other indication. Unannunciated checklists also have condition statements to help in understanding the reason for the checklist.

Some checklists have objective statements. The objective statement briefly describes the expected result of doing the checklist or briefly describes the reason for steps in the checklist.

Checklists can have both memory and reference items. Memory items are critical steps that must be done before reading the checklist. The last memory item is followed by a dashed horizontal line. Reference items are actions to be done while reading the checklist.

Some checklists have additional information at the end of the checklist. The additional information provides data the flight crew may wish to consider. The additional information does not need to be read.

Checklists that need a quick response are listed in the Quick Action Index. In each system section, Quick Action Index checklists are listed first, followed by checklists that are not in the Quick Action Index. The titles of Quick Action Index checklists are printed in **bold** type. Checklist titles in upper case (such as AUTO BRAKE DISARM) are annunciated by a light, alert, or other indication. Checklist titles in upper and lower case (such as Window Damage) are not annunciated.

Non-Normal Checklist Operation

Non-normal checklists start with steps to correct the situation. If needed, information for planning the rest of the flight is included. When special items are needed to configure the airplane for landing, the items are included in the Deferred Items section of the checklist. Flight patterns for some engine-out situations are located in the Maneuvers chapter and show the sequence of configuration changes.

While every attempt is made to supply needed non-normal checklists, it is not possible to develop checklists for all conceivable situations. In some smoke, fire or fumes situations, the flight crew may need to move between the Smoke, Fire or Fumes checklist and the Smoke or Fumes Removal checklist. In some multiple failure situations, the flight crew may need to perform several checklists or combine the elements of more than one checklist. Consider doing memory items first followed by reference steps. Upon completion of a non-normal checklist, review all warning lights, caution lights and other alerts to determine the need to do other non-normal checklists. In all situations, the captain must assess the situation and use good judgment to determine the safest course of action.

It should be noted that, in determining the safest course of action, troubleshooting (taking steps beyond published non-normal checklist steps) may cause further loss of system function or system failure. Troubleshooting should only be considered when completion of the published non-normal checklist results in an unacceptable situation.

There are some situations where the flight crew must land at the nearest suitable airport. These situations include, but are not limited to, conditions where:

- the non-normal checklist includes the item “Plan to land at the nearest suitable airport.”
- fire or smoke continues
- only one AC power source remains (engine or APU generator)
- only one hydraulic system remains (the standby system is considered a hydraulic system)
- any other situation determined by the flight crew to have a significant adverse effect on safety if the flight is continued.

It must be stressed that for smoke that continues or a fire that cannot be positively confirmed to be completely extinguished, the earliest possible descent, landing, and evacuation must be done.

If a smoke, fire or fumes situation becomes uncontrollable, the flight crew should consider an immediate landing. Immediate landing implies immediate diversion to a runway. However, in a severe situation, the flight crew should consider an overweight landing, a tailwind landing, an off-airport landing, or a ditching.

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Checklists directing an engine shutdown must be evaluated by the captain to determine whether an actual shutdown or operation at reduced thrust is the safest course of action. Consideration must be given to the probable effects of running the engine at reduced thrust.

There are no non-normal checklists for the loss of an engine indication or automatic display of the secondary engine indications. Automatic display of secondary engine parameters due to engine oil quantity is for flight crew awareness and does not require pilot action. Continue normal engine operation unless a limit is exceeded.

Non-normal checklists also assume:

- During engine start and before takeoff, the associated non-normal checklist is done if a non-normal situation is identified. After completion of the checklist, the Dispatch Deviations Guide (DDG) or operator equivalent is consulted to determine if Minimum Equipment List (MEL) dispatch relief is available.
- System controls are in the normal configuration for the phase of flight before the start of the non-normal checklist.
- If the MASTER CAUTION and system annunciator lights illuminate, all related amber lights are reviewed to assist in recognizing the cause(s) of the alert.
- Aural alerts are silenced and the master caution system is reset by the flight crew as soon as the cause of the alert is recognized.
- The EMERGENCY position of the oxygen regulator is used when needed to supply positive pressure in the masks and goggles to remove contaminants. The 100% position of the oxygen regulator is used when positive pressure is not needed but contamination of the flight deck air exists. The Normal position of the oxygen regulator is used if prolonged use is needed and the situation allows. Normal boom microphone operation is restored when oxygen is no longer in use.
- Indicator lights are tested to verify suspected faults.

VP-BSA, VP-BSQ - VP-BZY

- In flight, reset of a tripped circuit breaker is not recommended. However, a tripped circuit breaker may be reset once, after a short cooling period (approximately 2 minutes), if in the judgment of the captain, the situation resulting from the circuit breaker trip has a significant adverse effect on safety. On the ground, flight crew reset of a tripped circuit breaker should only be done after maintenance has determined that it is safe to reset the circuit breaker.

VP-BDP - VP-BPY, VP-BSC - VP-BSP, VQ-BDC - VQ-BVY

- In flight, reset of a tripped circuit breaker is not recommended unless directed by a non-normal checklist. However, a tripped circuit breaker may be reset once, after a short cooling period (approximately 2 minutes), if in the judgment of the captain, the situation resulting from the circuit breaker trip has a significant adverse effect on safety. On the ground, flight crew reset of a tripped circuit breaker should only be done after maintenance has determined that it is safe to reset the circuit breaker.
- Flight crew cycling (pulling and resetting) of a circuit breaker to clear a non-normal condition is not recommended, unless directed by a non-normal checklist.

After engine start and before takeoff, illumination of a red warning light, an amber caution light, an alert or other indication requires completion of the associated checklist. In certain cases, amber caution lights illuminate during MASTER CAUTION recall to inform the flight crew of the failure of one element in a system with redundant elements. If system operation is maintained by a second element, the amber caution light will extinguish when MASTER CAUTION is reset. In these situations, the amber caution light alerts the flight crew that normal system operation will be affected if another element fails. If an amber caution light illuminates during MASTER CAUTION recall, but extinguishes after MASTER CAUTION reset, completion of the associated checklist is not required.

Non-Normal Checklist Use

If a checklist or a step in a checklist is not applicable to all airplanes, airplane effectivity information is included in the checklist. Airplane effectivity can be listed by airplane number, registry number, serial number or tabulation number. If a checklist is applicable to some but not all airplanes, airplane effectivity is centered below the checklist title. If a step in a checklist is applicable to some but not all airplanes, airplane effectivity is included above the step. If a checklist or a step in a checklist is applicable to all airplanes, airplane effectivity information is not included.

Non-normal checklist use starts when the airplane flight path and configuration are correctly established. Only a few situations need an immediate response (such as CABIN ALTITUDE WARNING or Rapid Depressurization). Usually, time is available to assess the situation before corrective action is started. All actions must then be coordinated under the captain's supervision and done in a deliberate, systematic manner. Flight path control must never be compromised.

When a non-normal situation occurs, at the direction of the pilot flying, both crewmembers do all memory items in their areas of responsibility without delay.

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The pilot flying calls for the checklist when:

- the flight path is under control
- the airplane is not in a critical phase of flight (such as takeoff or landing)
- all memory items are complete.

The pilot monitoring reads aloud:

- the checklist title
- the airplane effectivity (if applicable) as needed to verify the correct checklist
- as much of the condition statement as needed to verify that the correct checklist has been selected
- as much of the objective statement (if applicable) as needed to understand the expected result of doing the checklist.

The pilot flying does not need to repeat this information but must acknowledge that the information was heard and understood.

For checklists with memory items, the pilot monitoring first verifies that each memory item has been done. The checklist is normally read aloud during this verification. The pilot flying does not need to respond except for items that are not in agreement with the checklist. The item numbers do not need to be read.

Non-memory items are called reference items. The pilot monitoring reads aloud the reference items, including:

- the precaution (if any)
- the response or action
- any amplifying information.

The pilot flying does not need to repeat this information but must acknowledge that the information was heard and understood. The item numbers do not need to be read. It is recommended that the flight crew read aloud all the choices in a "Choose one" step before taking action, to reduce the chance of error.

The word "Confirm" is added to checklist items when both flight crewmembers must verbally agree before action is taken. During an inflight non-normal situation, verbal confirmation is required for:

- an engine thrust lever
- an engine start lever
- an engine, APU or cargo fire switch
- a generator drive disconnect switch
- an IRS mode selector, when only one IRS is failed
- a flight control switch

This does not apply to the Loss of Thrust on Both Engines checklist.

With the airplane stationary on the ground:

- the captain and the first officer take action based on preflight and postflight areas of responsibility
- during an evacuation, the first officer sets the flap lever to 40.

With the airplane in flight or in motion on the ground:

- the pilot flying and the pilot monitoring take action based on each crewmember's Areas of Responsibility.

After moving the control, the flight crewmember taking the action also states the checklist response.

The pilot flying may also direct reference checklists to be done by memory if no hazard is created by such action, or if the situation does not allow reference to the checklist.

Checklists include an Inoperative Items table only when the condition of the items is needed for planning the rest of the flight. The inoperative items, including the consequences (if any), are read aloud by the pilot monitoring. The pilot flying does not need to repeat this information but must acknowledge that the information was heard and understood.

After completion of the non-normal checklist, normal procedures are used to configure the airplane for each phase of flight.

When there are no deferred items, the DESCENT, APPROACH and LANDING normal checklists are used to verify that the configuration is correct for each phase of flight.

When there are deferred items, the non-normal checklist will include the item **"Checklist Complete Except Deferred Items."** The pilot flying is to be made aware when there are deferred items. These items are included in the Deferred Items section of the checklist and may be delayed until the usual point during descent, approach, or landing.

The deferred items are read aloud by the pilot monitoring. The pilot flying or the pilot monitoring takes action based on each crewmember's area of responsibility.

After moving the control, the flight crewmember taking the action also states the response.

The Deferred Items section of the non-normal checklist includes the Descent, Approach and Landing normal checklists if one or more of the following occur:

- an item is deferred until after the Descent Checklist
- a normal checklist response is changed as a result of the non-normal situation. (The changed response is printed in **bold** type.)

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Use these checklists instead of the usual DESCENT, APPROACH and LANDING normal checklists. The pilot flying or the pilot monitoring responds to the deferred normal checklist items based on each crewmember's area of responsibility. However, during the deferred Landing normal checklist, the pilot flying responds to all deferred normal checklist items.

Each checklist has a checklist complete symbol at the end.

The checklist complete symbol can also be in the body of the checklist. This only occurs when a checklist divides into two or more paths. Each path can have a checklist complete symbol at the end. The flight crew does not need to continue reading the checklist after the checklist complete symbol.

After completion of each non-normal checklist, the pilot monitoring states “ CHECKLIST COMPLETE.”

Additional information at the end of the checklist is not required to be read.

The flight crew must be aware that checklists cannot be created for all conceivable situations and are not intended to replace good judgment. In some situations, at the captain's discretion, deviation from a checklist can be needed.



Non-Normal Situation Guidelines Overview

This is a general overview of how non-normal situations should be conducted. Consider maximum use of autoflight system to reduce workload, if available and appropriate. If flight directors are selected on, ensure the proper flight director modes are selected.

Pilot Flying	Pilot Monitoring
Recognizes and announces the non-normal situation. Other pilot acknowledges.	
MAINTAIN AIRPLANE CONTROL Ensure that the flight path is under control.	Monitor the flight path.
	ANALYZE THE SITUATION Review all warning lights, caution lights and other alert lights to identify the non-normal situation. Prioritize alerts and recommend course of action.
Acknowledges and confirms the Pilot Monitoring’s identification or recommendation.	
TAKE THE PROPER ACTION Do the NNC memory items based on each crewmember’s area of responsibility.	
Call for the appropriate NNC.	
	Completes the NNC.
Review all warning lights, caution lights and other alerts, and do other NNCs as needed.	
EVALUATE THE NEED TO LAND Review options for diversion or continued flight.	

Non-Normal Checklist Legend

Redirection Symbol



- The redirection symbol is used in two ways:
- In the Table of Contents of a system section, to direct the flight crew to a different system section.
 - In a non-normal checklist, with the word “Go to”, to direct the flight crew to a different checklist or to a different step in the current checklist.

Separator Symbol



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The separator symbol is used in two ways:

- In the Table of Contents of a system section, to separate the Quick Action Index checklists from the checklists that are not in the Quick Action Index.
- In a non-normal checklist, to separate the memory items from the reference items.

Task Divider Symbol

The task divider symbol is used to indicate the end of one task and the beginning of another task.

Decision Symbol



The decision symbol is used to identify possible choices.

Precaution Symbol



The precaution symbol is used to identify information that the flight crew must consider before taking the action.

Checklist Complete Symbol



The checklist complete symbol indicates that the checklist is complete.

Intentionally
Blank



**Engine Fire on the Ground Checklist
is on the reverse side of this page.**



Engine Fire on the Ground

Condition: One or more of these occur on the ground:

- An engine fire is observed
- Engine fire warning

- 1 Thrust levers (both) Close C
- 2 PARKING BRAKE. Set C
- 3 Advise the cabin. C or F/O
- 4 Engine start lever (affected engine) CUTOFF C
- 5 Engine fire switch (affected engine) Pull F/O

To manually unlock the engine fire switch, press the override and pull.

- 6 Engine fire switch (affected engine) Rotate to the stop and hold for 1 second F/O

7 Choose one:

◆Evacuation is **needed**:

▶▶Go to the Evacuation checklist on page Back Cover.4

■ ■ ■ ■

◆Evacuation is **not** needed:

▶▶Go to step 8

- 8 Advise the cabin. C
- 9 Advise ATC. F/O





737 NNC

**Evacuation Checklist is on the
reverse side of this page.**

Evacuation

Condition: Evacuation is needed.

- 1

PARKING BRAKE. Set

C
- 2

Speedbrake lever DOWN

C
- 3

FLAP lever 40

F/O
- 4

Pressurization mode selector MAN

F/O
- 5

Outflow VALVE
switch Hold in OPEN
until the outflow VALVE
indication shows fully open
to depressurize the airplane

F/O
- 6

If time allows, verify that the flaps are 40
before the engine start levers are moved
to CUTOFF.

C
- 7

Engine start levers (both) CUTOFF

C
- 8

Advise the cabin to evacuate.

C
- 9

Advise ATC.

F/O
- 10

Engine and APU
fire switches (all) Override and pull

F/O
- 11

If an engine or APU fire is **observed or indicated**:
Related fire
switch Rotate to the stop
and hold for 1 second

F/O

